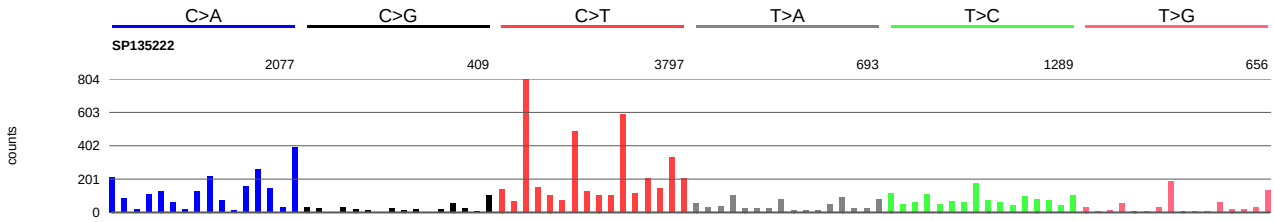
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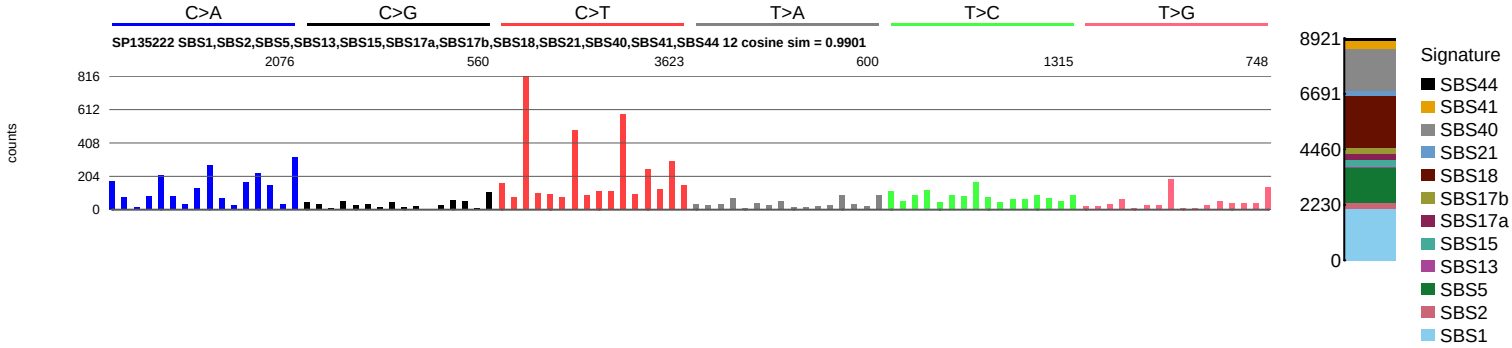
SP135222
2077
409
3797
693
1289
656
counts
804
603
402
201
0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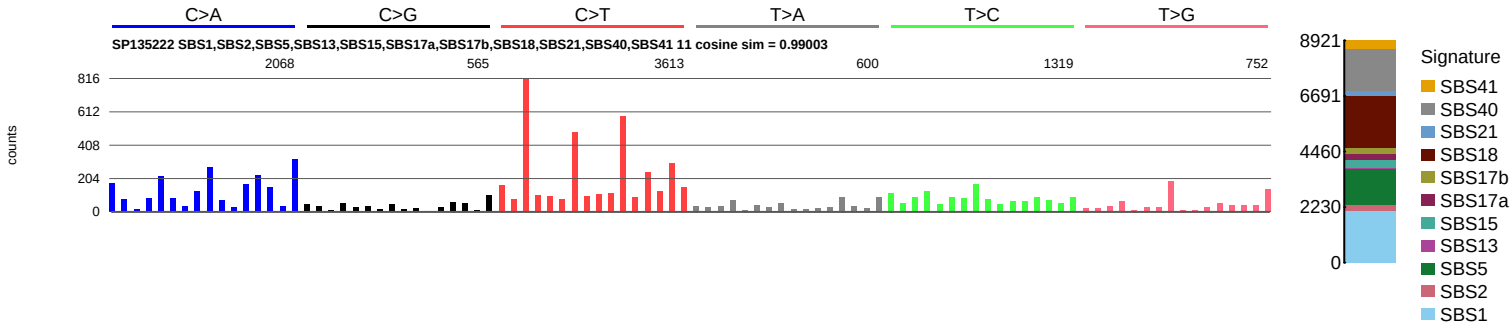
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.9901
2076
560
3623
600
1315
748
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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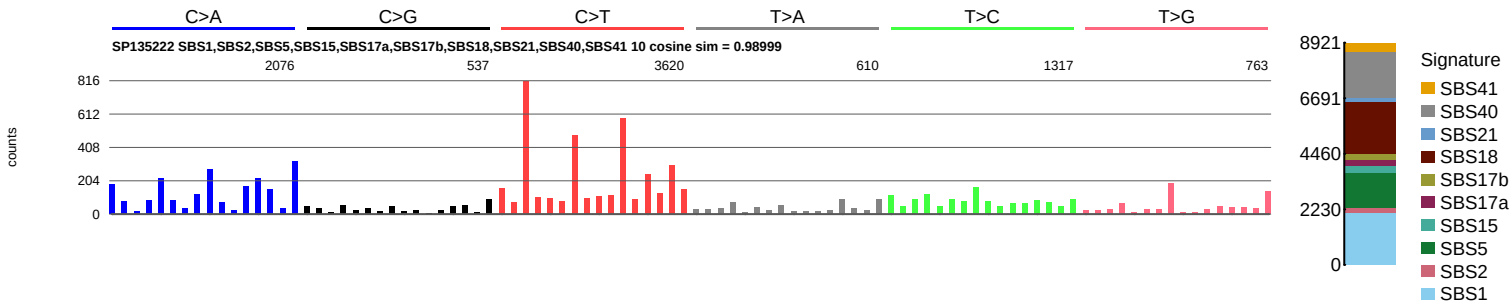
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.99003
2068
565
3613
600
1319
752
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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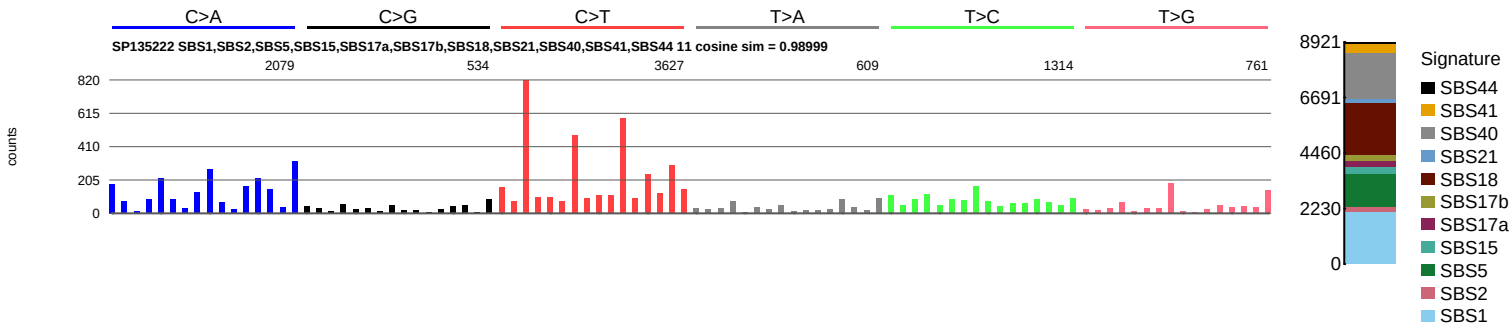
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98999
2076
537
3620
610
1317
763
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
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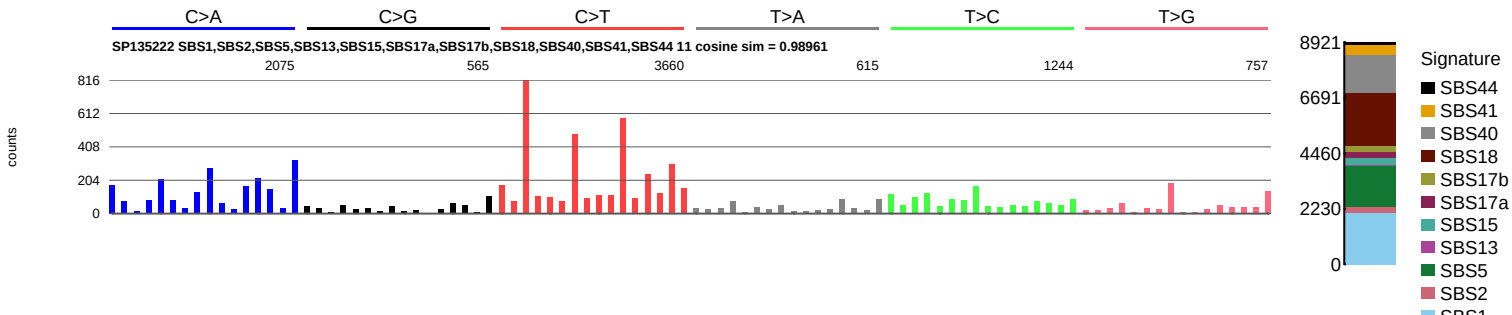
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98999
2079
534
3627
609
1314
761
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
C>T
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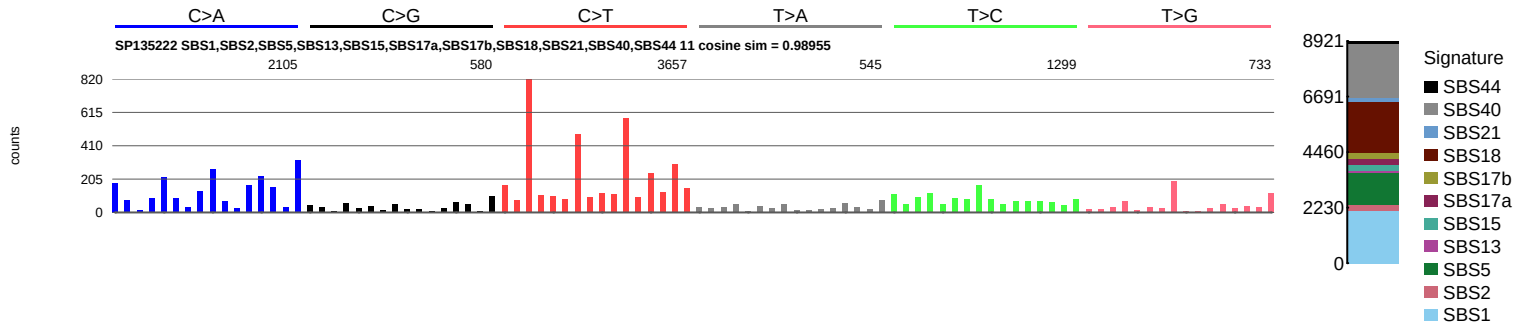
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98961
2075
565
3660
615
1244
757
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2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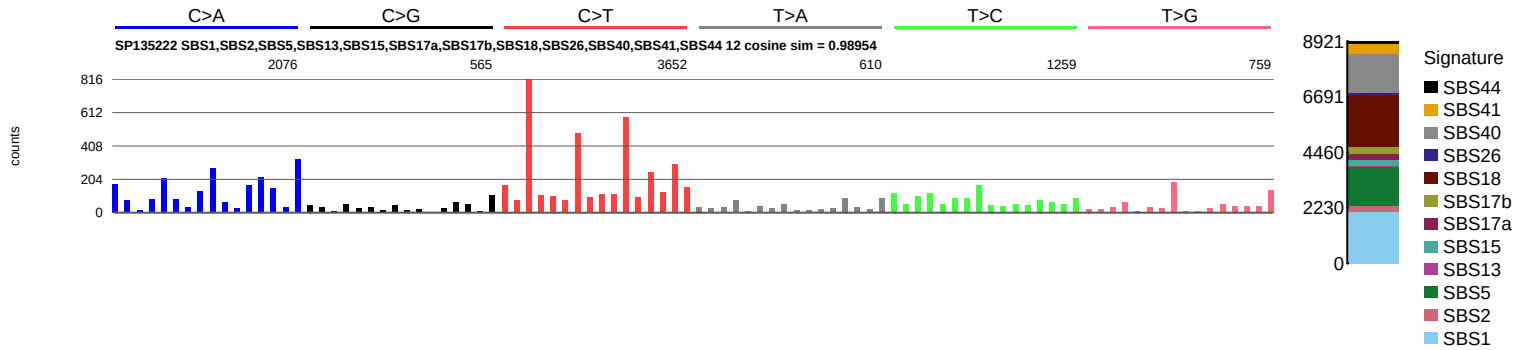
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98955
2105
580
3657
545
1299
733
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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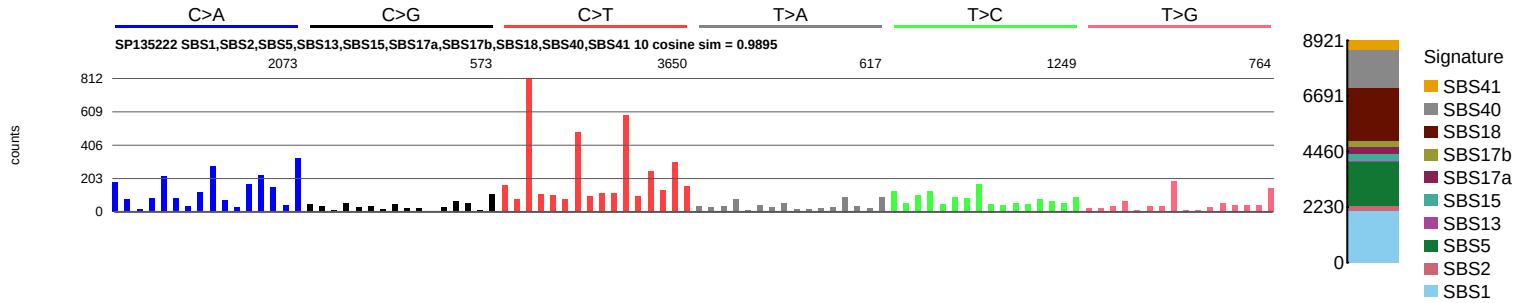
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98954
2076
565
3652
610
1259
759
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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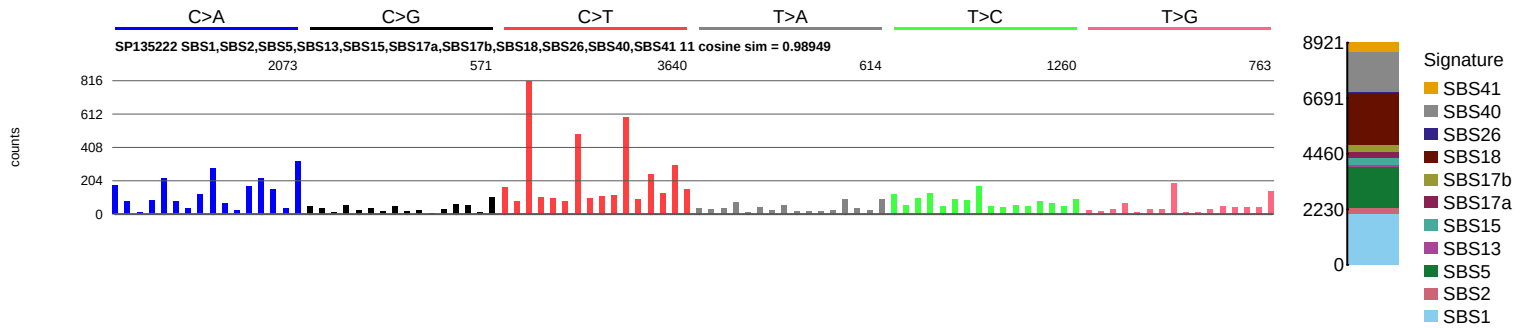
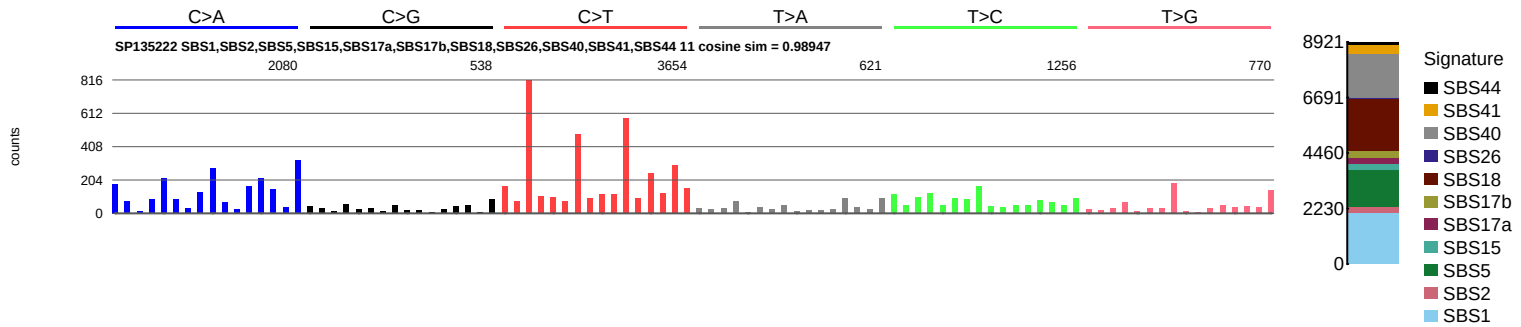
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.9895
2073
573
3650
617
1249
764
counts
812
609
406
203
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.98949
2073
571
3640
614
1260
763
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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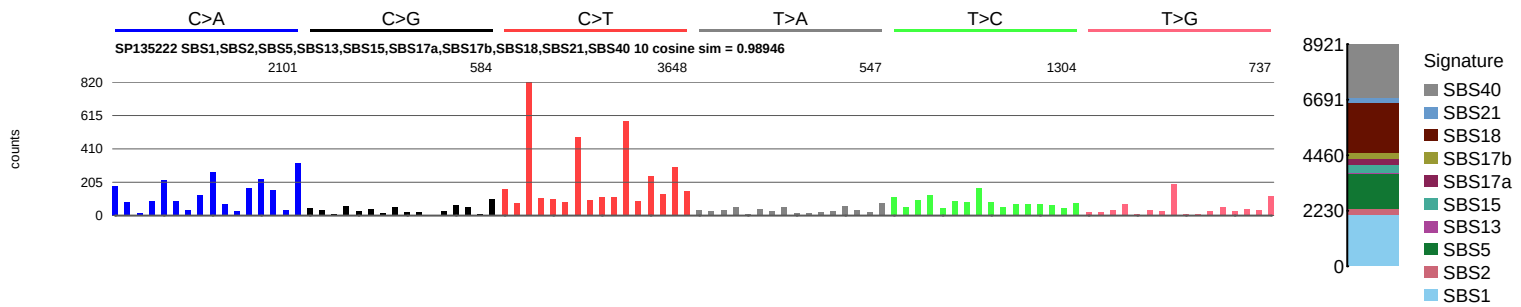
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98947
2080
538
3654
621
1256
770
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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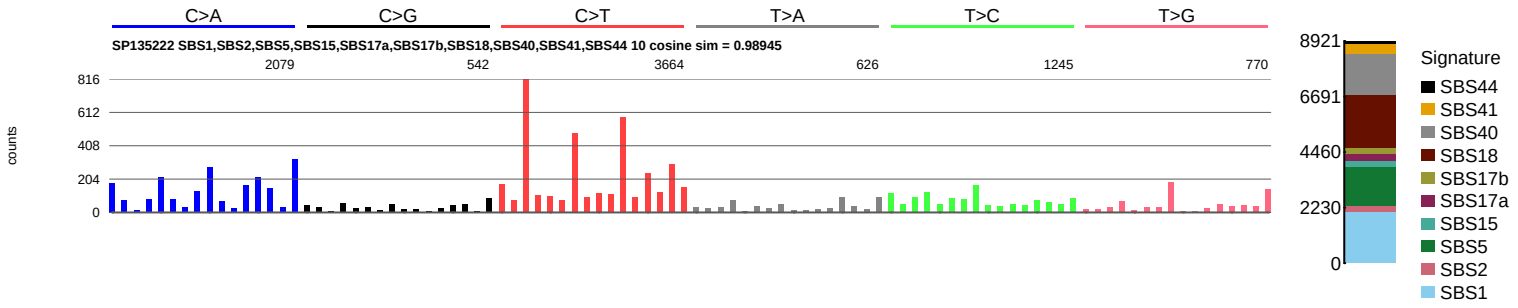
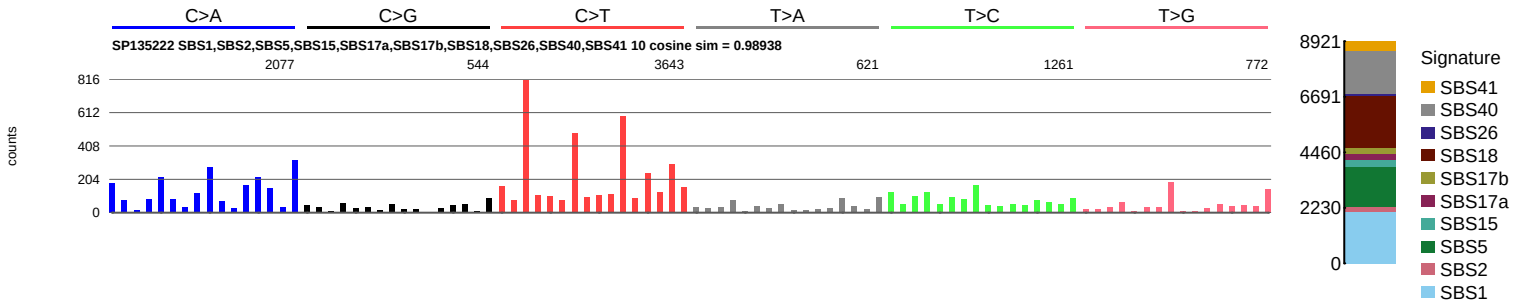
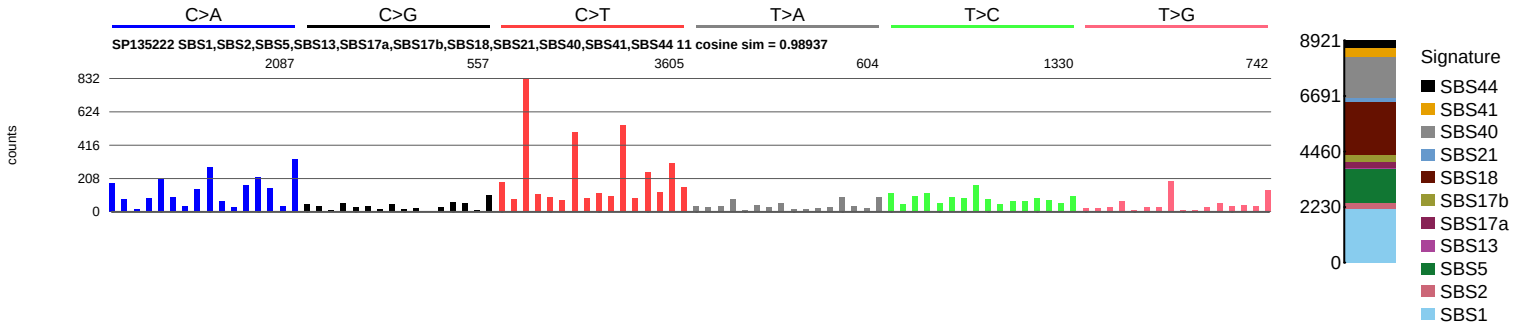
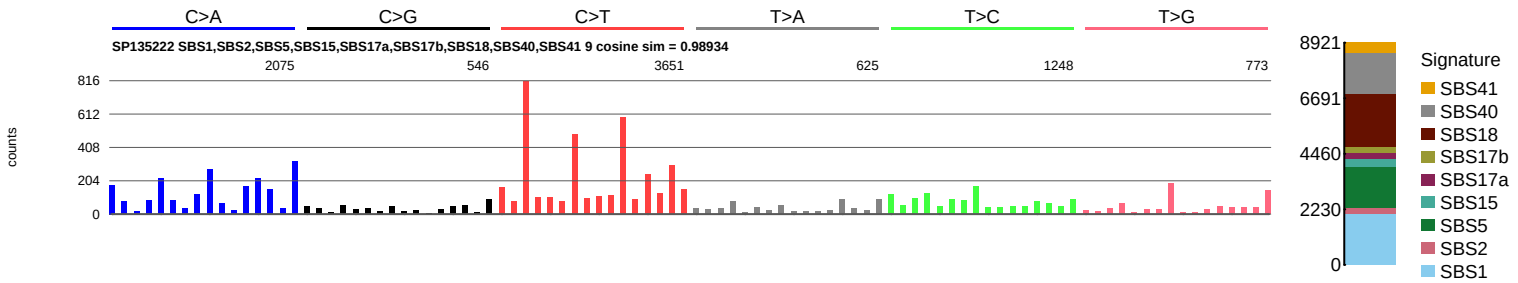
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98946
2101
584
3648
547
1304
737
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98945
2079
542
3664
626
1245
770
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98938
2077
544
3643
621
1261
772
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98937
2087
557
3605
604
1330
742
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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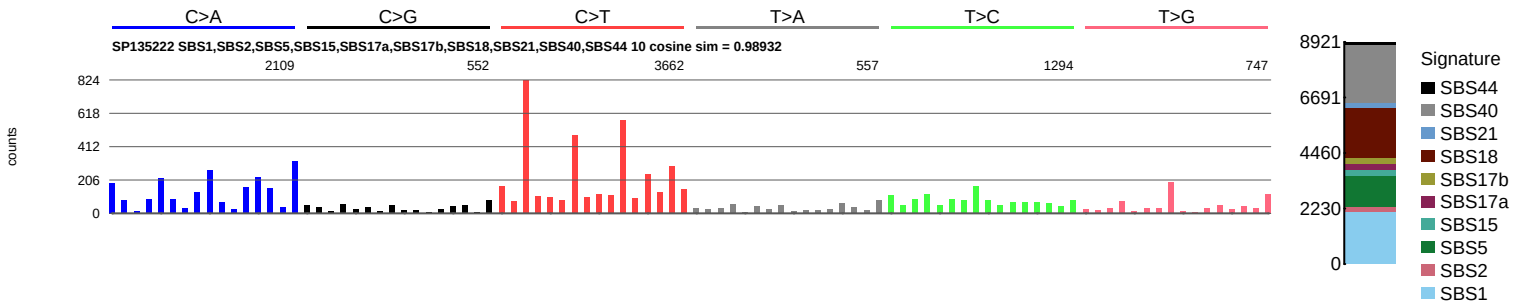
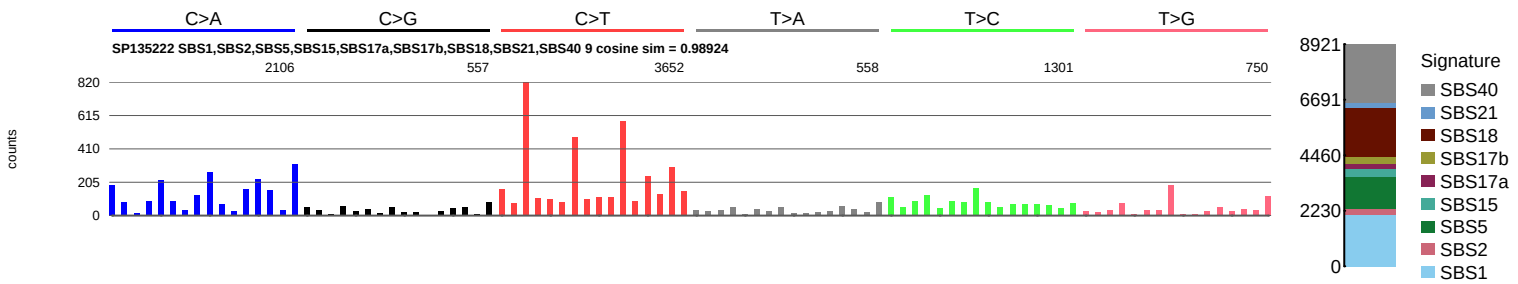
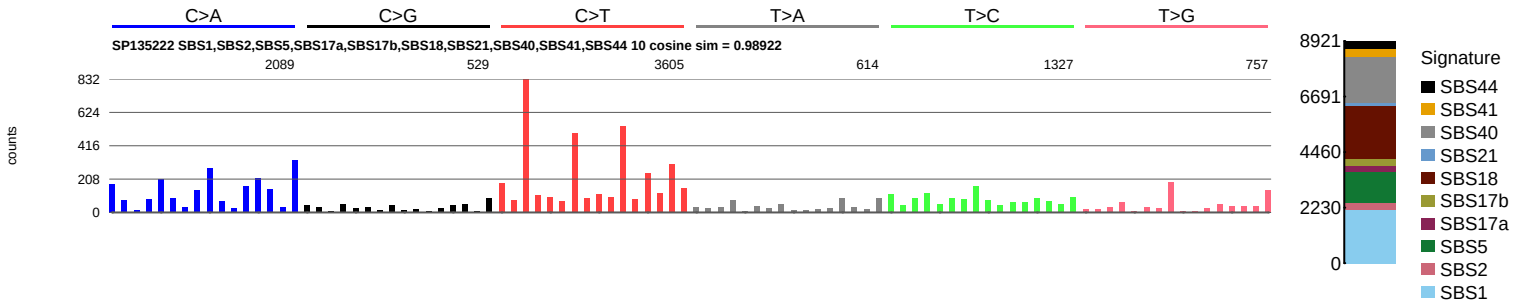
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98934
2075
546
3651
625
1248
773
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98932
2109
552
3662
557
1294
747
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98924
2106
557
3652
558
1301
750
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98922
2089
529
3605
614
1327
757
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
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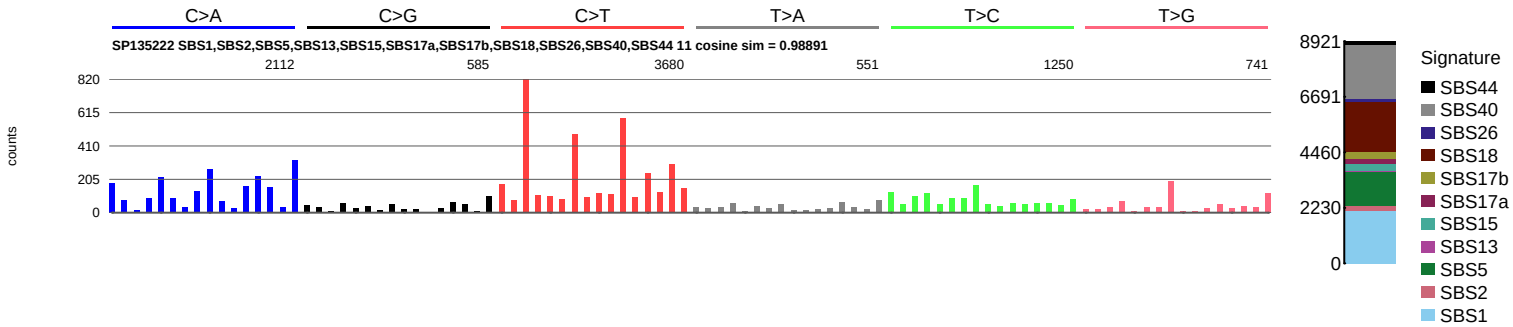
SBS2
SBS1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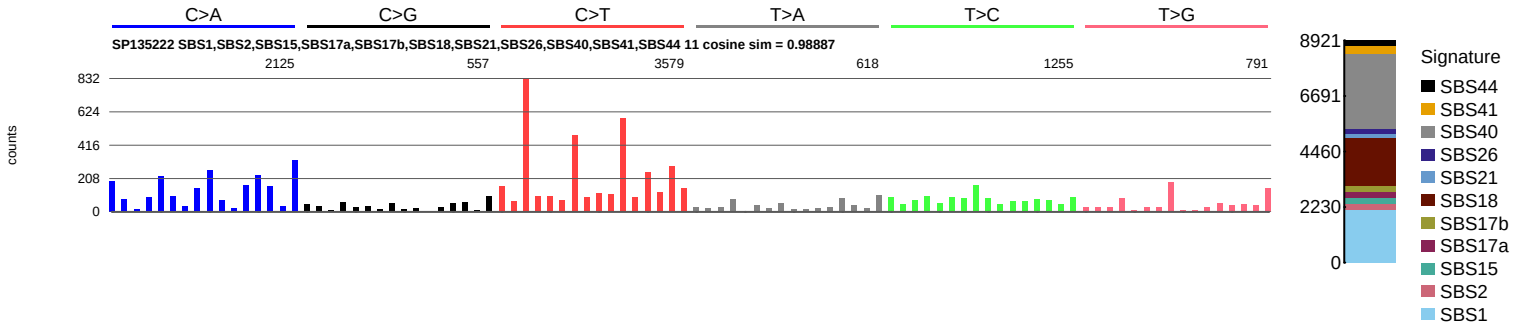
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98891
2112
585
3680
551
1250
741
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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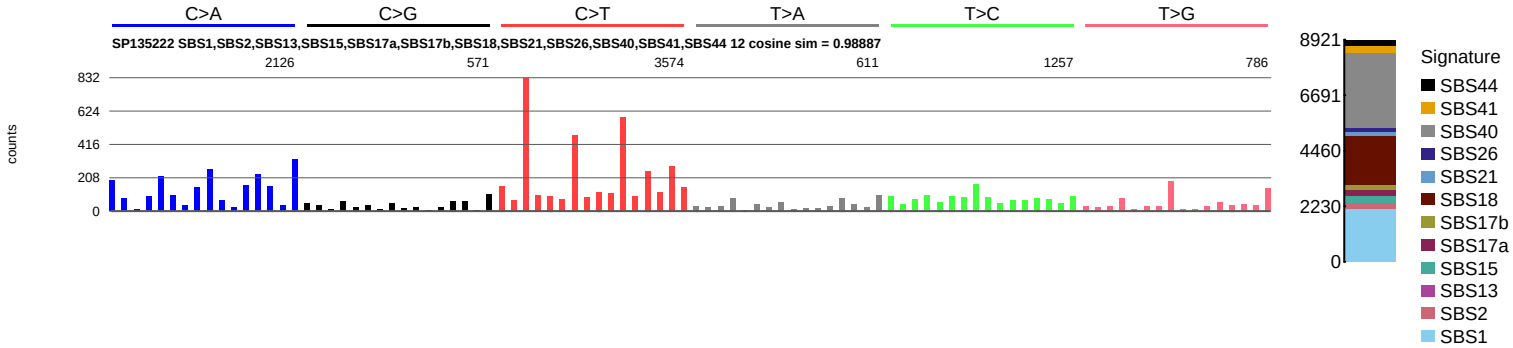
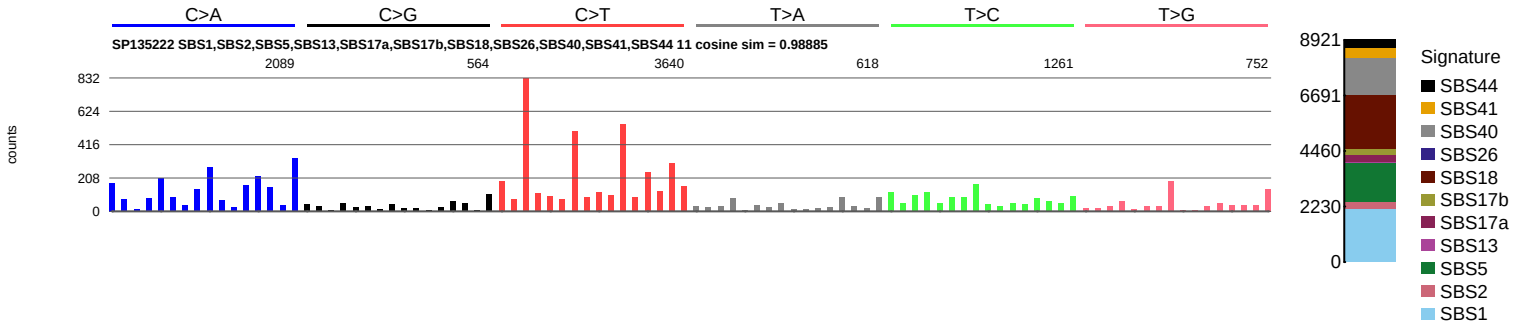
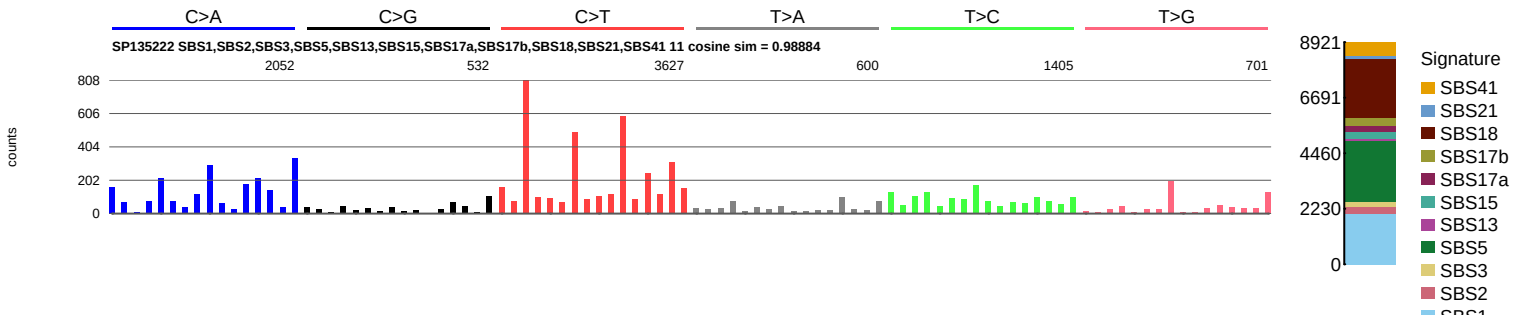
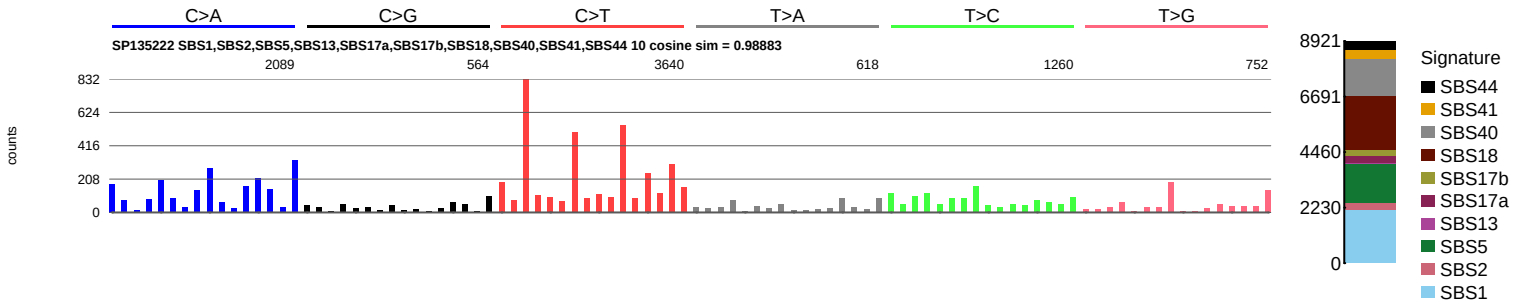
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98887
2125
557
3579
618
1255
791
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98887
2126
571
3574
611
1257
786
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98885
2089
564
3640
618
1261
752
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.98884
2052
532
3627
600
1405
701
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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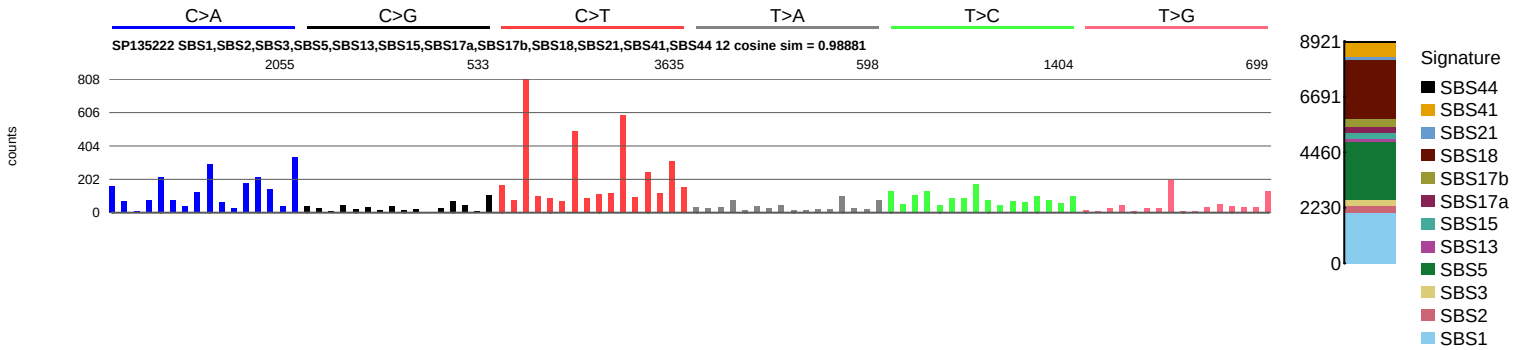
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98883
2089
564
3640
618
1260
752
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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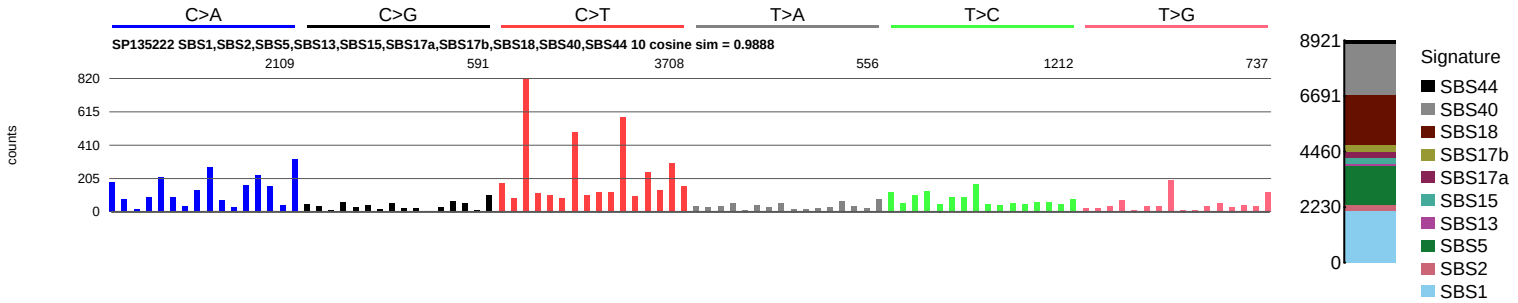
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 12 cosine sim = 0.98881
2055
533
3635
598
1404
699
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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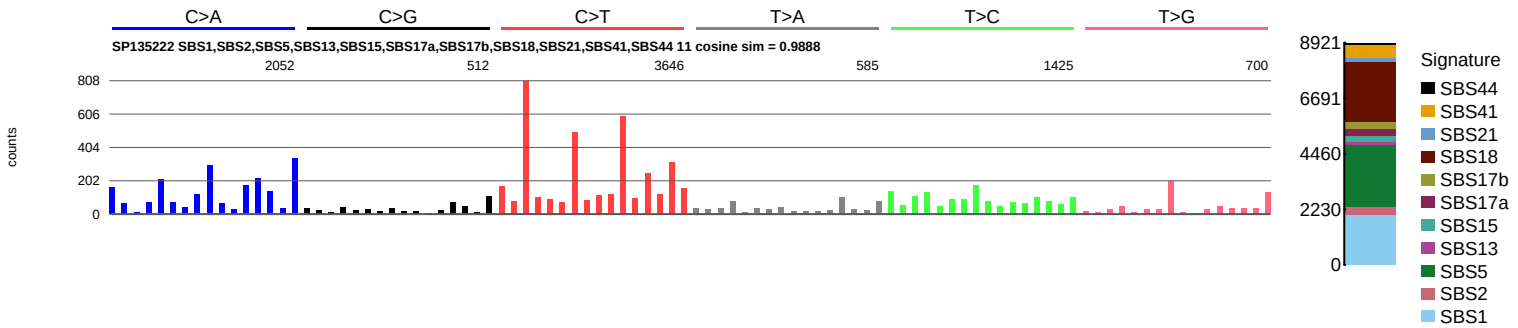
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.9888
2109
591
3708
556
1212
737
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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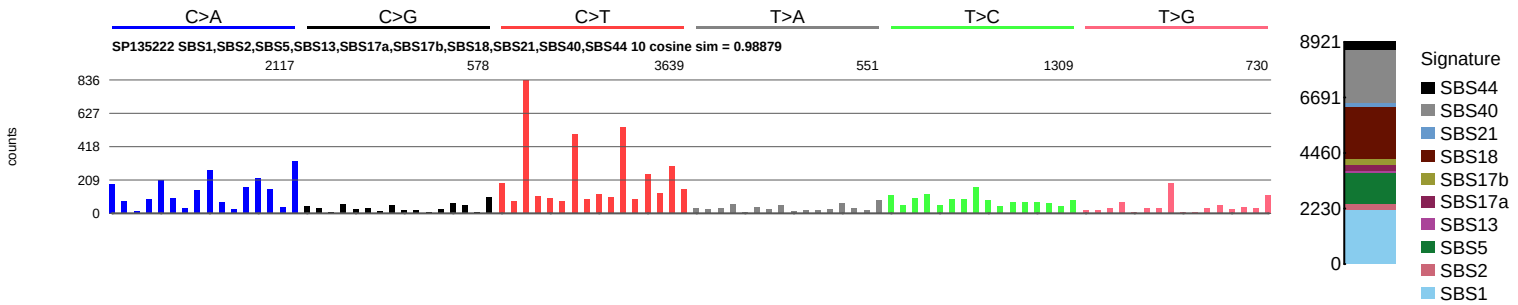
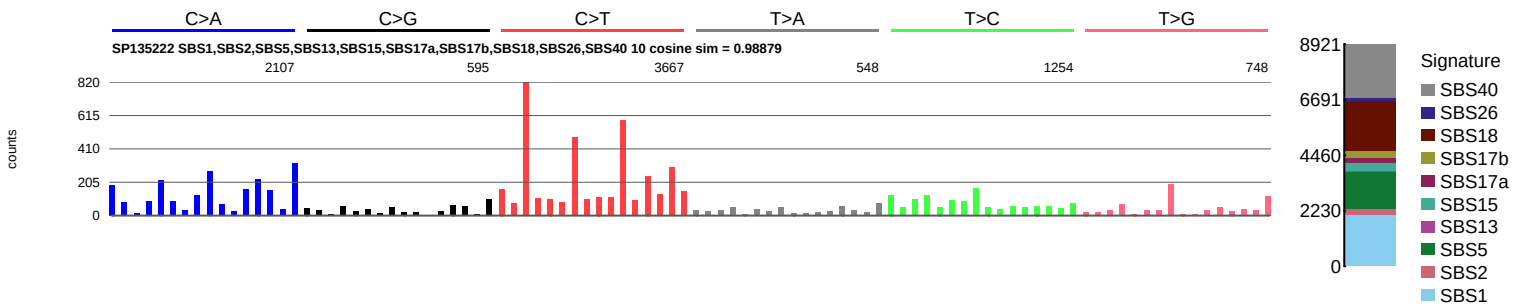
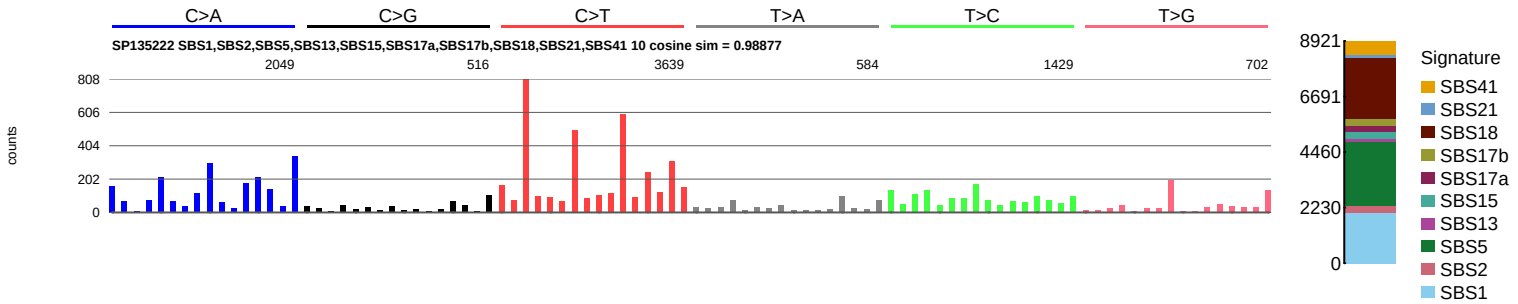
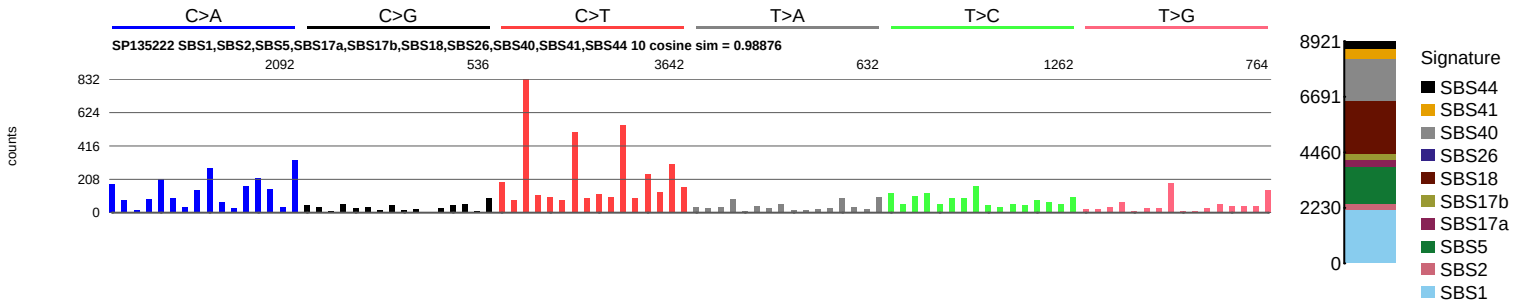
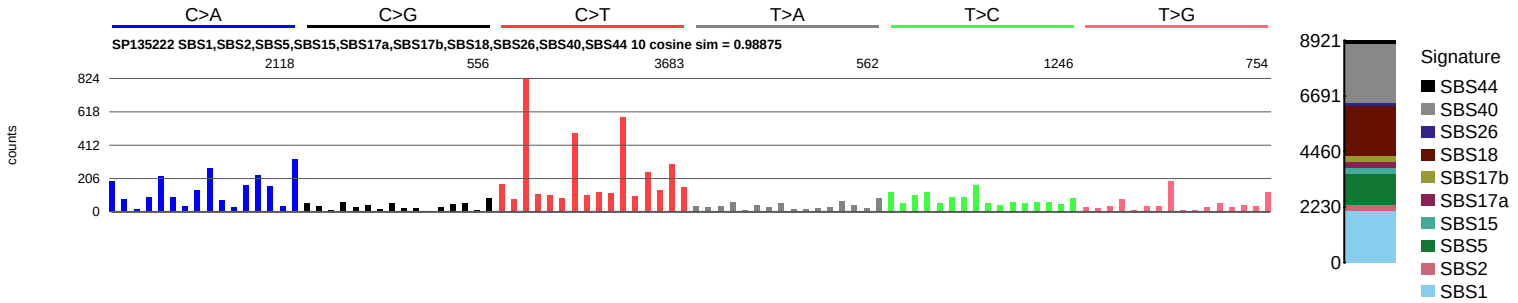
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.9888
2052
512
3646
585
1425
700
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

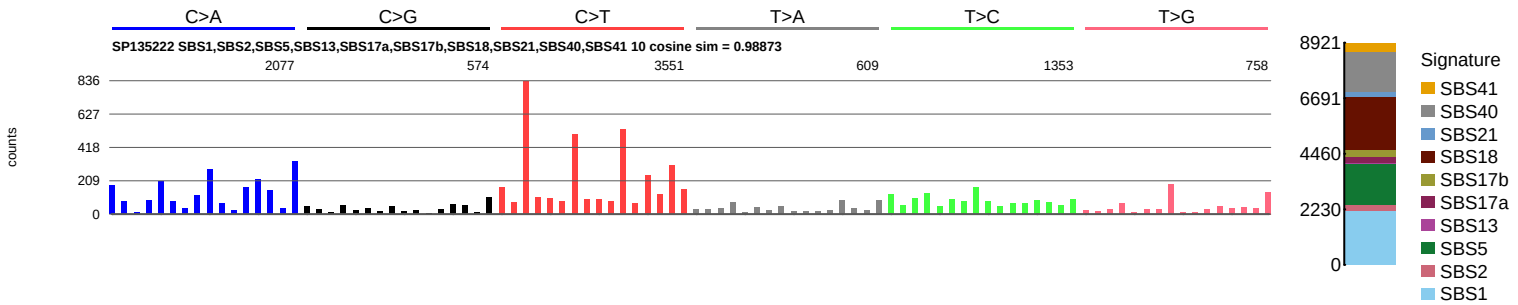

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98879
2117
578
3639
551
1309
730
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

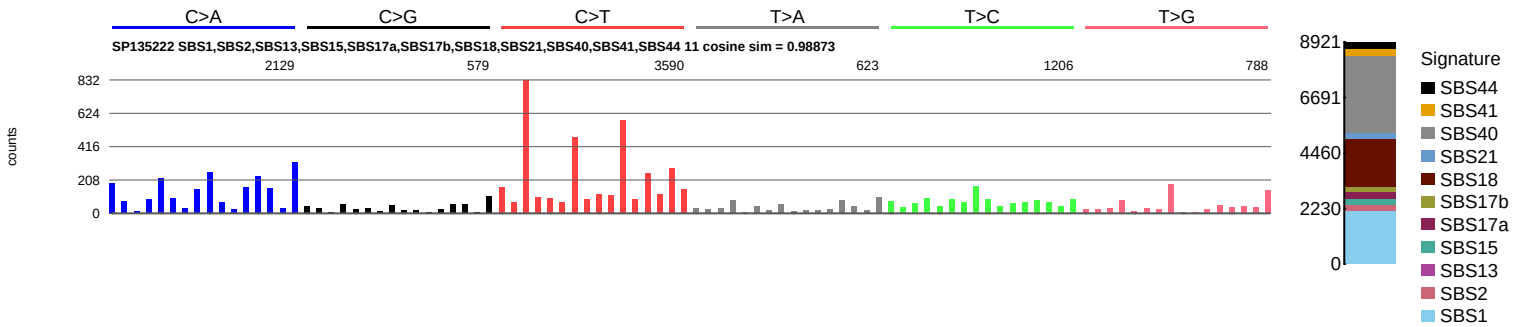

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98879
2107
595
3667
548
1254
748
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

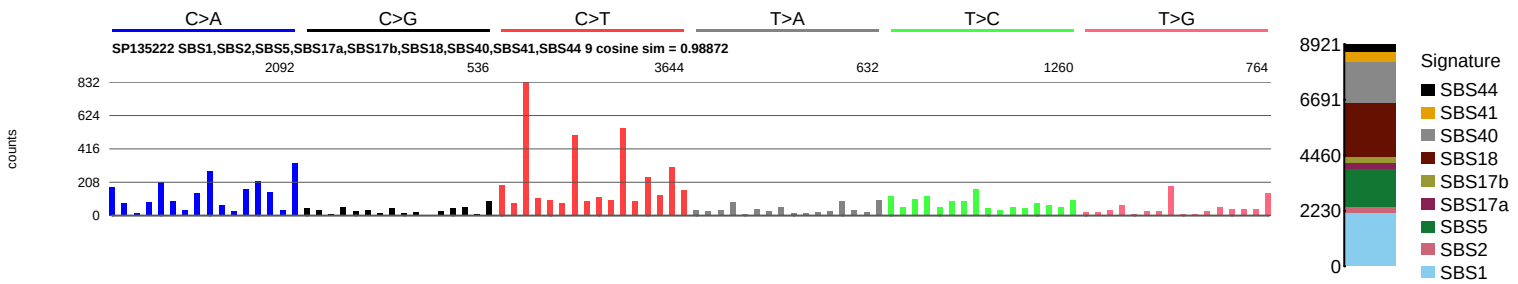

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98877
2049
516
3639
584
1429
702
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

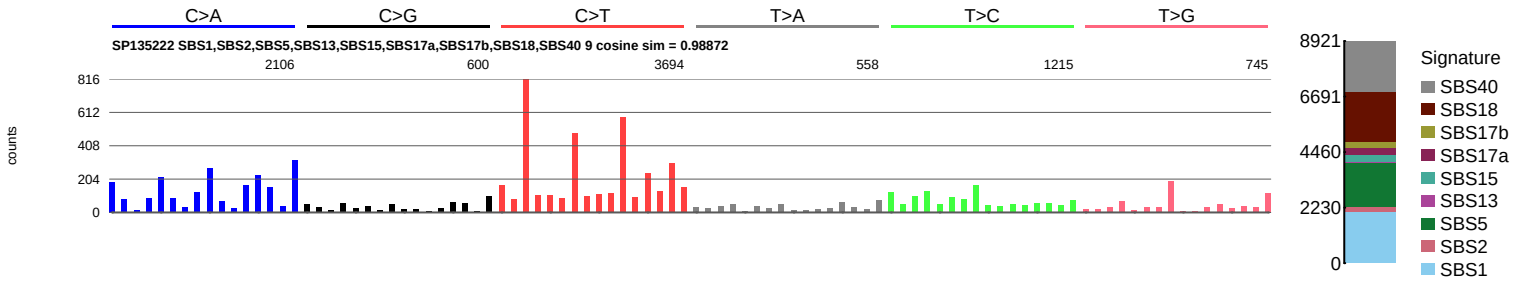

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98876
2092
536
3642
632
1262
764
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98875
2118
556
3683
562
1246
754
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98873
2077
574
3551
609
1353
758
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98873
2129
579
3590
623
1206
788
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98872
2092
536
3644
632
1260
764
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98872
2106
600
3694
558
1215
745
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
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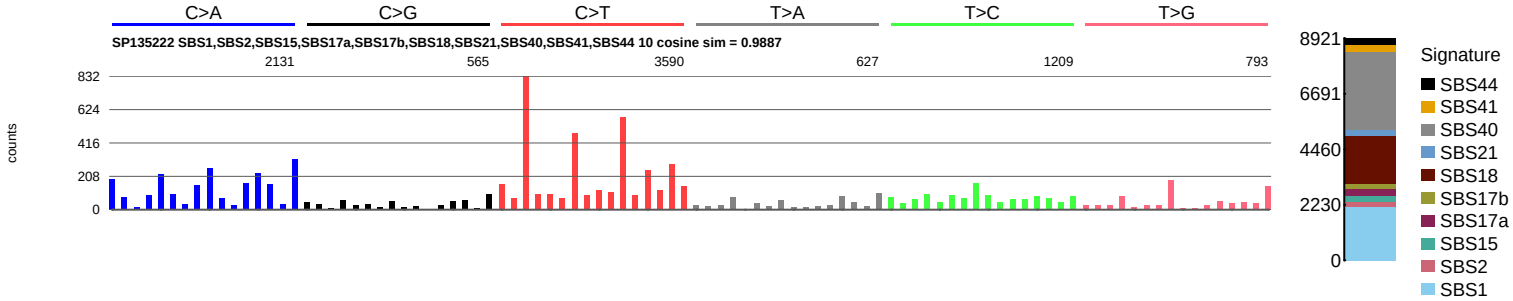
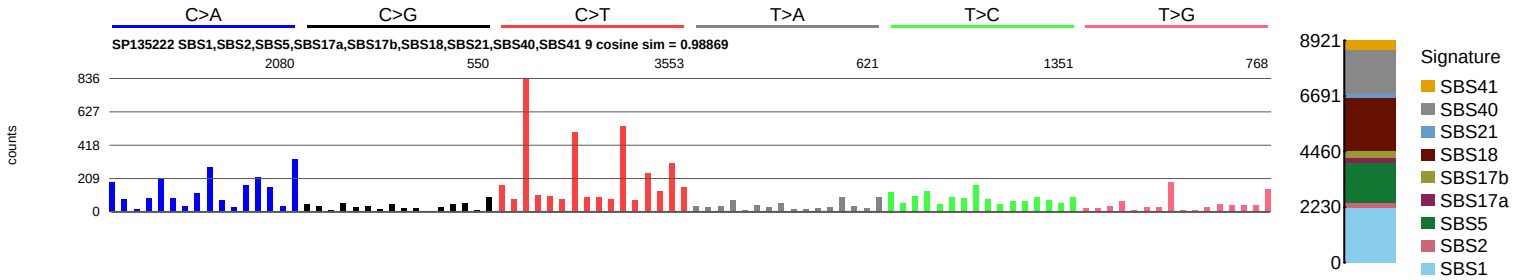
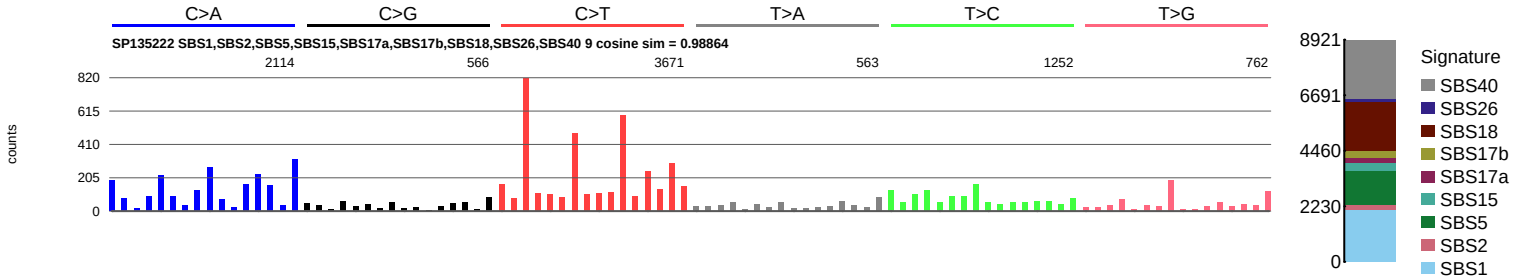
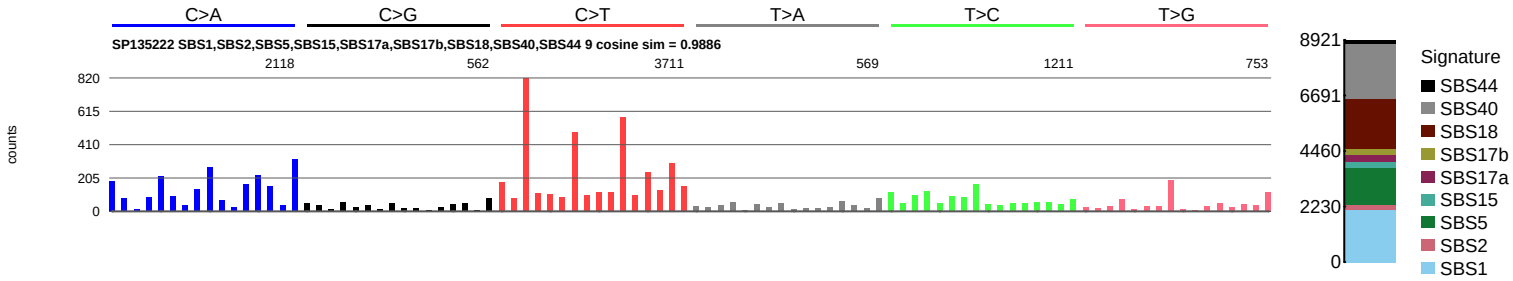
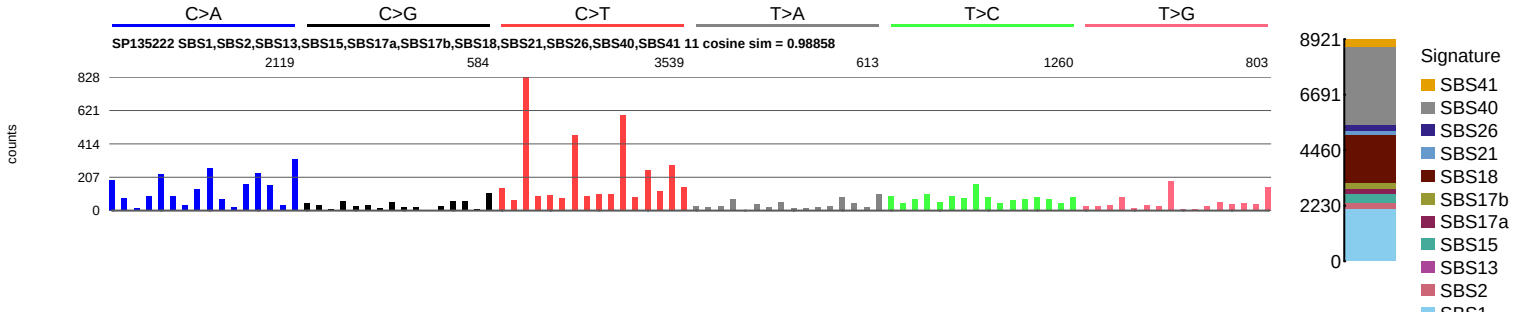
SBS5
SBS2
SBS1

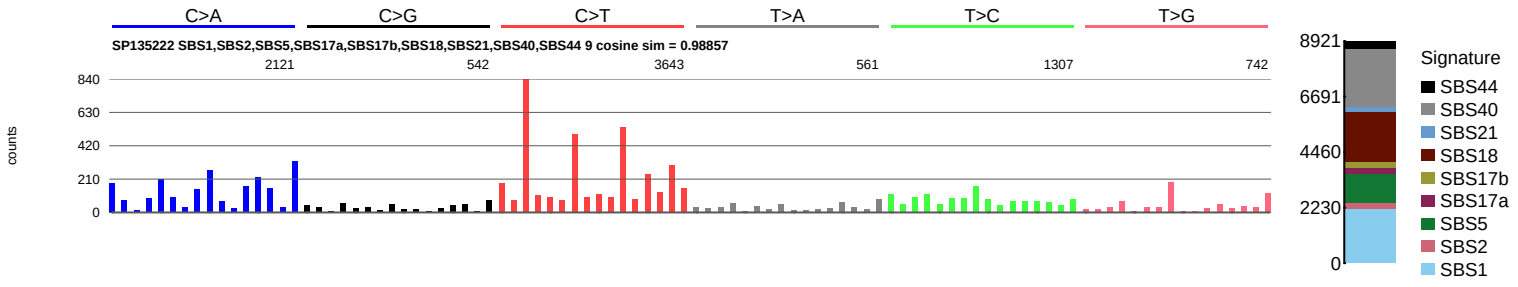

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.9887
2131
565
3590
627
1209
793
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98869
2080
550
3553
621
1351
768
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98864
2114
566
3671
563
1252
762
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.9886
2118
562
3711
569
1211
753
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98858
2119
584
3539
613
1260
803
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98857
2121
542
3643
561
1307
742
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
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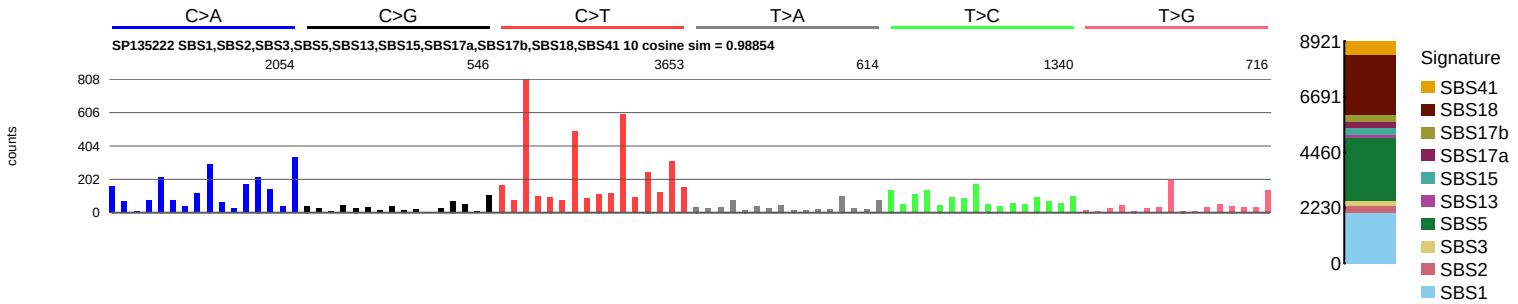
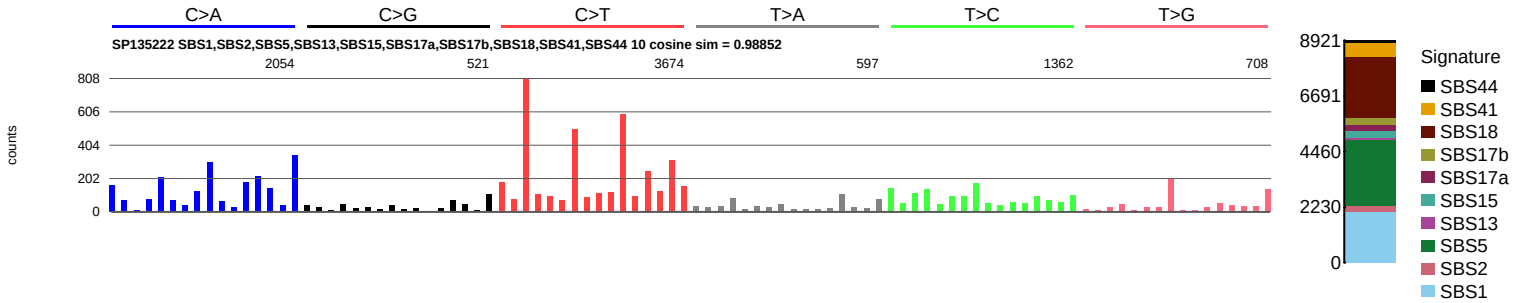
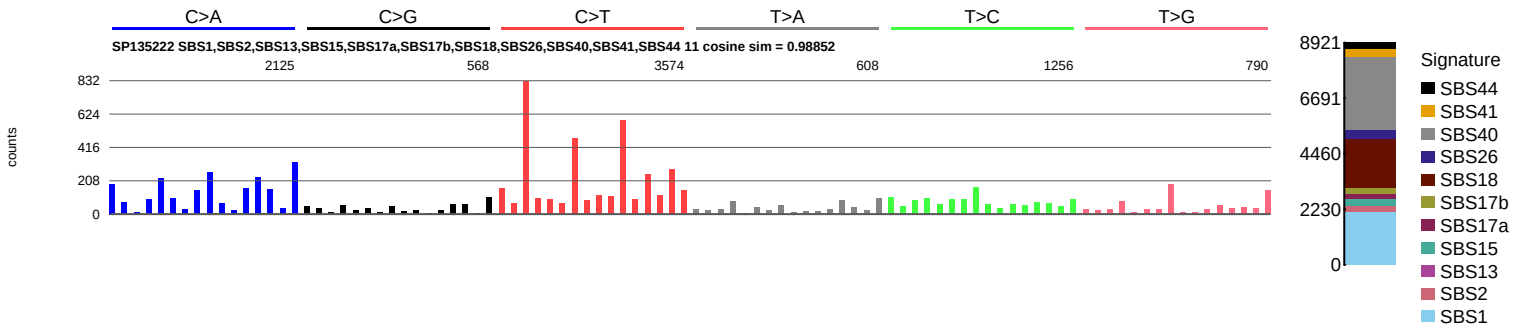
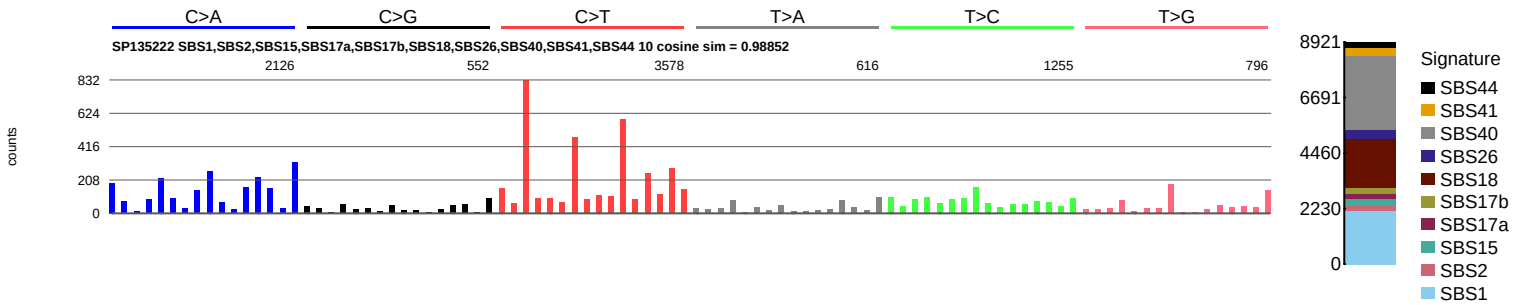
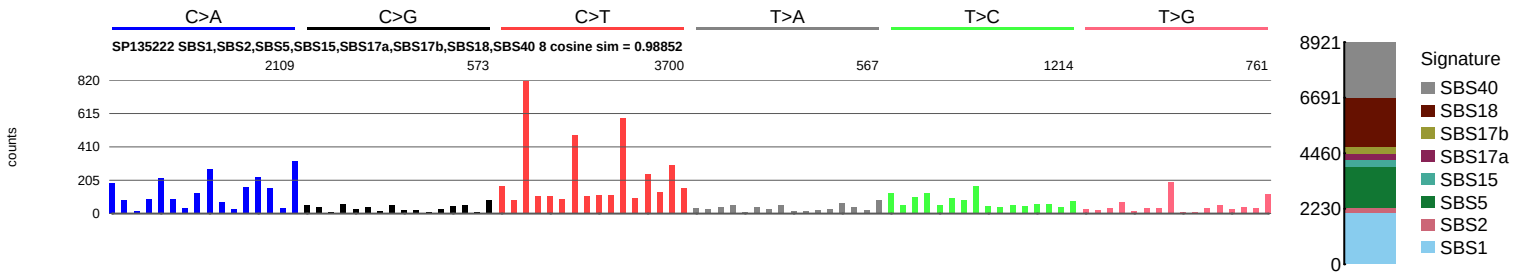
SBS5
SBS2
SBS1

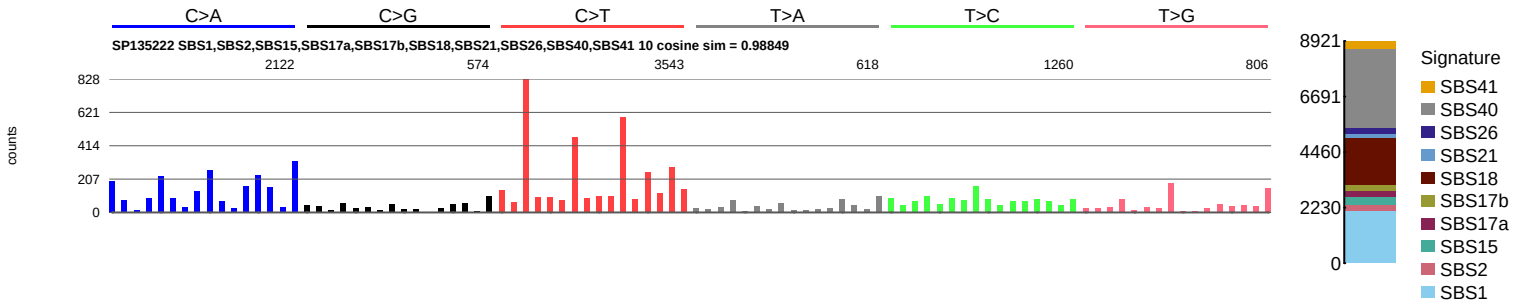

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.98854
2054
546
3653
614
1340
716
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98852
2054
521
3674
597
1362
708
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98852
2125
568
3574
608
1256
790
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98852
2126
552
3578
616
1255
796
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98852
2109
573
3700
567
1214
761
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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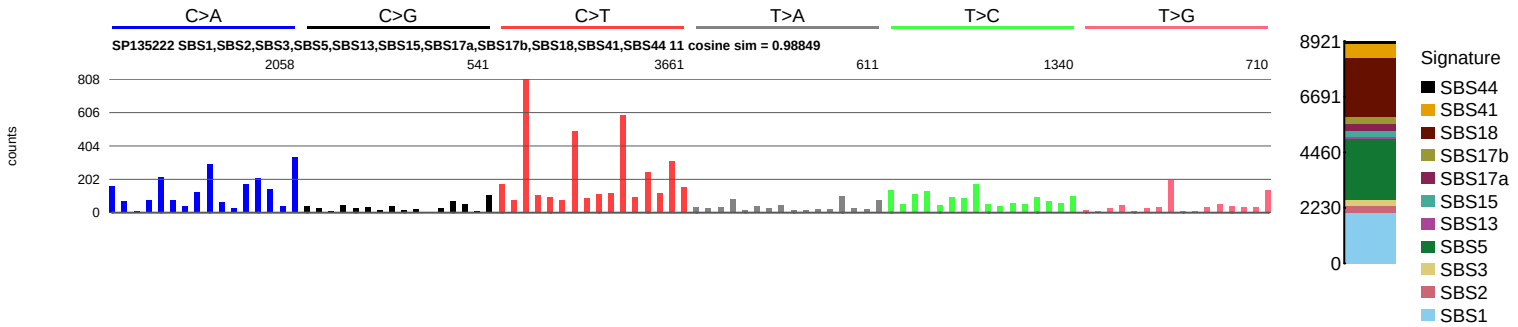
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98849
2122
574
3543
618
1260
806
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
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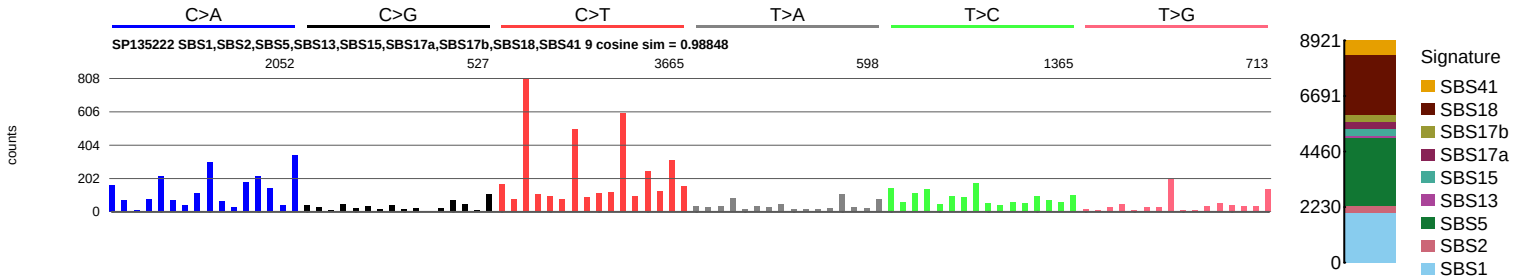
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 11 cosine sim = 0.98849
2058
541
3661
611
1340
710
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98848
2052
527
3665
598
1365
713
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
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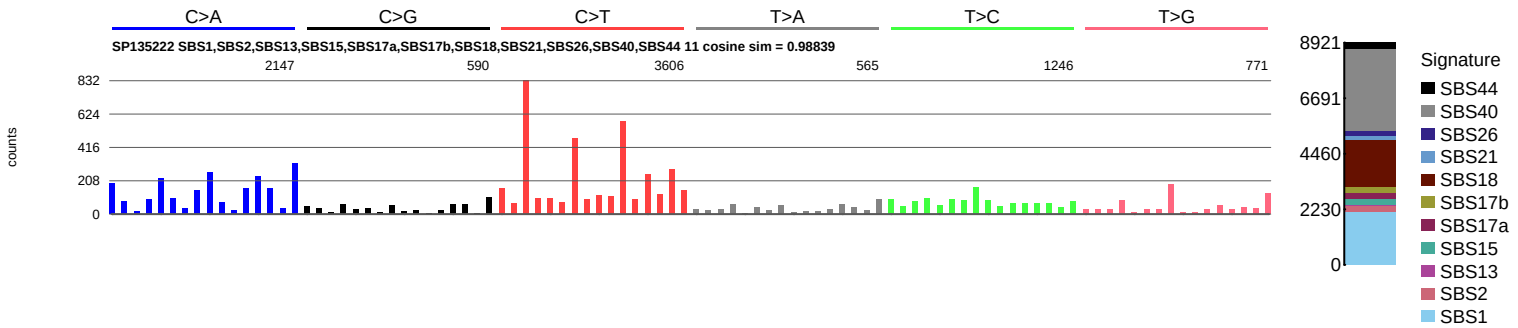
SBS2
SBS1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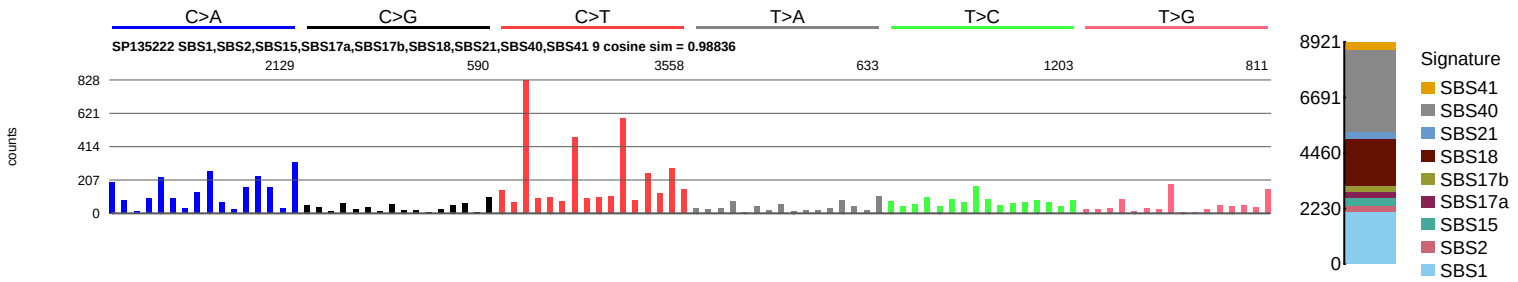
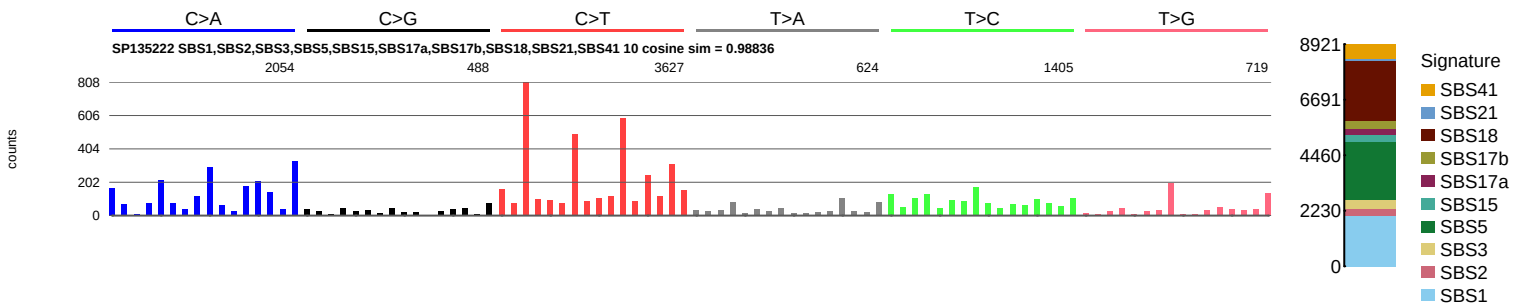
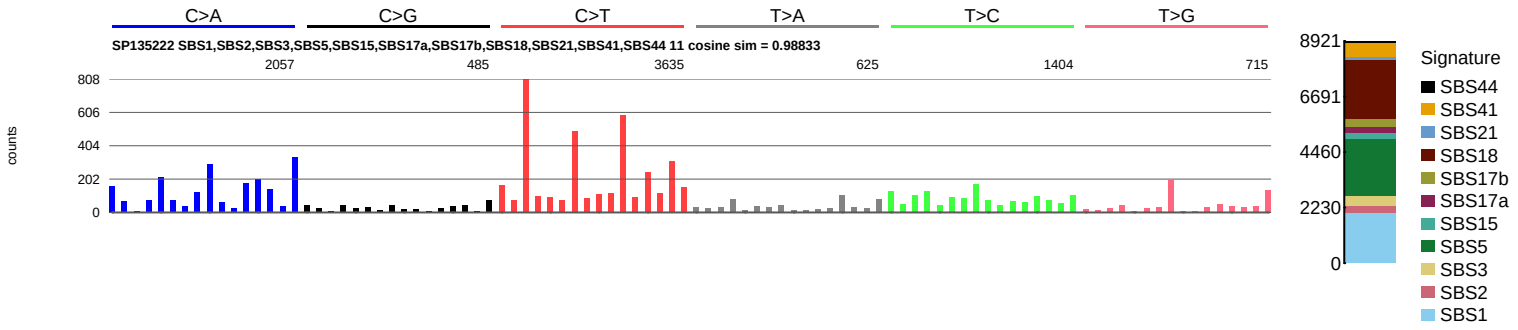
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98839
2147
590
3606
565
1246
771
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98836
2129
590
3558
633
1203
811
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98836
2054
488
3627
624
1405
719
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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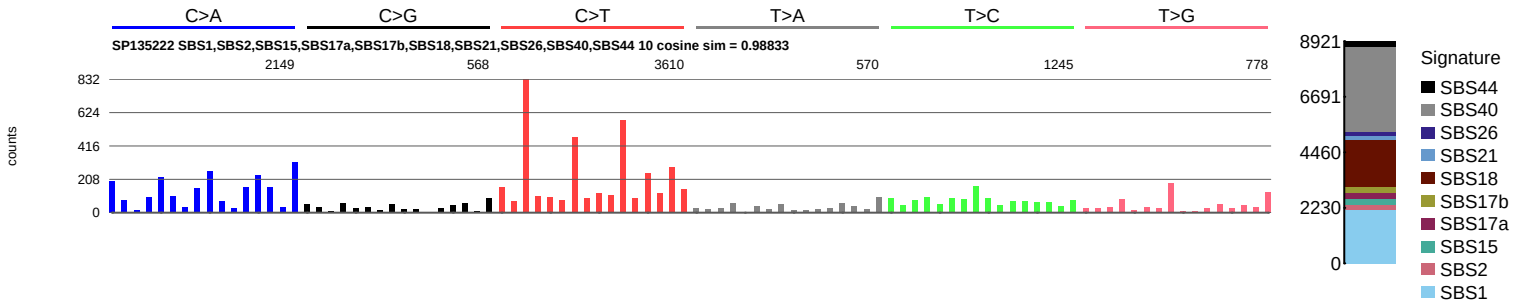
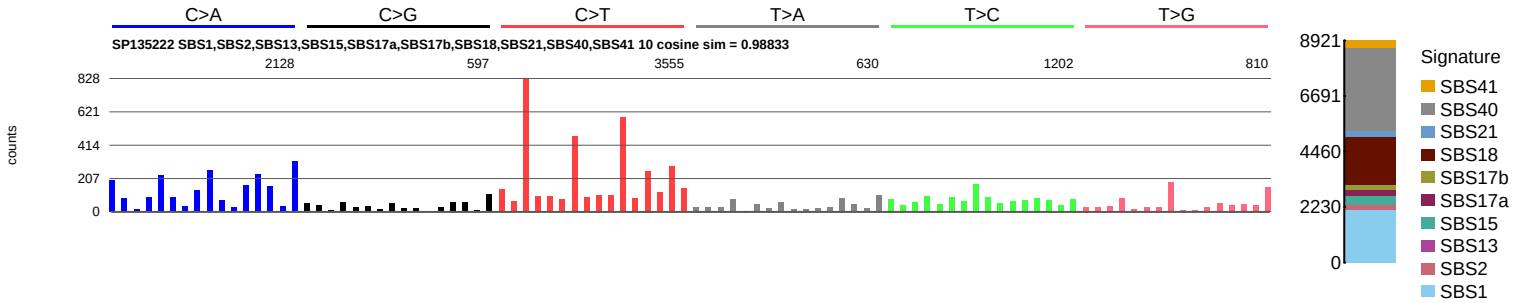
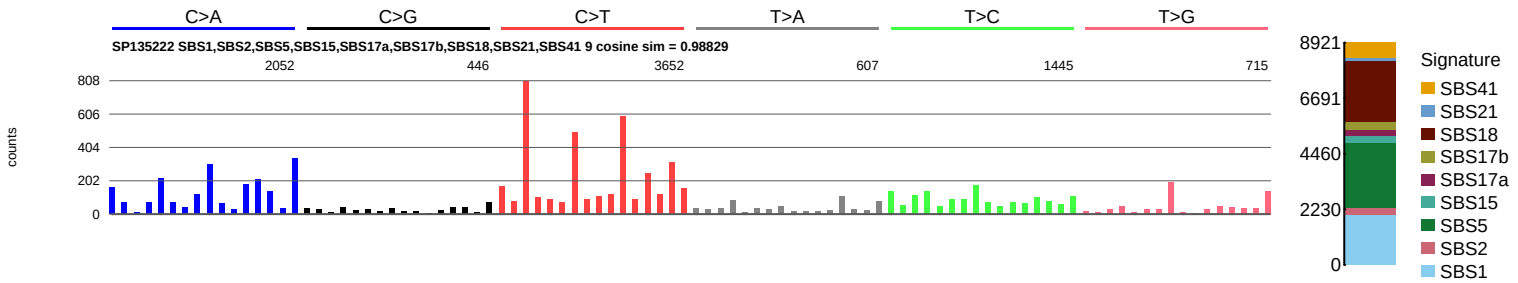
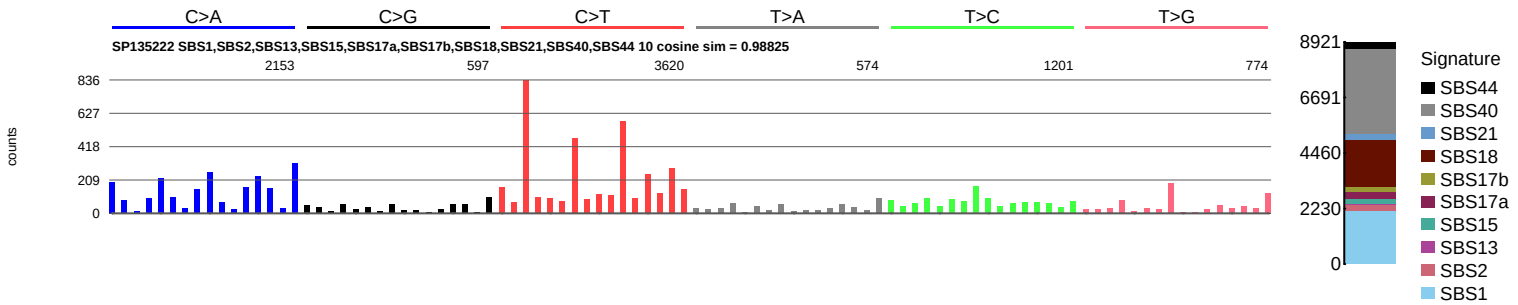
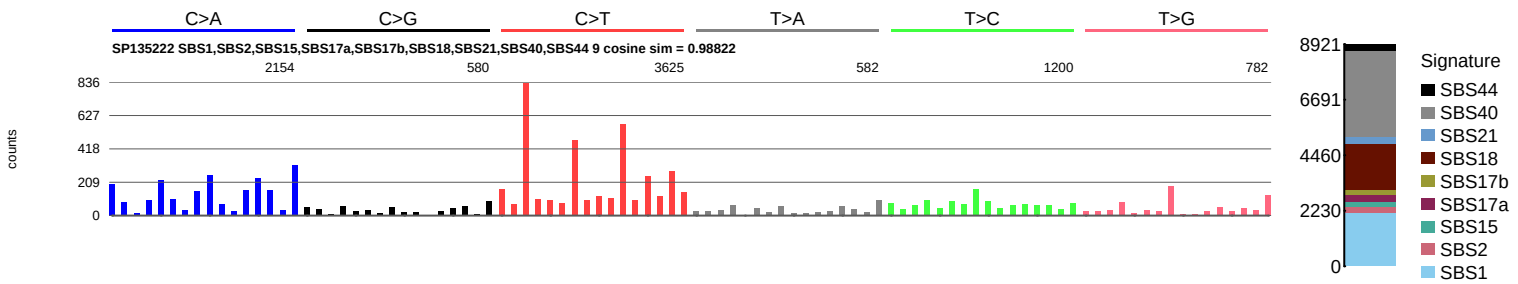
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98833
2057
485
3635
625
1404
715
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

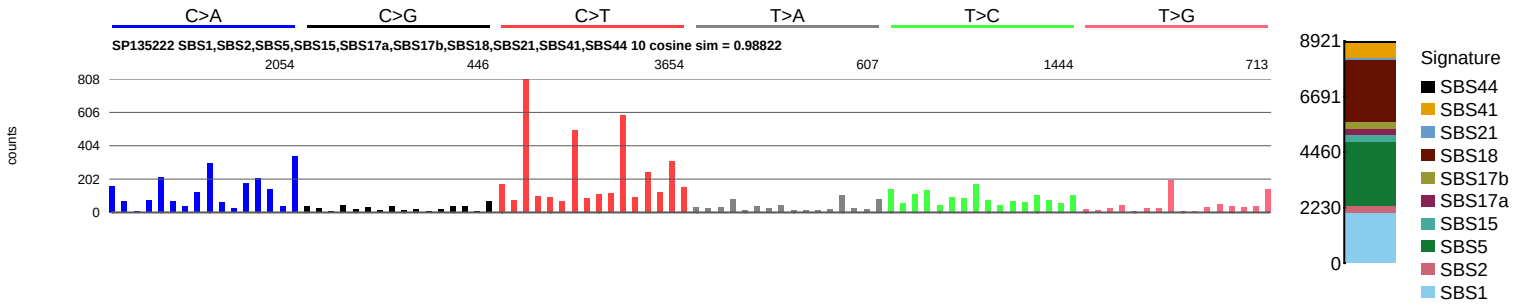

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98833
2149
568
3610
570
1245
778
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

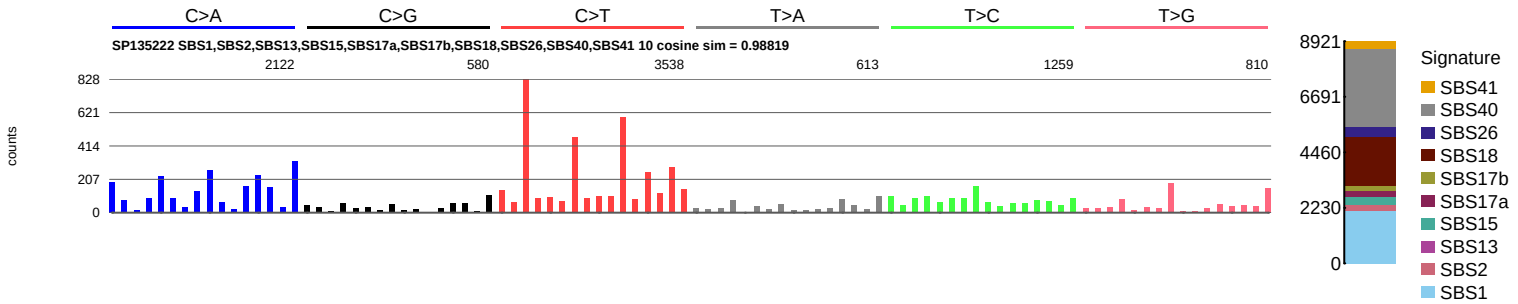

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98833
2128
597
3555
630
1202
810
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

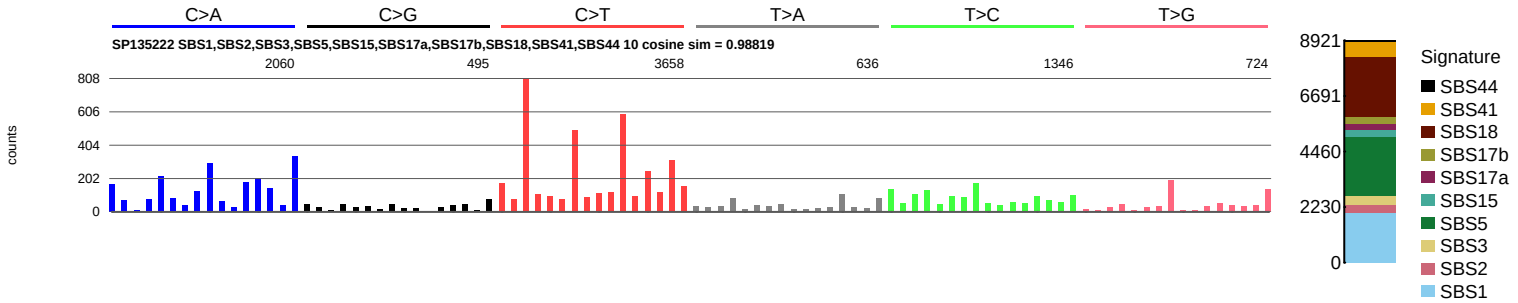

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98829
2052
446
3652
607
1445
715
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98825
2153
597
3620
574
1201
774
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98822
2154
580
3625
582
1200
782
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98822
2054
446
3654
607
1444
713
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98819
2122
580
3538
613
1259
810
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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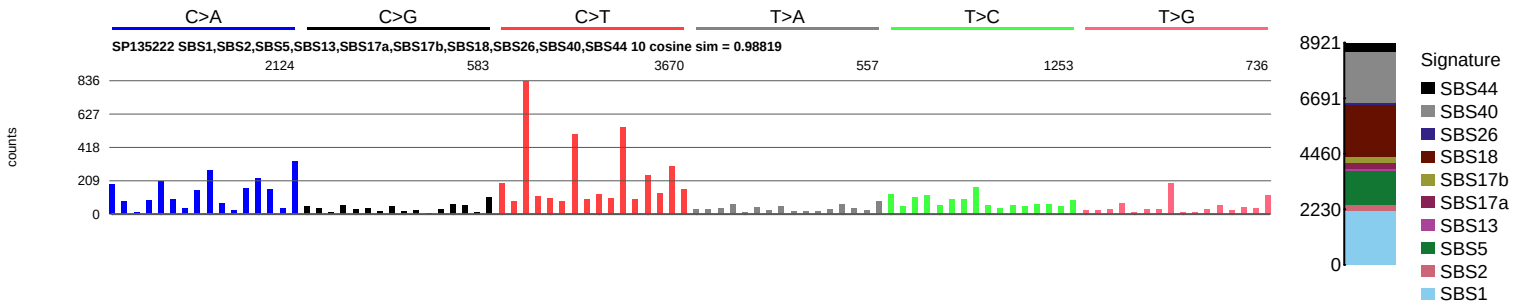
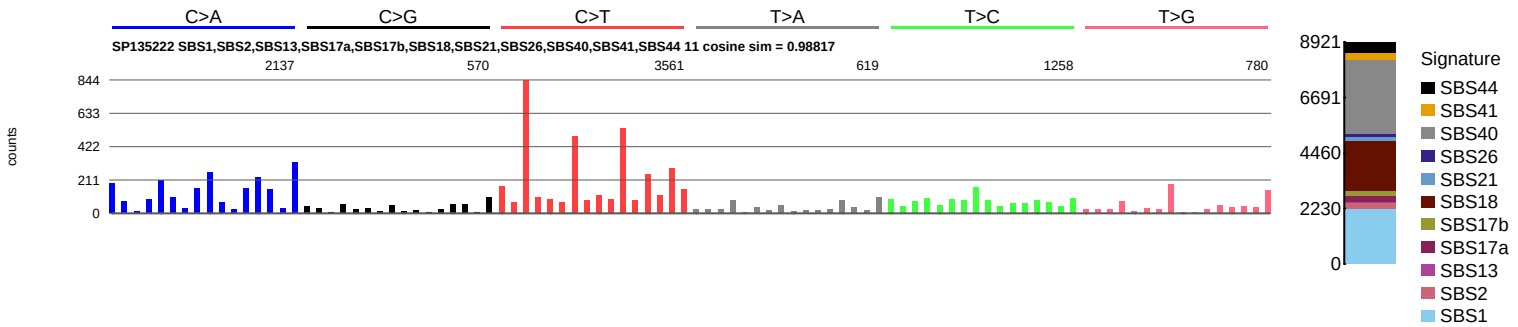
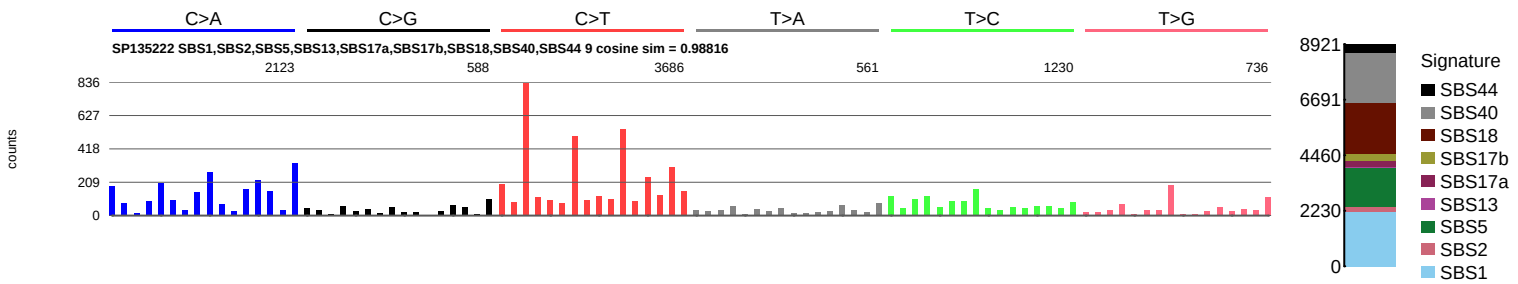
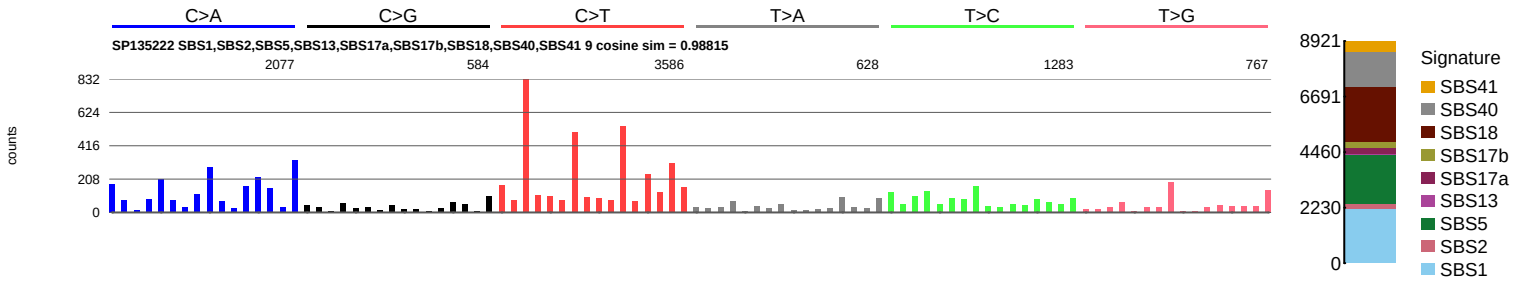
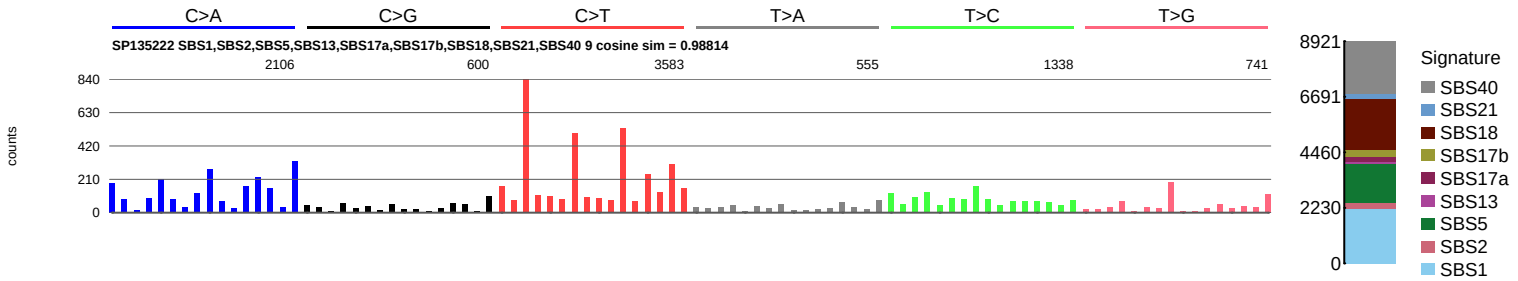
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98819
2060
495
3658
636
1346
724
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

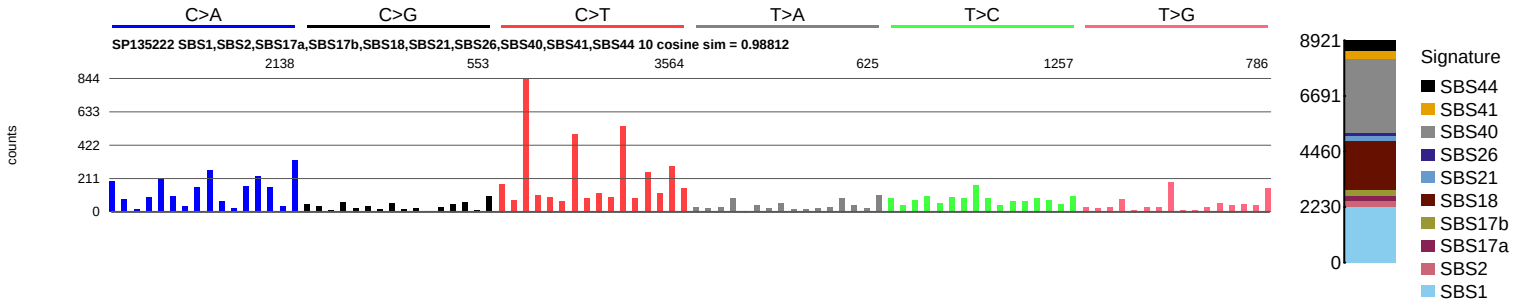

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98819
2124
583
3670
557
1253
736
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

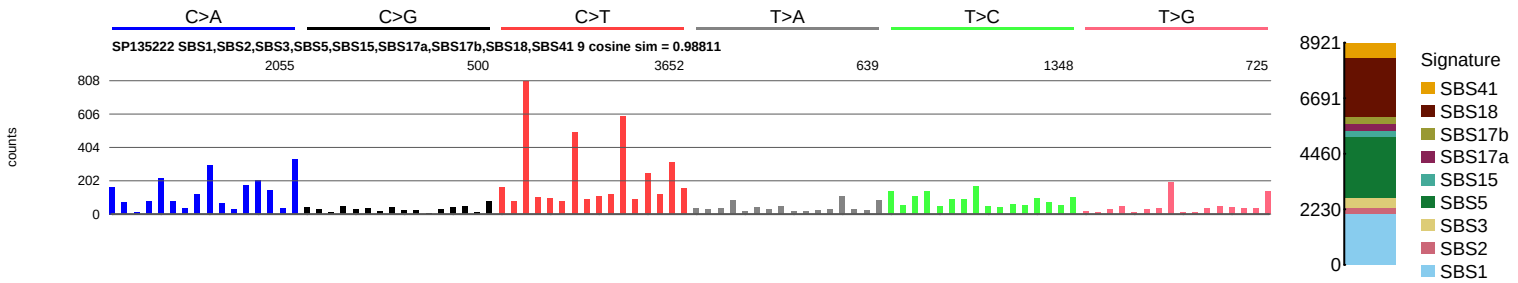

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98817
2137
570
3561
619
1258
780
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

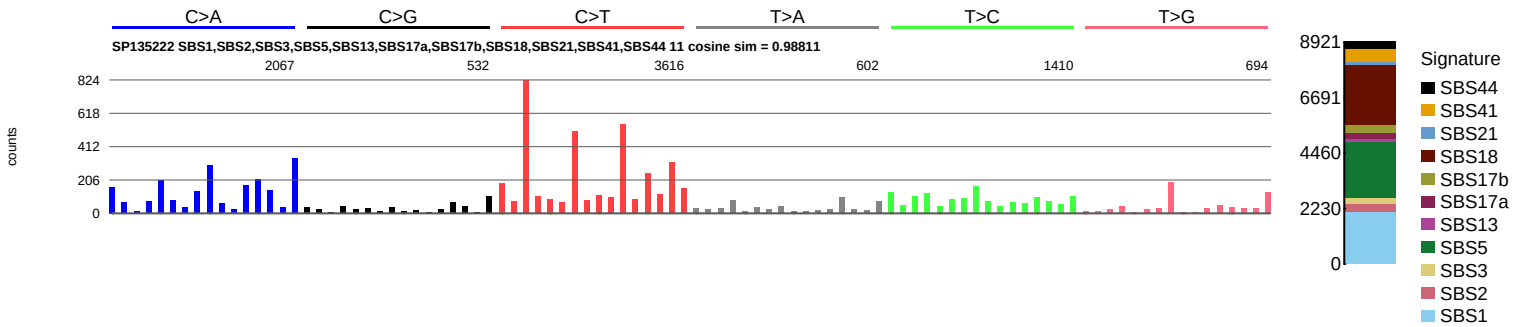

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98816
2123
588
3686
561
1230
736
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

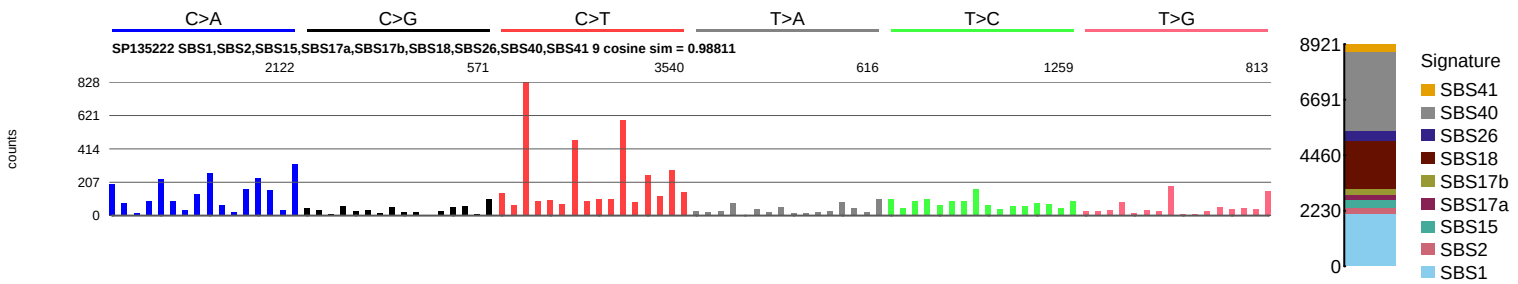

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98815
2077
584
3586
628
1283
767
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98814
2106
600
3583
555
1338
741
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98812
2138
553
3564
625
1257
786
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98811
2055
500
3652
639
1348
725
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98811
2067
532
3616
602
1410
694
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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SP135222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98811
2122
571
3540
616
1259
813
counts
828
621
414
207
0
8921
6691
4460
2230
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Signature
SBS41
SBS40
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SBS2
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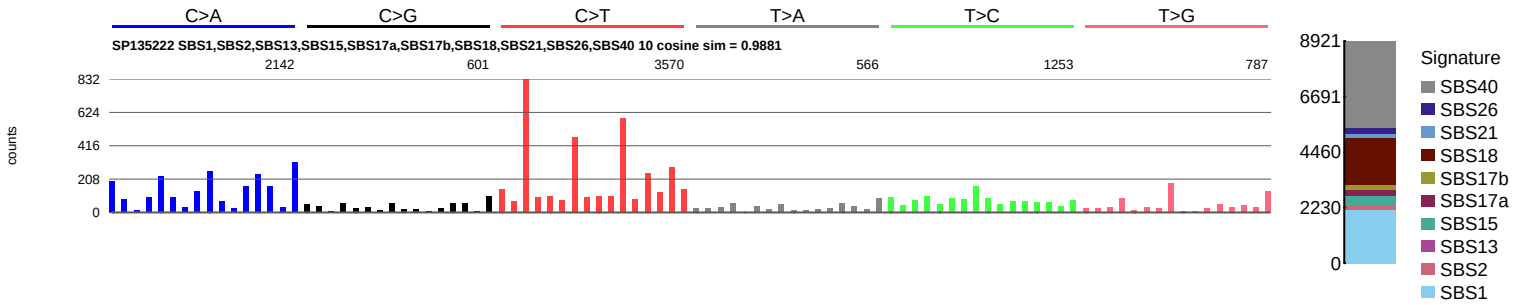
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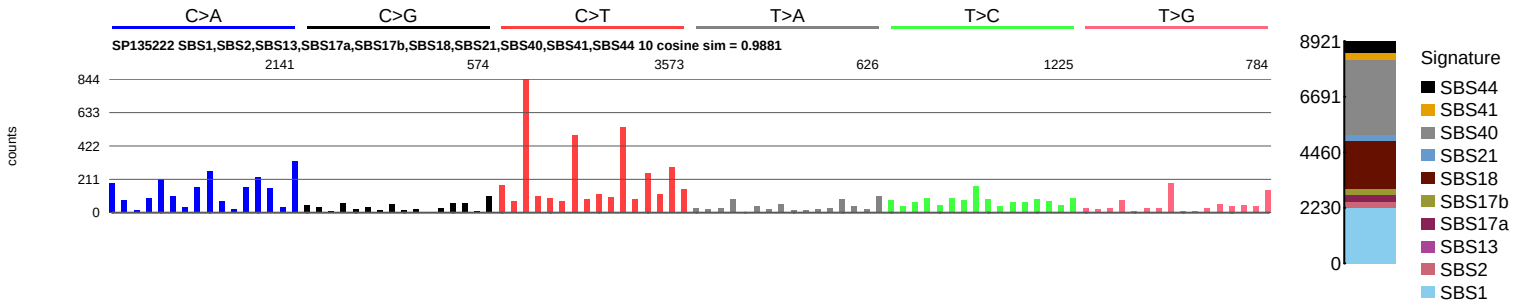
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.9881
2142
601
3570
566
1253
787
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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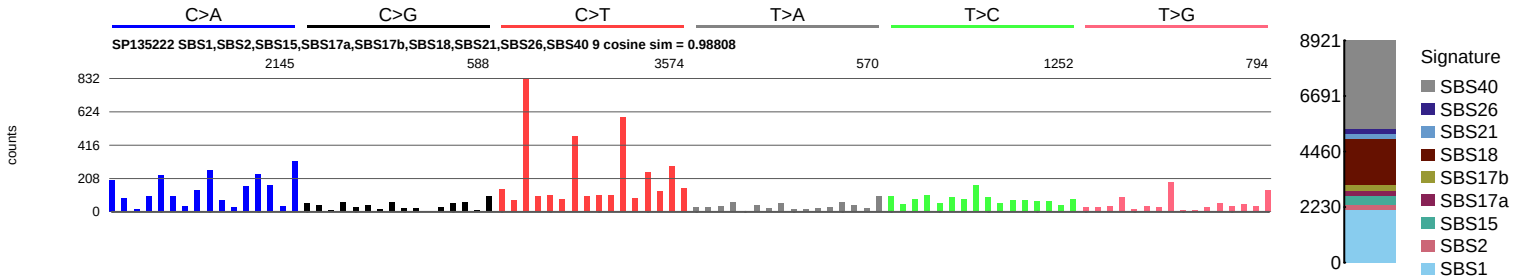
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.9881
2141
574
3573
626
1225
784
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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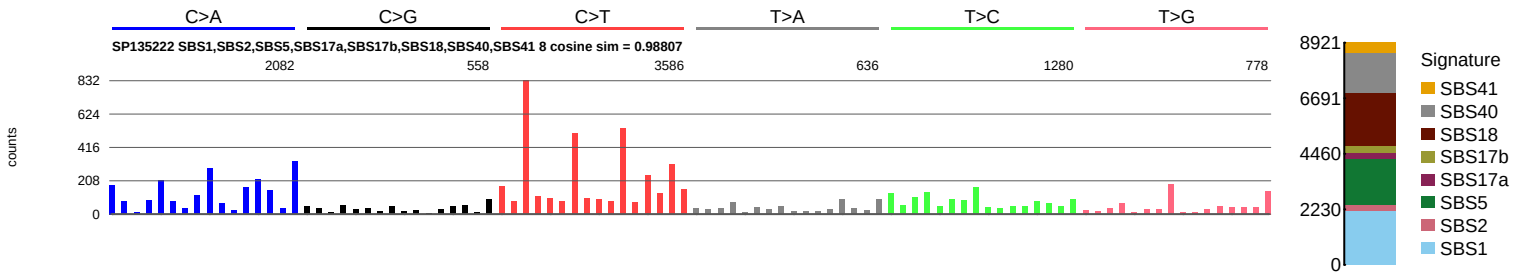
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98808
2145
588
3574
570
1252
794
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98807
2082
558
3586
636
1280
778
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
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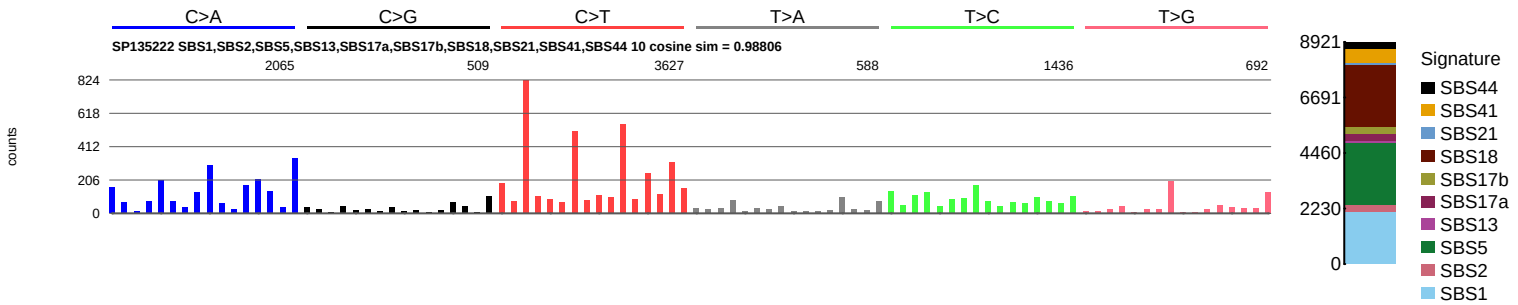
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
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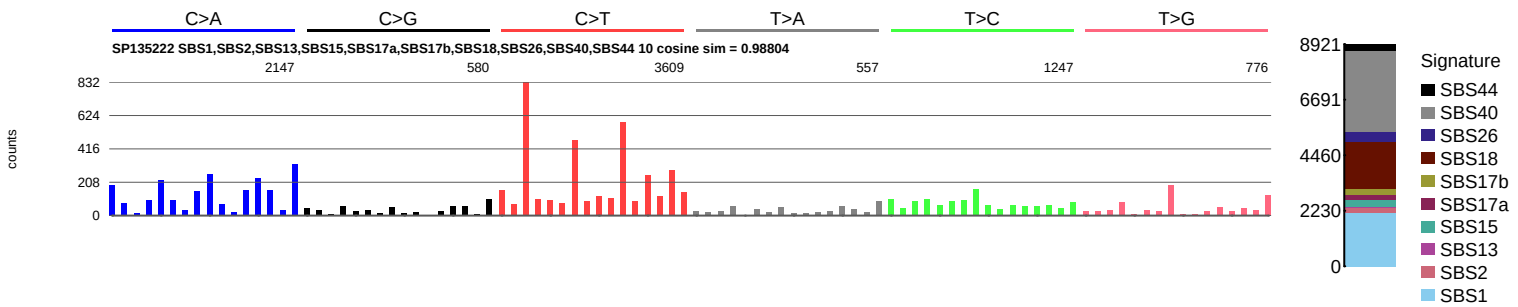
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98806
2065
509
3627
588
1436
692
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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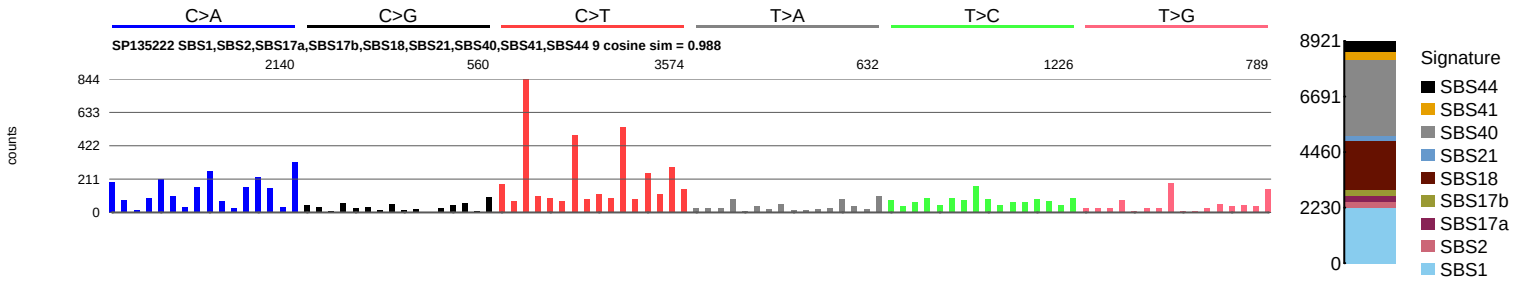
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98804
2147
580
3609
557
1247
776
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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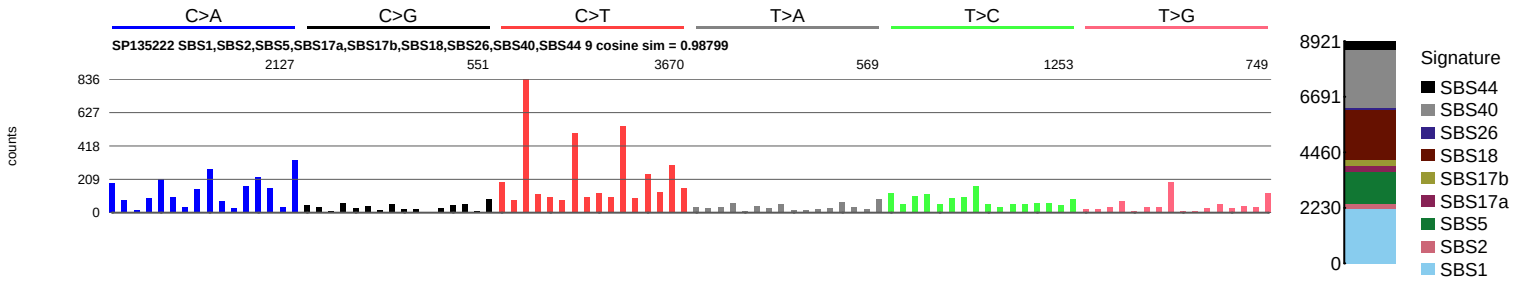
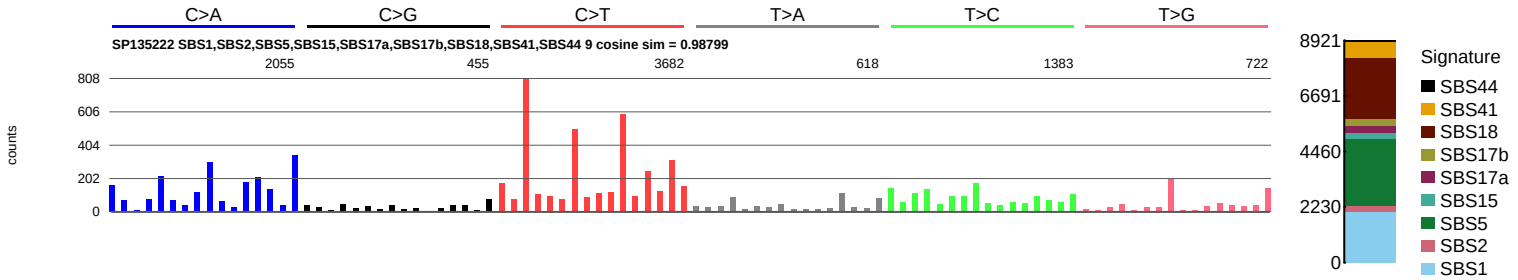
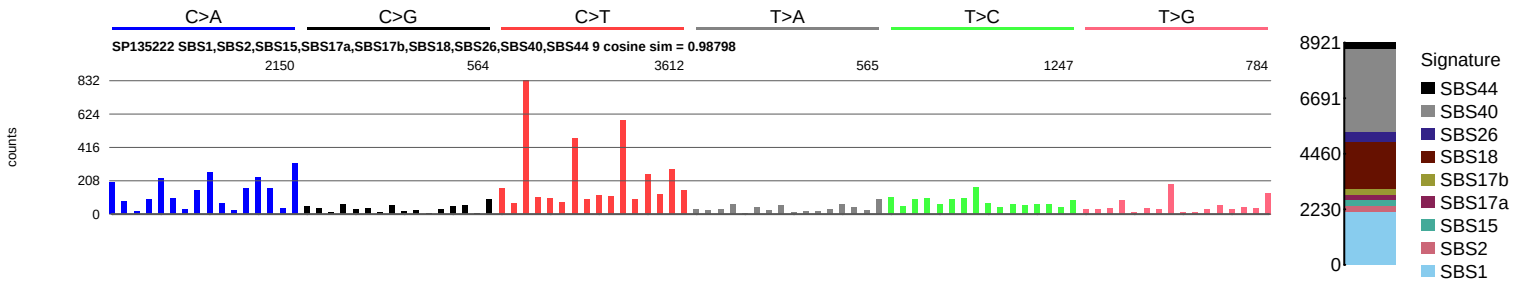
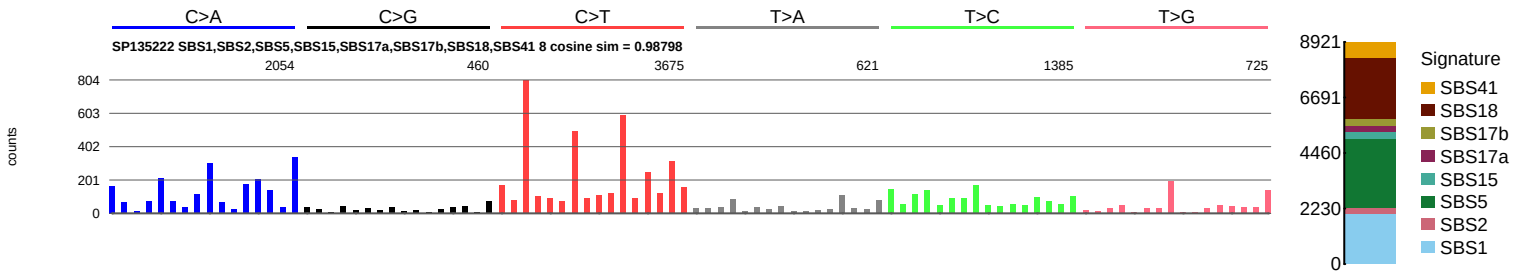
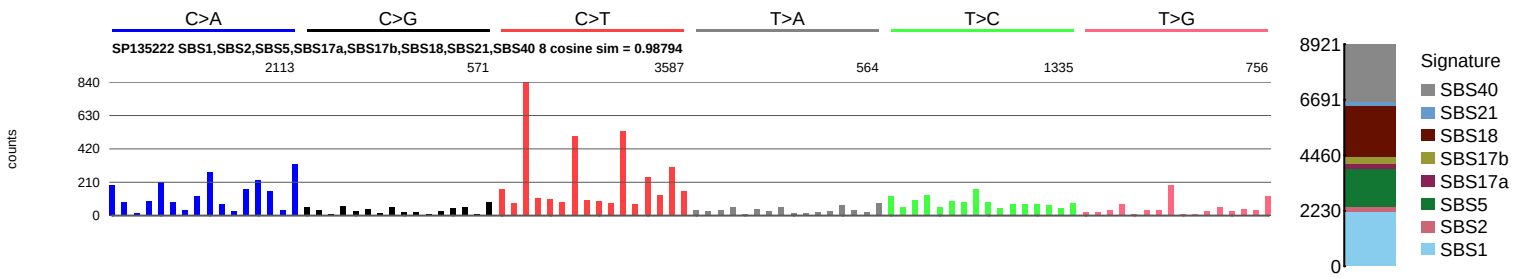
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.988
2140
560
3574
632
1226
789
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

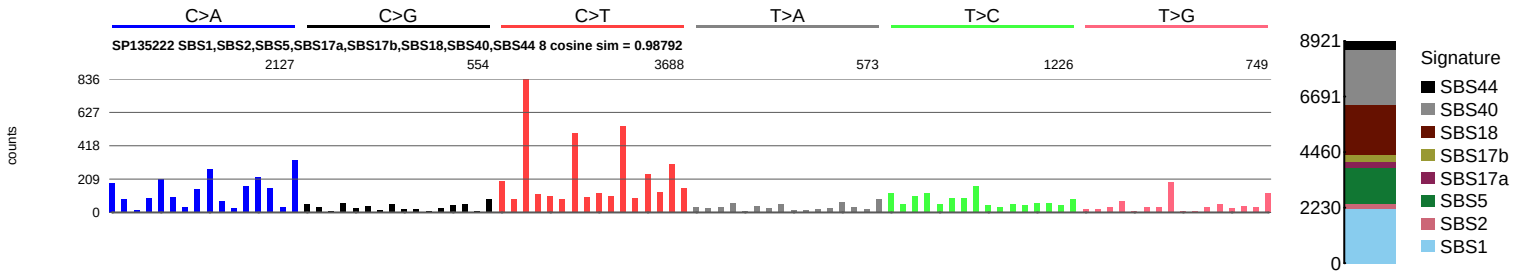

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98799
2127
551
3670
569
1253
749
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

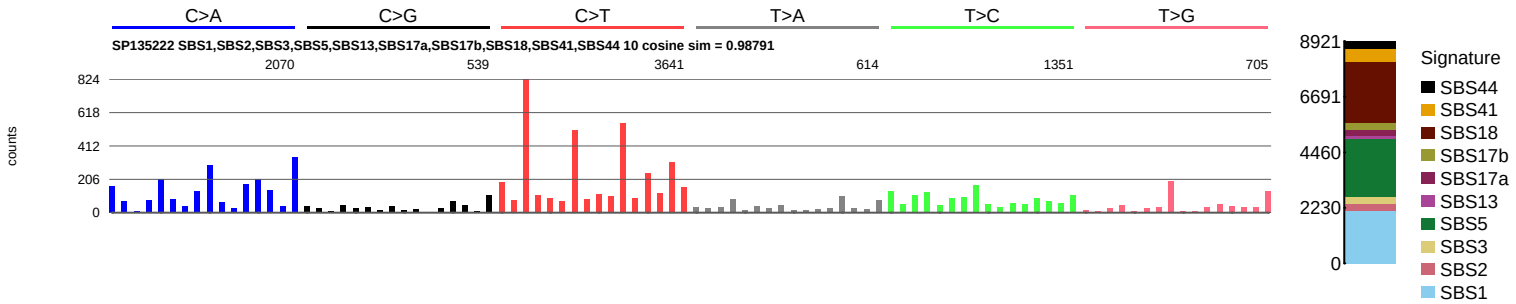

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98799
2055
455
3682
618
1383
722
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98798
2150
564
3612
565
1247
784
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98798
2054
460
3675
621
1385
725
804
603
402
201
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98794
2113
571
3587
564
1335
756
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98792
2127
554
3688
573
1226
749
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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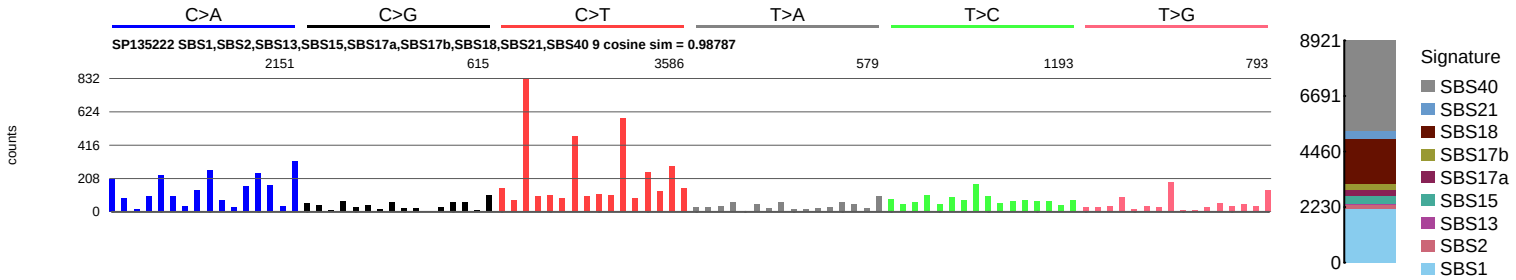
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98791
2070
539
3641
614
1351
705
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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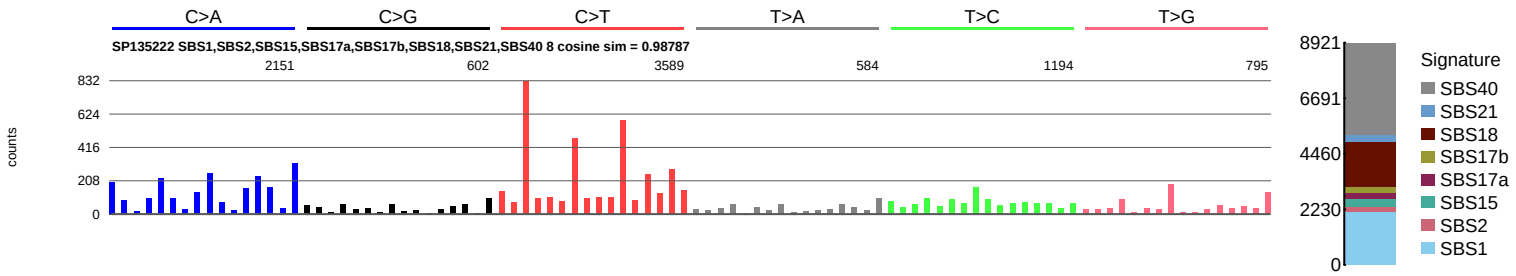
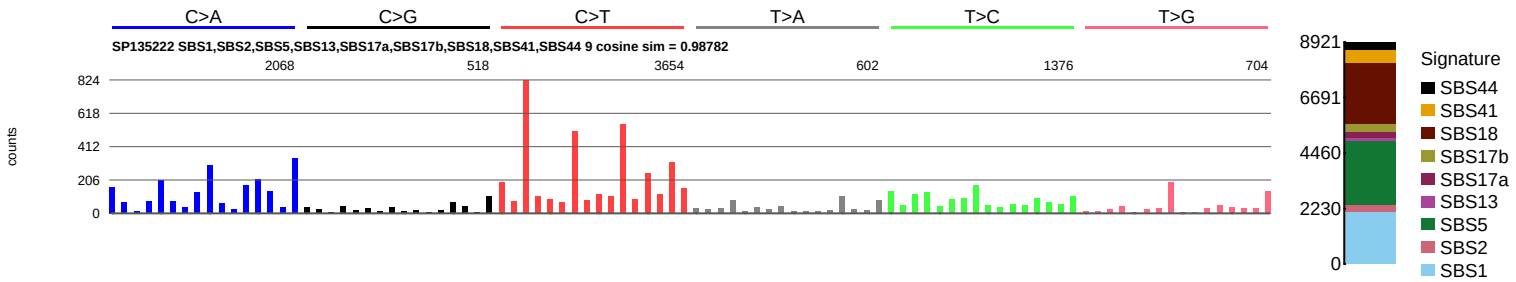
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98787
2151
615
3586
579
1193
793
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98787
2151
602
3589
584
1194
795
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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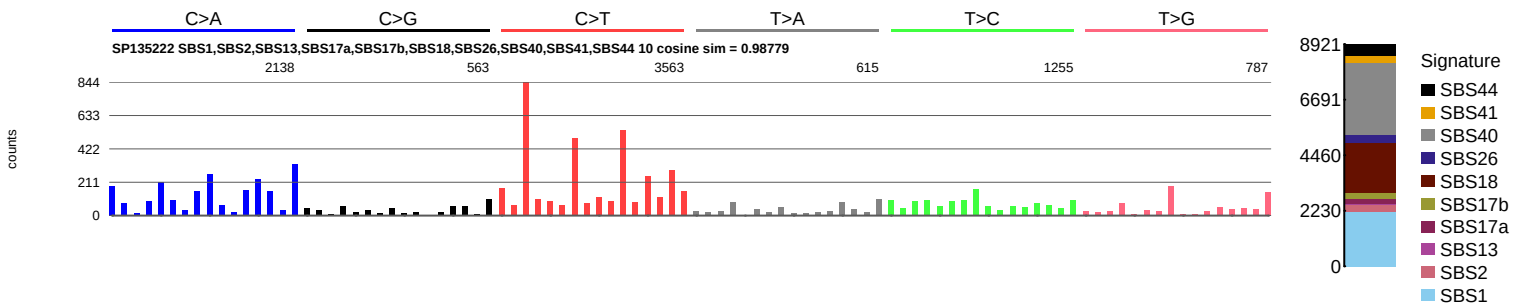
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98782
2068
518
3654
602
1376
704
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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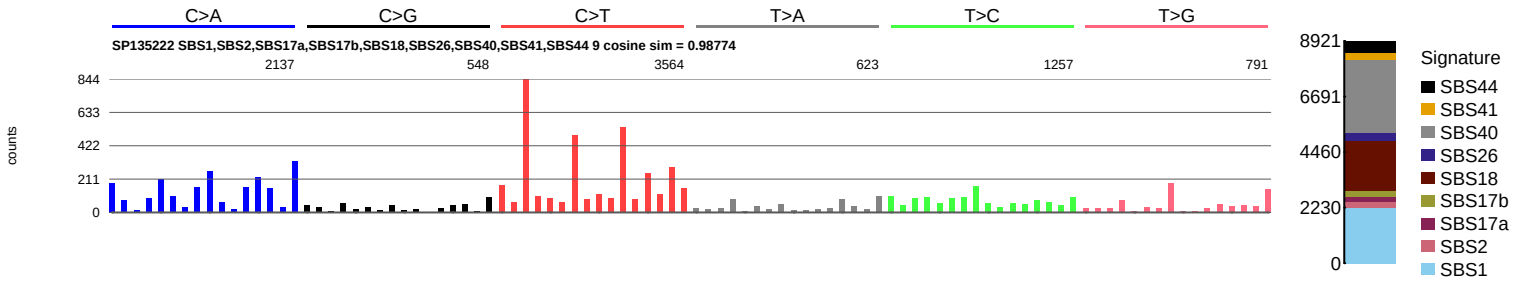
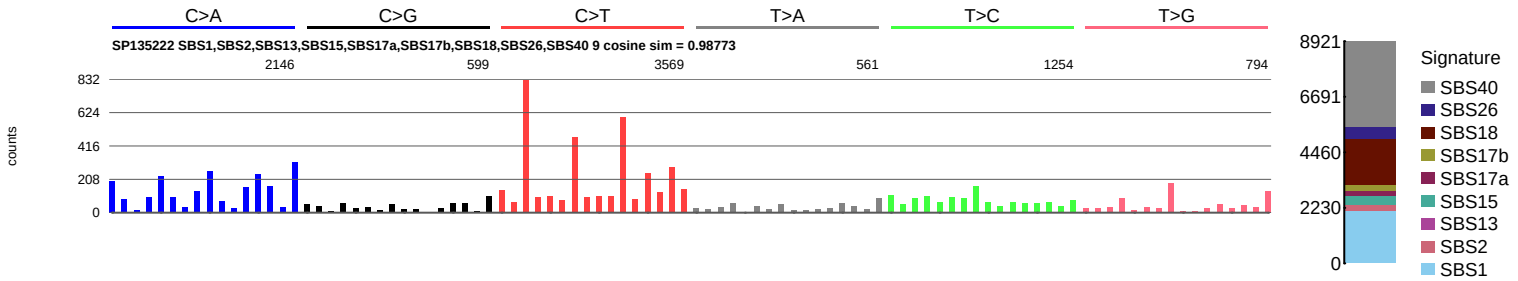
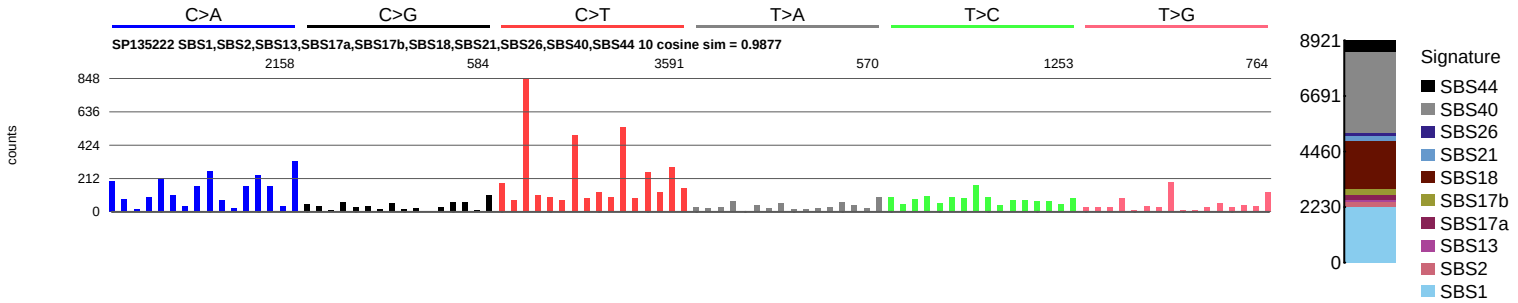
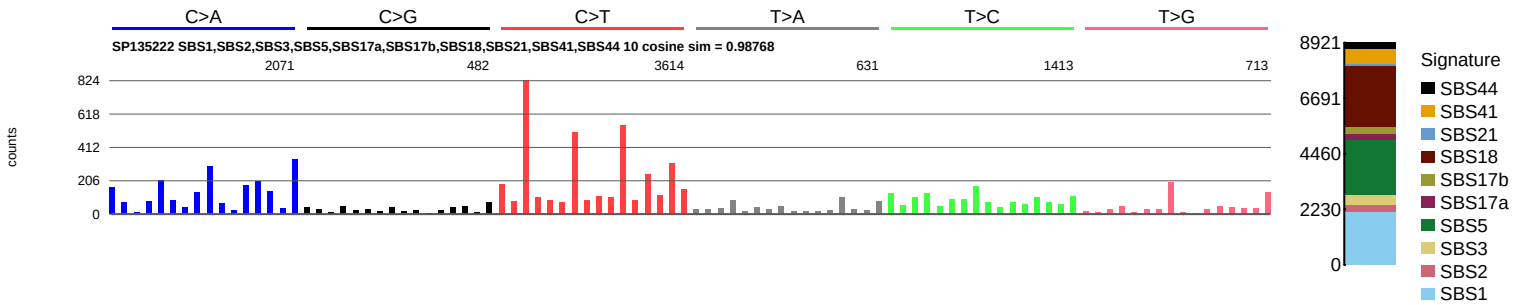
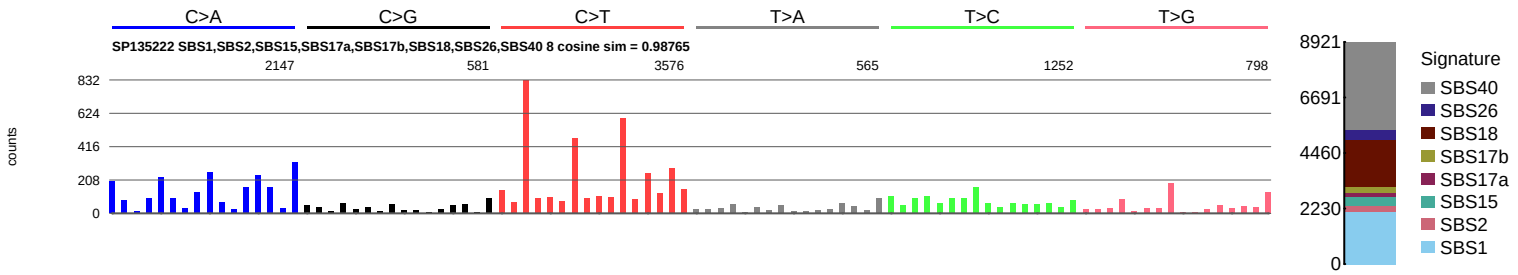
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98779
2138
563
3563
615
1255
787
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

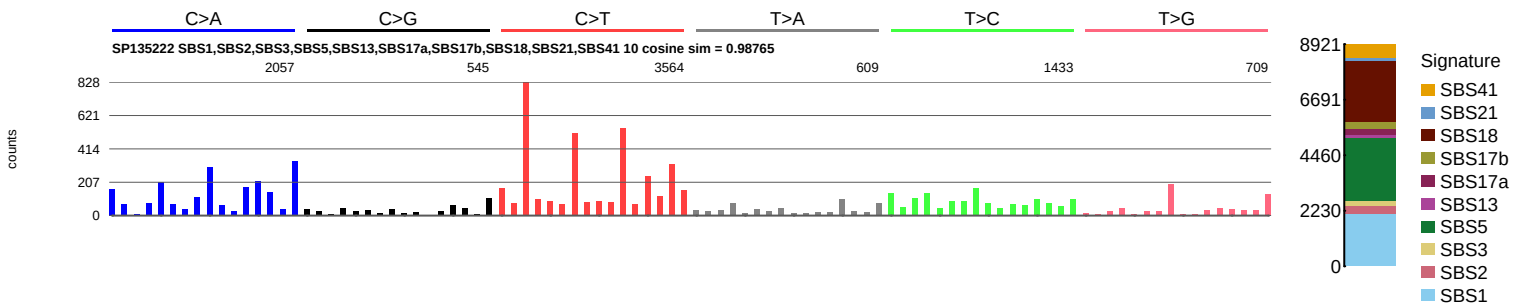

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98774
2137
548
3564
623
1257
791
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98773
2146
599
3569
561
1254
794
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.9877
2158
584
3591
570
1253
764
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98768
2071
482
3614
631
1413
713
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98765
2147
581
3576
565
1252
798
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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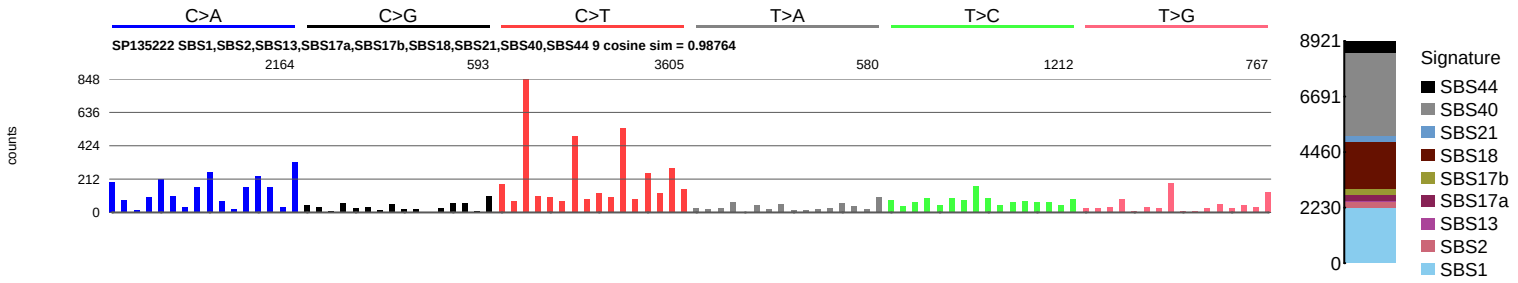
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98765
2057
545
3564
609
1433
709
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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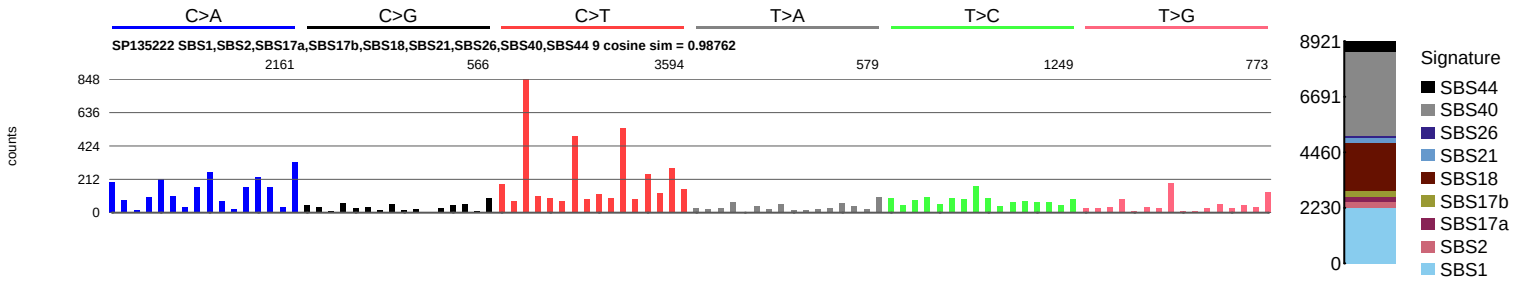
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98764
2164
593
3605
580
1212
767
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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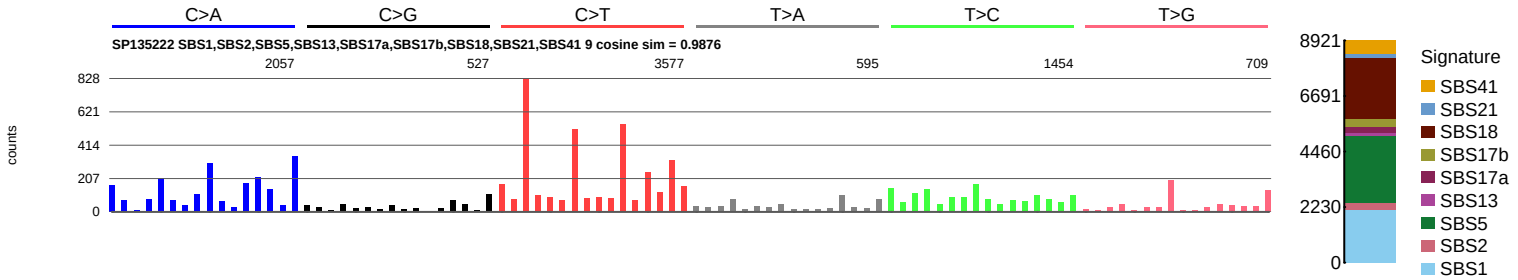
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98762
2161
566
3594
579
1249
773
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
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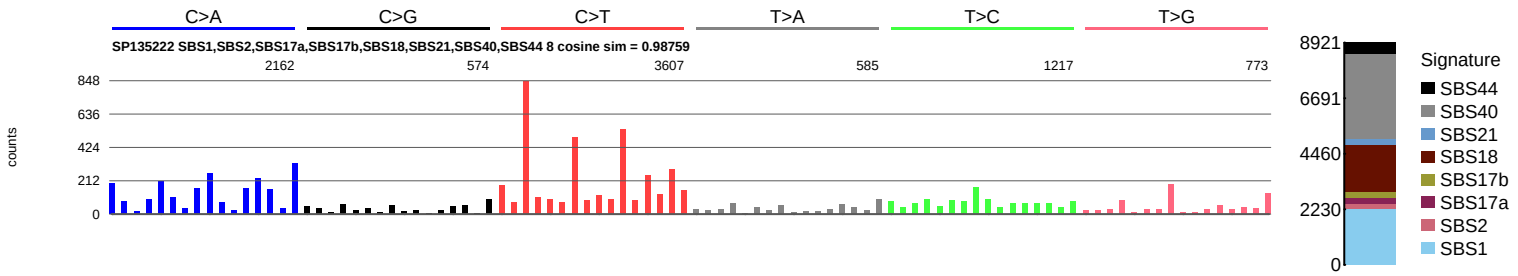
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9876
2057
527
3577
595
1454
709
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98759
2162
574
3607
585
1217
773
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
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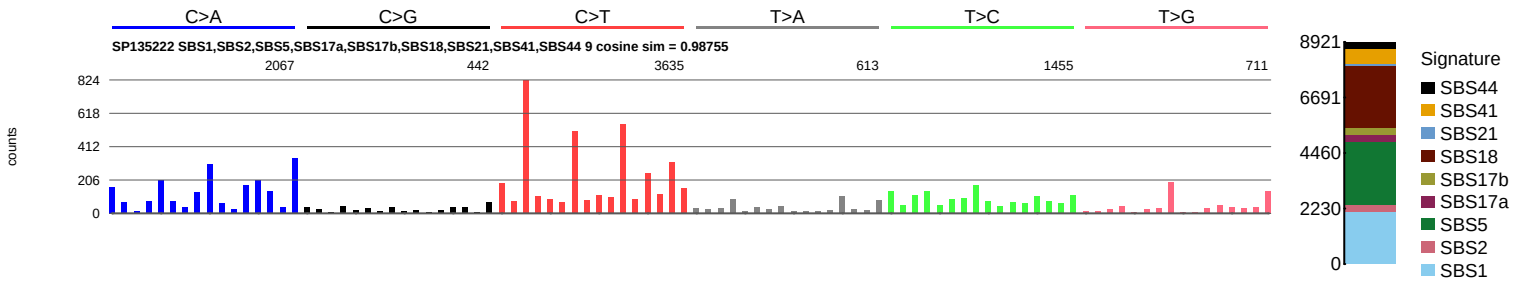
SBS2
SBS1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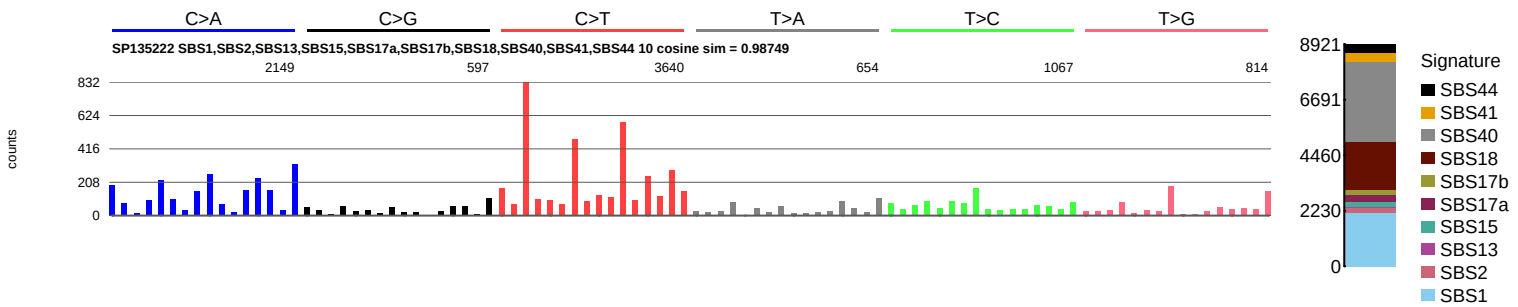
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98755
2067
442
3635
613
1455
711
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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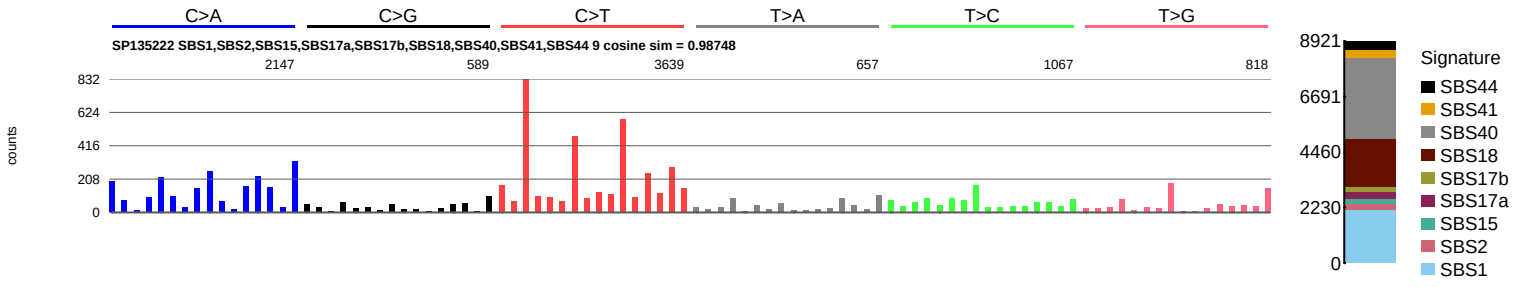
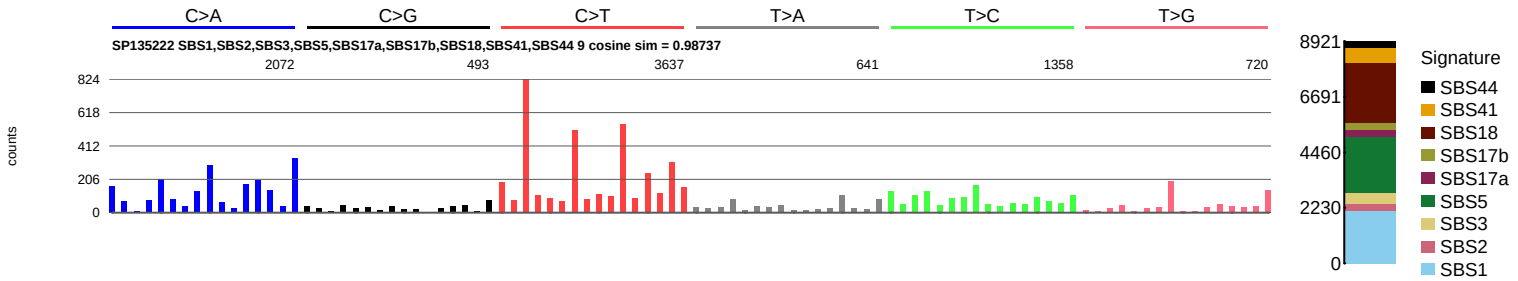
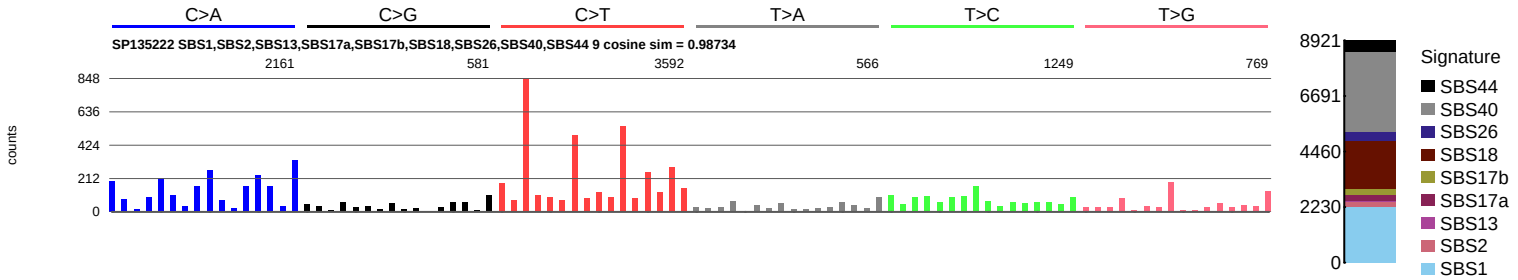
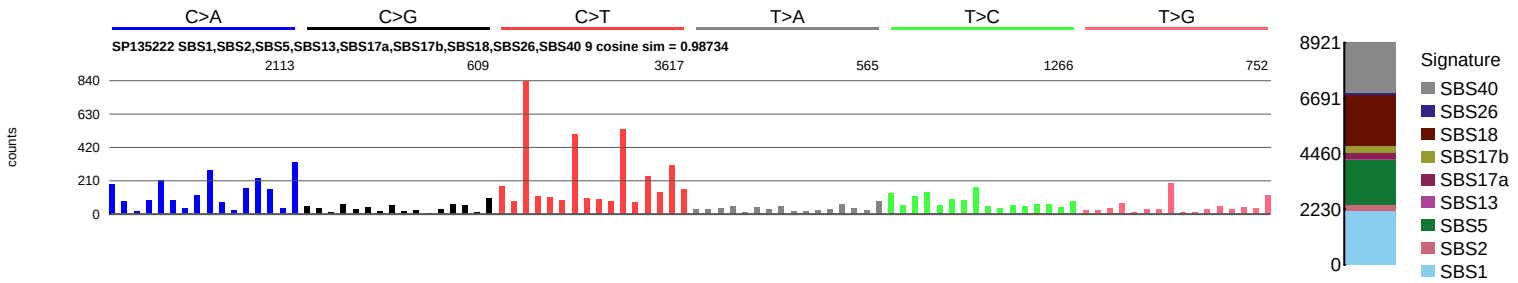
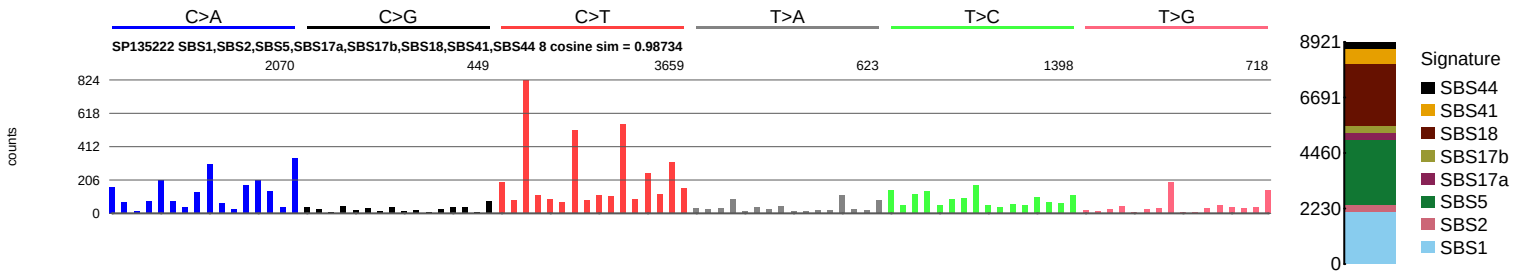
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98749
2149
597
3640
654
1067
814
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

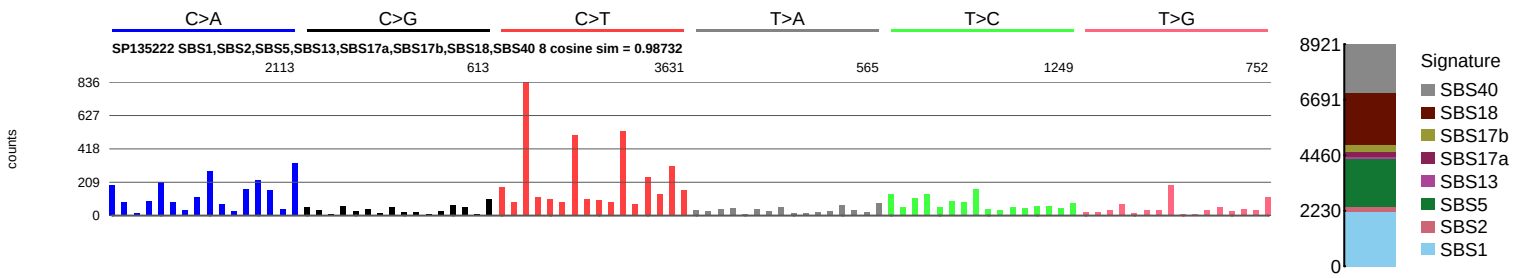

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98748
2147
589
3639
657
1067
818
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98737
2072
493
3637
641
1358
720
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98734
2161
581
3592
566
1249
769
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98734
2113
609
3617
565
1266
752
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98734
2070
449
3659
623
1398
718
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98732
2113
613
3631
565
1249
752
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
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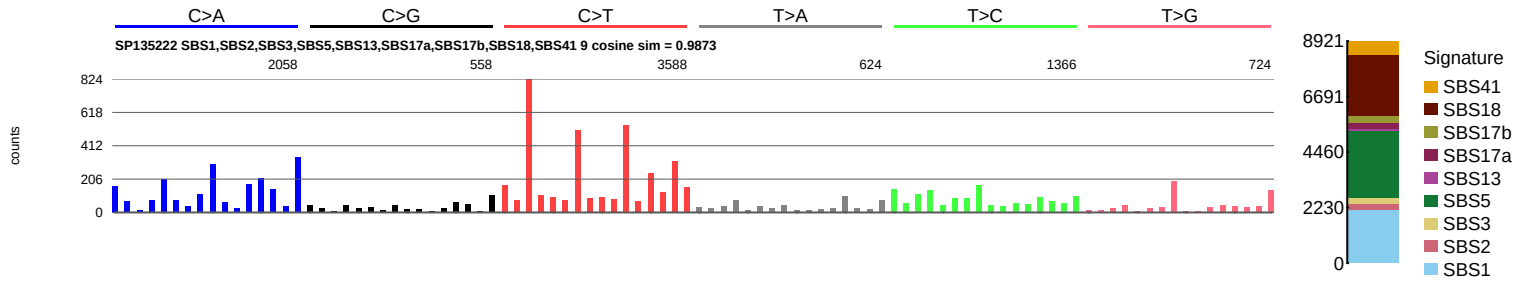
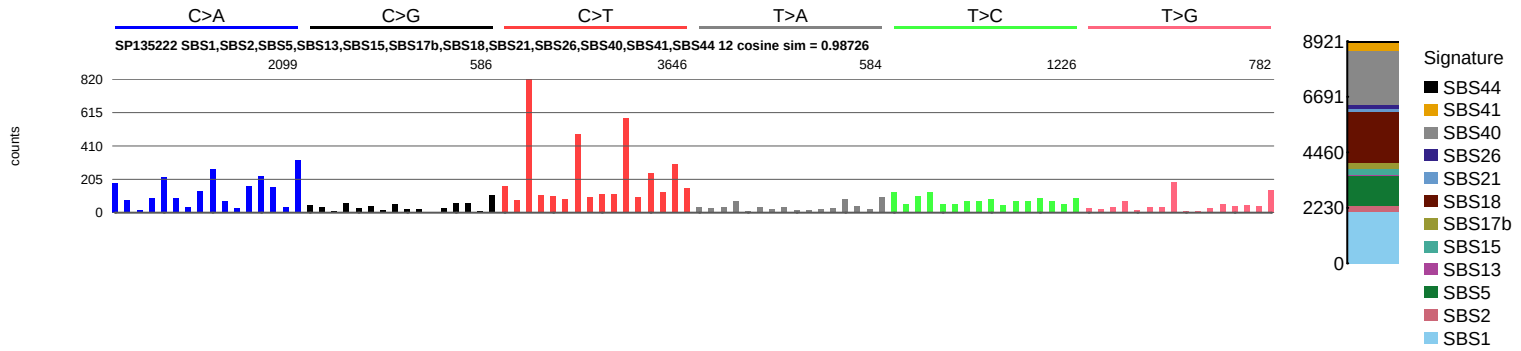
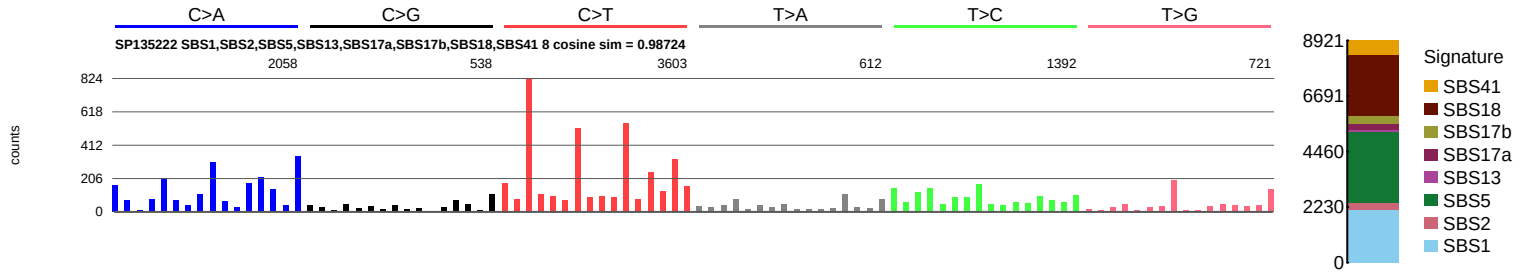
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.9873
2058
558
3588
624
1366
724
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98726
2099
586
3646
584
1226
782
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98724
2058
538
3603
612
1392
721
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS41
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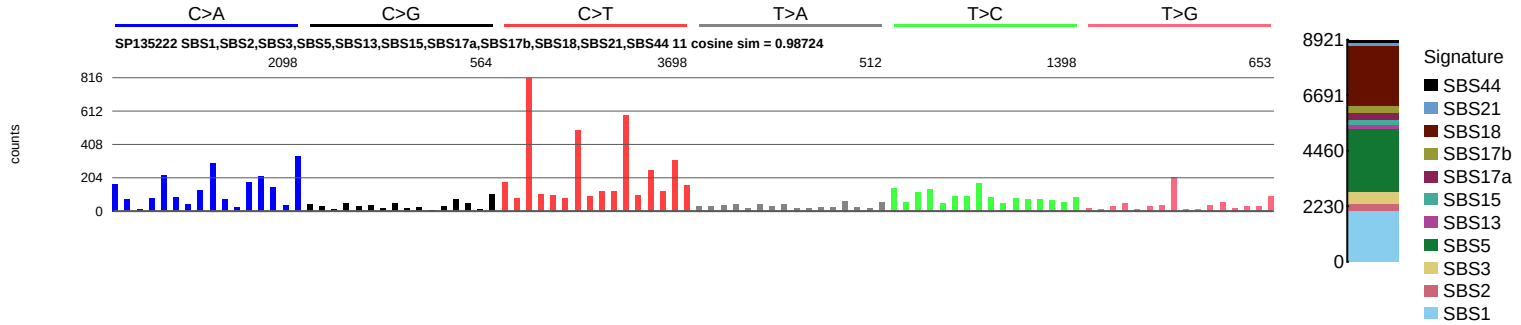
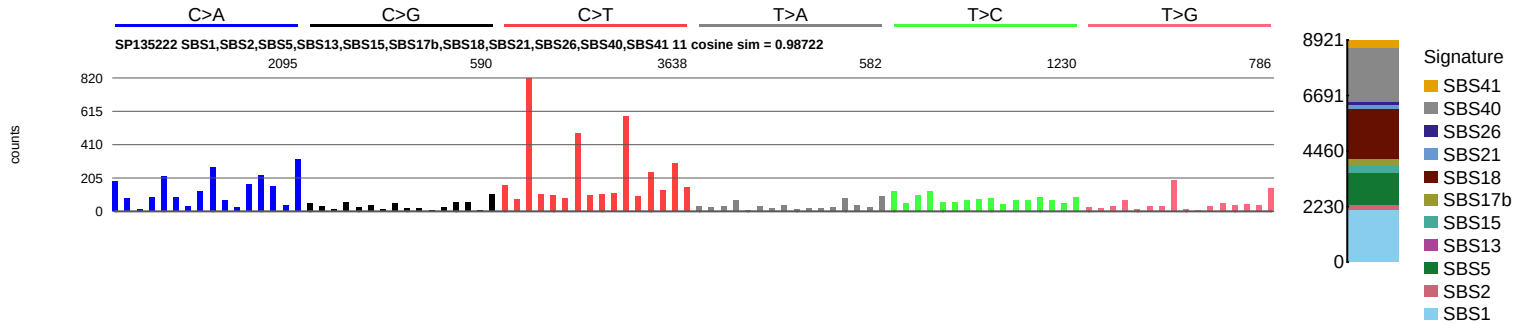
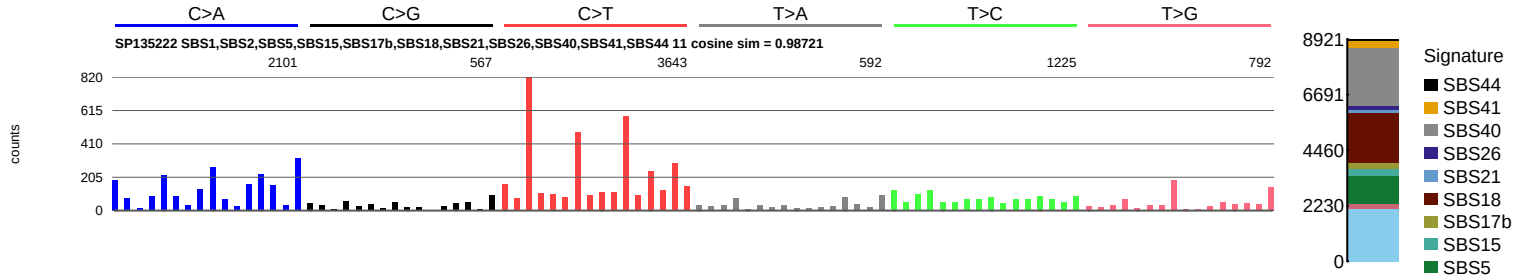
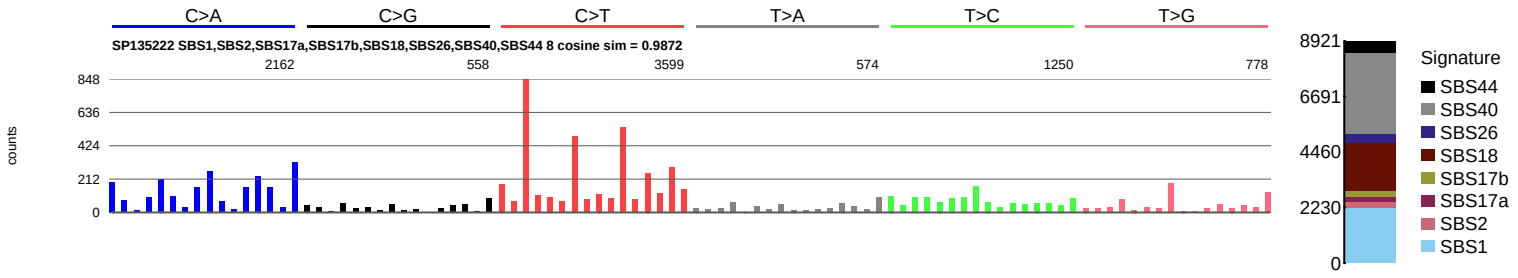
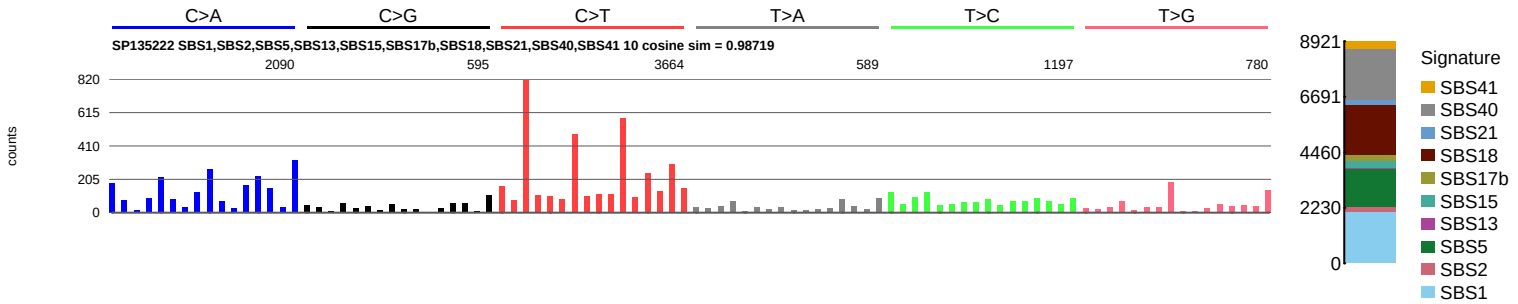
SBS5
SBS2
SBS1

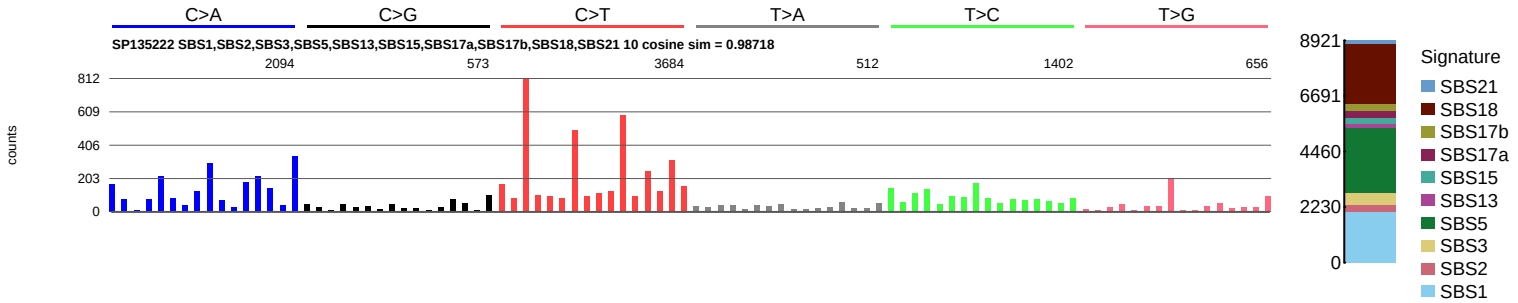

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 11 cosine sim = 0.98724
2098
564
3698
512
1398
653
816
612
408
204
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

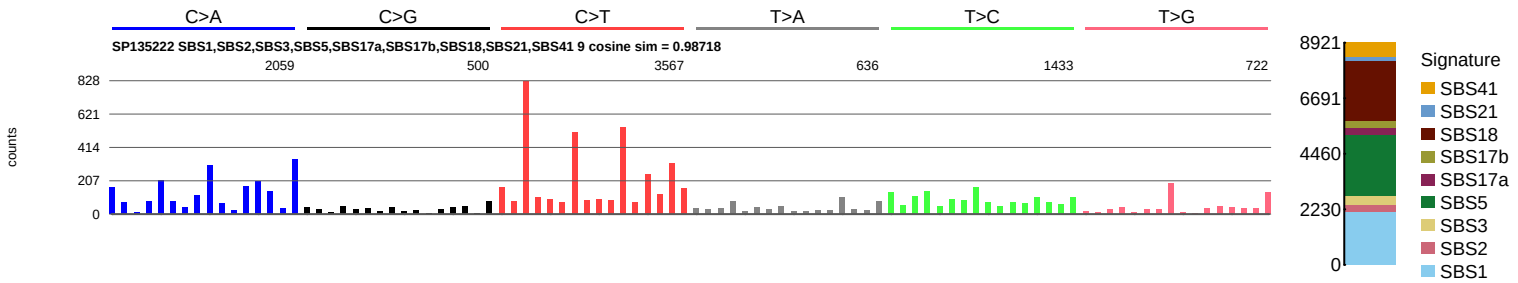

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98722
2095
590
3638
582
1230
786
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

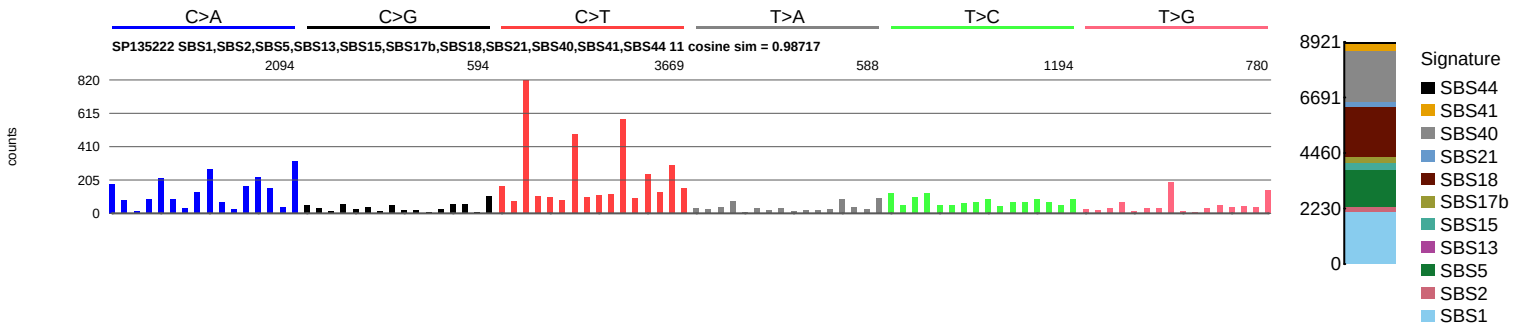

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98721
2101
567
3643
592
1225
792
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2

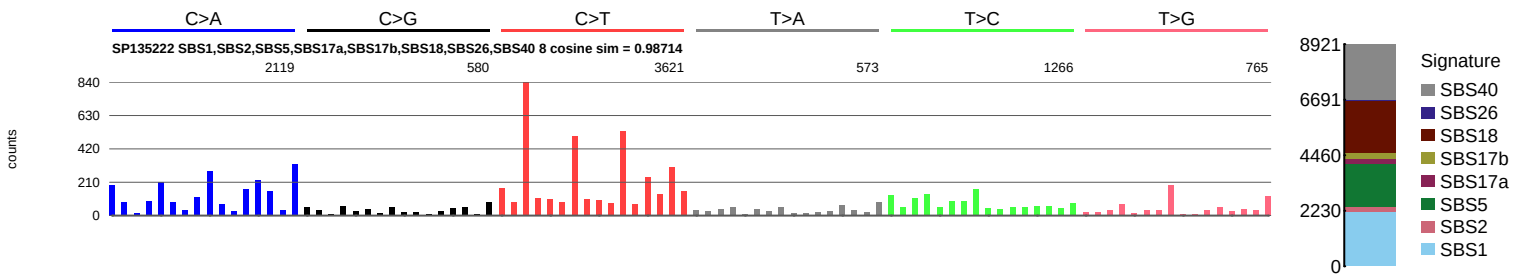

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.9872
2162
558
3599
574
1250
778
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98719
2090
595
3664
589
1197
780
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.98718
2094
573
3684
512
1402
656
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98718
2059
500
3567
636
1433
722
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98717
2094
594
3669
588
1194
780
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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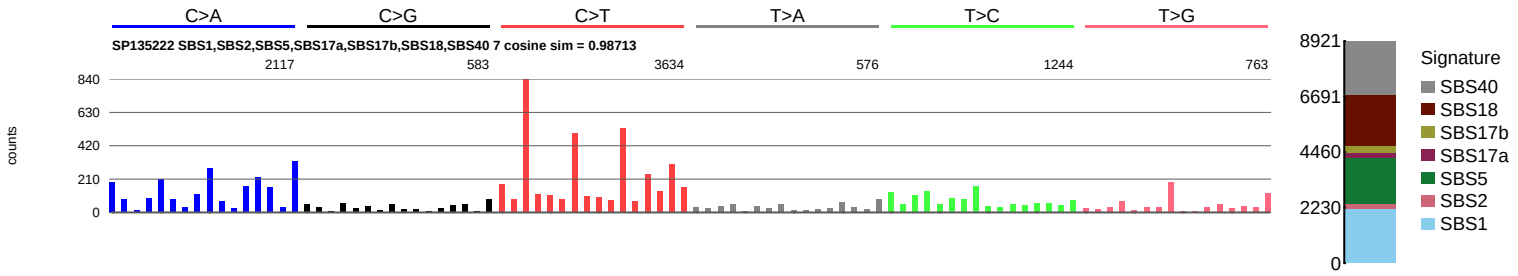
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98714
2119
580
3621
573
1266
765
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
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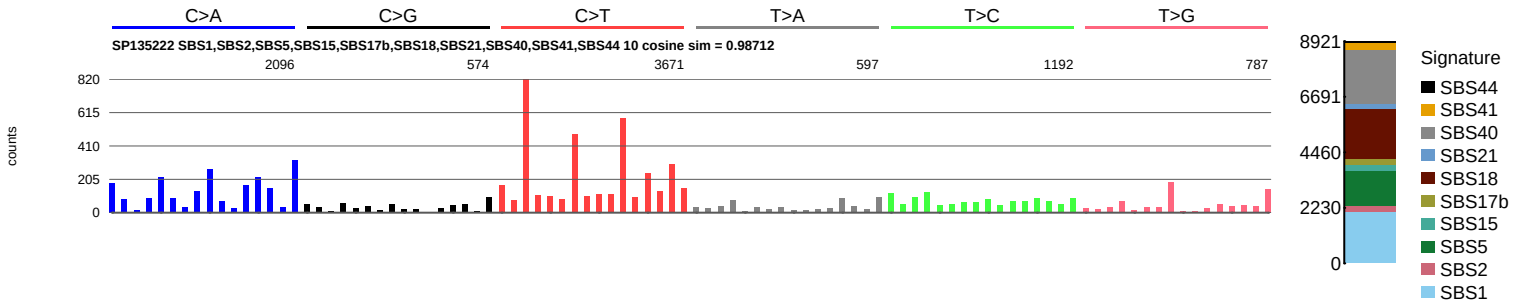
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98713
2117
583
3634
576
1244
763
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98712
2096
574
3671
597
1192
787
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS44
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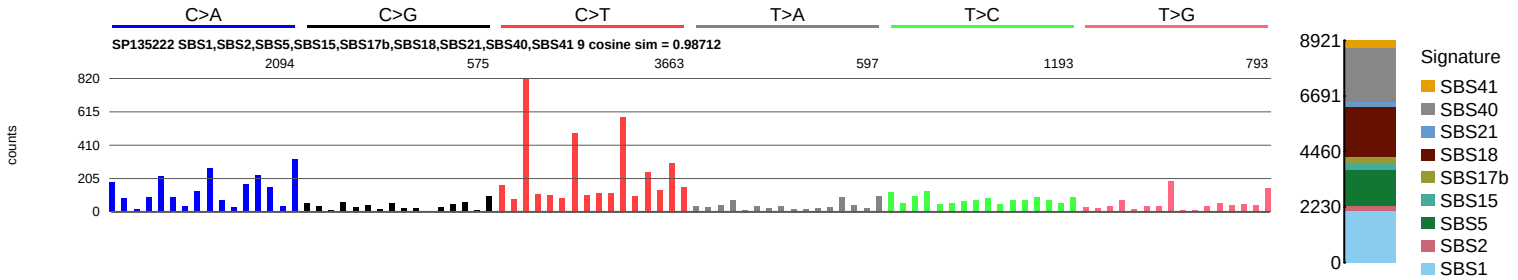
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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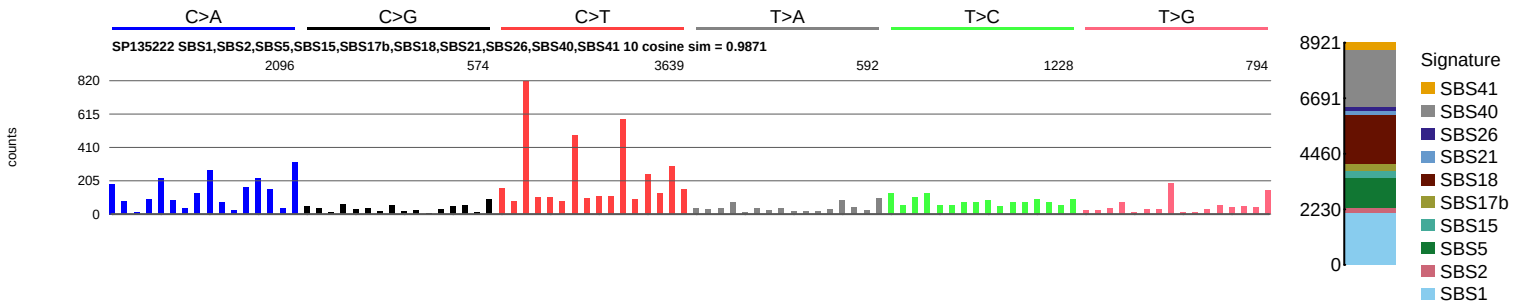
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98712
2094
575
3663
597
1193
793
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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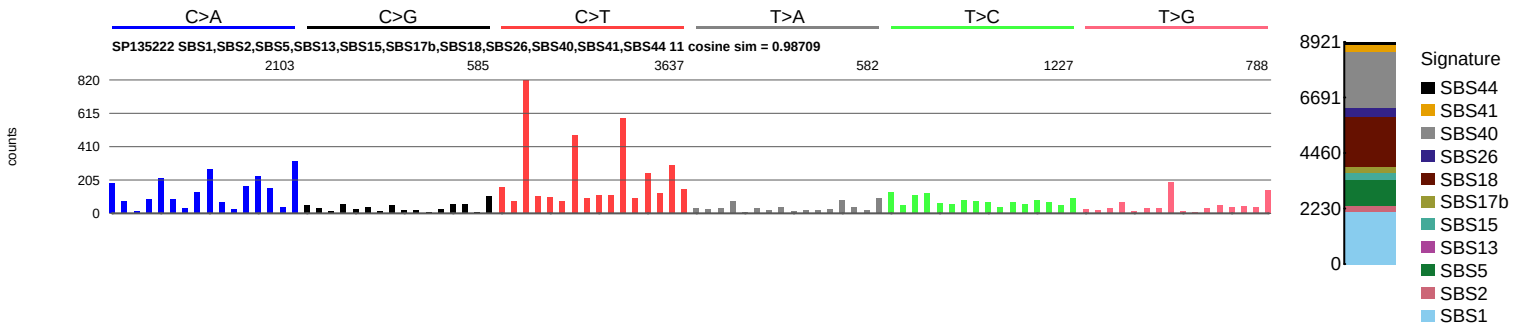
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.9871
2096
574
3639
592
1228
794
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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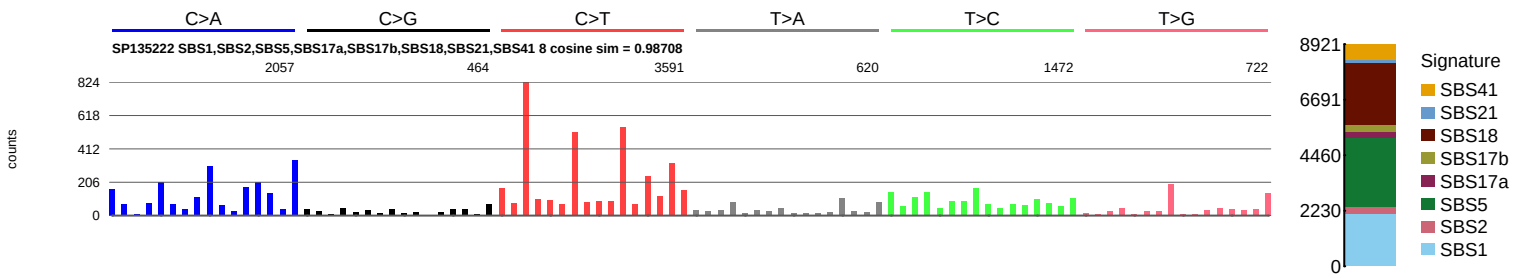
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98709
2103
585
3637
582
1227
788
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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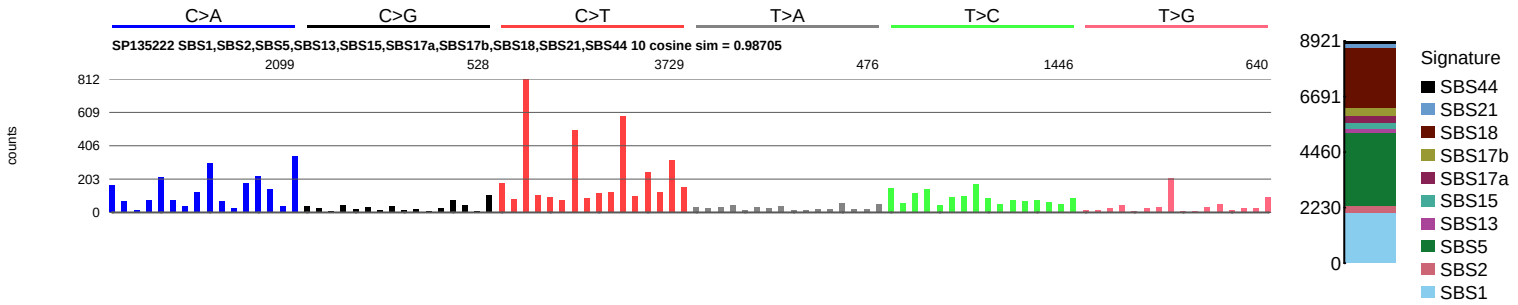
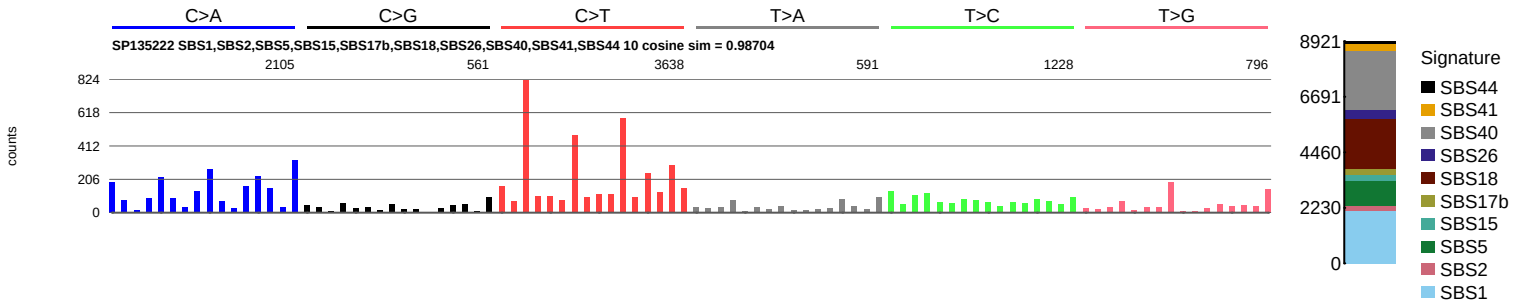
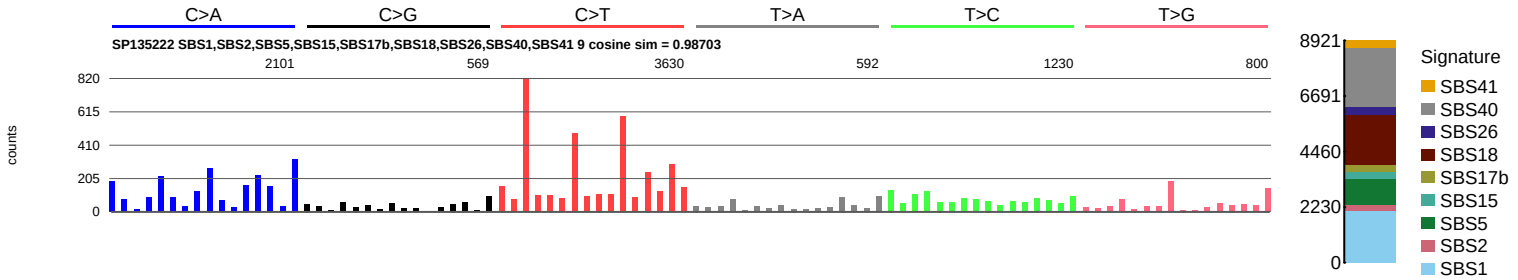
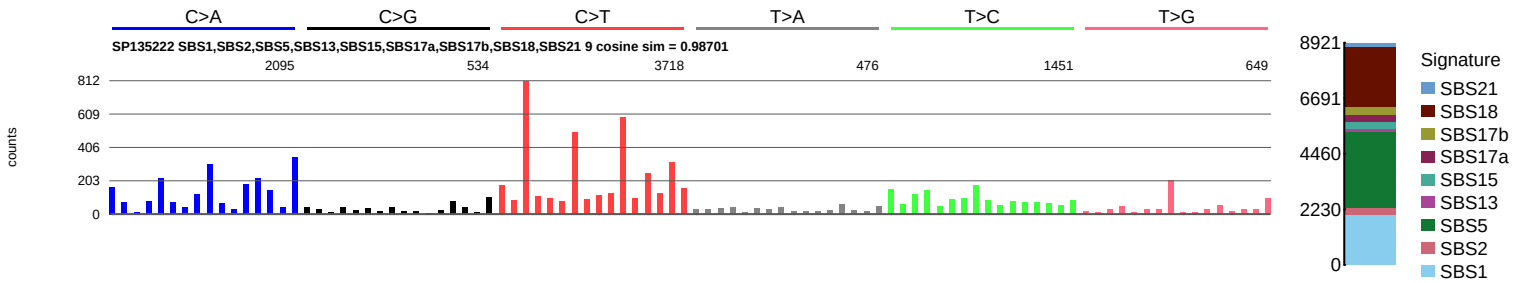
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98708
2057
464
3591
620
1472
722
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98705
2099
528
3729
476
1446
640
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98704
2105
561
3638
591
1228
796
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98703
2101
569
3630
592
1230
800
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98701
2095
534
3718
476
1451
649
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS21
SBS18
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SBS15
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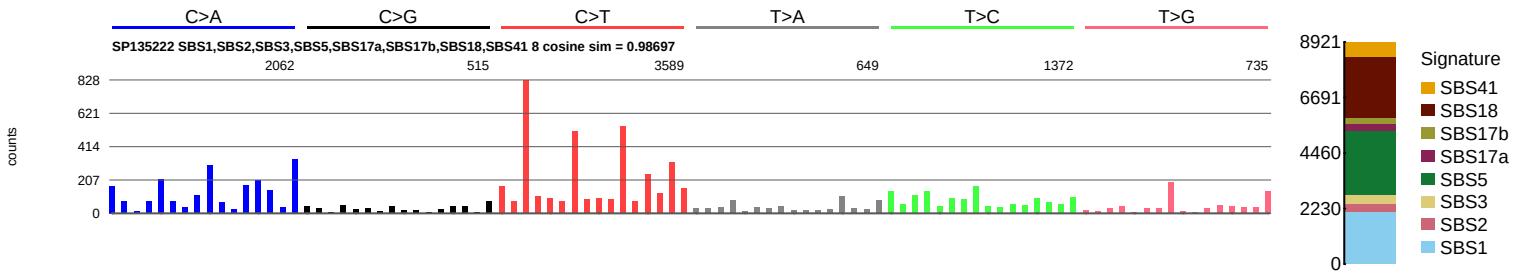
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98697
2062
515
3589
649
1372
735
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
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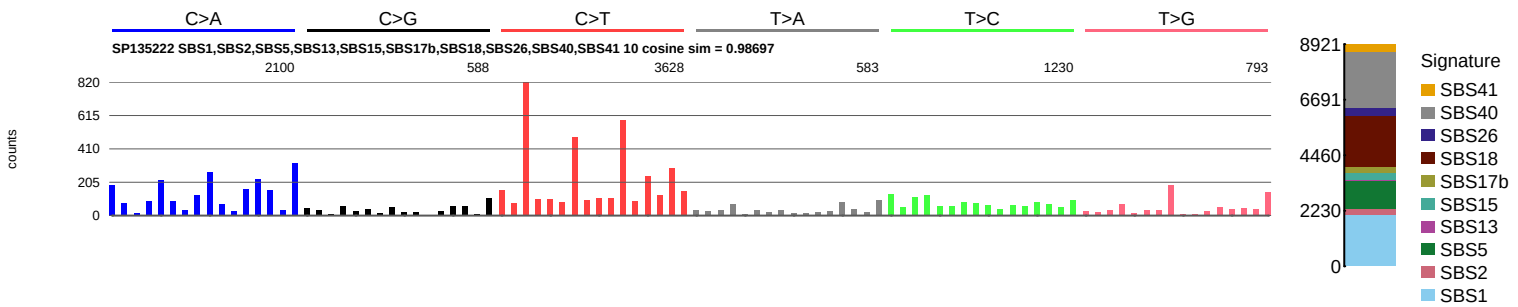
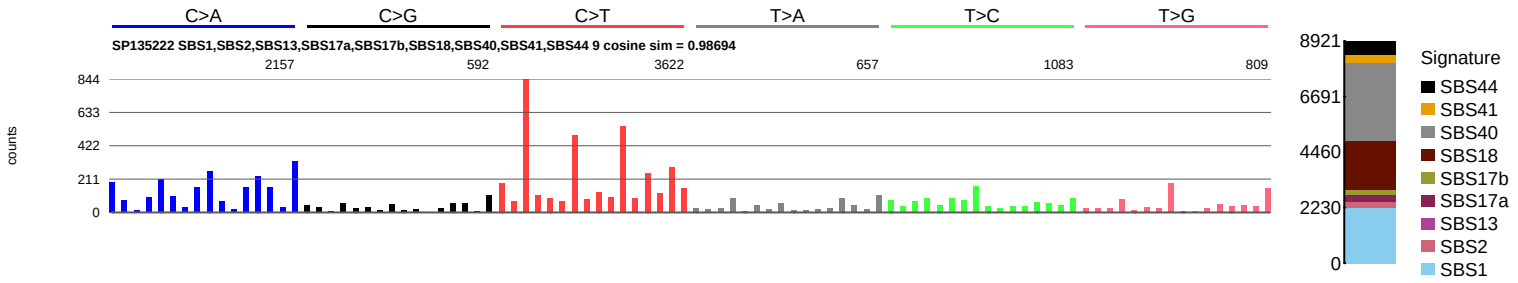
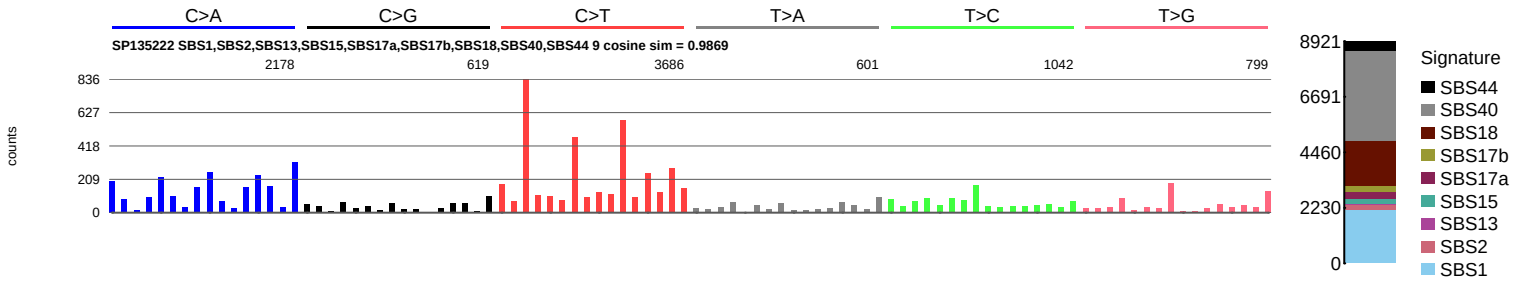
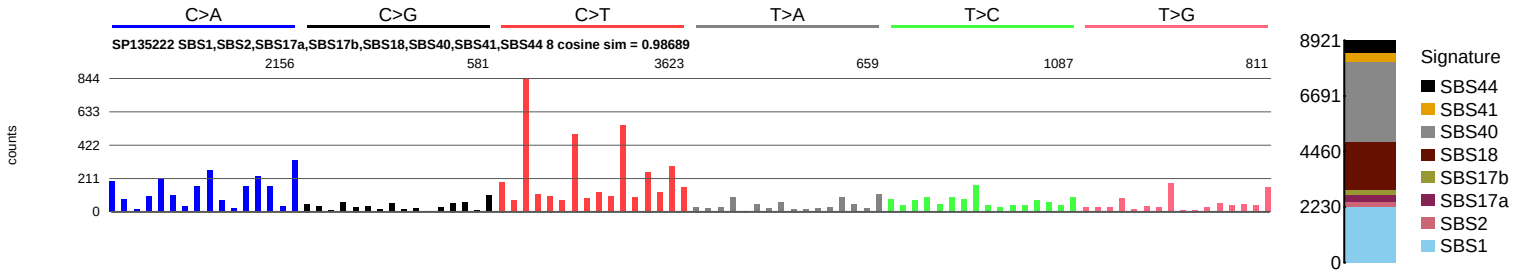
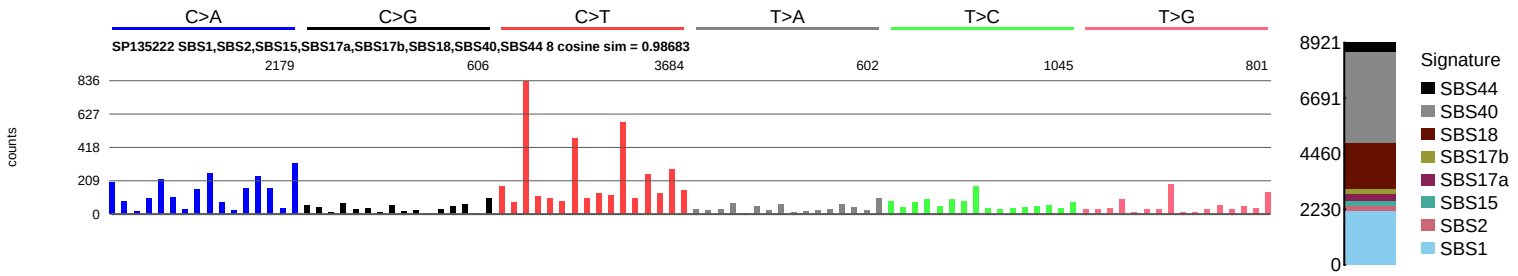
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98697
2100
588
3628
583
1230
793
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98694
2157
592
3622
657
1083
809
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.9869
2178
619
3686
601
1042
799
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98689
2156
581
3623
659
1087
811
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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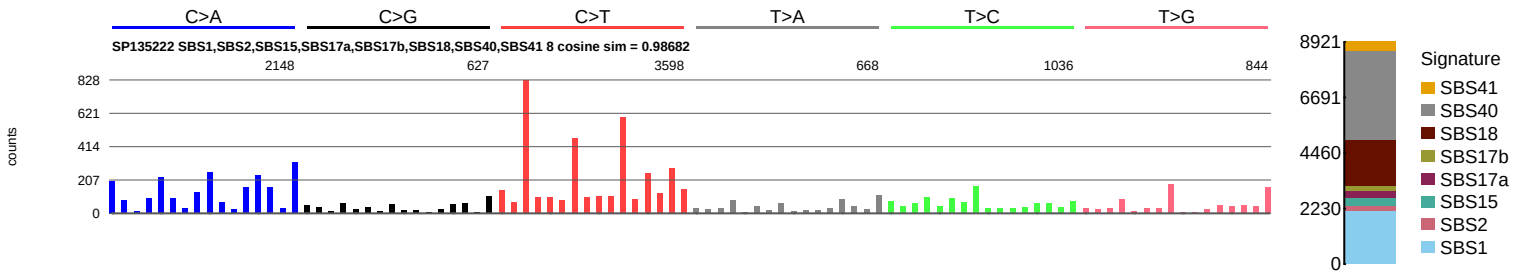
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98683
2179
606
3684
602
1045
801
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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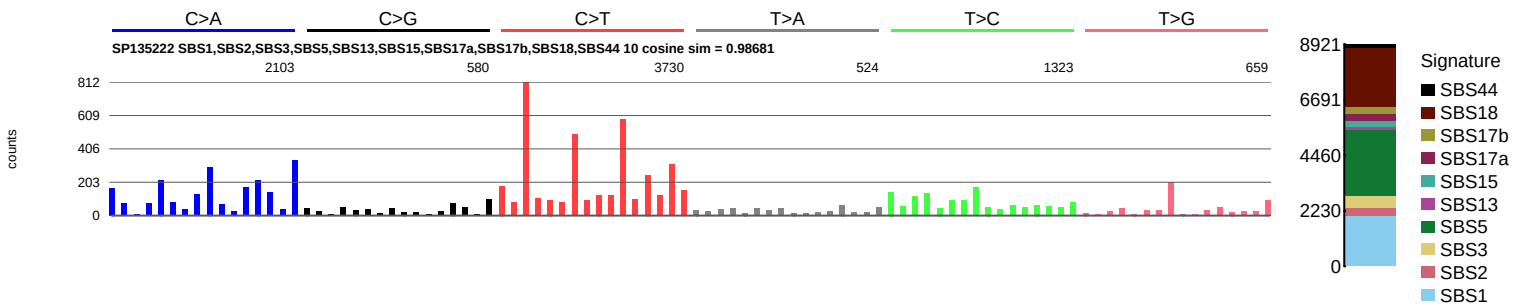
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98682
2148
627
3598
668
1036
844
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.98681
2103
580
3730
524
1323
659
counts
812
609
406
203
0
8921
6691
4460
2230
0
Signature
SBS44
SBS18
SBS17b
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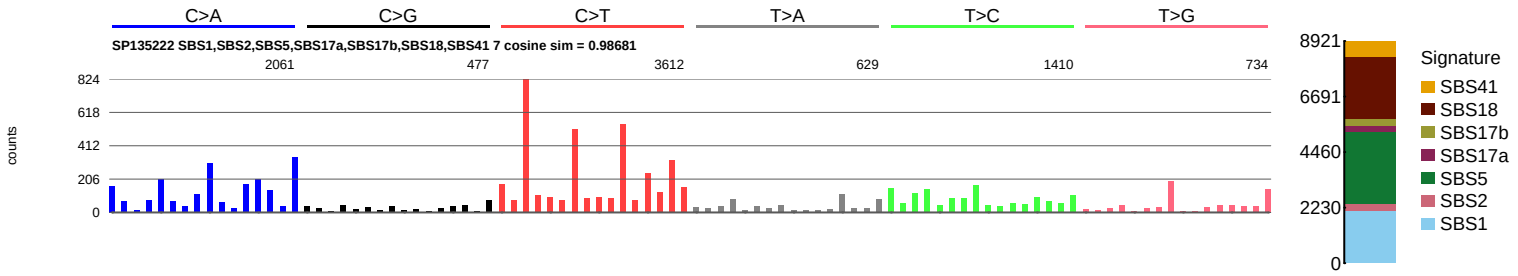
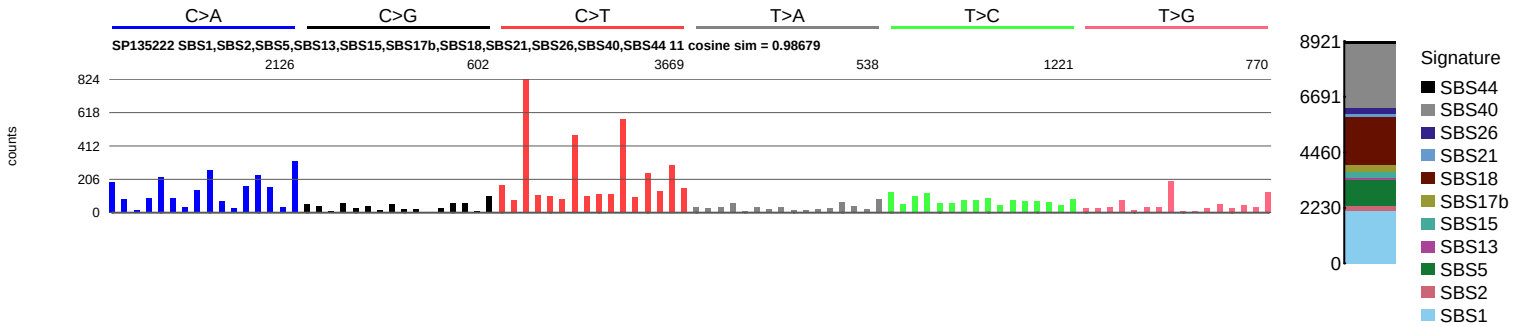
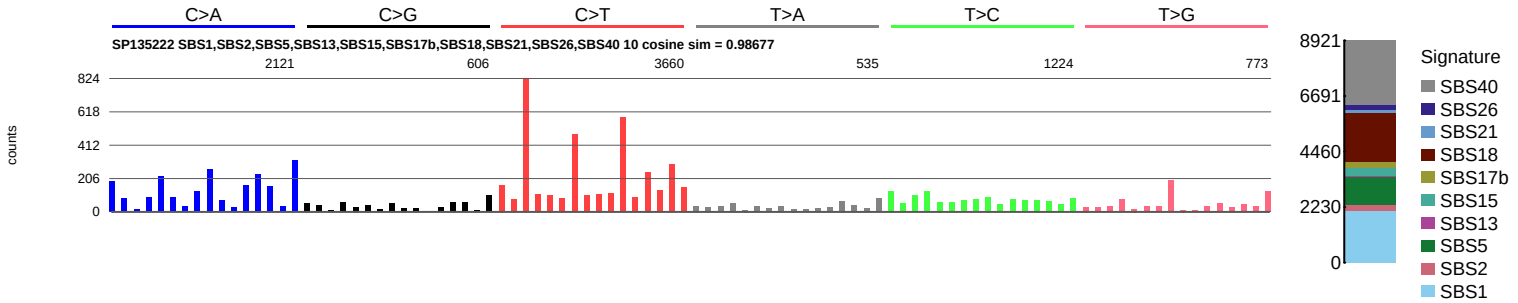
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98681
2061
477
3612
629
1410
734
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98679
2126
602
3669
538
1221
770
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98677
2121
606
3660
535
1224
773
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
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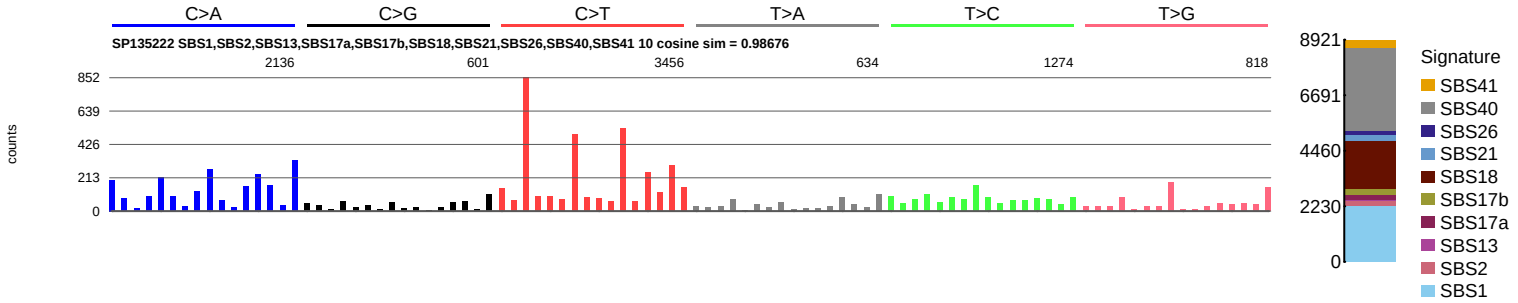
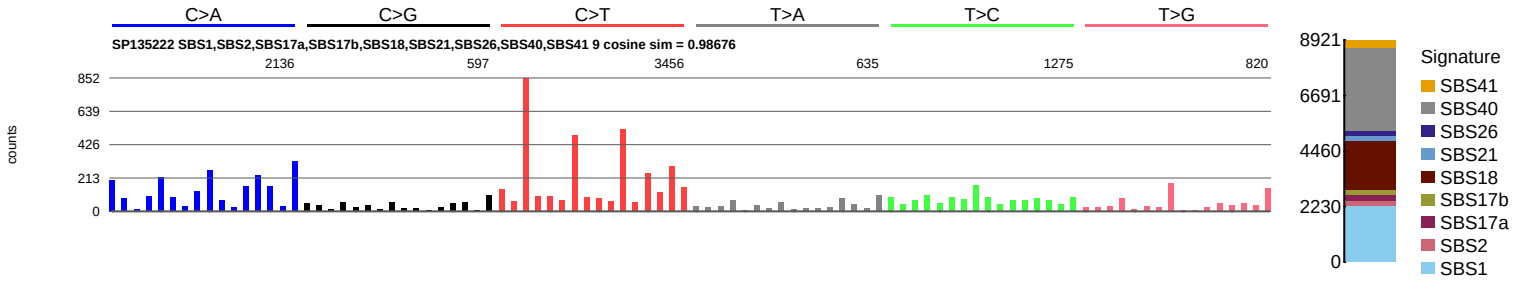
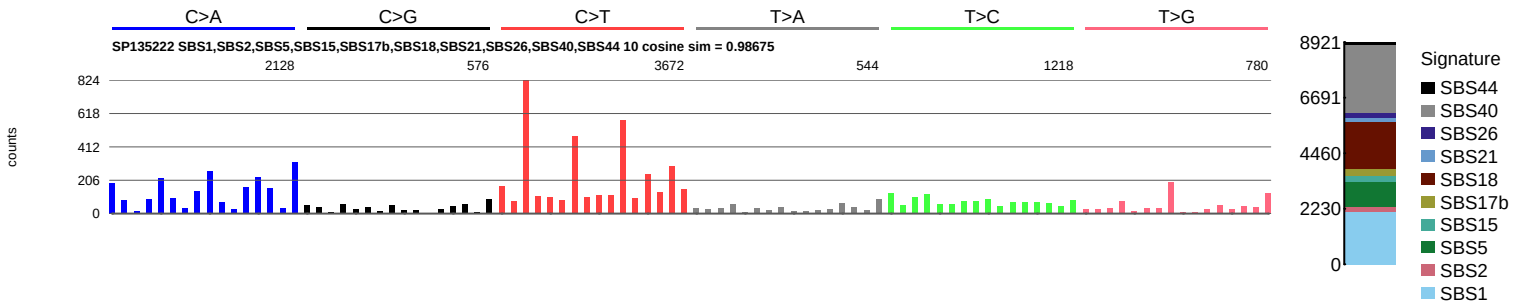
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98676
2136
601
3456
634
1274
818
counts
852
639
426
213
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98676
2136
597
3456
635
1275
820
counts
852
639
426
213
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>T
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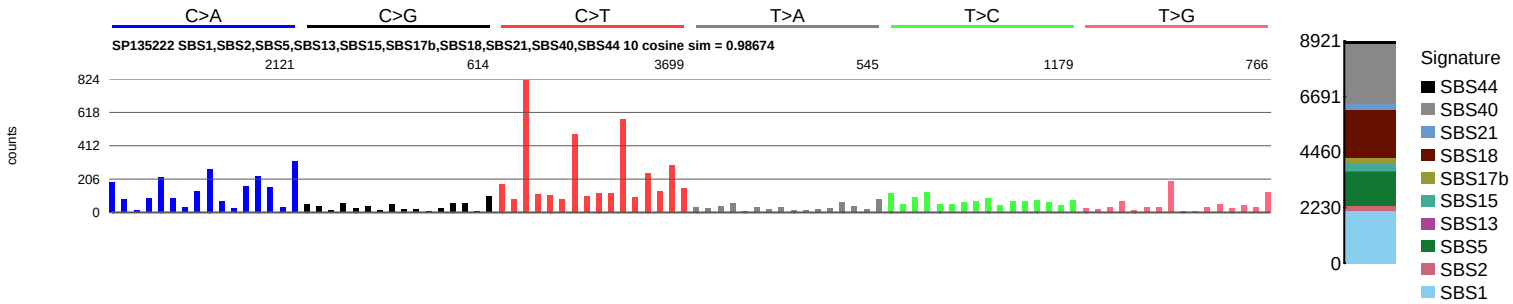
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98675
2128
576
3672
544
1218
780
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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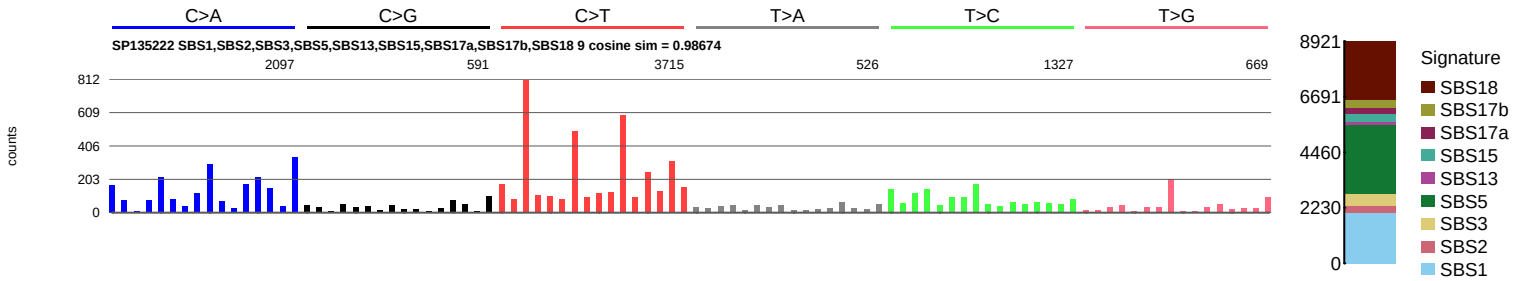
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98674
2121
614
3699
545
1179
766
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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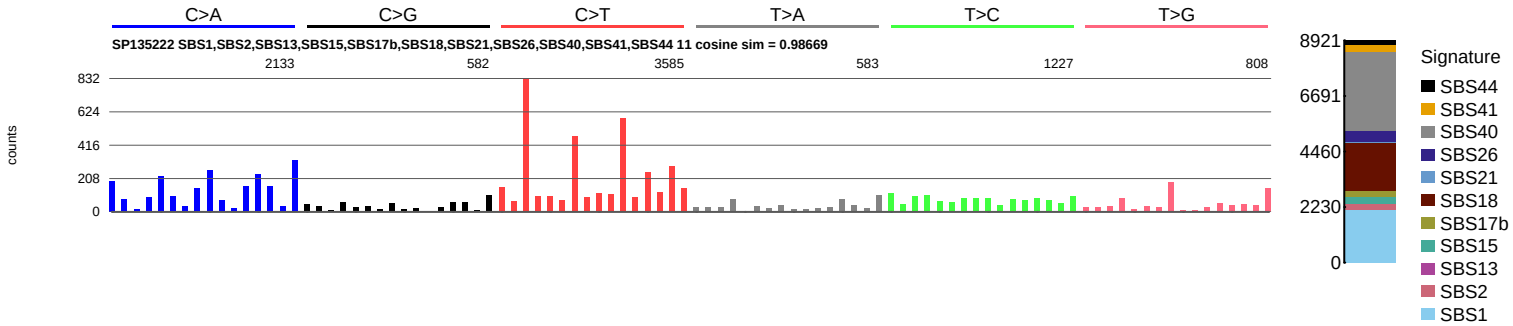
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.98674
2097
591
3715
526
1327
669
counts
812
609
406
203
0
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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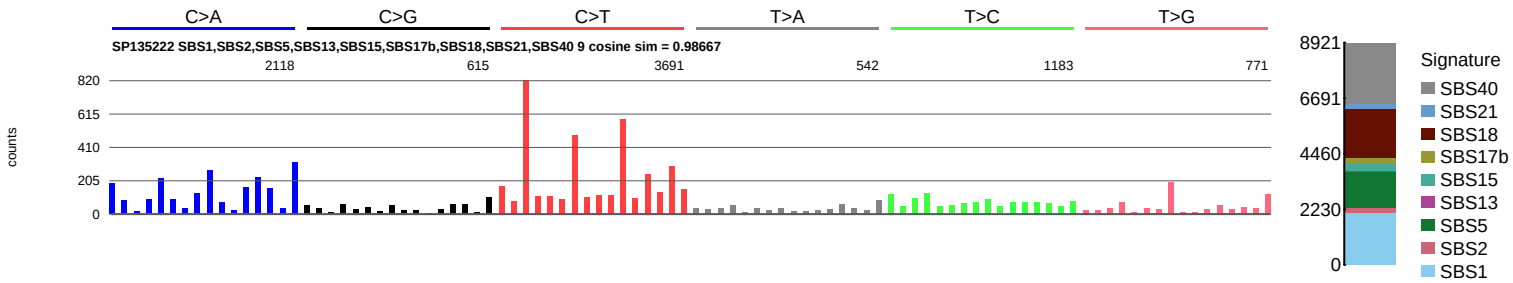
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98669
2133
582
3585
583
1227
808
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1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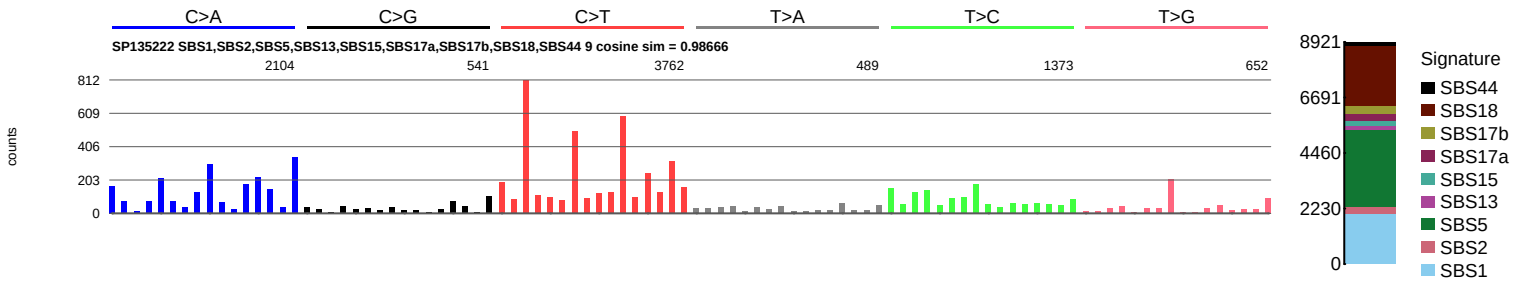
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98667
2118
615
3691
542
1183
771
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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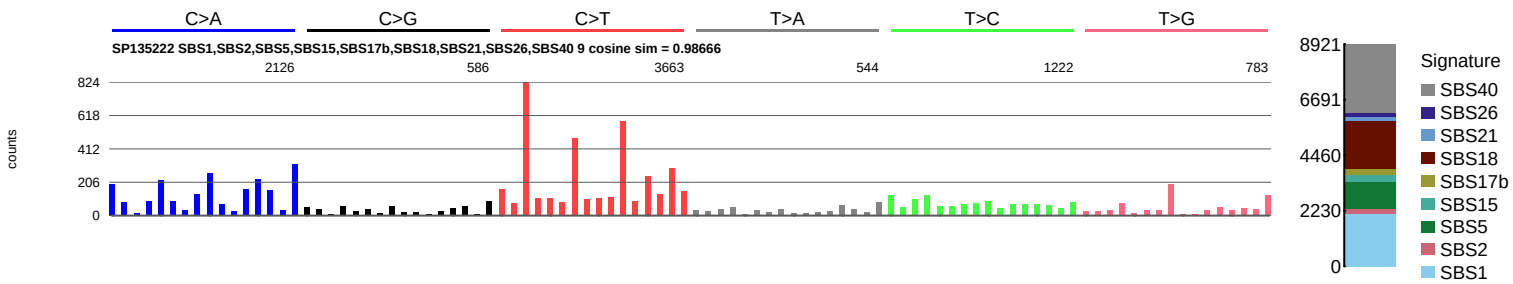
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98666
2104
541
3762
489
1373
652
counts
812
609
406
203
0
8921
6691
4460
2230
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98666
2126
586
3663
544
1222
783
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
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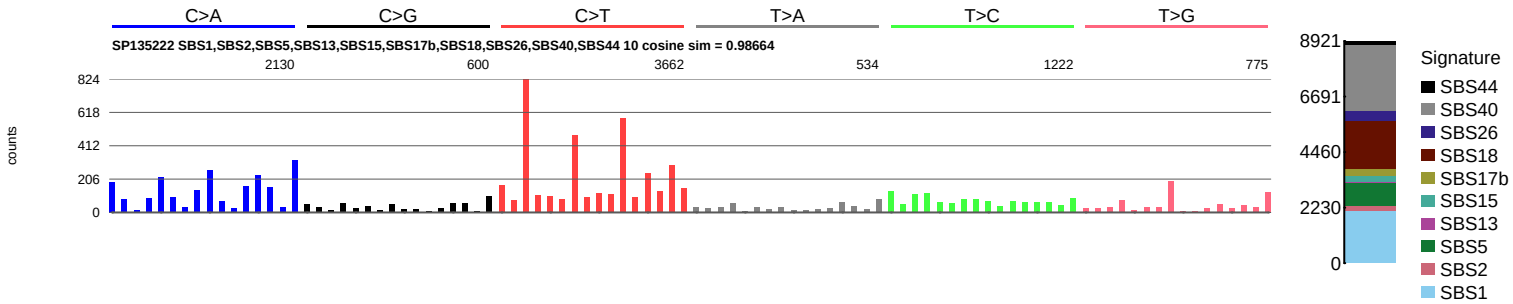
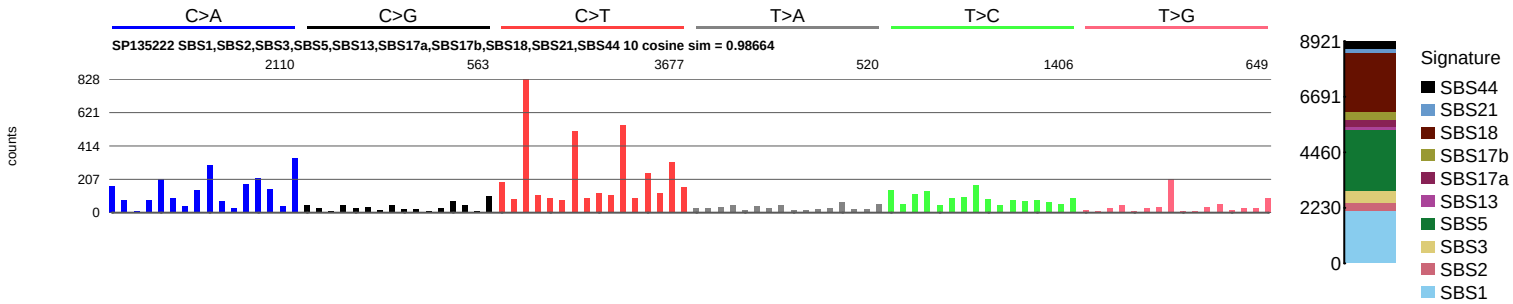
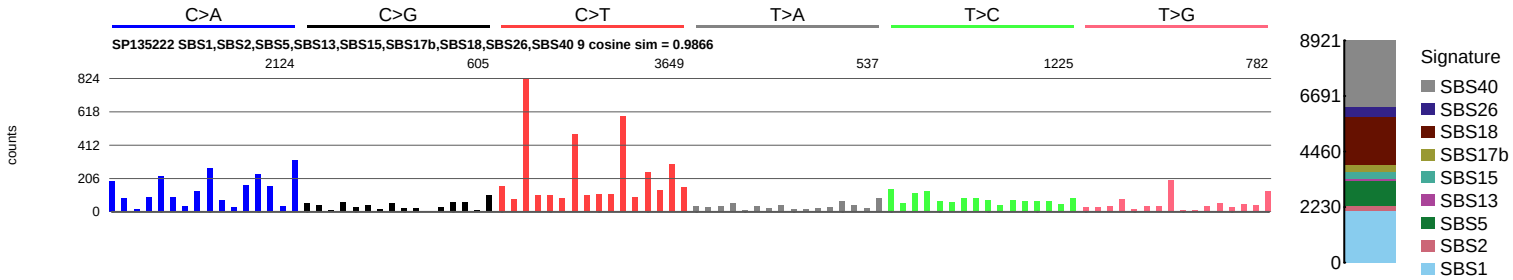
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98664
2130
600
3662
534
1222
775
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98664
2110
563
3677
520
1406
649
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.9866
2124
605
3649
537
1225
782
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
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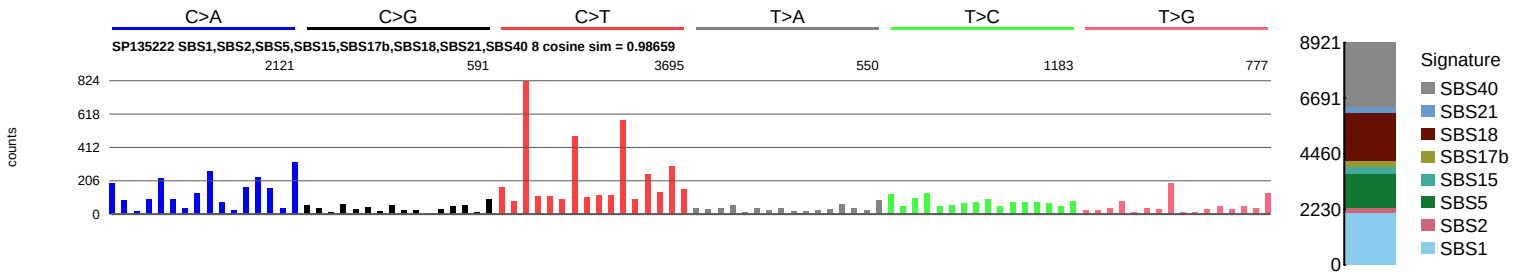
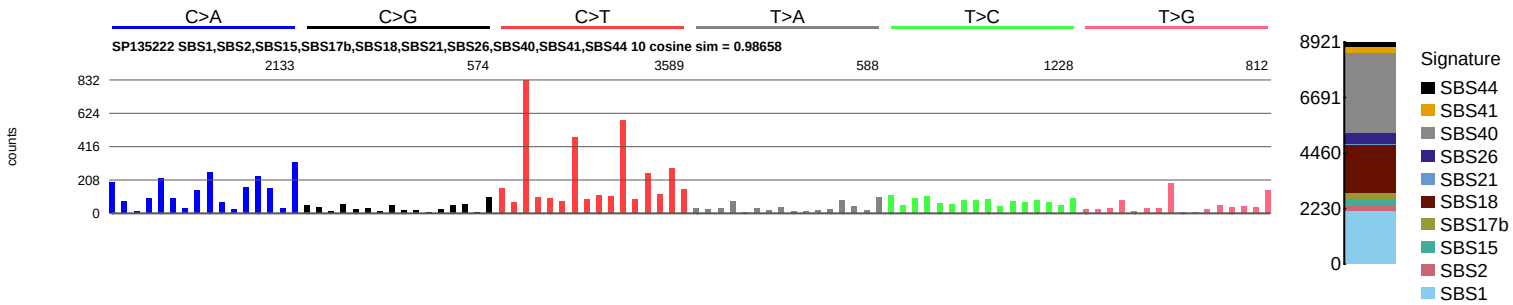
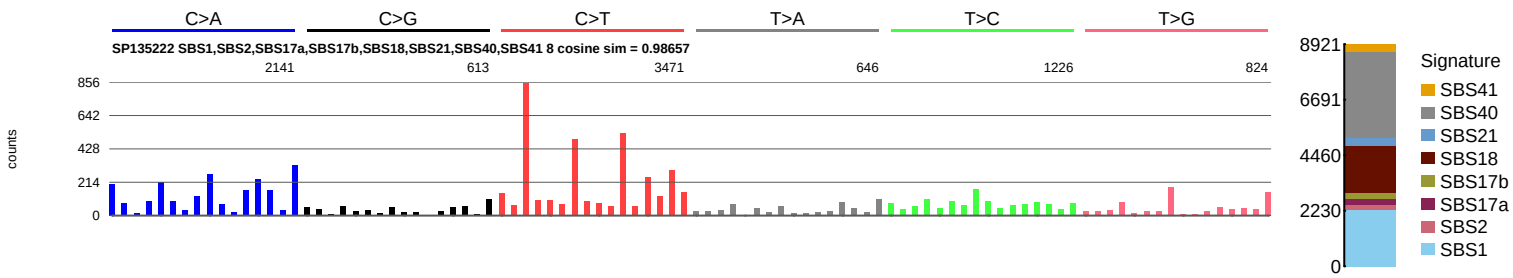
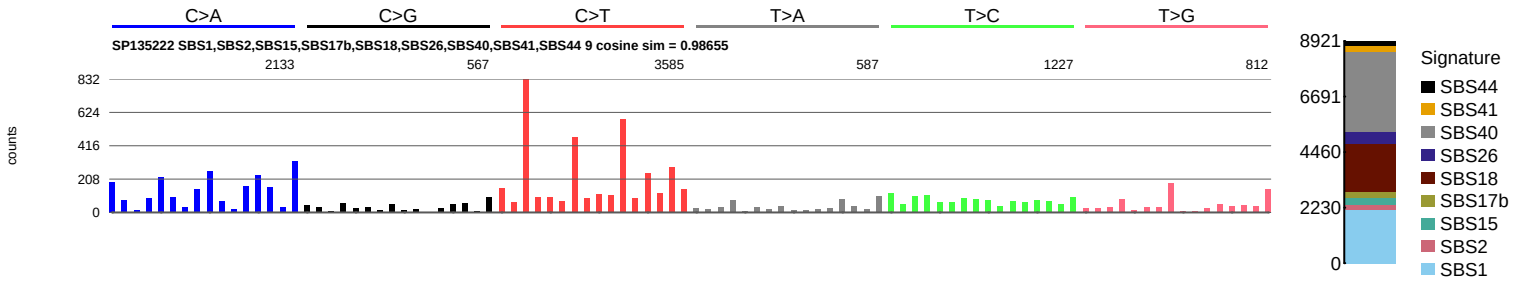
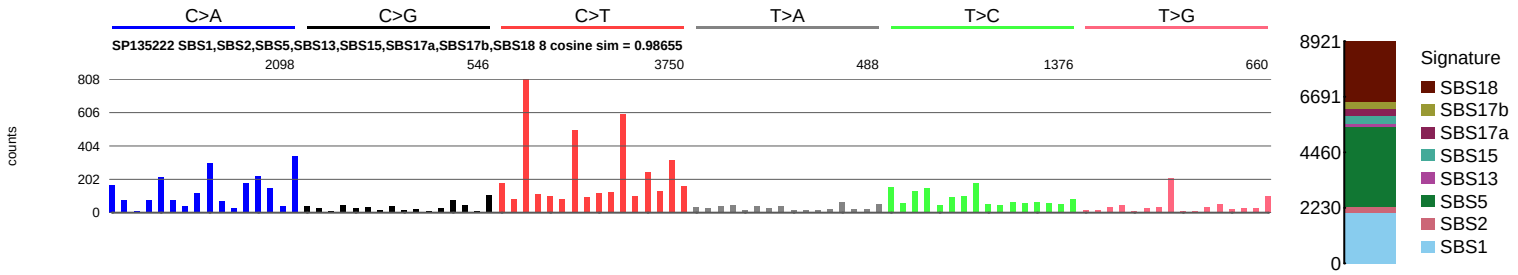
SBS5
SBS2
SBS1

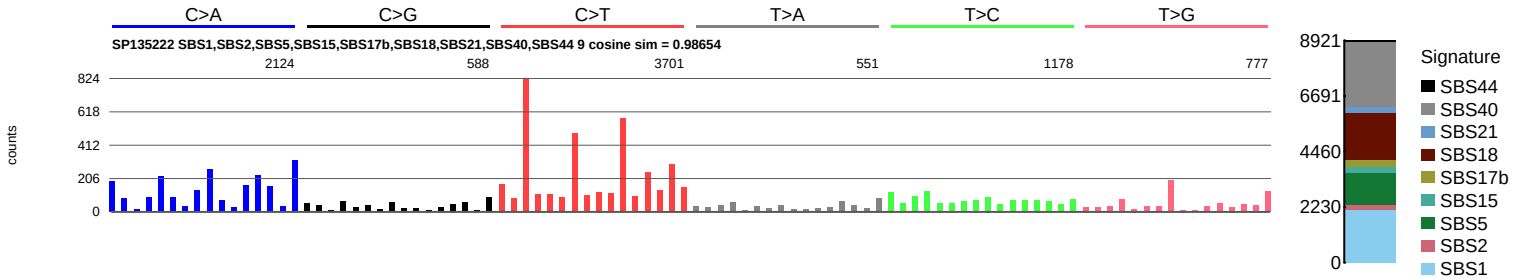

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98659
2121
591
3695
550
1183
777
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98658
2133
574
3589
588
1228
812
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98657
2141
613
3471
646
1226
824
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98655
2133
567
3585
587
1227
812
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98655
2098
546
3750
488
1376
660
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98654
2124
588
3701
551
1178
777
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
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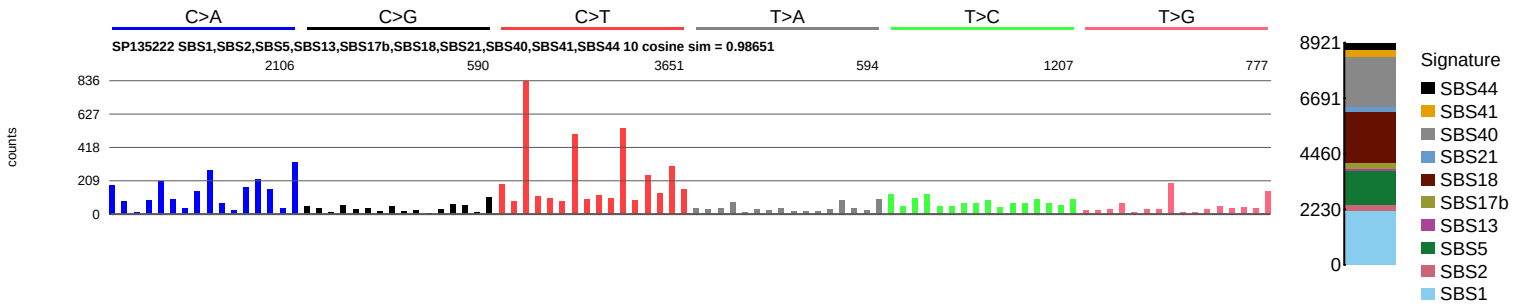
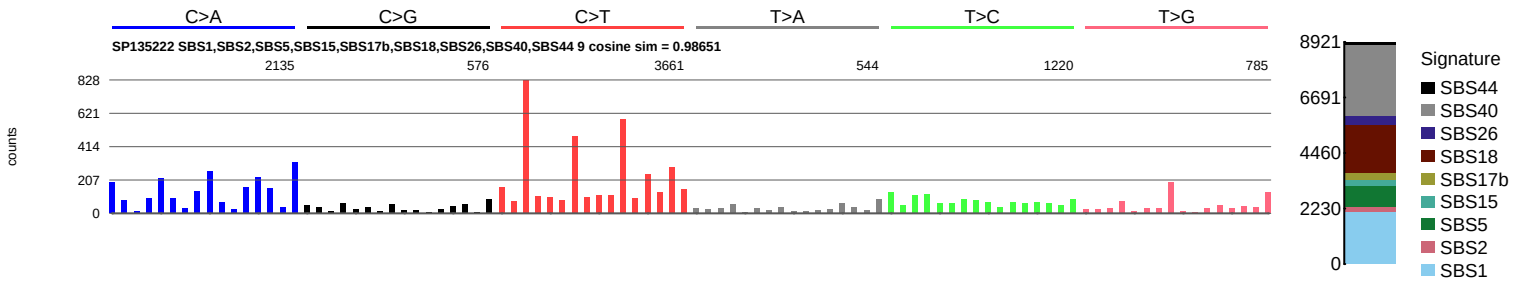
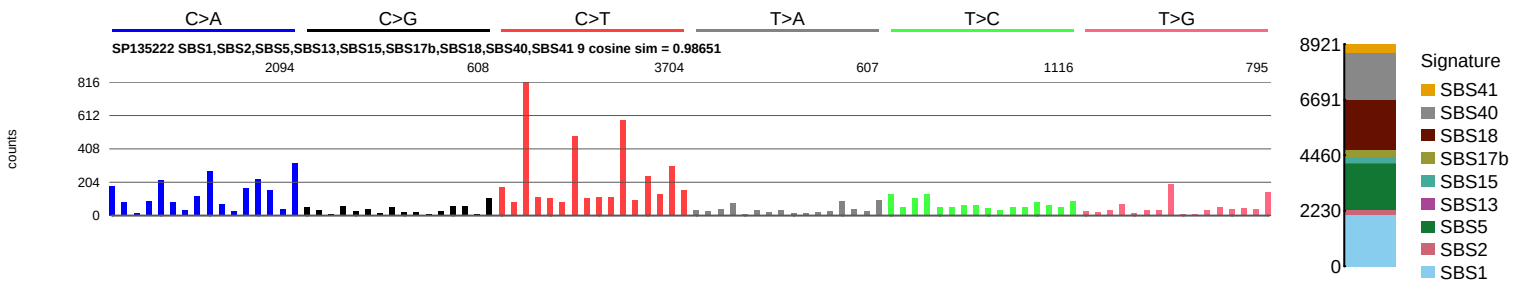
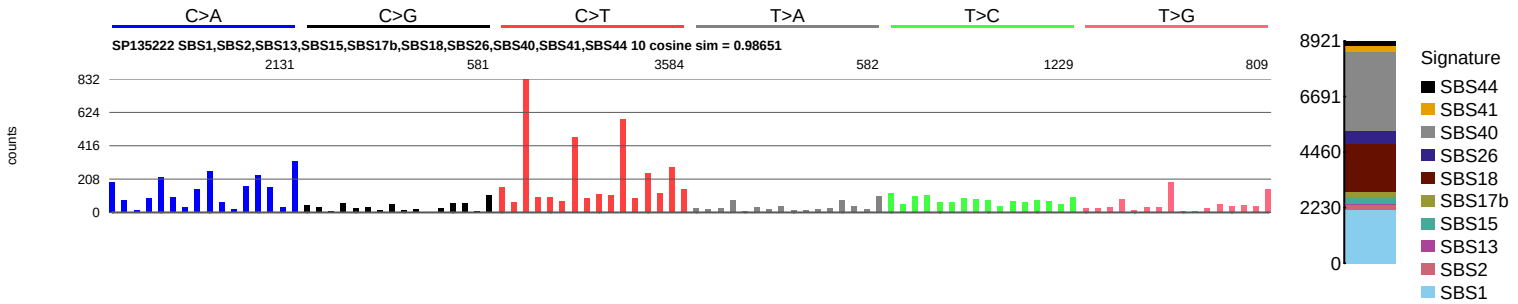
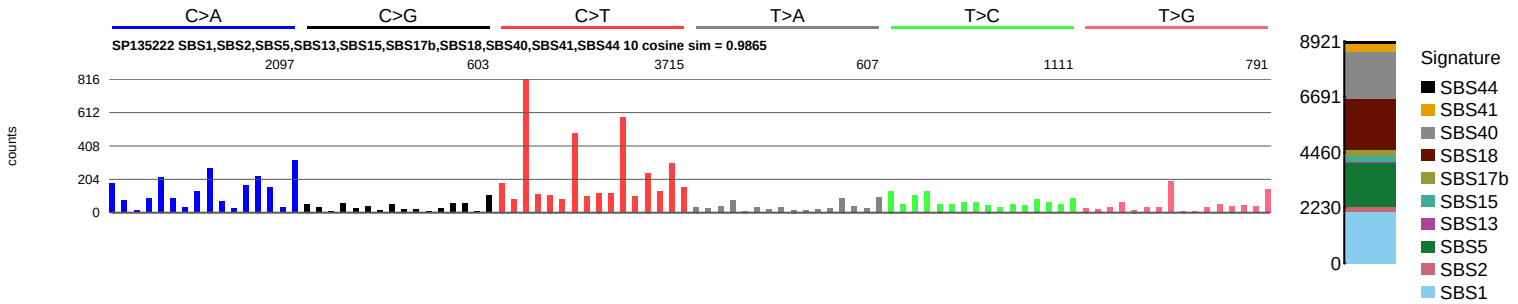
SBS5
SBS2
SBS1

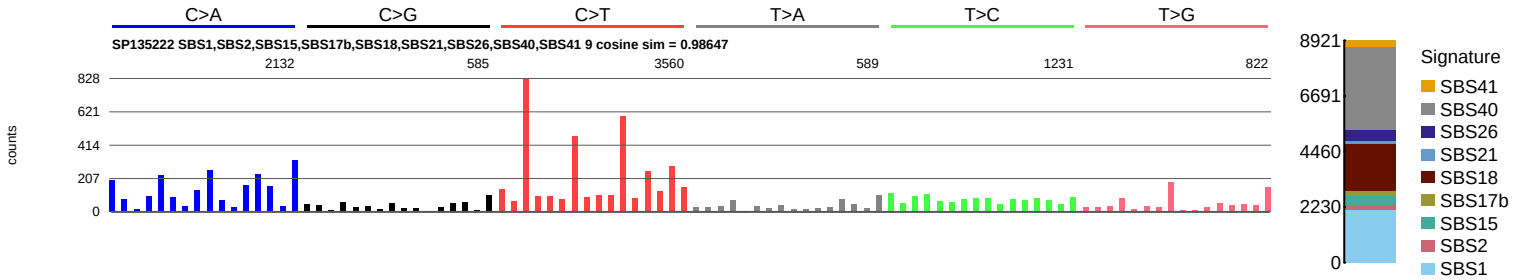

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98651
2106
590
3651
594
1207
777
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

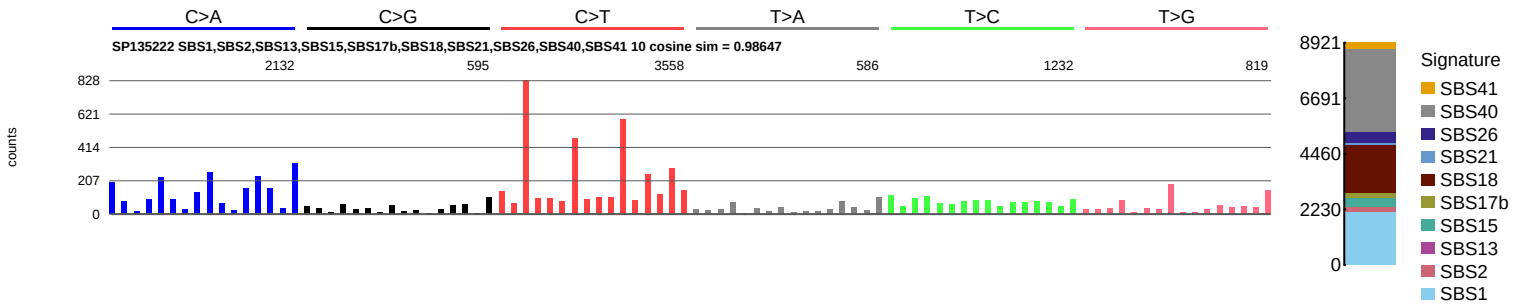

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98651
2135
576
3661
544
1220
785
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

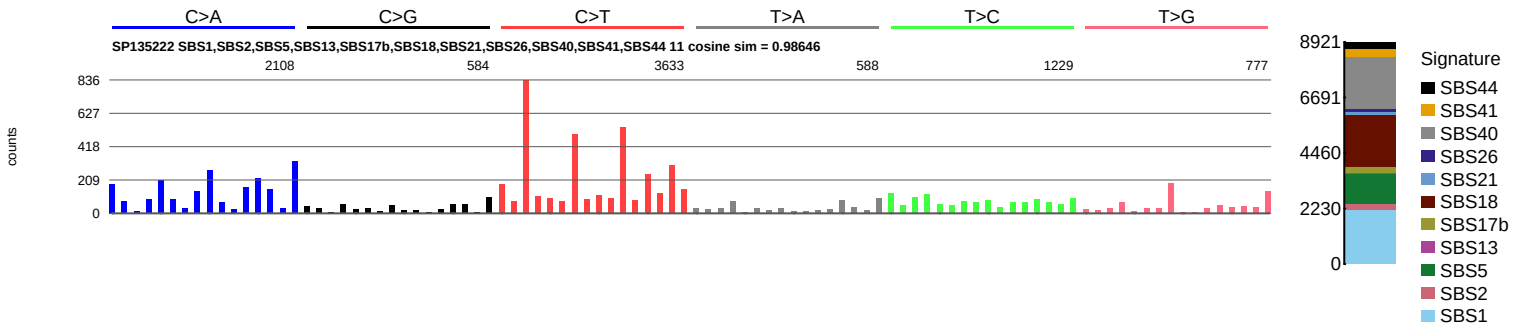

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98651
2094
608
3704
607
1116
795
816
612
408
204
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

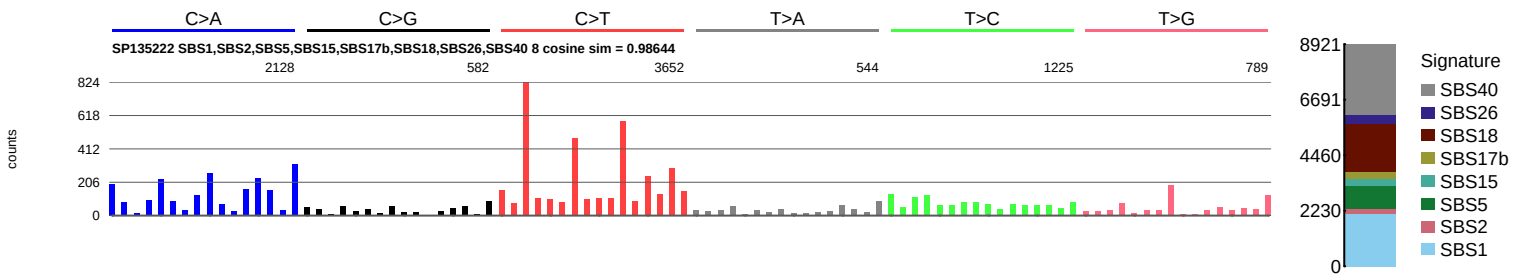

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98651
2131
581
3584
582
1229
809
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.9865
2097
603
3715
607
1111
791
816
612
408
204
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98647
2132
585
3560
589
1231
822
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98647
2132
595
3558
586
1232
819
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98646
2108
584
3633
588
1229
777
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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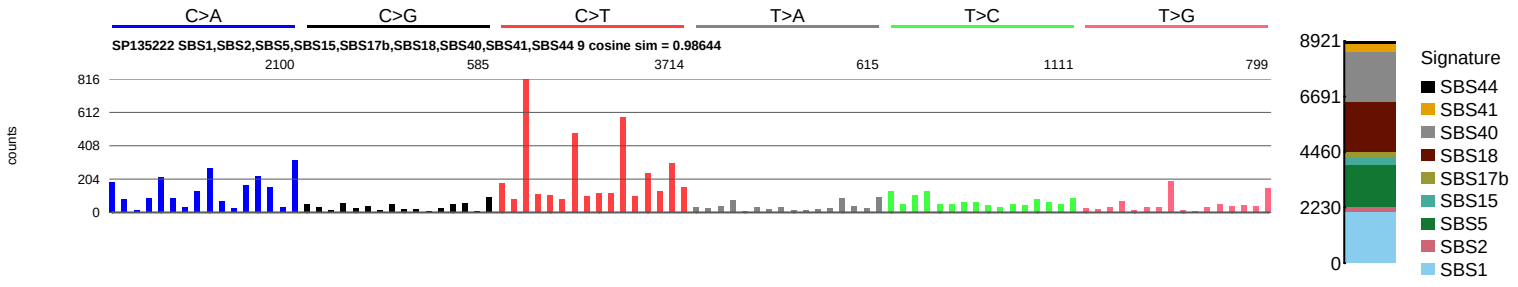
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98644
2128
582
3652
544
1225
789
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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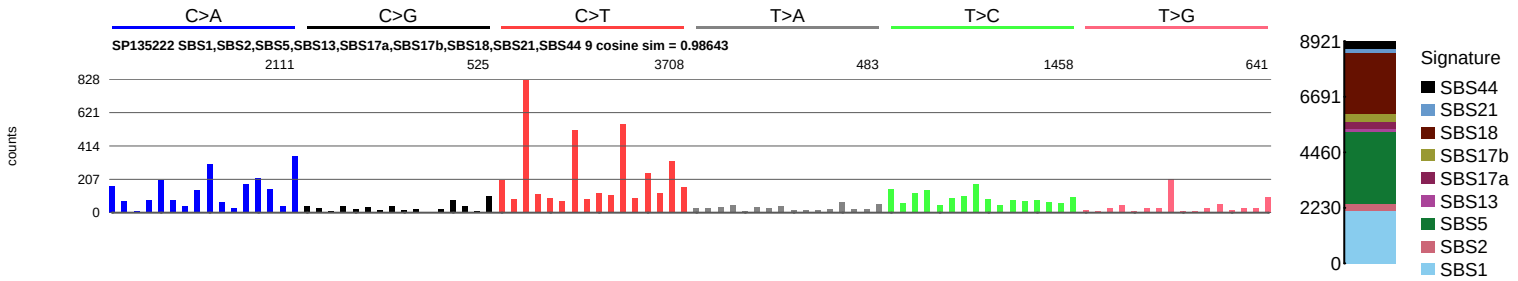
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98644
2100
585
3714
615
1111
799
816
612
408
204
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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C>T
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T>C
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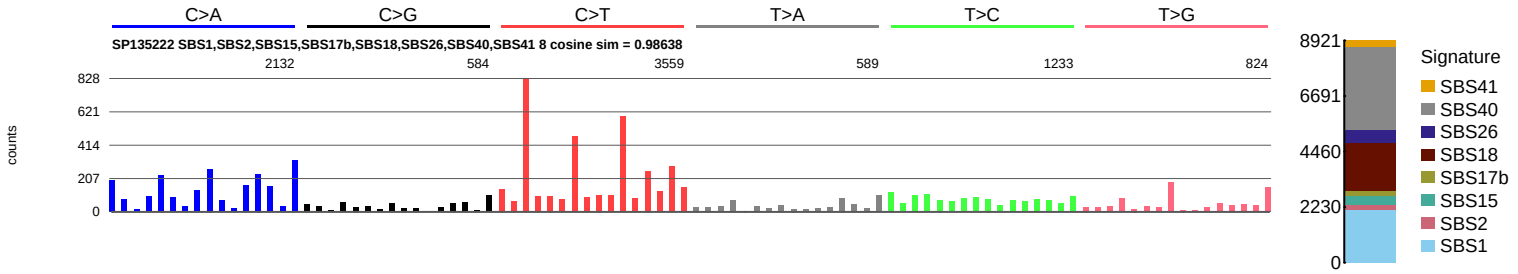
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98643
2111
525
3708
483
1458
641
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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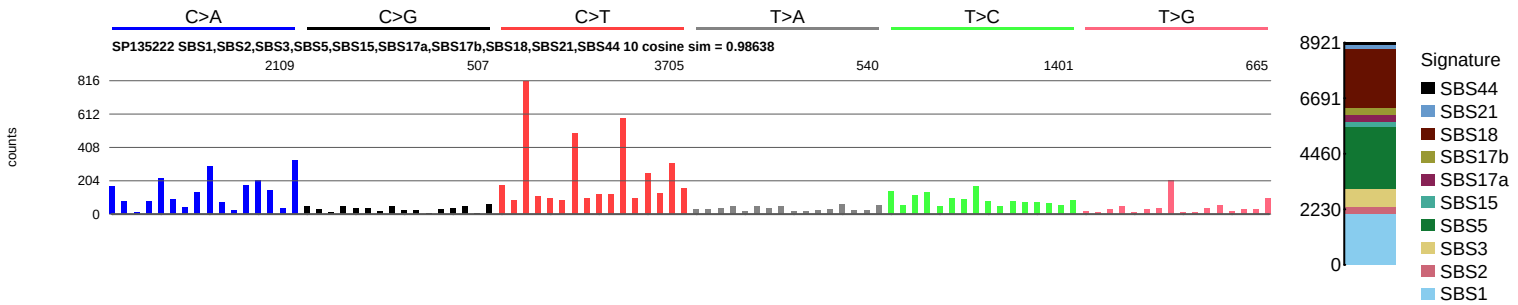
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98638
2132
584
3559
589
1233
824
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1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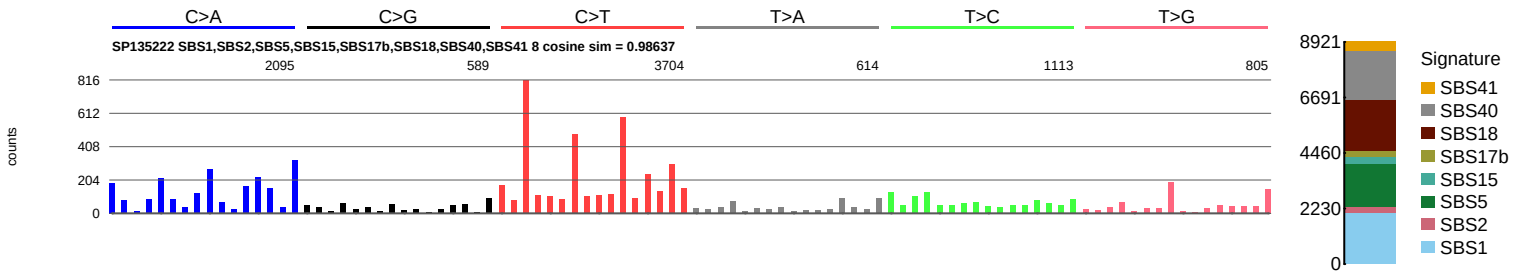
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98638
2109
507
3705
540
1401
665
816
612
408
204
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98637
2095
589
3704
614
1113
805
816
612
408
204
0
counts
8921
6691
4460
2230
0
Signature
SBS41
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SBS18
SBS17b
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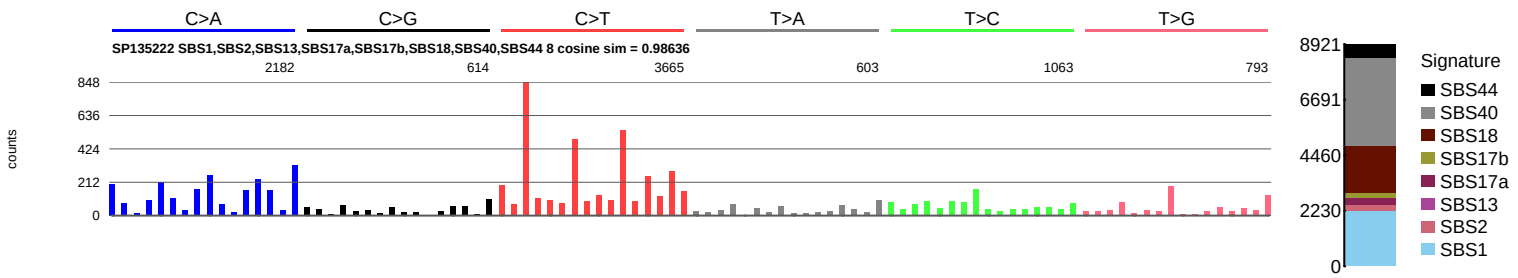
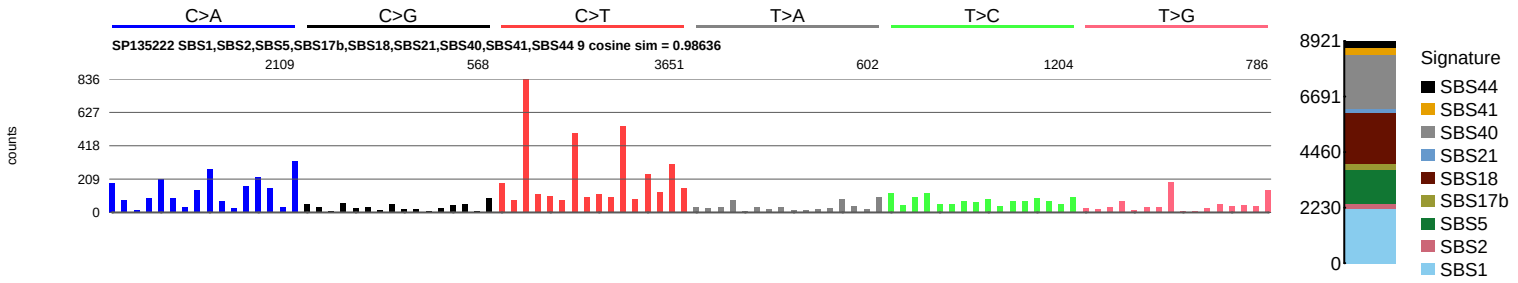
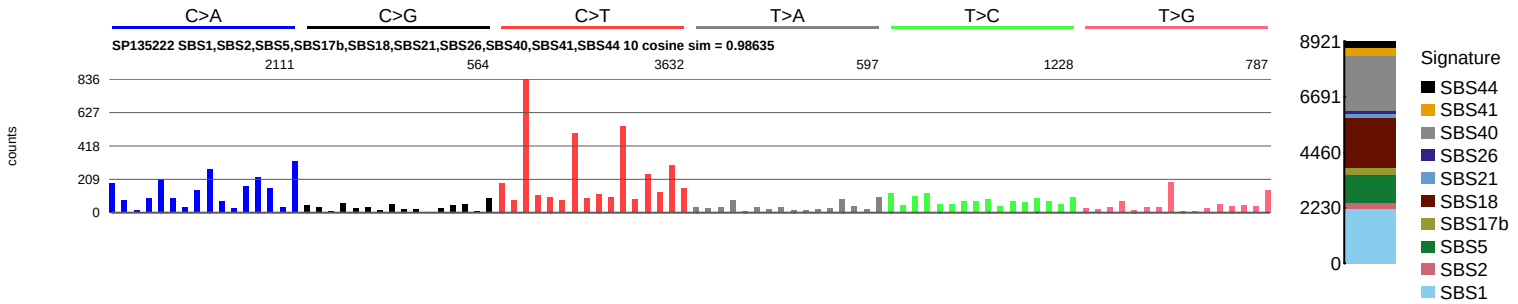
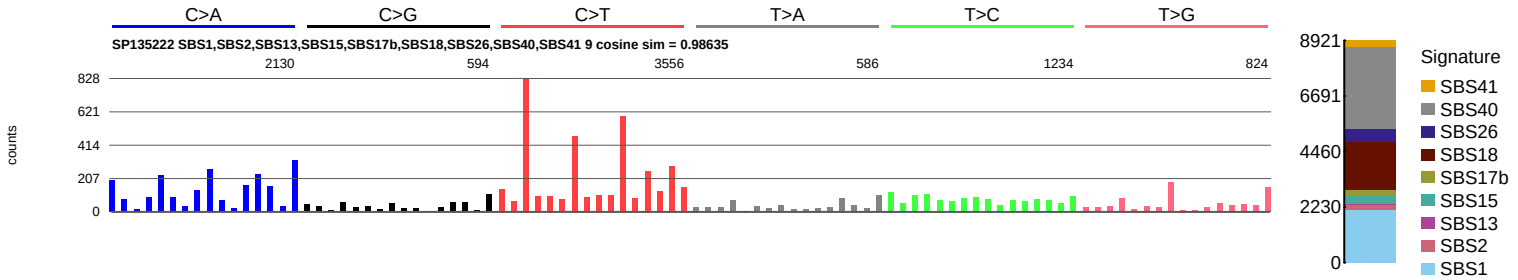
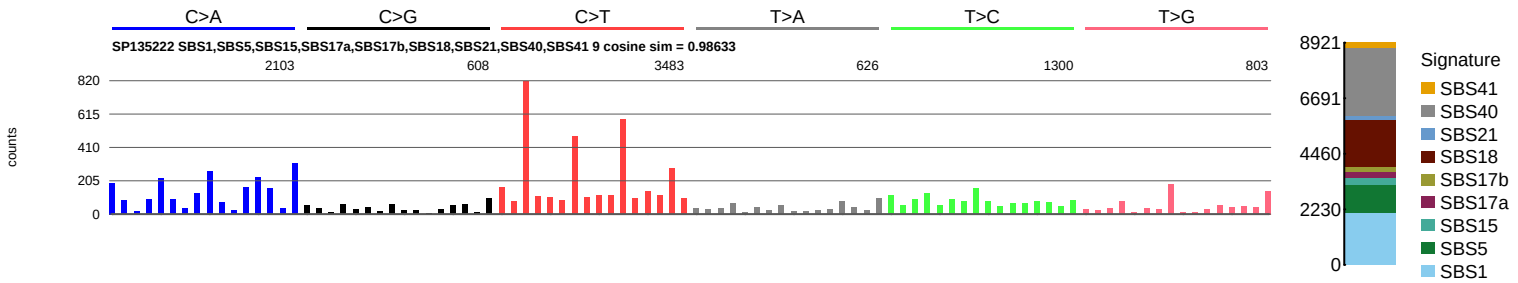
SBS5
SBS2
SBS1

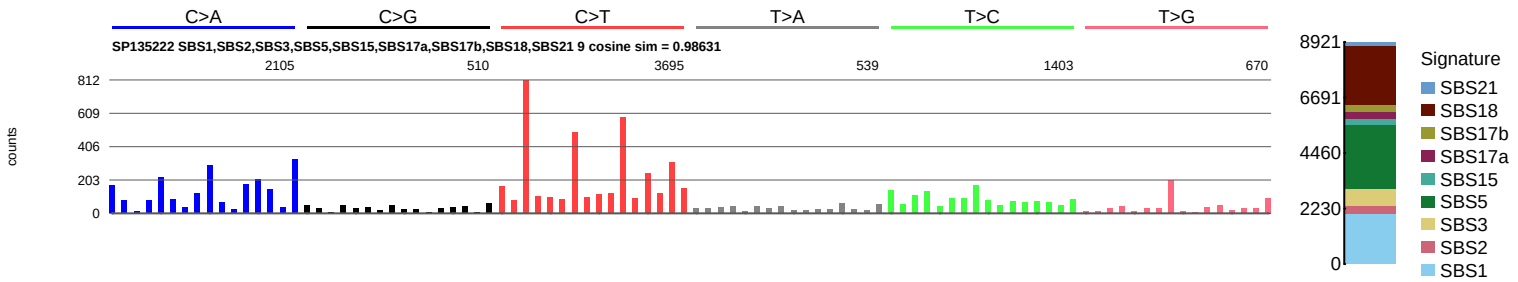

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98636
2182
614
3665
603
1063
793
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

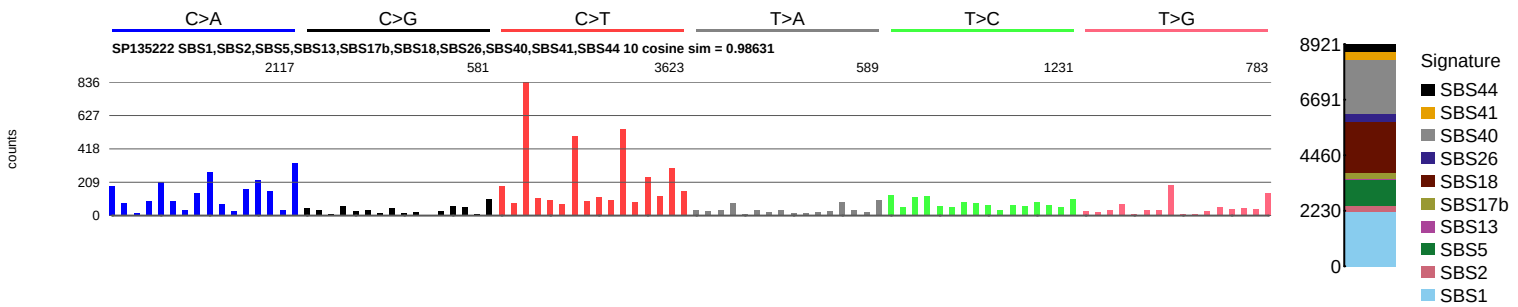

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98636
2109
568
3651
602
1204
786
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1

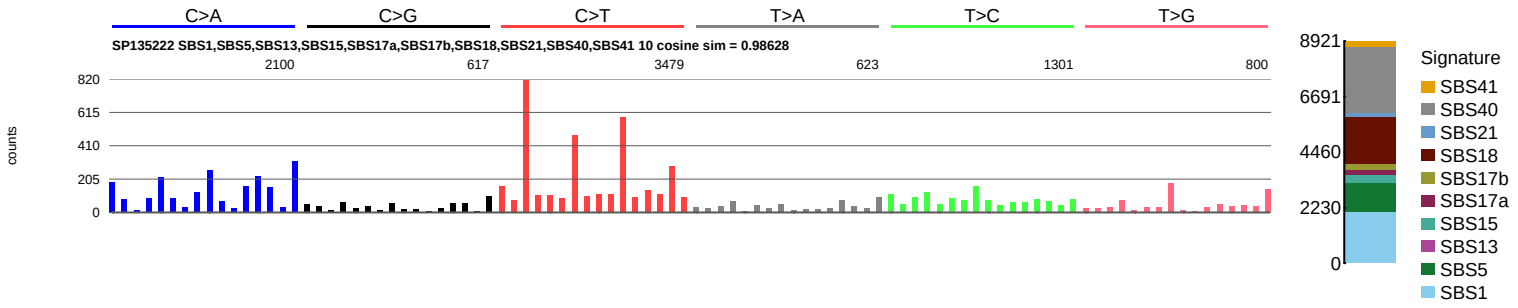

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98635
2111
564
3632
597
1228
787
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1

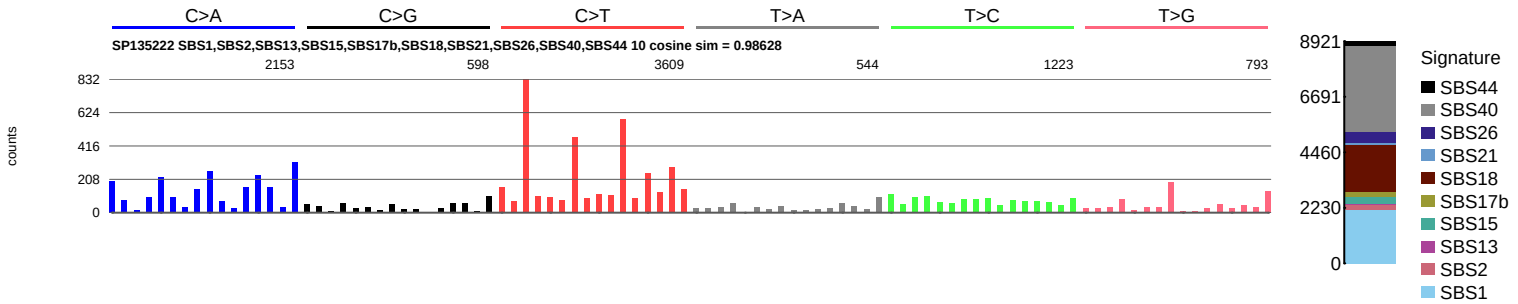

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98635
2130
594
3556
586
1234
824
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

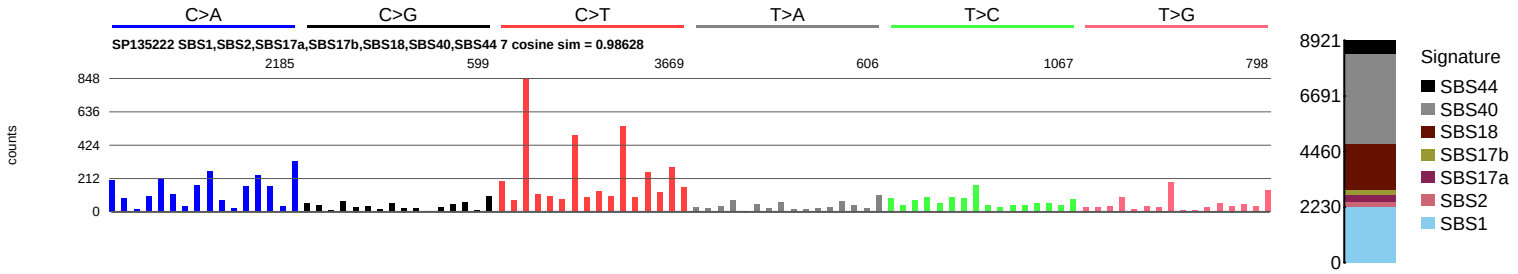

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98633
2103
608
3483
626
1300
803
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

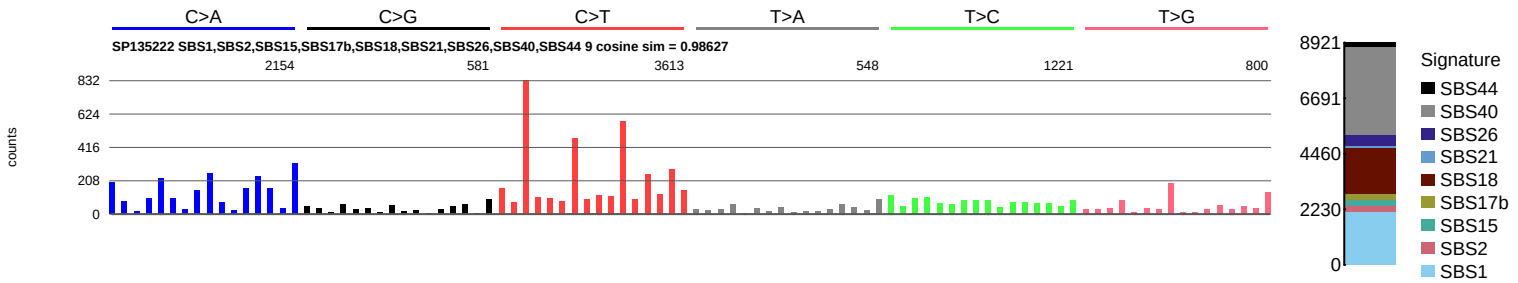

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98631
2105
510
3695
539
1403
670
counts
812
609
406
203
0
8921
6691
4460
2230
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

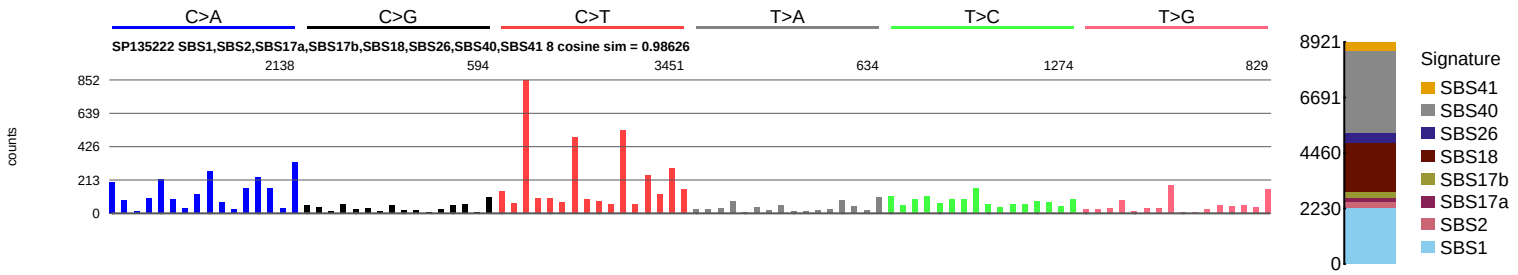

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98631
2117
581
3623
589
1231
783
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

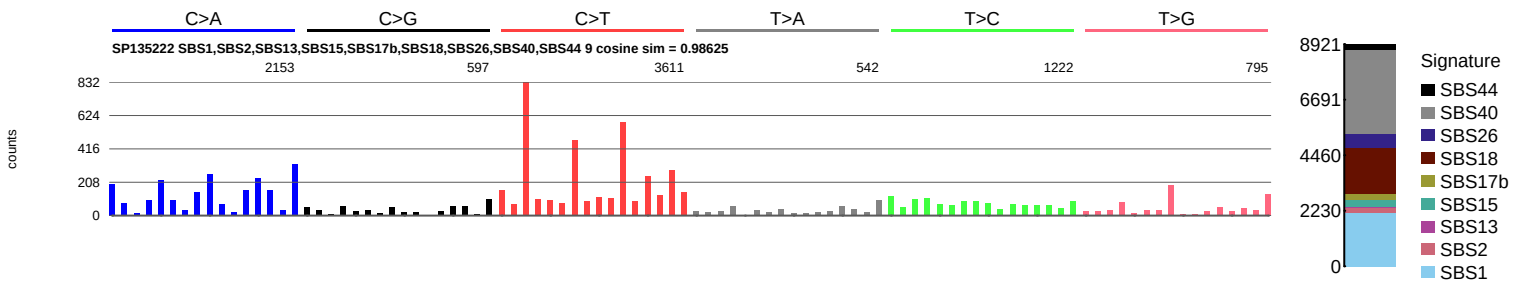

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98628
2100
617
3479
623
1301
800
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98628
2153
598
3609
544
1223
793
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98628
2185
599
3669
606
1067
798
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98627
2154
581
3613
548
1221
800
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98626
2138
594
3451
634
1274
829
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98625
2153
597
3611
542
1222
795
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
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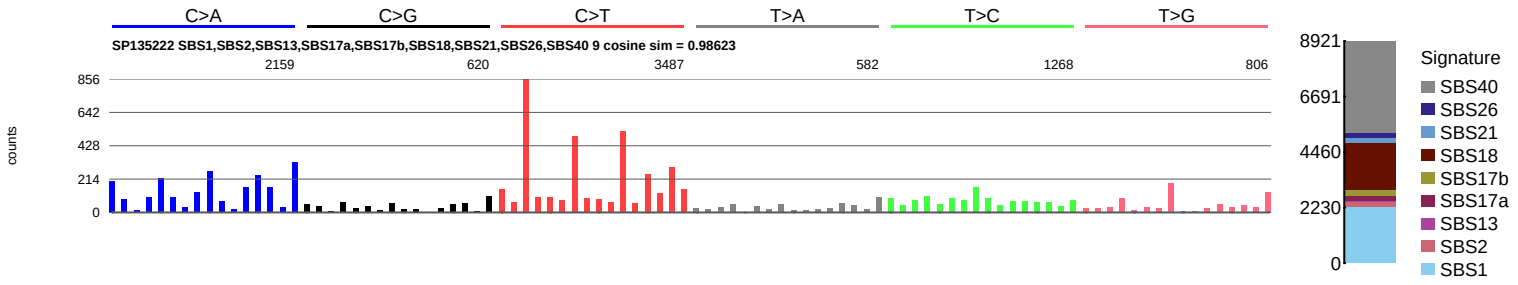
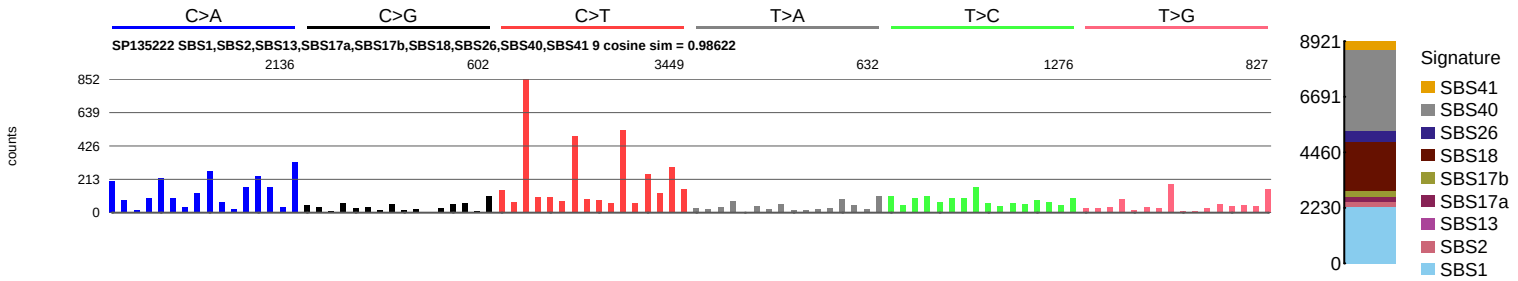
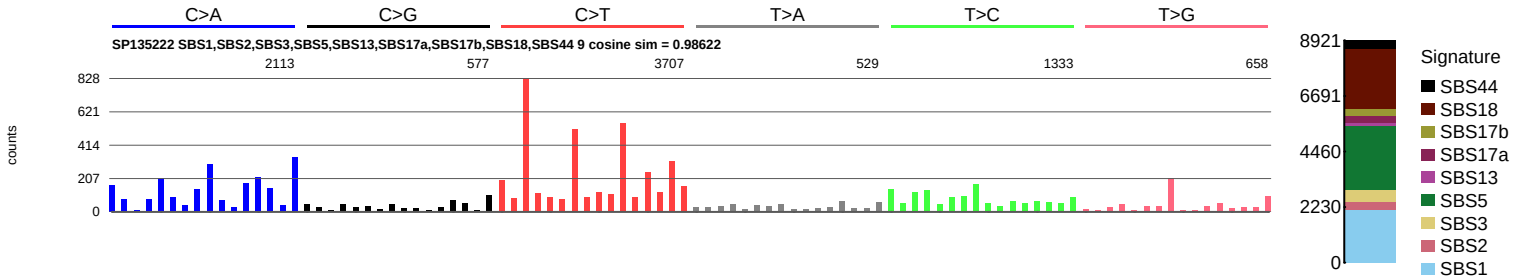
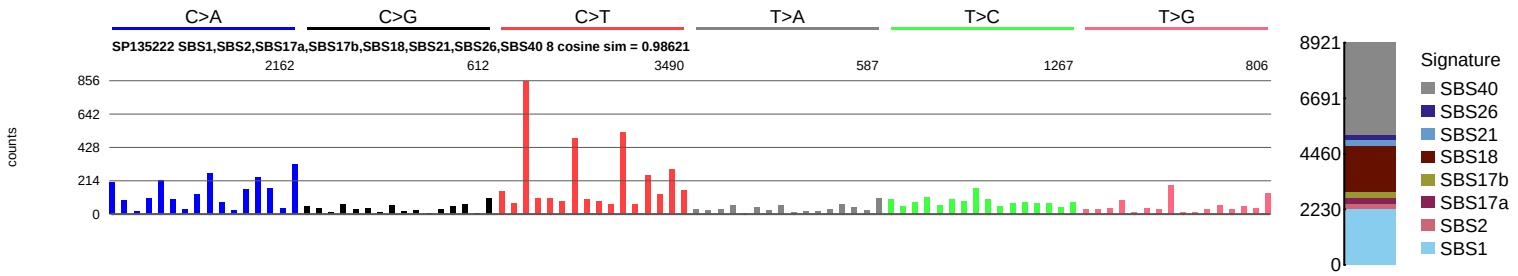
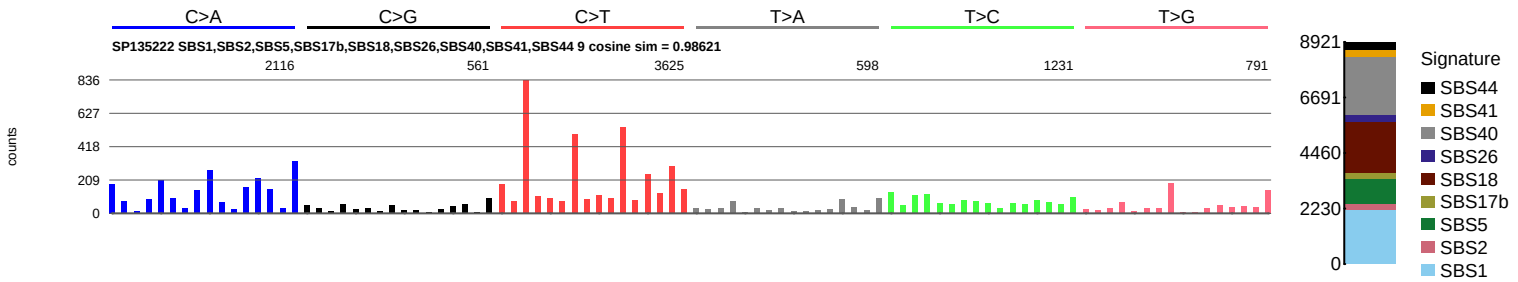
SBS13
SBS2
SBS1

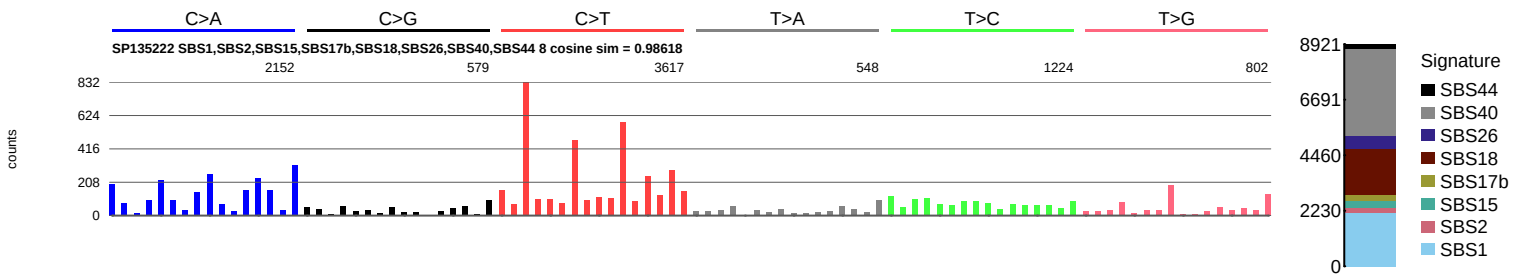

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98623
2159
620
3487
582
1268
806
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

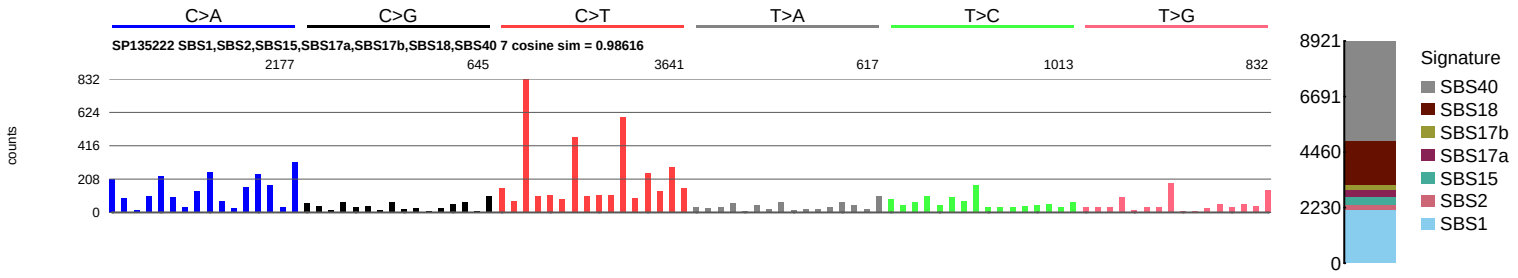

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98622
2136
602
3449
632
1276
827
counts
852
639
426
213
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

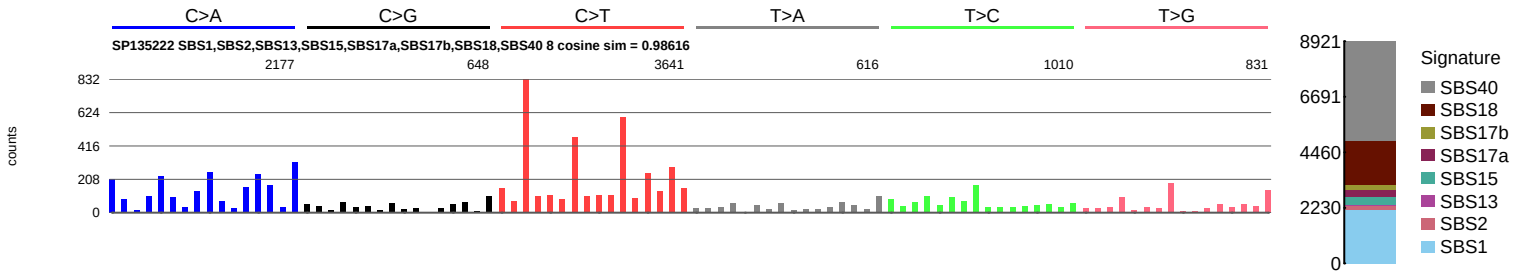

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98622
2113
577
3707
529
1333
658
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98621
2162
612
3490
587
1267
806
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98621
2116
561
3625
598
1231
791
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98618
2152
579
3617
548
1224
802
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98616
2177
645
3641
617
1013
832
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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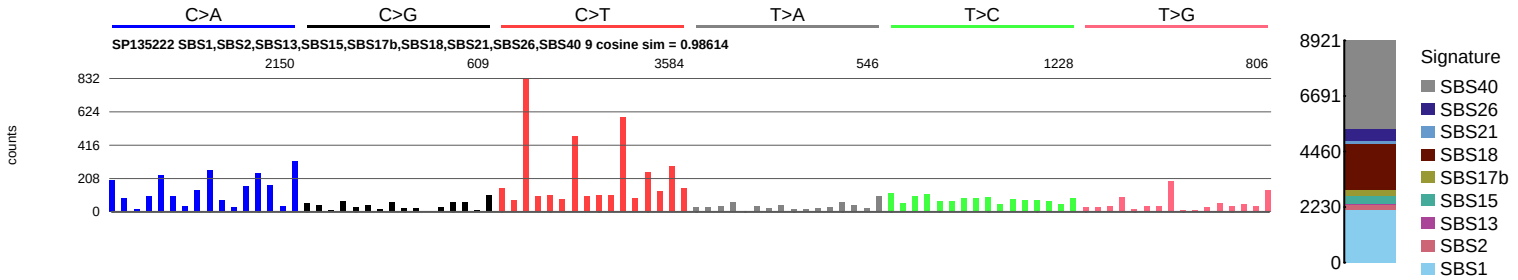
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98616
2177
648
3641
616
1010
831
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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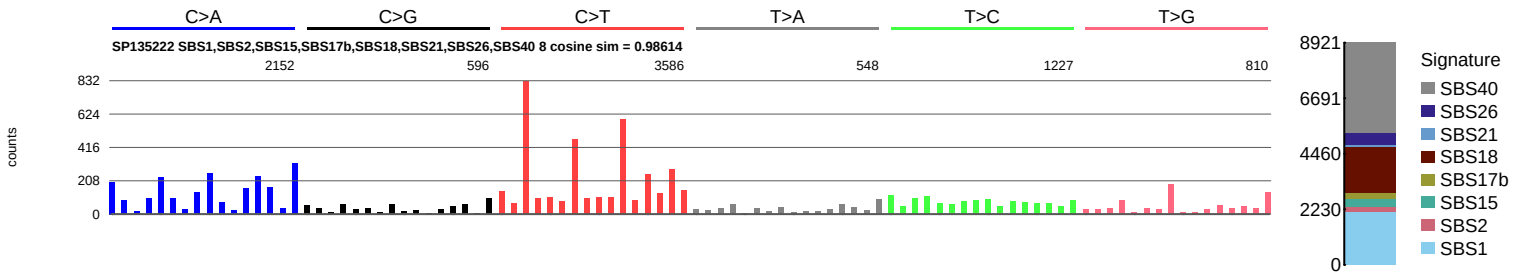
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98614
2150
609
3584
546
1228
806
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1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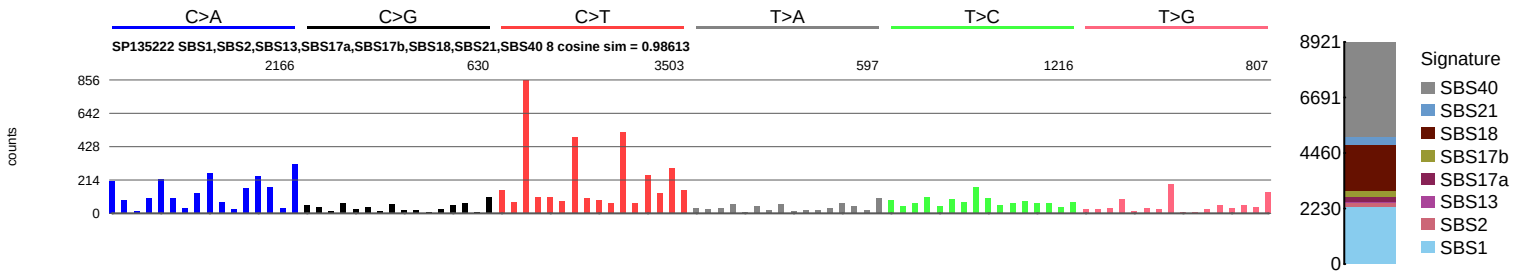
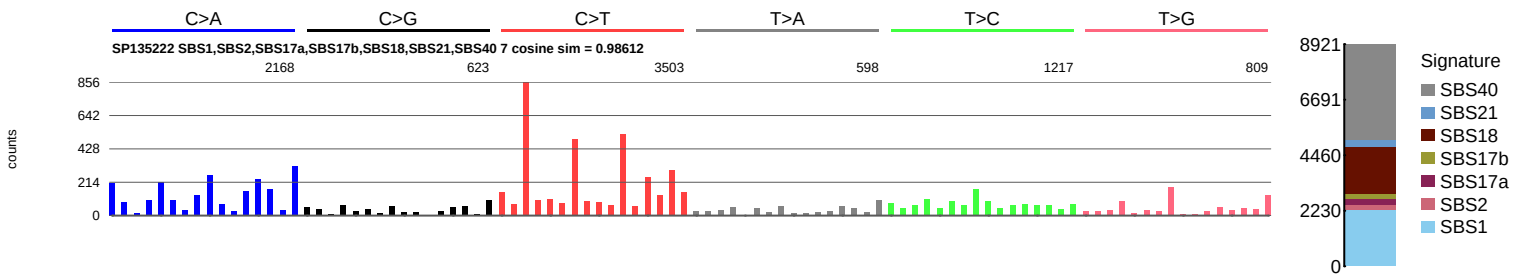
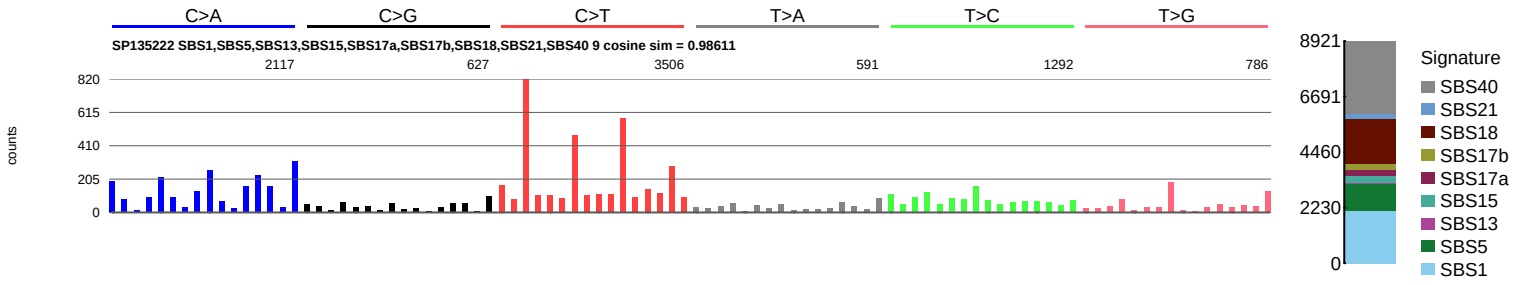
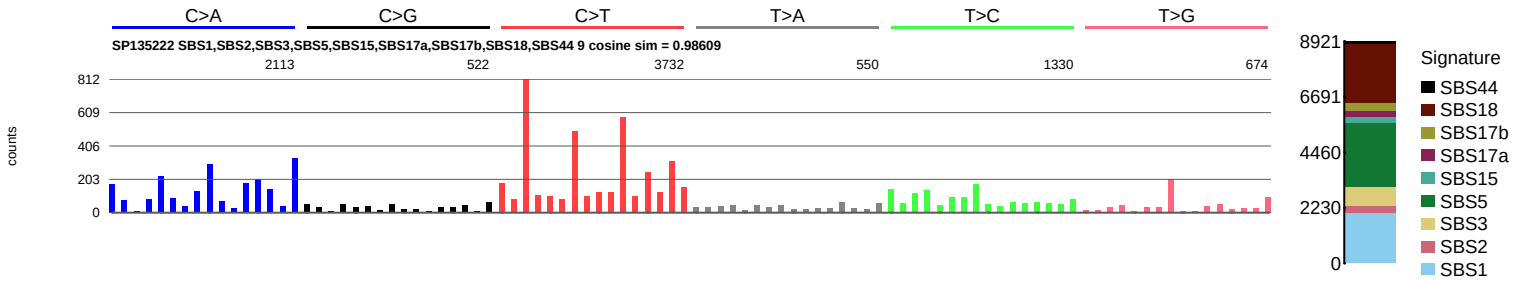
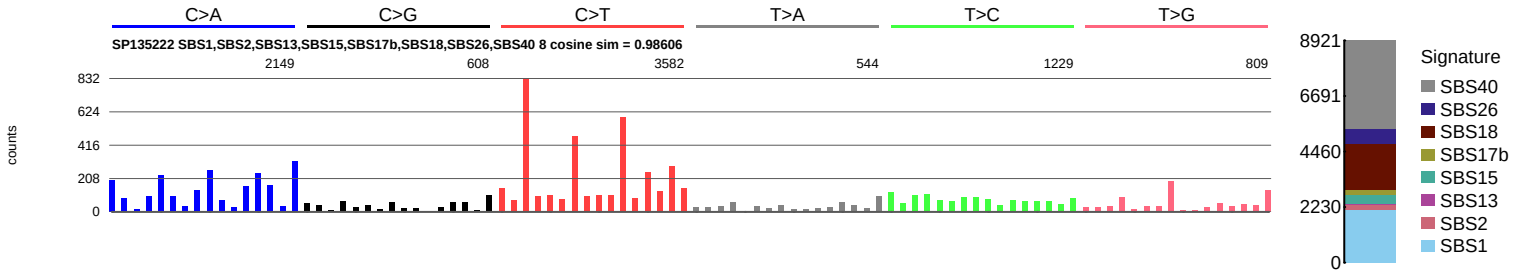
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98614
2152
596
3586
548
1227
810
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1

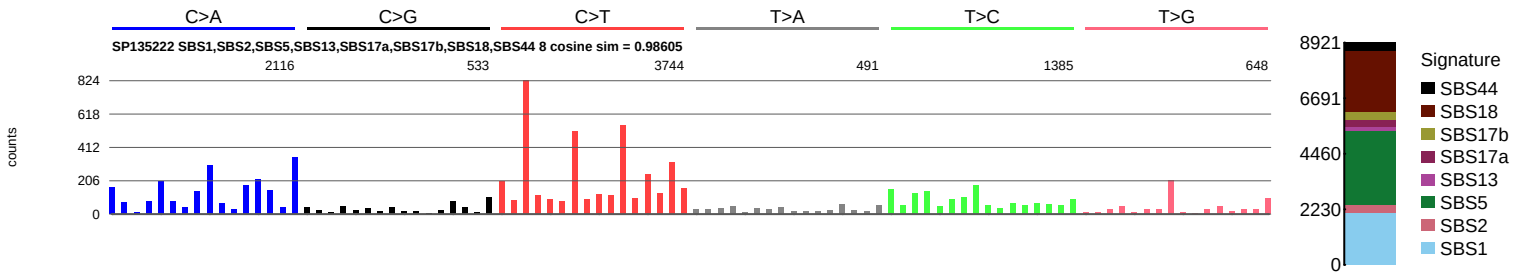

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98613
2166
630
3503
597
1216
807
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

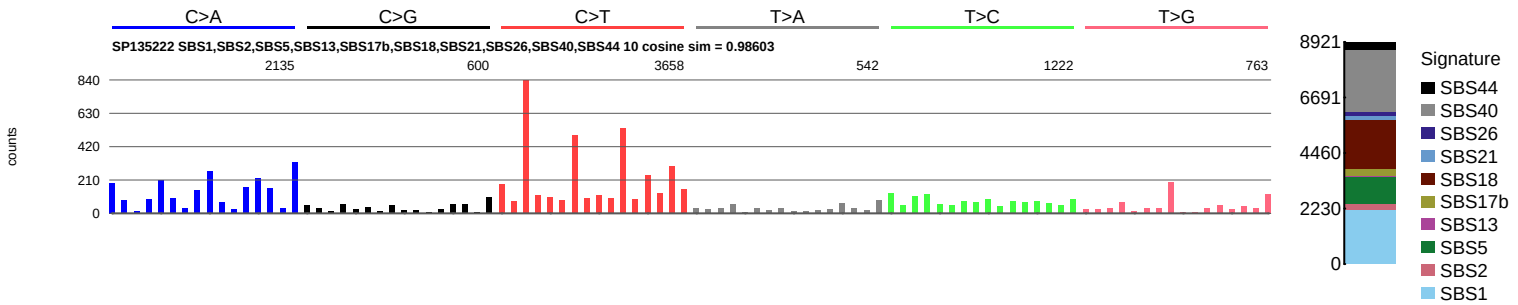

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98612
2168
623
3503
598
1217
809
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

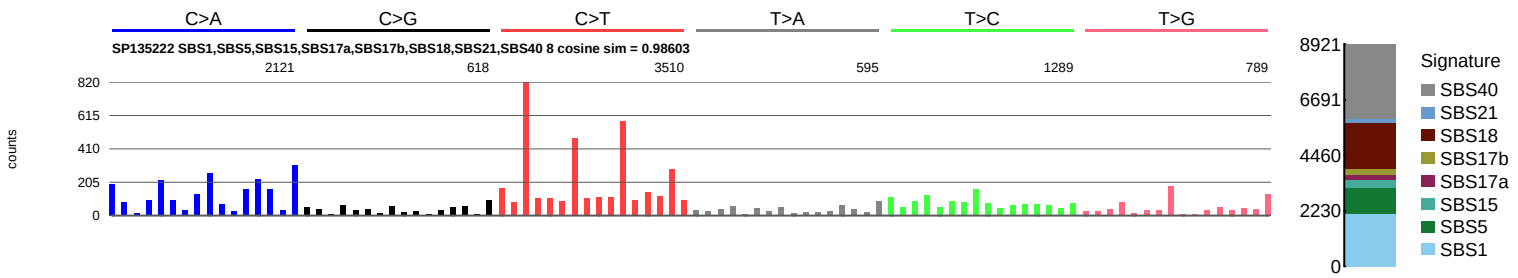

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98611
2117
627
3506
591
1292
786
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

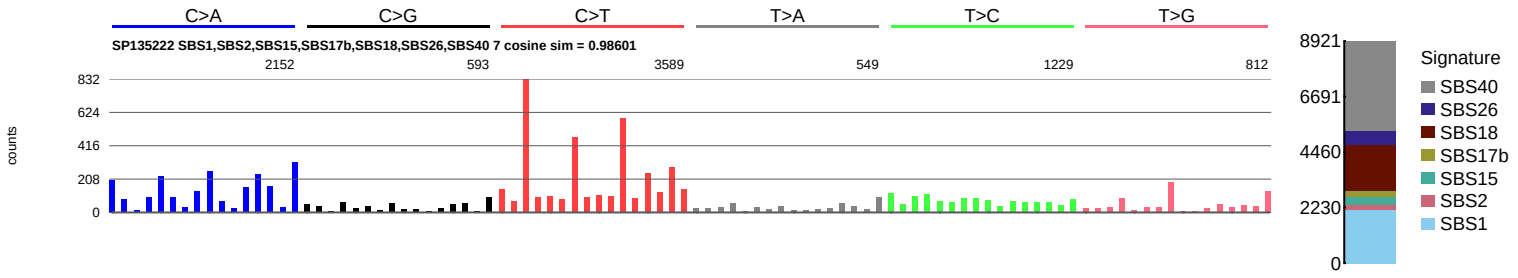

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98609
2113
522
3732
550
1330
674
counts
812
609
406
203
0
8921
6691
4460
2230
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

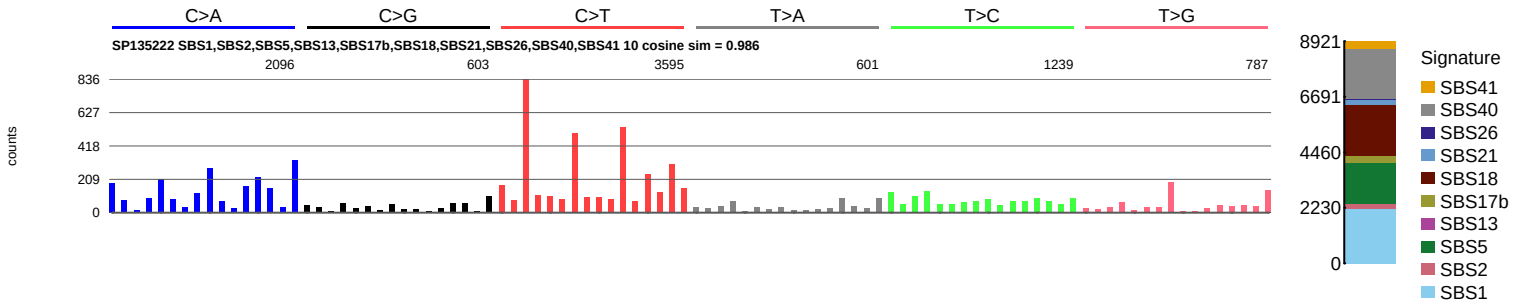

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98606
2149
608
3582
544
1229
809
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

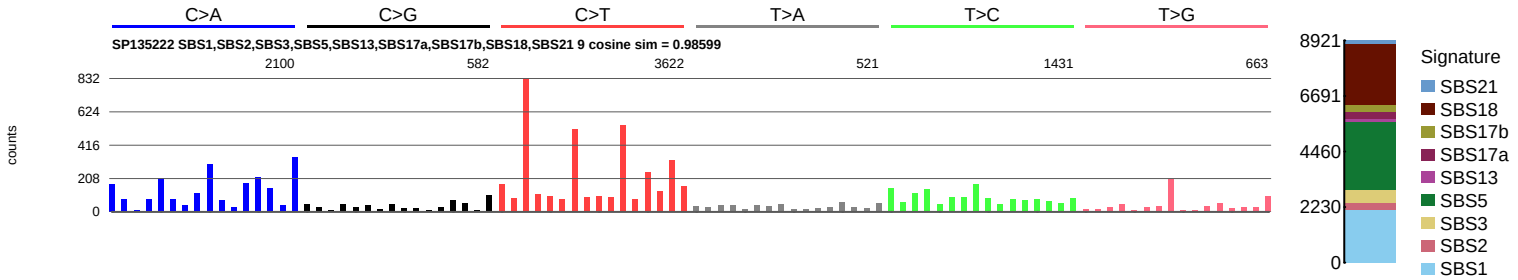

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98605
2116
533
3744
491
1385
648
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

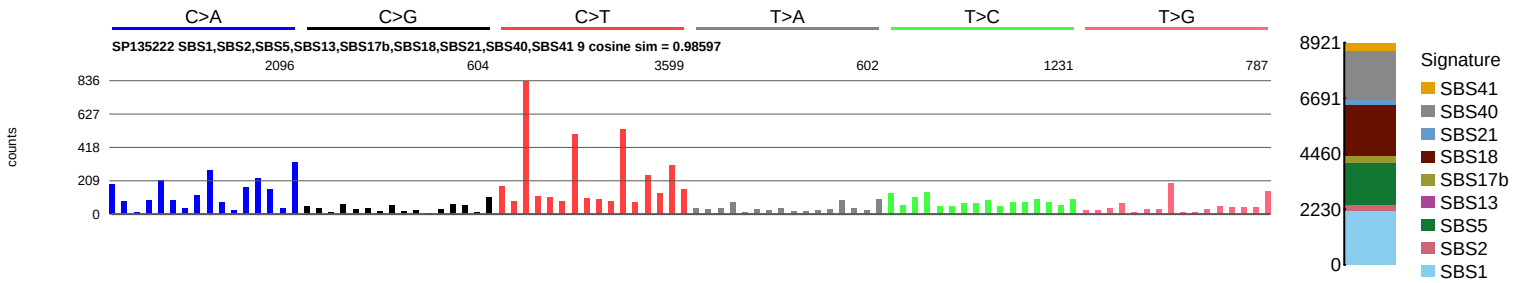

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98603
2135
600
3658
542
1222
763
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

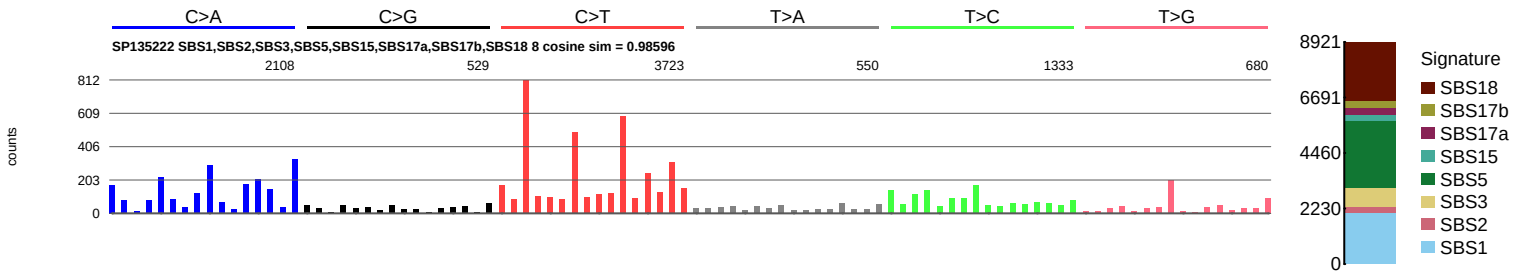

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98603
2121
618
3510
595
1289
789
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

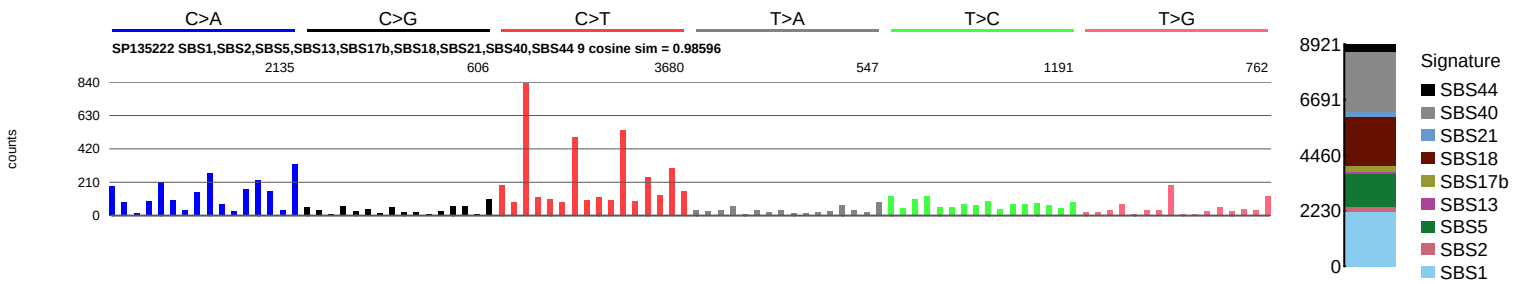

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98601
2152
593
3589
549
1229
812
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS2
SBS1

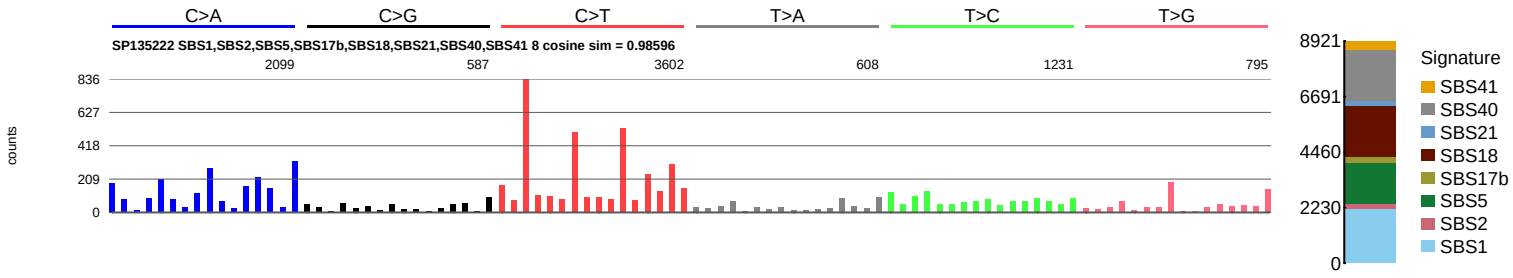

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.986
2096
603
3595
601
1239
787
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

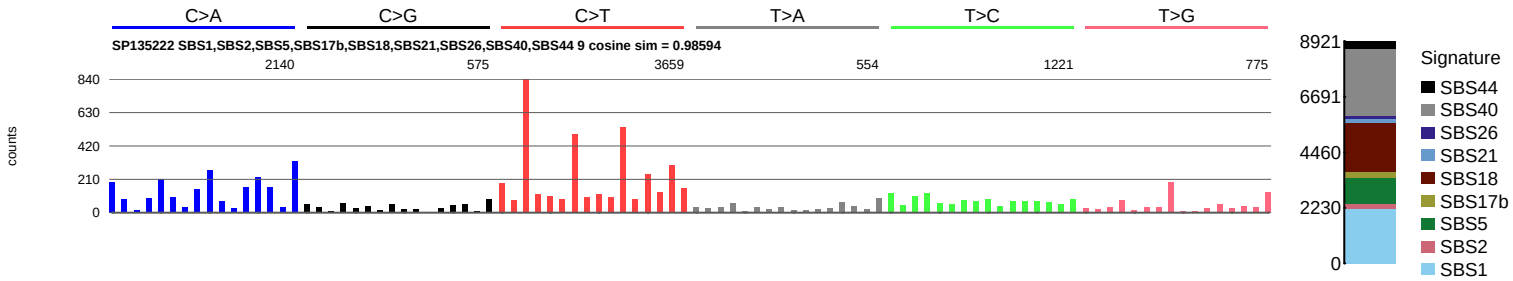

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98599
2100
582
3622
521
1431
663
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

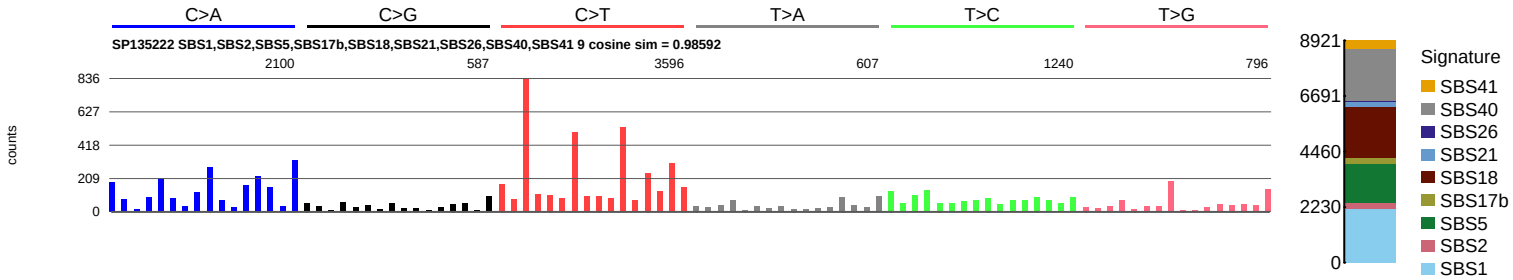

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98597
2096
604
3599
602
1231
787
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

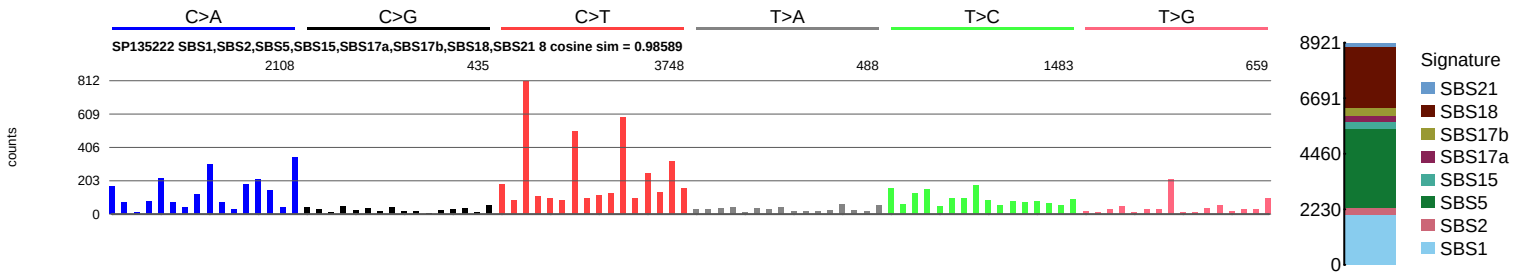

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98596
2108
529
3723
550
1333
680
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

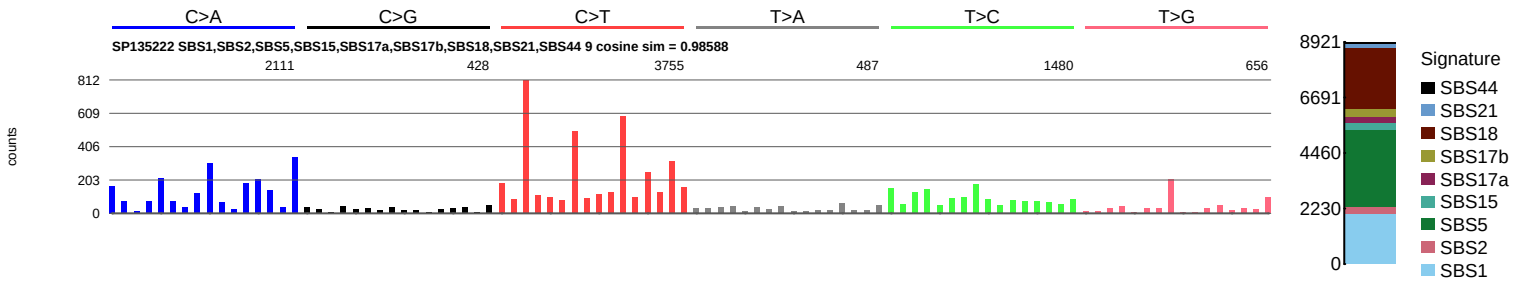

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98596
2135
606
3680
547
1191
762
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

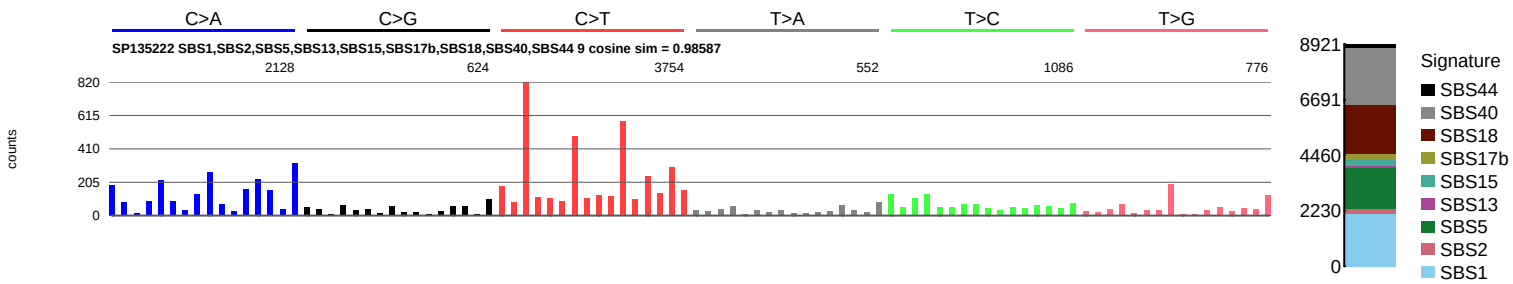

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98596
2099
587
3602
608
1231
795
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1

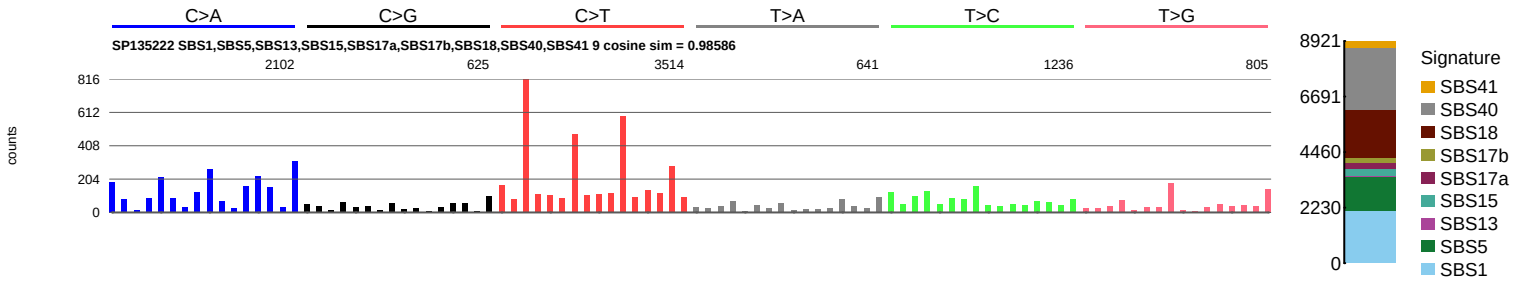

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98594
2140
575
3659
554
1221
775
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98592
2100
587
3596
607
1240
796
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98589
2108
435
3748
488
1483
659
counts
812
609
406
203
0
8921
6691
4460
2230
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98588
2111
428
3755
487
1480
656
counts
812
609
406
203
0
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98587
2128
624
3754
552
1086
776
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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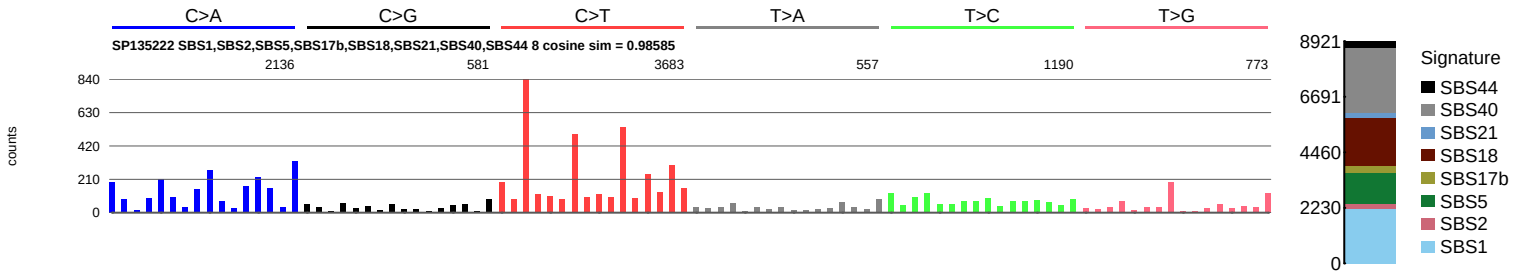
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98586
2102
625
3514
641
1236
805
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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C>G
C>T
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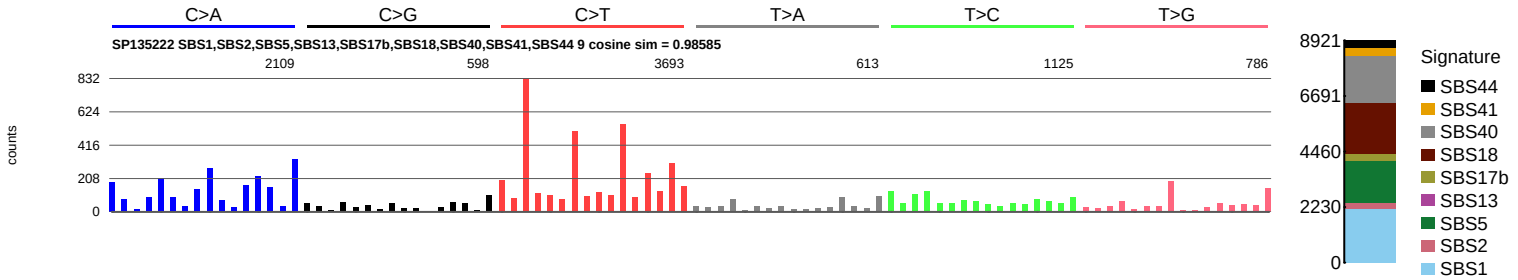
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98585
2136
581
3683
557
1190
773
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1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C>T
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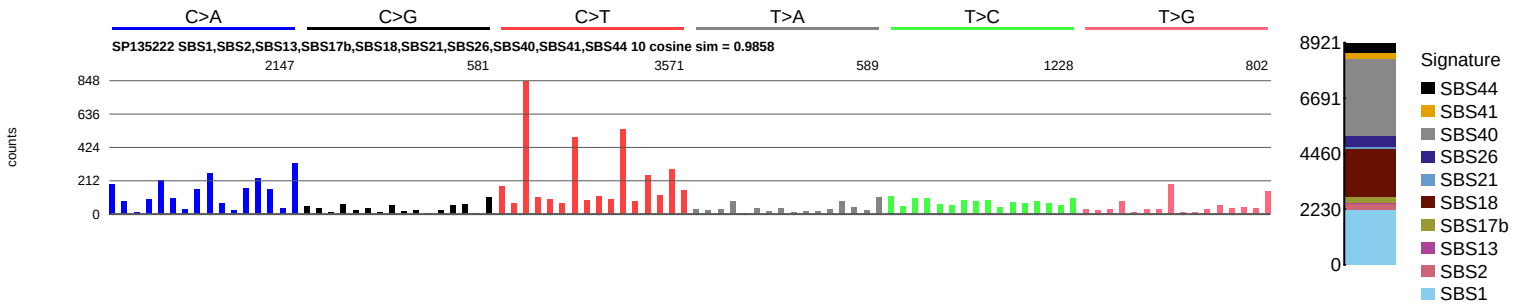
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98585
2109
598
3693
613
1125
786
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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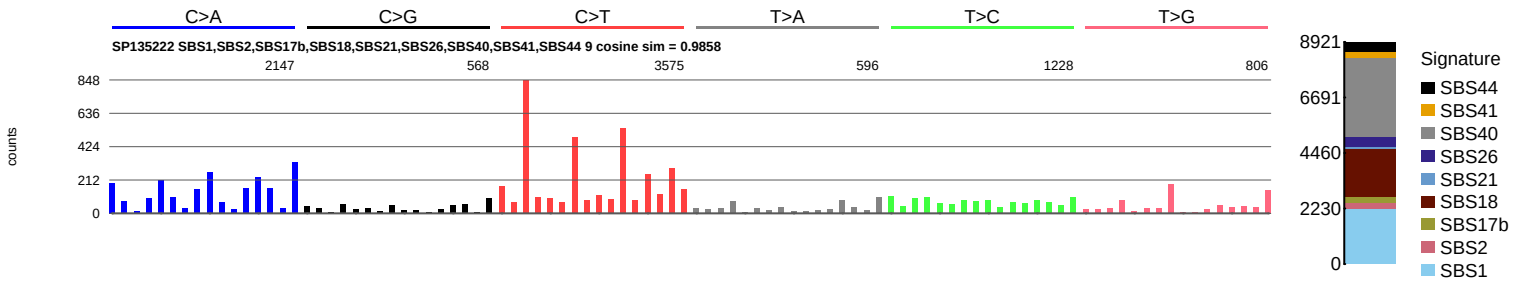
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9858
2147
581
3571
589
1228
802
counts
848
636
424
212
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1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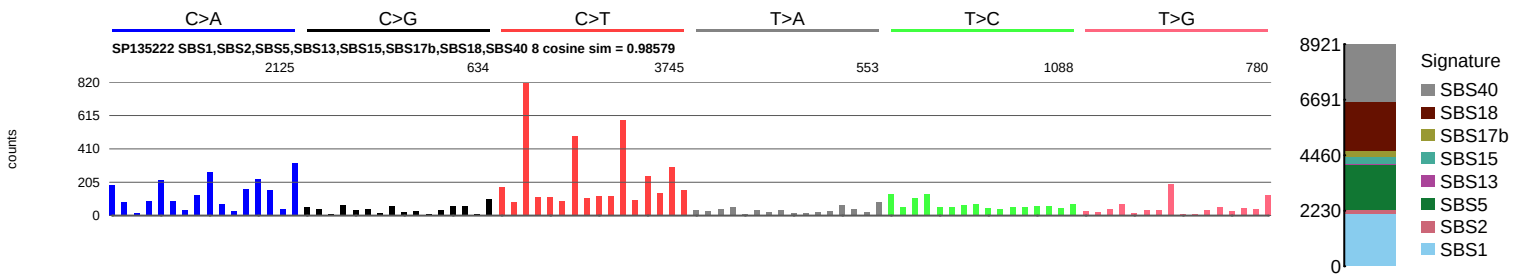
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9858
2147
568
3575
596
1228
806
counts
848
636
424
212
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS2
SBS1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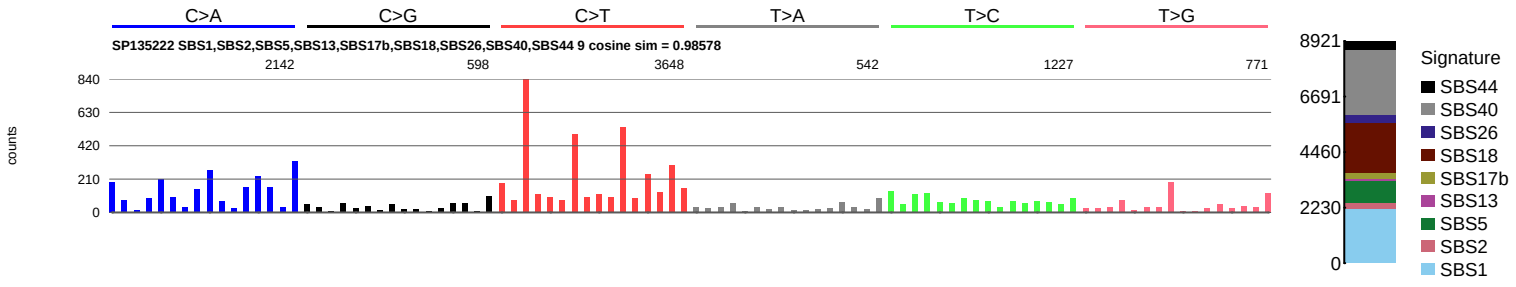
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 8 cosine sim = 0.98579
2125
634
3745
553
1088
780
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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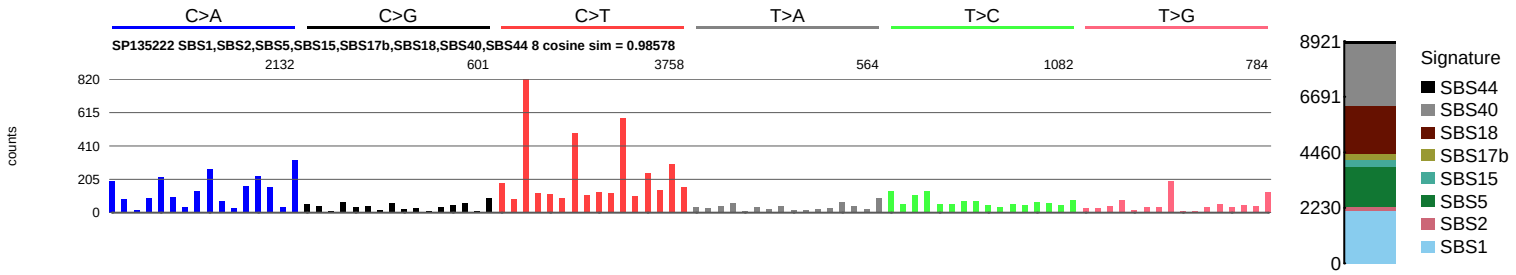
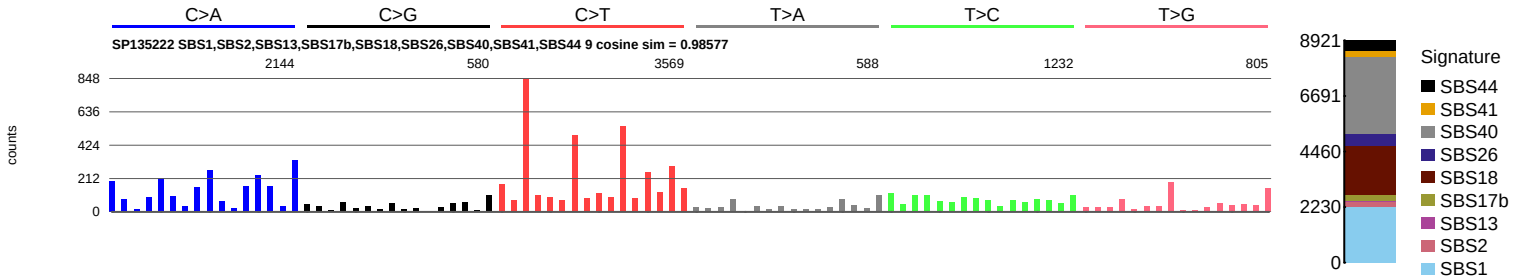
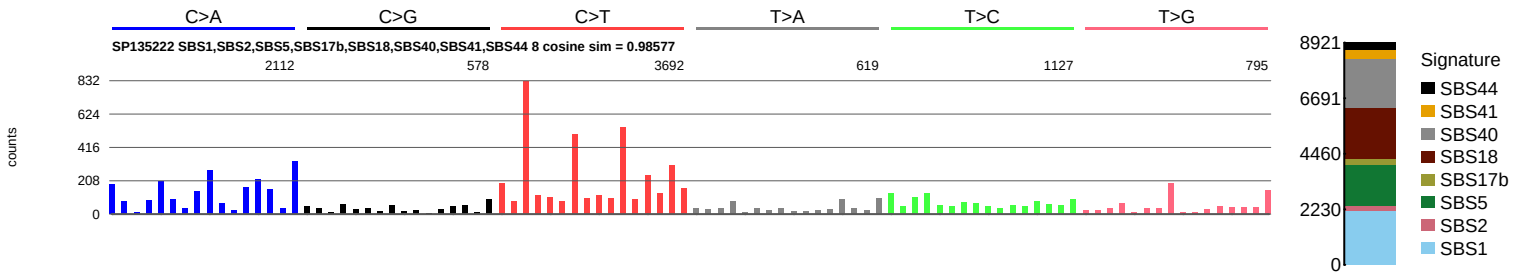
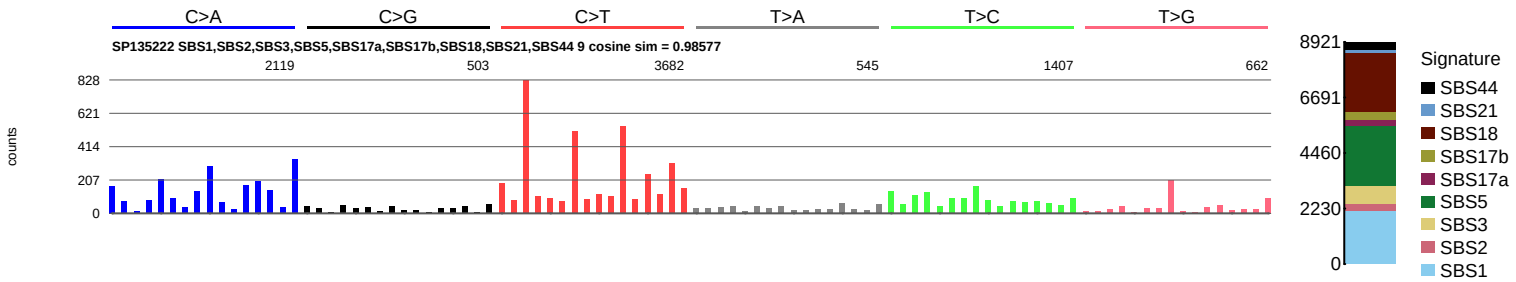
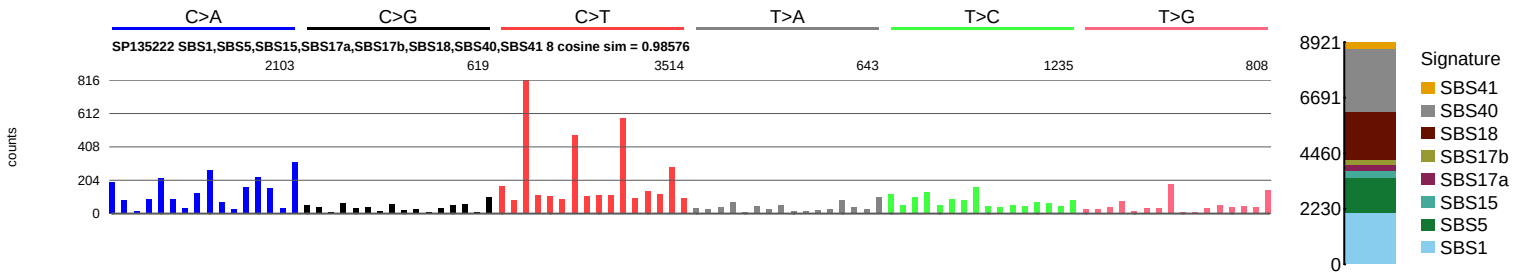
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98578
2142
598
3648
542
1227
771
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

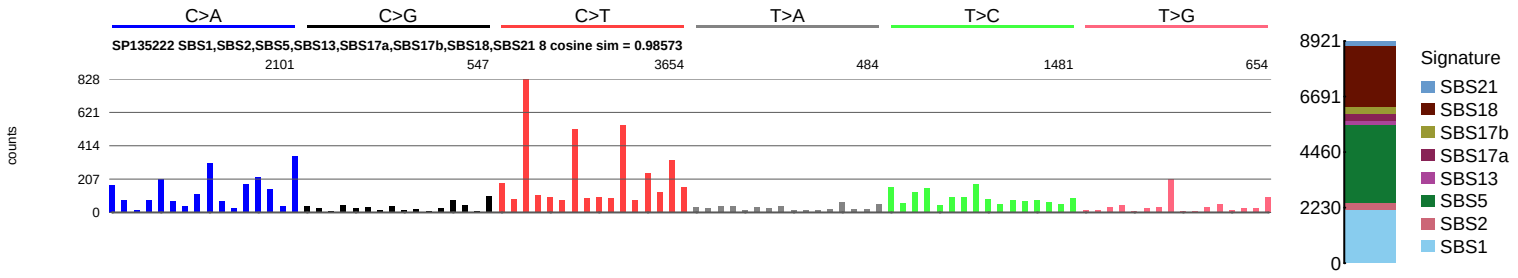

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98578
2132
601
3758
564
1082
784
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

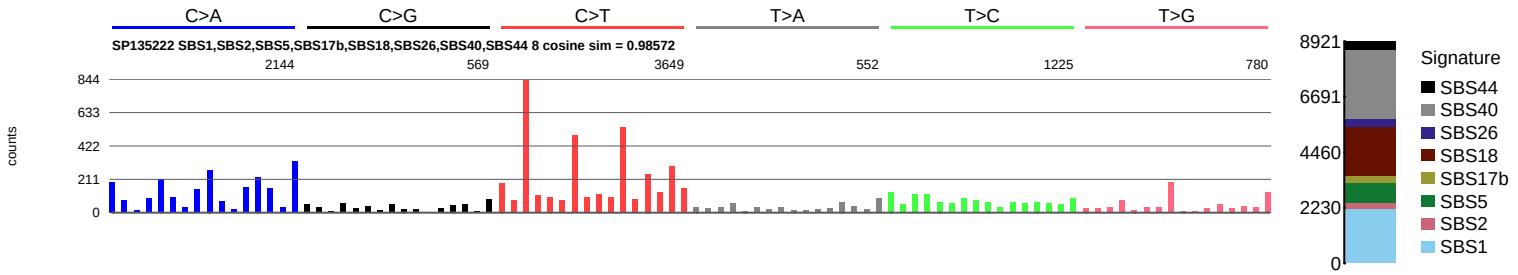

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98577
2144
580
3569
588
1232
805
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1

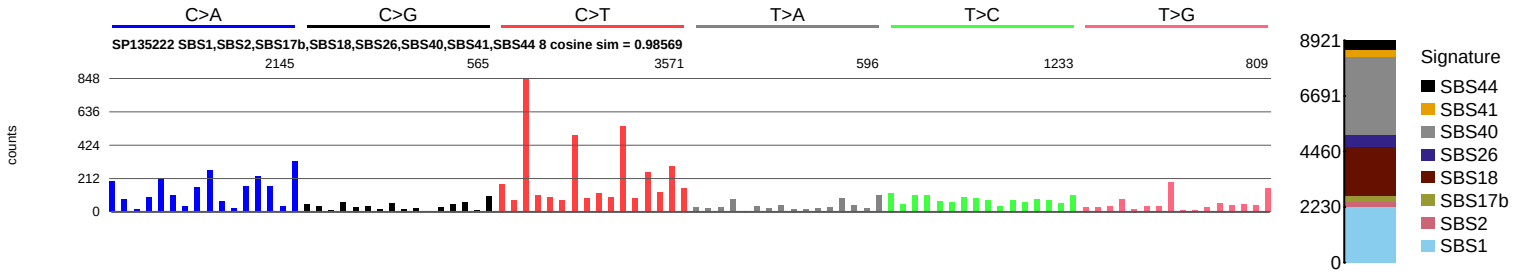

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98577
2112
578
3692
619
1127
795
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1

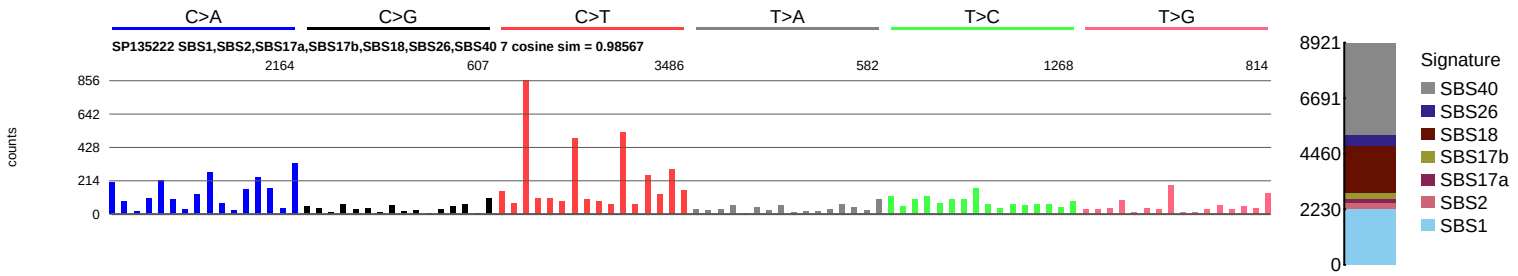

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98577
2119
503
3682
545
1407
662
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

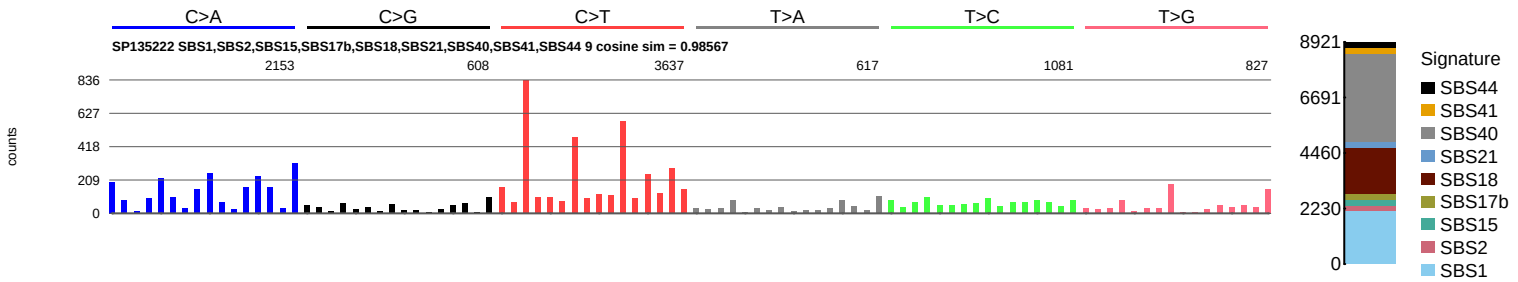

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98576
2103
619
3514
643
1235
808
816
612
408
204
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

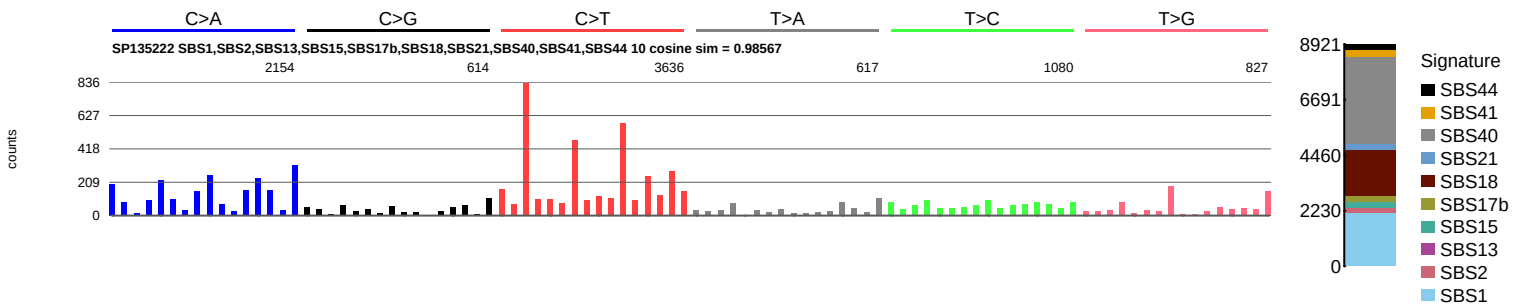

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98573
2101
547
3654
484
1481
654
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

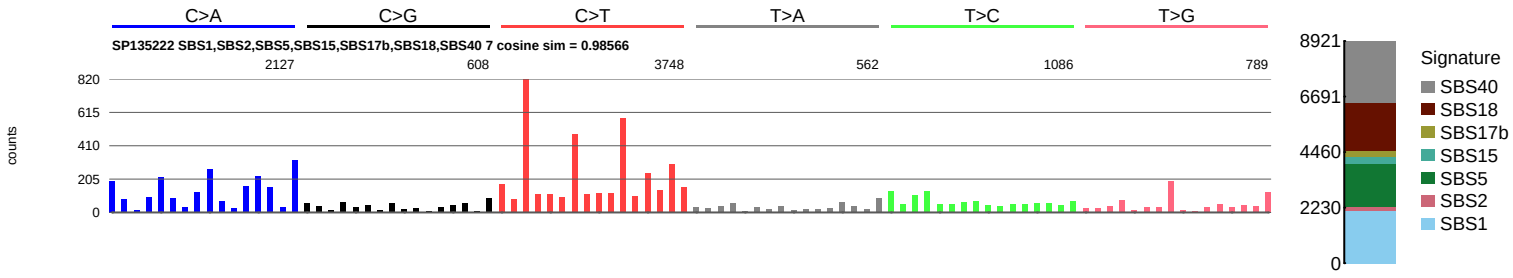

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98572
2144
569
3649
552
1225
780
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1

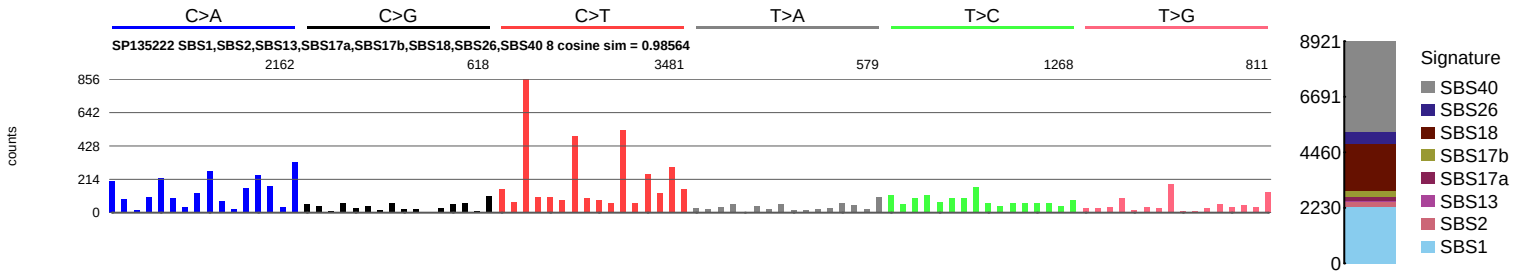

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98569
2145
565
3571
596
1233
809
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1

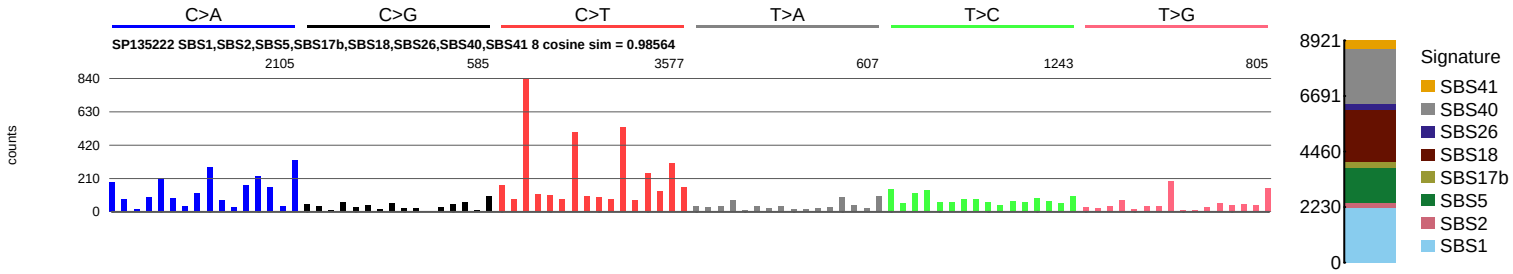

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98567
2164
607
3486
582
1268
814
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

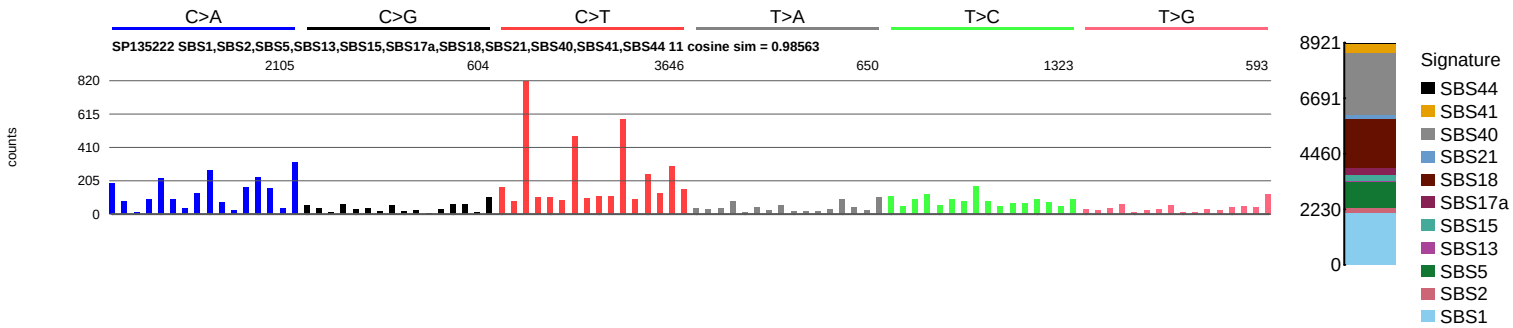

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98567
2153
608
3637
617
1081
827
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1

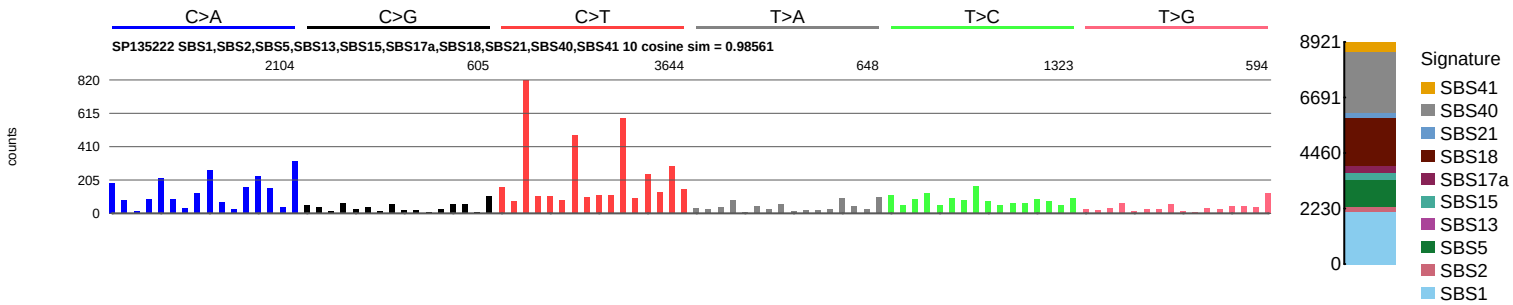

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98567
2154
614
3636
617
1080
827
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS40 7 cosine sim = 0.98566
2127
608
3748
562
1086
789
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98564
2162
618
3481
579
1268
811
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98564
2105
585
3577
607
1243
805
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98563
2105
604
3646
650
1323
593
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98561
2104
605
3644
648
1323
594
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
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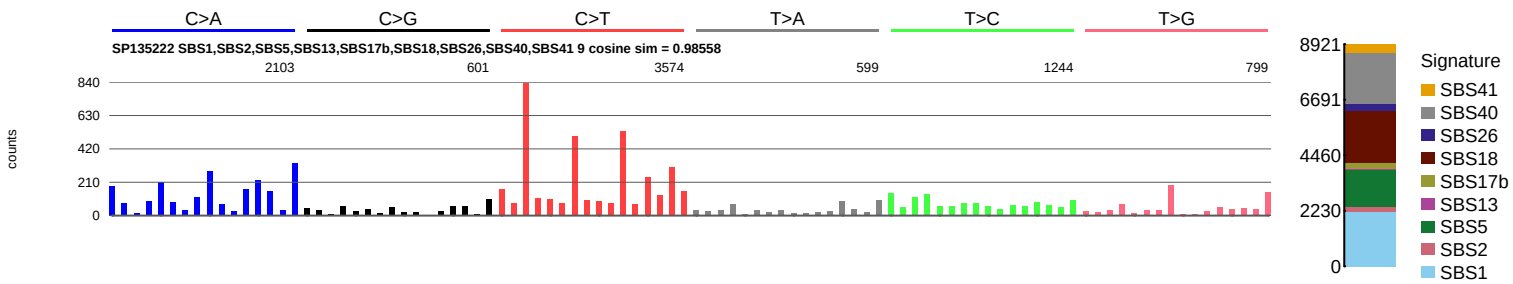
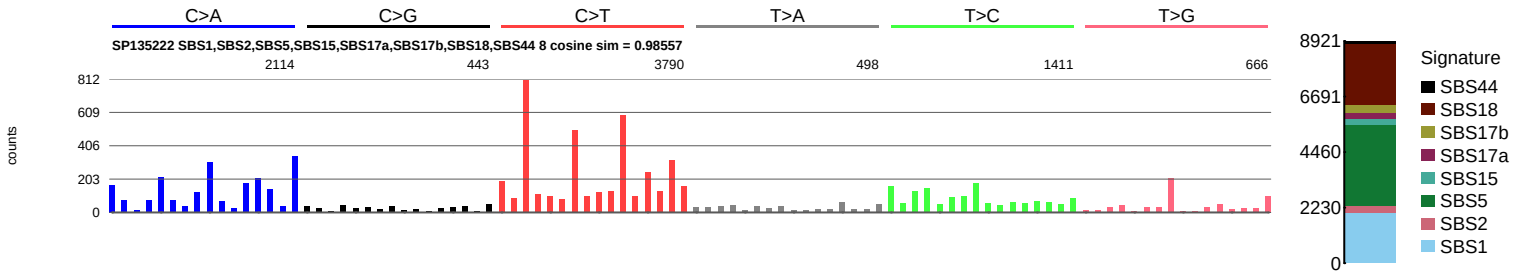
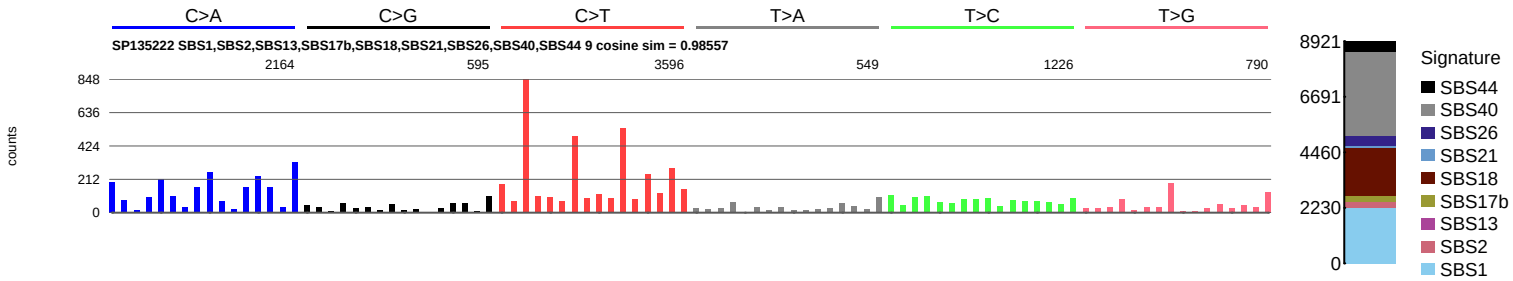
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98558
2103
601
3574
599
1244
799
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98557
2114
443
3790
498
1411
666
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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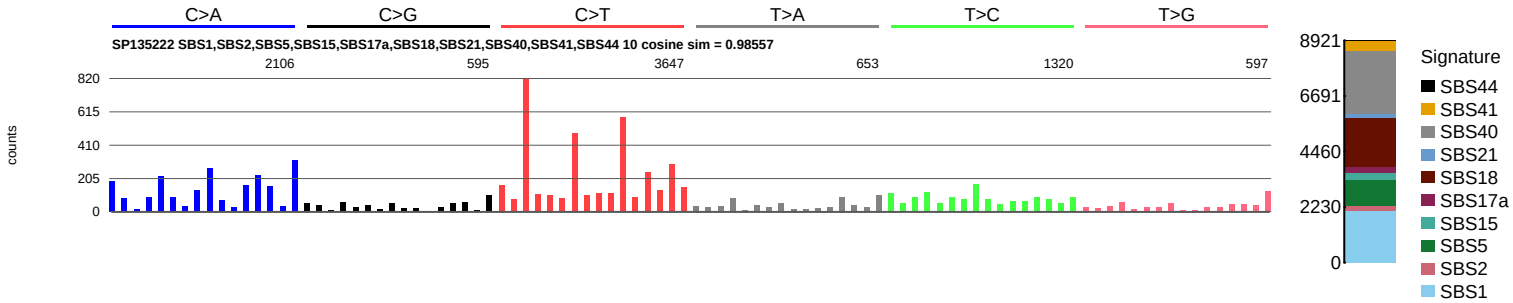
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98557
2164
595
3596
549
1226
790
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1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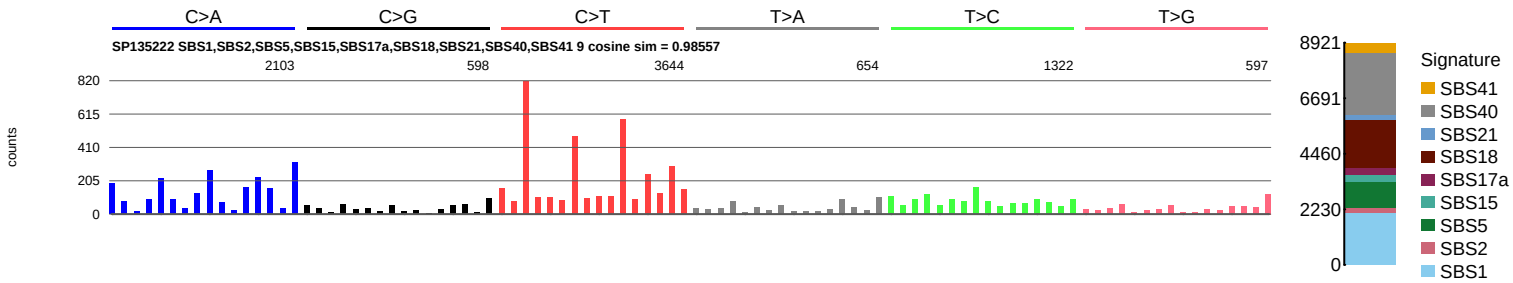
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98557
2106
595
3647
653
1320
597
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98557
2103
598
3644
654
1322
597
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
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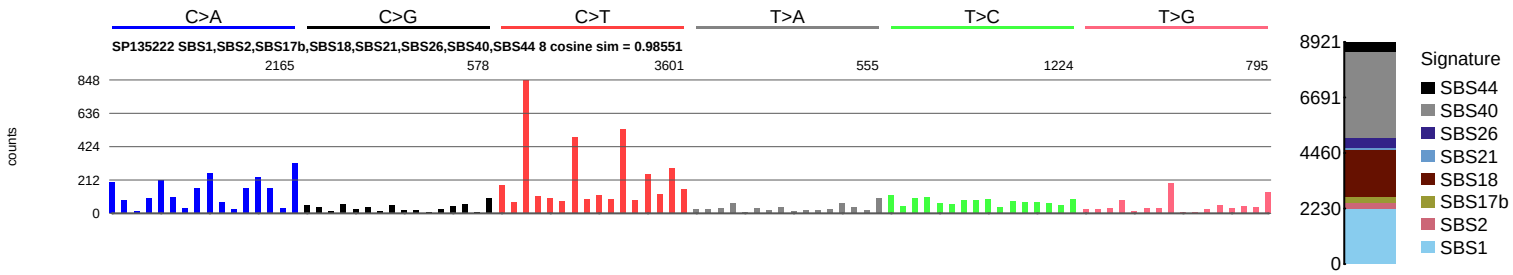
SBS5
SBS2
SBS1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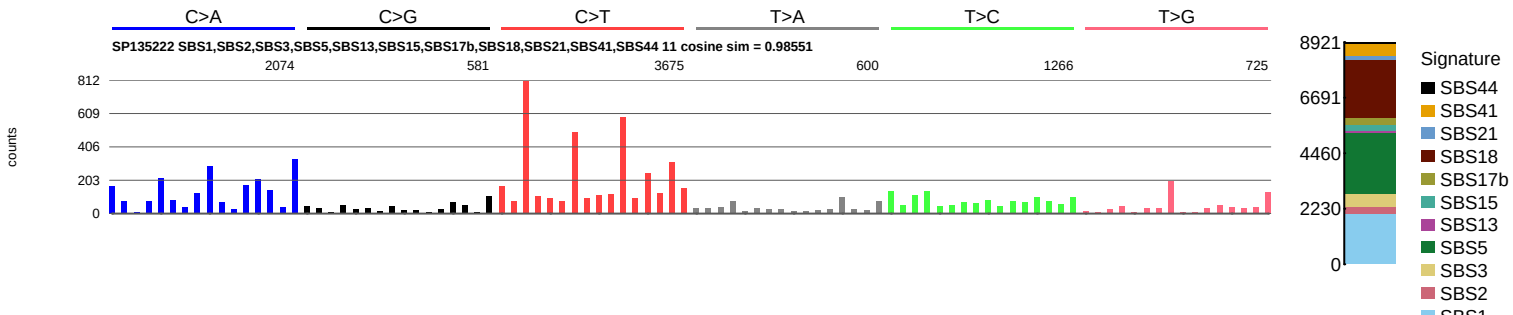
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98551
2165
578
3601
555
1224
795
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS2
SBS1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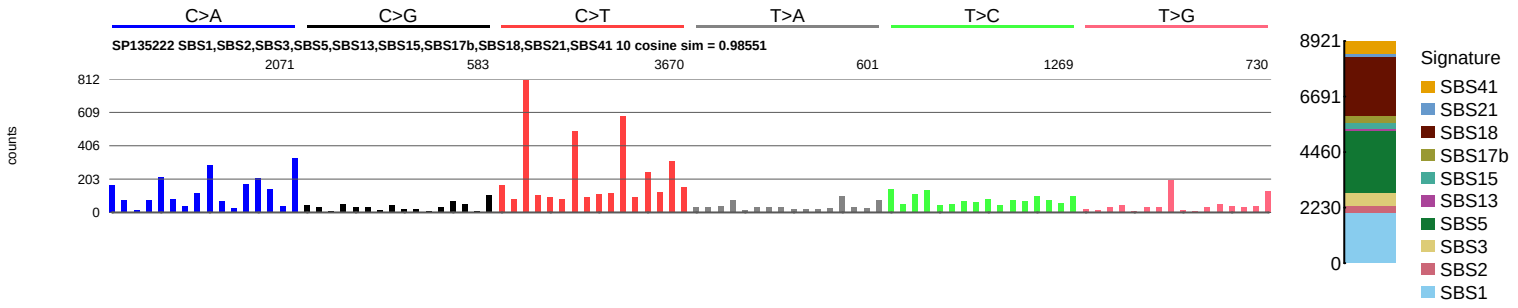
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98551
2074
581
3675
600
1266
725
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98551
2071
583
3670
601
1269
730
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
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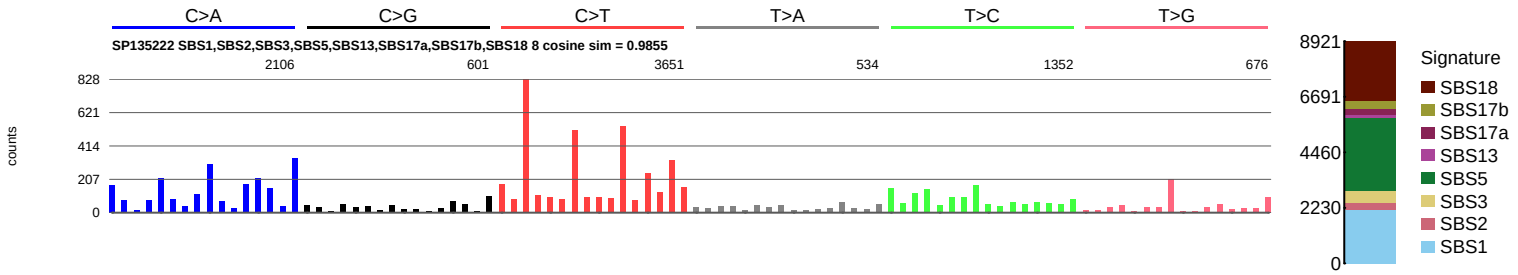
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.9855
2106
601
3651
534
1352
676
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS17a
SBS13
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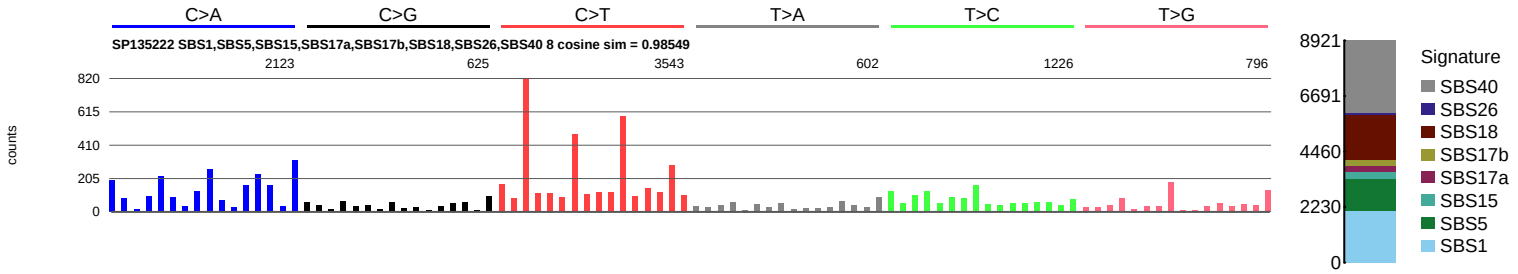
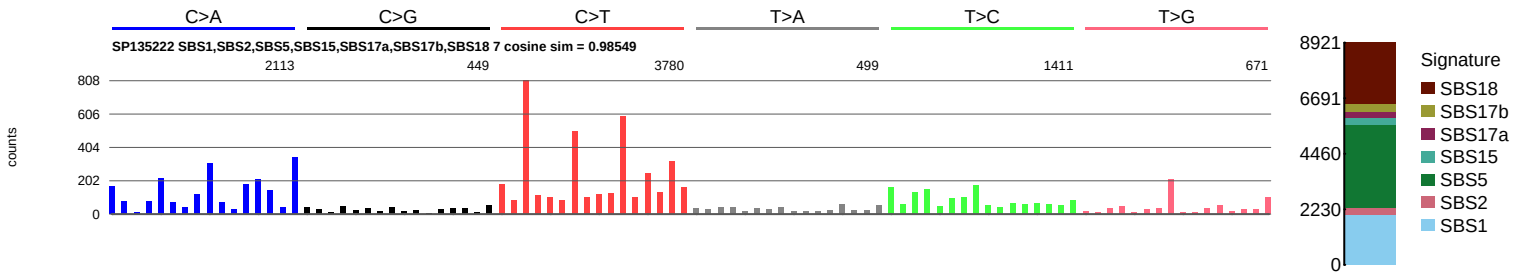
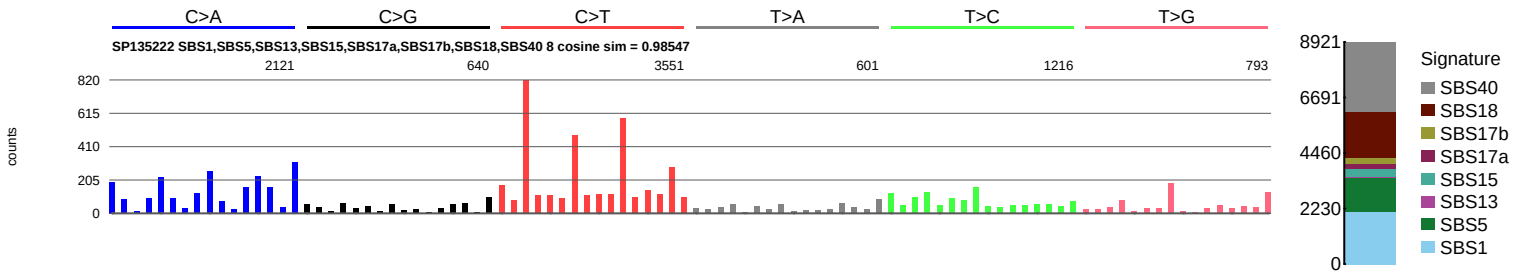
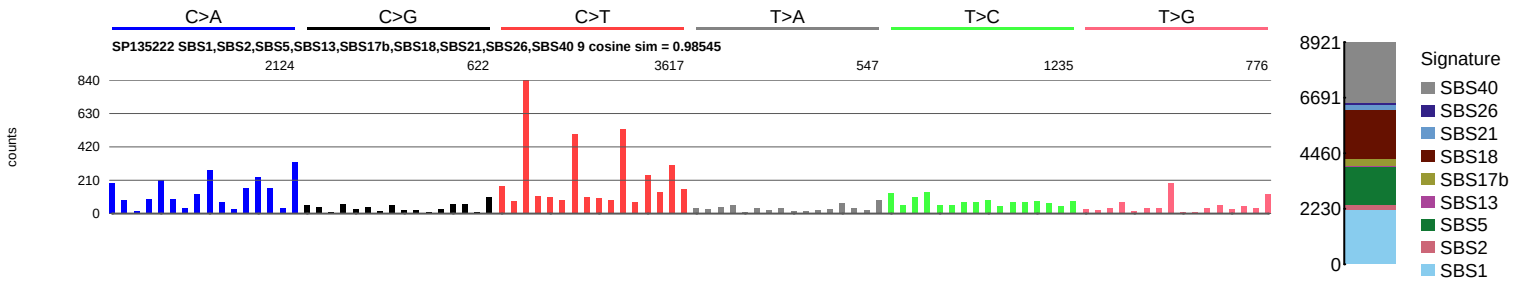
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98549
2123
625
3543
602
1226
796
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98549
2113
449
3780
499
1411
671
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98547
2121
640
3551
601
1216
793
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98545
2124
622
3617
547
1235
776
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
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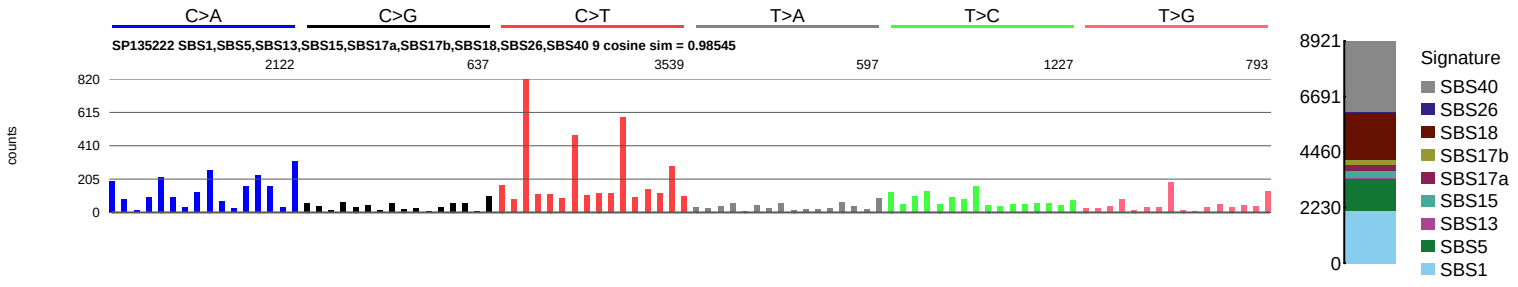
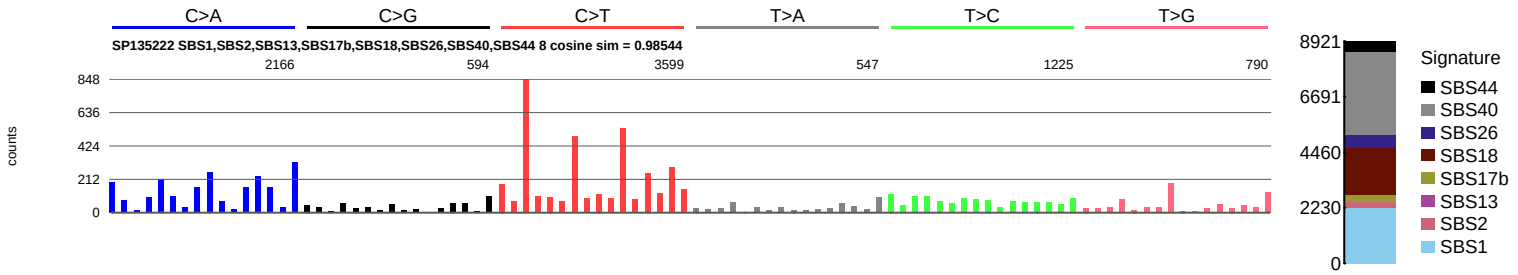
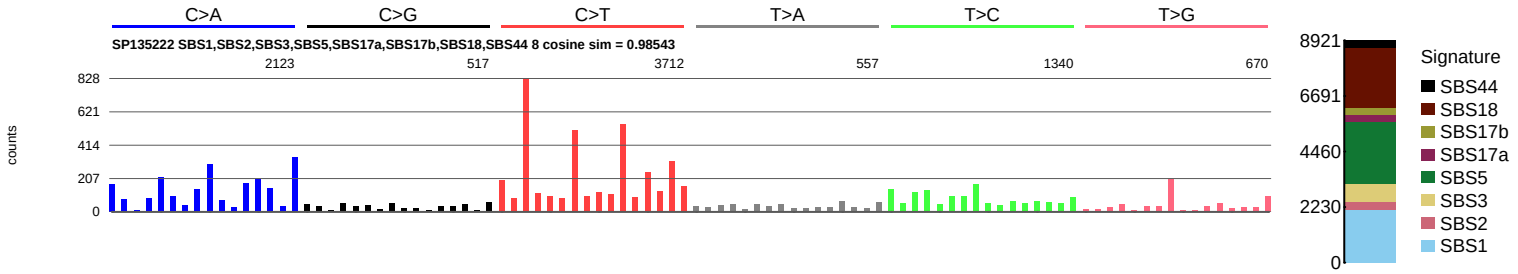
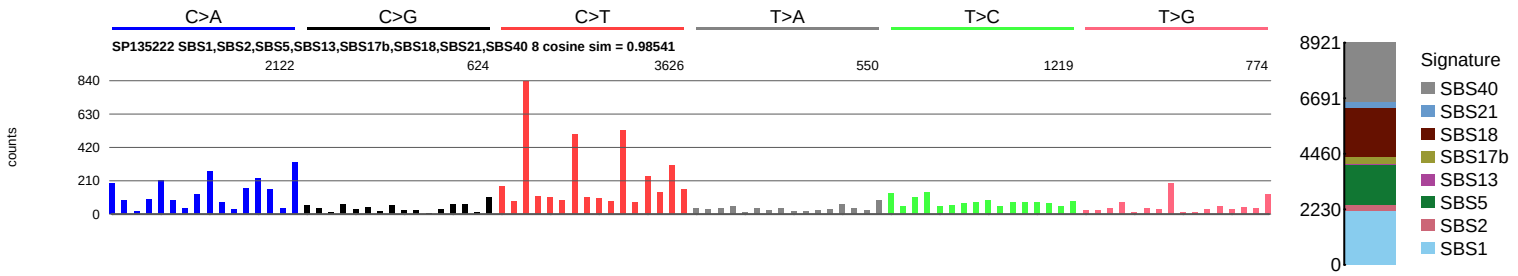
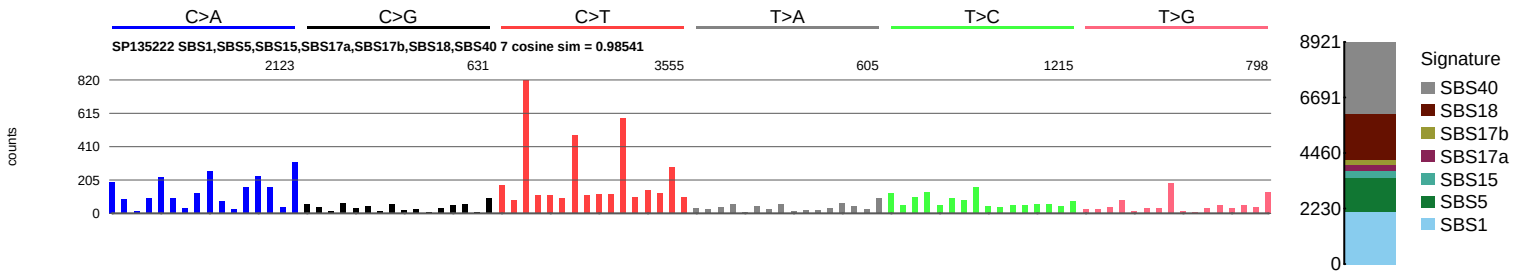
SBS5
SBS2
SBS1

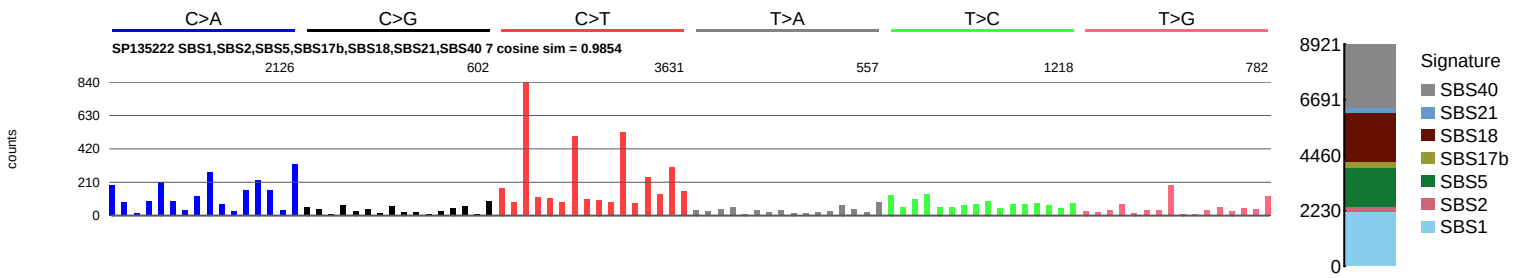

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98545
2122
637
3539
597
1227
793
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

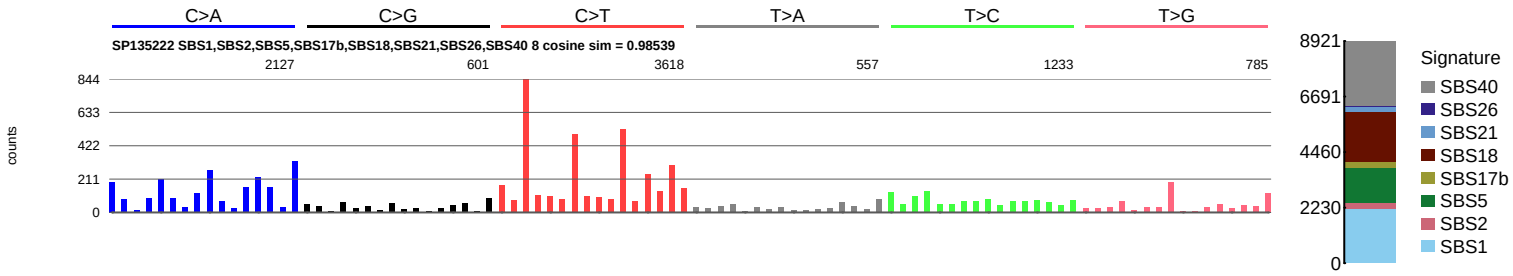

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98544
2166
594
3599
547
1225
790
counts
848
636
424
212
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1

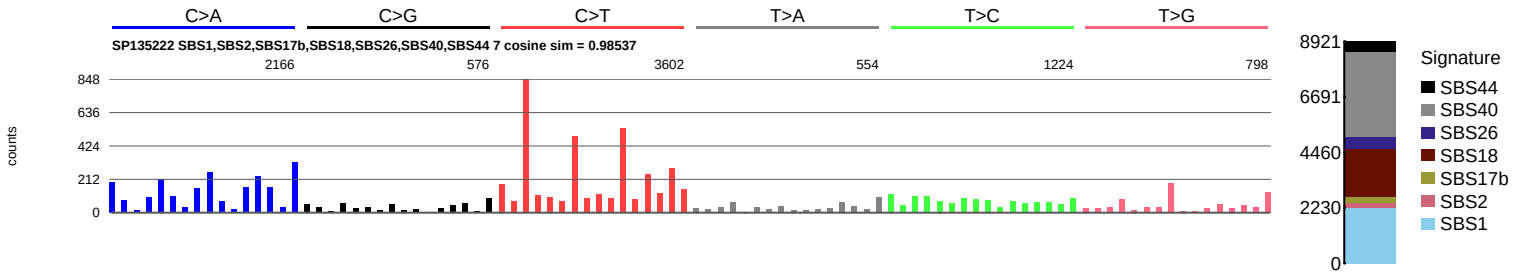

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98543
2123
517
3712
557
1340
670
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

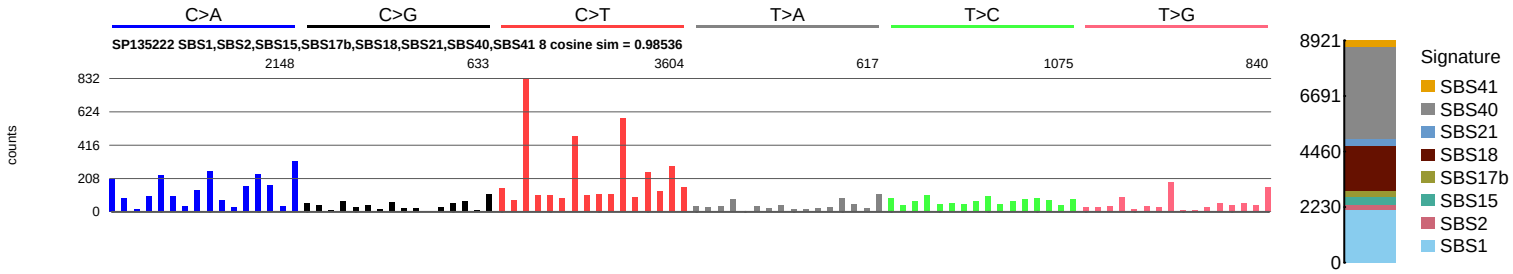

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98541
2122
624
3626
550
1219
774
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

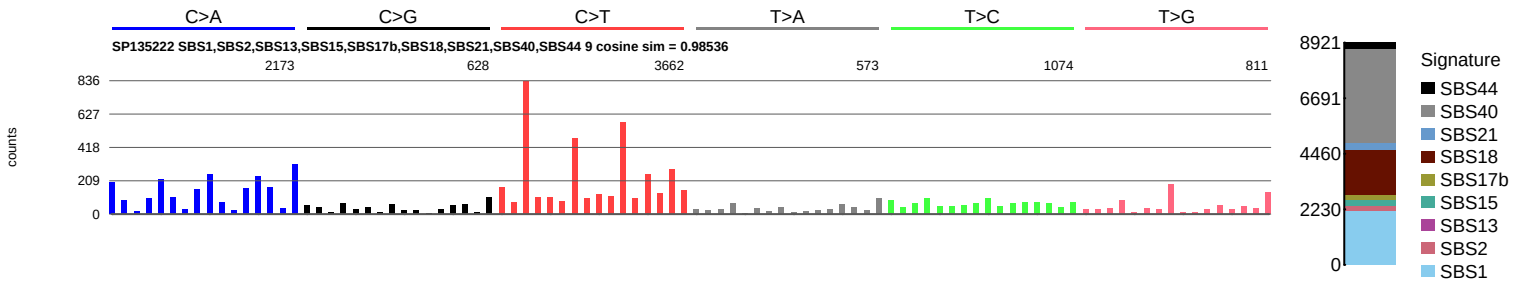

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98541
2123
631
3555
605
1215
798
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

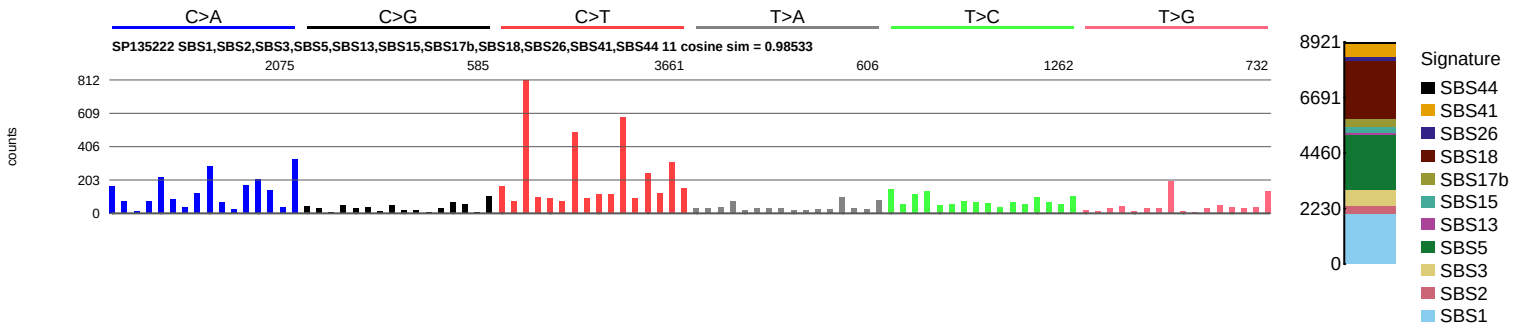

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.9854
2126
602
3631
557
1218
782
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1

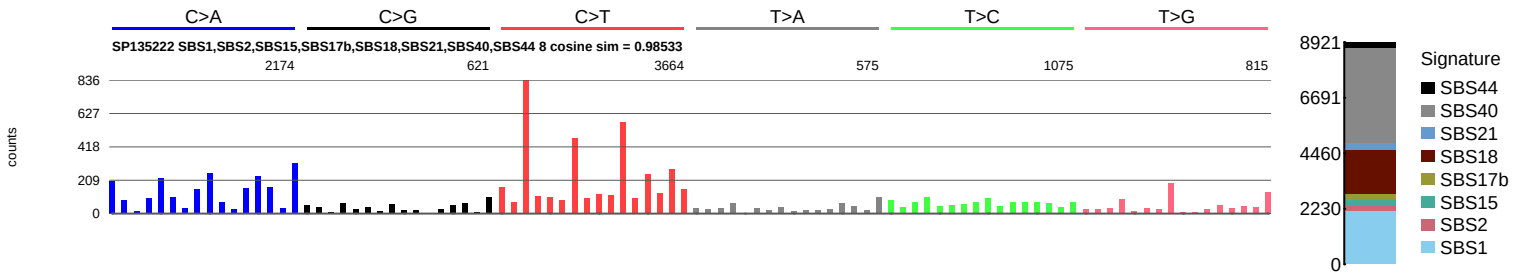

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98539
2127
601
3618
557
1233
785
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1

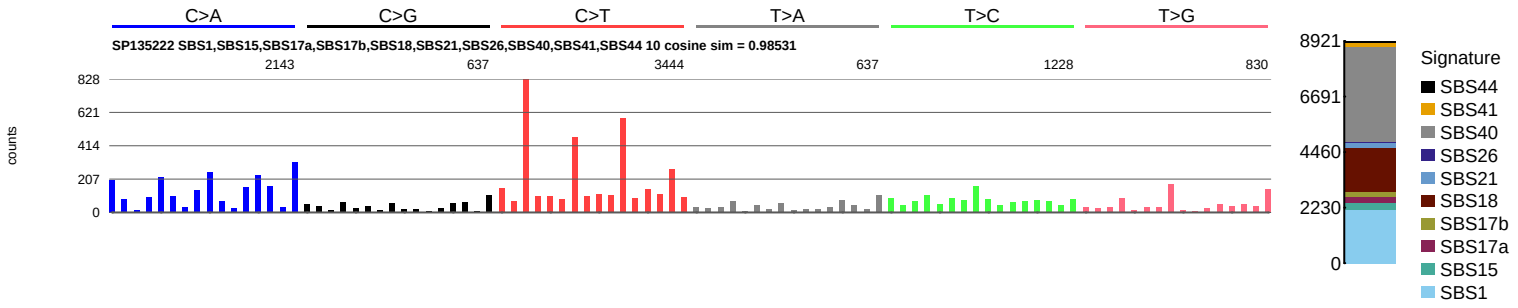

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98537
2166
576
3602
554
1224
798
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1

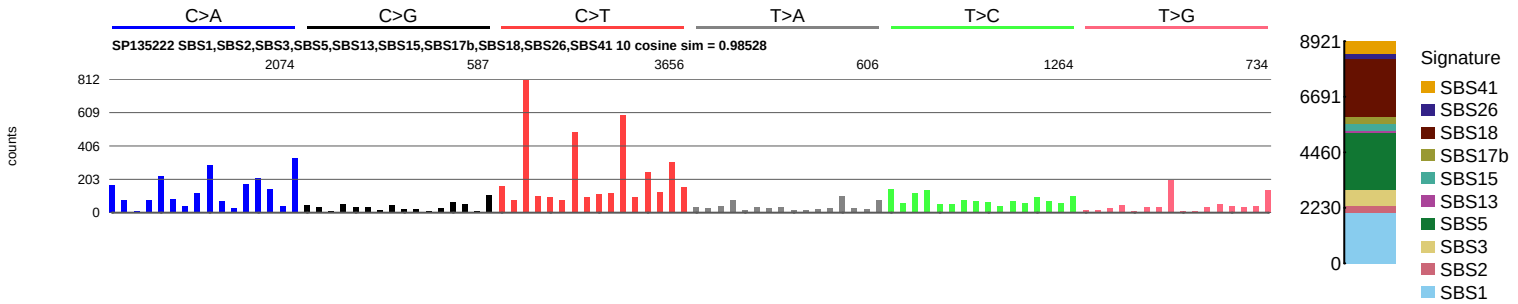

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98536
2148
633
3604
617
1075
840
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1

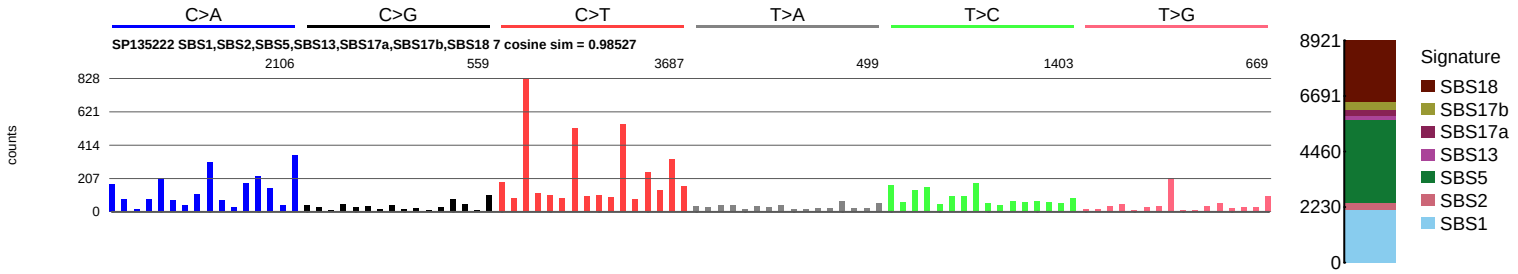

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98536
2173
628
3662
573
1074
811
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

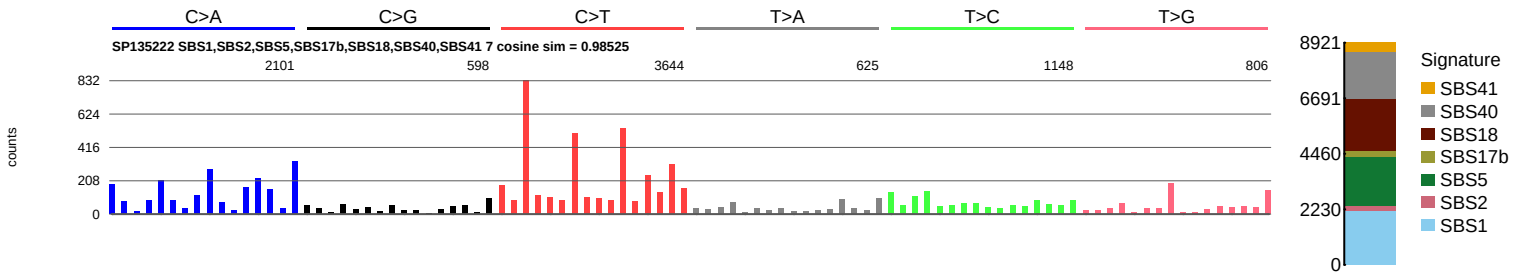

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.98533
2075
585
3661
606
1262
732
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

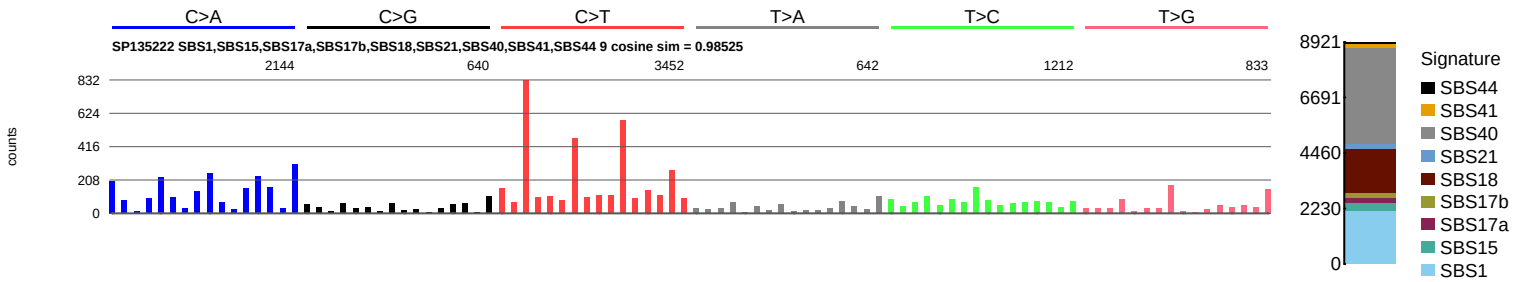

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98533
2174
621
3664
575
1075
815
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1

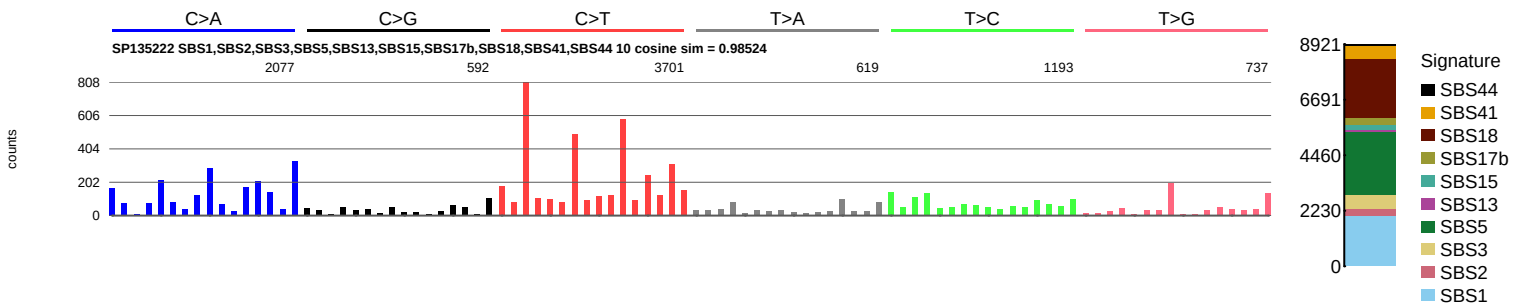

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98531
2143
637
3444
637
1228
830
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

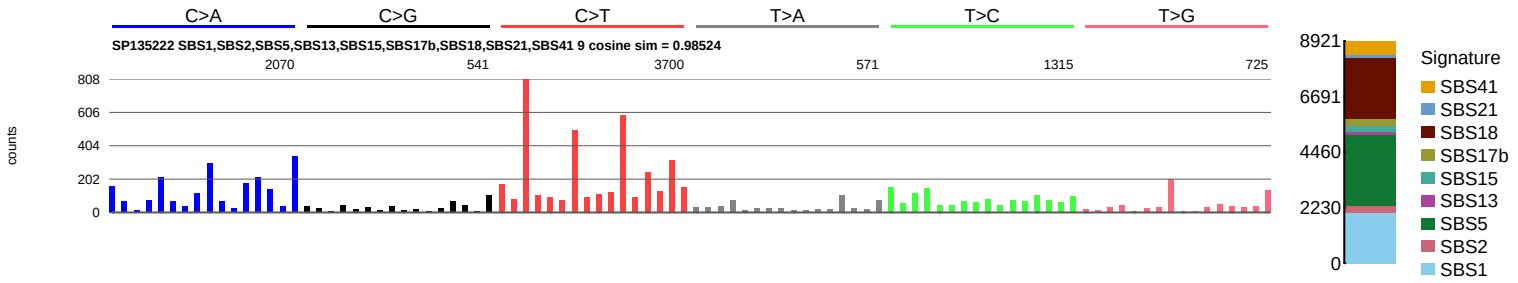

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98528
2074
587
3656
606
1264
734
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98527
2106
559
3687
499
1403
669
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98525
2101
598
3644
625
1148
806
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98525
2144
640
3452
642
1212
833
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98524
2077
592
3701
619
1193
737
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98524
2070
541
3700
571
1315
725
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS41
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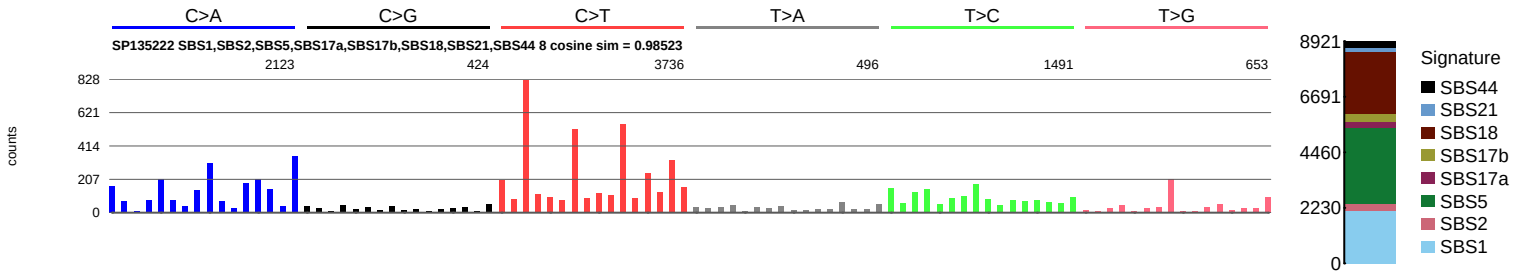
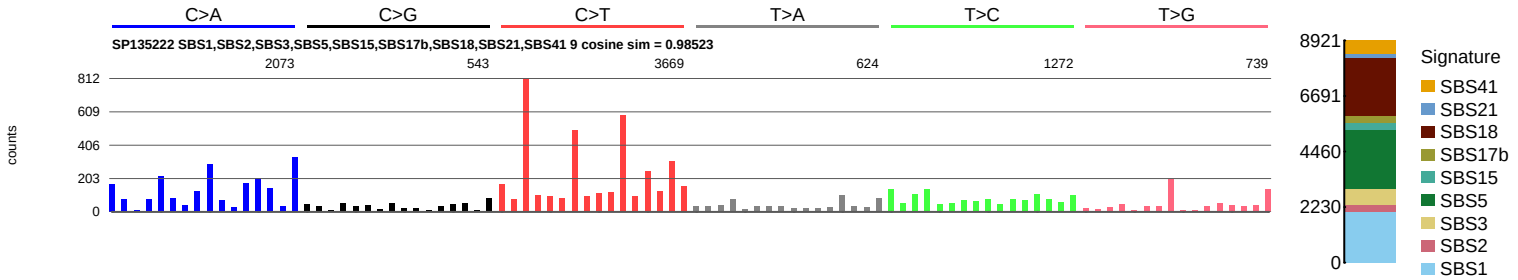
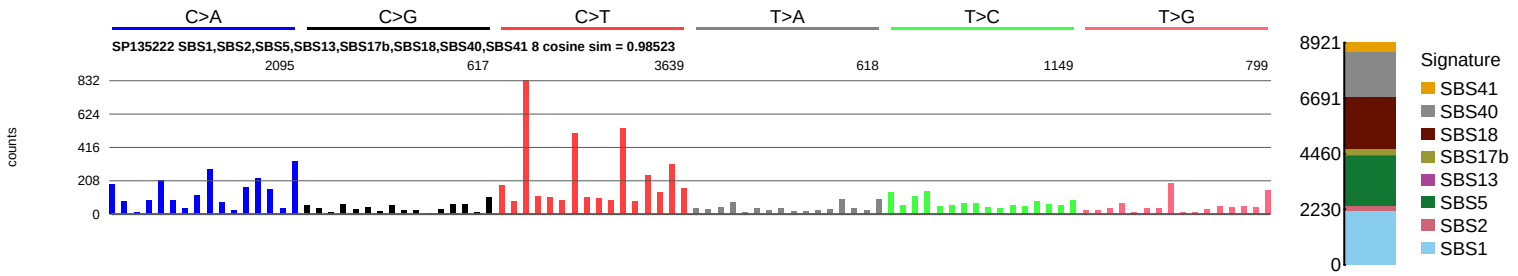
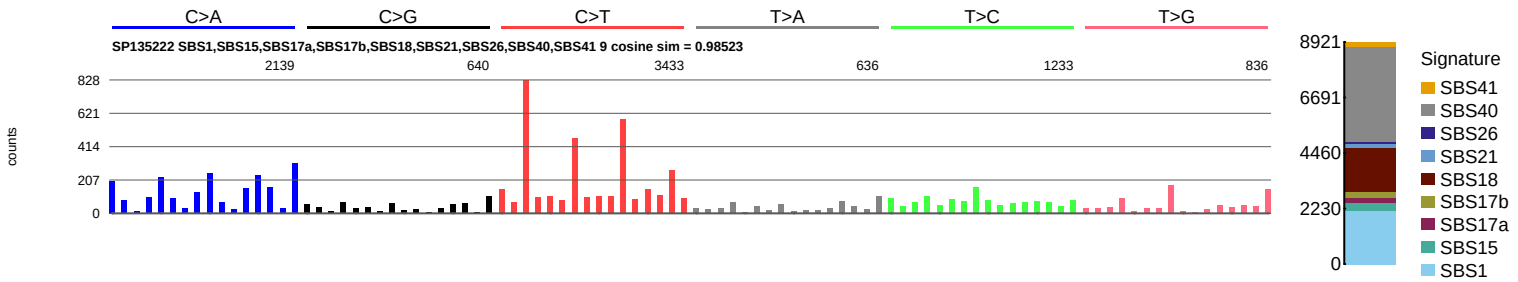
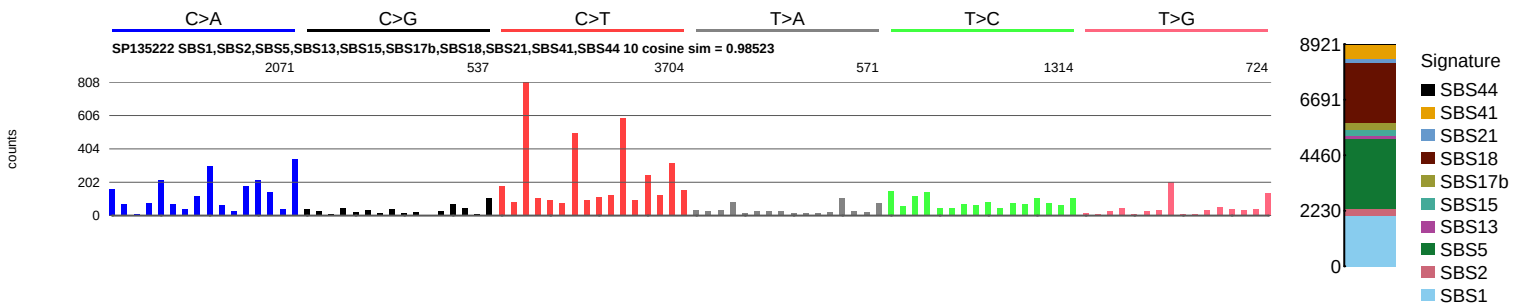
SBS2
SBS1

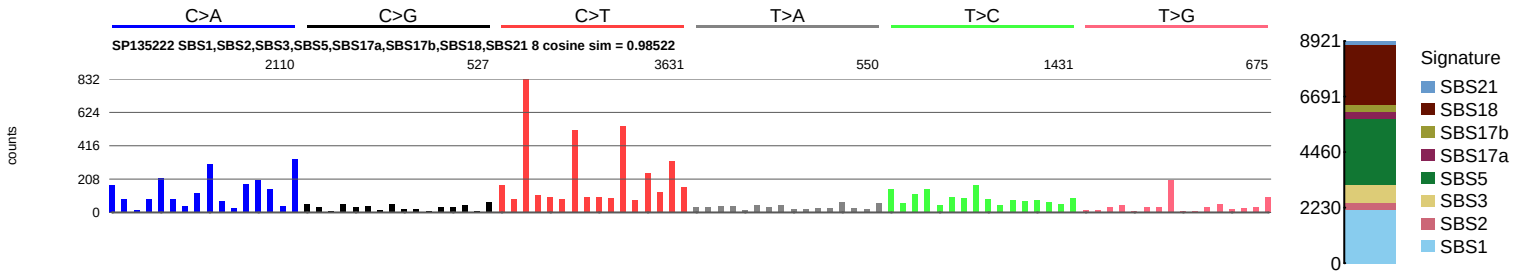

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98523
2123
424
3736
496
1491
653
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

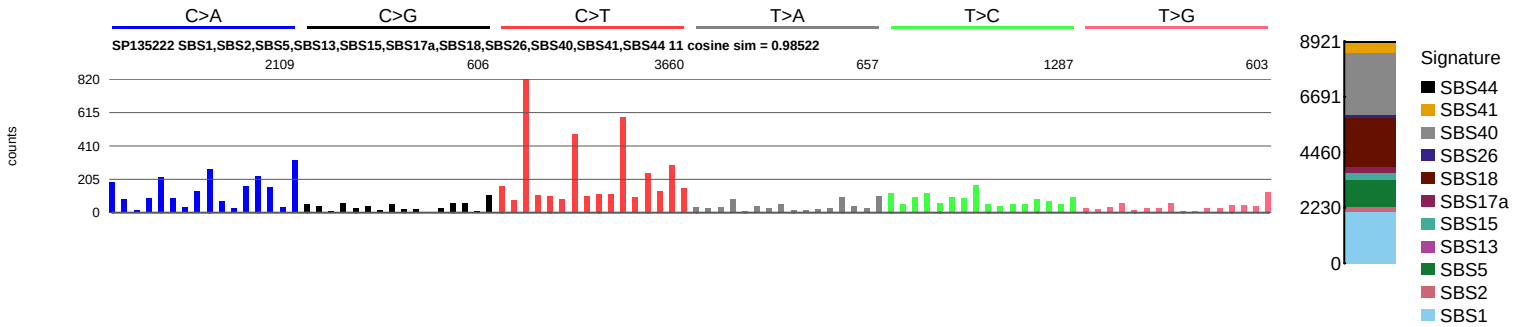

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98523
2073
543
3669
624
1272
739
counts
812
609
406
203
0
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

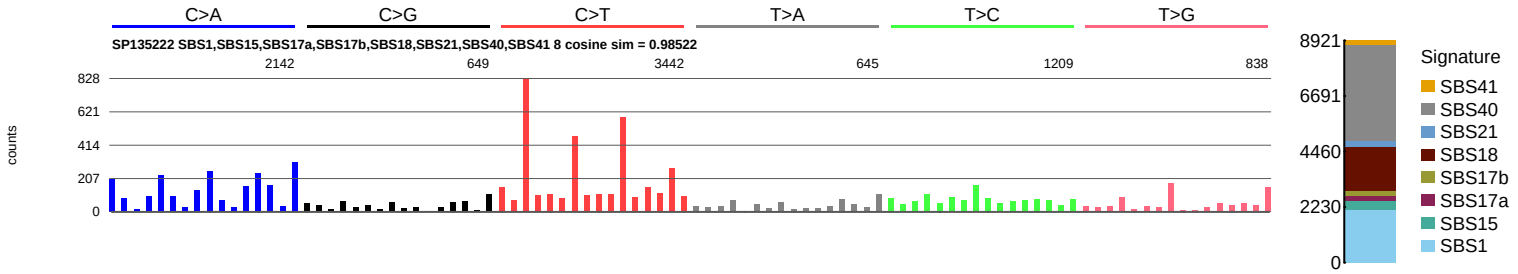

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98523
2095
617
3639
618
1149
799
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

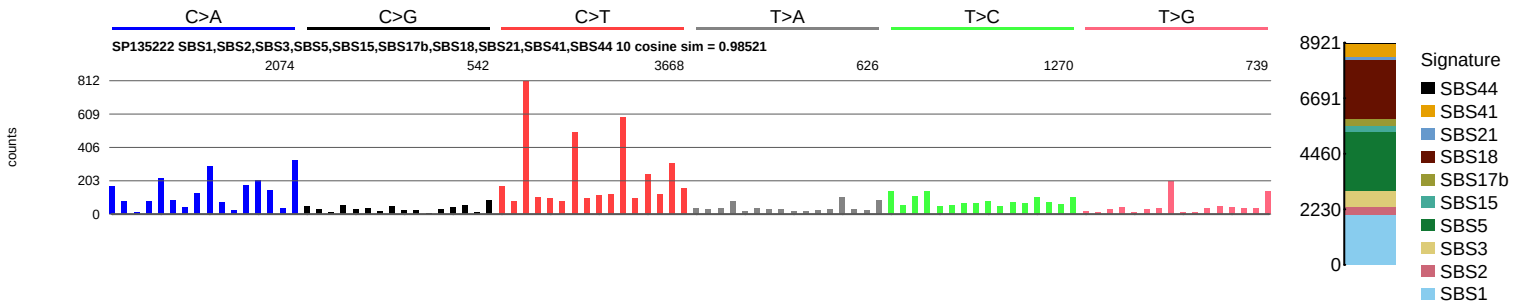

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98523
2139
640
3433
636
1233
836
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

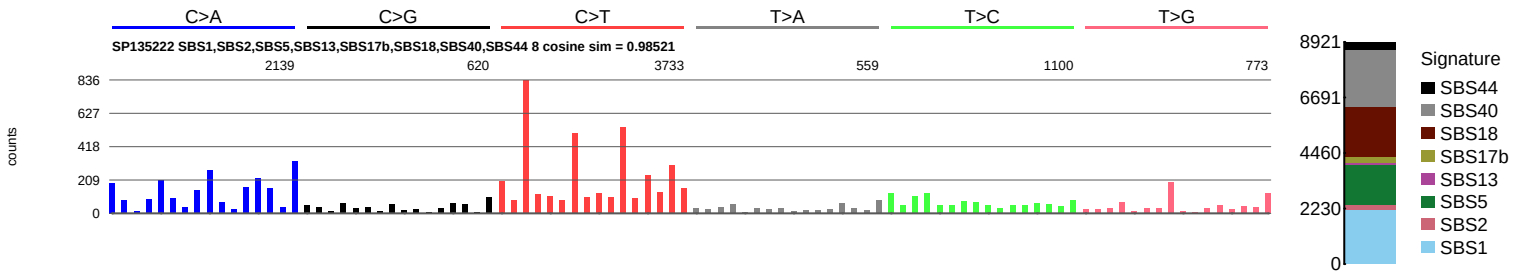

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98523
2071
537
3704
571
1314
724
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1

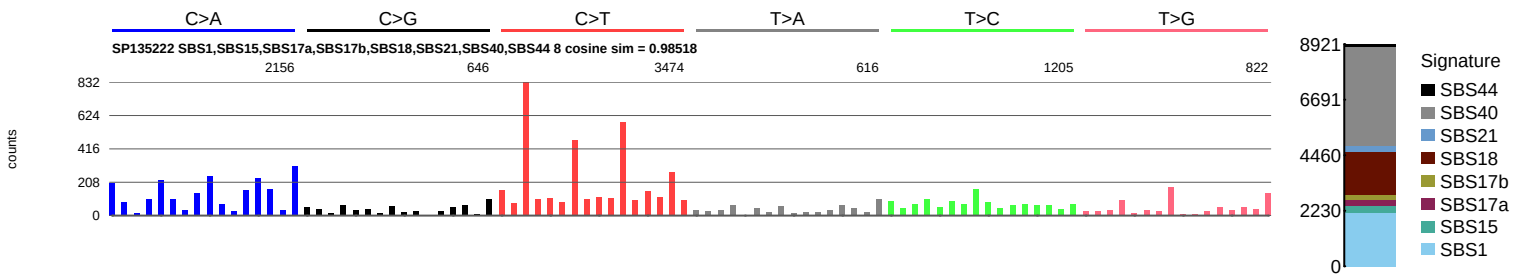

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98522
2110
527
3631
550
1431
675
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

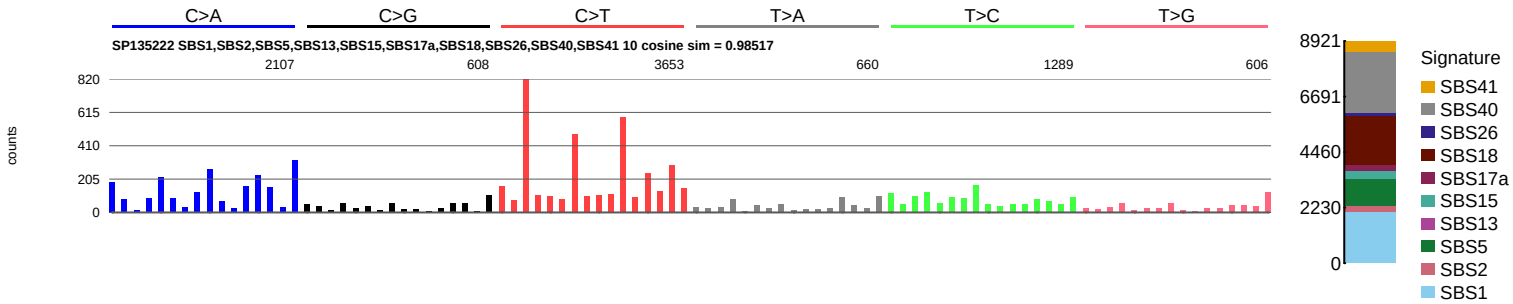

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98522
2109
606
3660
657
1287
603
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

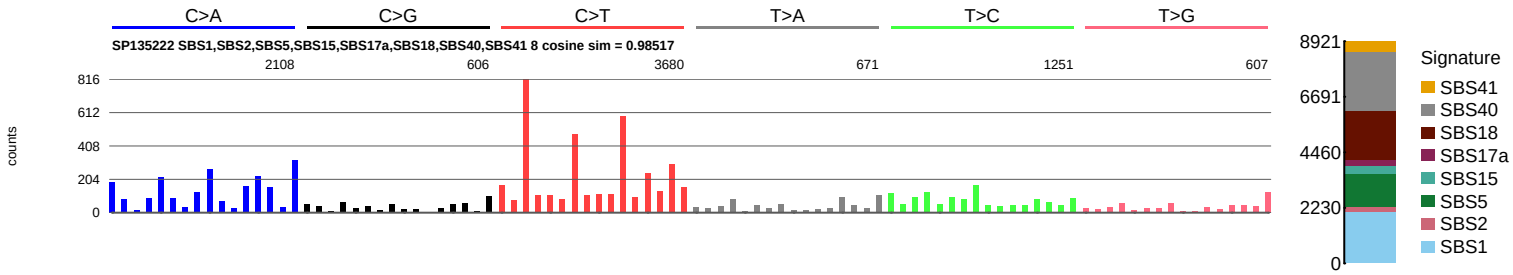

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98522
2142
649
3442
645
1209
838
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

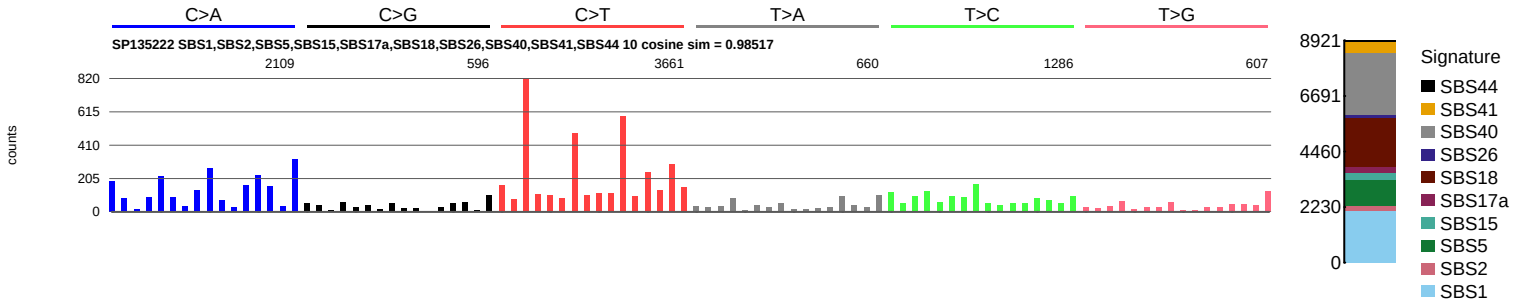

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98521
2074
542
3668
626
1270
739
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

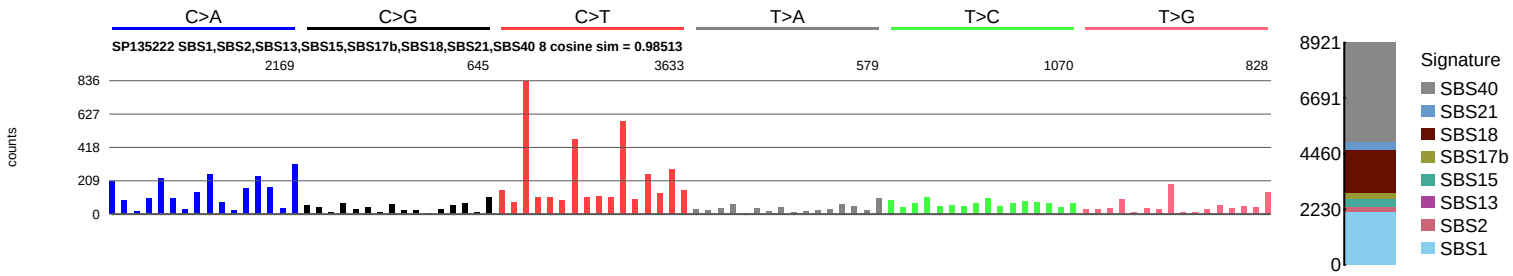

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98521
2139
620
3733
559
1100
773
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

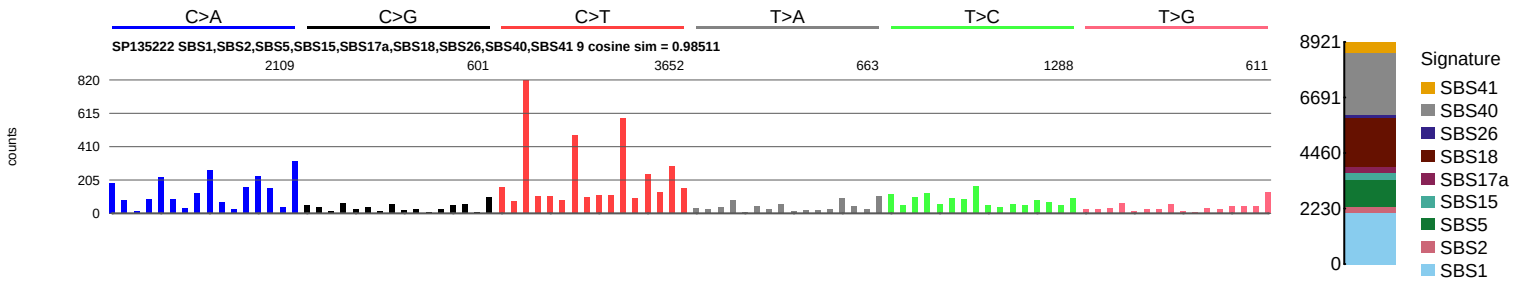

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98518
2156
646
3474
616
1205
822
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

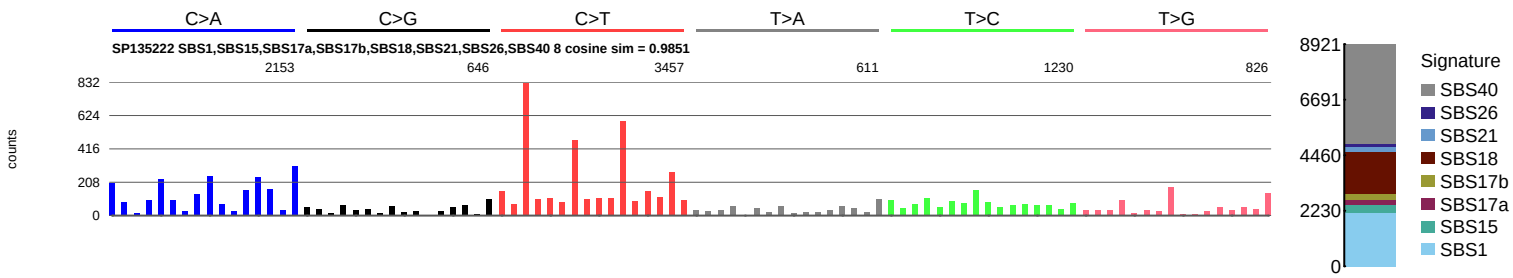

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98517
2107
608
3653
660
1289
606
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

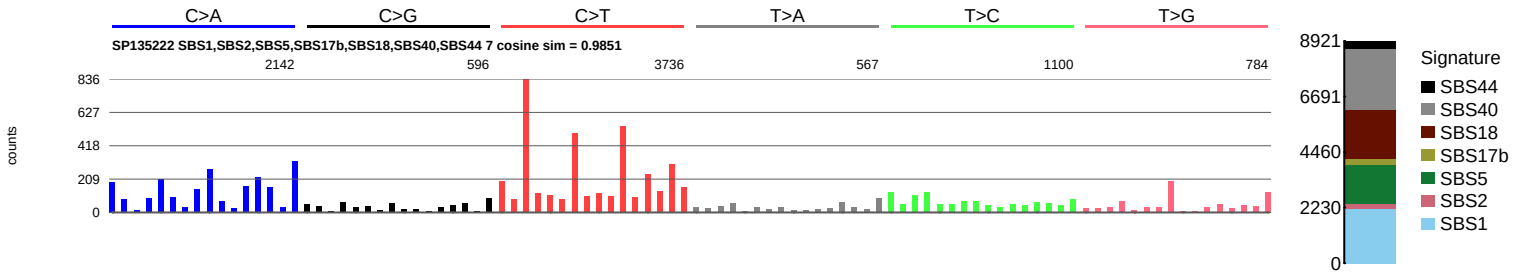

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98517
2108
606
3680
671
1251
607
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1

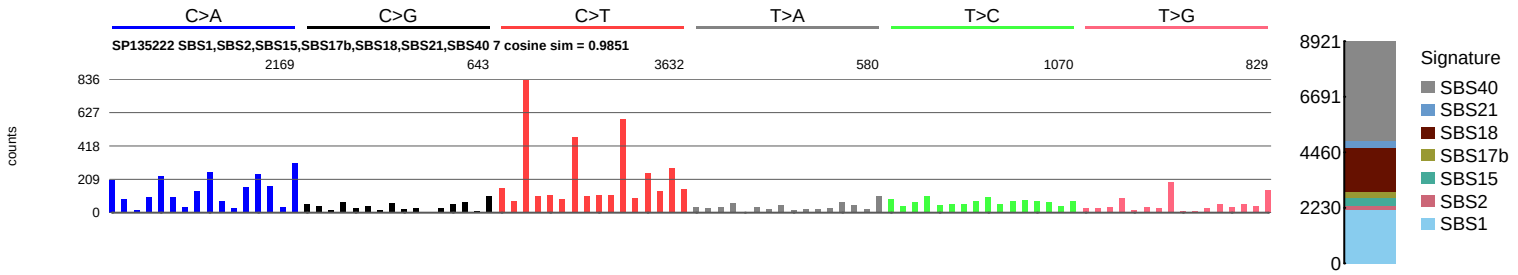

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98517
2109
596
3661
660
1286
607
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1

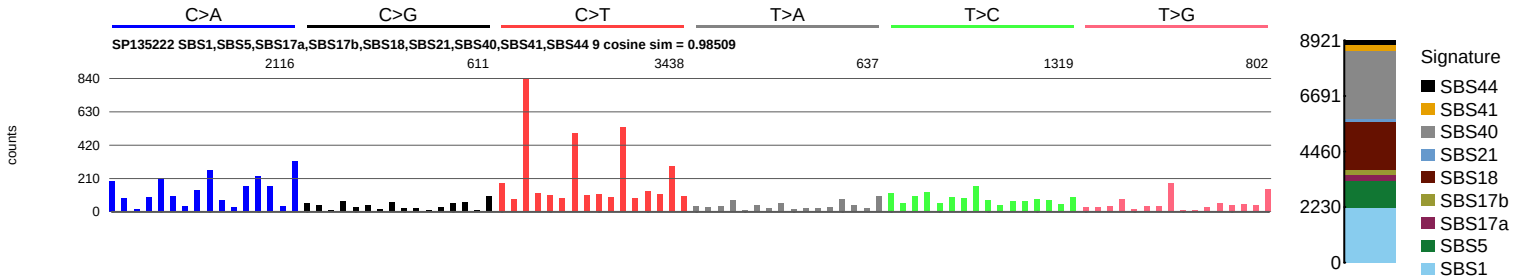

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98513
2169
645
3633
579
1070
828
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98511
2109
601
3652
663
1288
611
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.9851
2153
646
3457
611
1230
826
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.9851
2142
596
3736
567
1100
784
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.9851
2169
643
3632
580
1070
829
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS2
SBS1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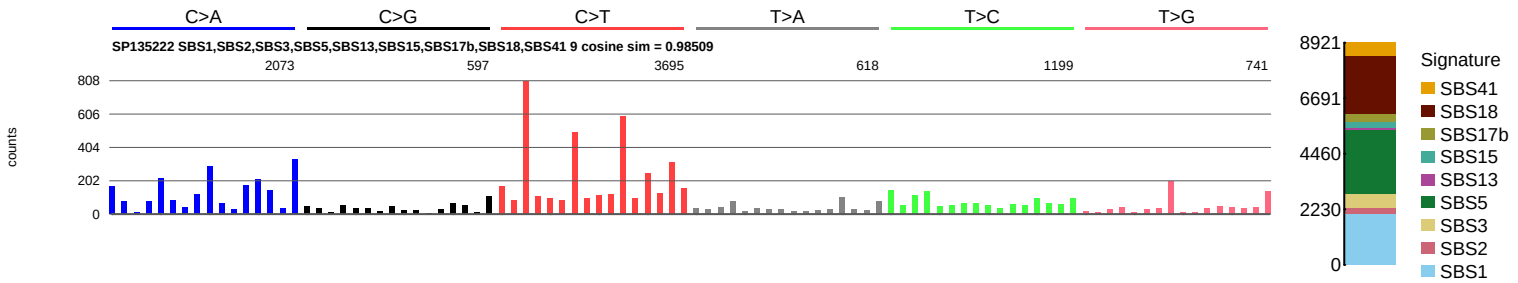
T>G
SP135222 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98509
2116
611
3438
637
1319
802
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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T>C
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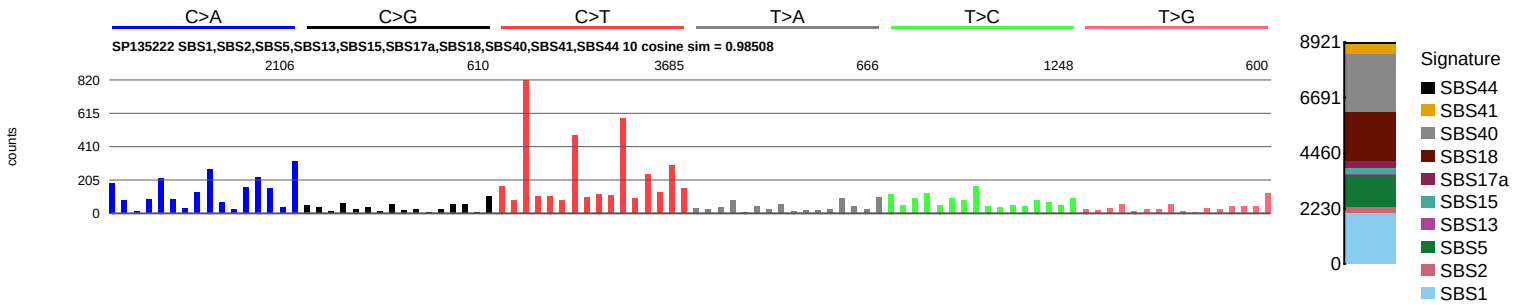
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 9 cosine sim = 0.98509
2073
597
3695
618
1199
741
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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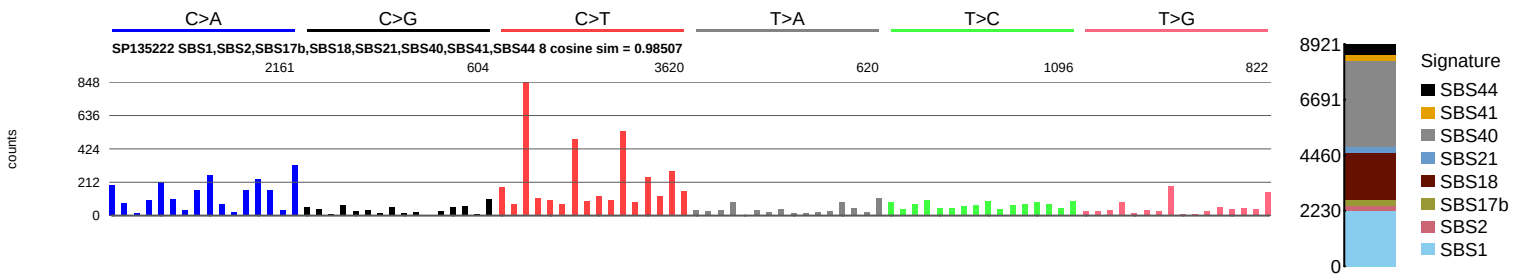
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98508
2106
610
3685
666
1248
600
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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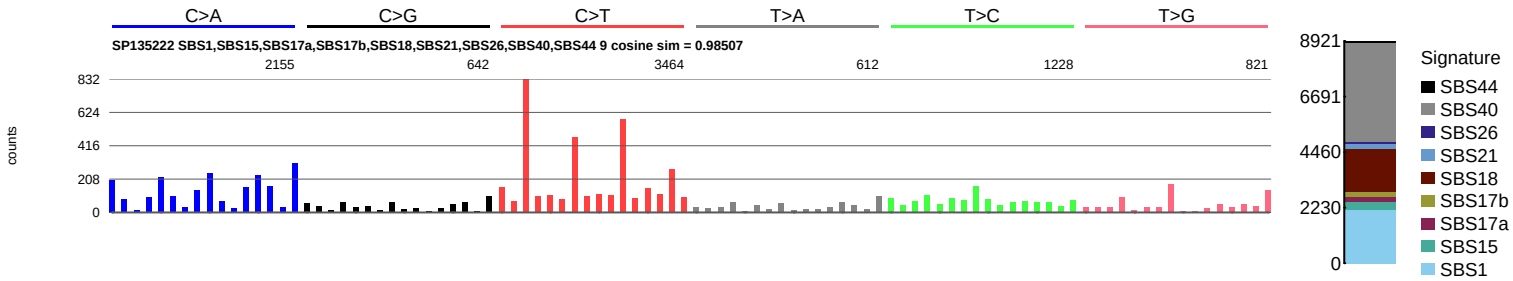
SP135222 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98507
2161
604
3620
620
1096
822
counts
848
636
424
212
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1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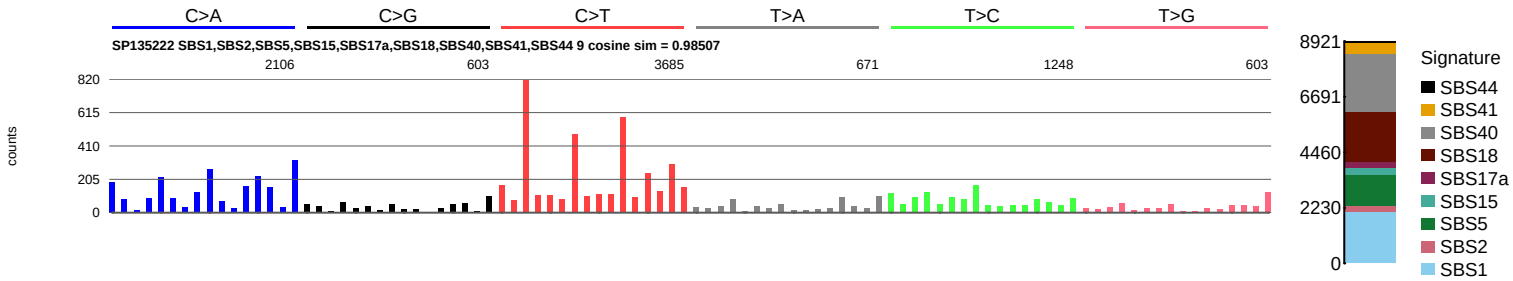
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98507
2155
642
3464
612
1228
821
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98507
2106
603
3685
671
1248
603
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS44
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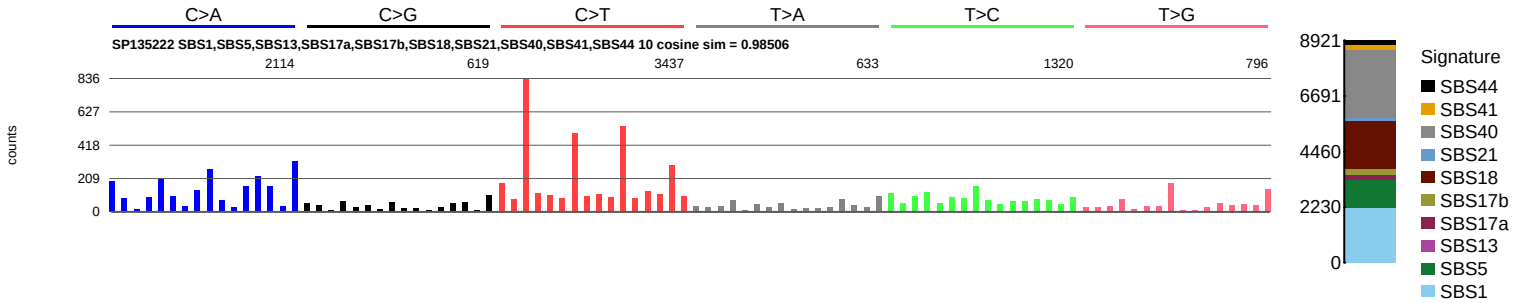
SBS5
SBS2
SBS1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C>G
C>T
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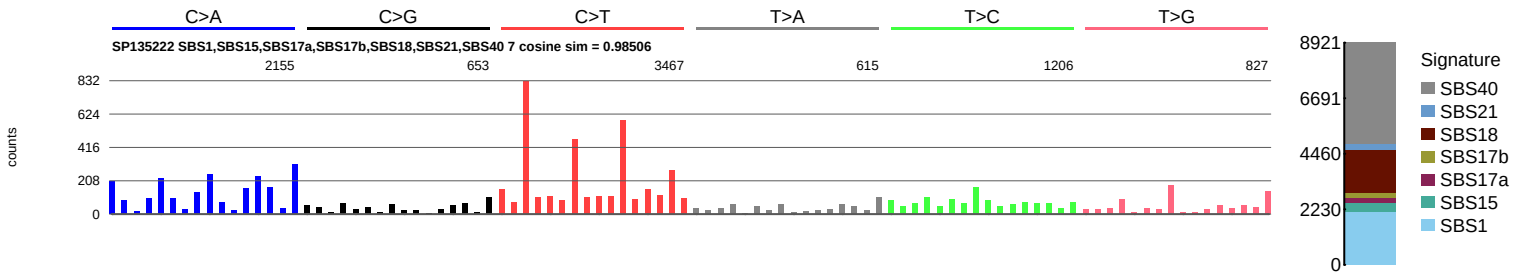
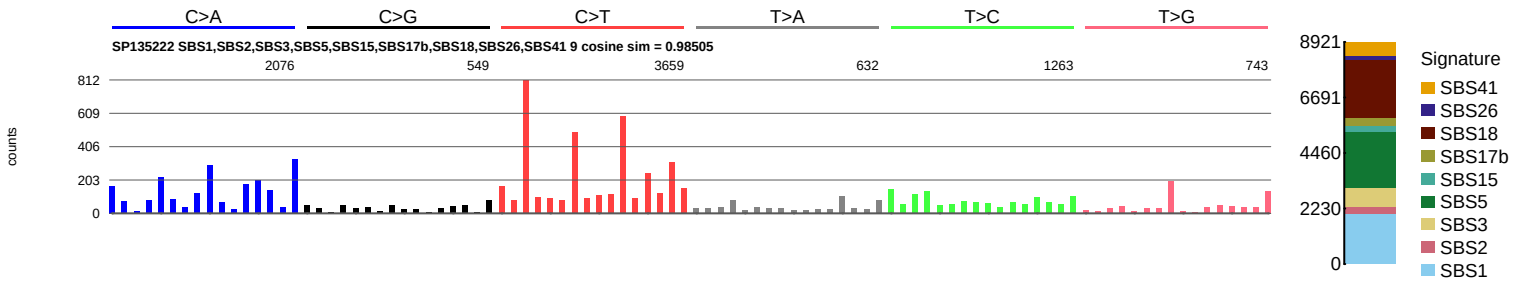
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98506
2114
619
3437
633
1320
796
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98506
2155
653
3467
615
1206
827
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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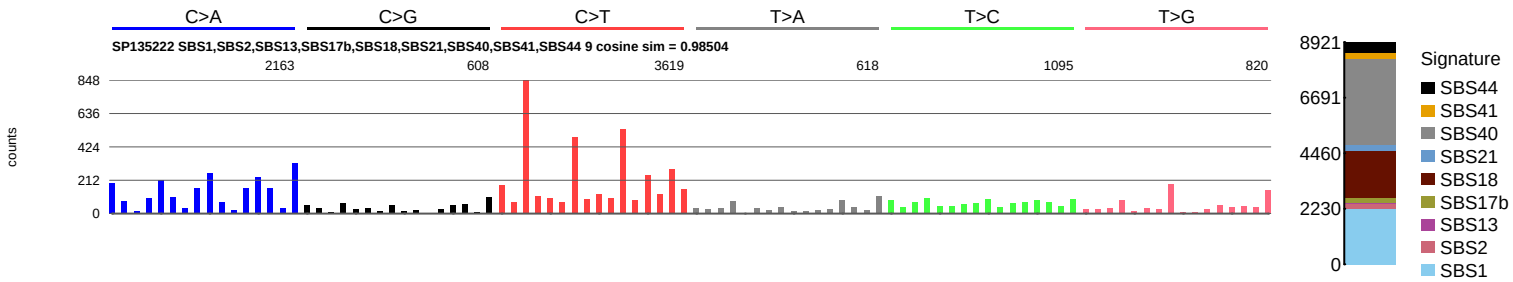
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98505
2076
549
3659
632
1263
743
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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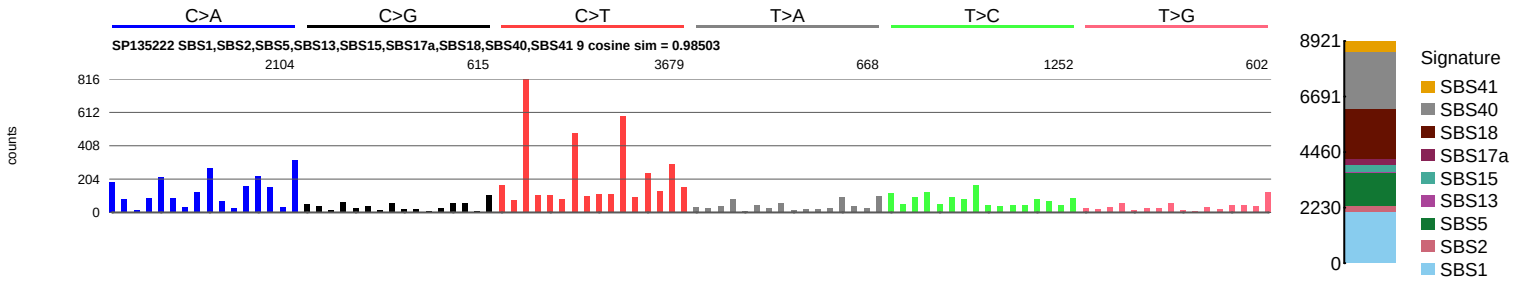
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98504
2163
608
3619
618
1095
820
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS41 9 cosine sim = 0.98503
2104
615
3679
668
1252
602
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS41
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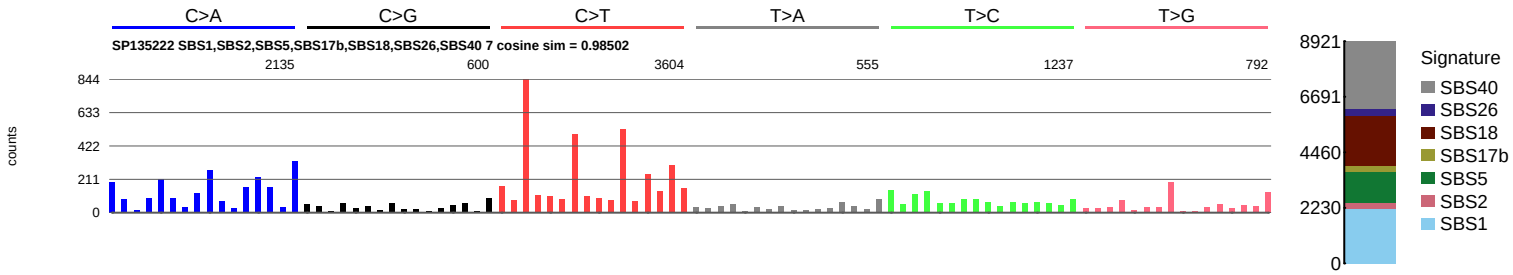
SBS2
SBS1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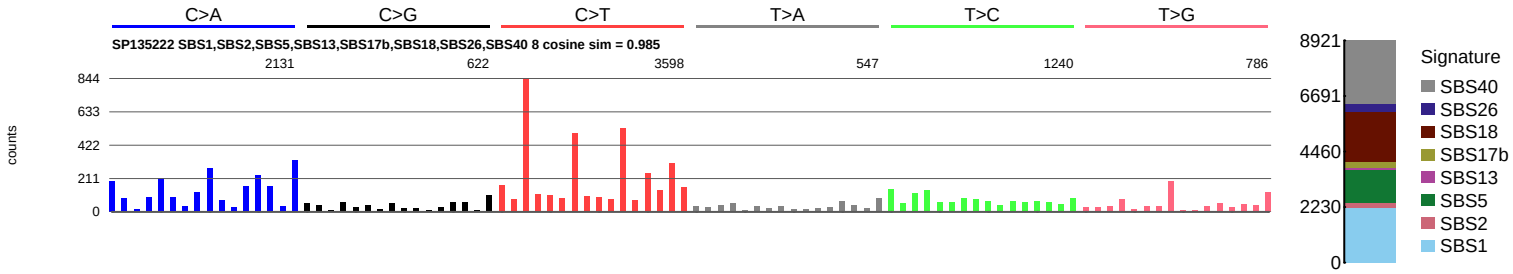
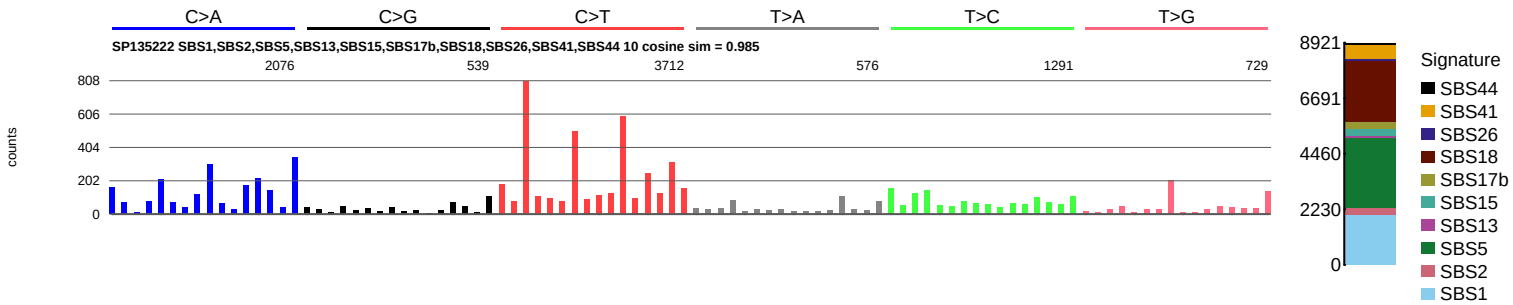
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98502
2135
600
3604
555
1237
792
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.985
2131
622
3598
547
1240
786
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.985
2076
539
3712
576
1291
729
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
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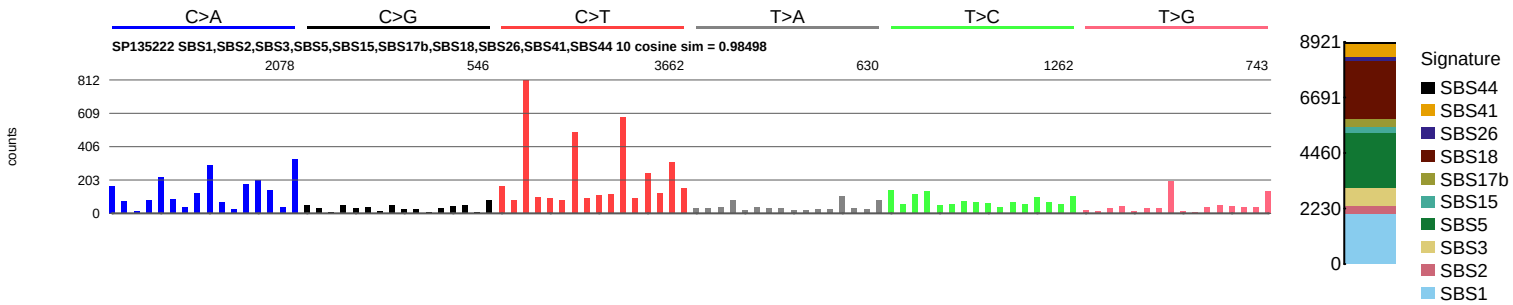
SBS2
SBS1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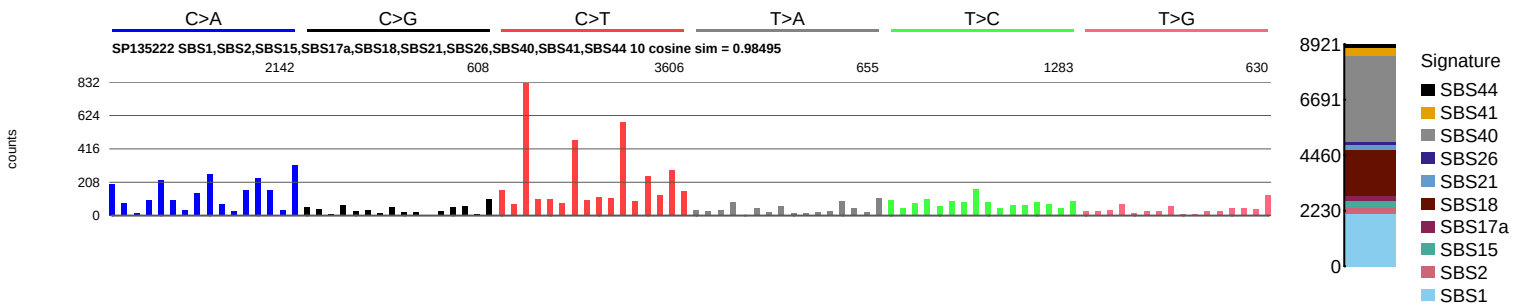
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98498
2078
546
3662
630
1262
743
counts
812
609
406
203
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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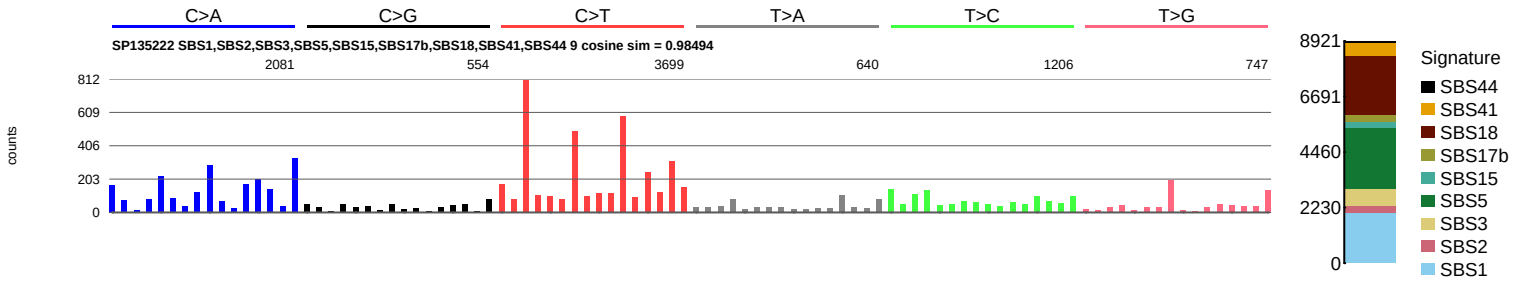
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98495
2142
608
3606
655
1283
630
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1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C>G
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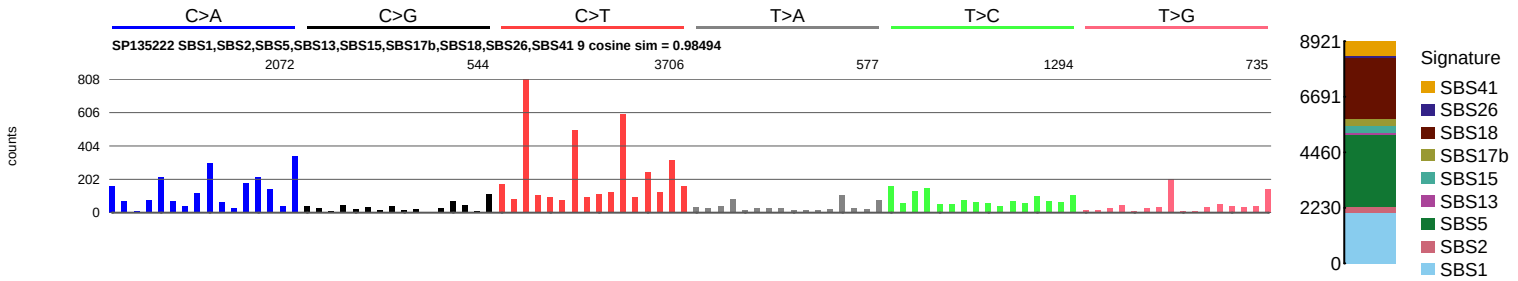
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98494
2081
554
3699
640
1206
747
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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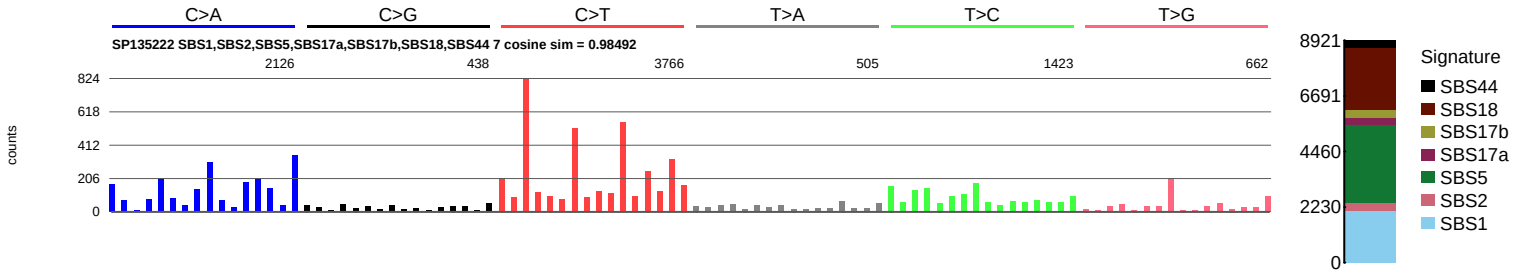
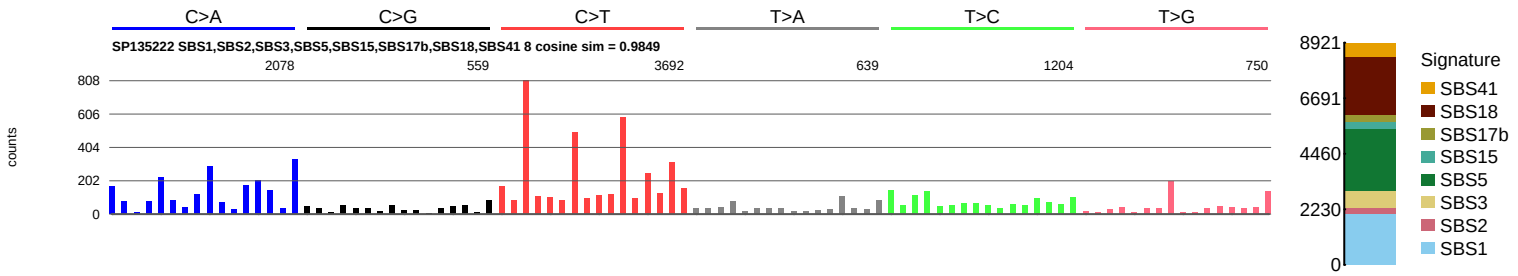
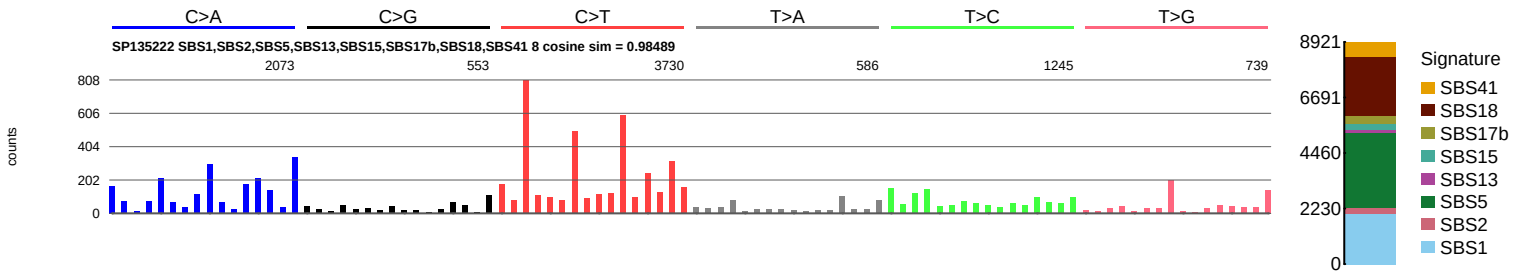
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98494
2072
544
3706
577
1294
735
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98492
2126
438
3766
505
1423
662
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.9849
2078
559
3692
639
1204
750
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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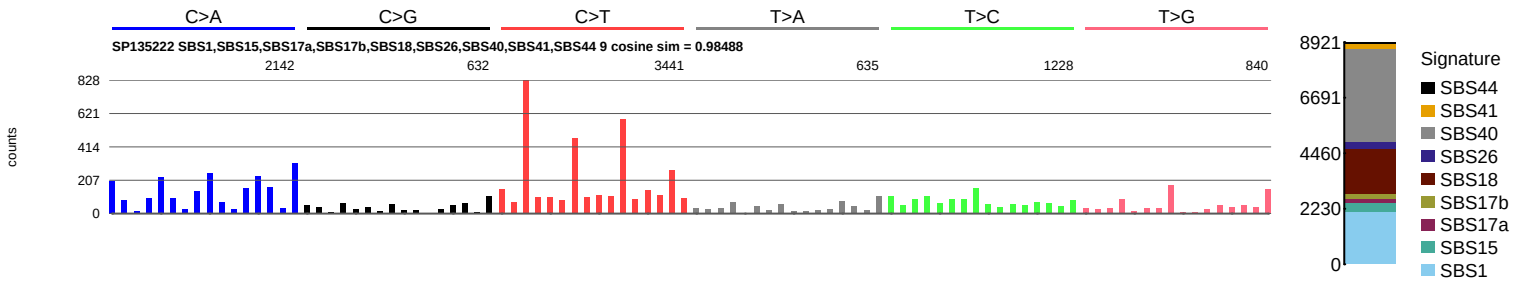
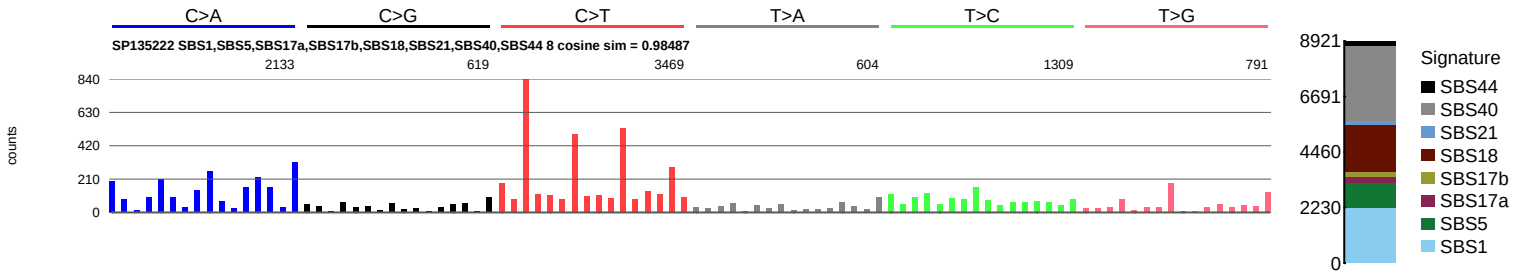
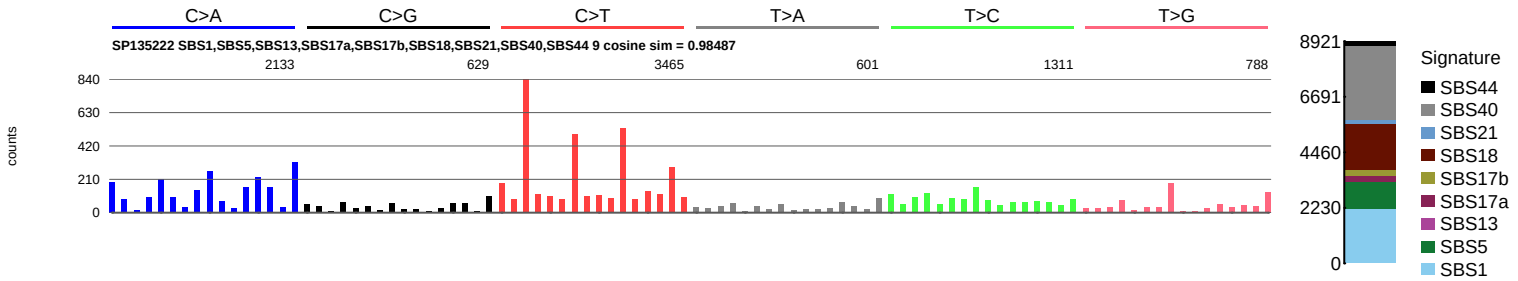
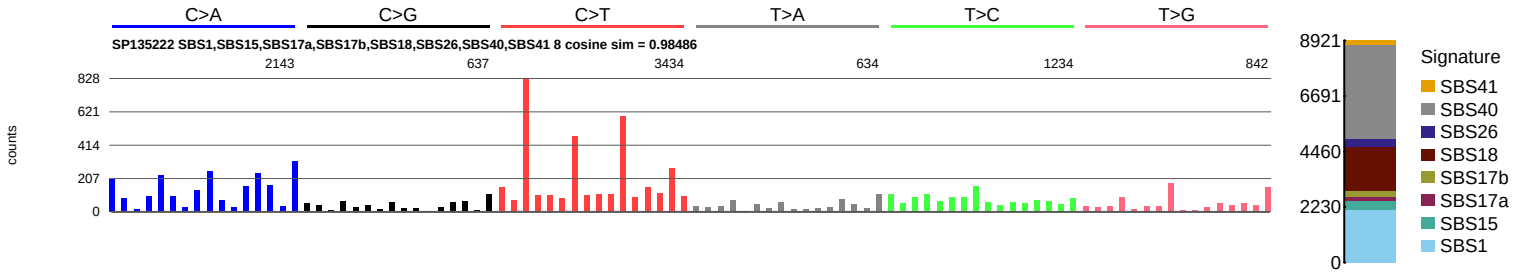
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41 8 cosine sim = 0.98489
2073
553
3730
586
1245
739
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98488
2142
632
3441
635
1228
840
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98487
2133
619
3469
604
1309
791
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98487
2133
629
3465
601
1311
788
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98486
2143
637
3434
634
1234
842
828
621
414
207
0
counts
8921
6691
4460
2230
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Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
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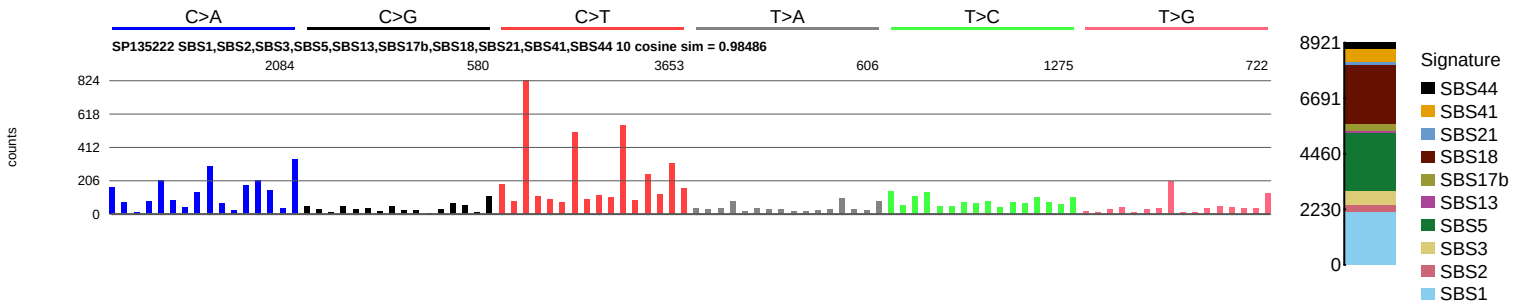
SBS15
SBS1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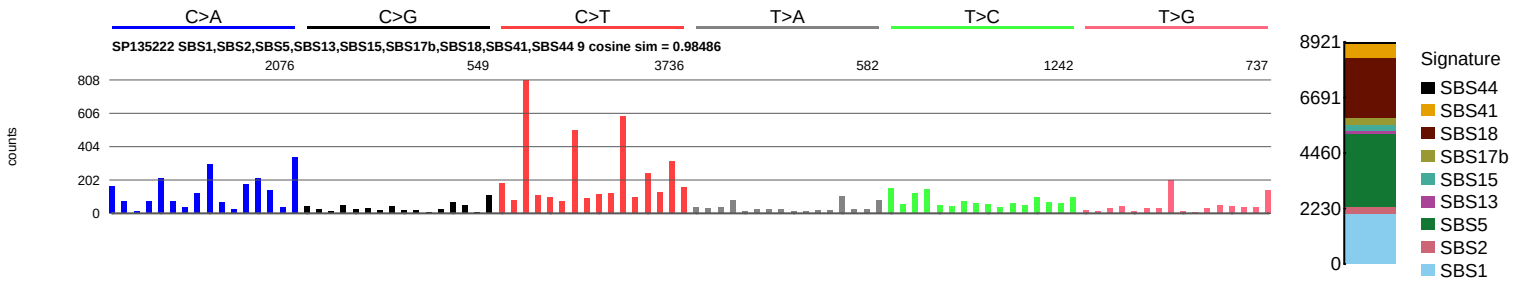
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98486
2084
580
3653
606
1275
722
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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C>T
T>A
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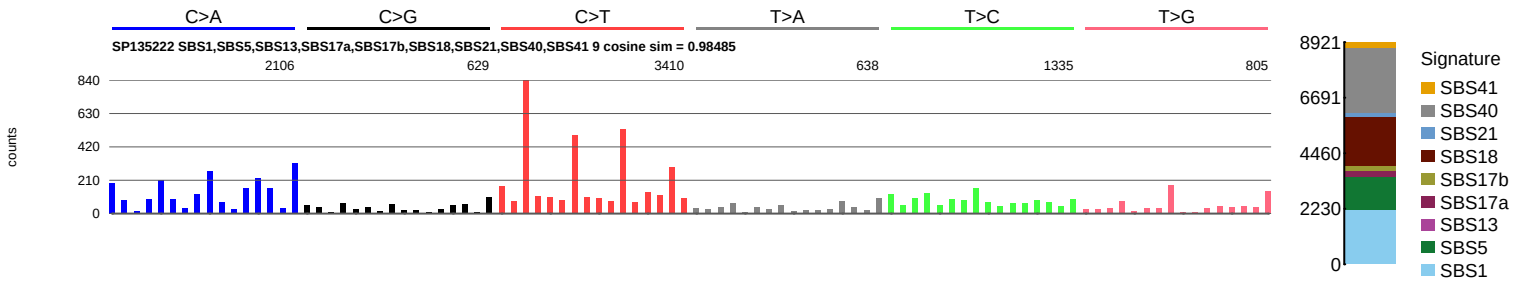
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98486
2076
549
3736
582
1242
737
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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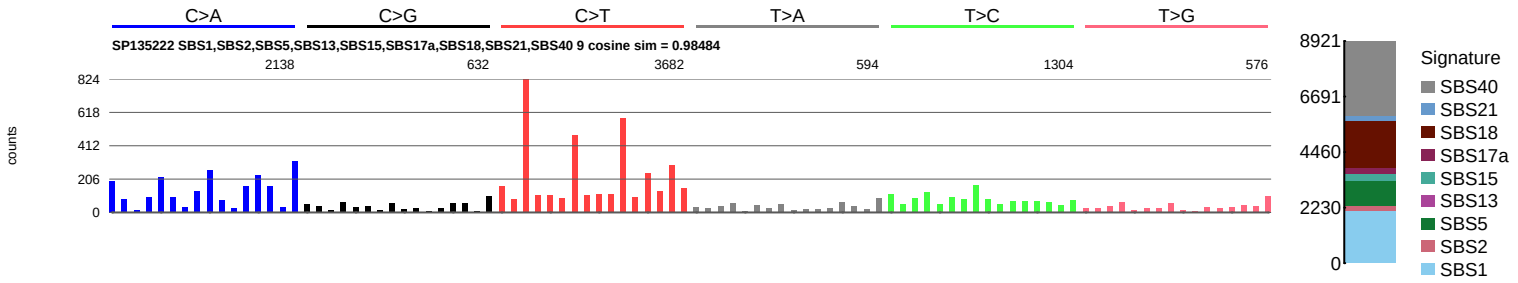
T>G
SP135222 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98485
2106
629
3410
638
1335
805
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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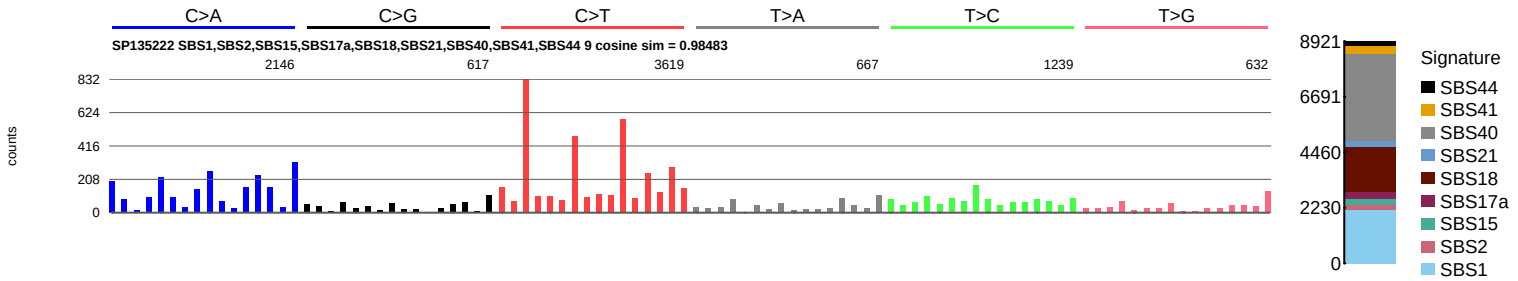
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40 9 cosine sim = 0.98484
2138
632
3682
594
1304
576
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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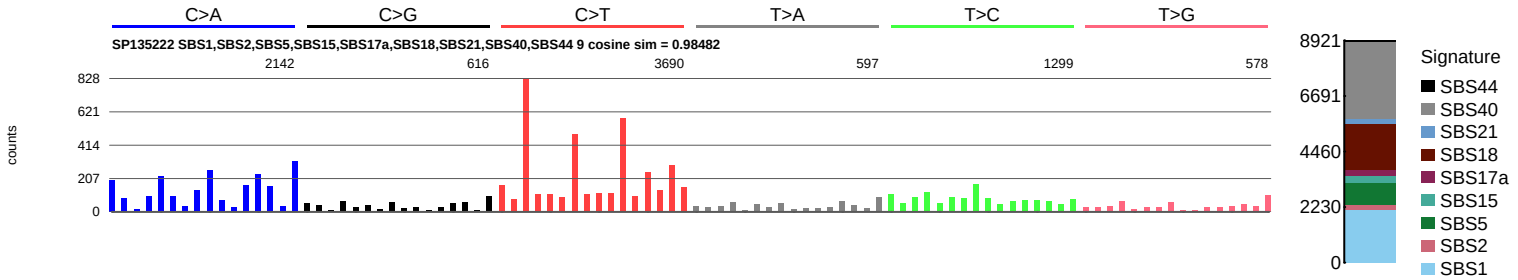
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98483
2146
617
3619
667
1239
632
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1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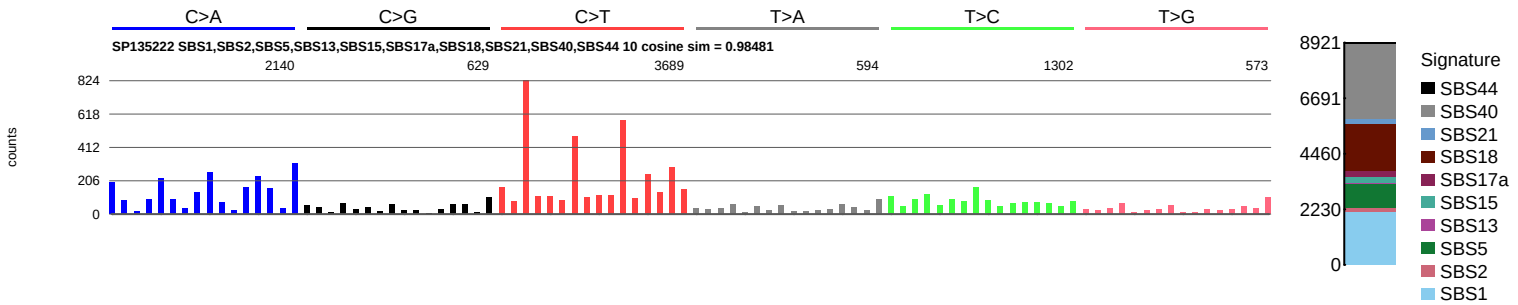
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98482
2142
616
3690
597
1299
578
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1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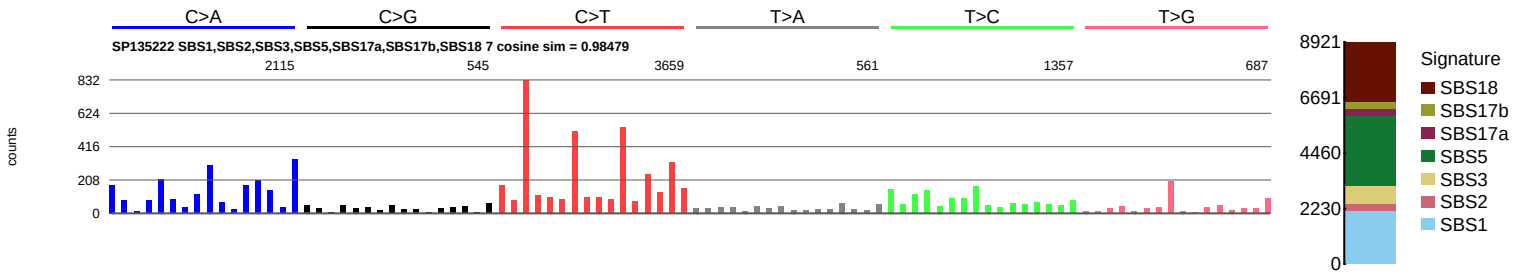
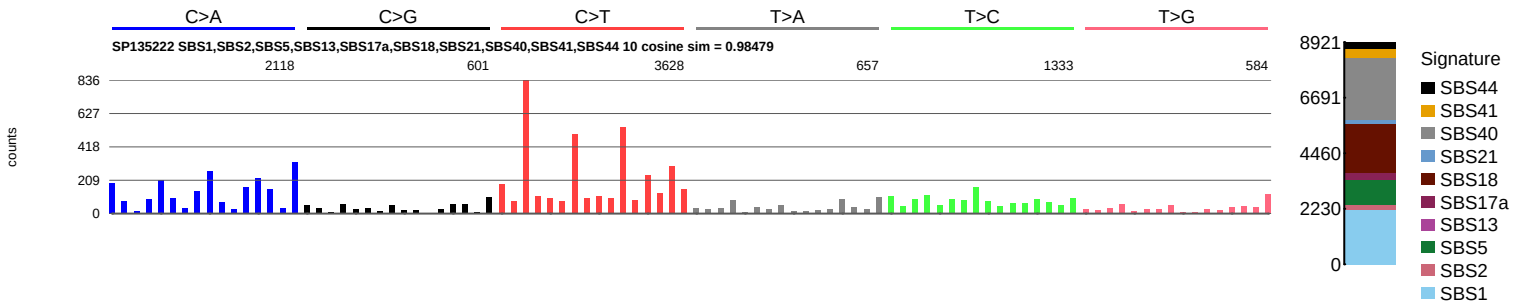
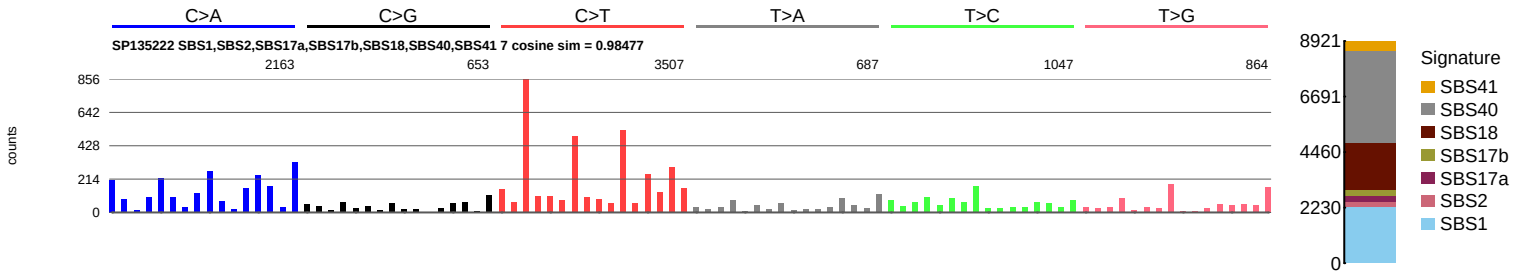
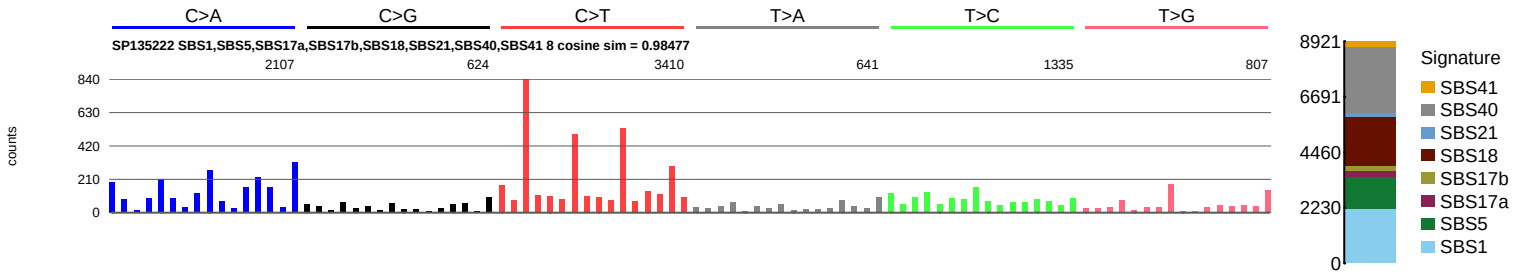
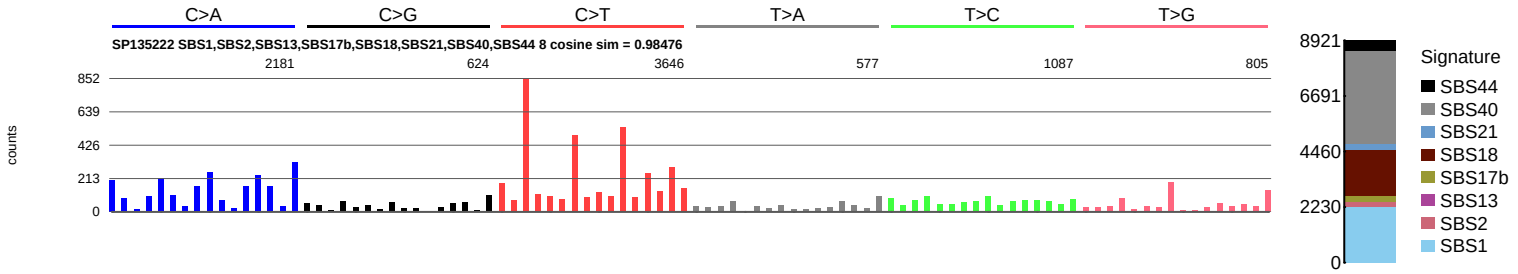
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98481
2140
629
3689
594
1302
573
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98479
2115
545
3659
561
1357
687
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98479
2118
601
3628
657
1333
584
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98477
2163
653
3507
687
1047
864
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98477
2107
624
3410
641
1335
807
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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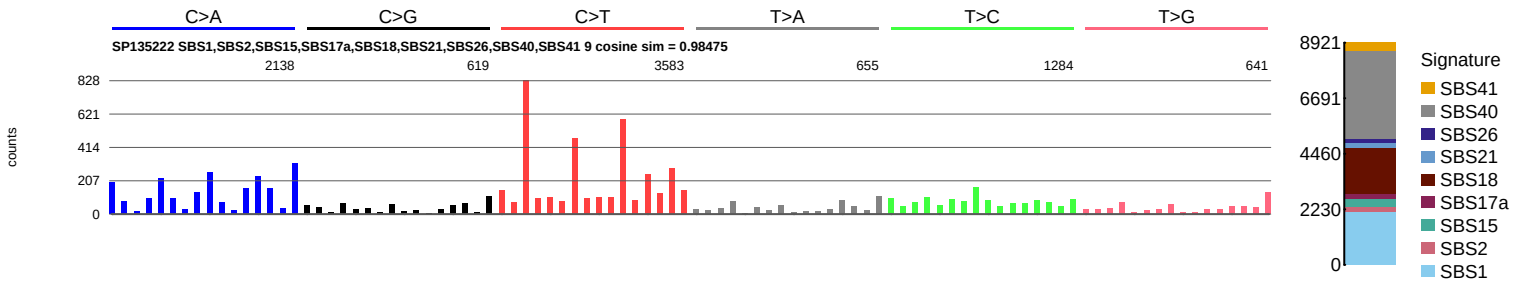
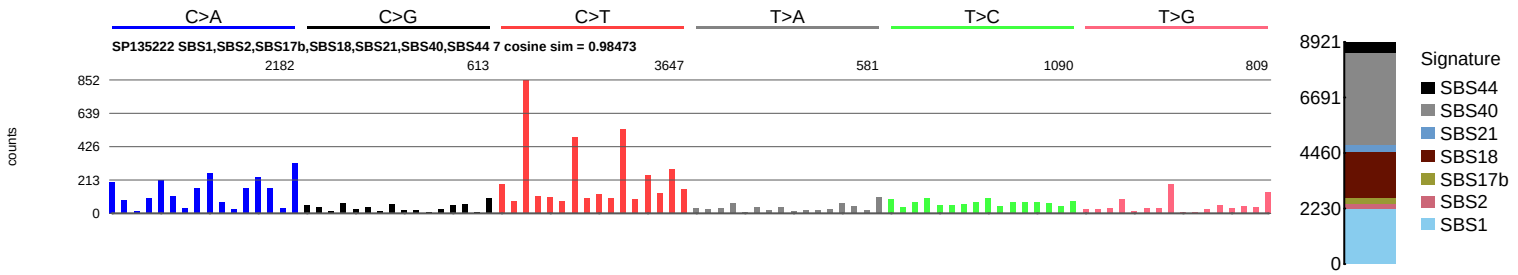
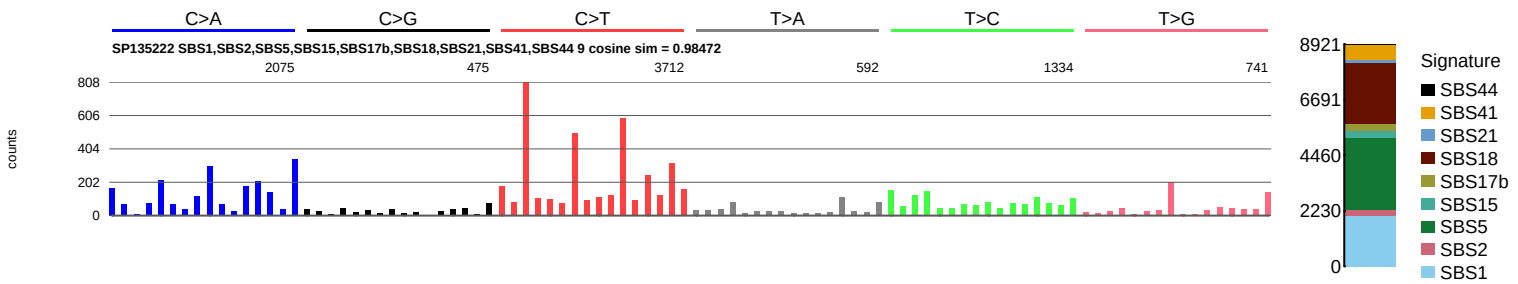
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98476
2181
624
3646
577
1087
805
counts
852
639
426
213
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98475
2138
619
3583
655
1284
641
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98473
2182
613
3647
581
1090
809
counts
852
639
426
213
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1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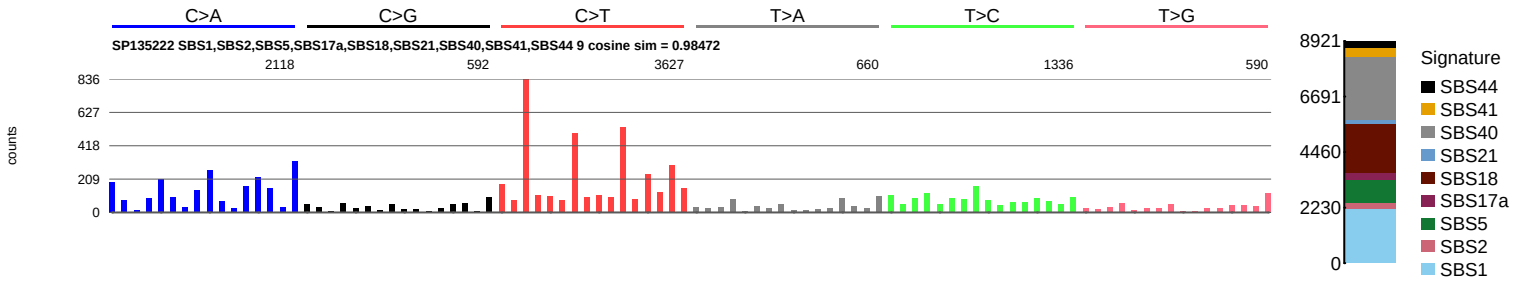
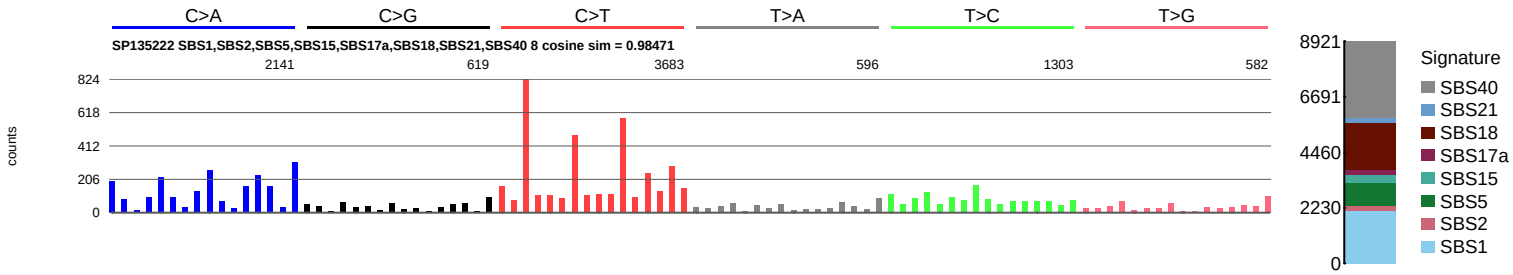
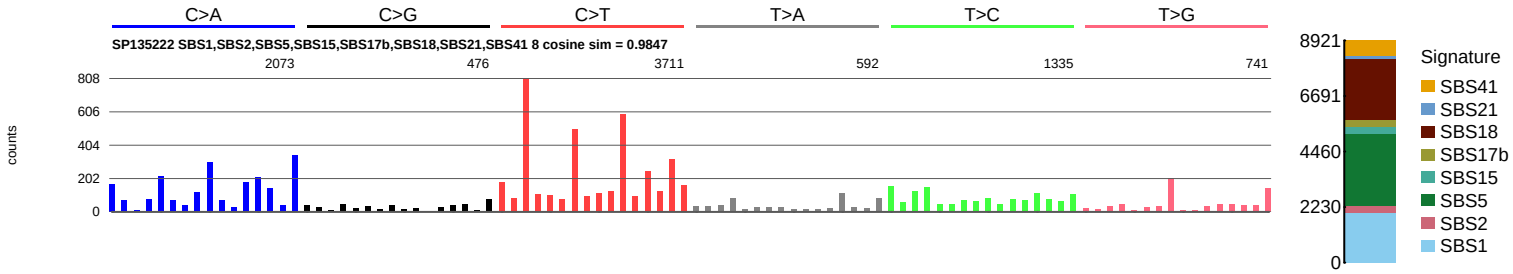
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98472
2075
475
3712
592
1334
741
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98472
2118
592
3627
660
1336
590
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.98471
2141
619
3683
596
1303
582
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.9847
2073
476
3711
592
1335
741
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS41
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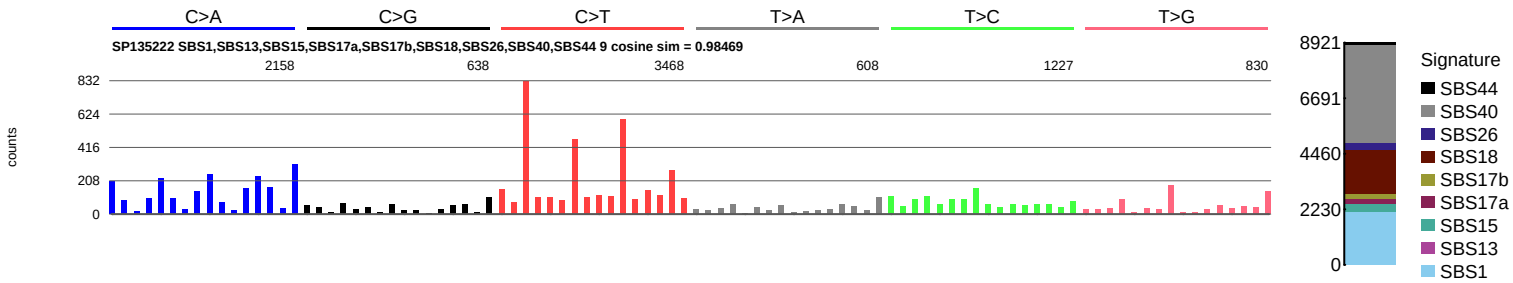
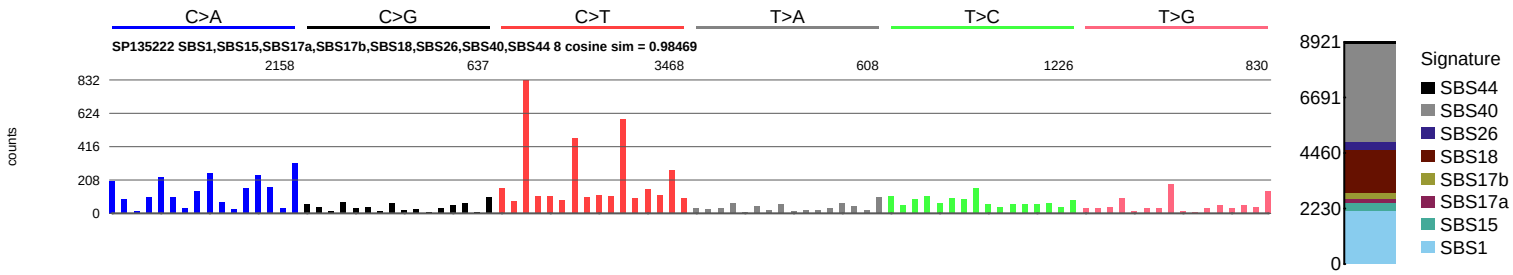
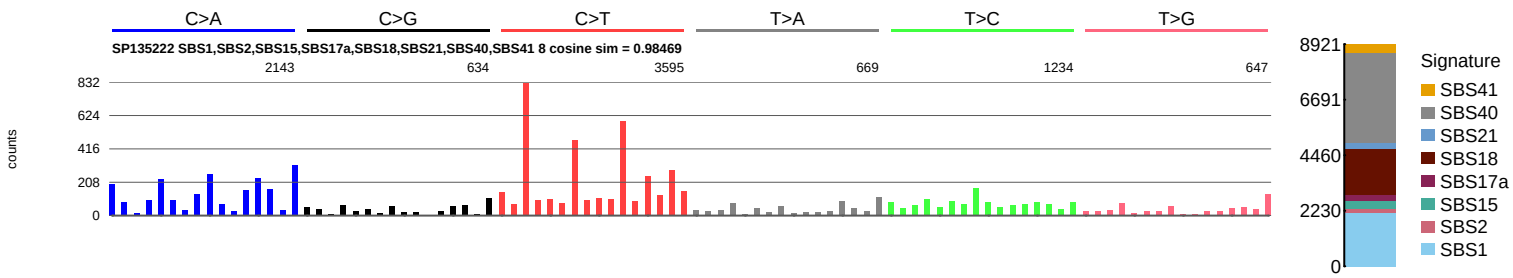
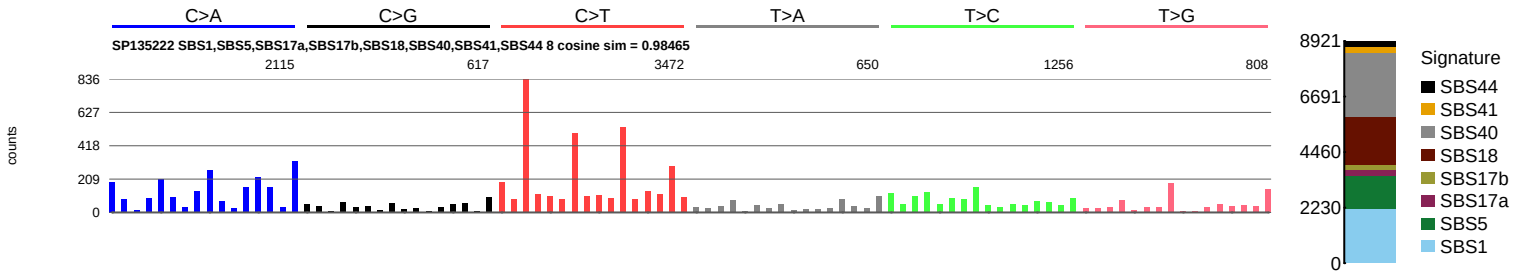
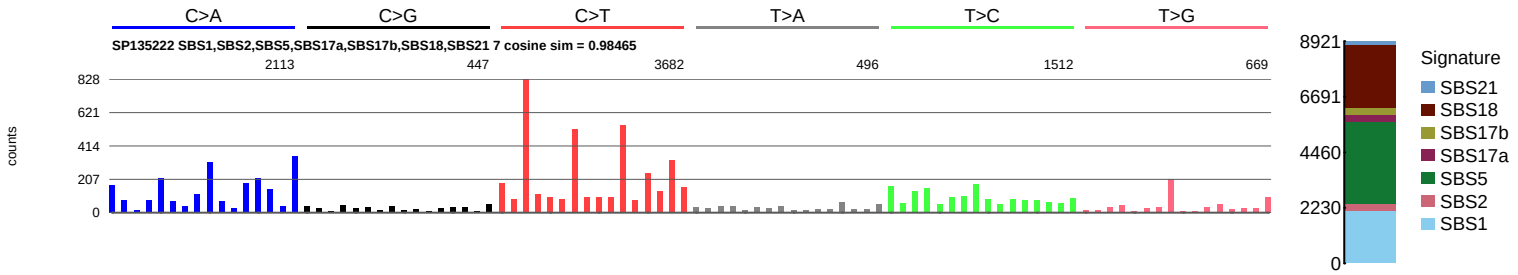
SBS2
SBS1

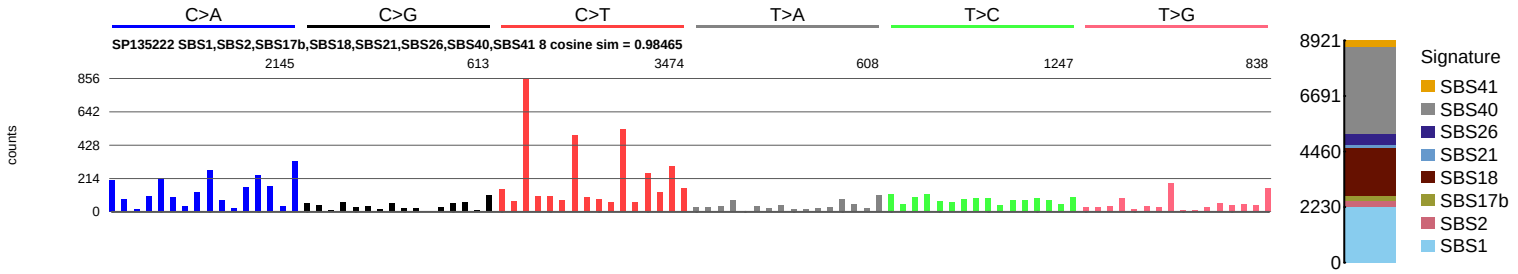

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98469
2158
638
3468
608
1227
830
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

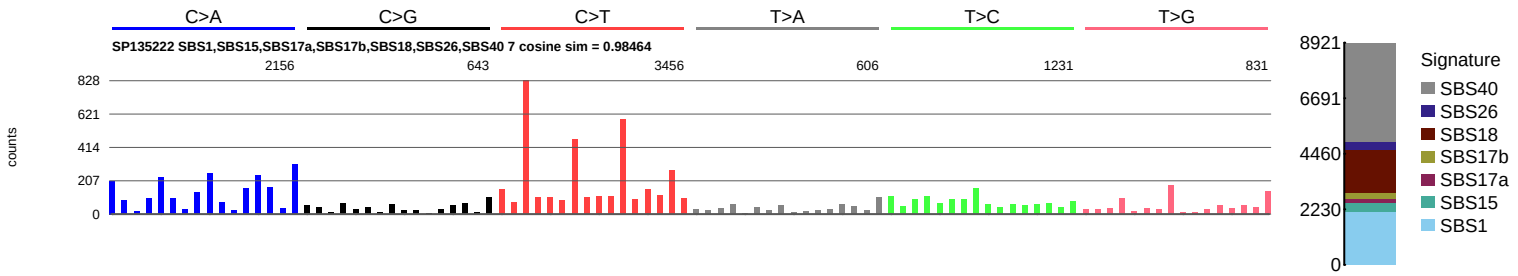

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98469
2158
637
3468
608
1226
830
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

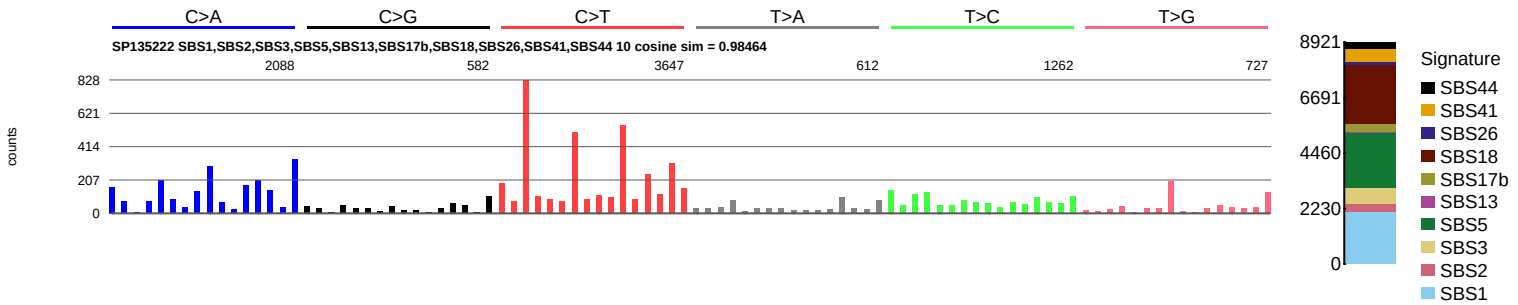

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98469
2143
634
3595
669
1234
647
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1

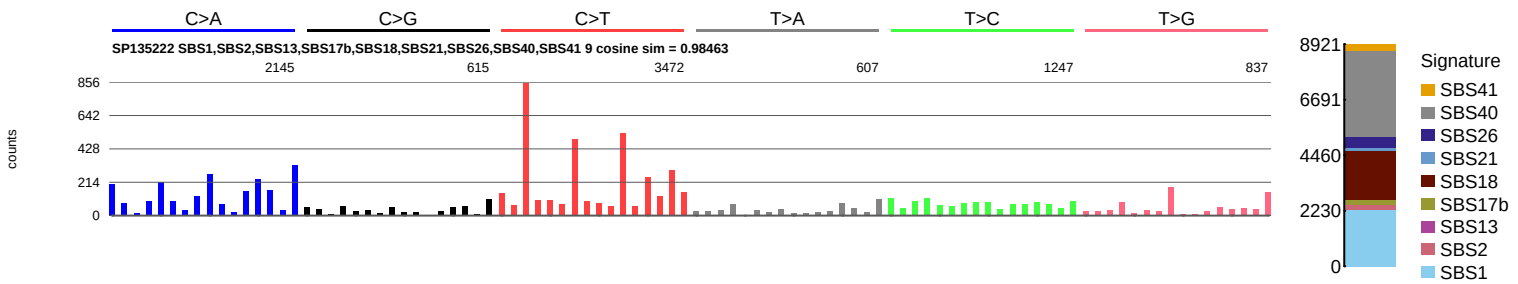

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98465
2115
617
3472
650
1256
808
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98465
2113
447
3682
496
1512
669
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98465
2145
613
3474
608
1247
838
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98464
2156
643
3456
606
1231
831
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98464
2088
582
3647
612
1262
727
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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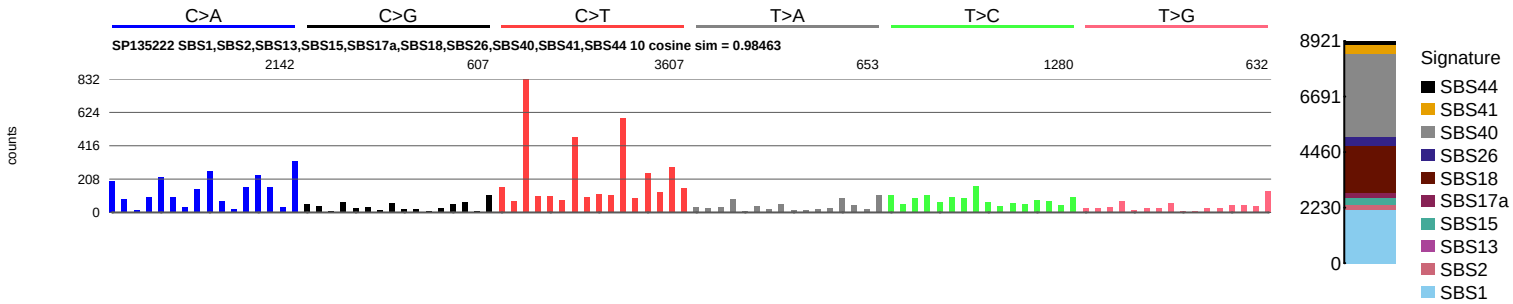
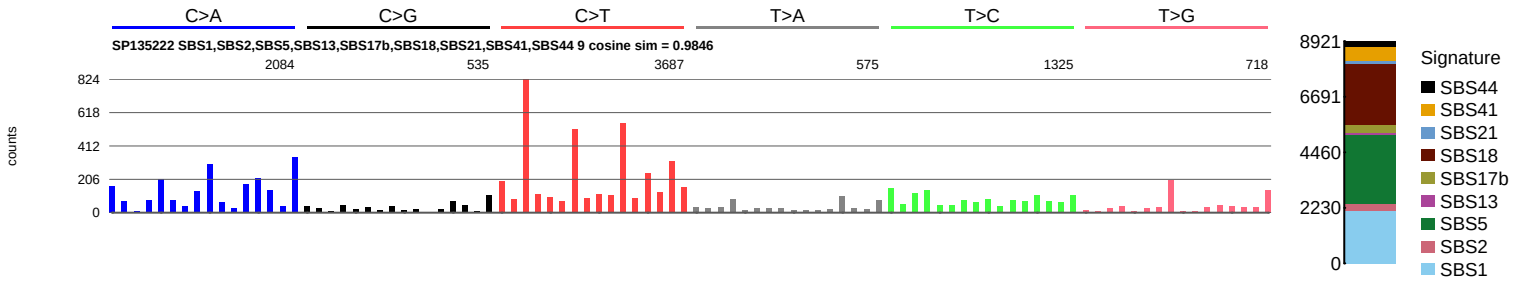
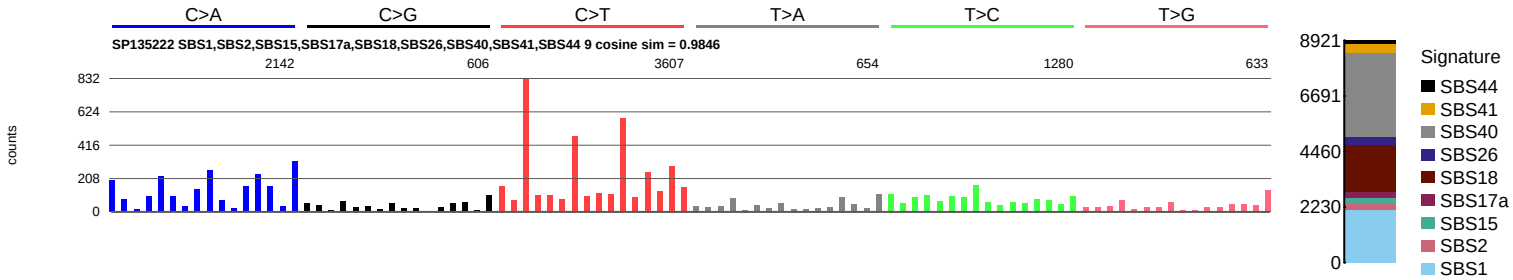
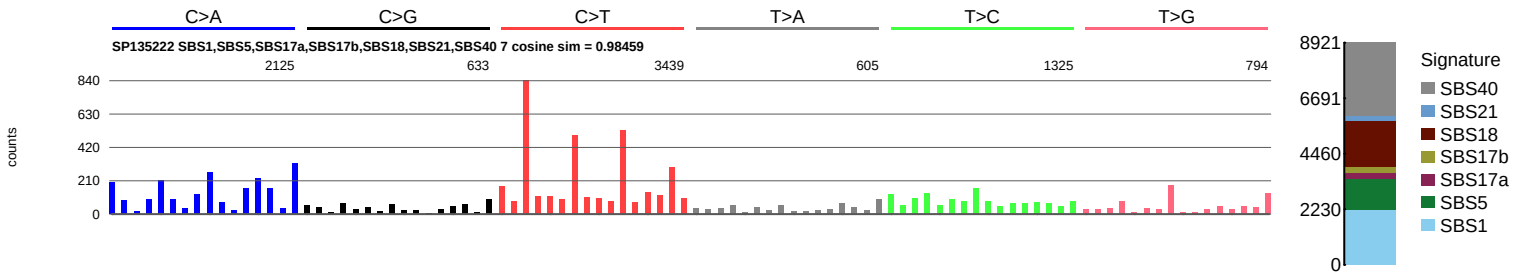
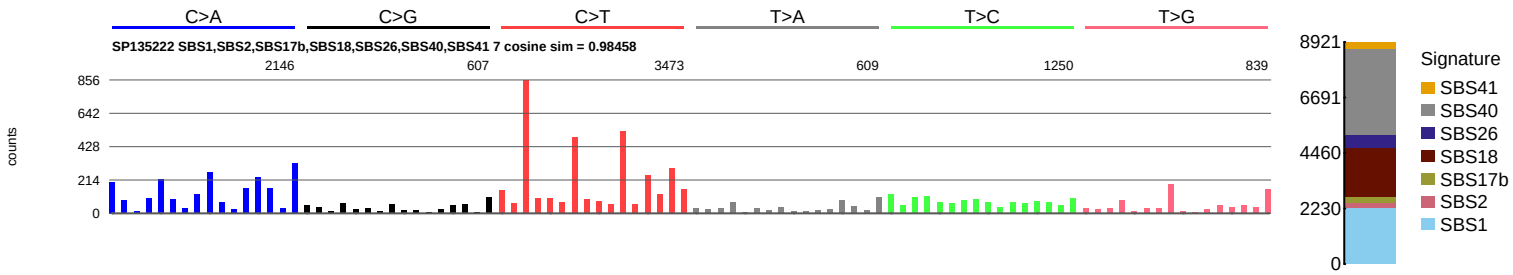
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98463
2145
615
3472
607
1247
837
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1

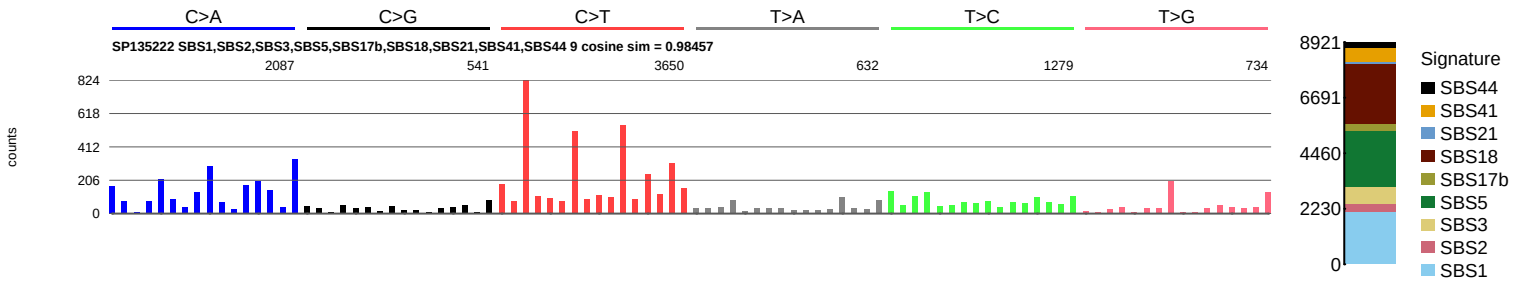

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98463
2142
607
3607
653
1280
632
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

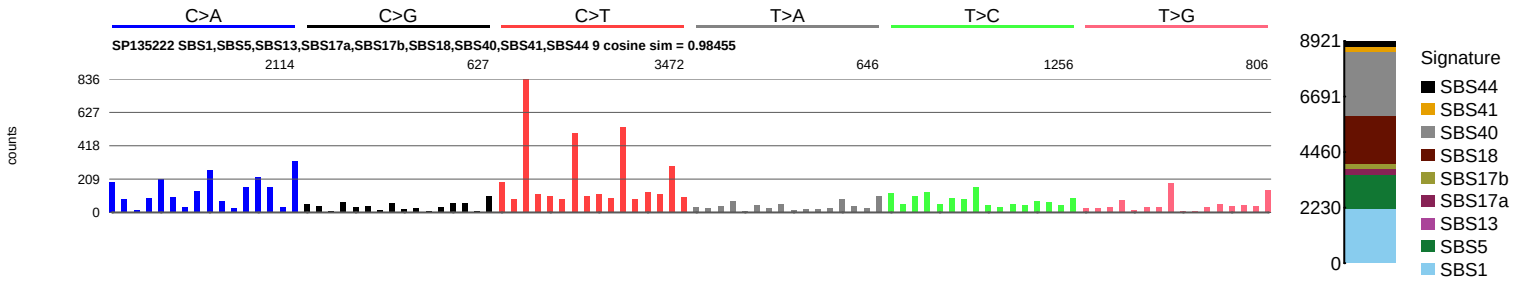

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.9846
2084
535
3687
575
1325
718
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

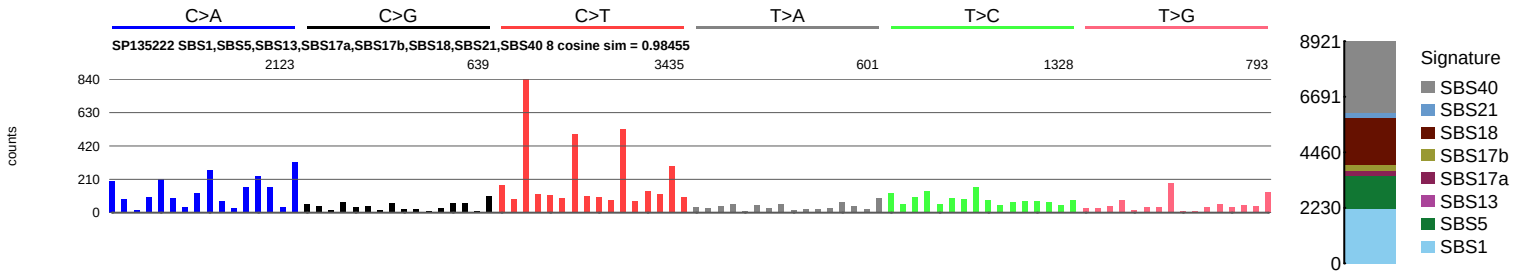

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9846
2142
606
3607
654
1280
633
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS2
SBS1

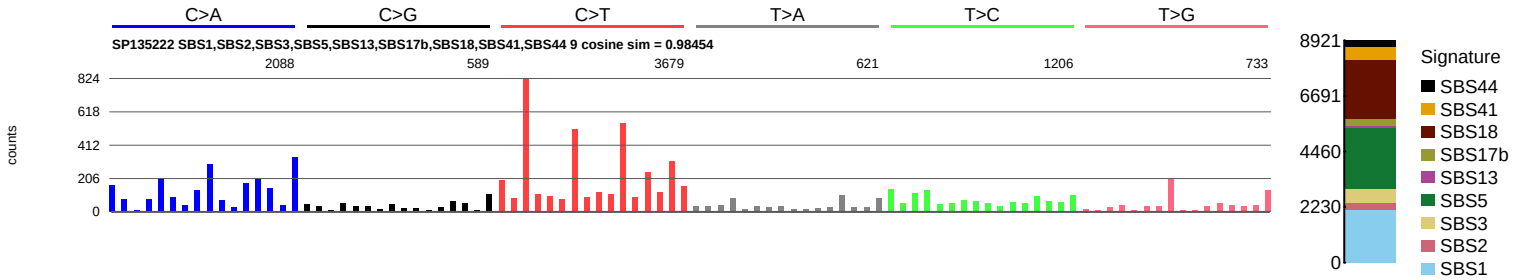

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98459
2125
633
3439
605
1325
794
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

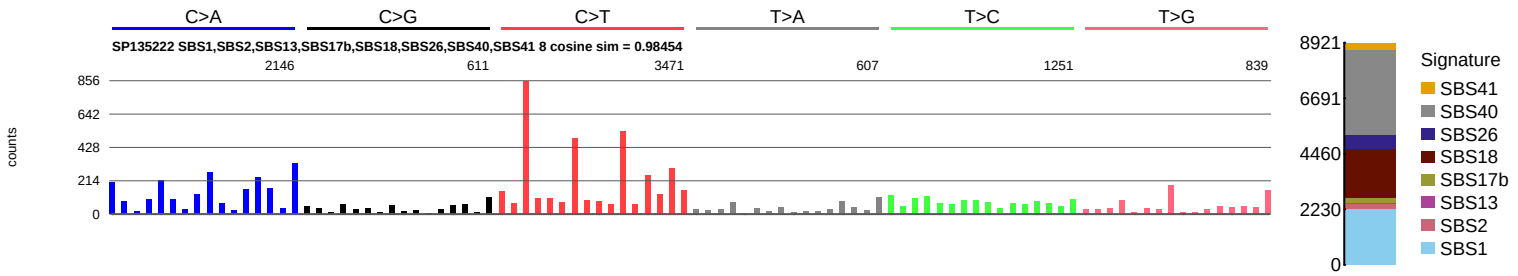

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98458
2146
607
3473
609
1250
839
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1

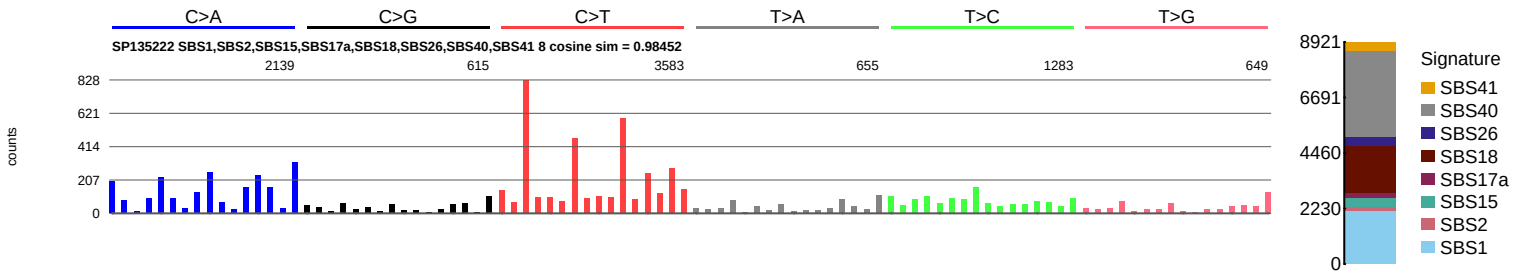

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98457
2087
541
3650
632
1279
734
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

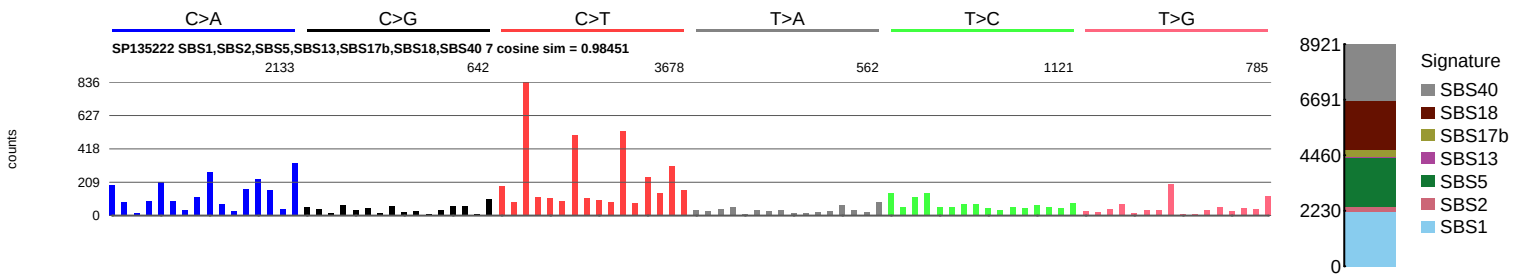

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98455
2114
627
3472
646
1256
806
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

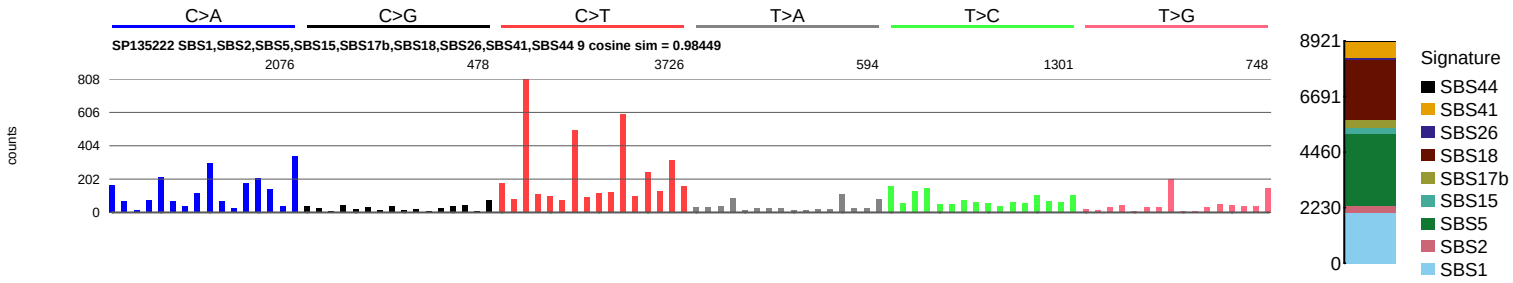

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98455
2123
639
3435
601
1328
793
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

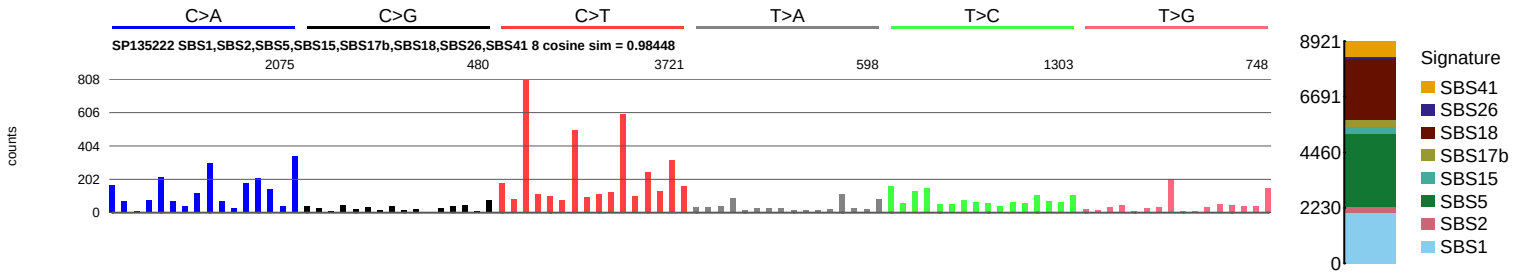

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98454
2088
589
3679
621
1206
733
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

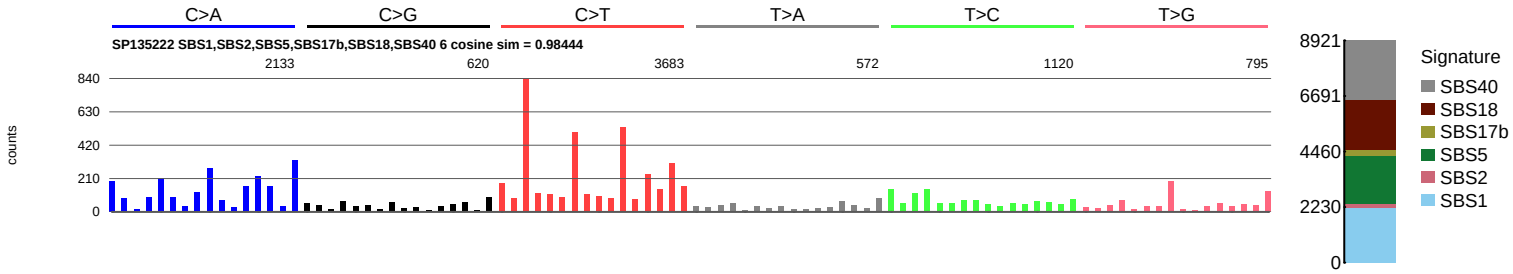

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98454
2146
611
3471
607
1251
839
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1

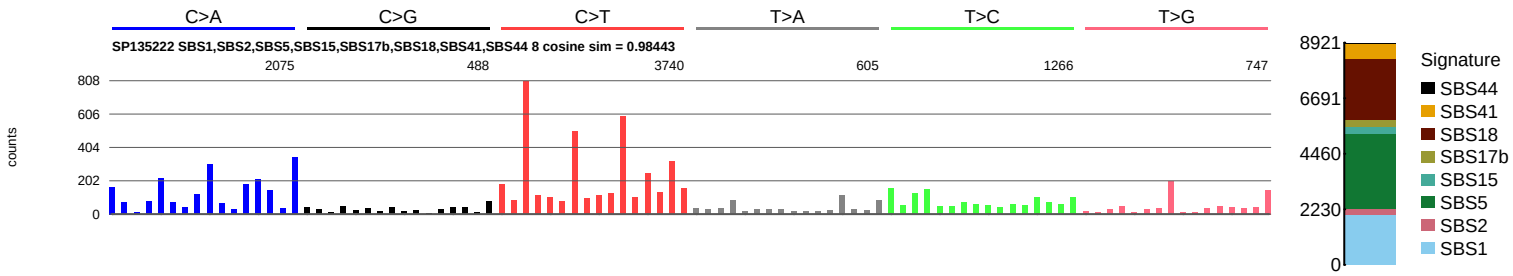

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98452
2139
615
3583
655
1283
649
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS40 7 cosine sim = 0.98451
2133
642
3678
562
1121
785
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98449
2076
478
3726
594
1301
748
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98448
2075
480
3721
598
1303
748
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS40 6 cosine sim = 0.98444
2133
620
3683
572
1120
795
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98443
2075
488
3740
605
1266
747
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
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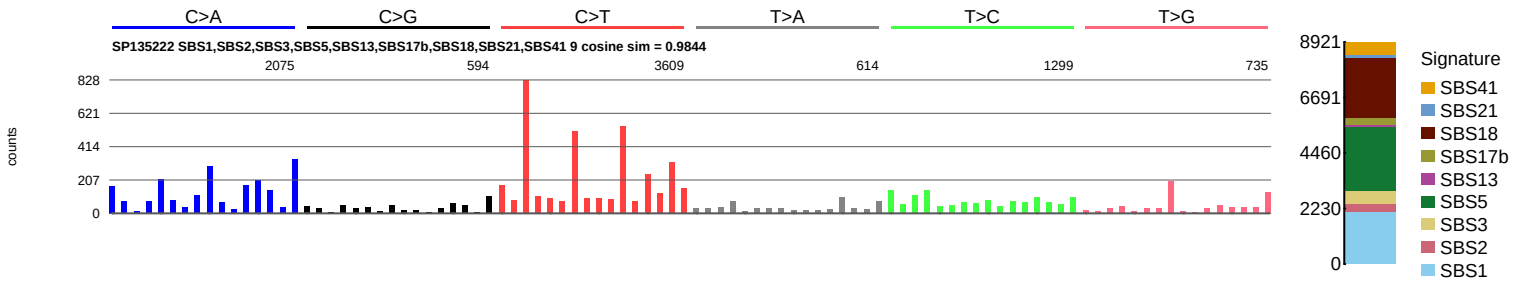
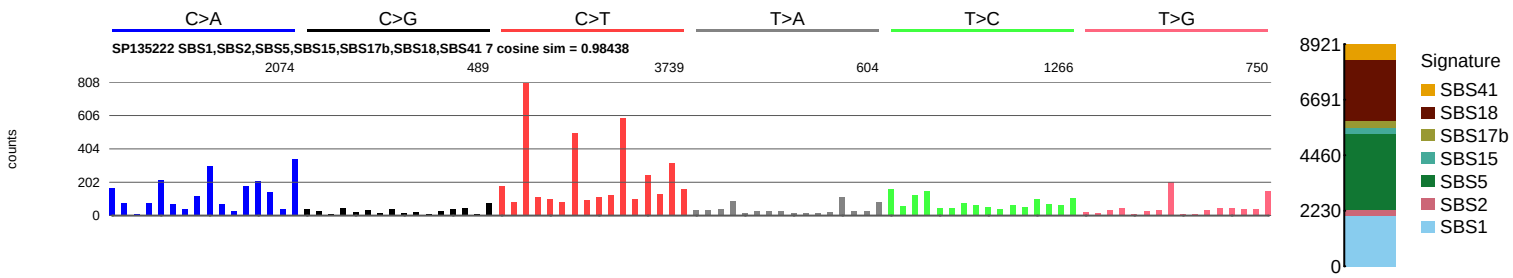
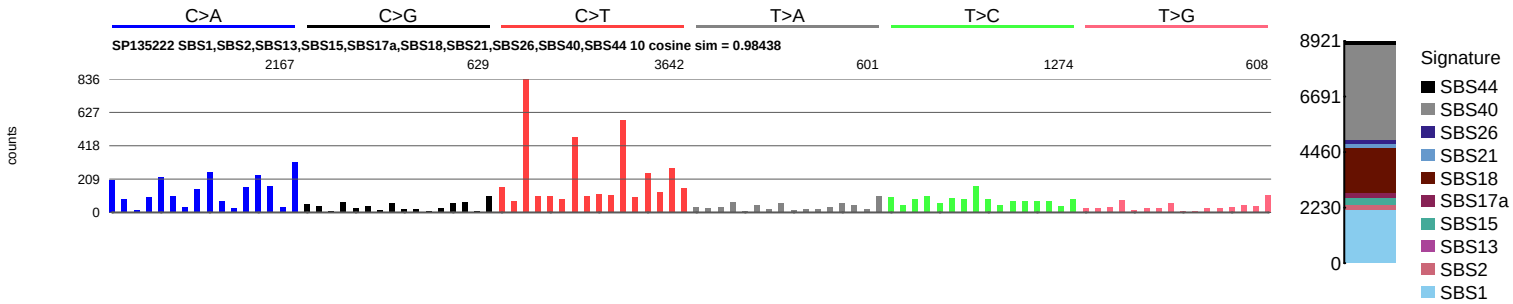
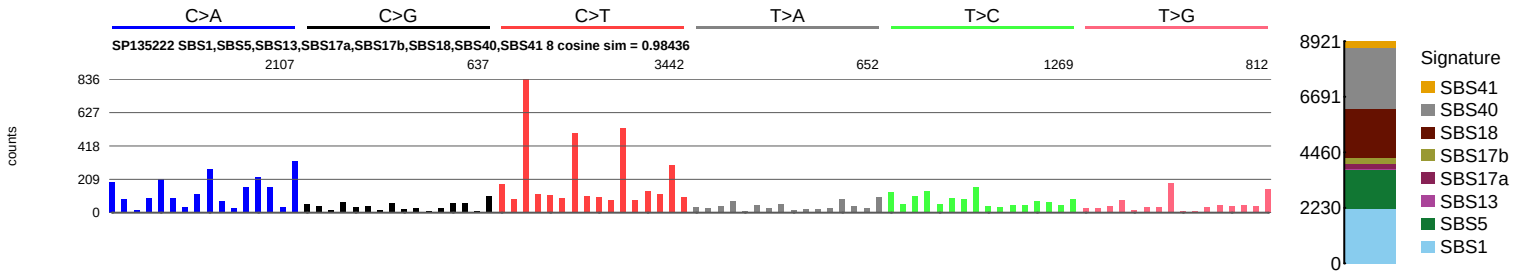
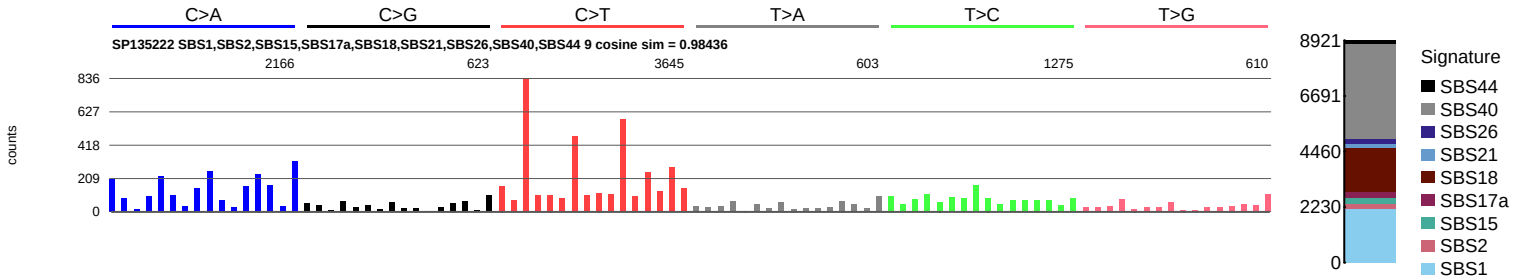
SBS5
SBS2
SBS1

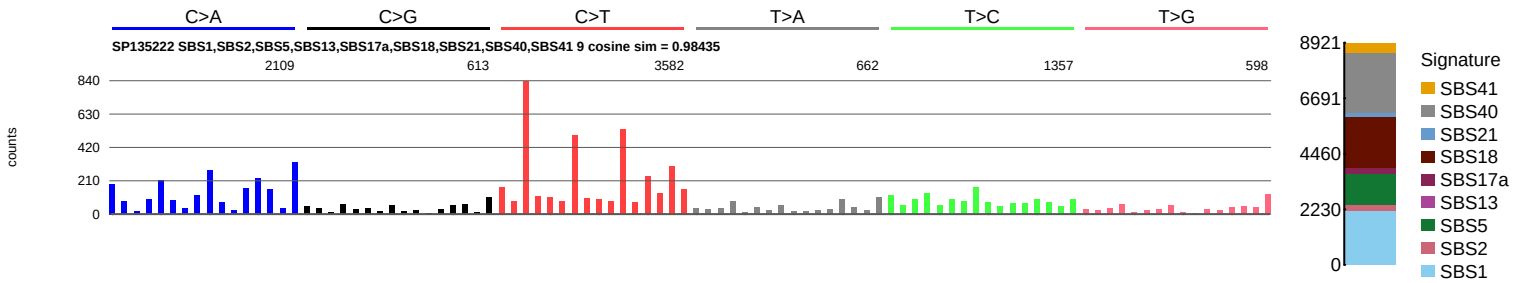

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.9844
2075
594
3609
614
1299
735
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

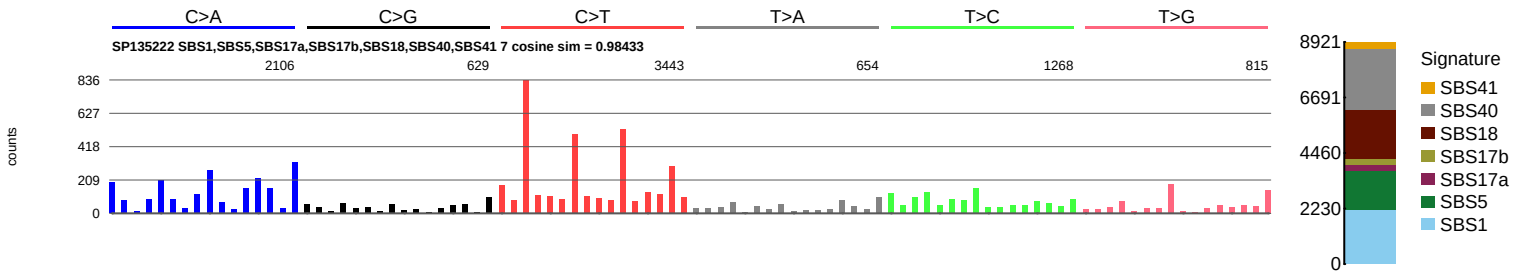

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS41 7 cosine sim = 0.98438
2074
489
3739
604
1266
750
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

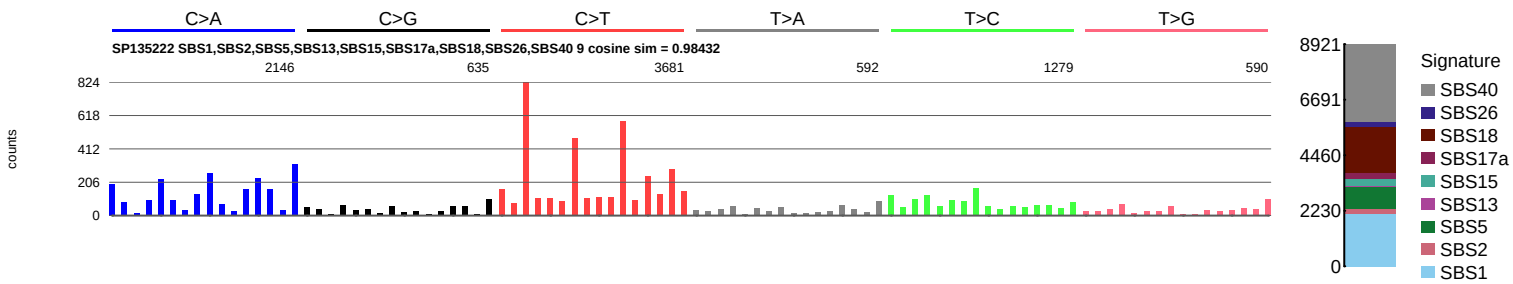

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98438
2167
629
3642
601
1274
608
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1

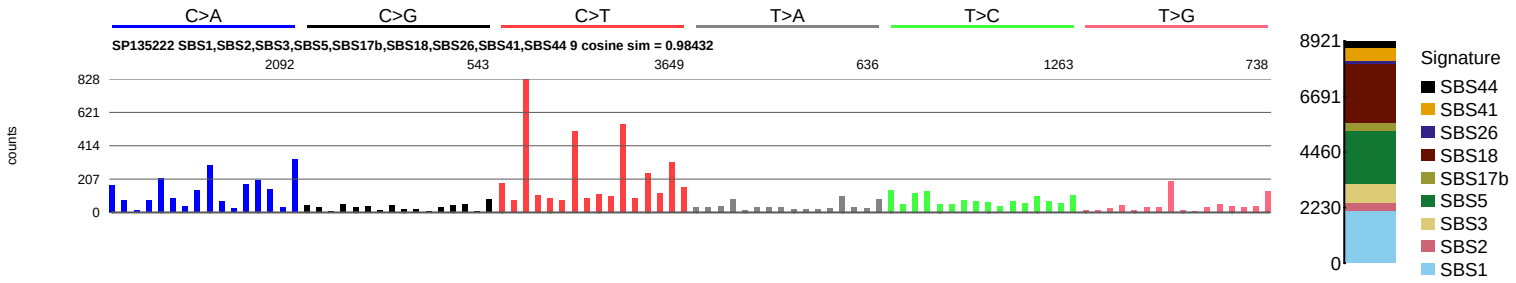

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98436
2107
637
3442
652
1269
812
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

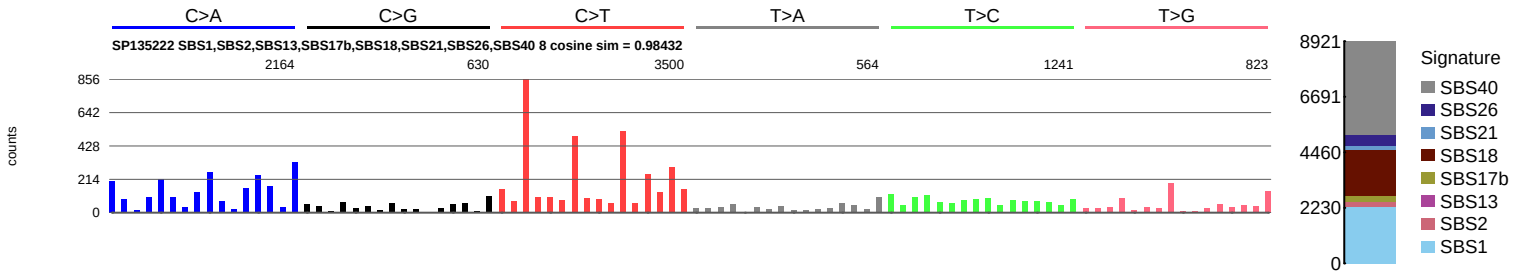

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98436
2166
623
3645
603
1275
610
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1

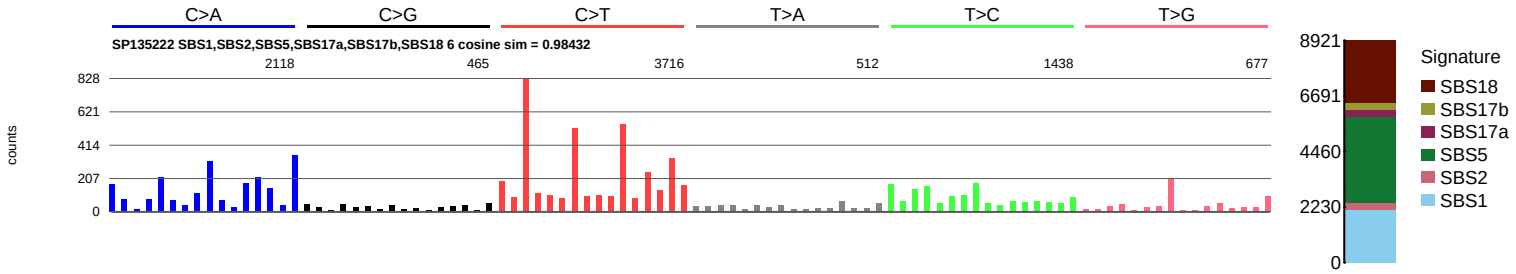

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98435
2109
613
3582
662
1357
598
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1

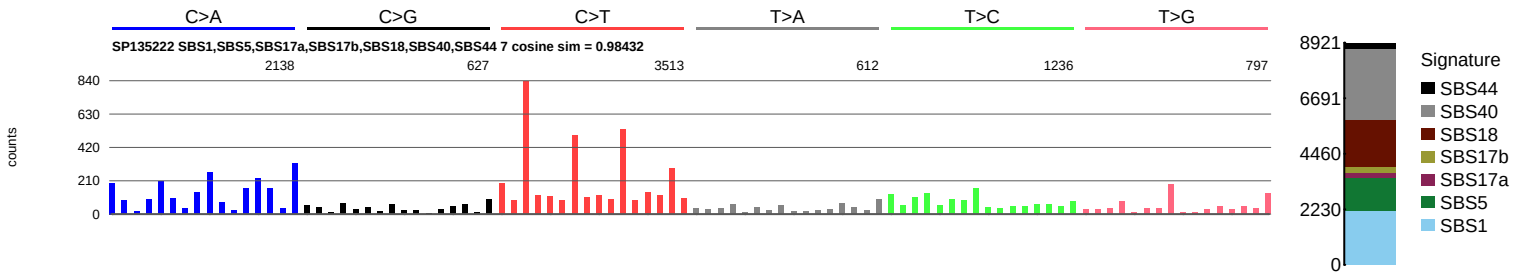

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98433
2106
629
3443
654
1268
815
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1

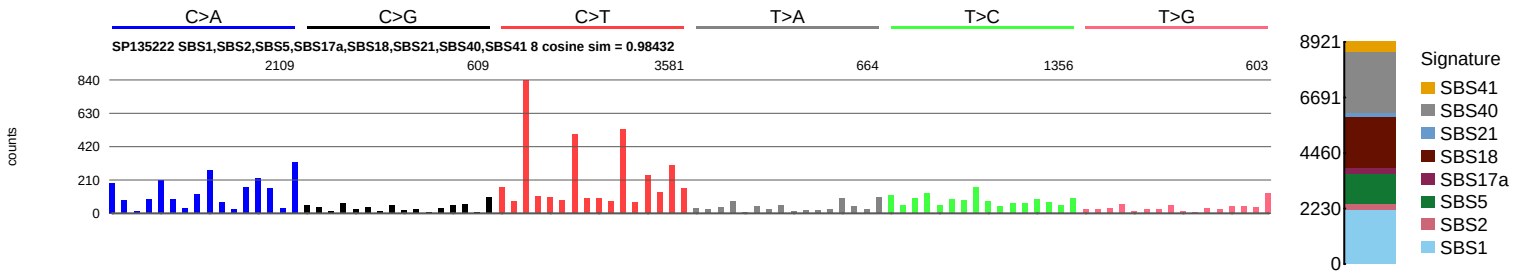

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40 9 cosine sim = 0.98432
2146
635
3681
592
1279
590
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

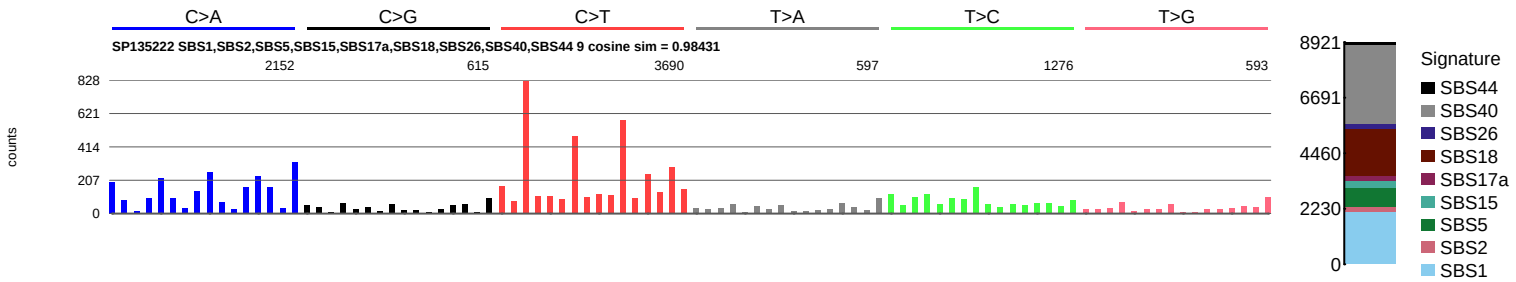

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98432
2092
543
3649
636
1263
738
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

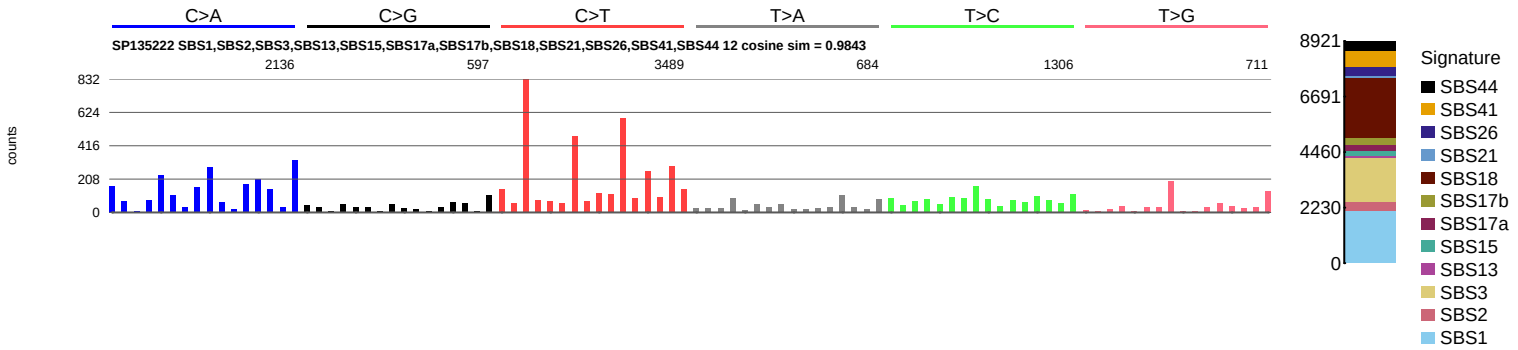

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98432
2164
630
3500
564
1241
823
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS13
SBS2
SBS1

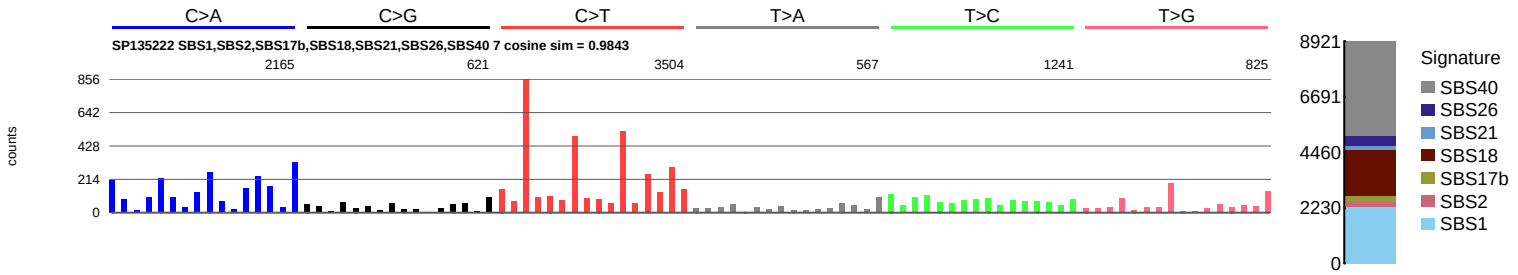

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98432
2118
465
3716
512
1438
677
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

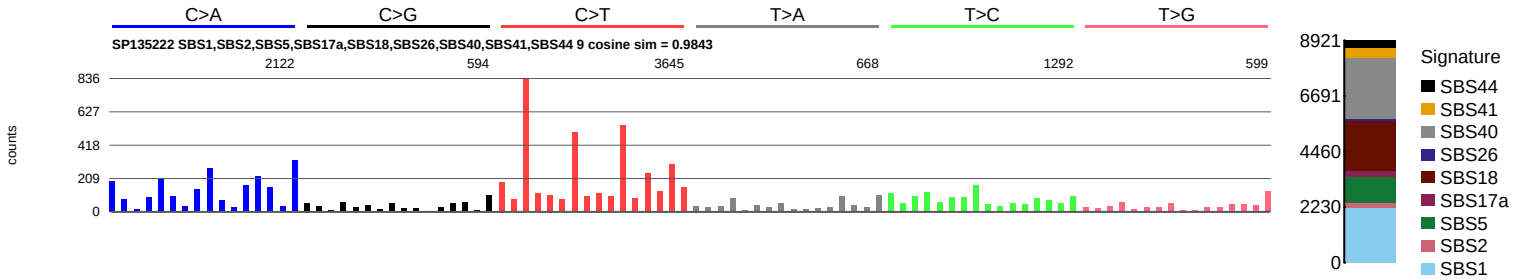

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98432
2138
627
3513
612
1236
797
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1

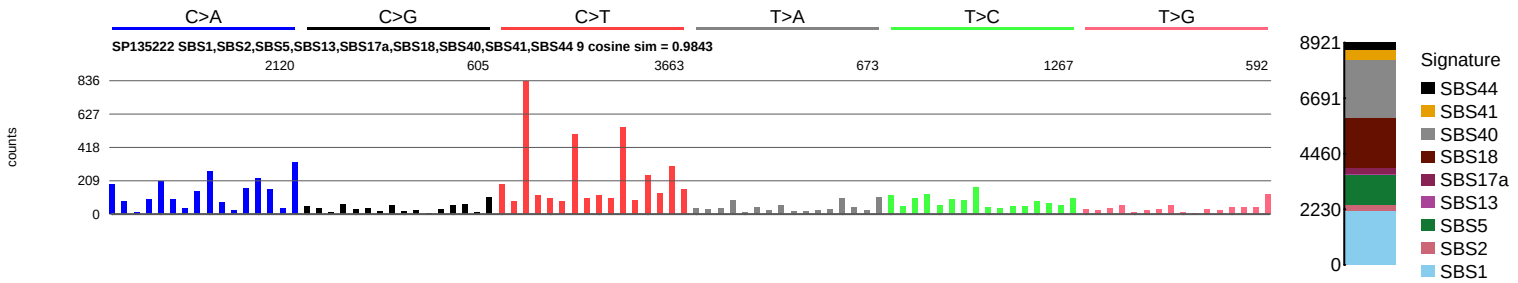

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98432
2109
609
3581
664
1356
603
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98431
2152
615
3690
597
1276
593
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 12 cosine sim = 0.9843
2136
597
3489
684
1306
711
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.9843
2165
621
3504
567
1241
825
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9843
2122
594
3645
668
1292
599
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.9843
2120
605
3663
673
1267
592
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
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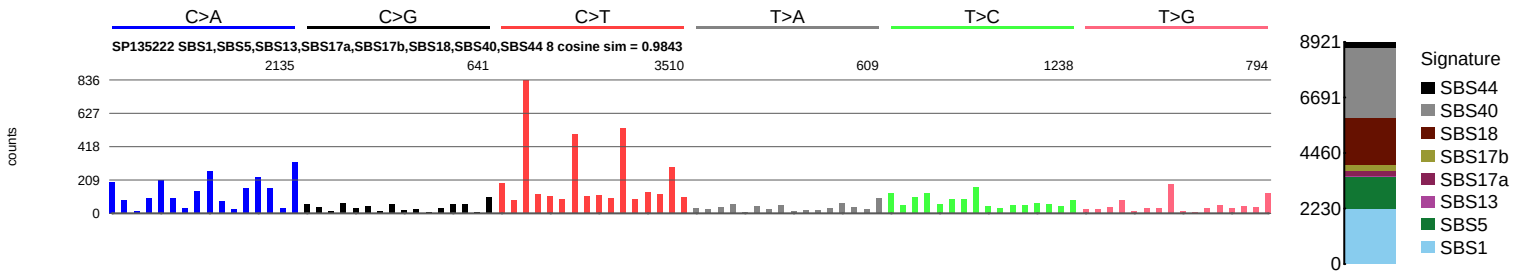
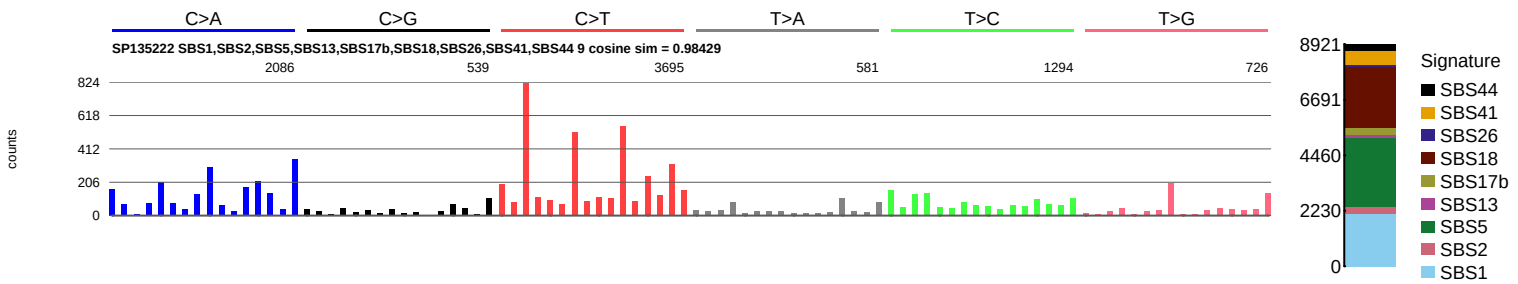
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.9843
2135
641
3510
609
1238
794
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98429
2086
539
3695
581
1294
726
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
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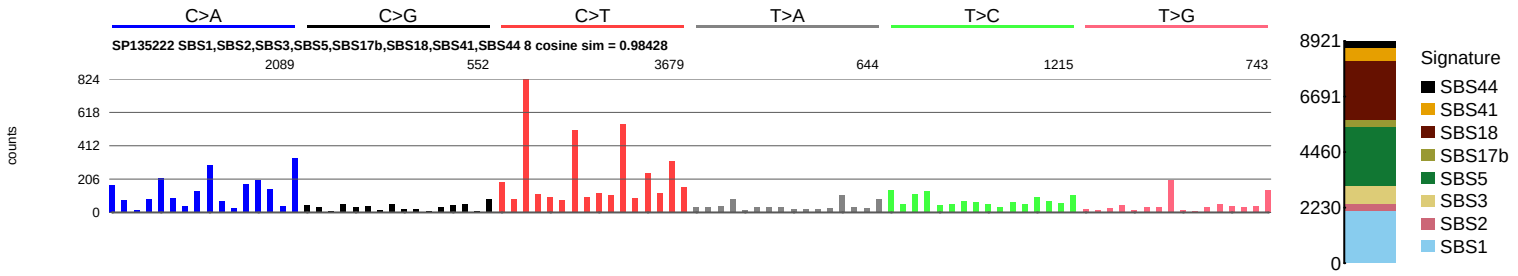
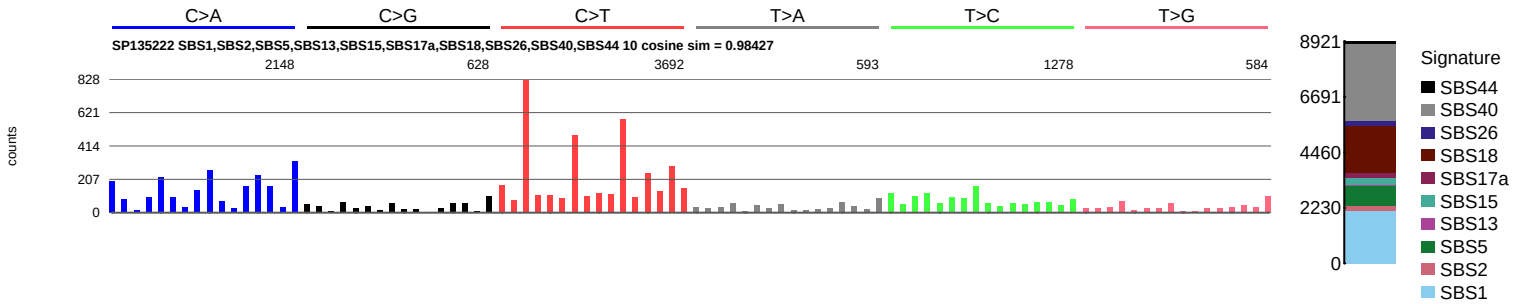
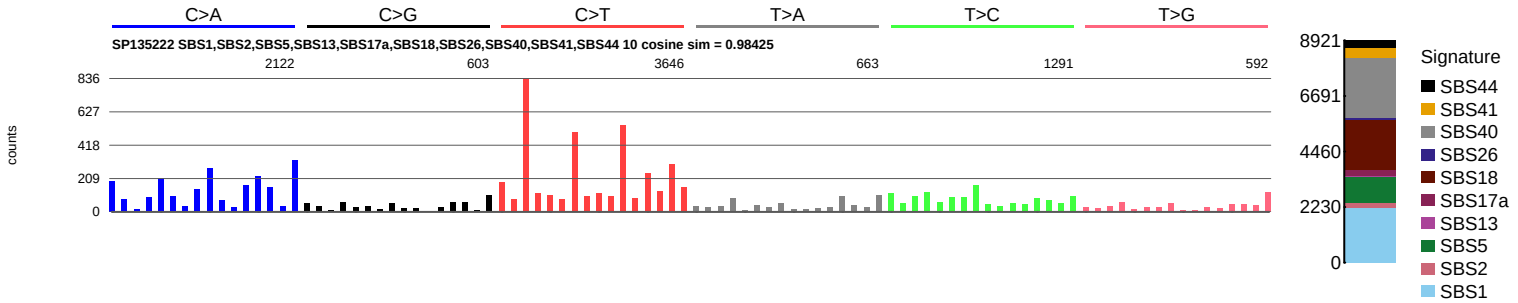
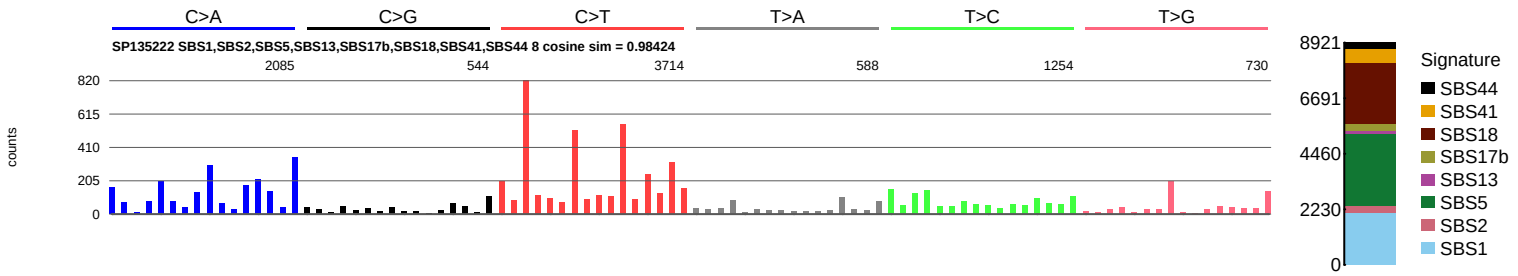
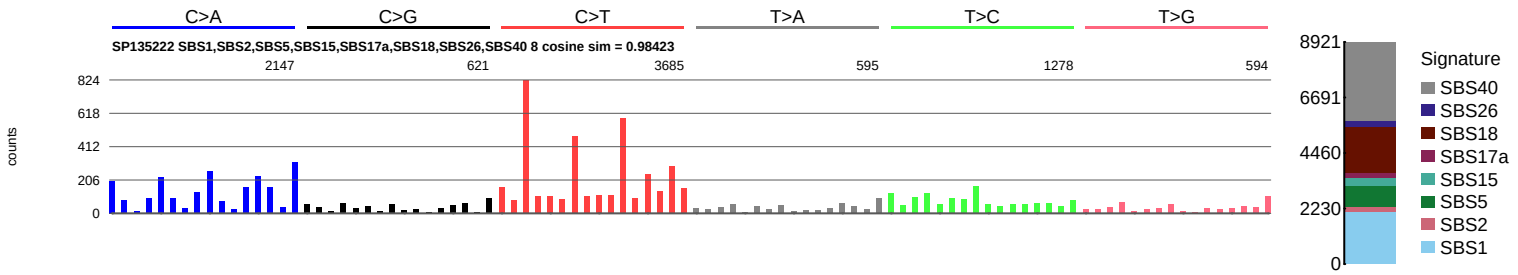
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98428
2089
552
3679
644
1215
743
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98427
2148
628
3692
593
1278
584
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98425
2122
603
3646
663
1291
592
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98424
2085
544
3714
588
1254
730
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40 8 cosine sim = 0.98423
2147
621
3685
595
1278
594
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
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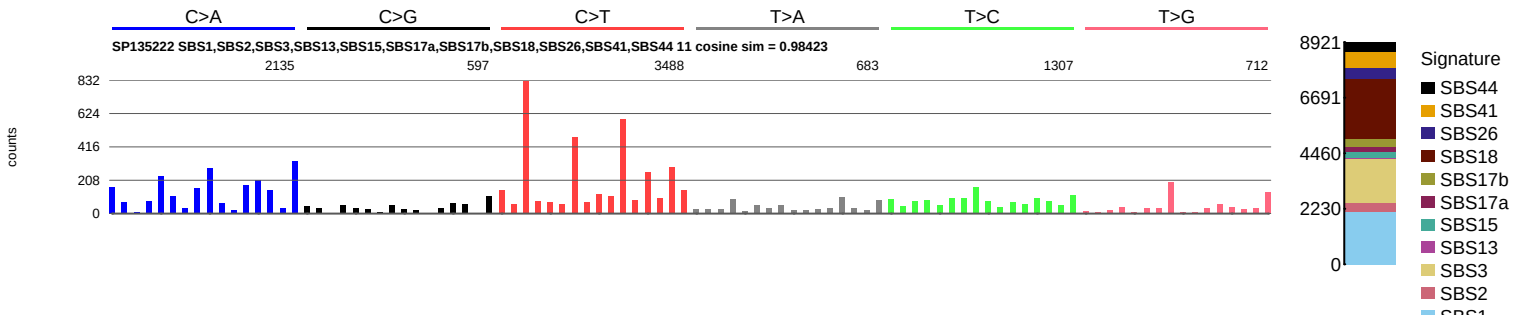
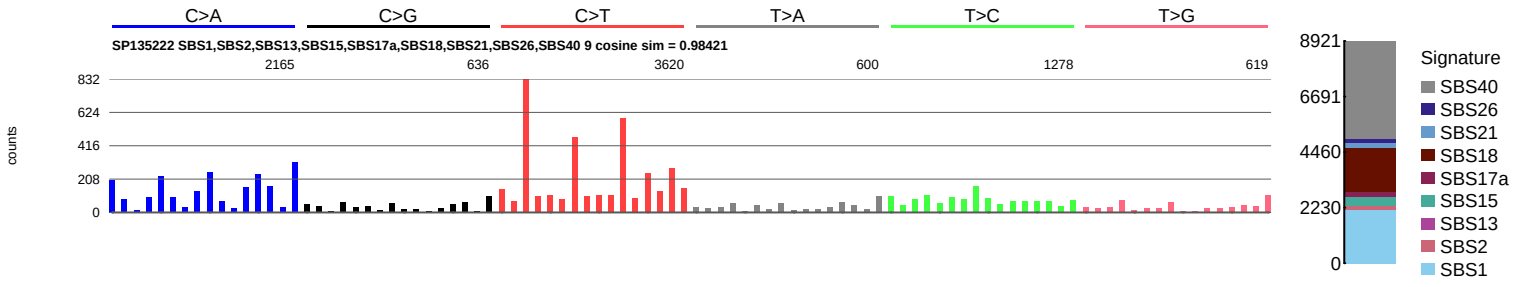
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.98423
2135
597
3488
683
1307
712
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98421
2165
636
3620
600
1278
619
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
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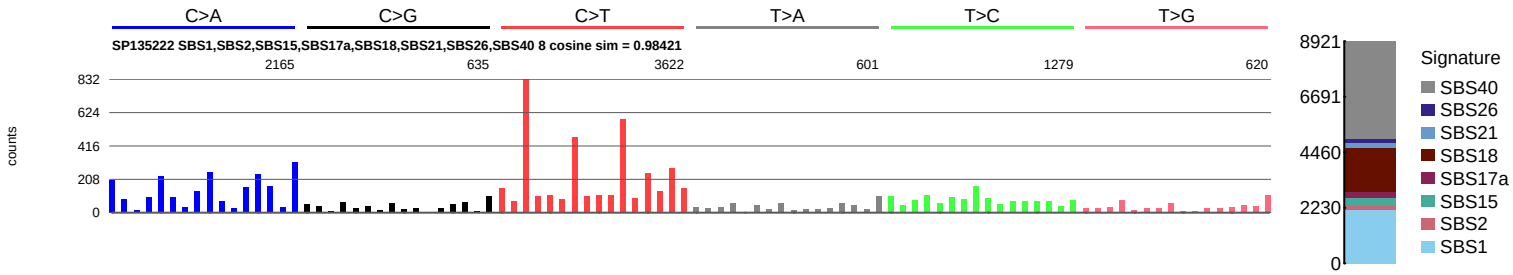
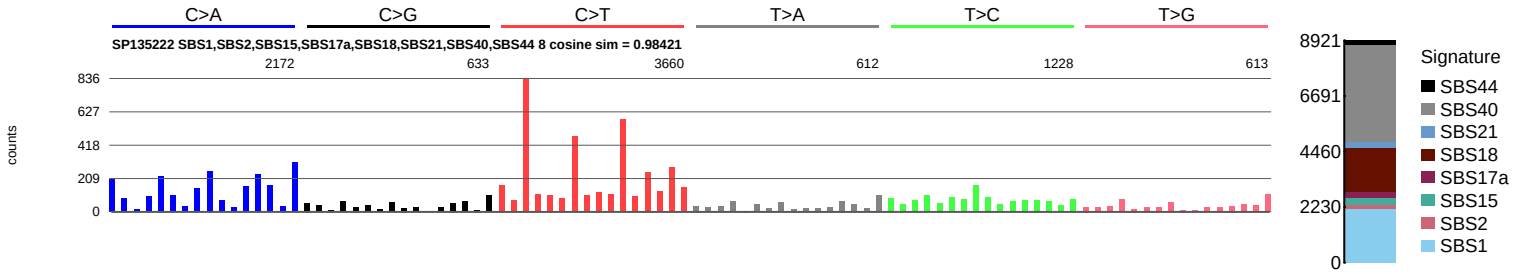
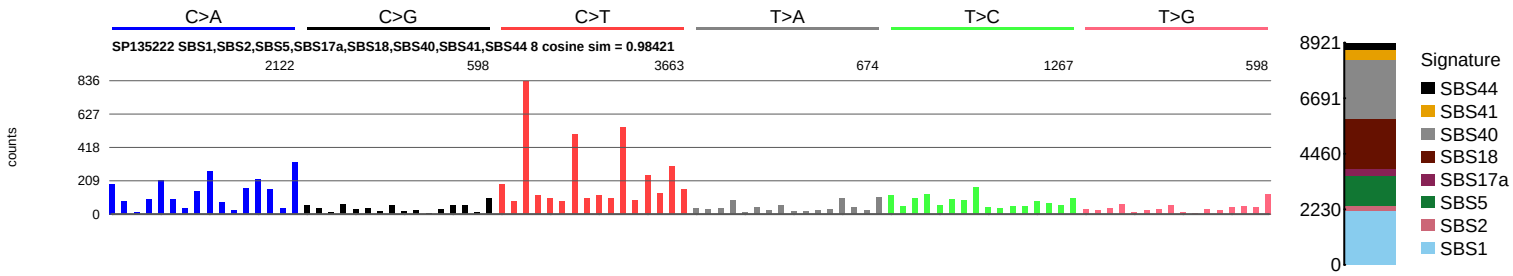
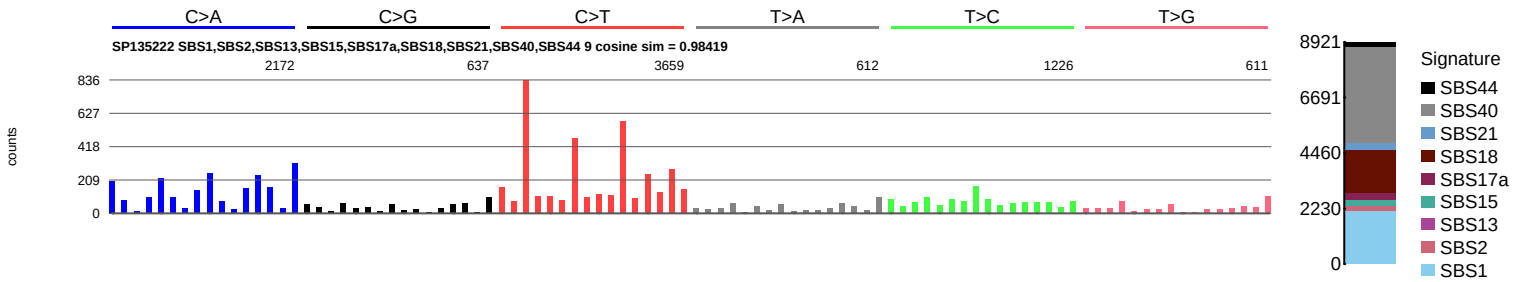
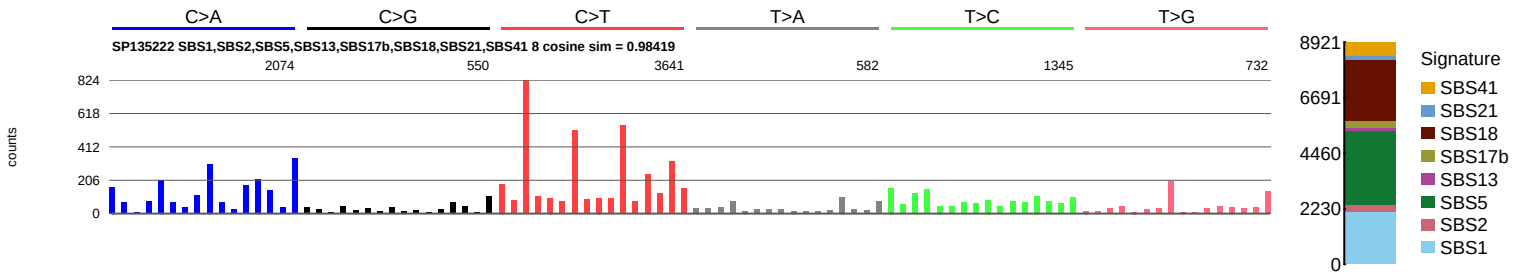
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98421
2165
635
3622
601
1279
620
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98421
2172
633
3660
612
1228
613
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98421
2122
598
3663
674
1267
598
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98419
2172
637
3659
612
1226
611
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
C>T
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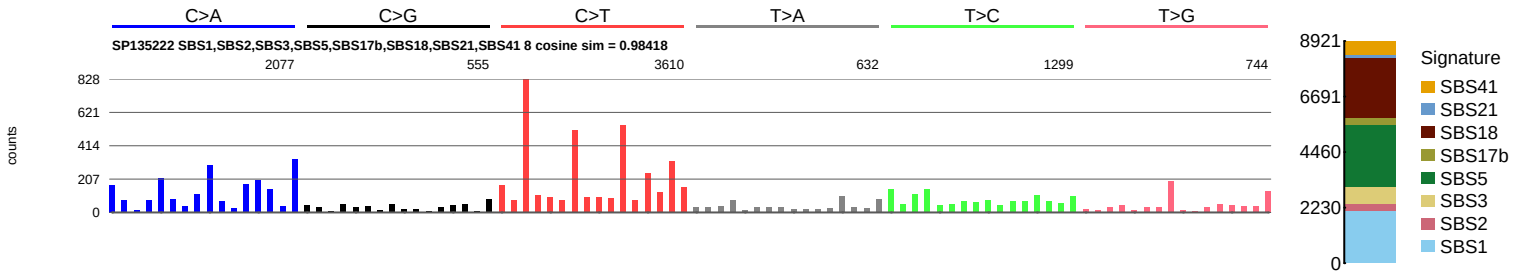
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98419
2074
550
3641
582
1345
732
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1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C>T
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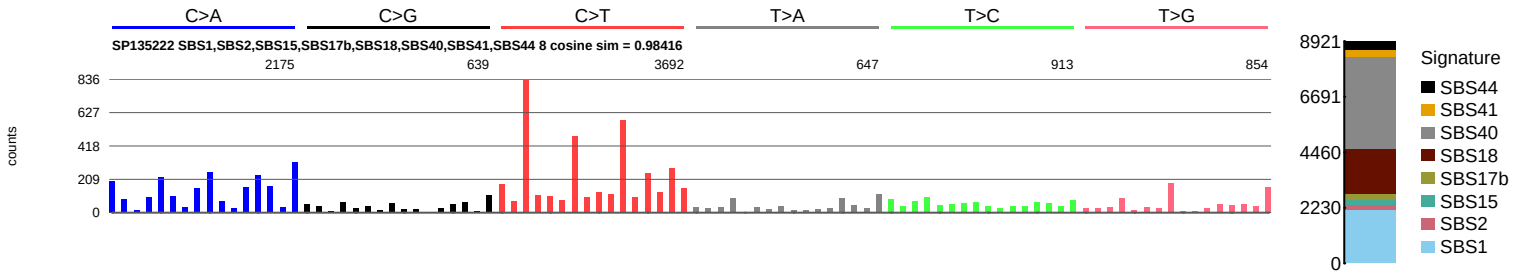
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98418
2077
555
3610
632
1299
744
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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C>T
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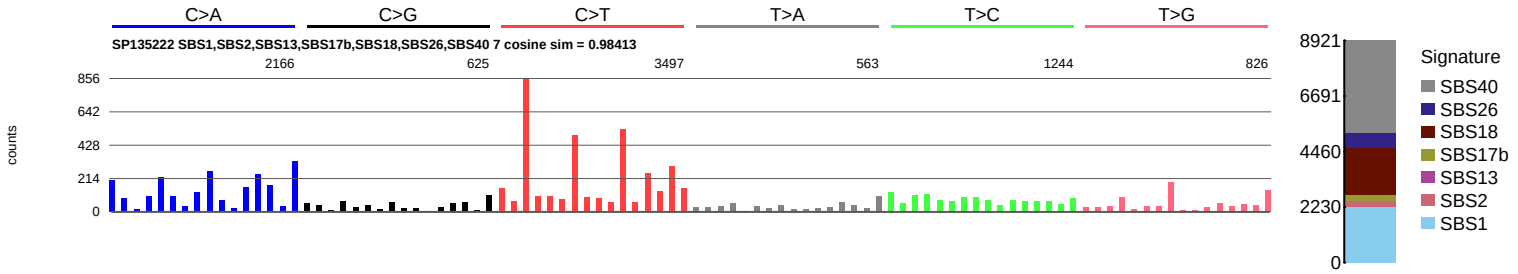
SP135222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98416
2175
639
3692
647
913
854
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1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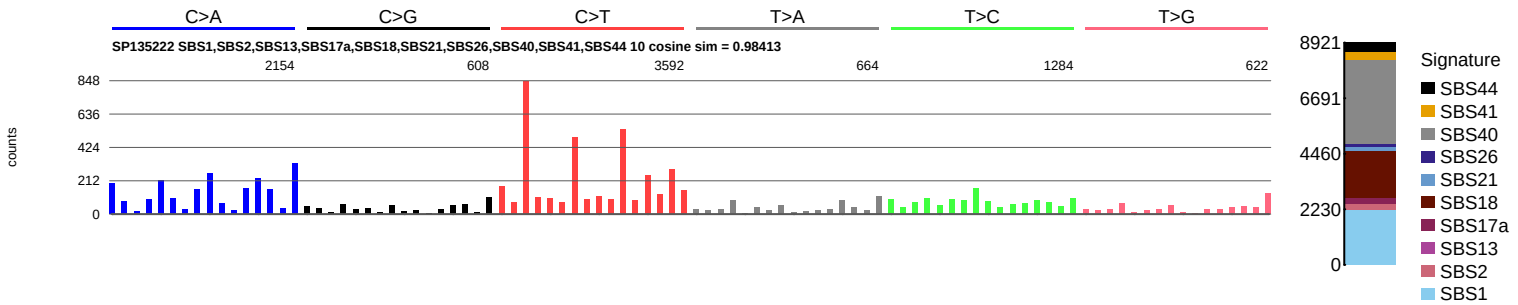
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98413
2166
625
3497
563
1244
826
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS2
SBS1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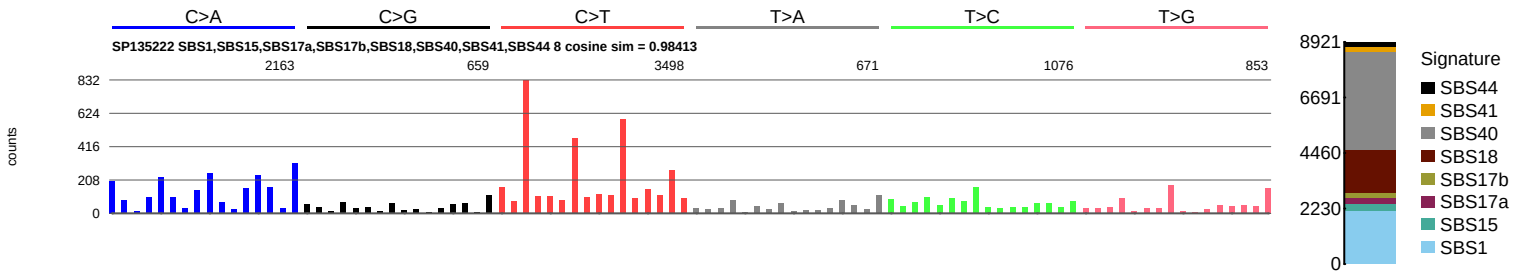
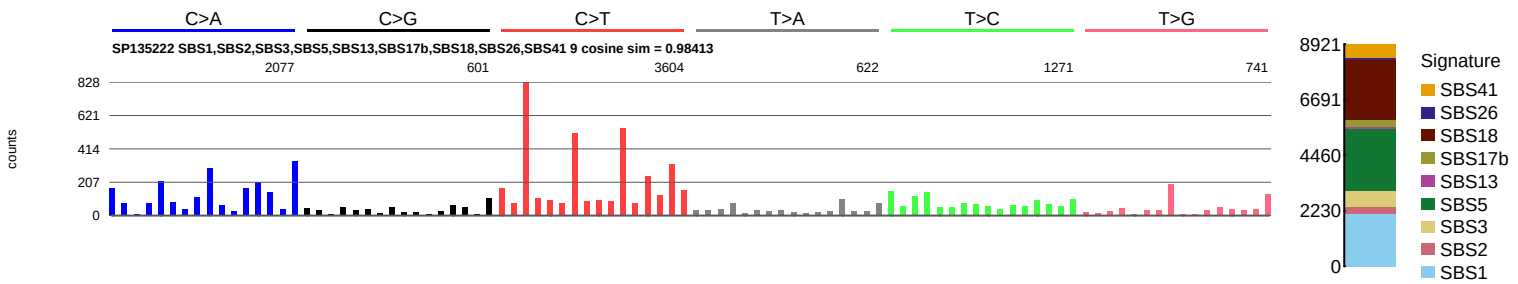
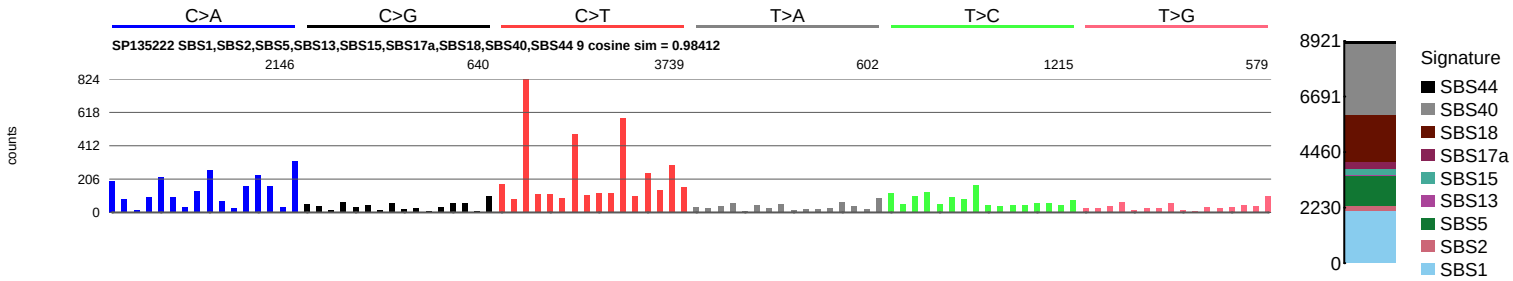
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98413
2154
608
3592
664
1284
622
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98413
2163
659
3498
671
1076
853
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98413
2077
601
3604
622
1271
741
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1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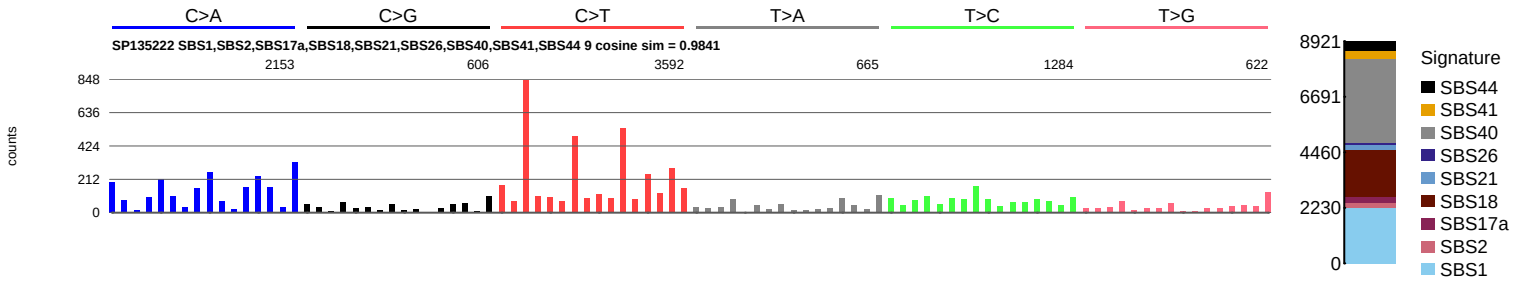
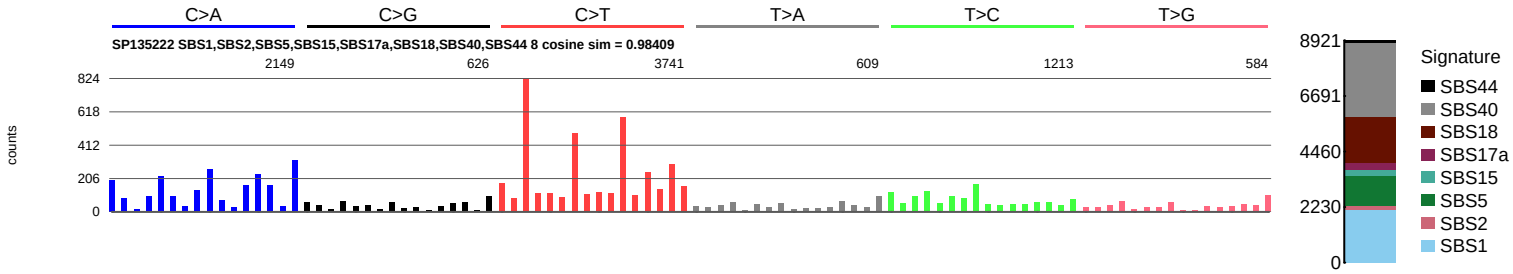
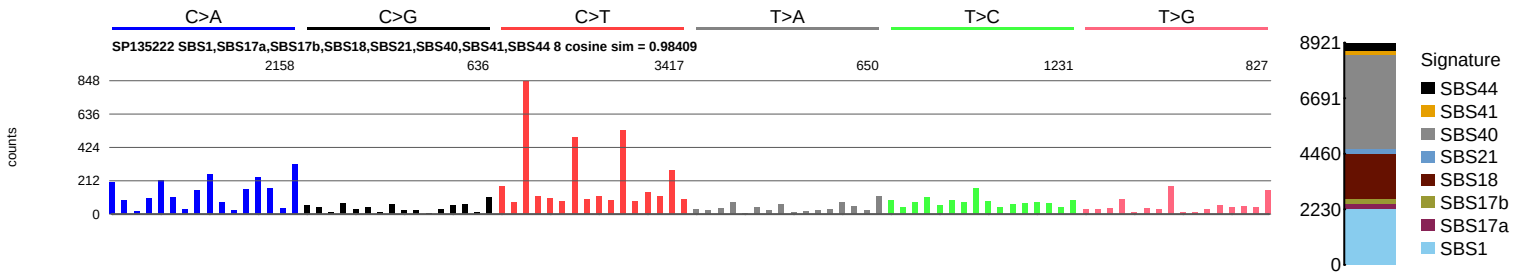
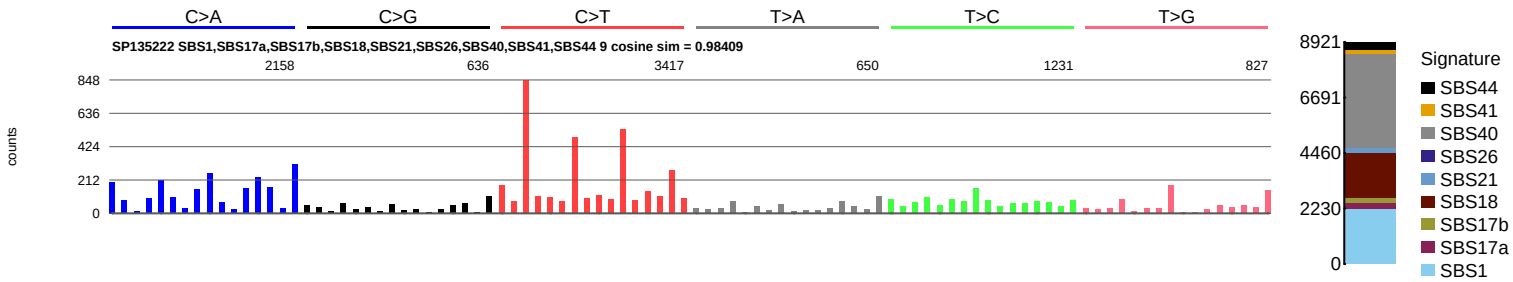
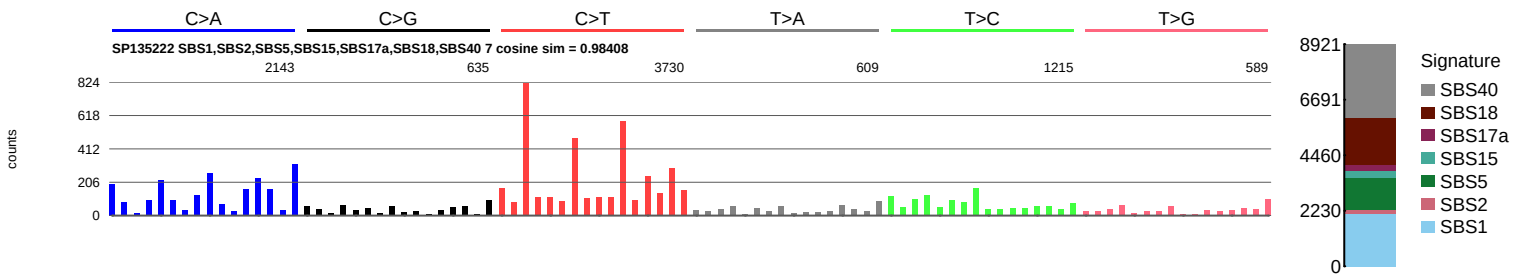
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40,SBS44 9 cosine sim = 0.98412
2146
640
3739
602
1215
579
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9841
2153
606
3592
665
1284
622
counts
848
636
424
212
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.98409
2149
626
3741
609
1213
584
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98409
2158
636
3417
650
1231
827
counts
848
636
424
212
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98409
2158
636
3417
650
1231
827
counts
848
636
424
212
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS40 7 cosine sim = 0.98408
2143
635
3730
609
1215
589
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS40
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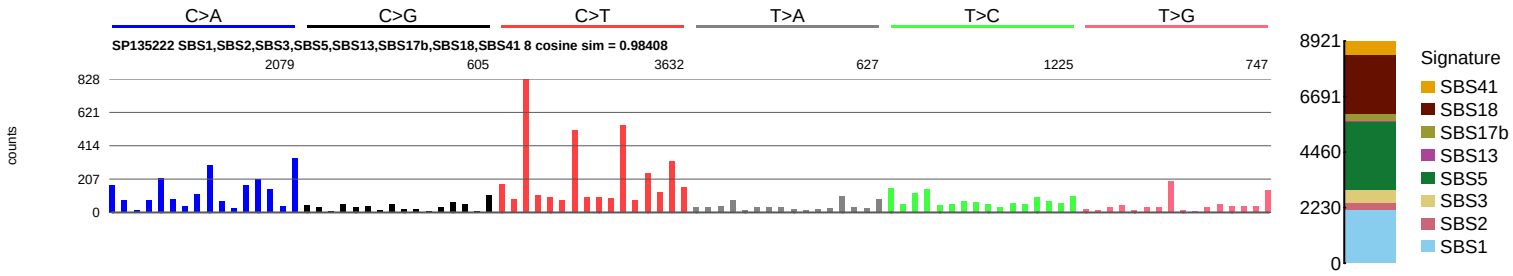
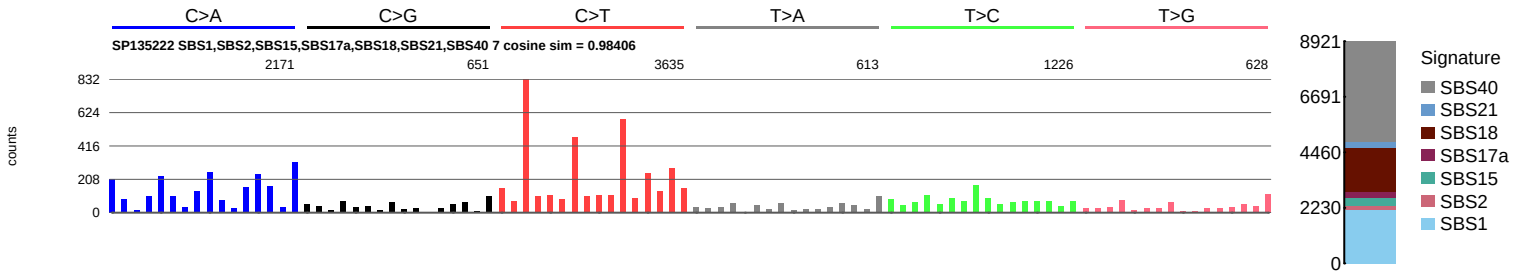
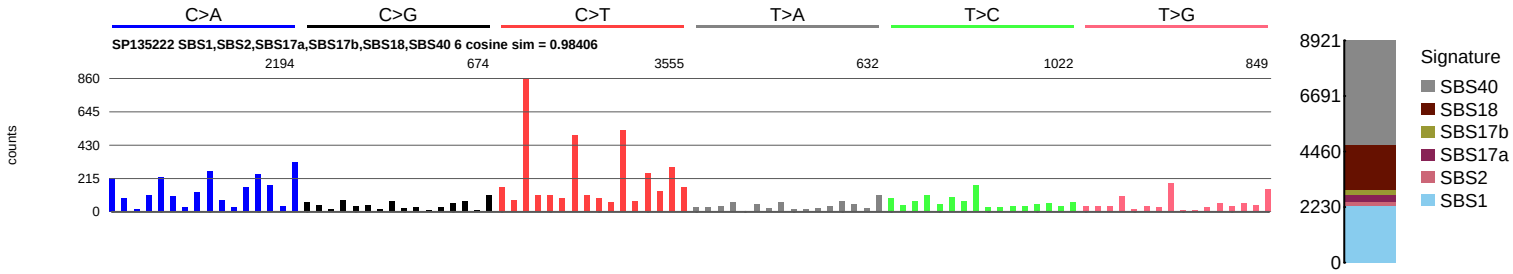
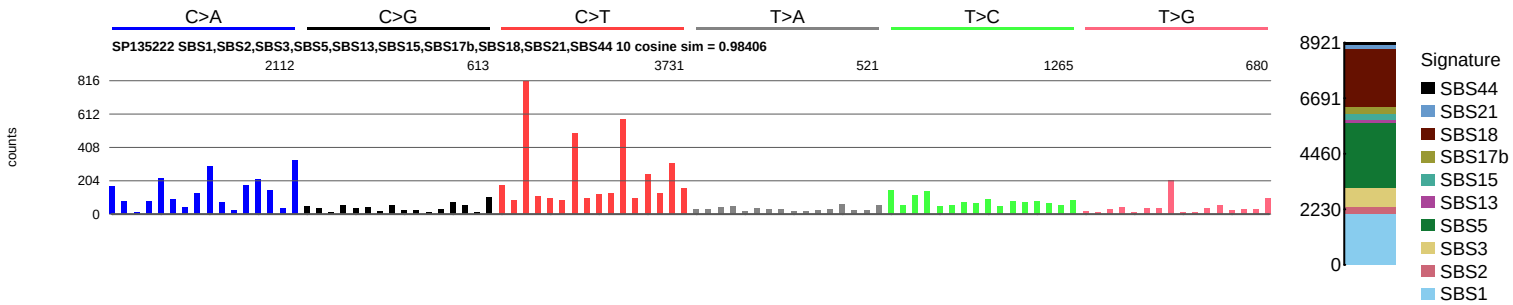
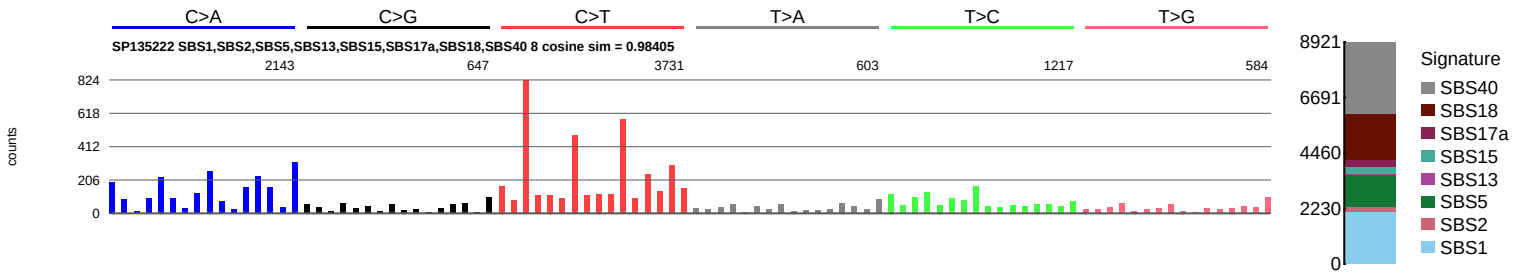
SBS5
SBS2
SBS1

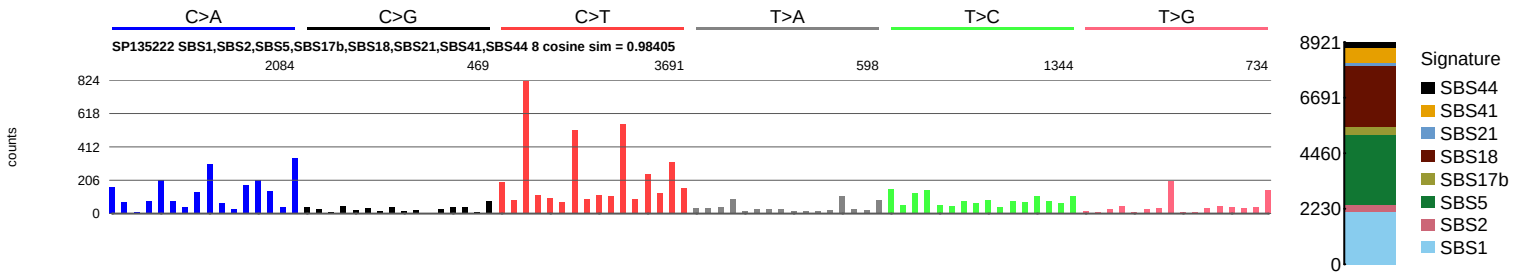

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS41 8 cosine sim = 0.98408
2079
605
3632
627
1225
747
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

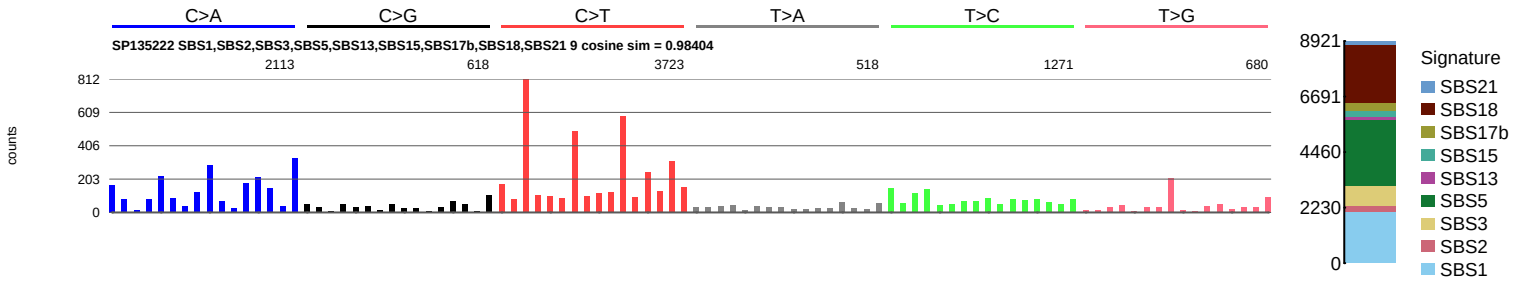

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS21,SBS40 7 cosine sim = 0.98406
2171
651
3635
613
1226
628
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS2
SBS1

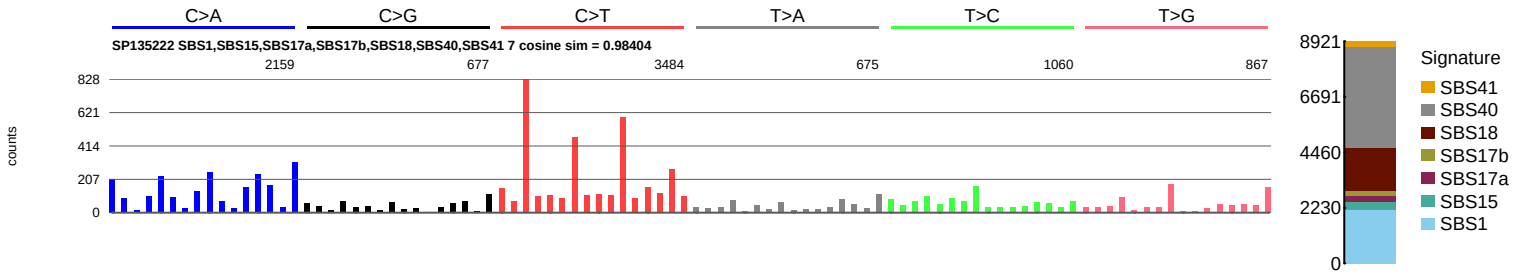

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98406
2194
674
3555
632
1022
849
860
645
430
215
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1

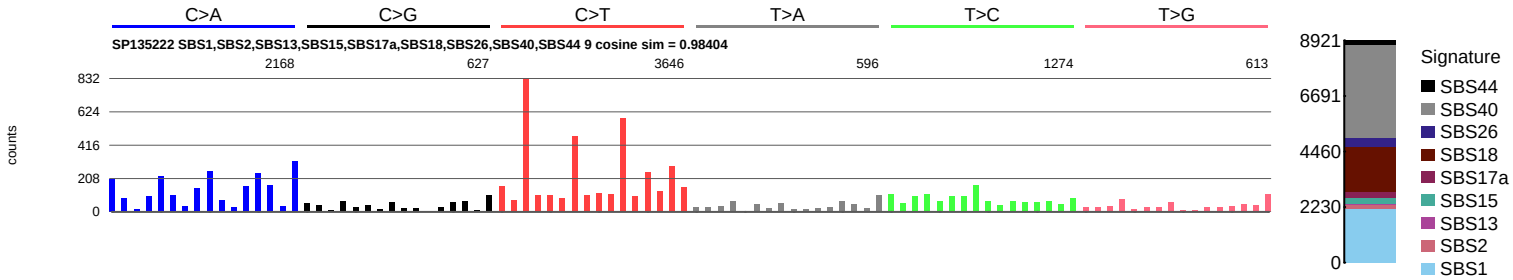

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98406
2112
613
3731
521
1265
680
816
612
408
204
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

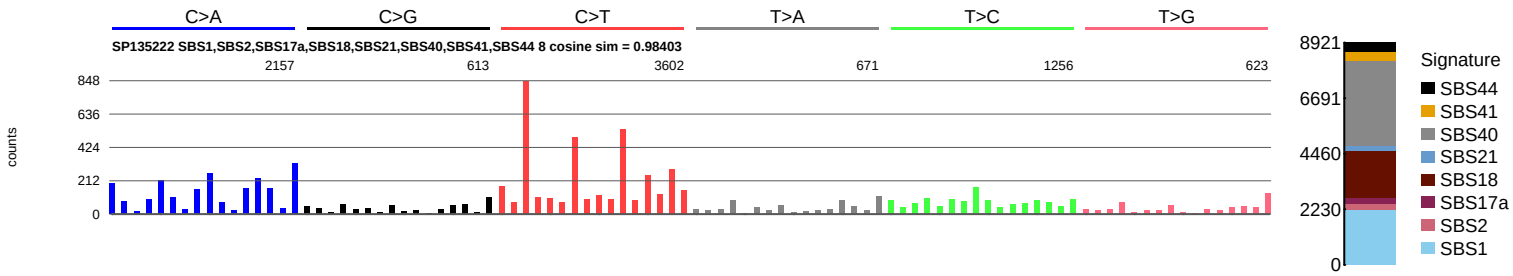

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS40 8 cosine sim = 0.98405
2143
647
3731
603
1217
584
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

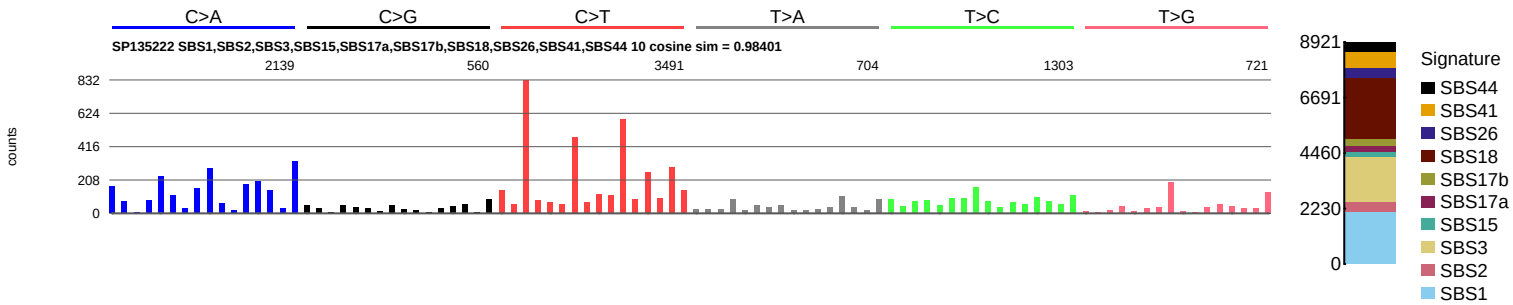

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98405
2084
469
3691
598
1344
734
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1

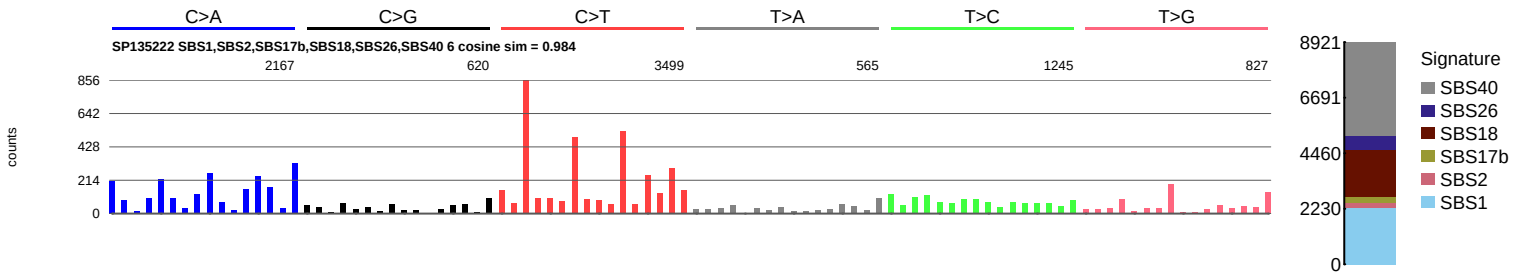

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 9 cosine sim = 0.98404
2113
618
3723
518
1271
680
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

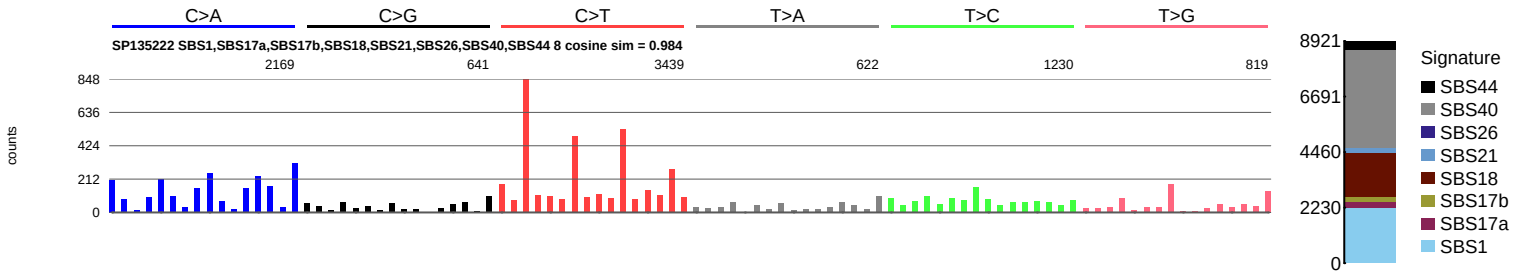

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98404
2159
677
3484
675
1060
867
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98404
2168
627
3646
596
1274
613
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98403
2157
613
3602
671
1256
623
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98401
2139
560
3491
704
1303
721
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.984
2167
620
3499
565
1245
827
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS2
SBS1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C>T
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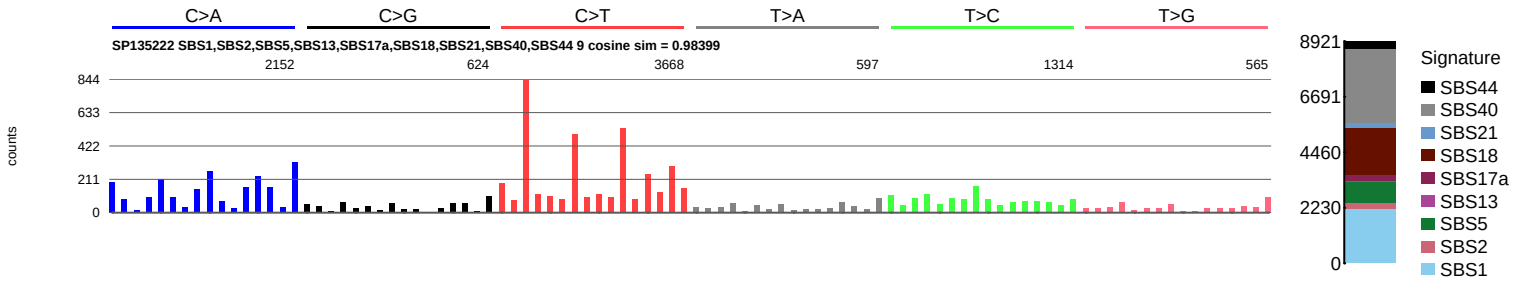
T>C
T>G
SP135222 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.984
2169
641
3439
622
1230
819
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1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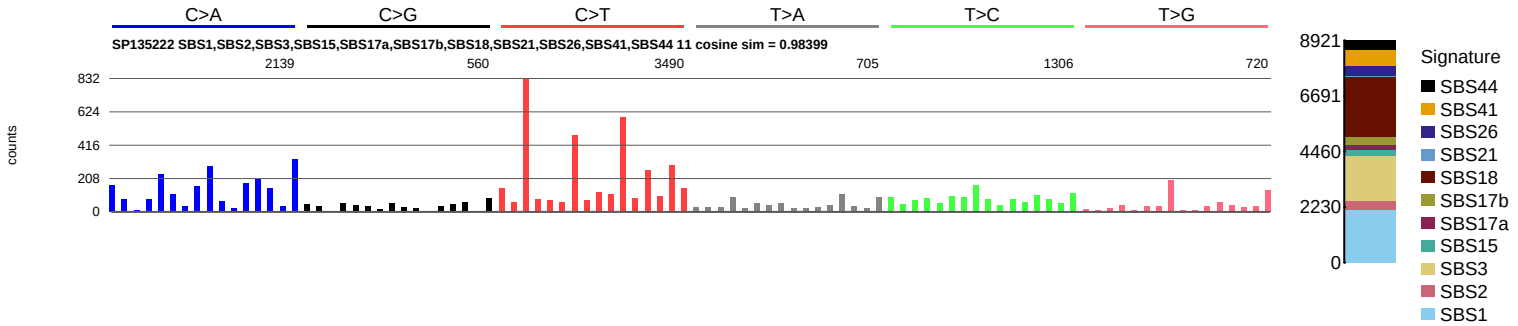
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98399
2152
624
3668
597
1314
565
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1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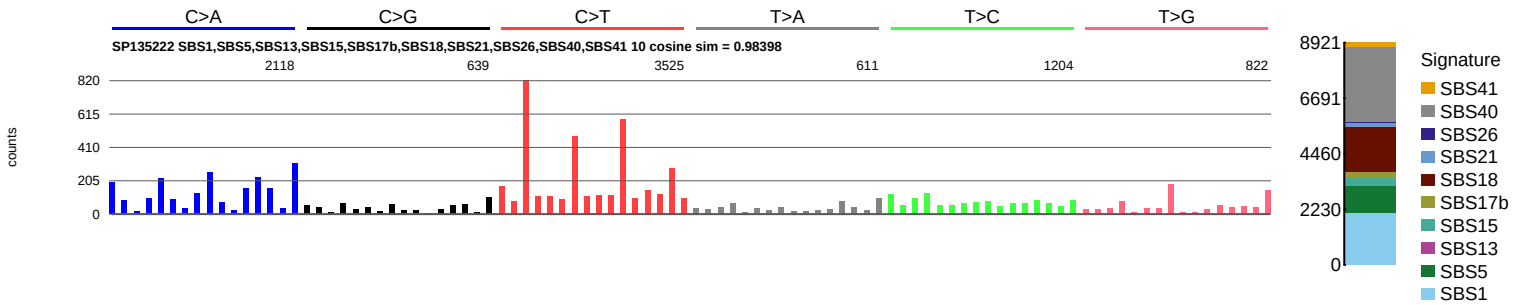
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.98399
2139
560
3490
705
1306
720
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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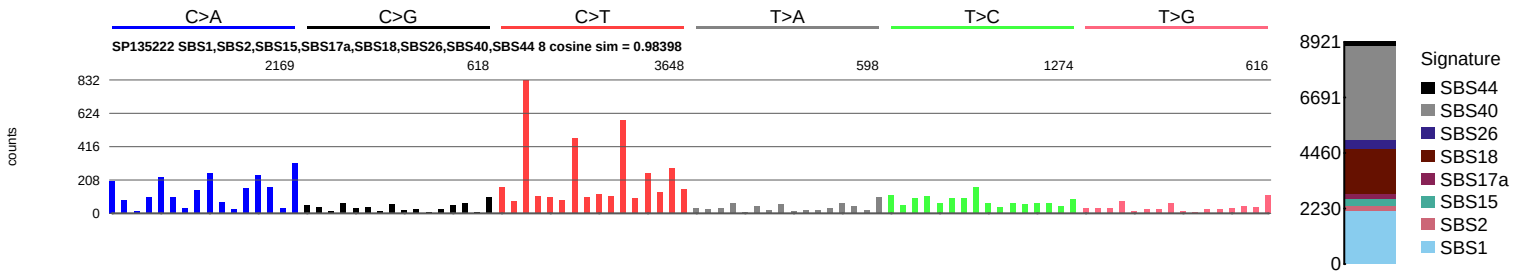
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98398
2118
639
3525
611
1204
822
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1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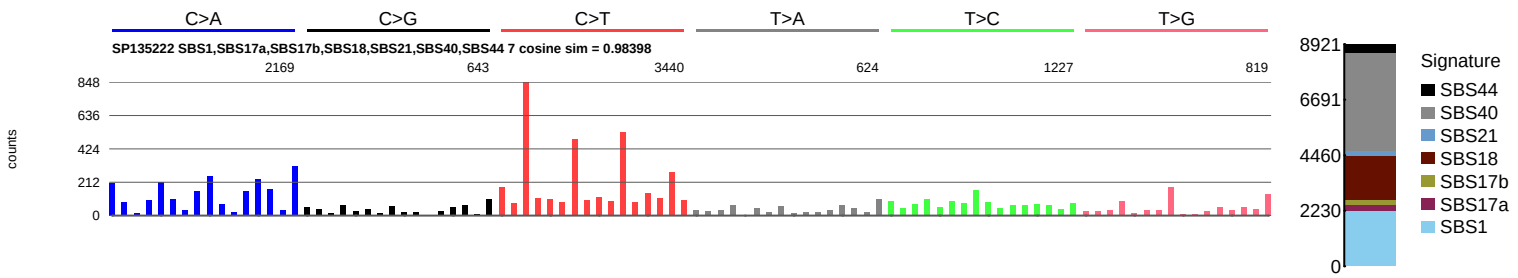
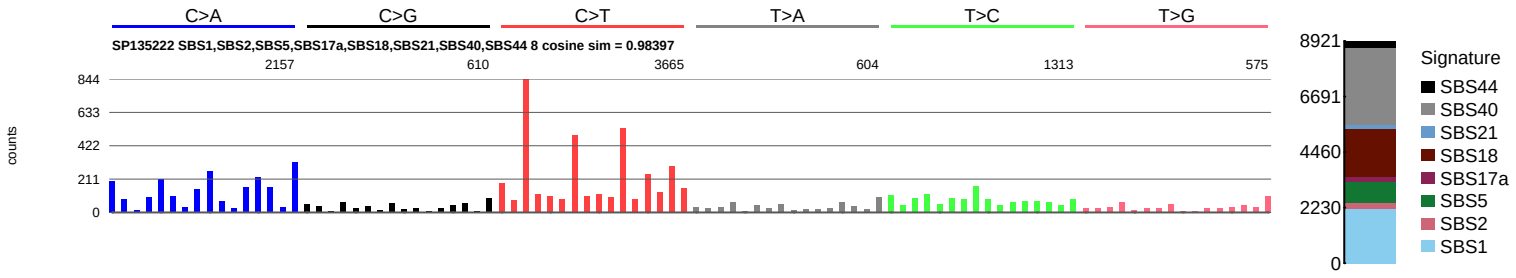
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98398
2169
618
3648
598
1274
616
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98398
2169
643
3440
624
1227
819
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1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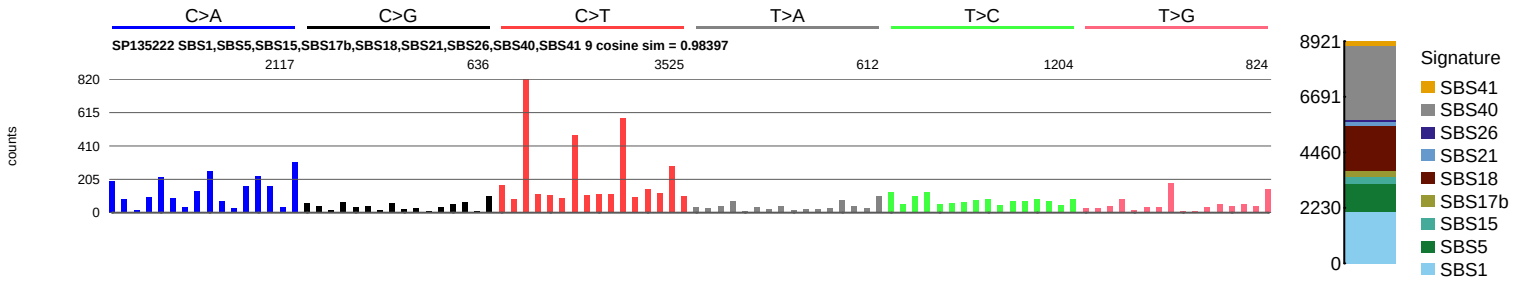
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98397
2157
610
3665
604
1313
575
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1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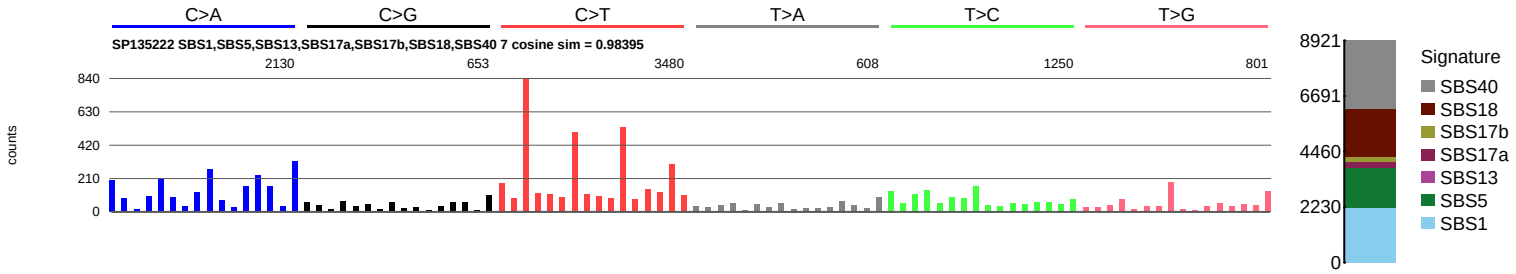
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98397
2117
636
3525
612
1204
824
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS1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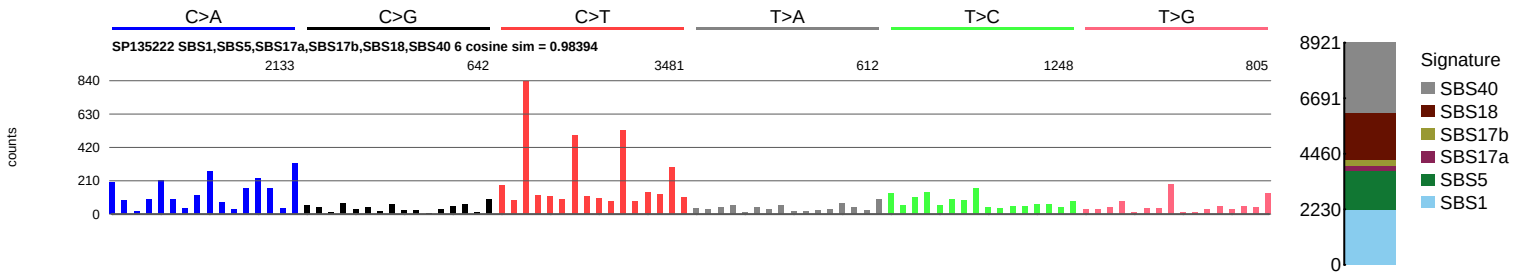
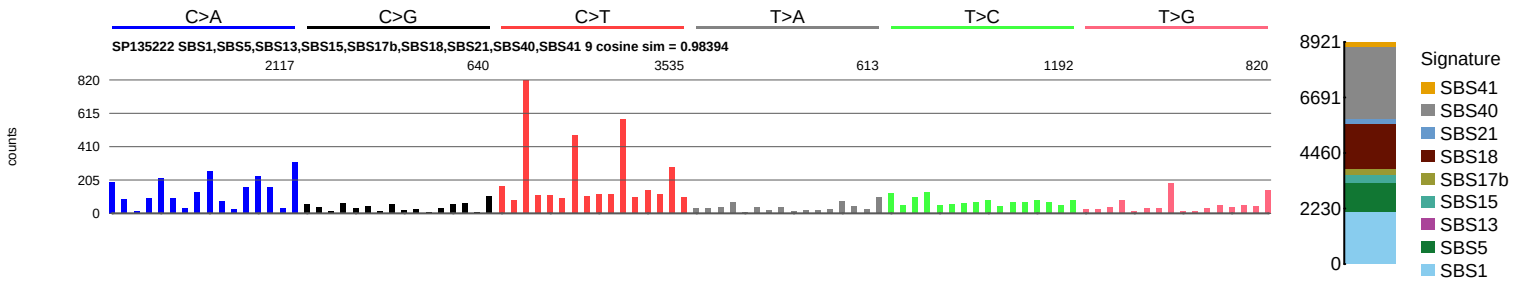
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98395
2130
653
3480
608
1250
801
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98394
2133
642
3481
612
1248
805
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98394
2117
640
3535
613
1192
820
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS41
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SBS21
SBS18
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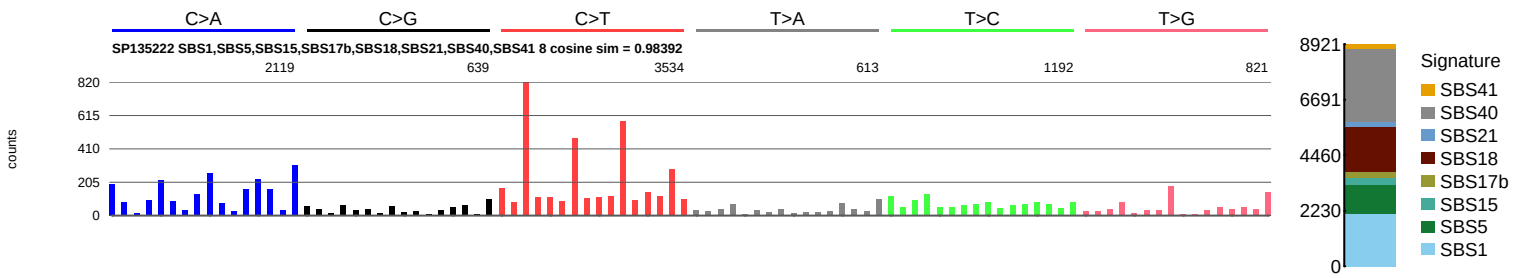
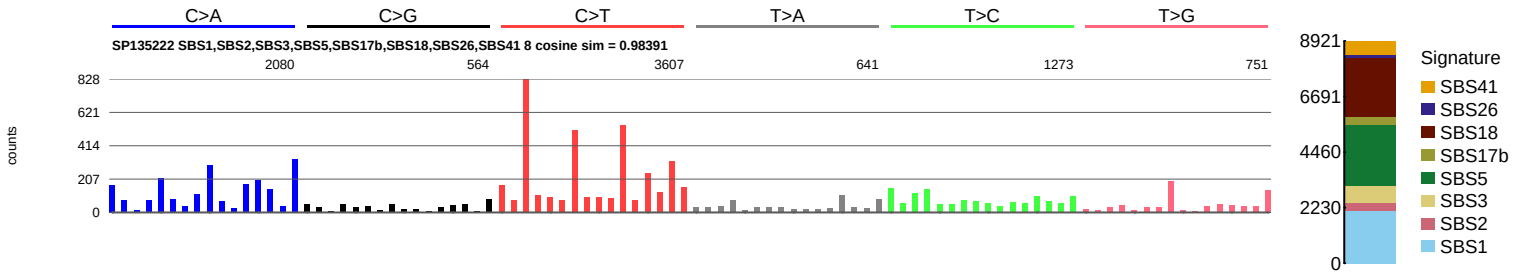
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98392
2119
639
3534
613
1192
821
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS1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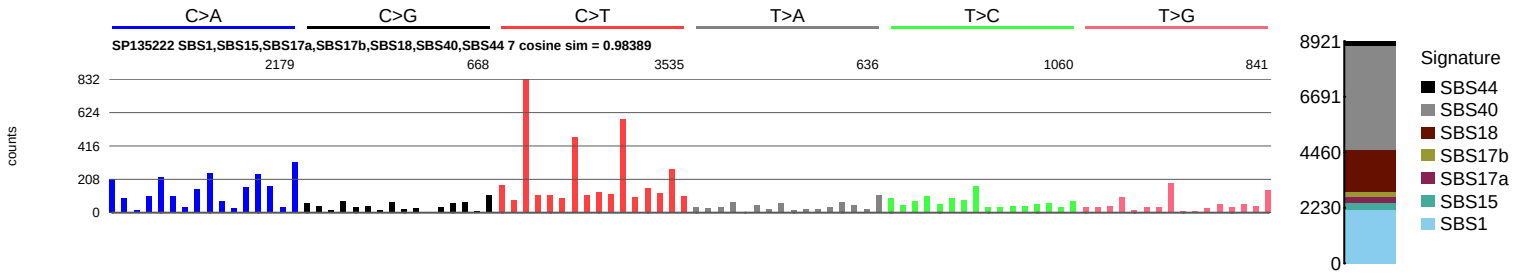
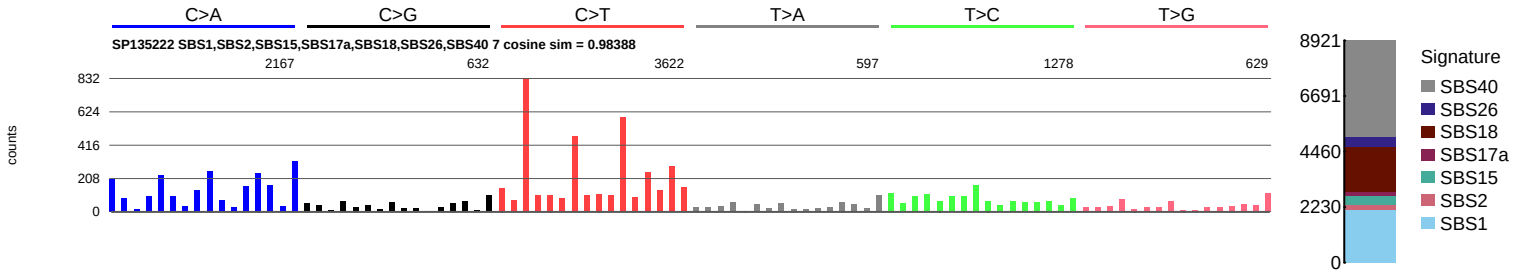
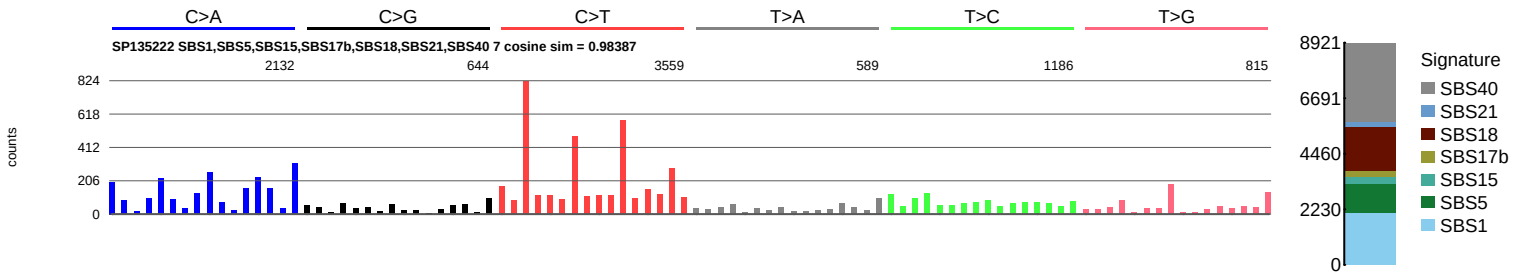
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98391
2080
564
3607
641
1273
751
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98389
2179
668
3535
636
1060
841
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS26,SBS40 7 cosine sim = 0.98388
2167
632
3622
597
1278
629
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS2
SBS1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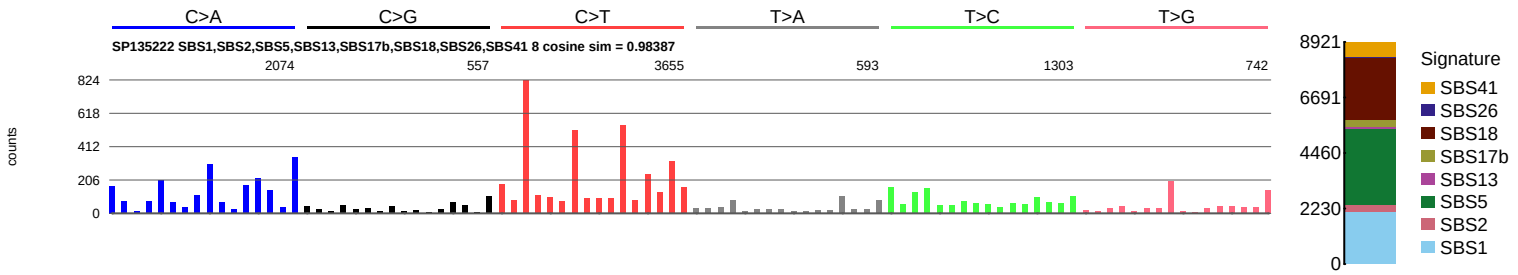
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98387
2132
644
3559
589
1186
815
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98387
2074
557
3655
593
1303
742
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS18
SBS17b
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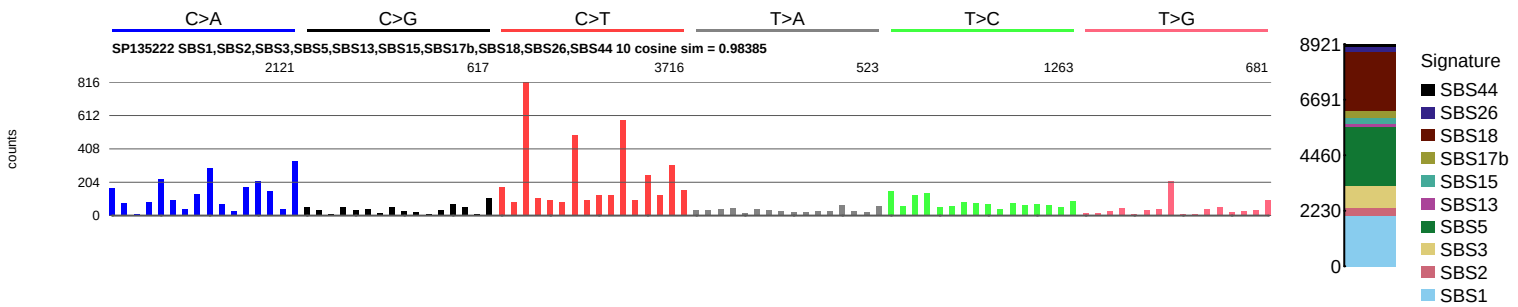
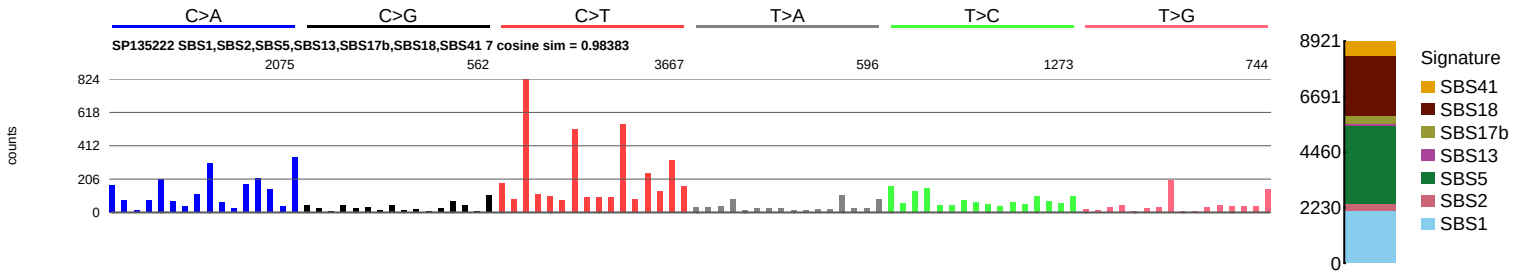
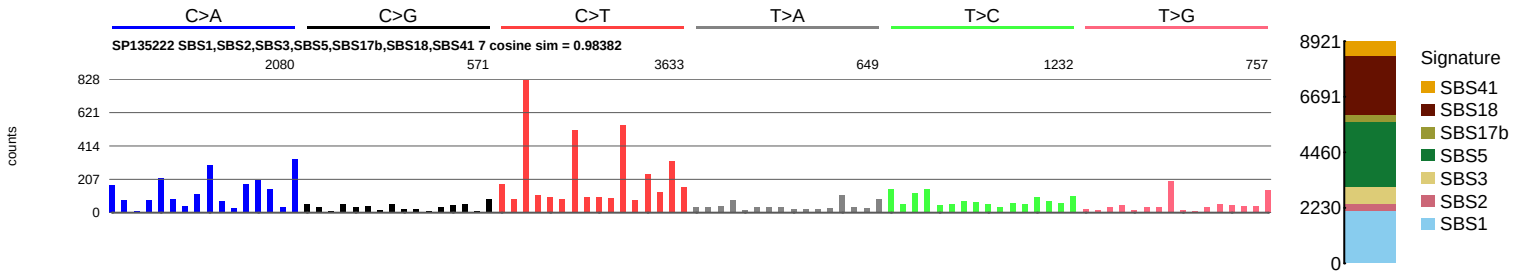
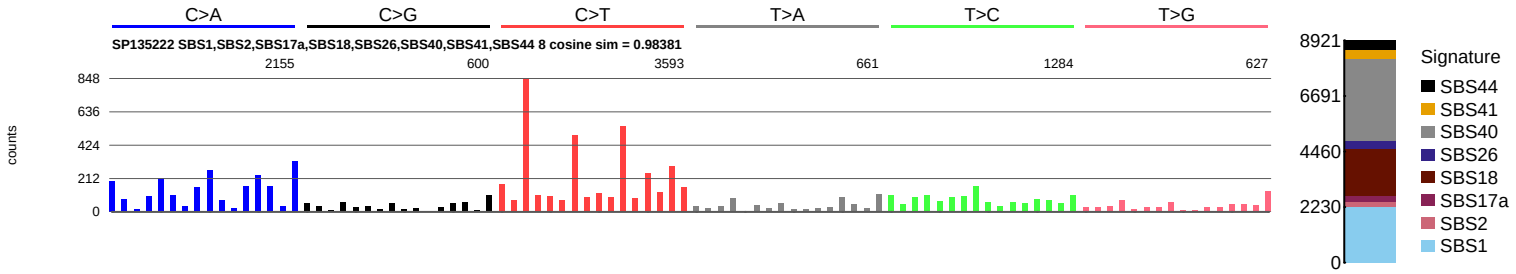
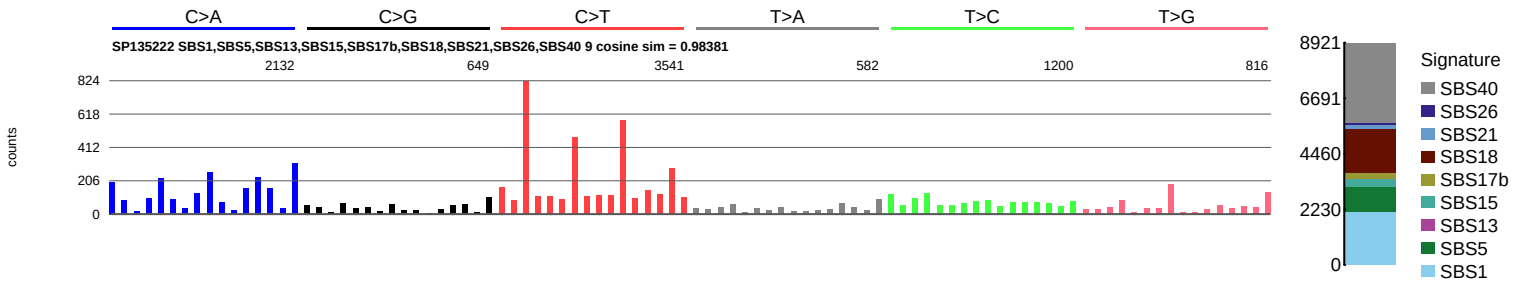
SBS5
SBS2
SBS1

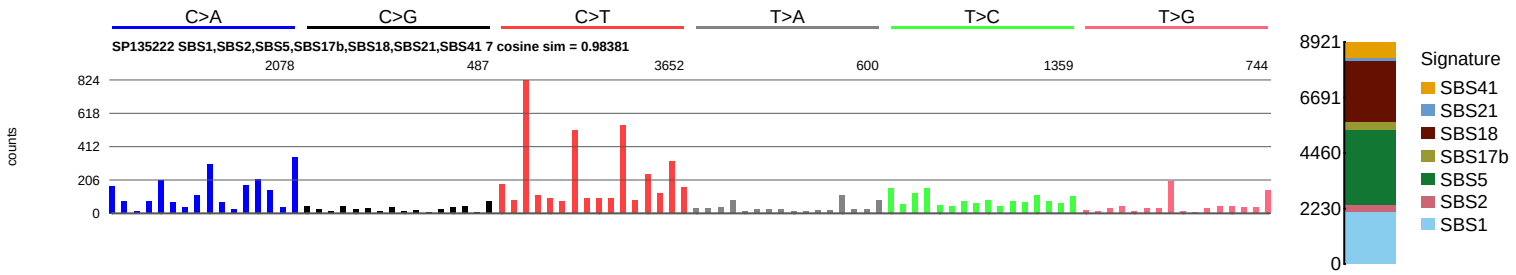

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98385
2121
617
3716
523
1263
681
816
612
408
204
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

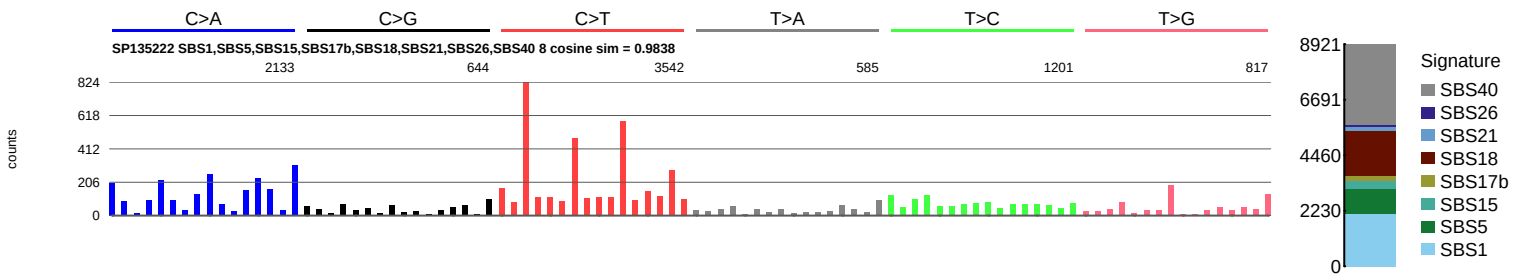

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS41 7 cosine sim = 0.98383
2075
562
3667
596
1273
744
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

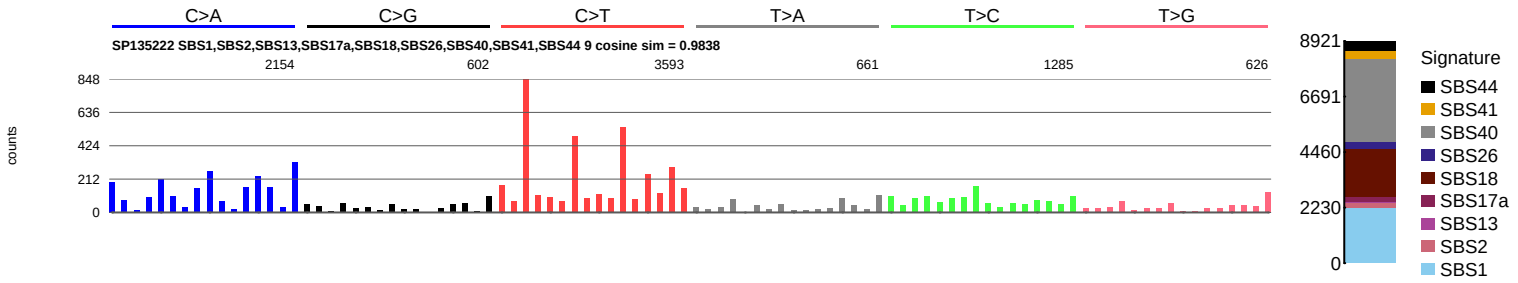

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS41 7 cosine sim = 0.98382
2080
571
3633
649
1232
757
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

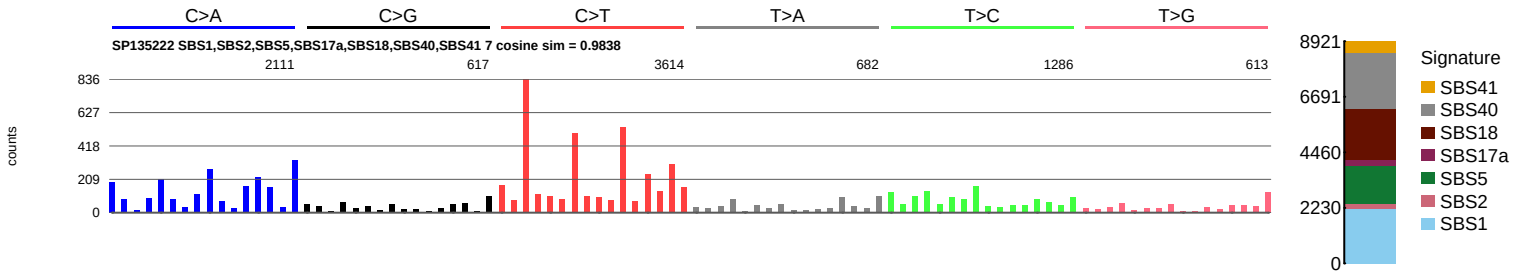

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98381
2155
600
3593
661
1284
627
counts
848
636
424
212
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98381
2132
649
3541
582
1200
816
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98381
2078
487
3652
600
1359
744
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.9838
2133
644
3542
585
1201
817
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9838
2154
602
3593
661
1285
626
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
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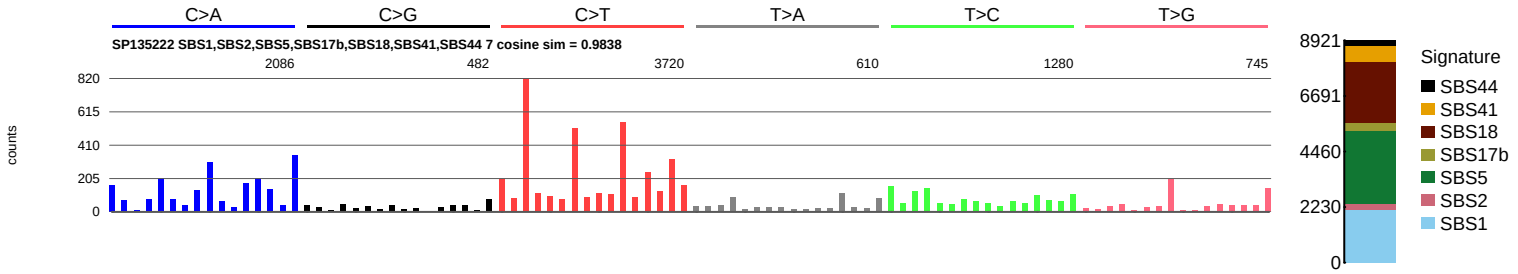
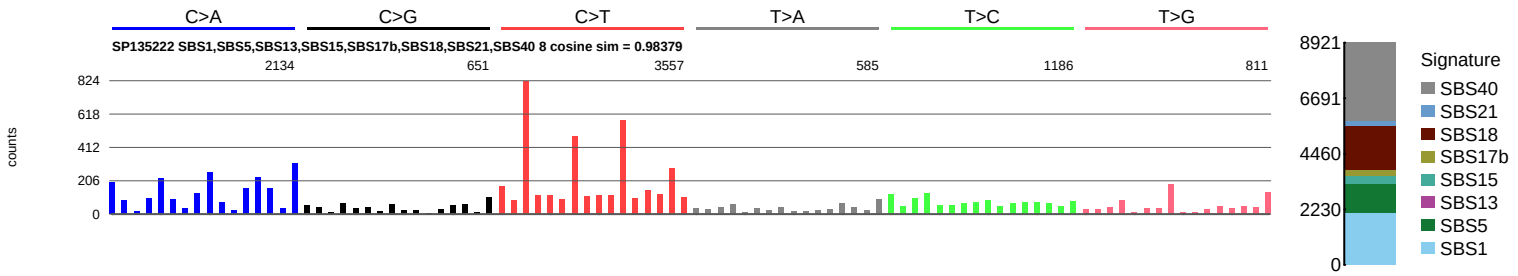
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS40,SBS41 7 cosine sim = 0.9838
2111
617
3614
682
1286
613
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.9838
2086
482
3720
610
1280
745
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS5
SBS2
SBS1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C>G
C>T
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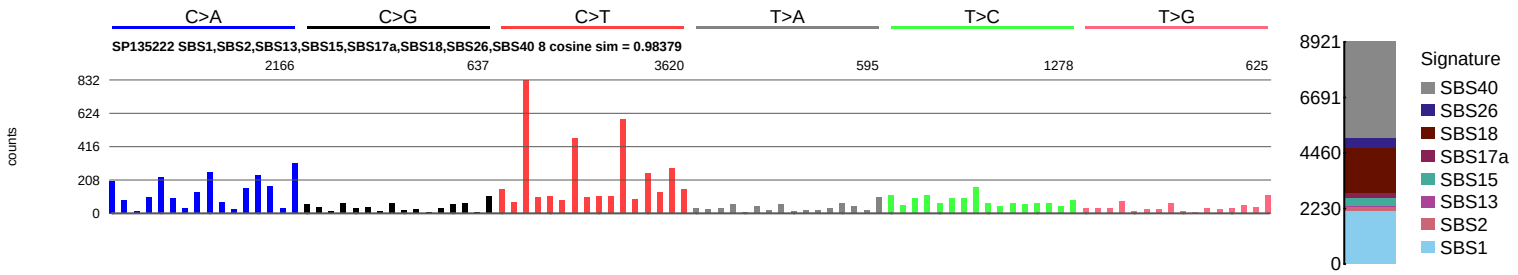
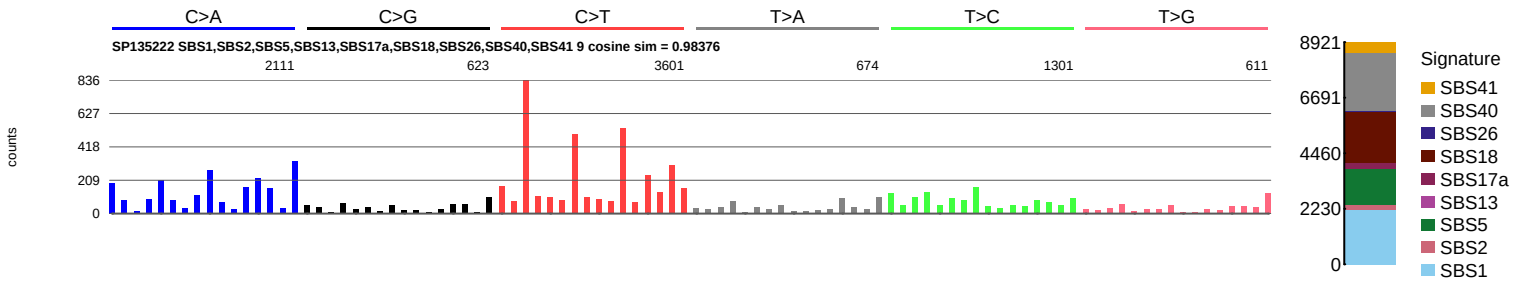
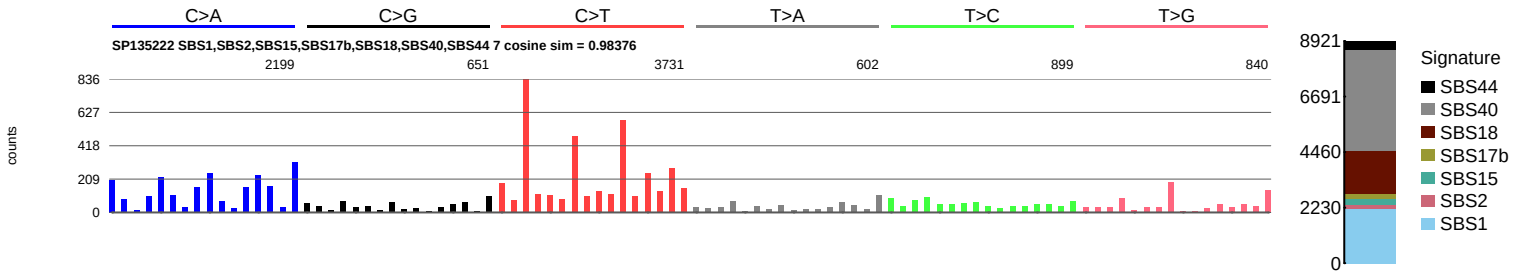
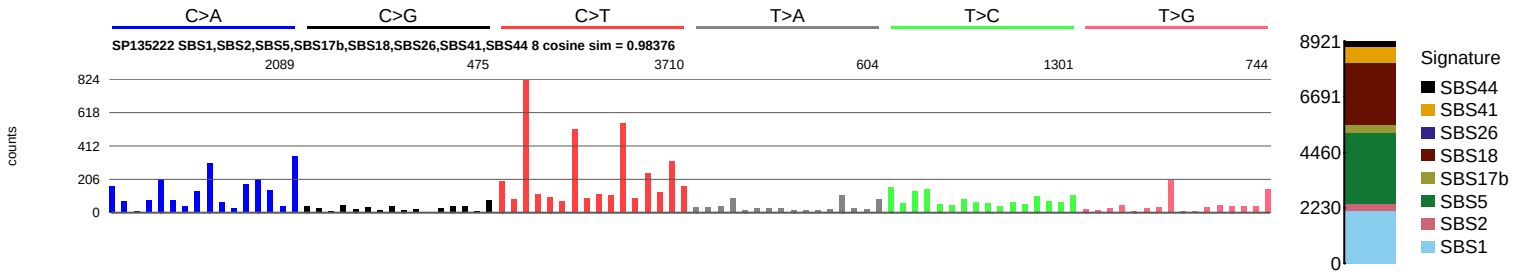
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98379
2134
651
3557
585
1186
811
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS18,SBS26,SBS40 8 cosine sim = 0.98379
2166
637
3620
595
1278
625
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98376
2111
623
3601
674
1301
611
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98376
2199
651
3731
602
899
840
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1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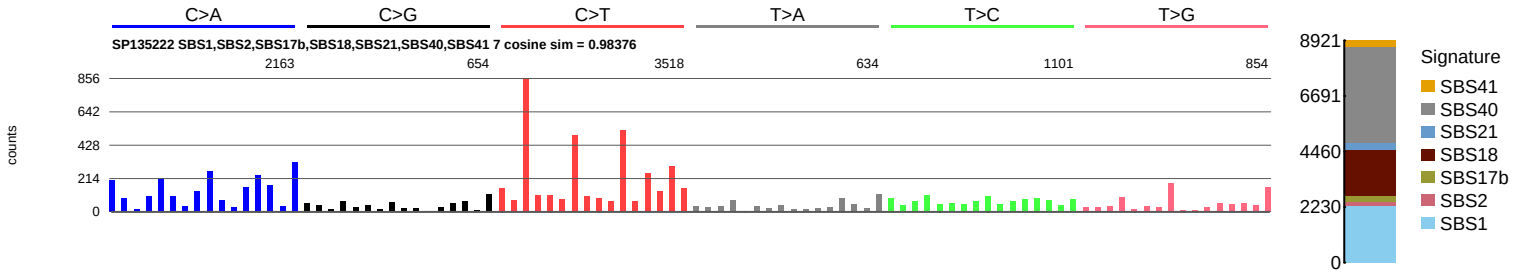
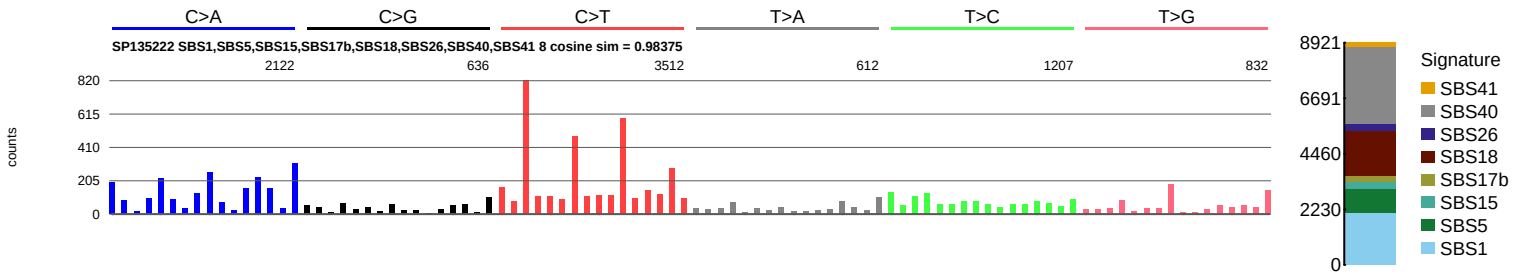
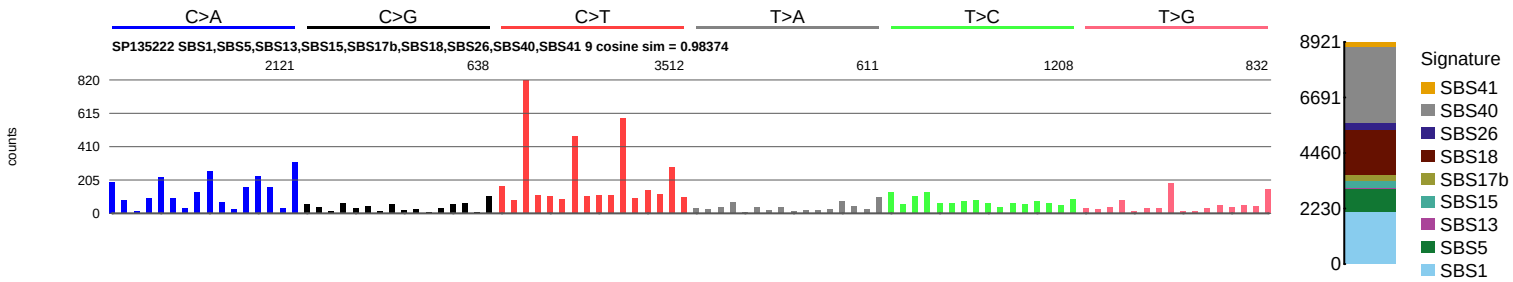
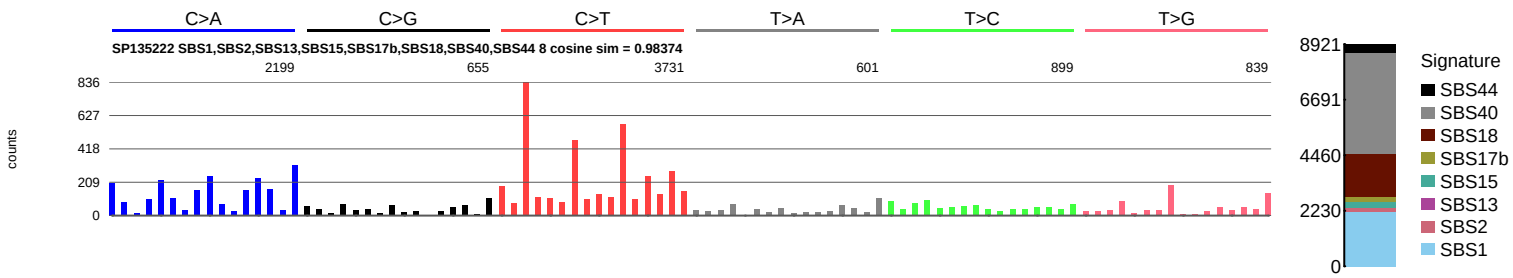
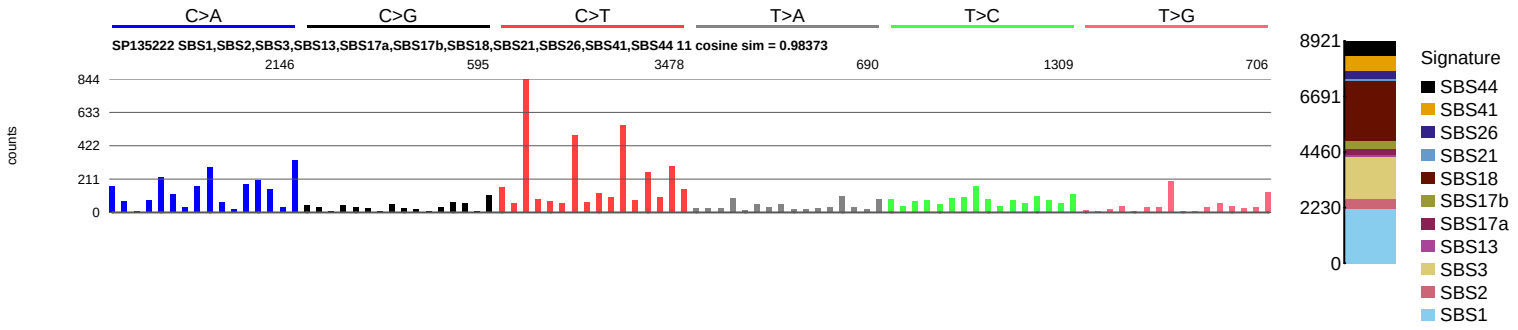
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98376
2089
475
3710
604
1301
744
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1

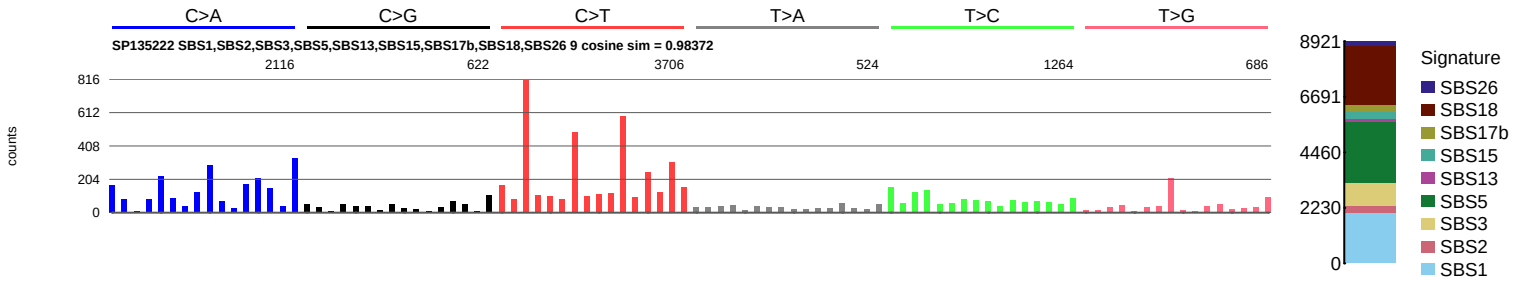

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98376
2163
654
3518
634
1101
854
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1

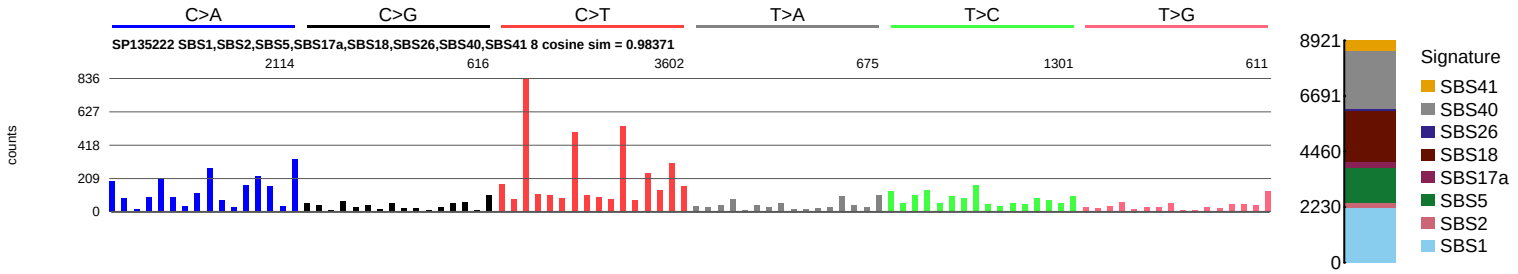

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98375
2122
636
3512
612
1207
832
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS1

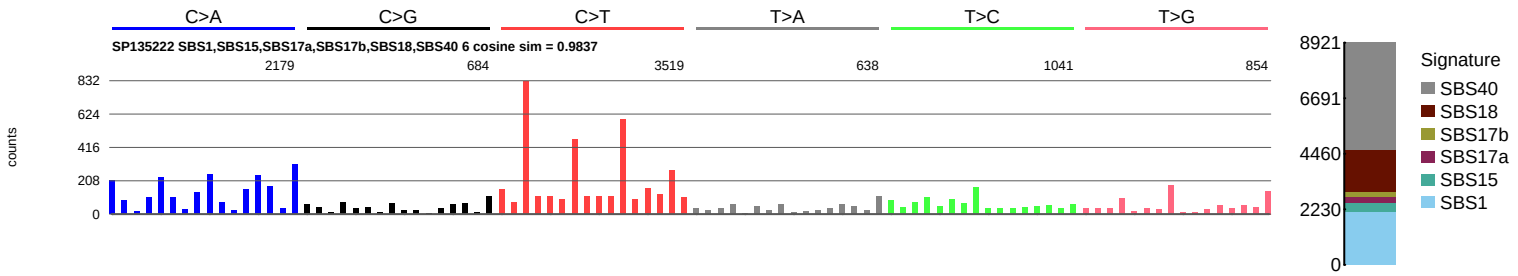

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98374
2121
638
3512
611
1208
832
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1

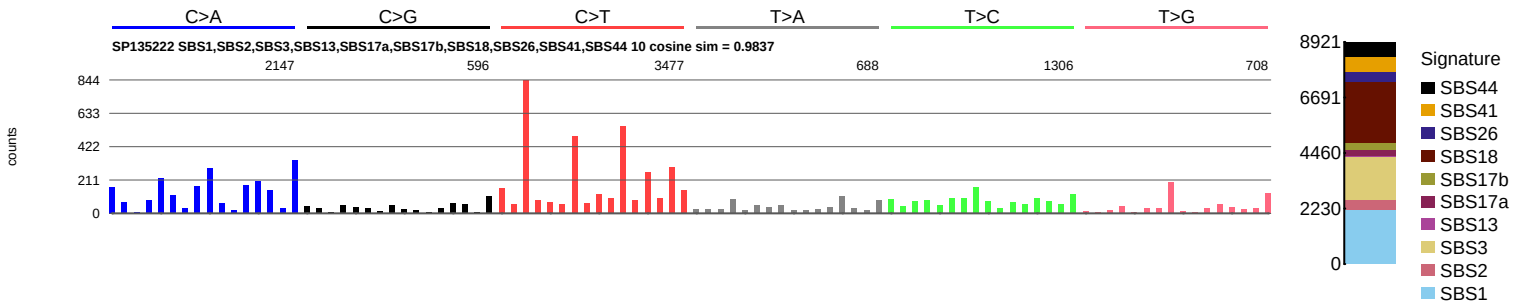

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98374
2199
655
3731
601
899
839
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS2
SBS1

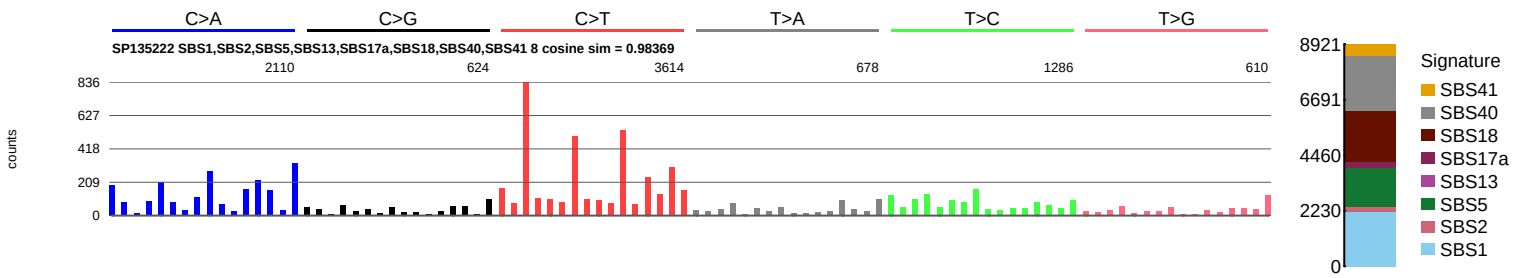

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.98373
2146
595
3478
690
1309
706
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

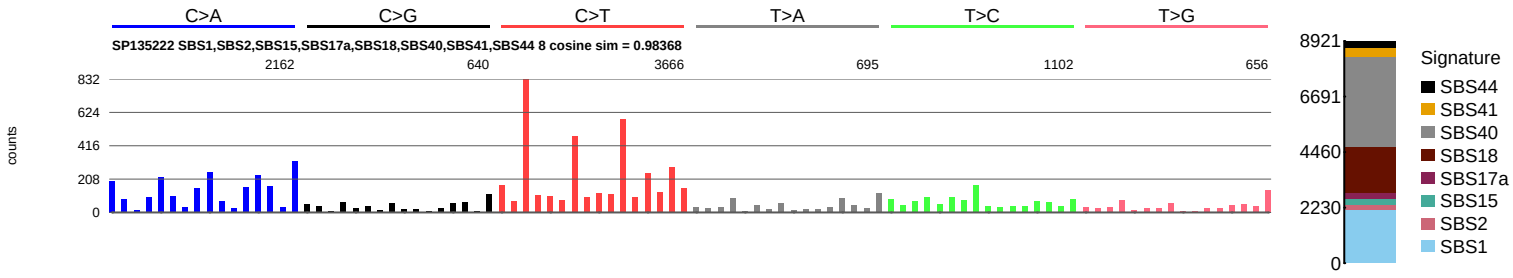

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26 9 cosine sim = 0.98372
2116
622
3706
524
1264
686
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

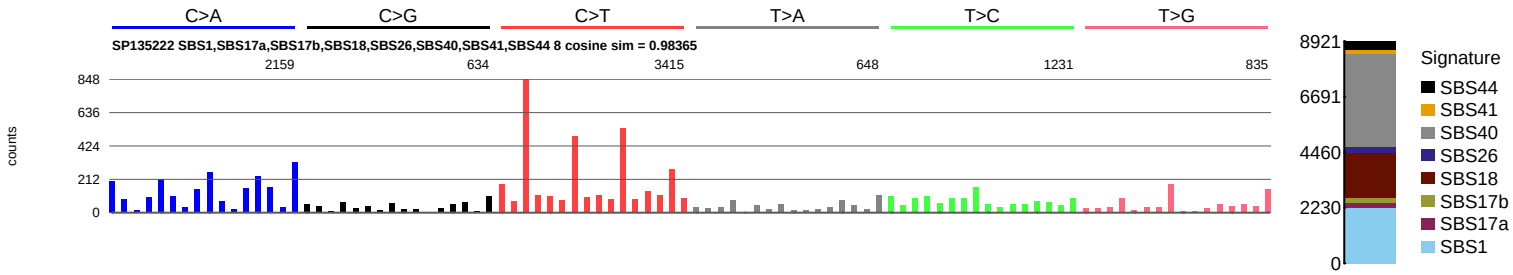

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98371
2114
616
3602
675
1301
611
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS5
SBS2
SBS1

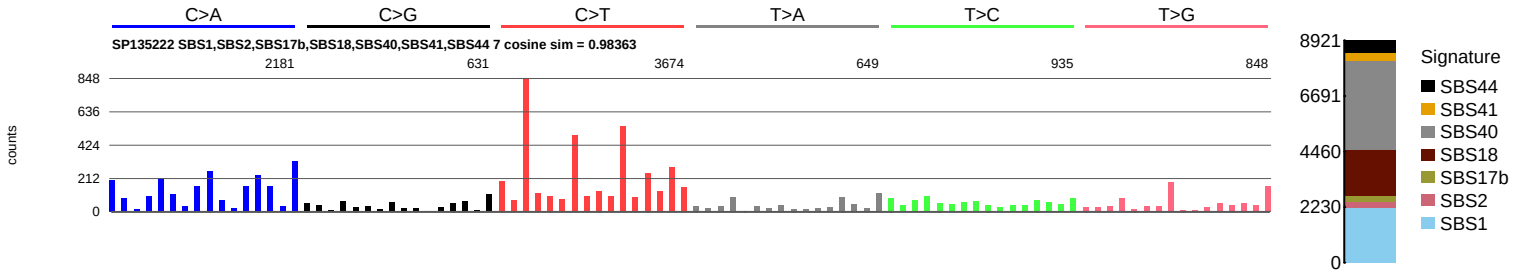

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.9837
2179
684
3519
638
1041
854
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1

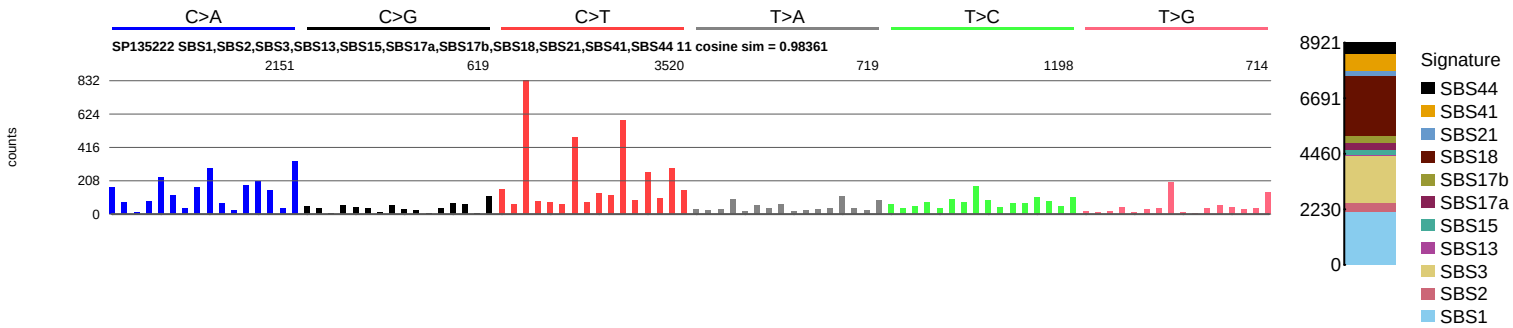

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.9837
2147
596
3477
688
1306
708
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

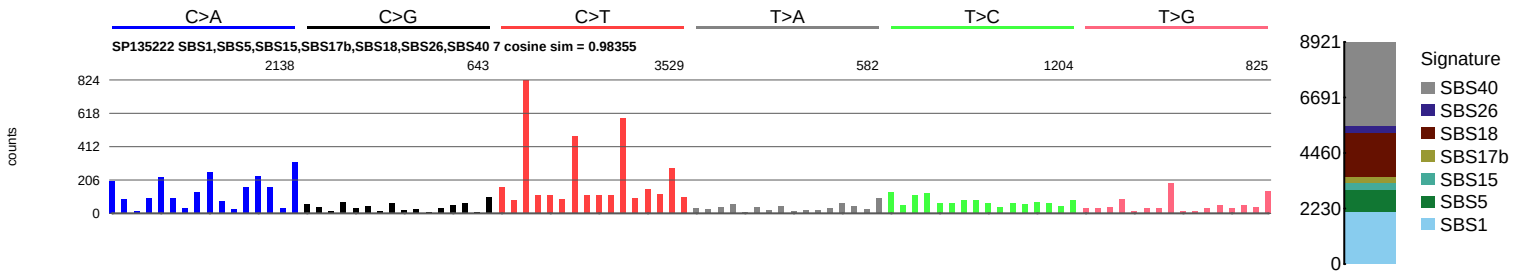

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS41 8 cosine sim = 0.98369
2110
624
3614
678
1286
610
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1

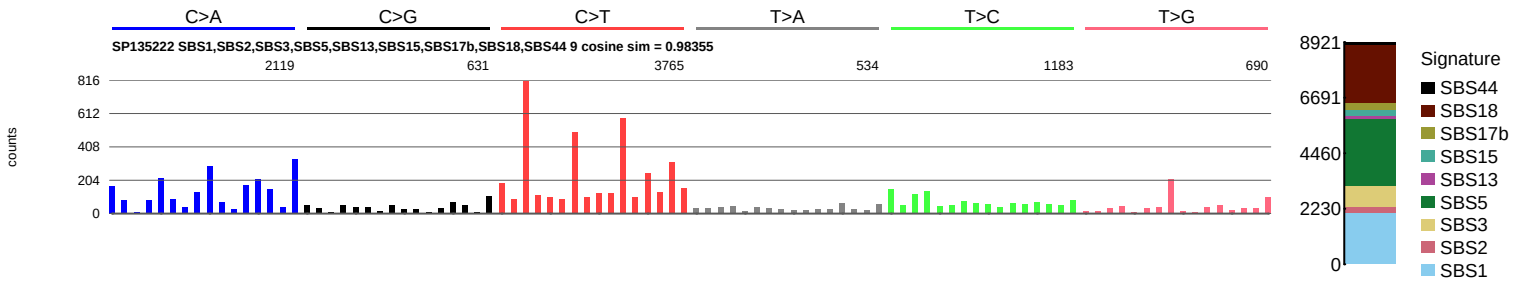

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98368
2162
640
3666
695
1102
656
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS2
SBS1

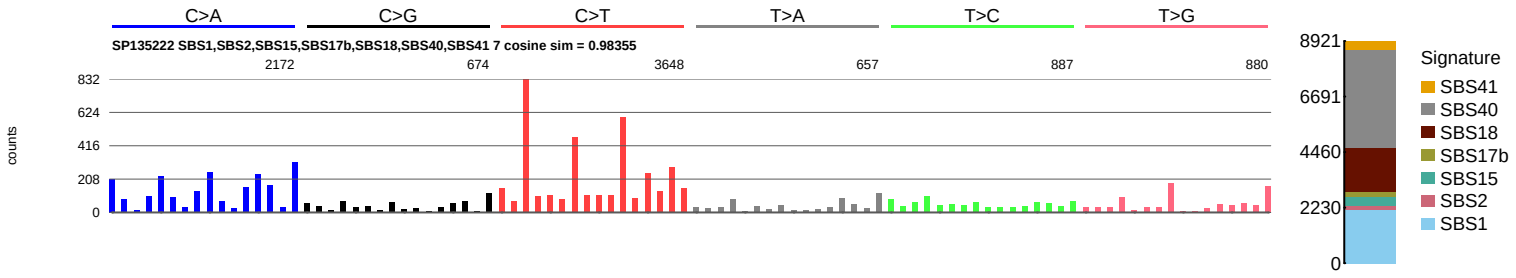

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98365
2159
634
3415
648
1231
835
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1

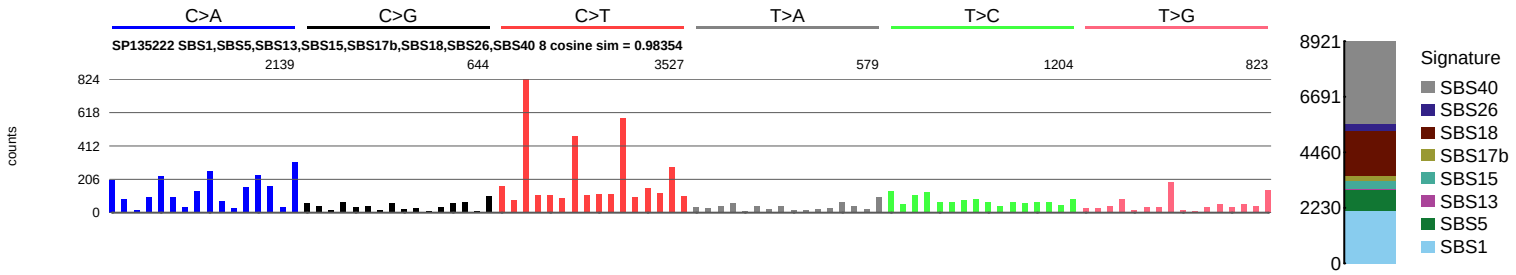

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98363
2181
631
3674
649
935
848
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS2
SBS1

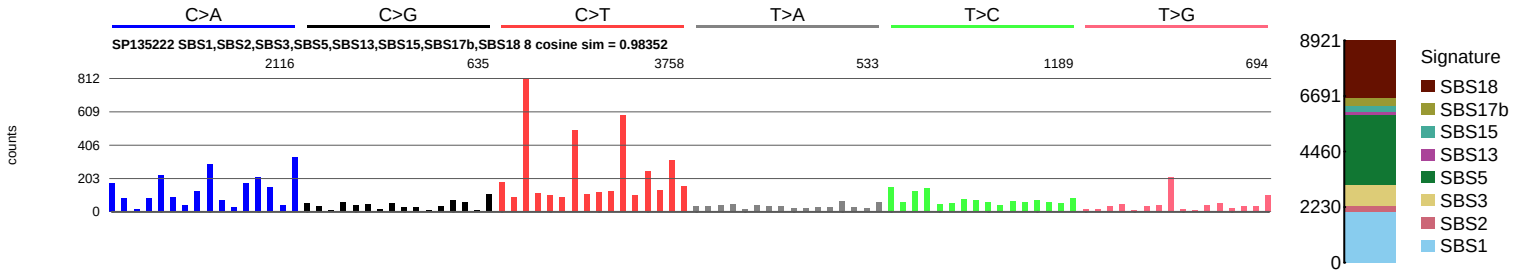

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98361
2151
619
3520
719
1198
714
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98355
2138
643
3529
582
1204
825
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS44 9 cosine sim = 0.98355
2119
631
3765
534
1183
690
816
612
408
204
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98355
2172
674
3648
657
887
880
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98354
2139
644
3527
579
1204
823
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1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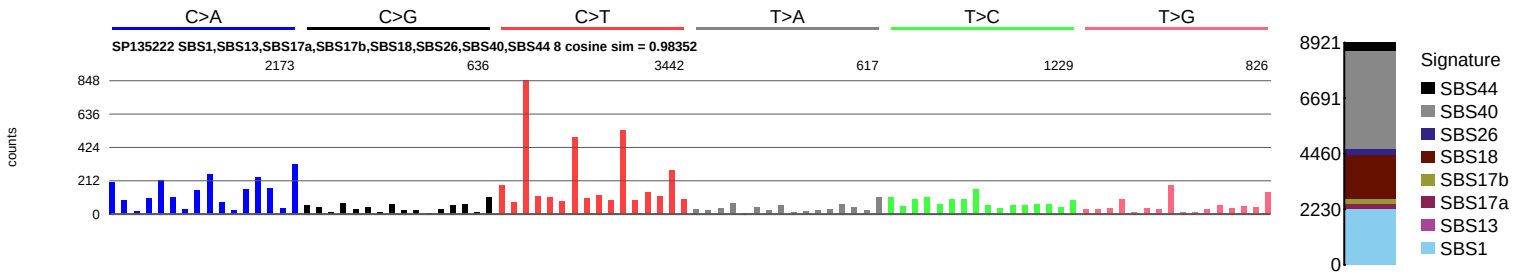
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17b,SBS18 8 cosine sim = 0.98352
2116
635
3758
533
1189
694
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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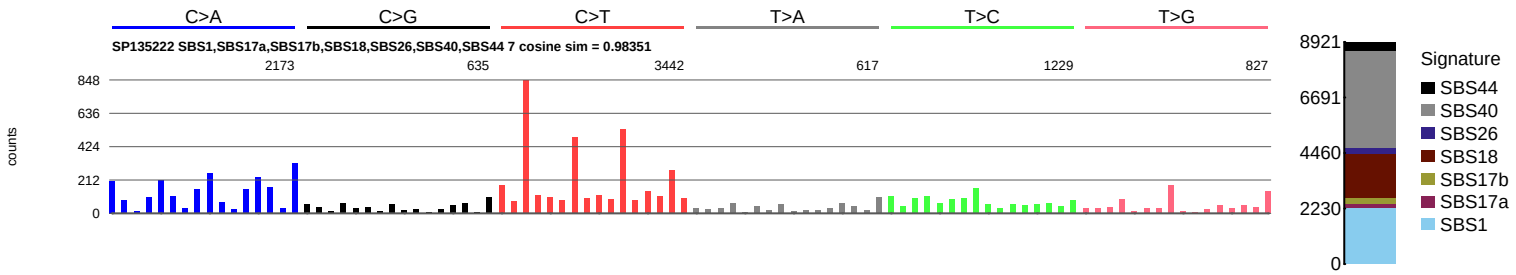
SP135222 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98352
2173
636
3442
617
1229
826
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1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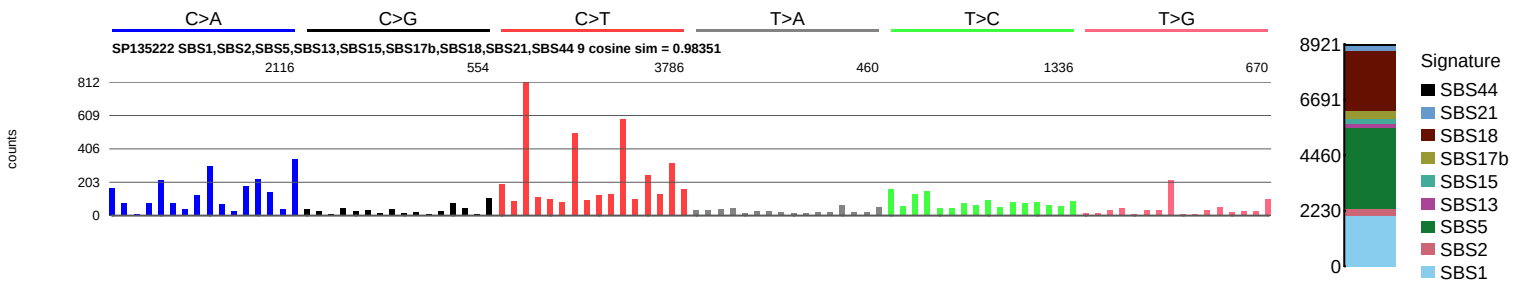
SP135222 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98351
2173
635
3442
617
1229
827
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1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C>T
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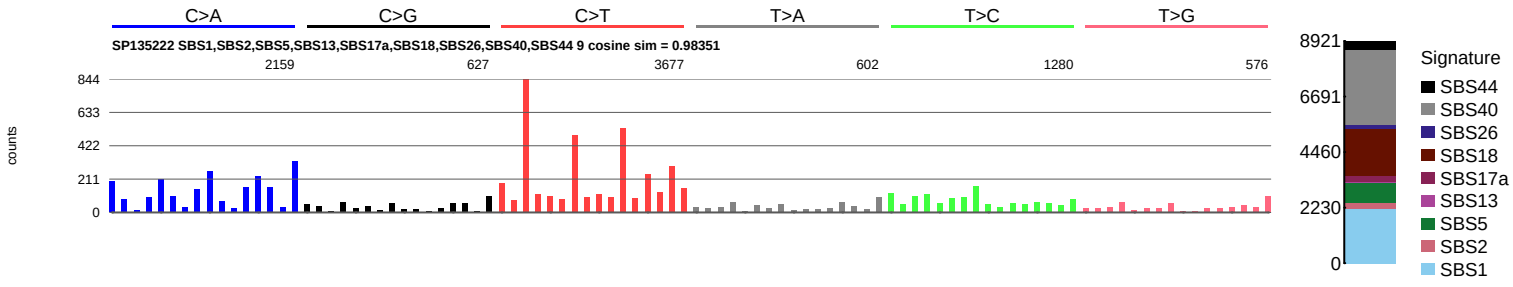
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98351
2116
554
3786
460
1336
670
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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C>T
T>A
T>C
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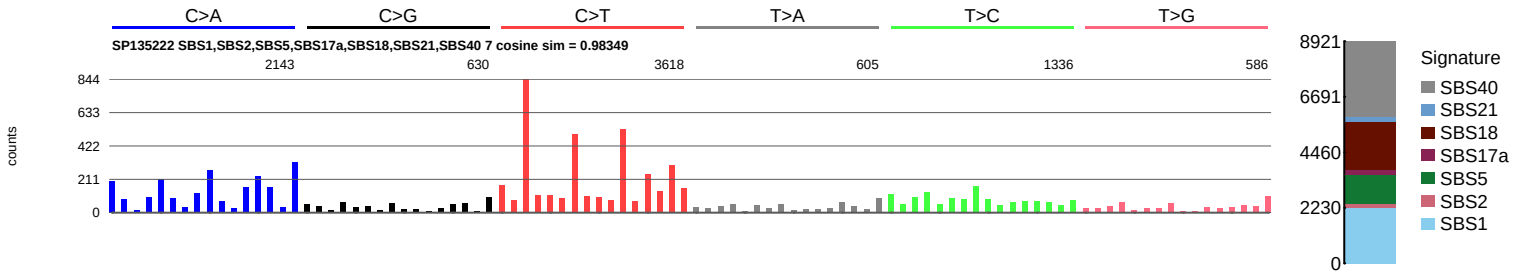
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98351
2159
627
3677
602
1280
576
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1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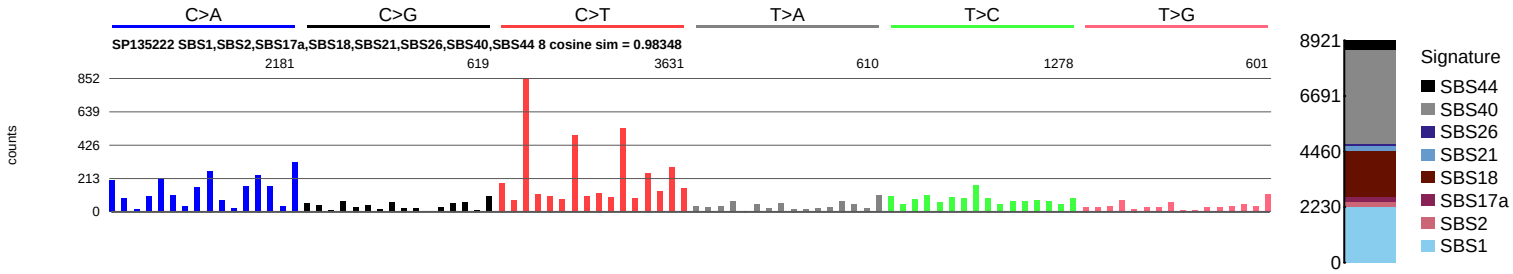
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS40 7 cosine sim = 0.98349
2143
630
3618
605
1336
586
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98348
2181
619
3631
610
1278
601
counts
852
639
426
213
0
8921
6691
4460
2230
0
Signature
SBS44
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SBS26
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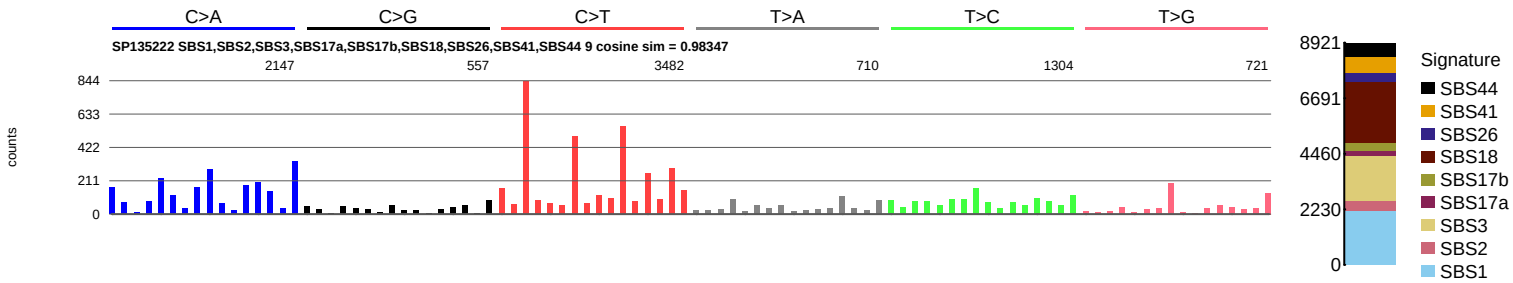
SBS17a
SBS2
SBS1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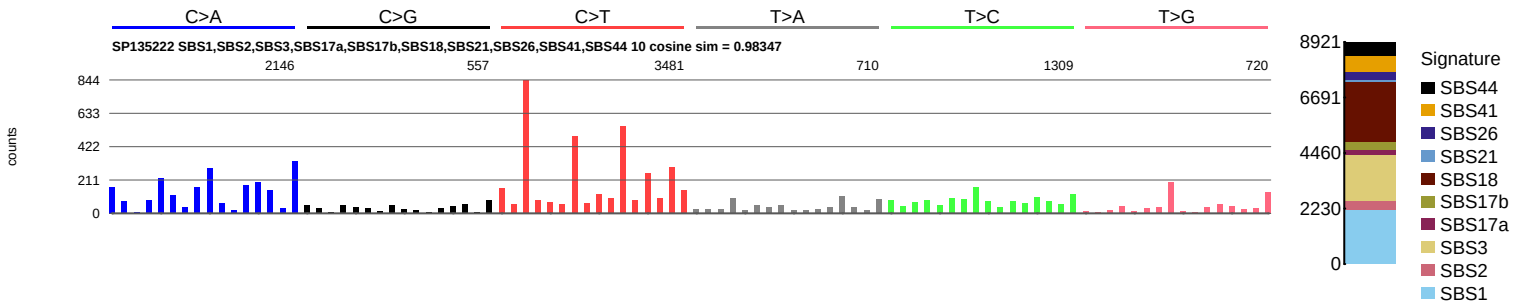
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98347
2147
557
3482
710
1304
721
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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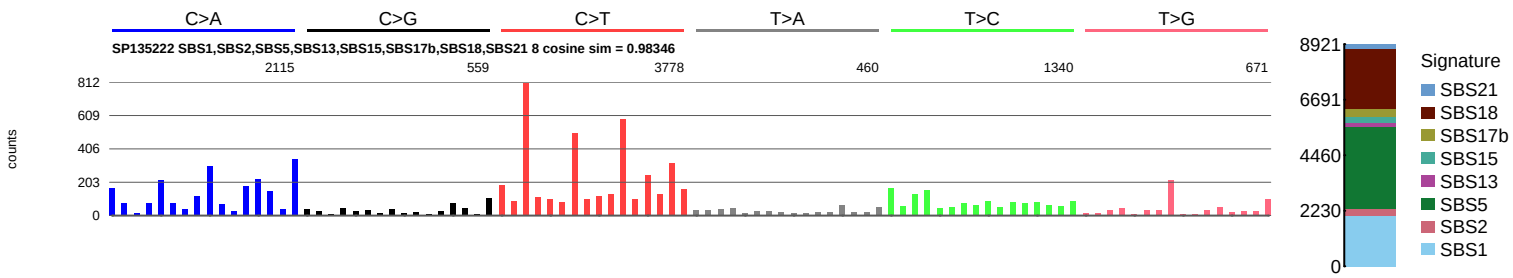
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98347
2146
557
3481
710
1309
720
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS21 8 cosine sim = 0.98346
2115
559
3778
460
1340
671
counts
812
609
406
203
0
8921
6691
4460
2230
0
Signature
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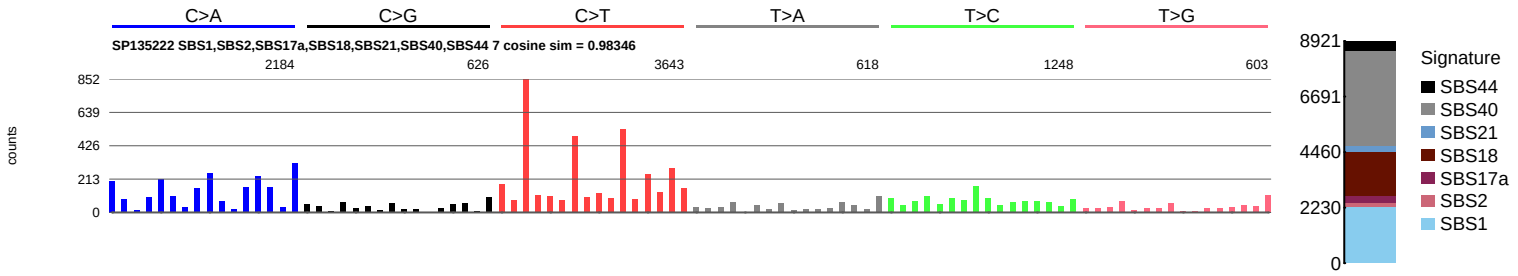
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98346
2184
626
3643
618
1248
603
counts
852
639
426
213
0
8921
6691
4460
2230
0
Signature
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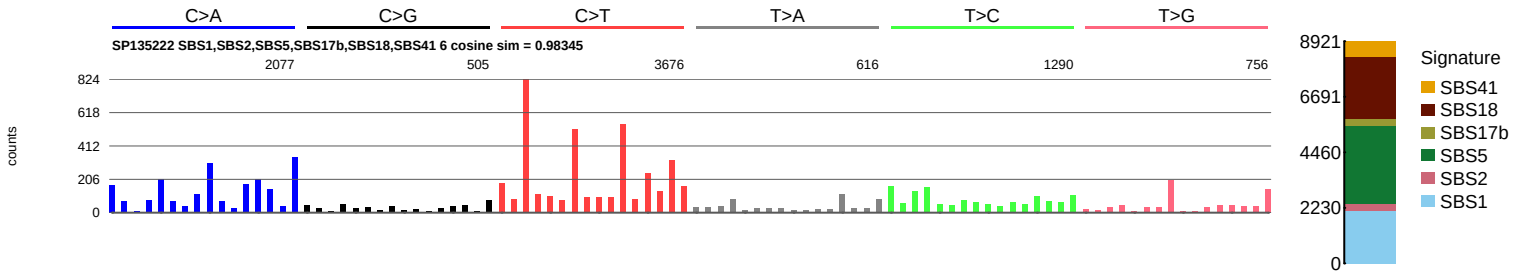
SBS17a
SBS2
SBS1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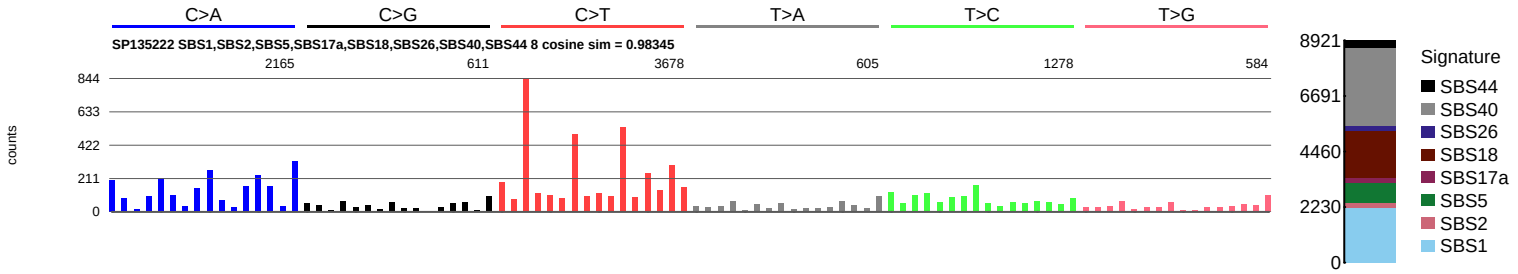
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS41 6 cosine sim = 0.98345
2077
505
3676
616
1290
756
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS5
SBS2
SBS1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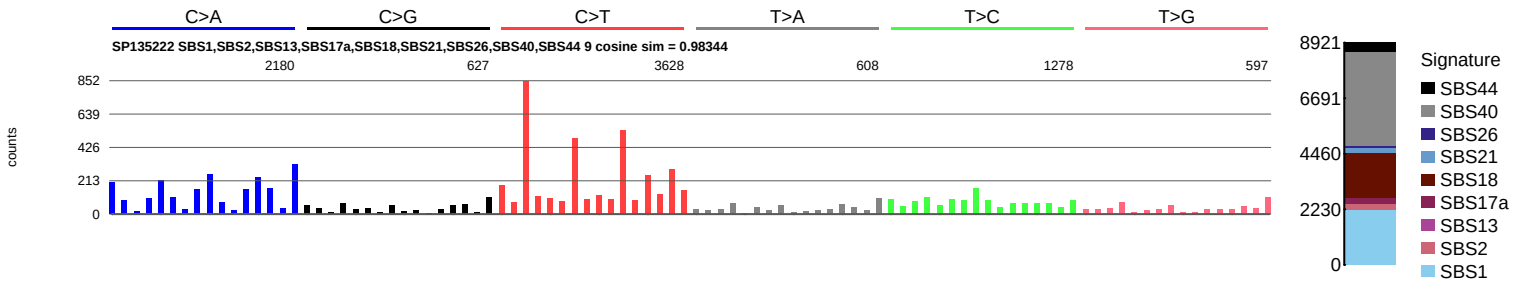
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98345
2165
611
3678
605
1278
584
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS5
SBS2
SBS1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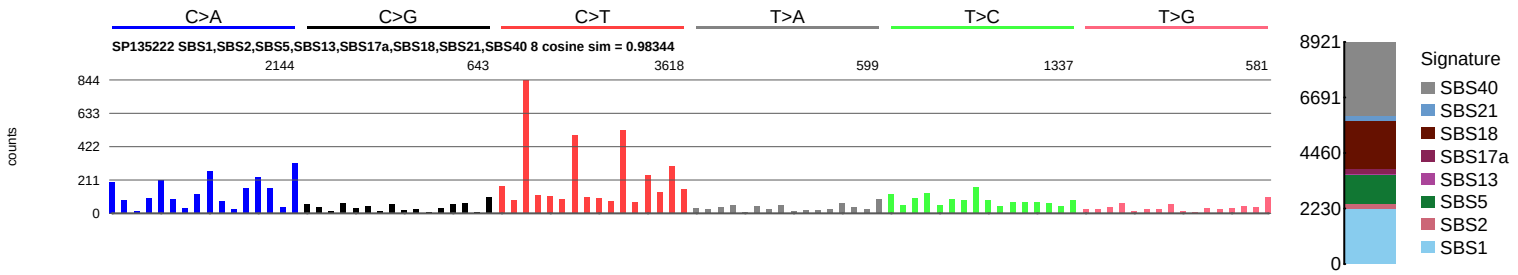
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98344
2180
627
3628
608
1278
597
counts
852
639
426
213
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1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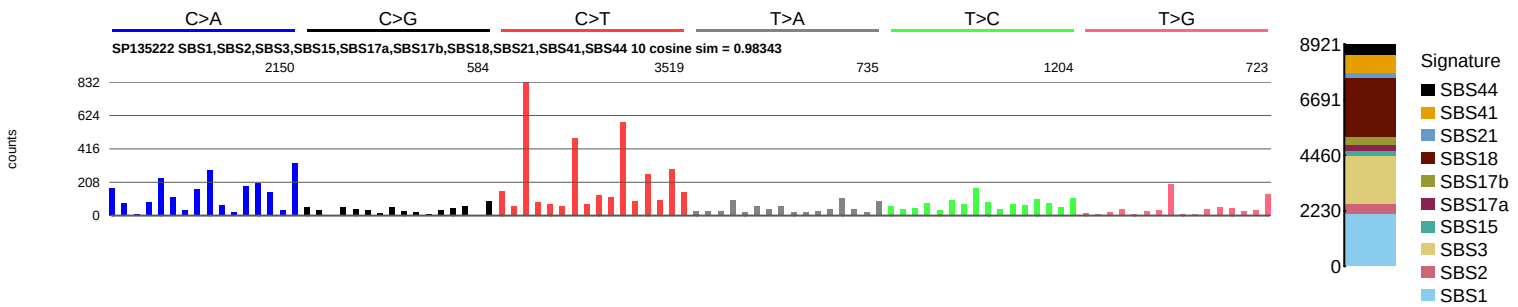
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS40 8 cosine sim = 0.98344
2144
643
3618
599
1337
581
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1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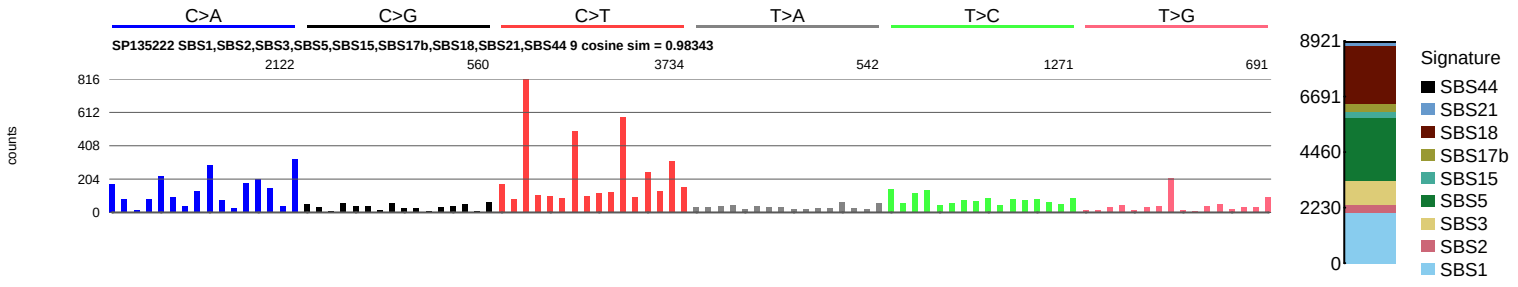
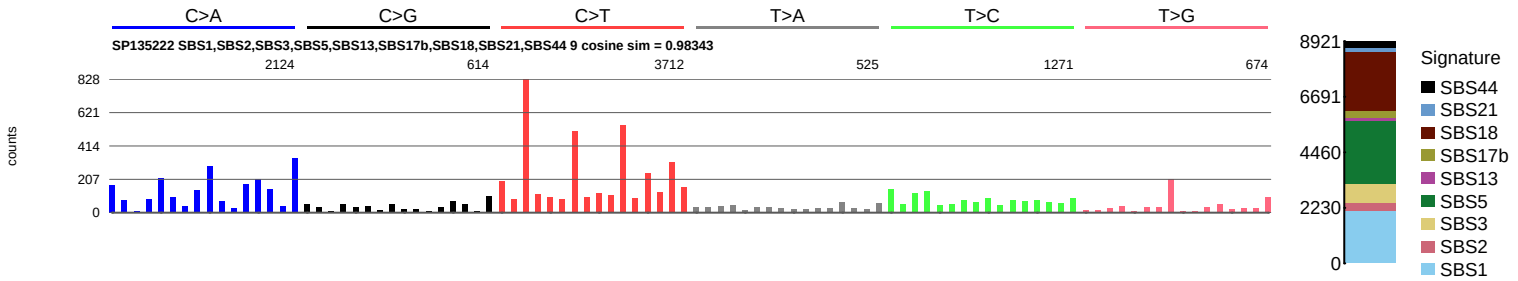
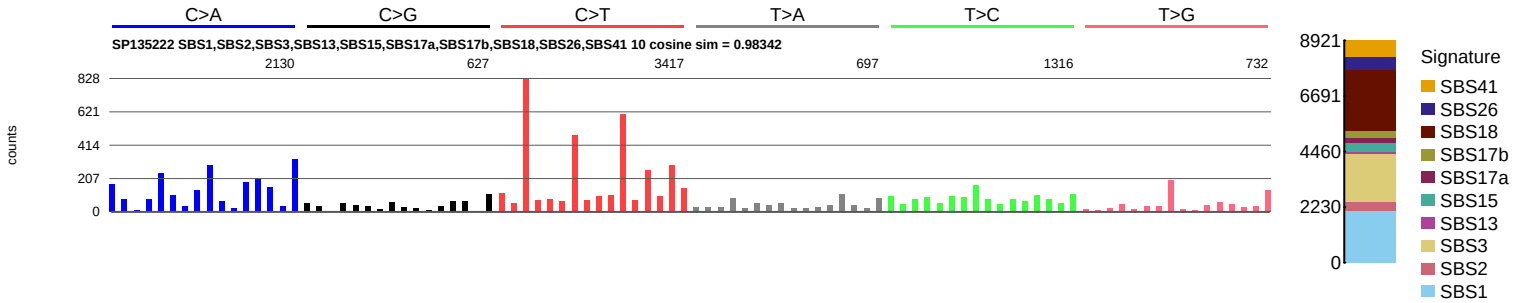
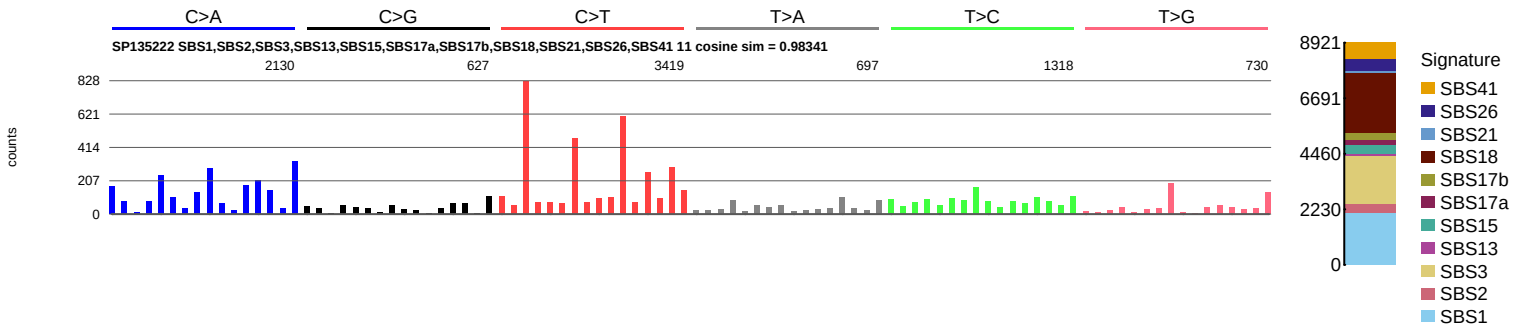
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98343
2150
584
3519
735
1204
723
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98343
2122
560
3734
542
1271
691
816
612
408
204
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98343
2124
614
3712
525
1271
674
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98342
2130
627
3417
697
1316
732
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 11 cosine sim = 0.98341
2130
627
3419
697
1318
730
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
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SBS17b
SBS17a
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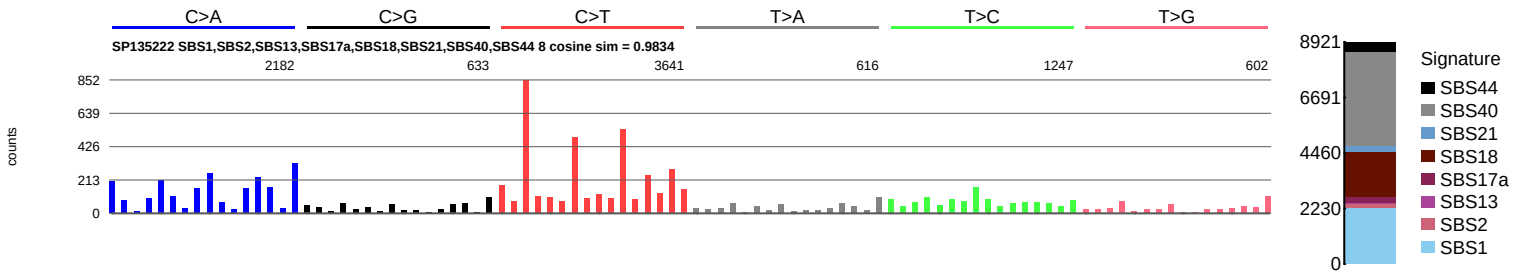
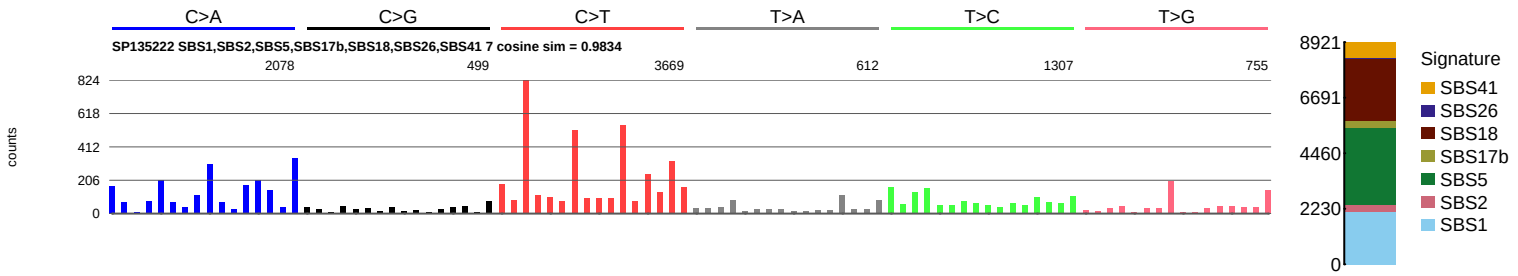
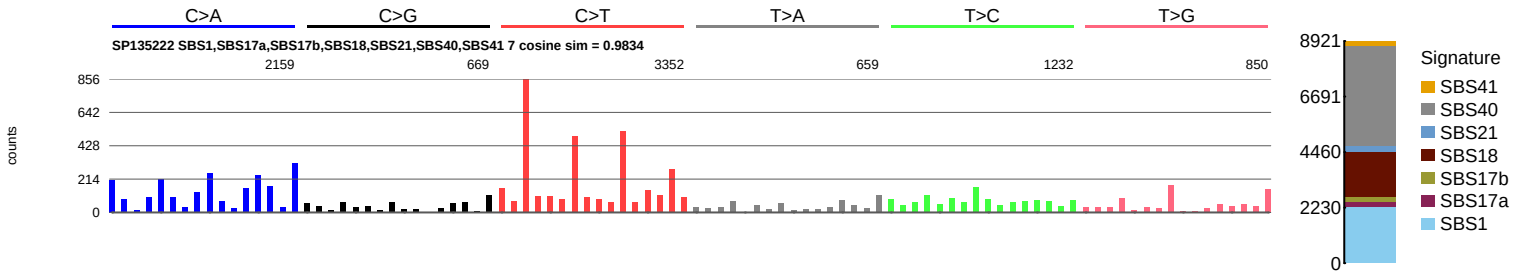
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.9834
2182
633
3641
616
1247
602
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.9834
2078
499
3669
612
1307
755
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.9834
2159
669
3352
659
1232
850
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS41
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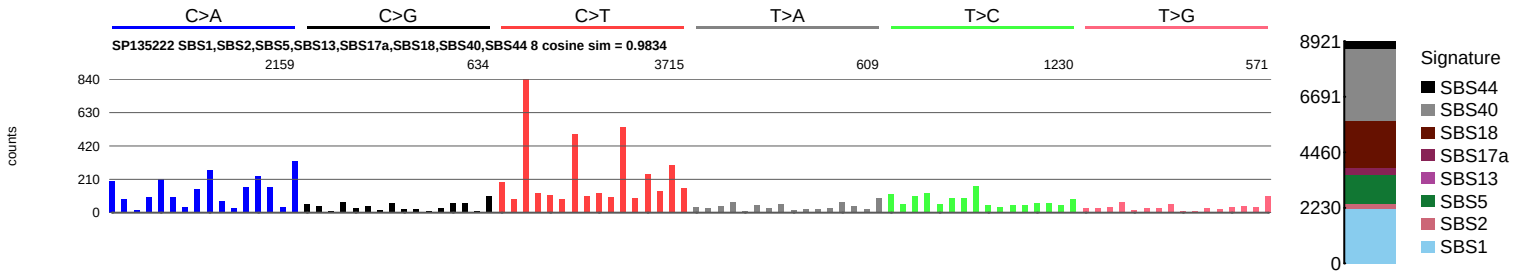
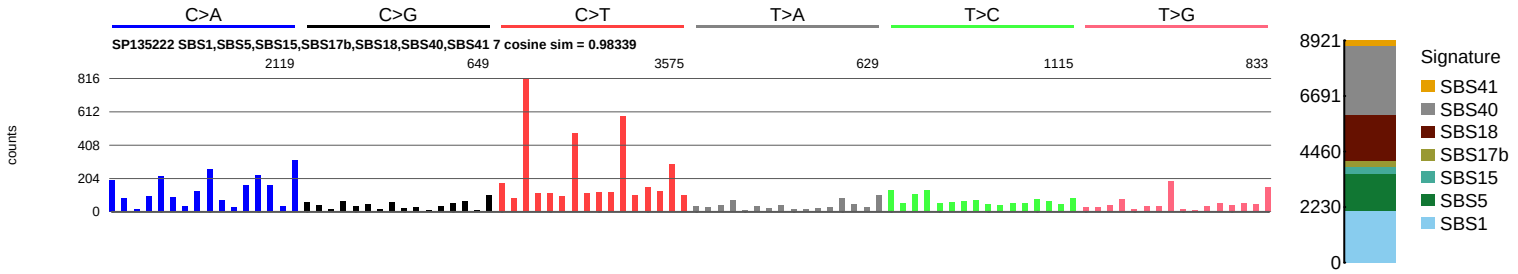
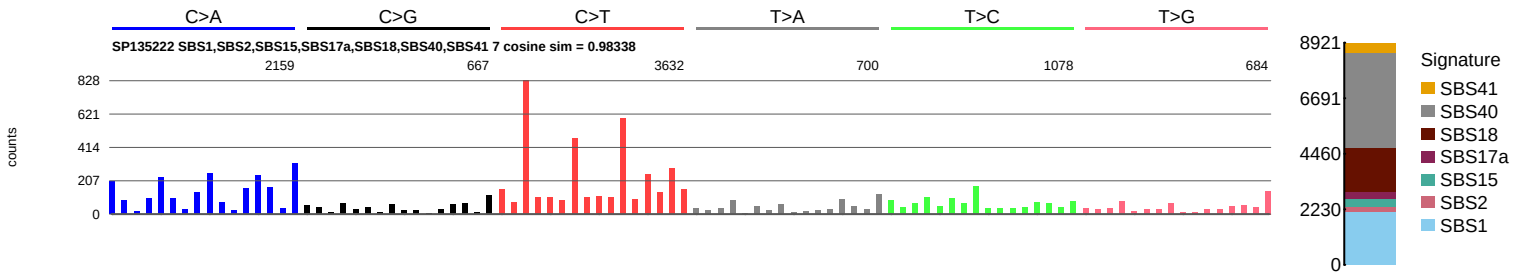
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40,SBS44 8 cosine sim = 0.9834
2159
634
3715
609
1230
571
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98339
2119
649
3575
629
1115
833
816
612
408
204
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS1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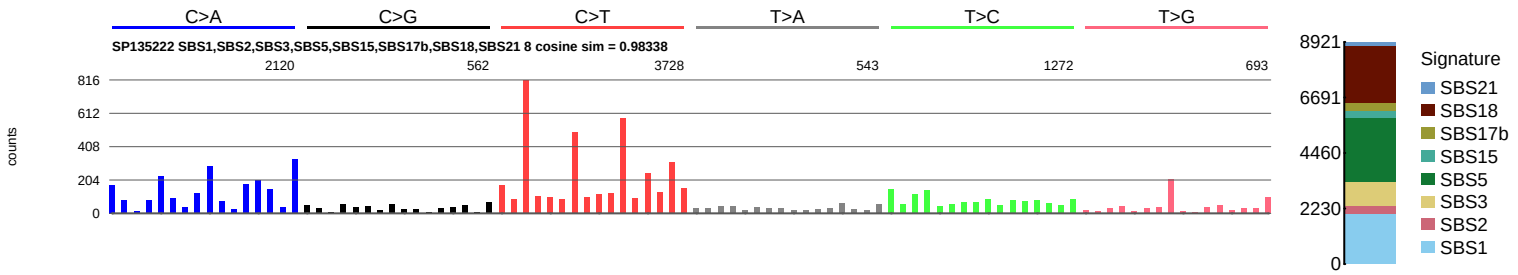
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS40,SBS41 7 cosine sim = 0.98338
2159
667
3632
700
1078
684
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS2
SBS1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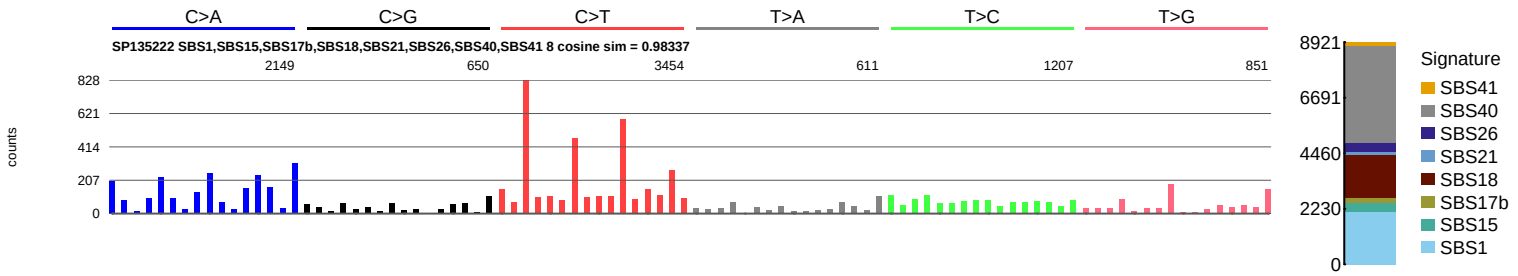
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS21 8 cosine sim = 0.98338
2120
562
3728
543
1272
693
816
612
408
204
0
counts
8921
6691
4460
2230
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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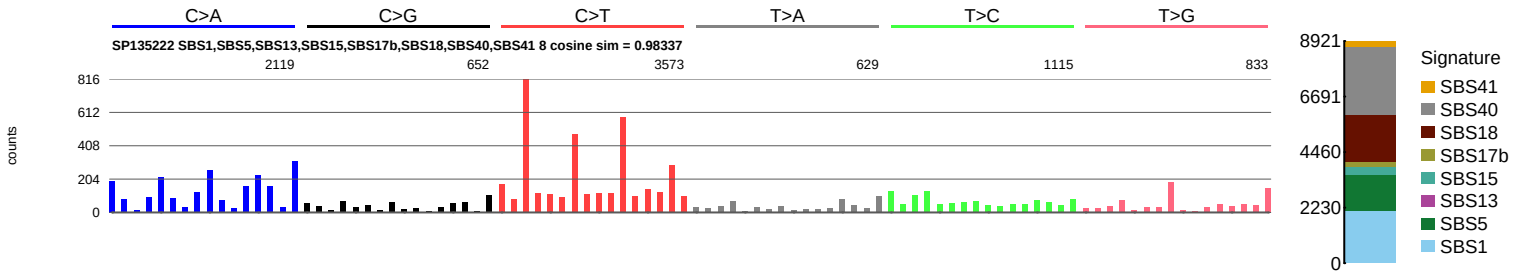
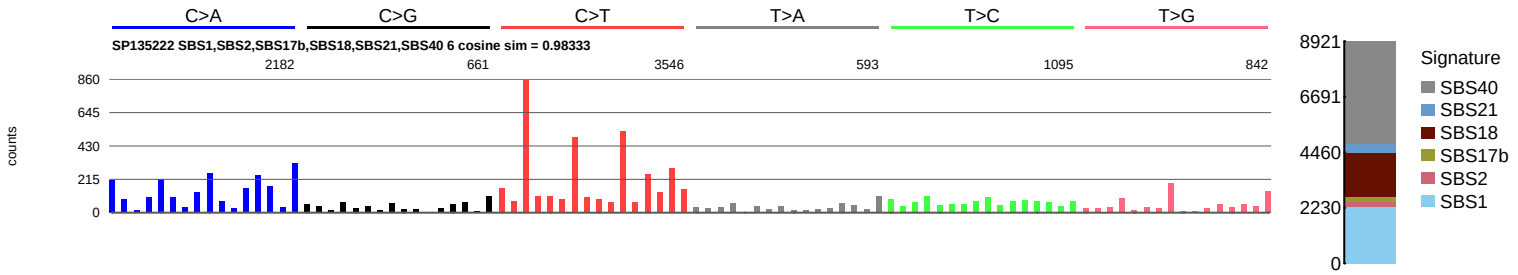
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98337
2149
650
3454
611
1207
851
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98337
2119
652
3573
629
1115
833
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1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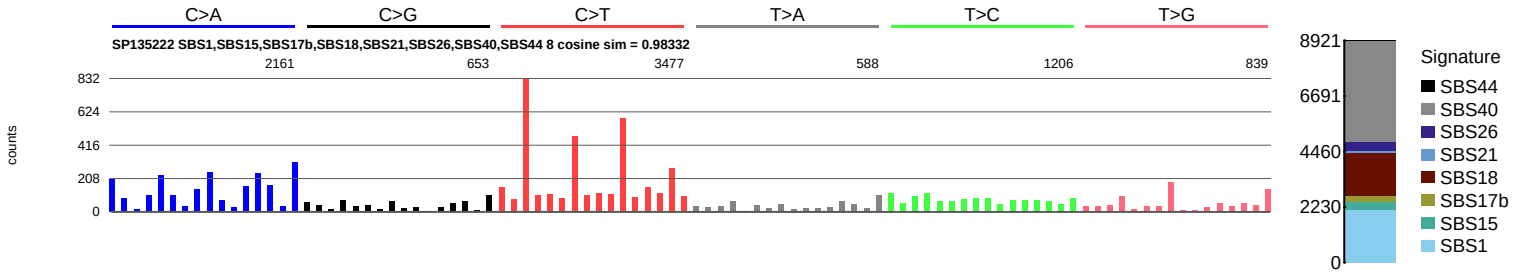
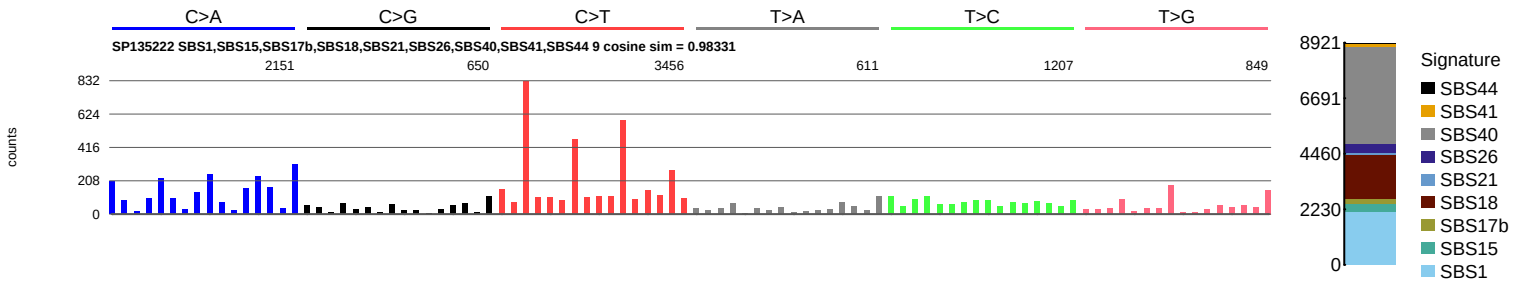
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98333
2182
661
3546
593
1095
842
counts
860
645
430
215
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98332
2161
653
3477
588
1206
839
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS1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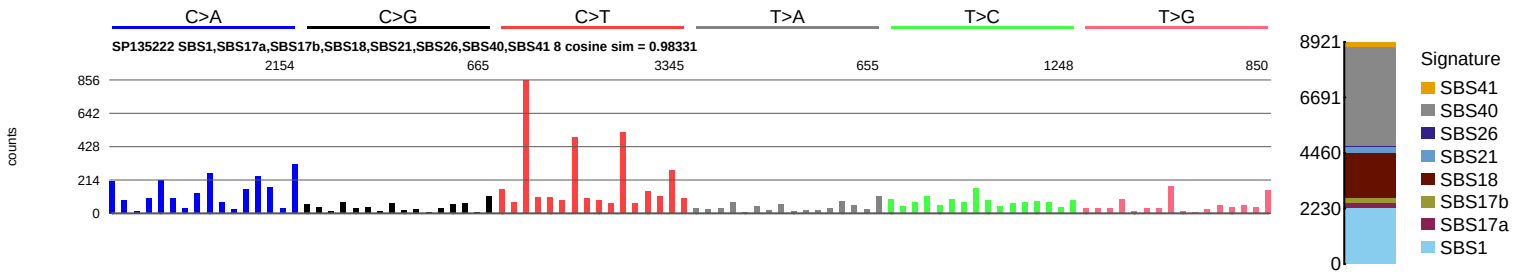
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98331
2151
650
3456
611
1207
849
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98331
2154
665
3345
655
1248
850
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
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SBS26
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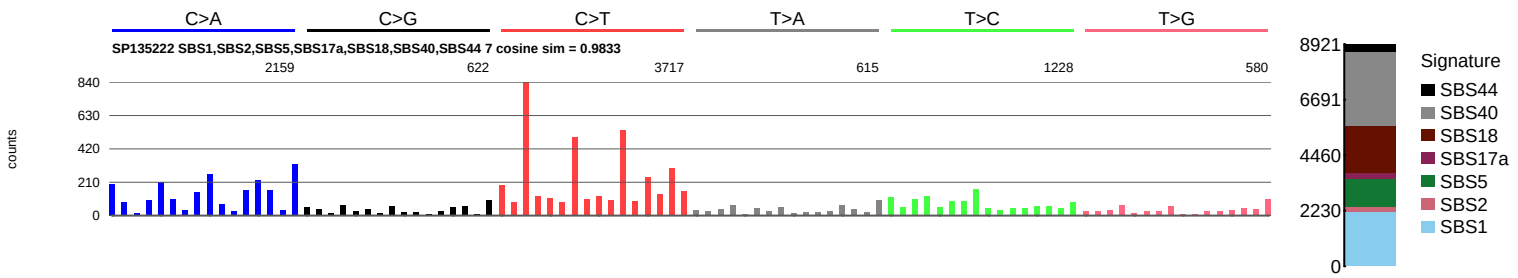
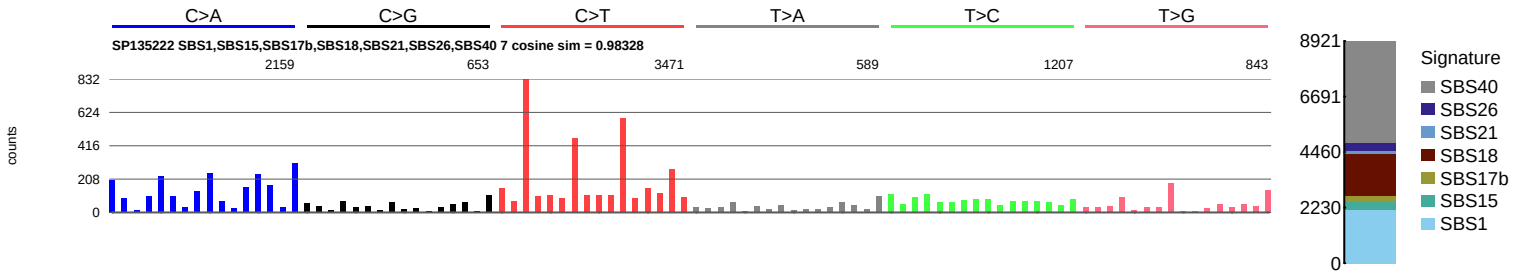
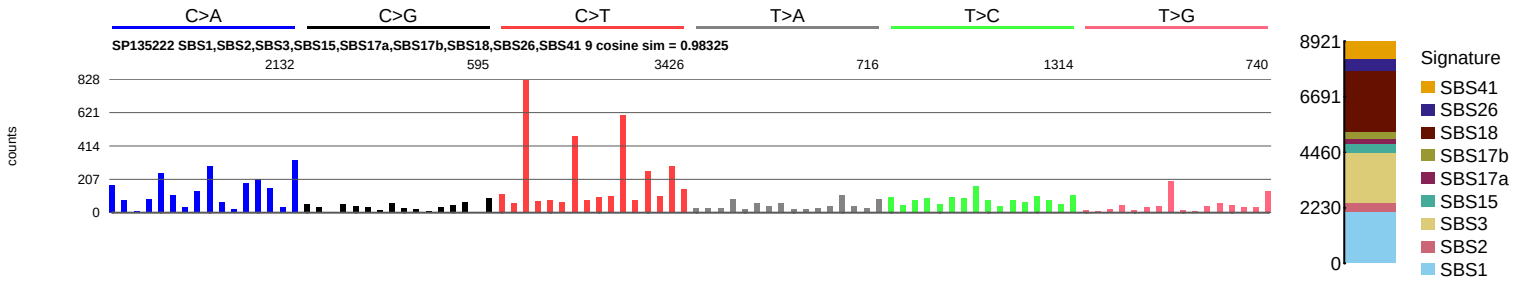
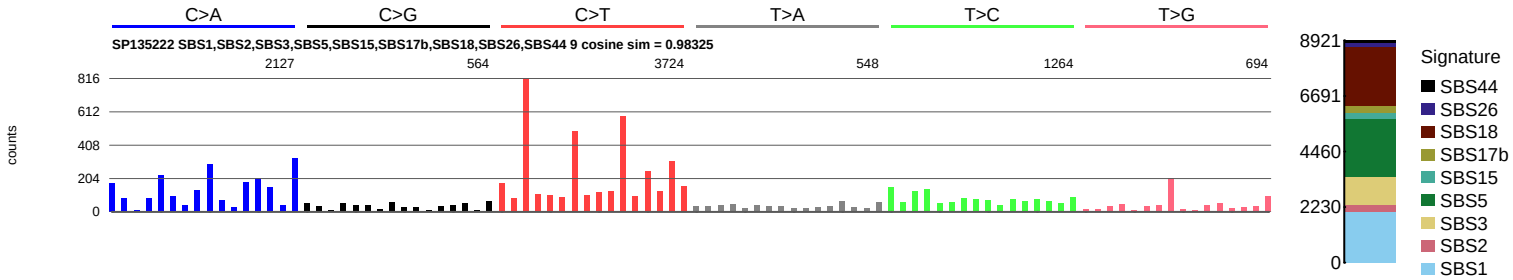
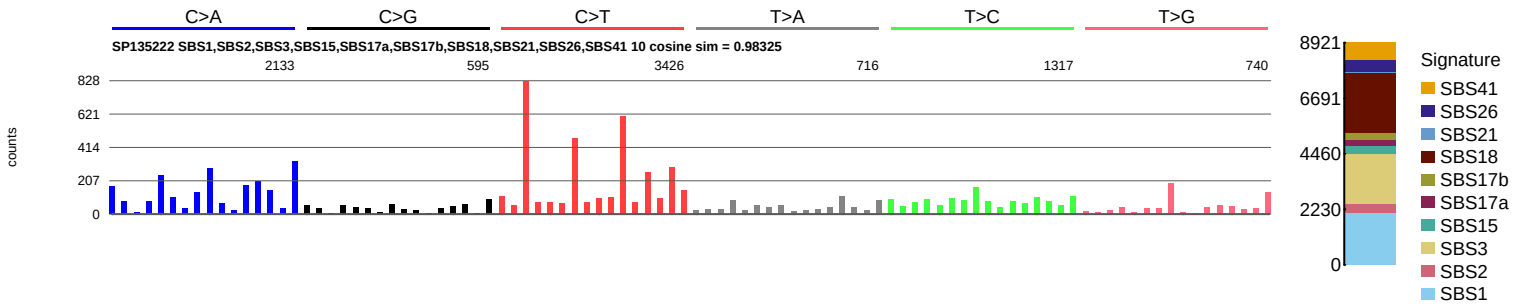
SBS18
SBS17b
SBS17a
SBS1

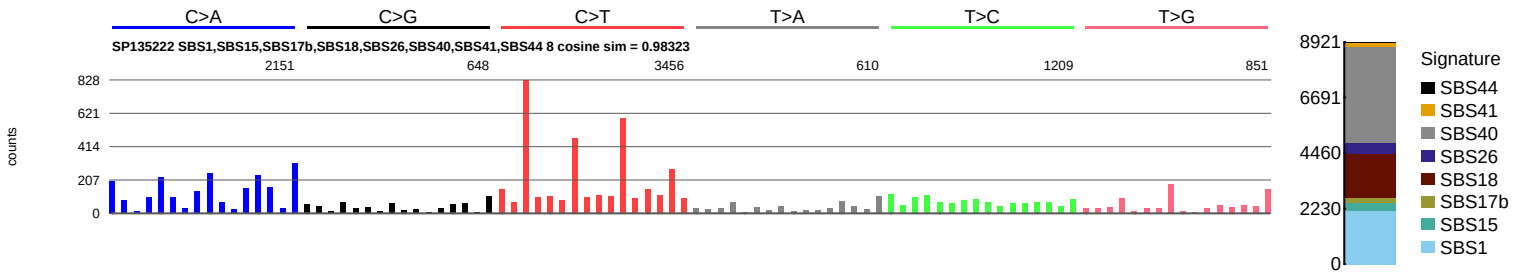

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS40,SBS44 7 cosine sim = 0.9833
2159
622
3717
615
1228
580
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS5
SBS2
SBS1

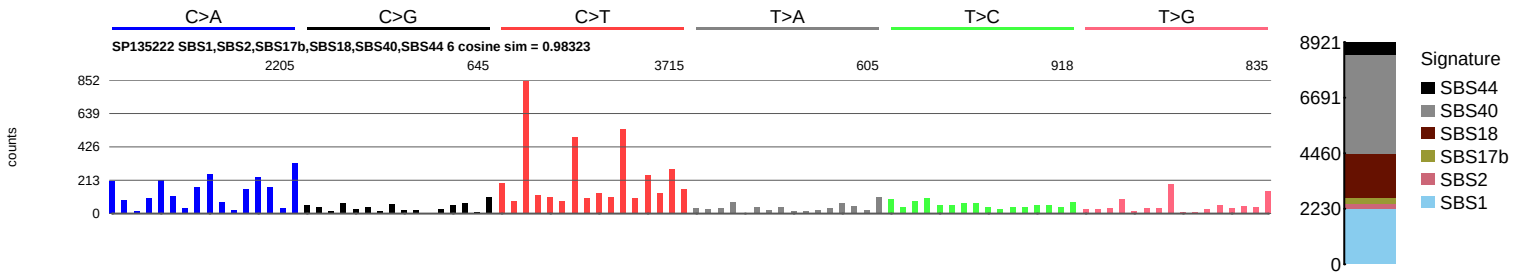

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17b,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98328
2159
653
3471
589
1207
843
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS15
SBS1

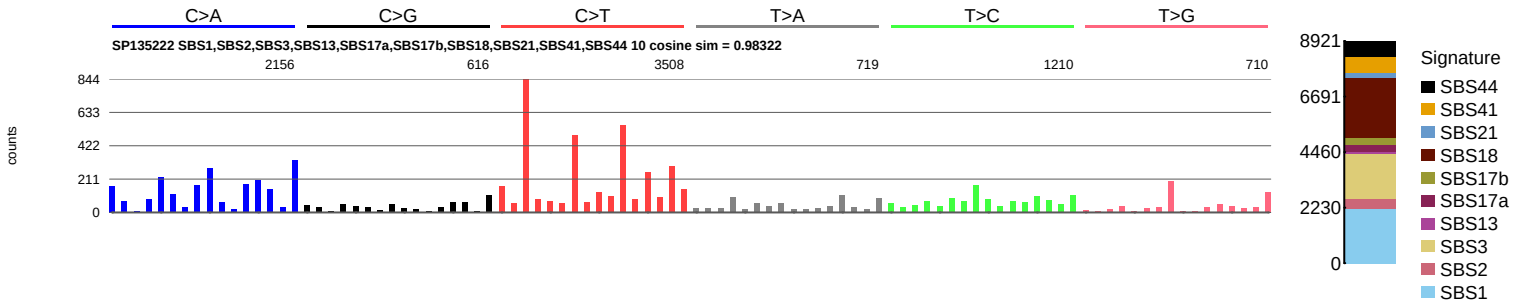

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98325
2132
595
3426
716
1314
740
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

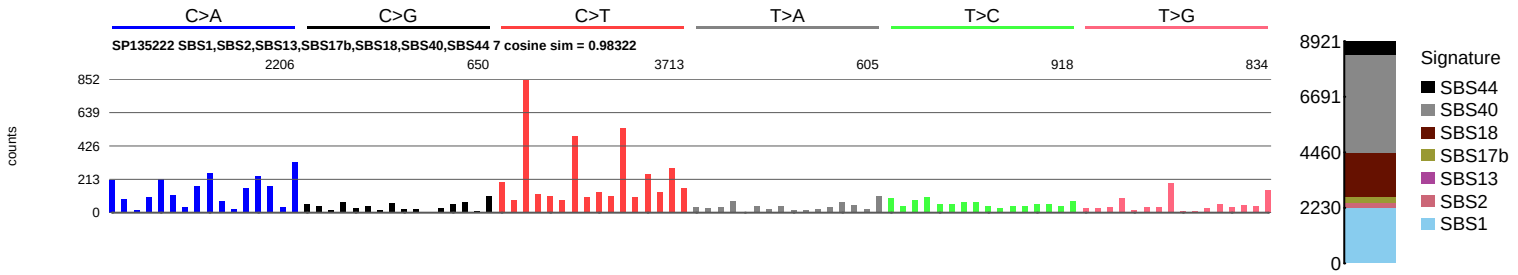

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98325
2127
564
3724
548
1264
694
816
612
408
204
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

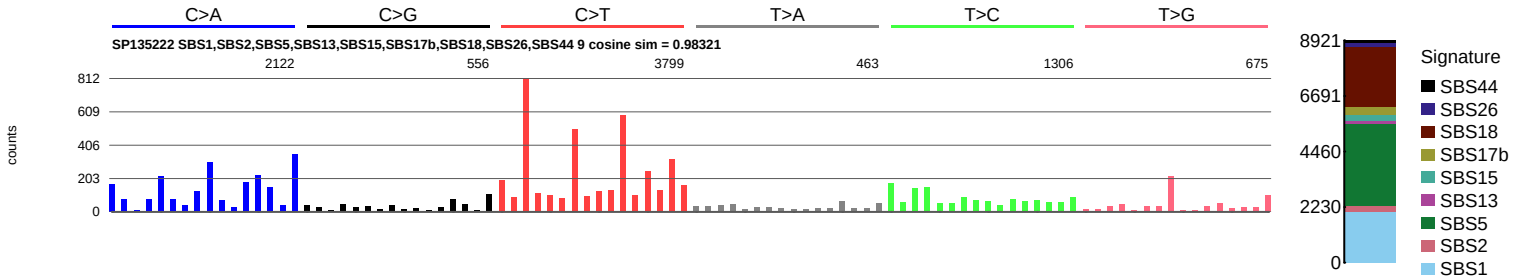

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98325
2133
595
3426
716
1317
740
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

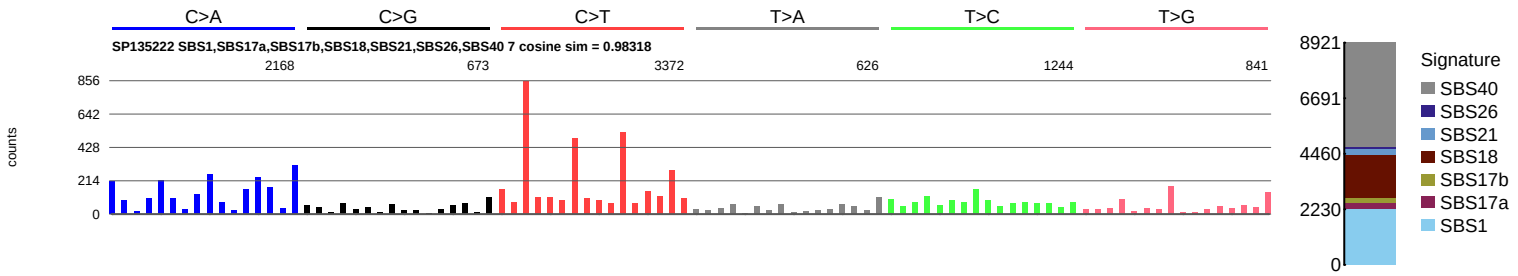

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98323
2151
648
3456
610
1209
851
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS1

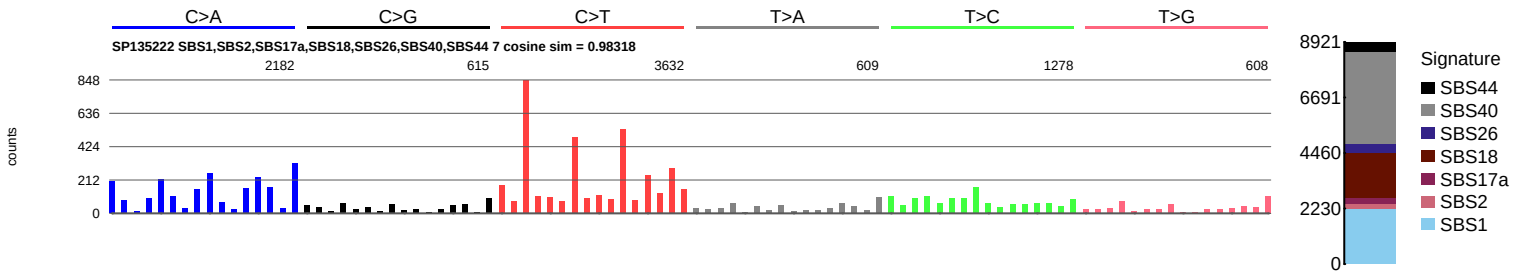

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98323
2205
645
3715
605
918
835
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98322
2156
616
3508
719
1210
710
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98322
2206
650
3713
605
918
834
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98321
2122
556
3799
463
1306
675
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98318
2168
673
3372
626
1244
841
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98318
2182
615
3632
609
1278
608
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
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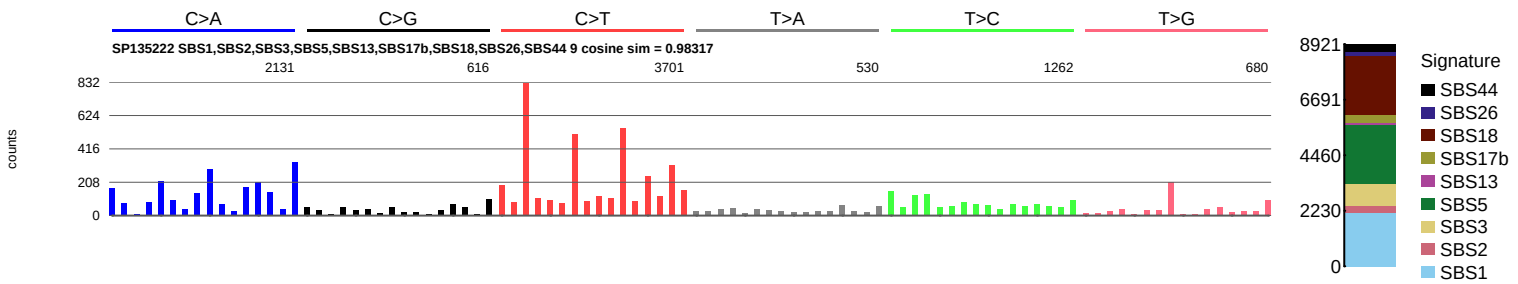
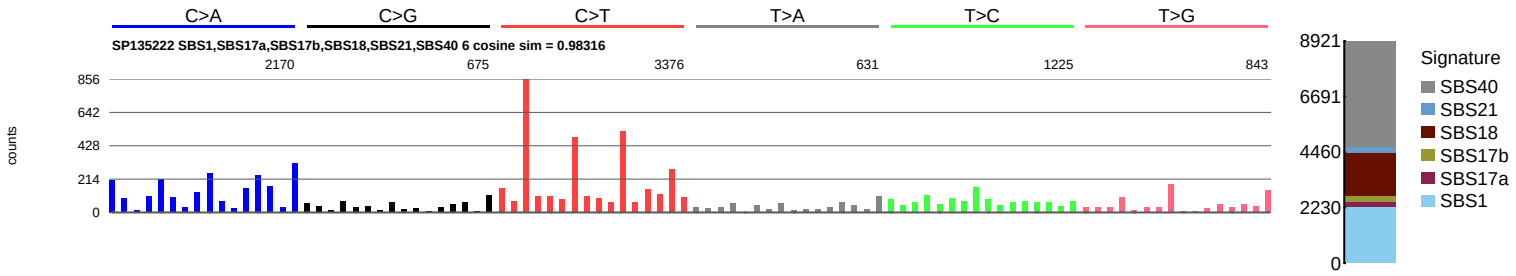
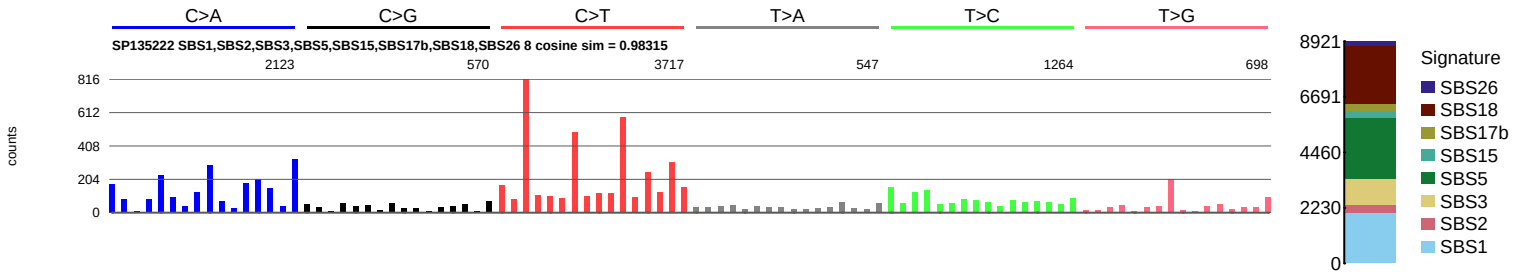
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98317
2131
616
3701
530
1262
680
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98316
2170
675
3376
631
1225
843
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS26 8 cosine sim = 0.98315
2123
570
3717
547
1264
698
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS26
SBS18
SBS17b
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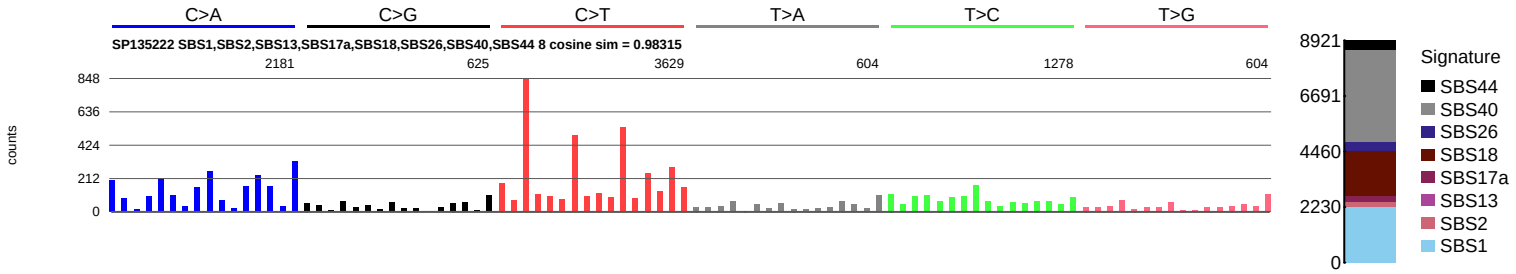
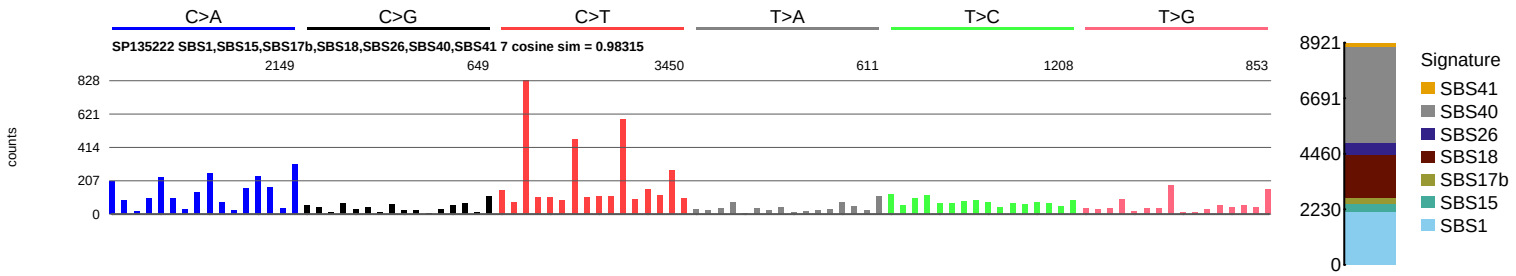
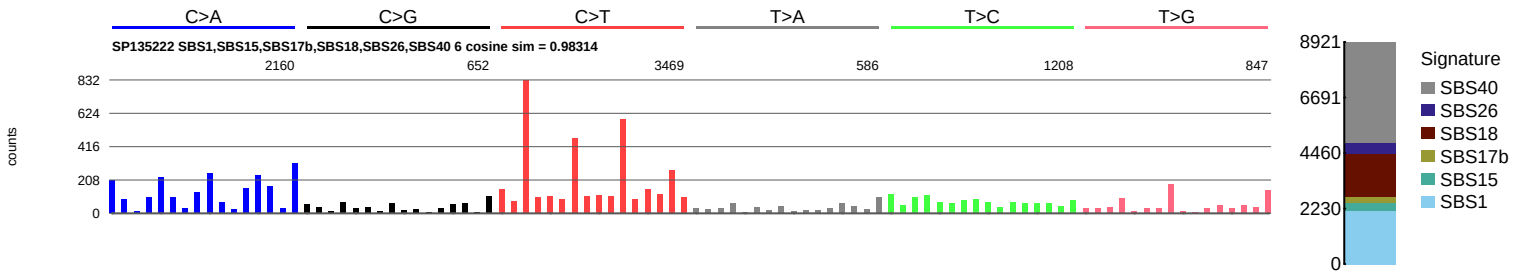
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98315
2181
625
3629
604
1278
604
counts
848
636
424
212
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98315
2149
649
3450
611
1208
853
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS1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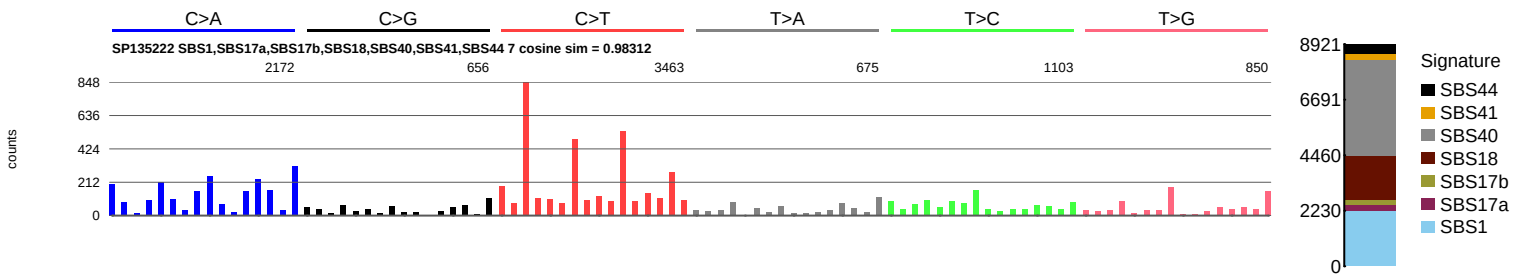
SP135222 SBS1,SBS15,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98314
2160
652
3469
586
1208
847
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS1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SP135222 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98312
2172
656
3463
675
1103
850
counts
848
636
424
212
0
8921
6691
4460
2230
0
Signature
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SBS41
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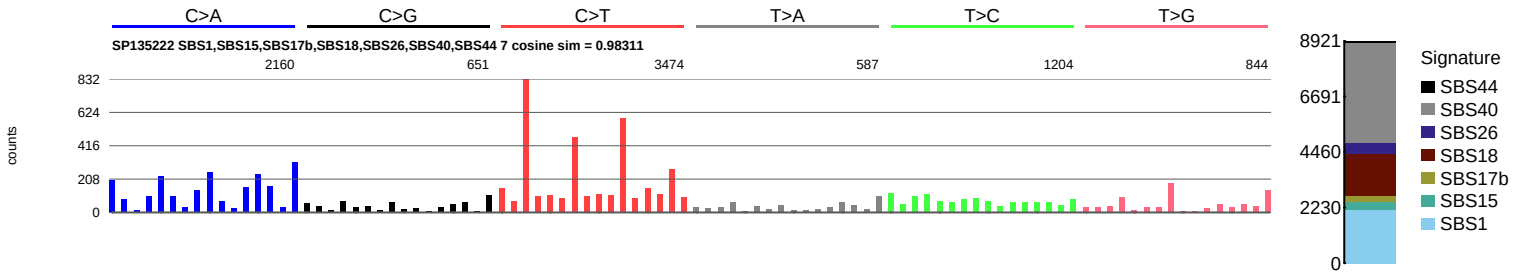
SBS17b
SBS17a
SBS1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C>G
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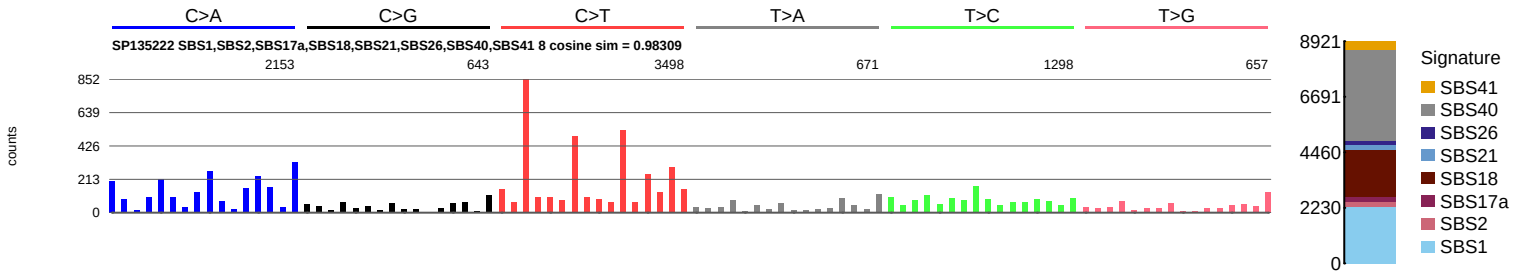
SP135222 SBS1,SBS15,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98311
2160
651
3474
587
1204
844
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS15
SBS1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C>G
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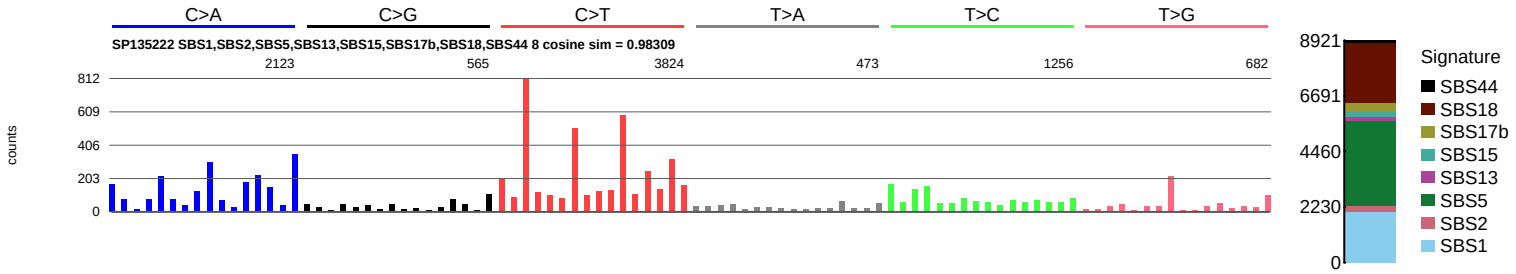
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98309
2153
643
3498
671
1298
657
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS2
SBS1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C>T
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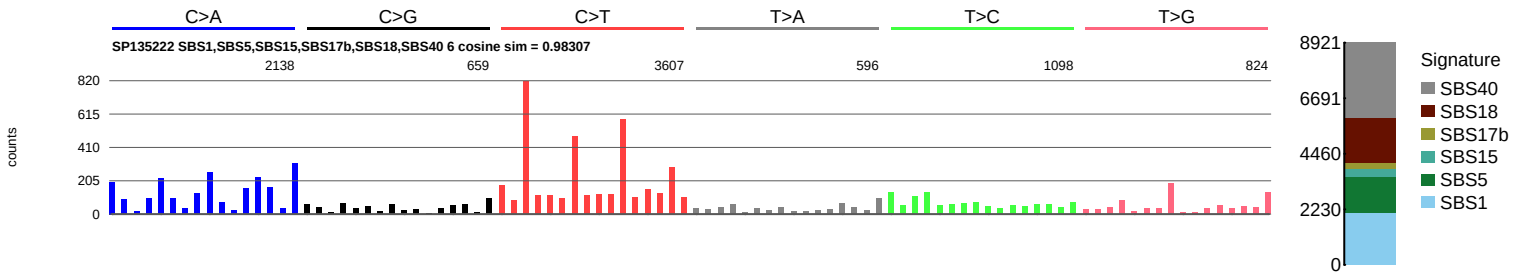
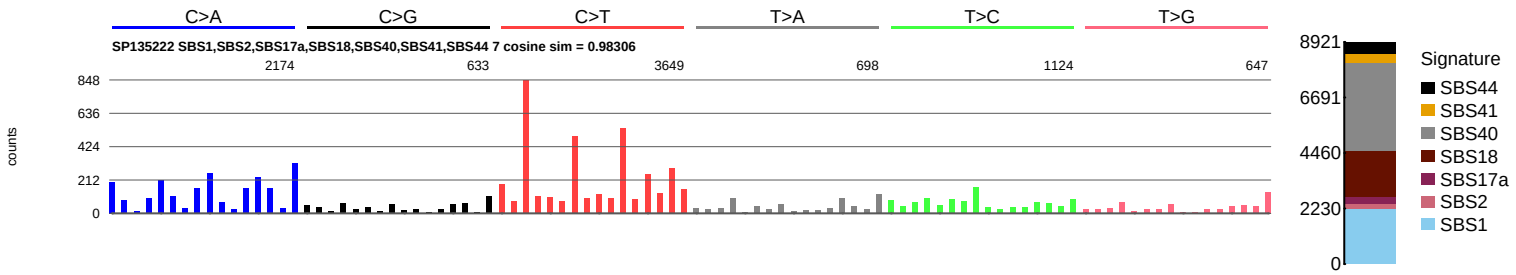
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS44 8 cosine sim = 0.98309
2123
565
3824
473
1256
682
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17b,SBS18,SBS40 6 cosine sim = 0.98307
2138
659
3607
596
1098
824
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS5
SBS1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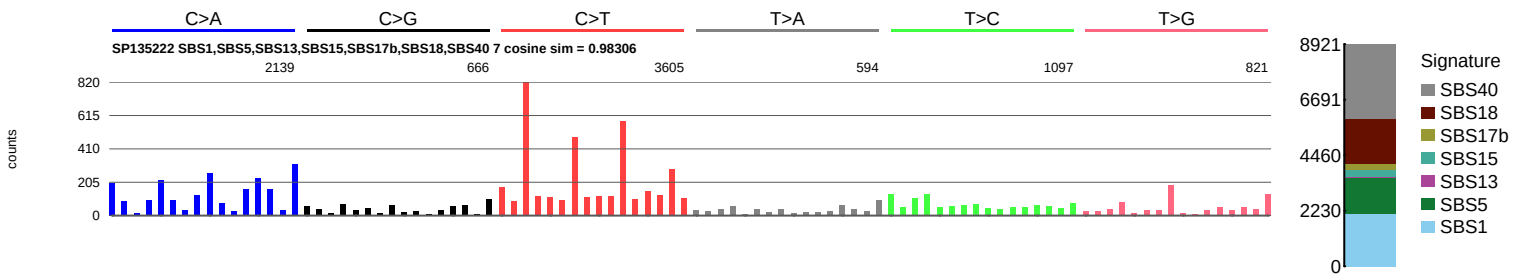
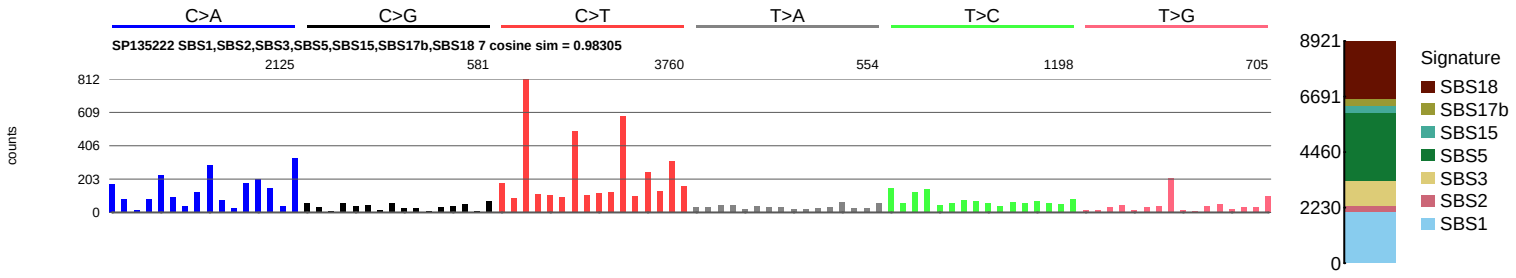
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98306
2174
633
3649
698
1124
647
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS40 7 cosine sim = 0.98306
2139
666
3605
594
1097
821
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS1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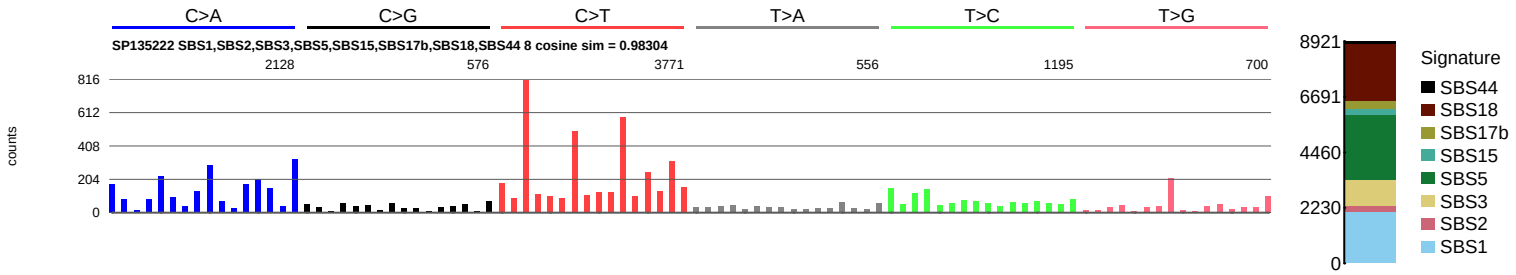
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18 7 cosine sim = 0.98305
2125
581
3760
554
1198
705
counts
812
609
406
203
0
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1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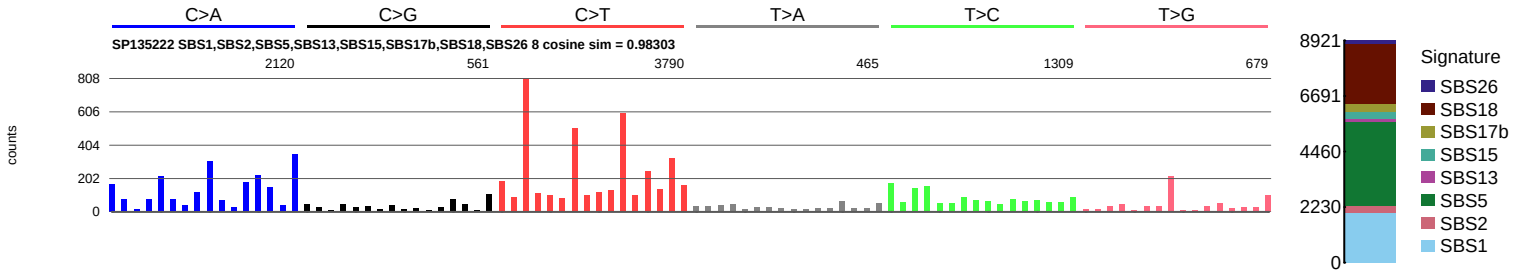
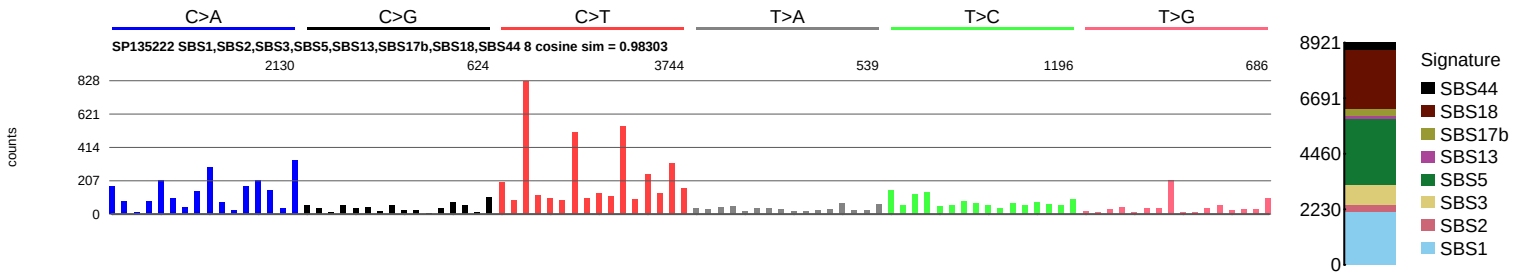
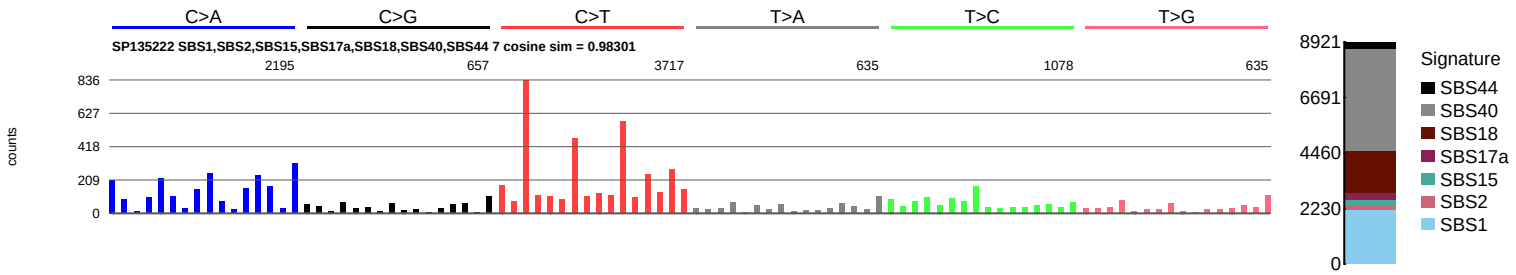
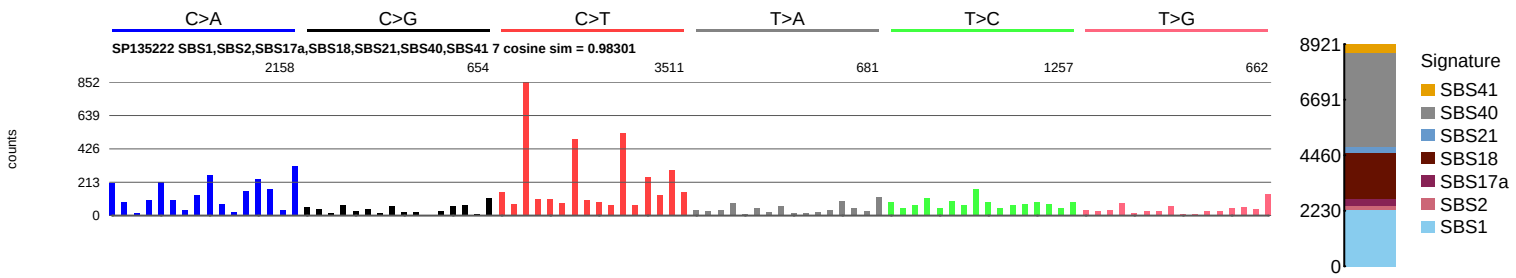
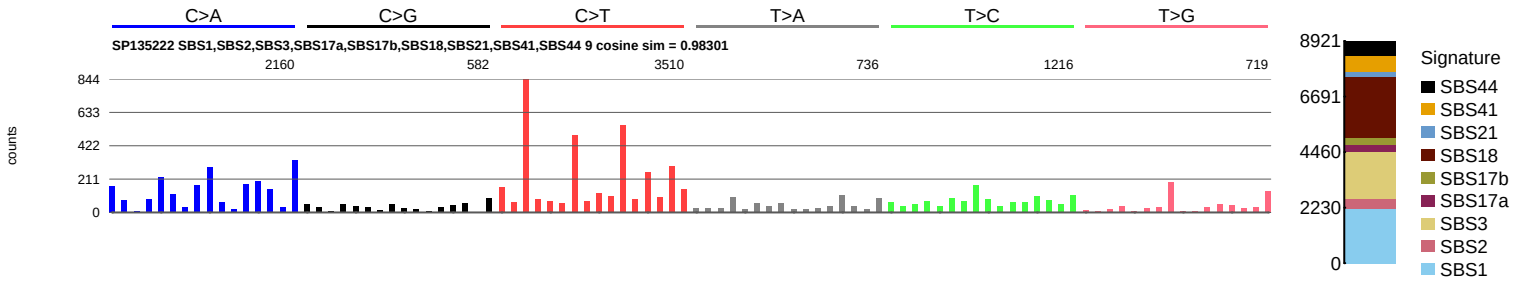
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17b,SBS18,SBS44 8 cosine sim = 0.98304
2128
576
3771
556
1195
700
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS44
SBS18
SBS17b
SBS15
SBS5
SBS3
SBS2
SBS1

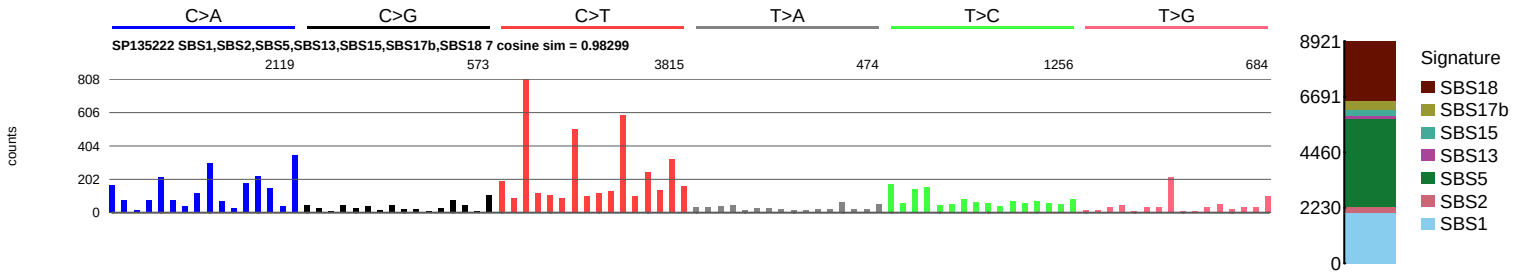

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18,SBS26 8 cosine sim = 0.98303
2120
561
3790
465
1309
679
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS44 8 cosine sim = 0.98303
2130
624
3744
539
1196
686
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS44
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS40,SBS44 7 cosine sim = 0.98301
2195
657
3717
635
1078
635
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98301
2158
654
3511
681
1257
662
counts
852
639
426
213
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98301
2160
582
3510
736
1216
719
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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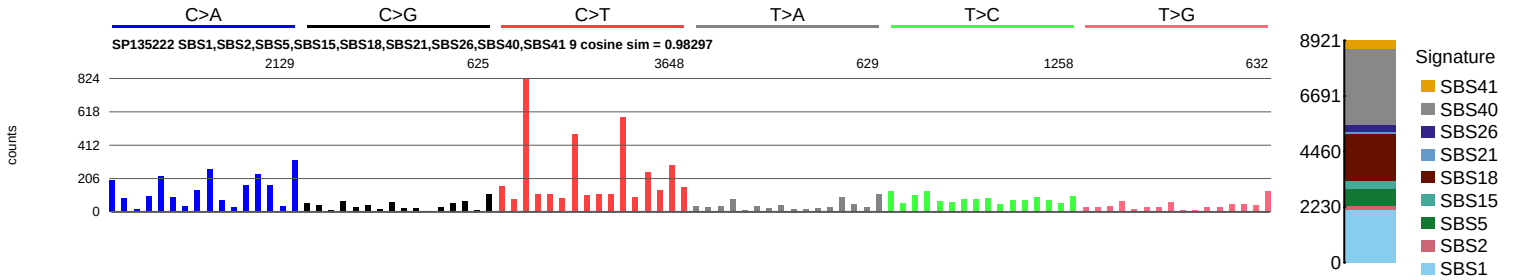
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17b,SBS18 7 cosine sim = 0.98299
2119
573
3815
474
1256
684
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS15
SBS13
SBS5
SBS2
SBS1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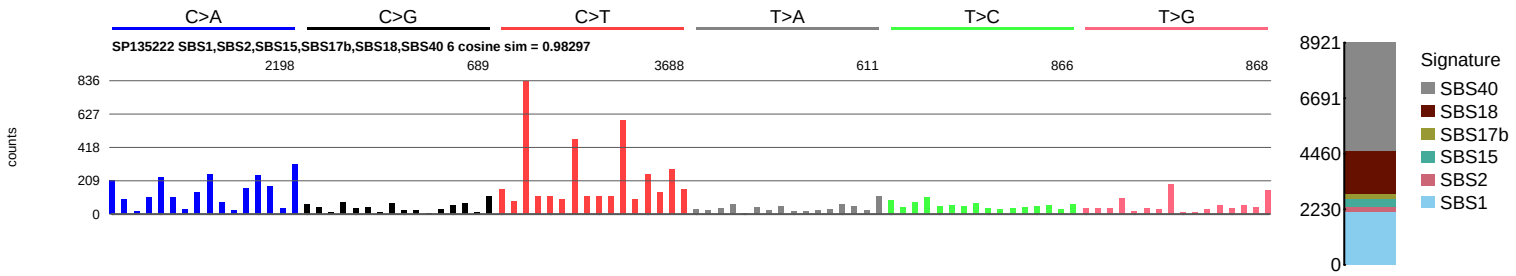
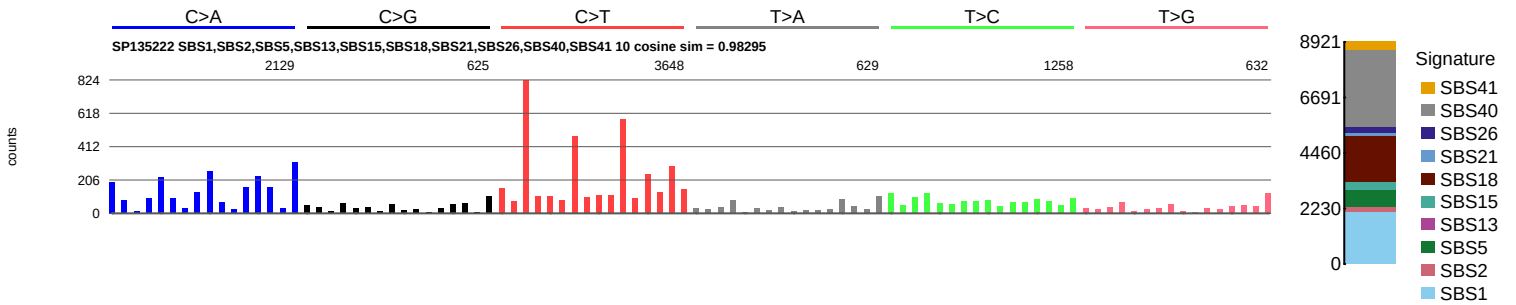
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98297
2129
625
3648
629
1258
632
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17b,SBS18,SBS40 6 cosine sim = 0.98297
2198
689
3688
611
866
868
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98295
2129
625
3648
629
1258
632
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
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SBS21
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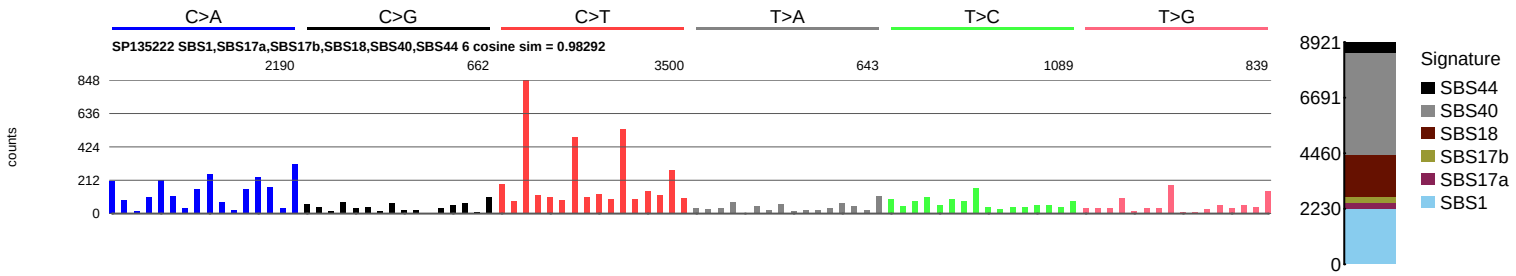
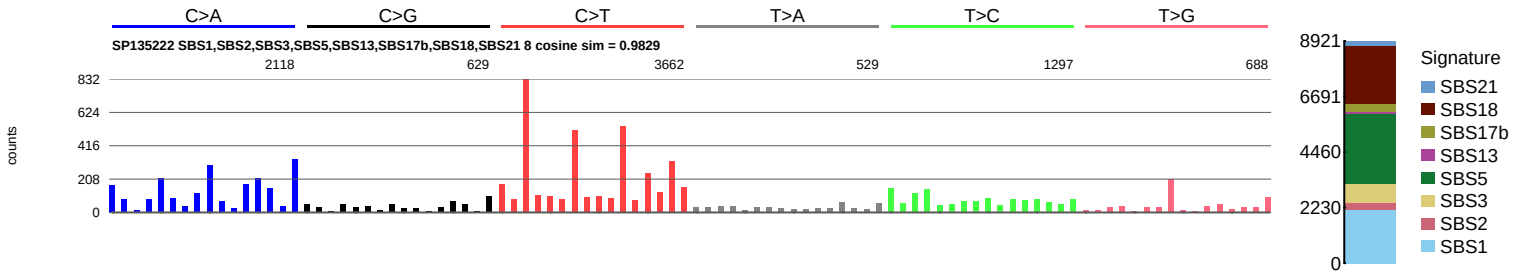
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98292
2190
662
3500
643
1089
839
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS21 8 cosine sim = 0.9829
2118
629
3662
529
1297
688
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
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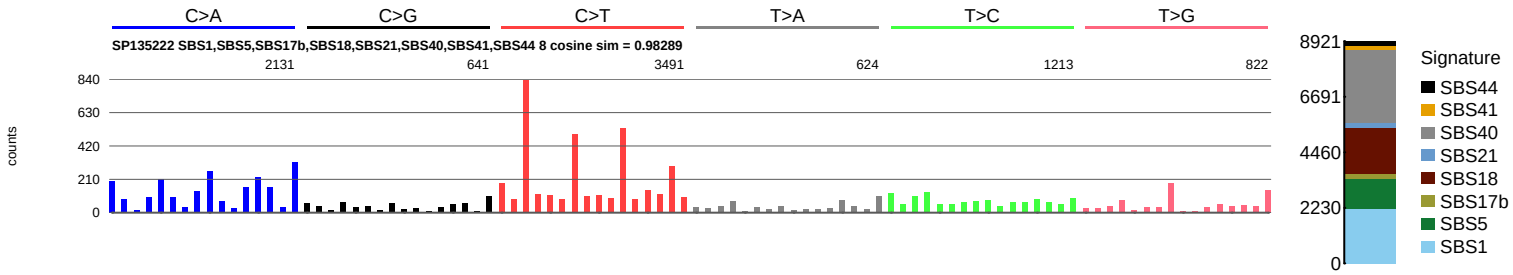
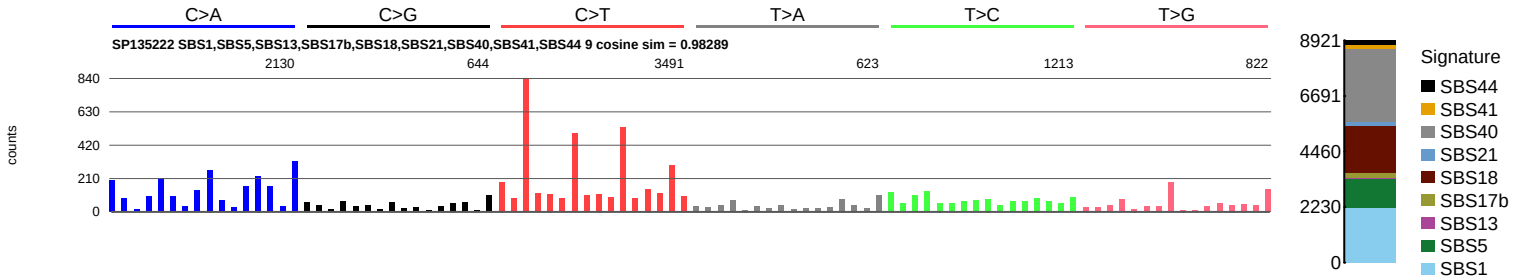
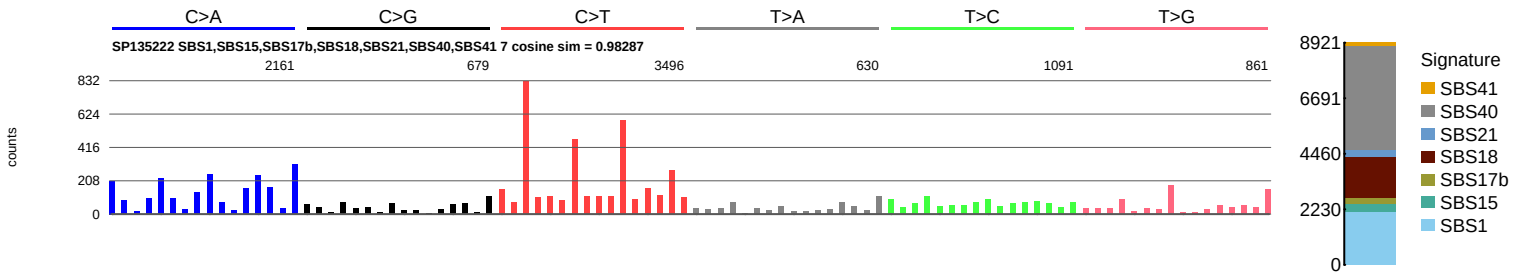
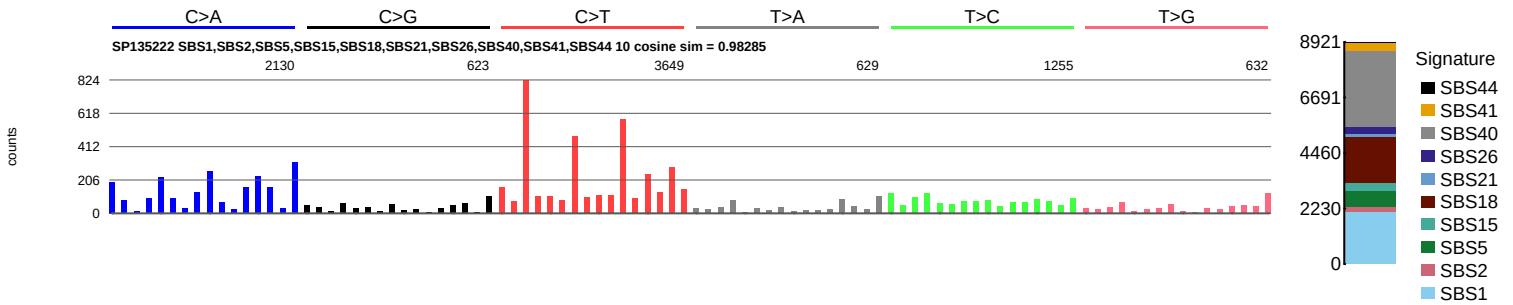
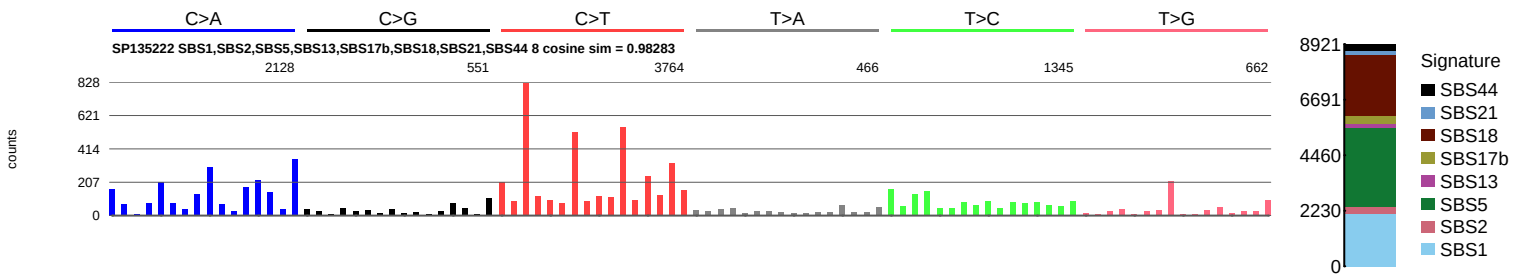
SBS2
SBS1

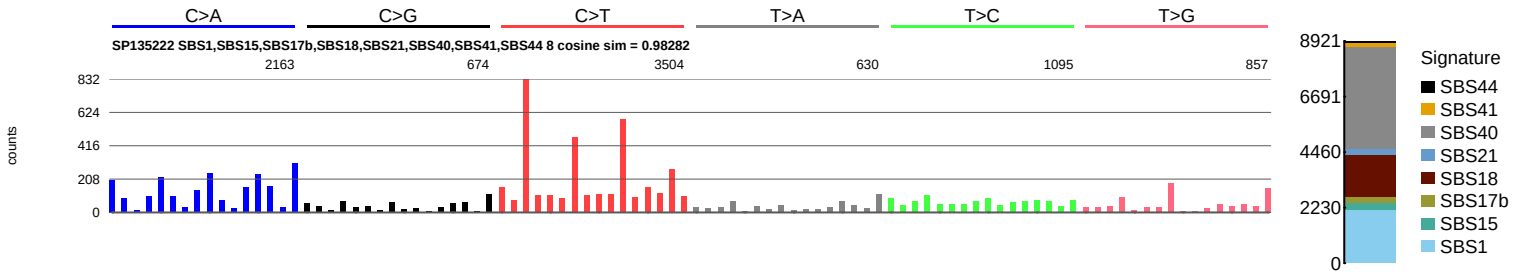

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98289
2131
641
3491
624
1213
822
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS1

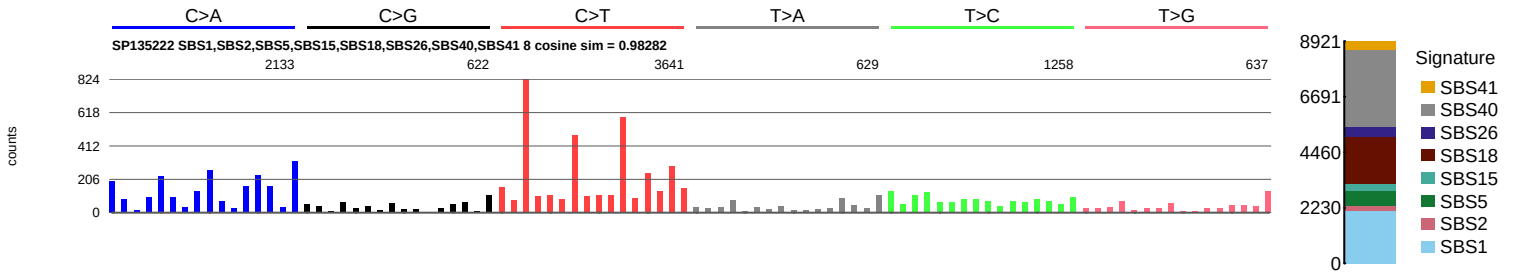

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98289
2130
644
3491
623
1213
822
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS1

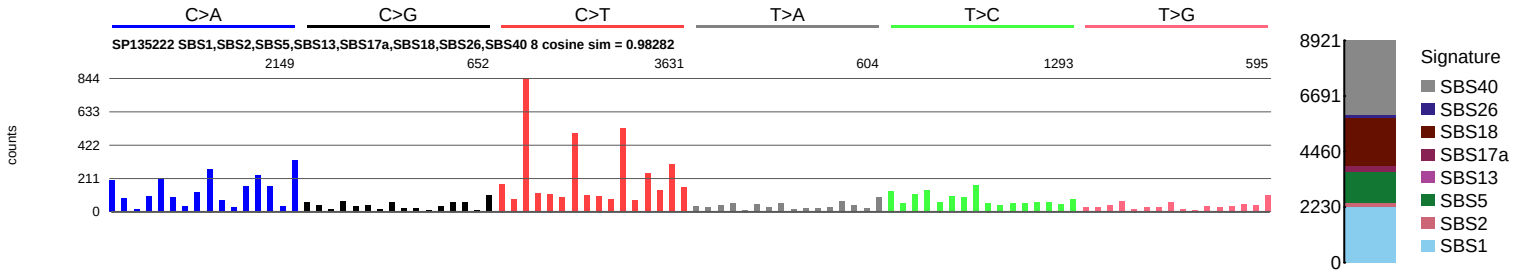

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98287
2161
679
3496
630
1091
861
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS1

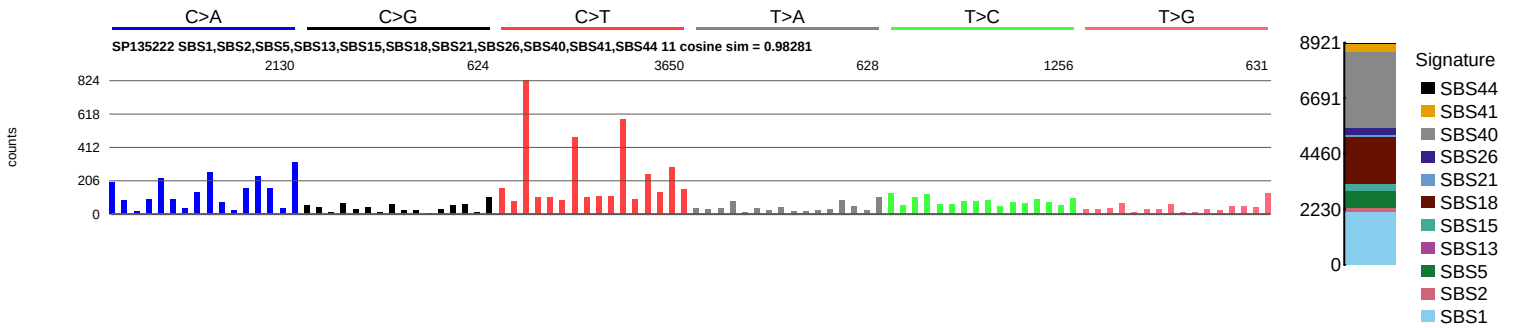

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98285
2130
623
3649
629
1255
632
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98283
2128
551
3764
466
1345
662
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98282
2163
674
3504
630
1095
857
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98282
2133
622
3641
629
1258
637
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS26,SBS40 8 cosine sim = 0.98282
2149
652
3631
604
1293
595
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98281
2130
624
3650
628
1256
631
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
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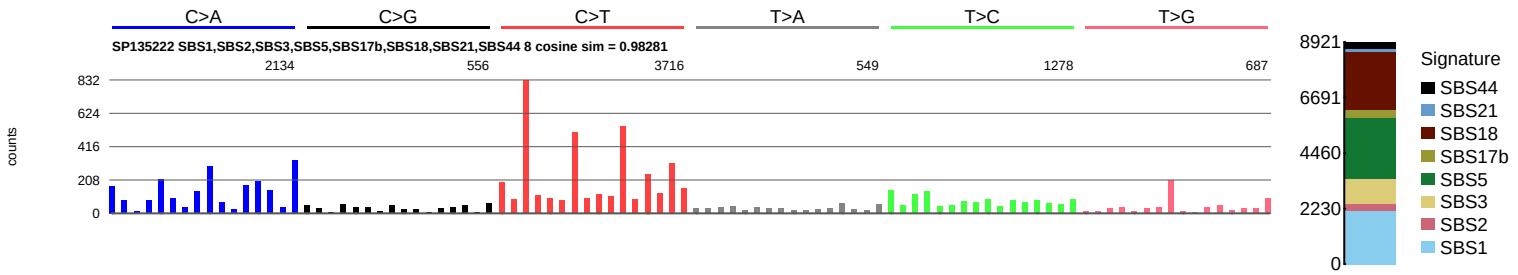
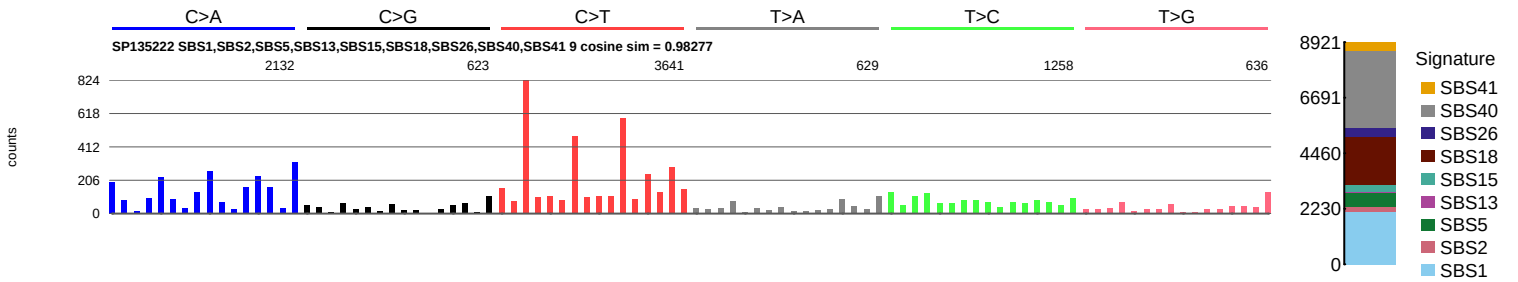
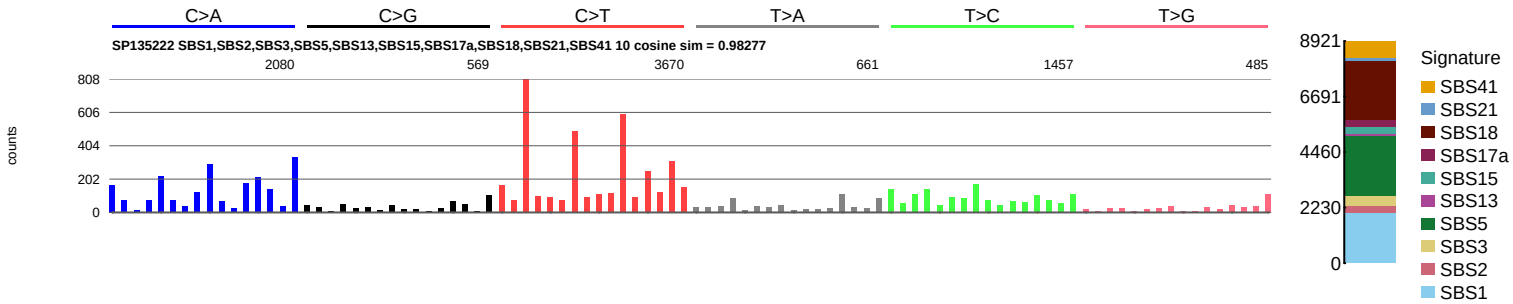
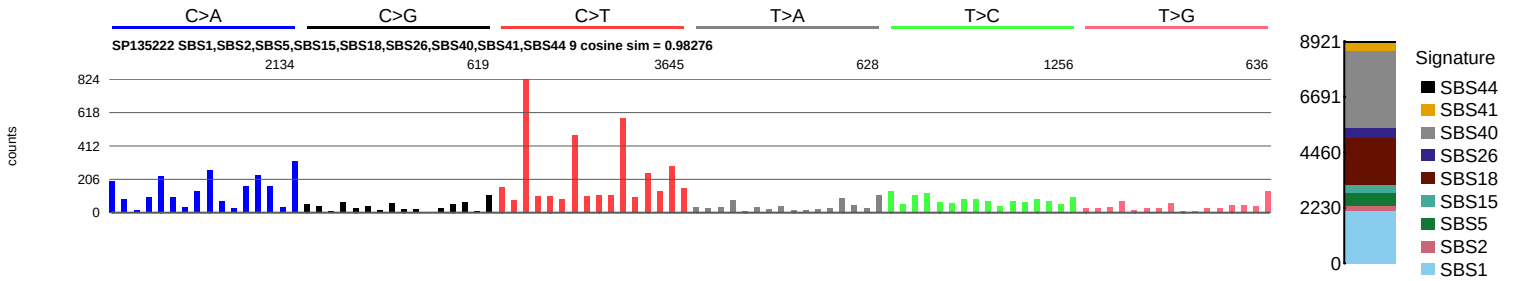
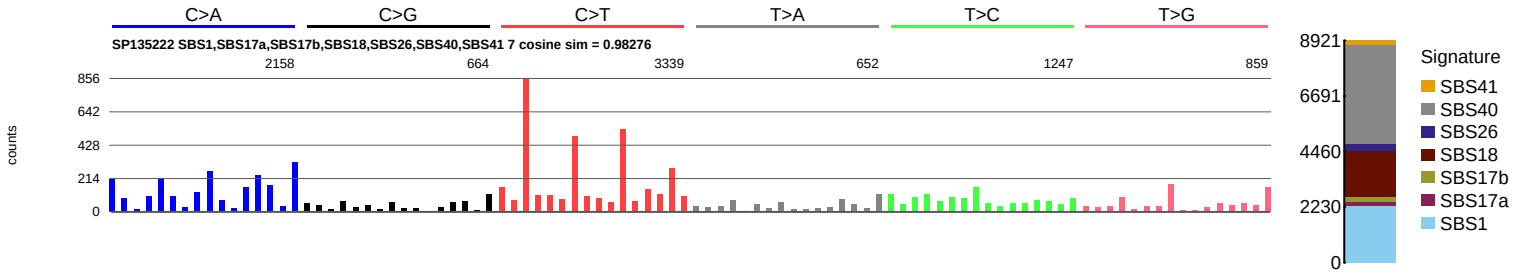
SBS2
SBS1

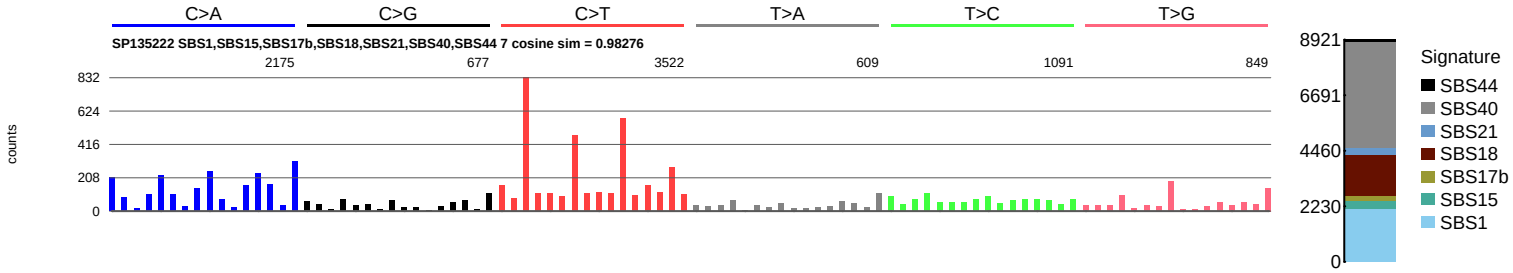

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98281
2134
556
3716
549
1278
687
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

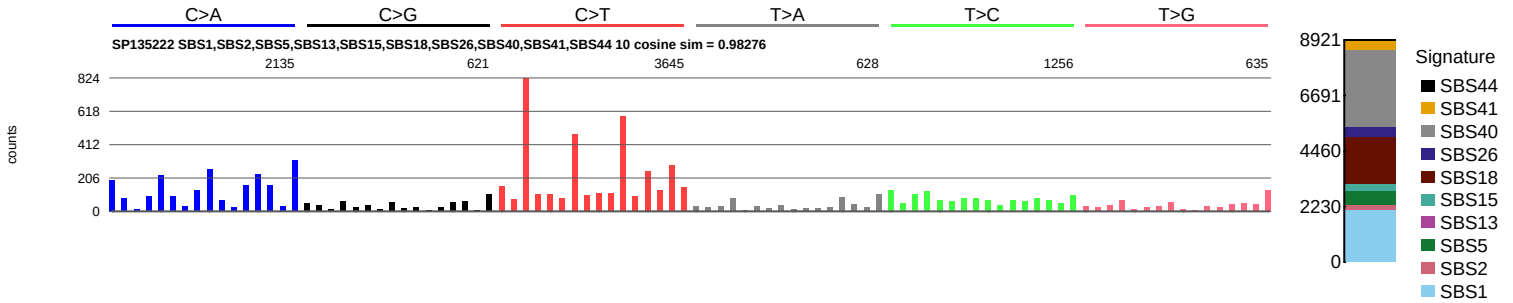

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98277
2132
623
3641
629
1258
636
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS41 10 cosine sim = 0.98277
2080
569
3670
661
1457
485
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98276
2134
619
3645
628
1256
636
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98276
2158
664
3339
652
1247
859
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98276
2175
677
3522
609
1091
849
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98276
2135
621
3645
628
1256
635
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
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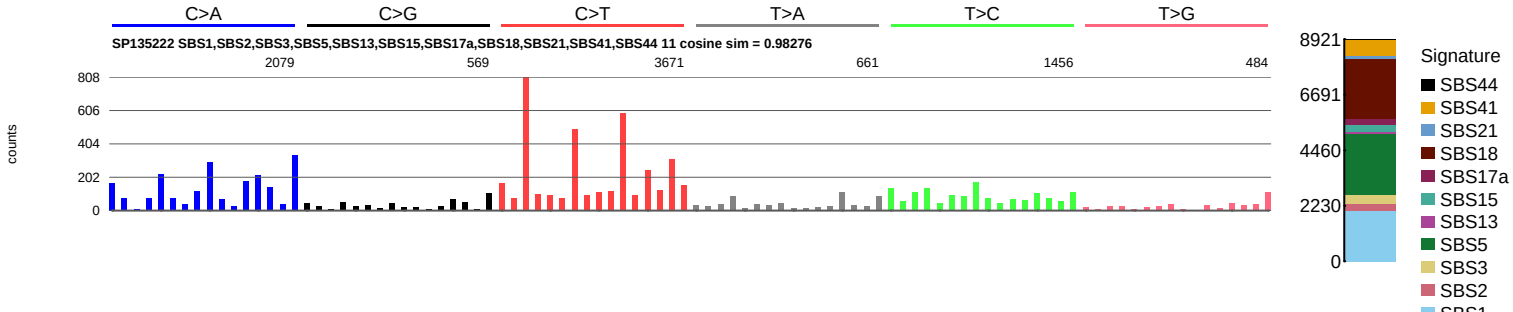
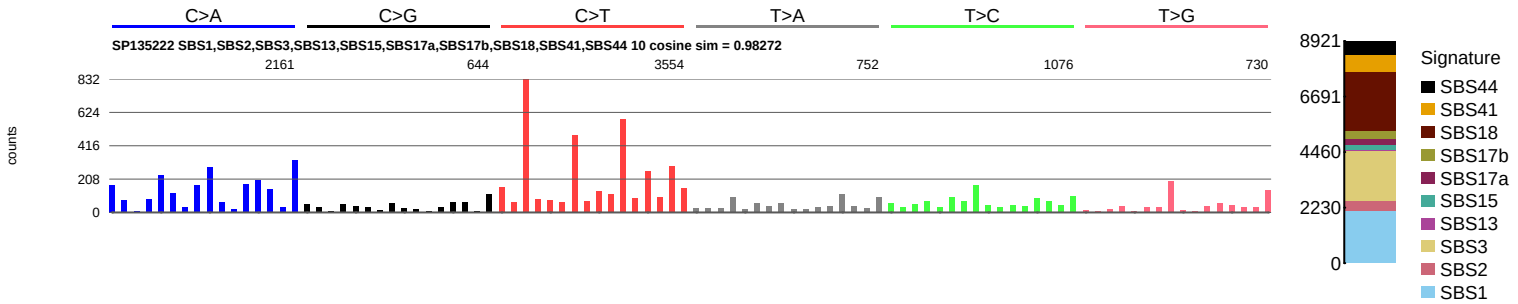
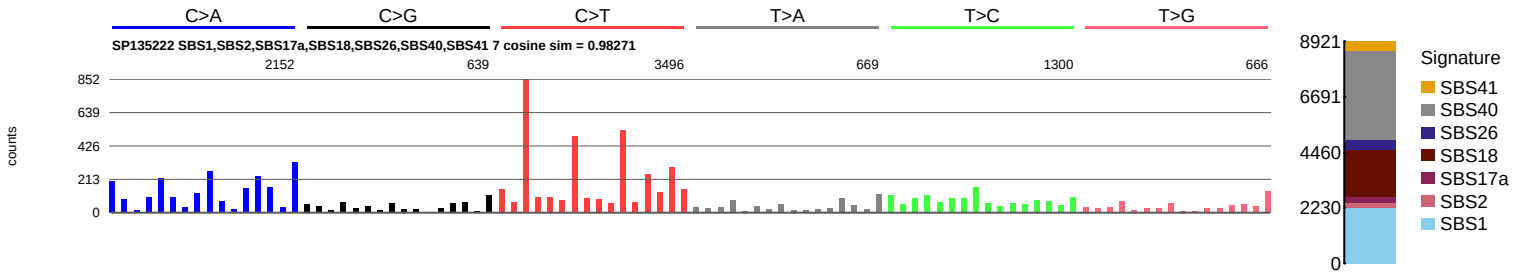
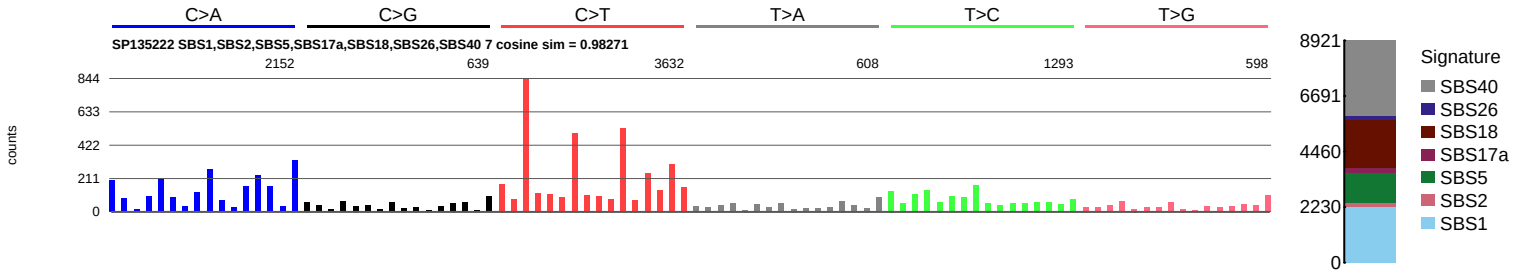
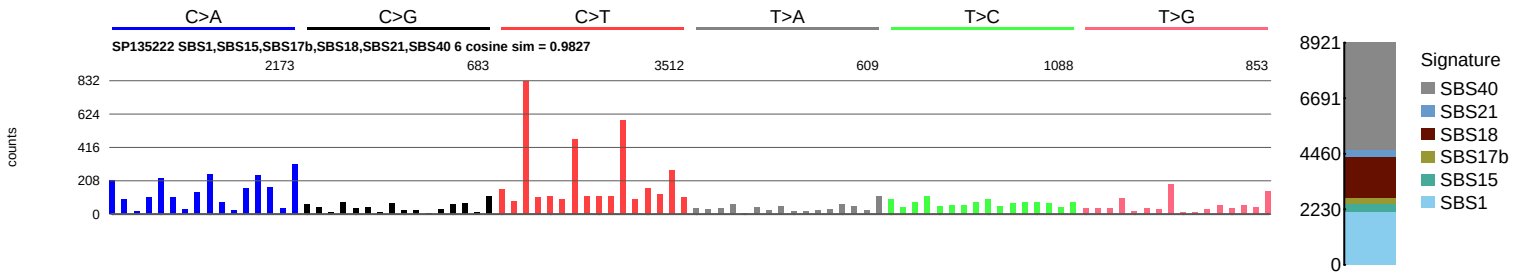
SBS5
SBS2
SBS1

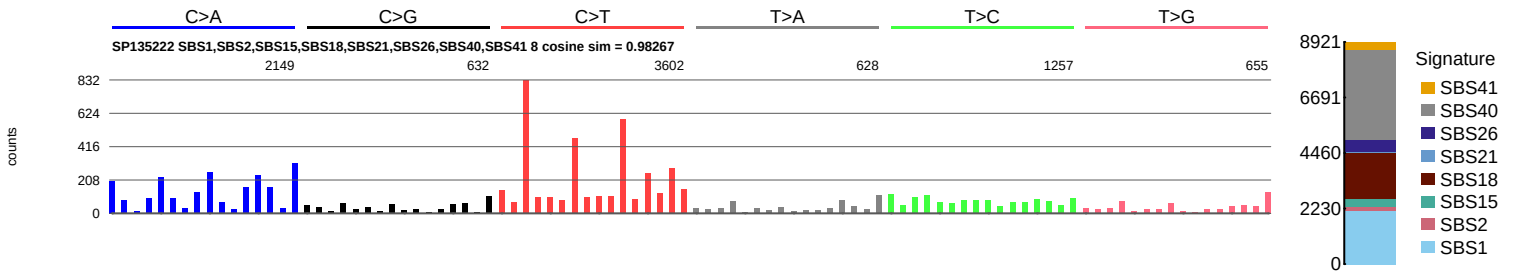

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98276
2079
569
3671
661
1456
484
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

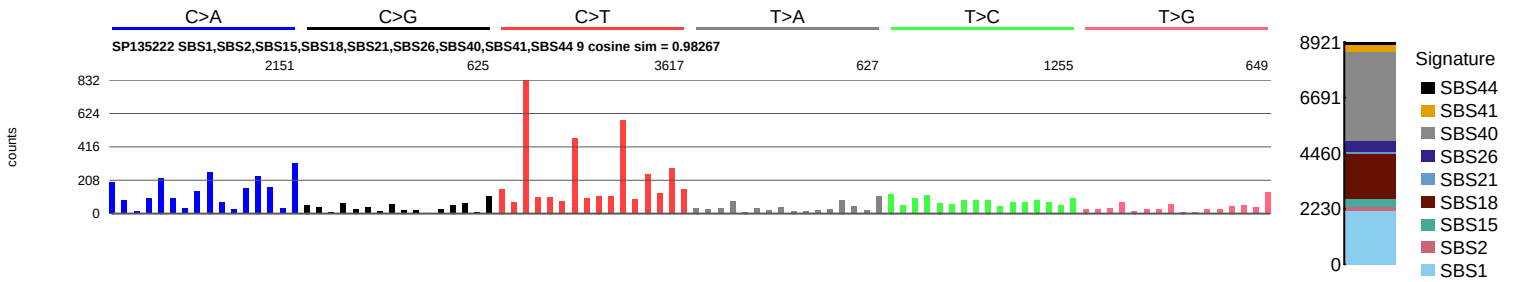

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98272
2161
644
3554
752
1076
730
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

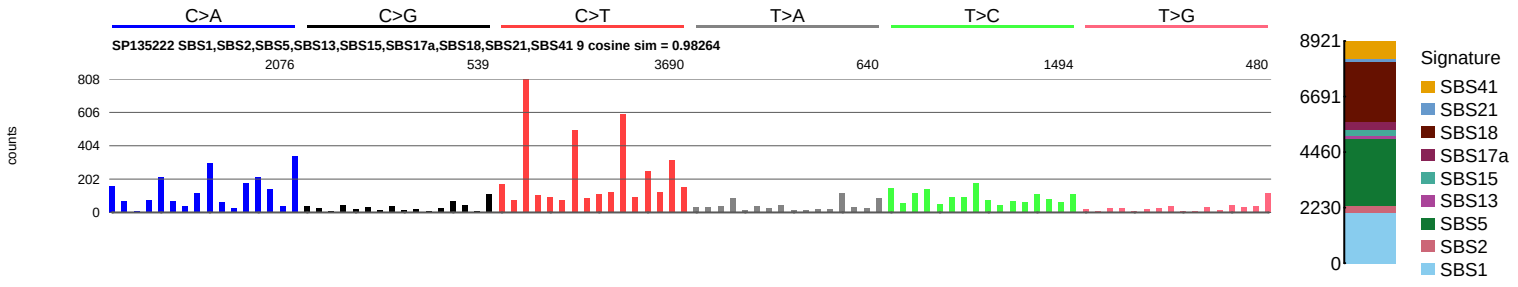

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98271
2152
639
3496
669
1300
666
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS26,SBS40 7 cosine sim = 0.98271
2152
639
3632
608
1293
598
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.9827
2173
683
3512
609
1088
853
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98267
2149
632
3602
628
1257
655
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98267
2151
625
3617
627
1255
649
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS41 9 cosine sim = 0.98264
2076
539
3690
640
1494
480
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17a
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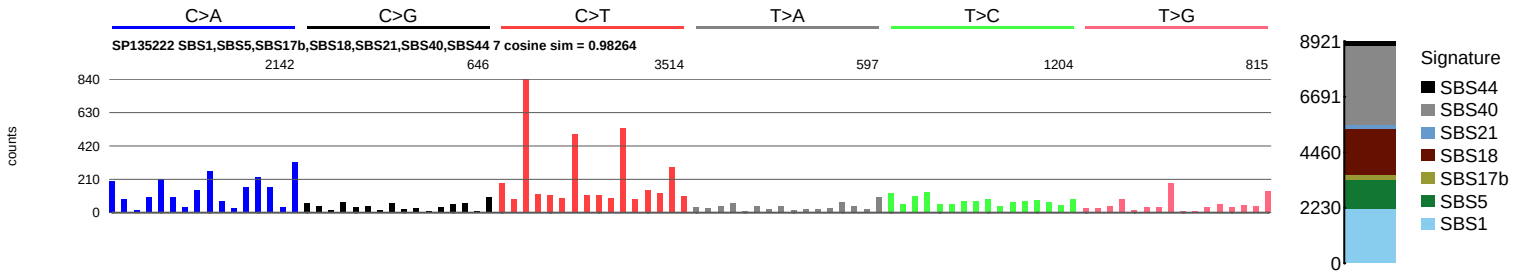
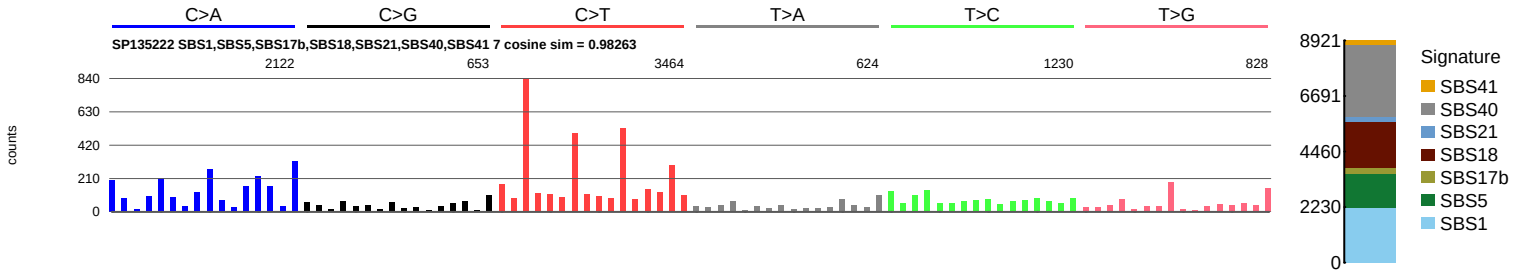
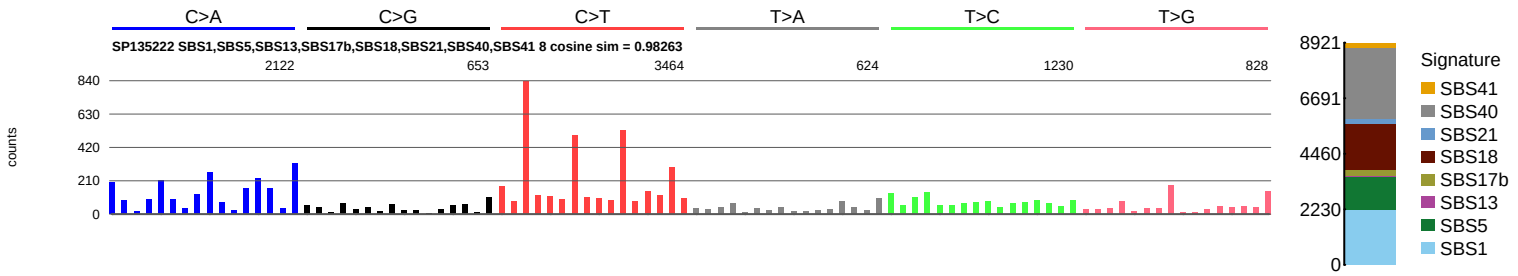
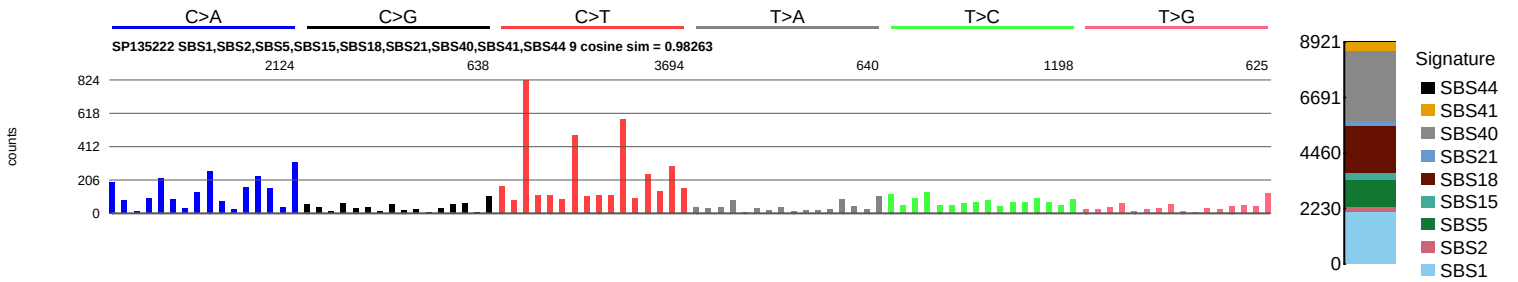
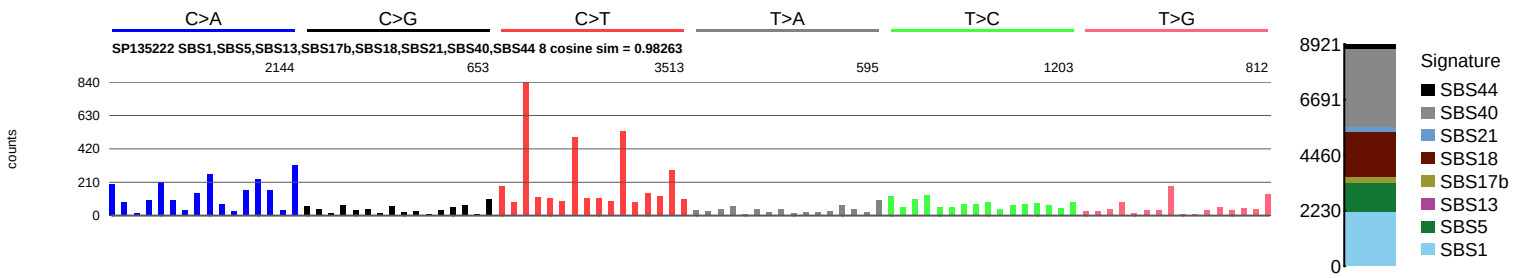
SBS2
SBS1

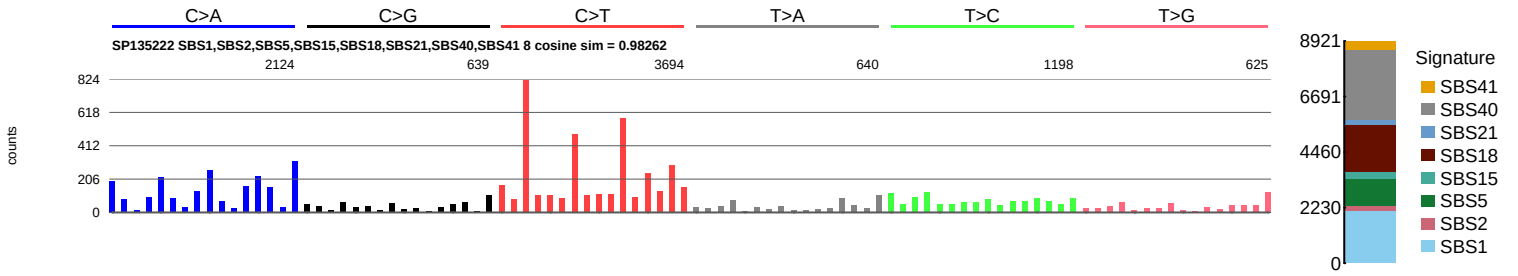

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98264
2142
646
3514
597
1204
815
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS1

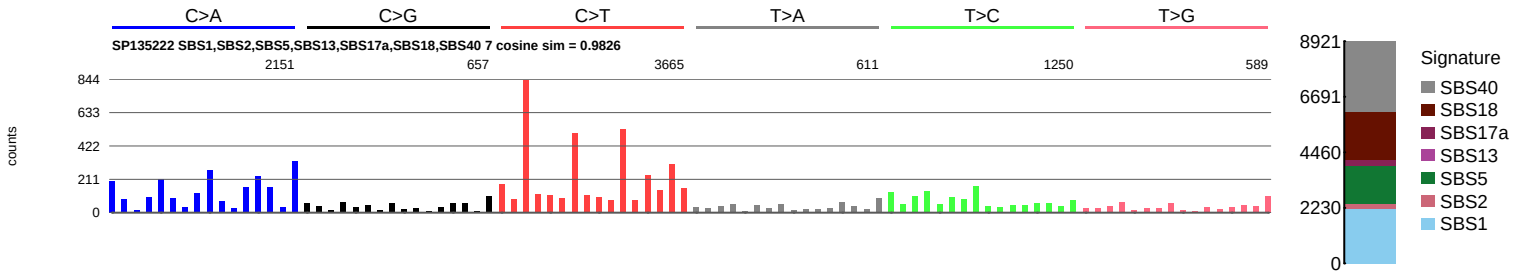

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98263
2122
653
3464
624
1230
828
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS1

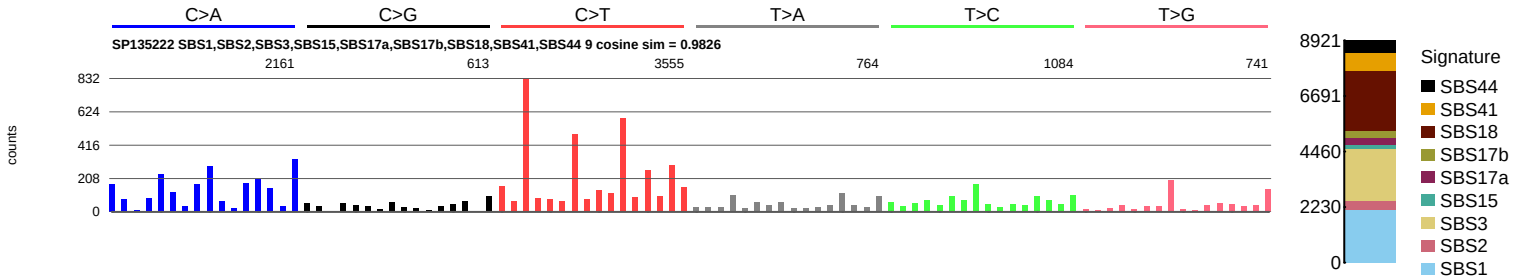

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98263
2122
653
3464
624
1230
828
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS1

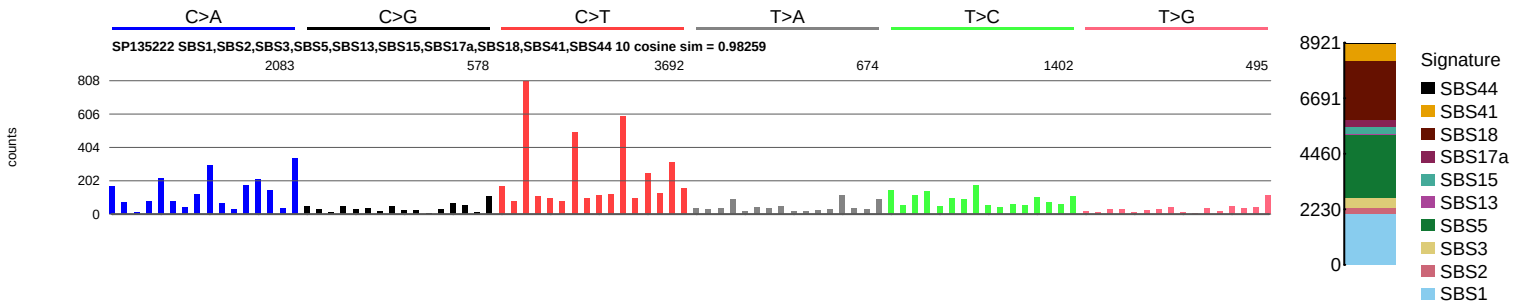

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98263
2124
638
3694
640
1198
625
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS5
SBS2
SBS1

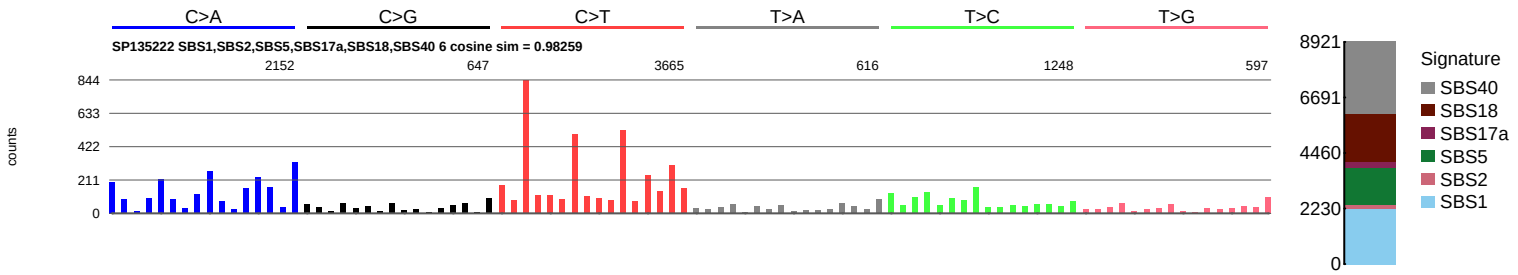

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98263
2144
653
3513
595
1203
812
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS1

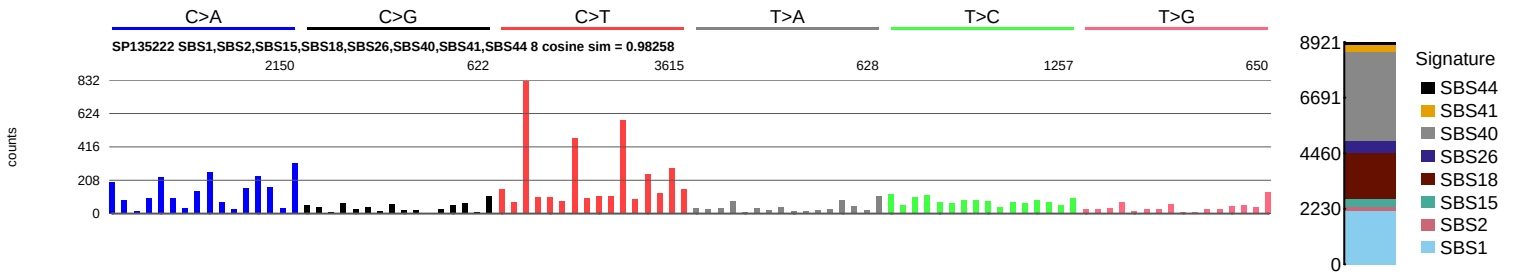

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98262
2124
639
3694
640
1198
625
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS40 7 cosine sim = 0.9826
2151
657
3665
611
1250
589
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.9826
2161
613
3555
764
1084
741
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41,SBS44 10 cosine sim = 0.98259
2083
578
3692
674
1402
495
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS40 6 cosine sim = 0.98259
2152
647
3665
616
1248
597
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98258
2150
622
3615
628
1257
650
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
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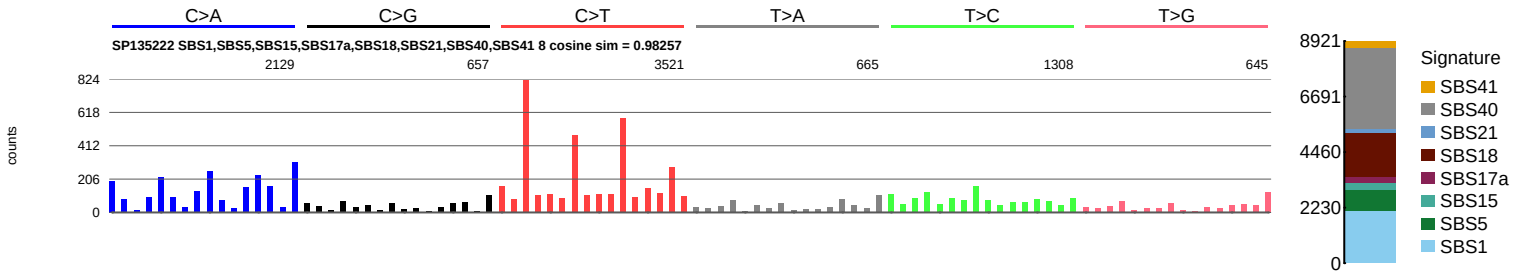
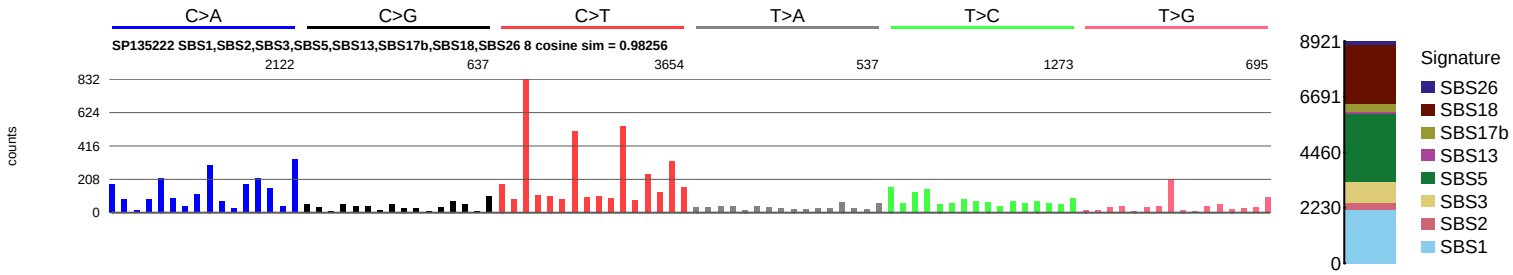
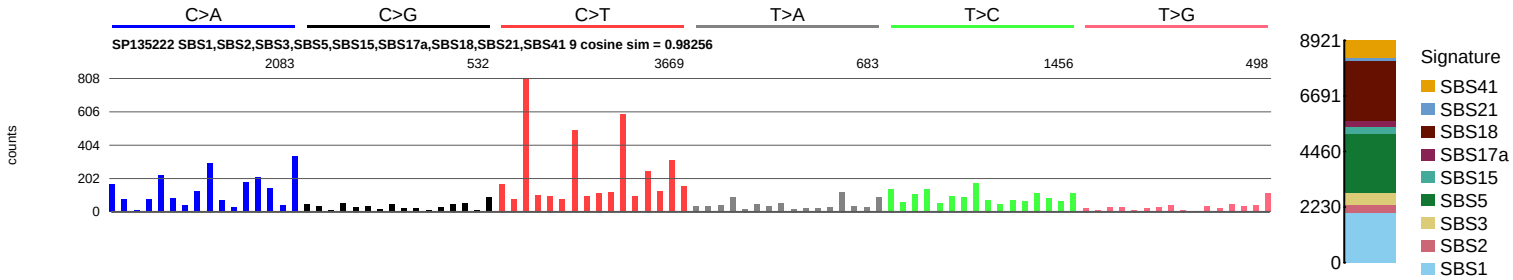
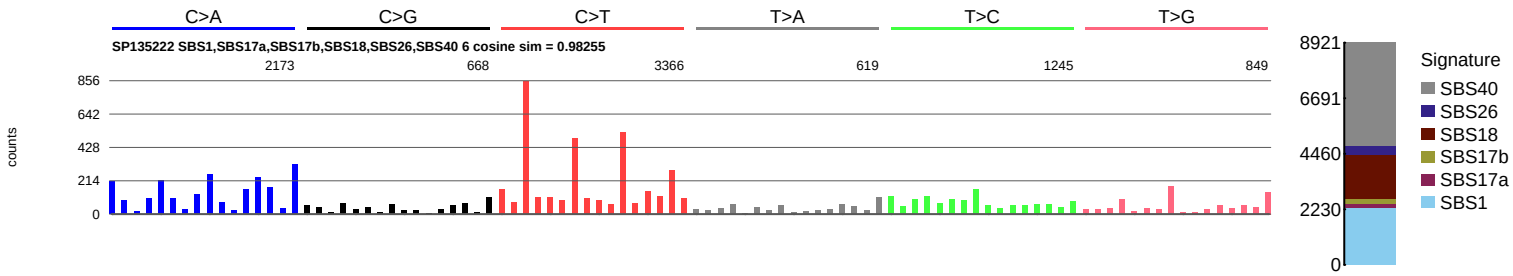
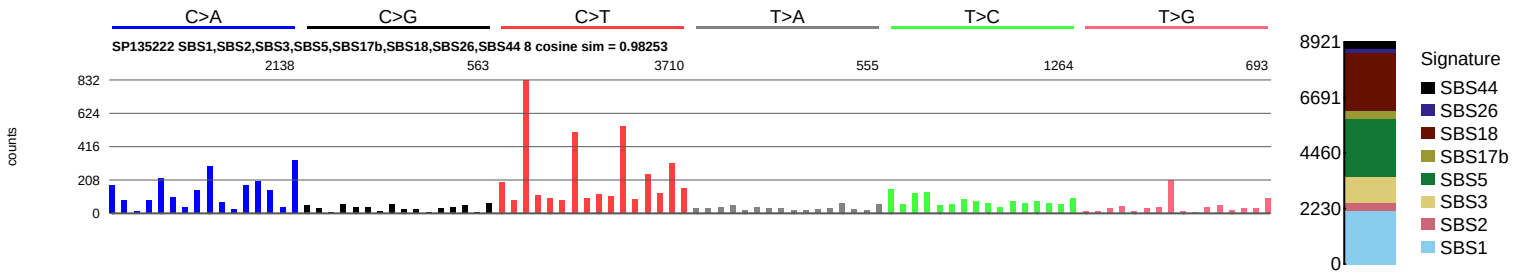
SBS15
SBS2
SBS1

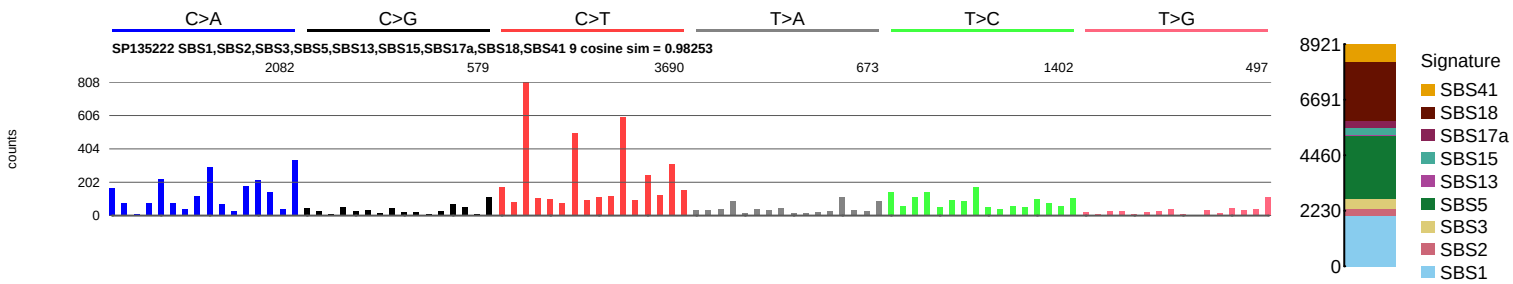

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98257
2129
657
3521
665
1308
645
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS1

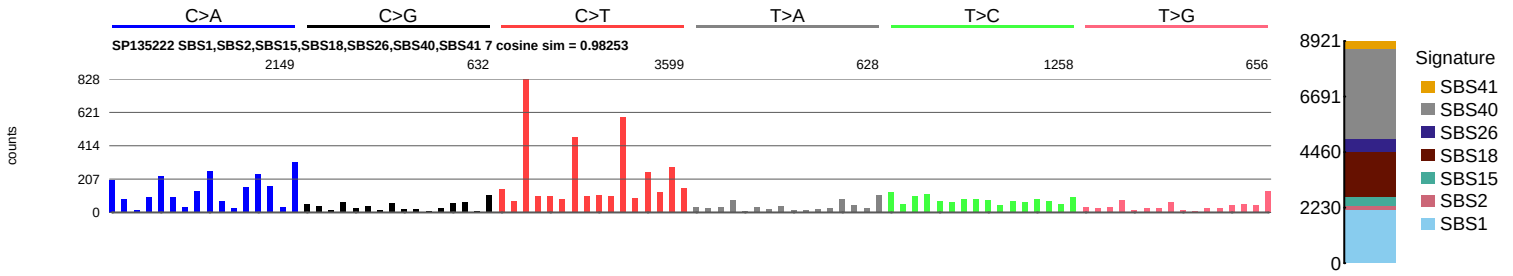

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18,SBS26 8 cosine sim = 0.98256
2122
637
3654
537
1273
695
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS3
SBS2
SBS1

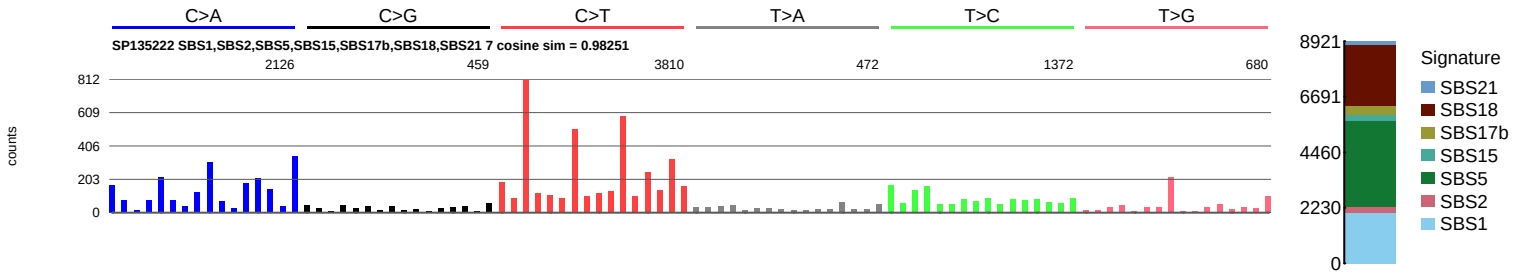

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS41 9 cosine sim = 0.98256
2083
532
3669
683
1456
498
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

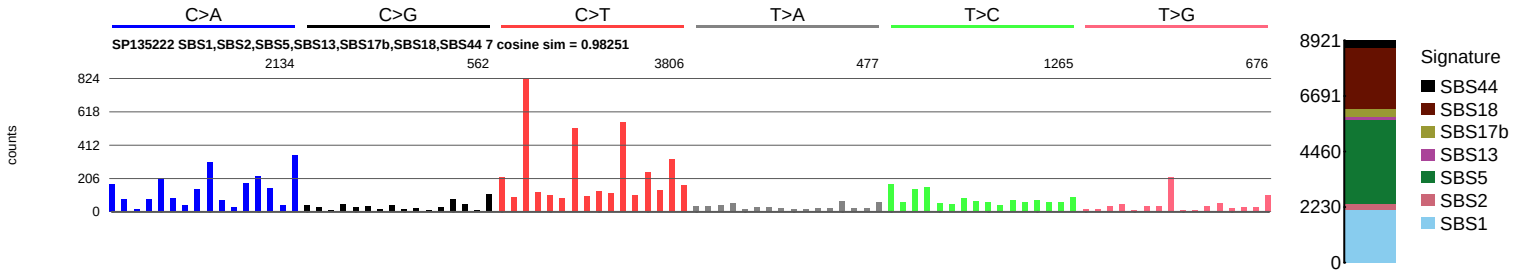

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98255
2173
668
3366
619
1245
849
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1

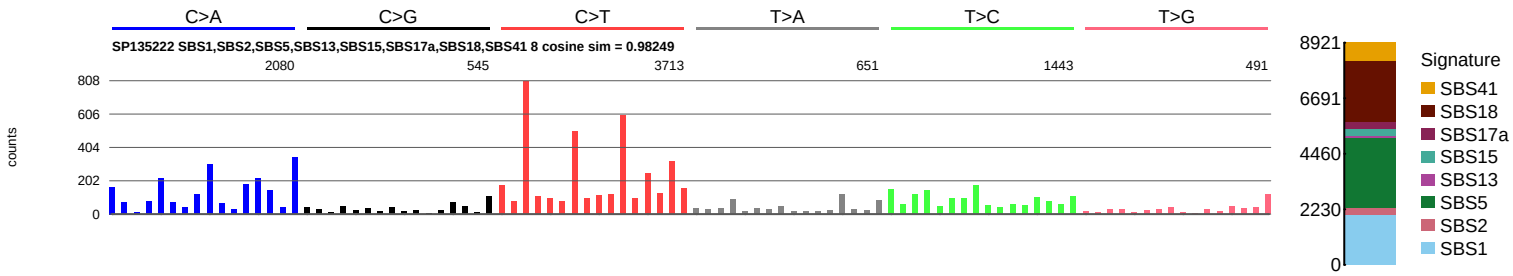

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98253
2138
563
3710
555
1264
693
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

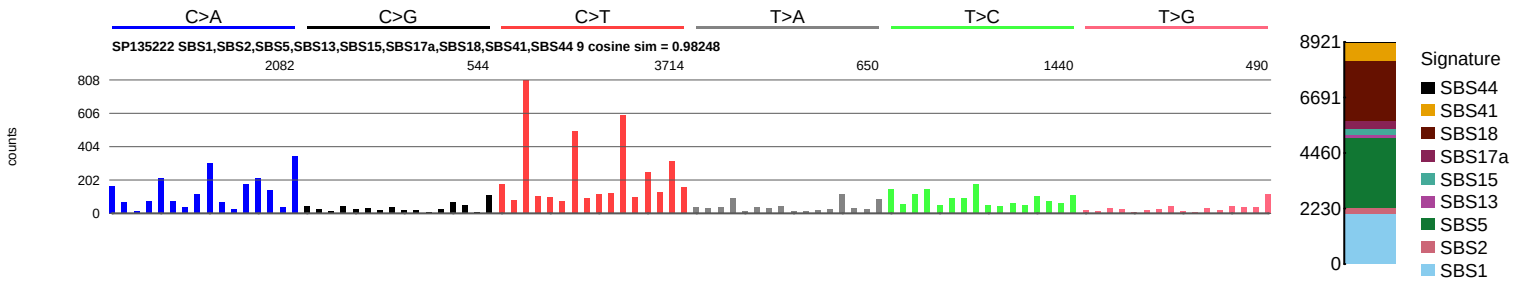

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41 9 cosine sim = 0.98253
2082
579
3690
673
1402
497
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

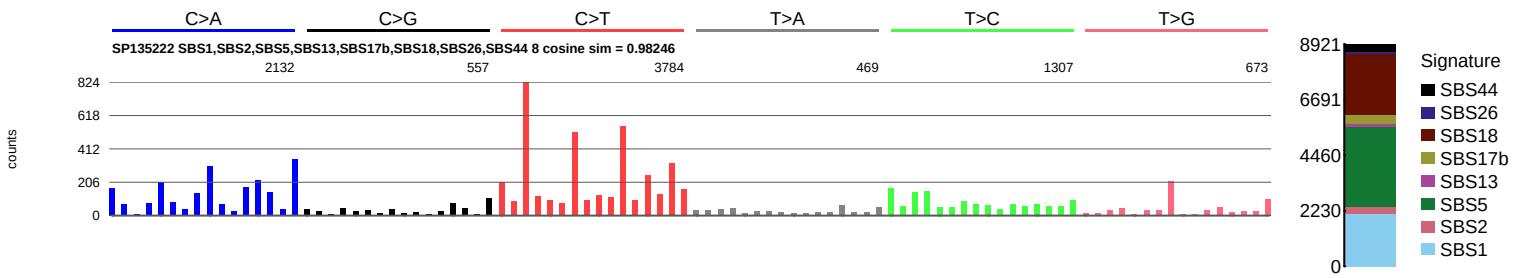

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98253
2149
632
3599
628
1258
656
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS2
SBS1

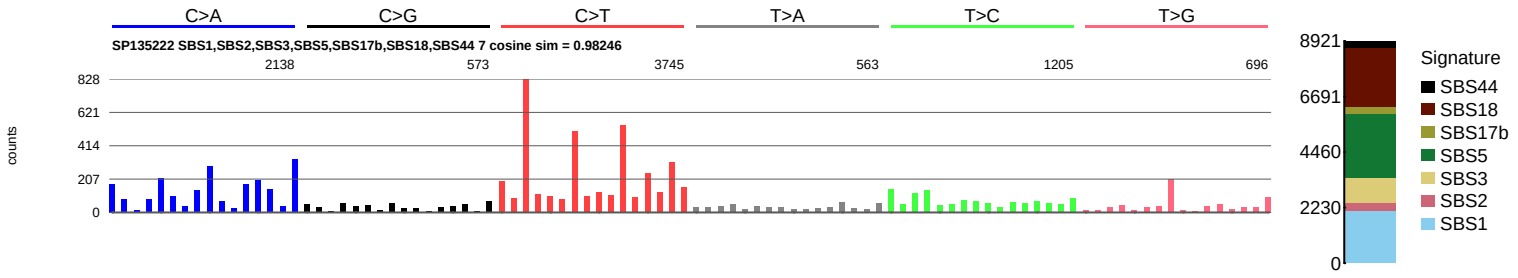

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21 7 cosine sim = 0.98251
2126
459
3810
472
1372
680
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

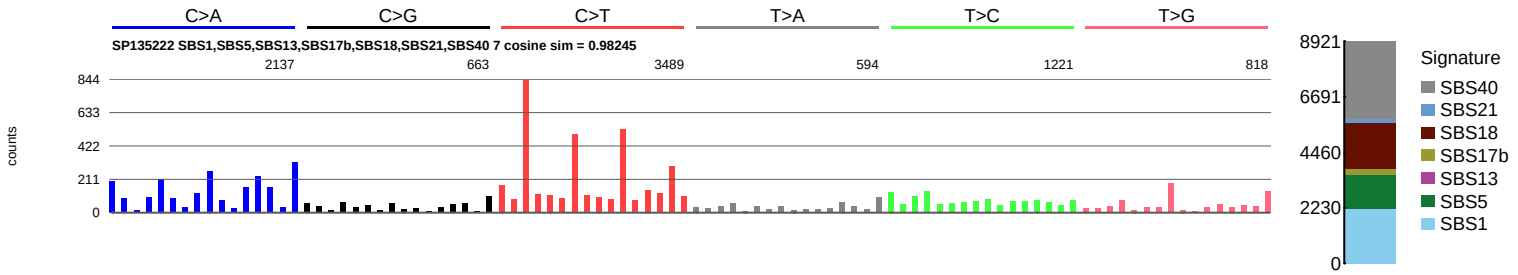

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS44 7 cosine sim = 0.98251
2134
562
3806
477
1265
676
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

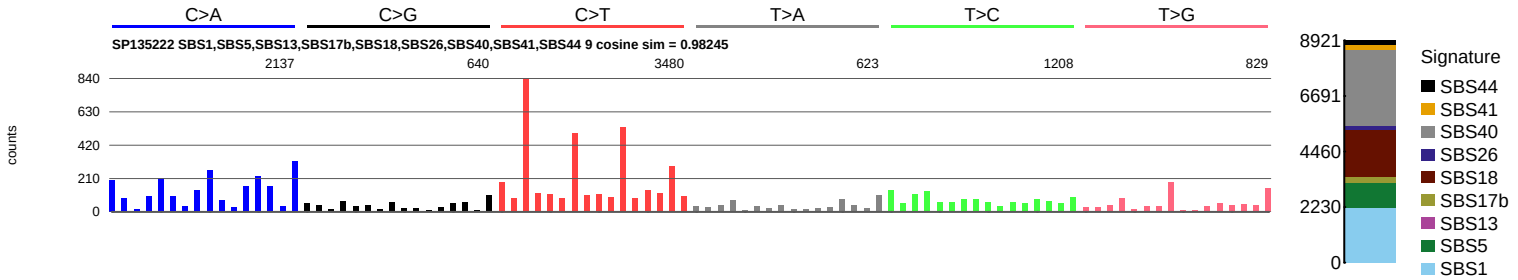

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41 8 cosine sim = 0.98249
2080
545
3713
651
1443
491
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

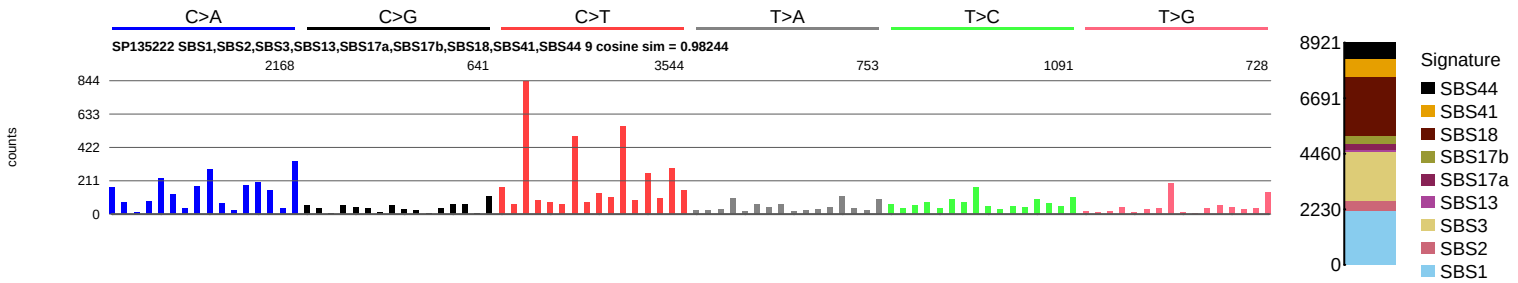

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS41,SBS44 9 cosine sim = 0.98248
2082
544
3714
650
1440
490
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

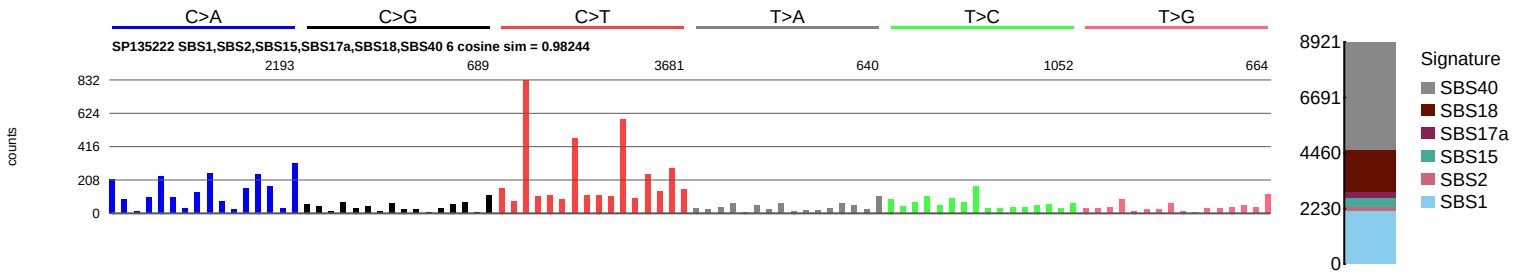

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98246
2132
557
3784
469
1307
673
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

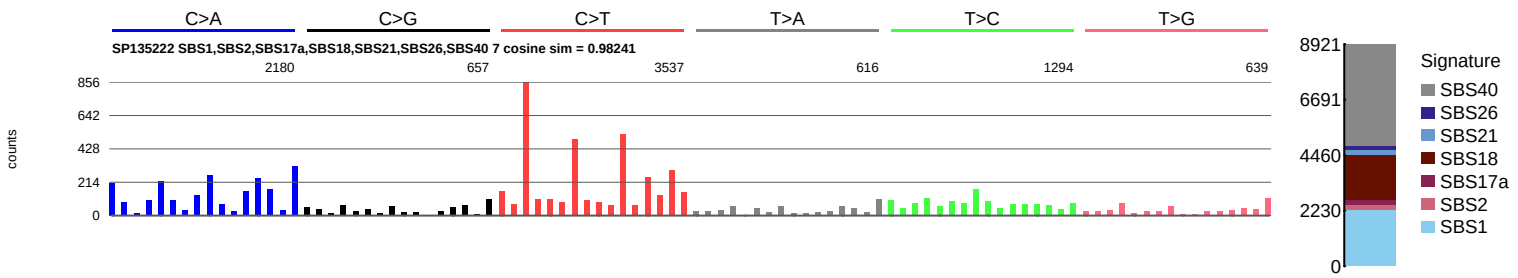

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS44 7 cosine sim = 0.98246
2138
573
3745
563
1205
696
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

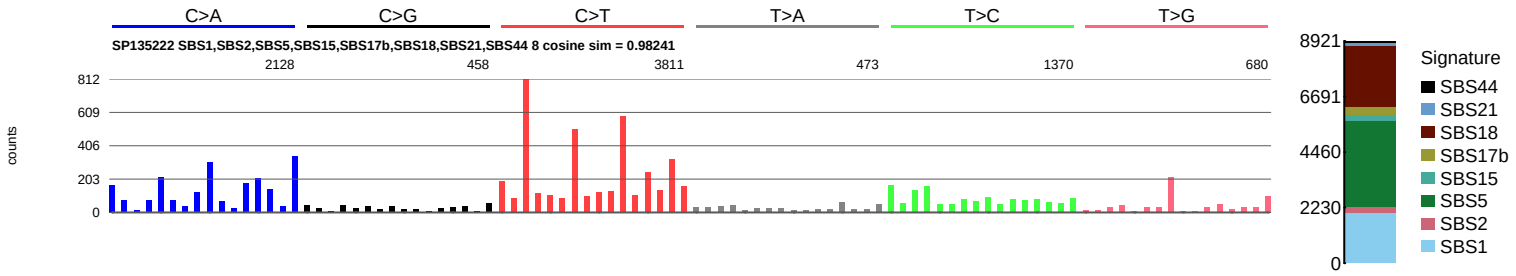

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98245
2137
663
3489
594
1221
818
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS13
SBS5
SBS1

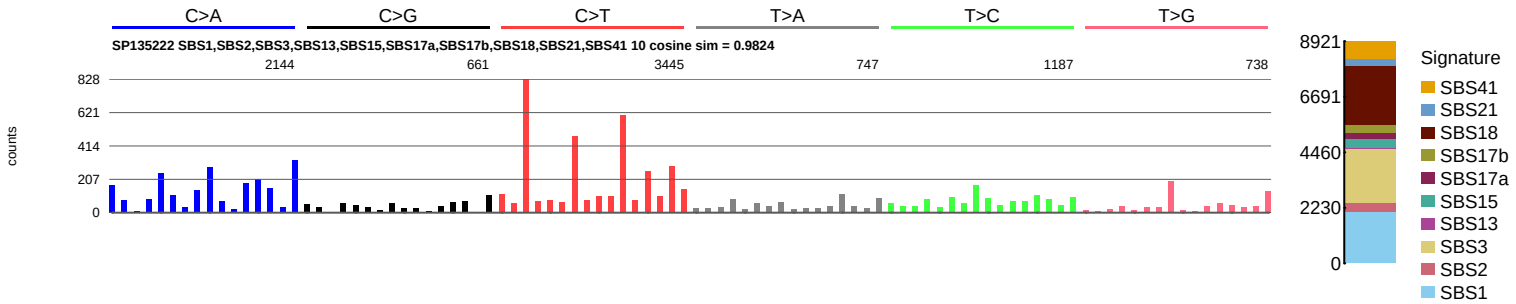

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98245
2137
640
3480
623
1208
829
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98244
2168
641
3544
753
1091
728
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS17a,SBS18,SBS40 6 cosine sim = 0.98244
2193
689
3681
640
1052
664
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98241
2180
657
3537
616
1294
639
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98241
2128
458
3811
473
1370
680
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.9824
2144
661
3445
747
1187
738
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
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SBS21
SBS18
SBS17b
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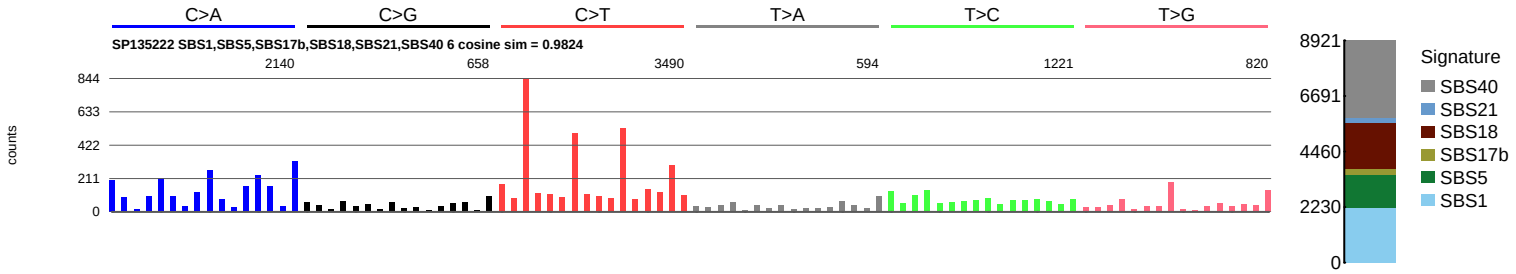
SBS1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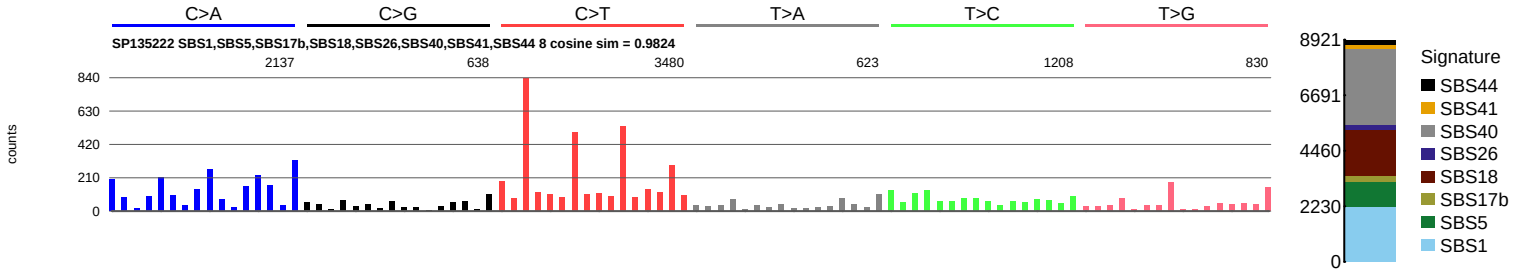
SP135222 SBS1,SBS5,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.9824
2140
658
3490
594
1221
820
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS5
SBS1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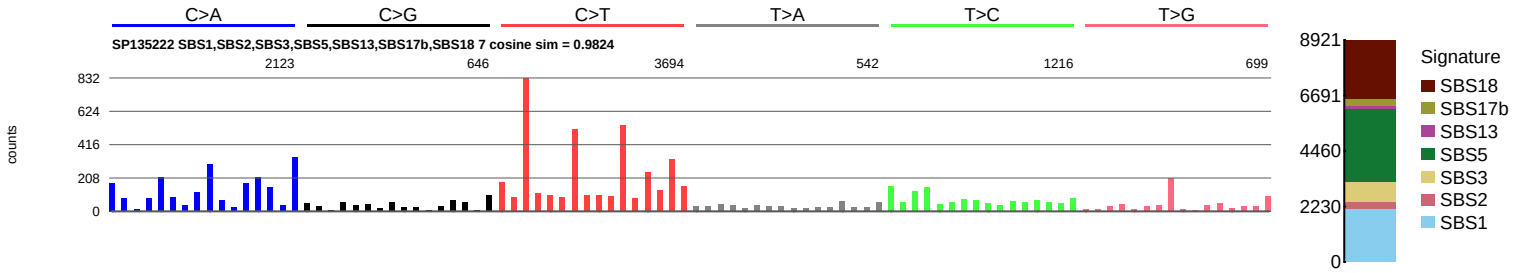
SP135222 SBS1,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.9824
2137
638
3480
623
1208
830
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS1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C>G
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T>C
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SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17b,SBS18 7 cosine sim = 0.9824
2123
646
3694
542
1216
699
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
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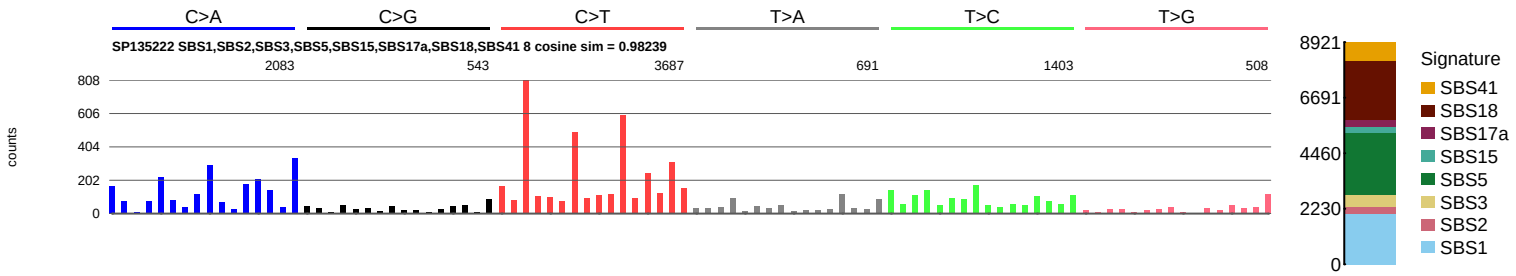
SBS2
SBS1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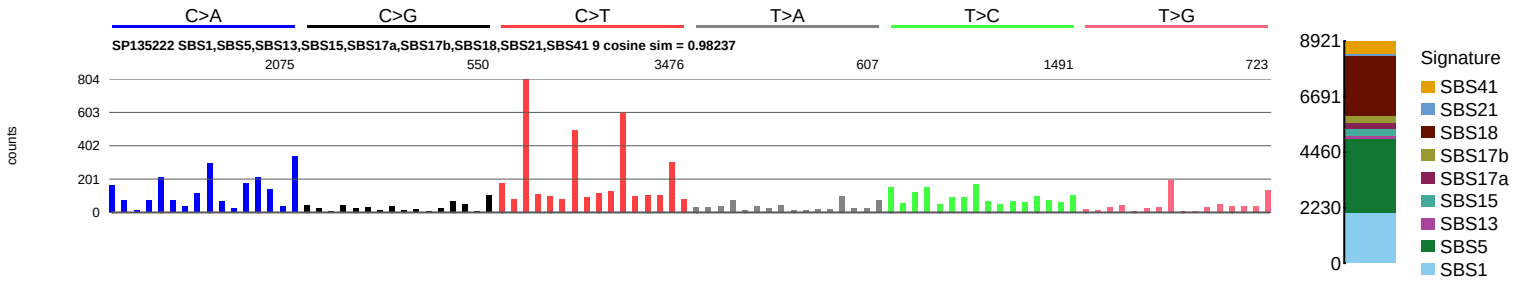
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS41 8 cosine sim = 0.98239
2083
543
3687
691
1403
508
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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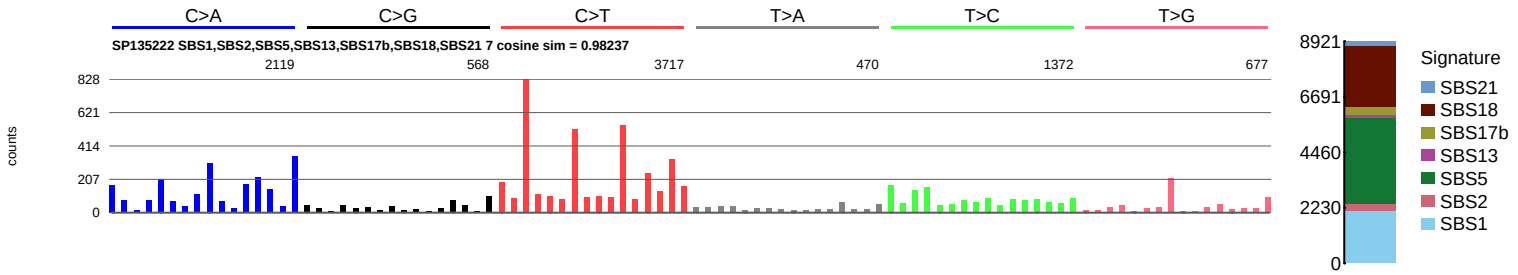
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98237
2075
550
3476
607
1491
723
counts
804
603
402
201
0
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS21 7 cosine sim = 0.98237
2119
568
3717
470
1372
677
counts
828
621
414
207
0
8921
6691
4460
2230
0
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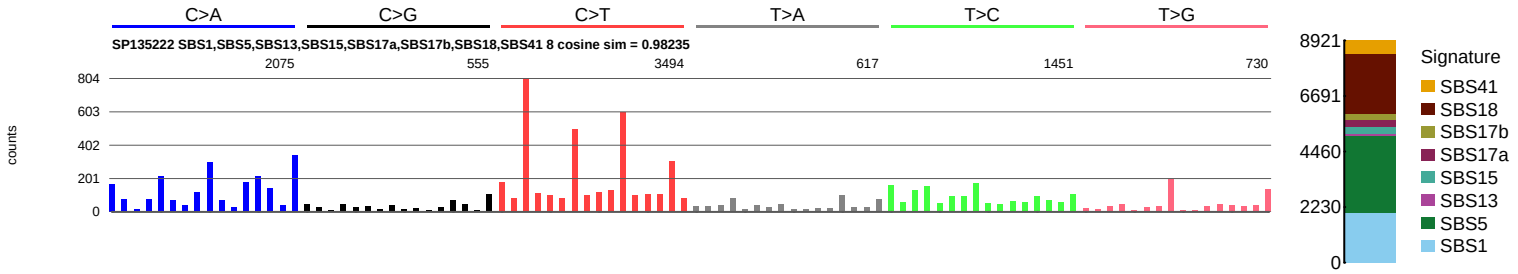
SBS2
SBS1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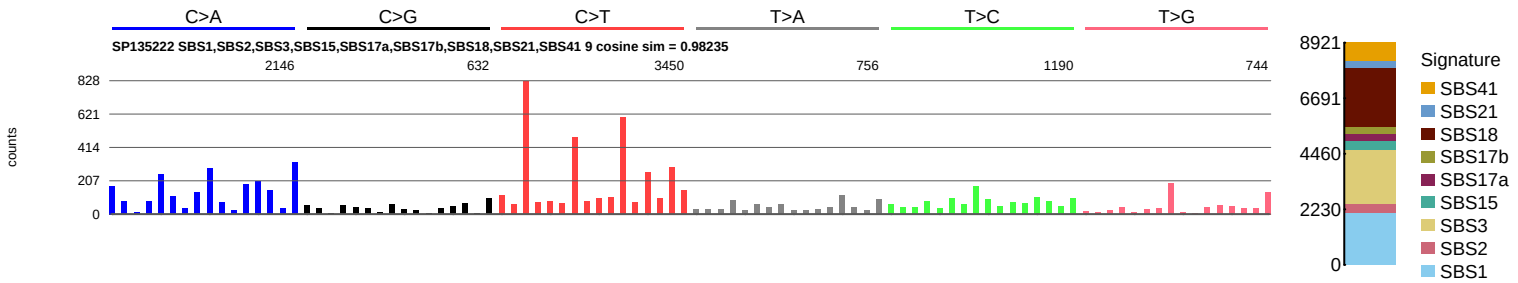
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98235
2075
555
3494
617
1451
730
counts
804
603
402
201
0
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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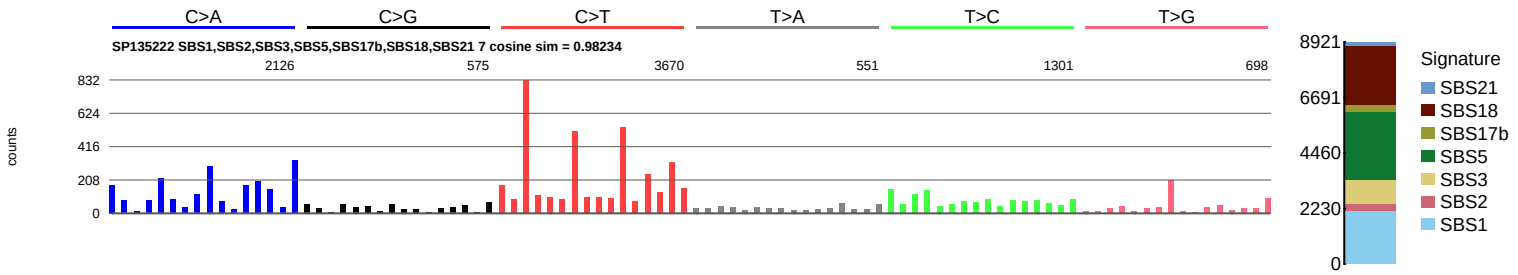
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98235
2146
632
3450
756
1190
744
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS21 7 cosine sim = 0.98234
2126
575
3670
551
1301
698
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS21
SBS18
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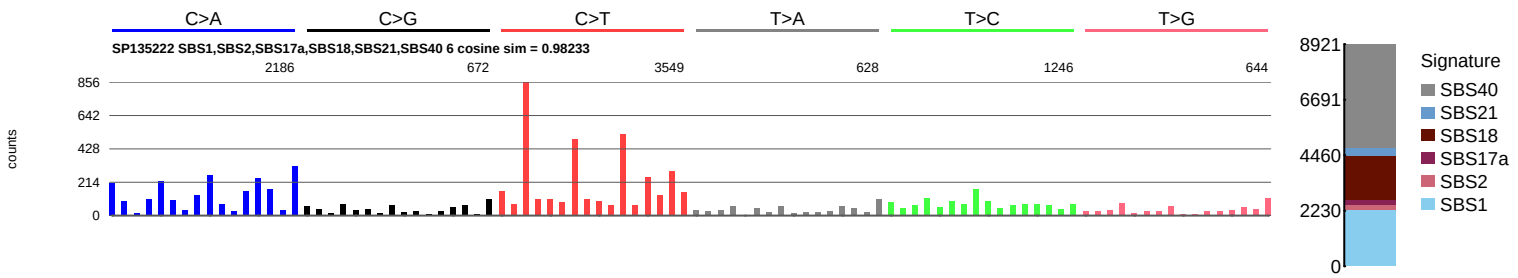
SBS3
SBS2
SBS1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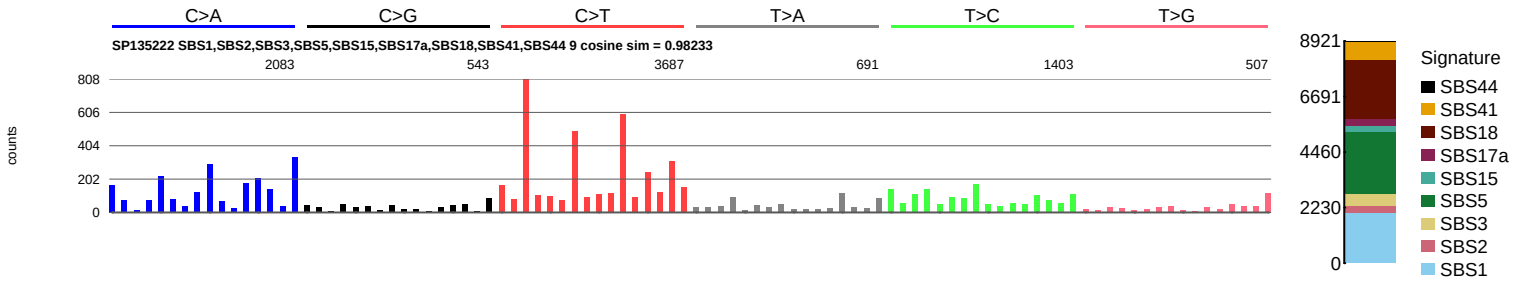
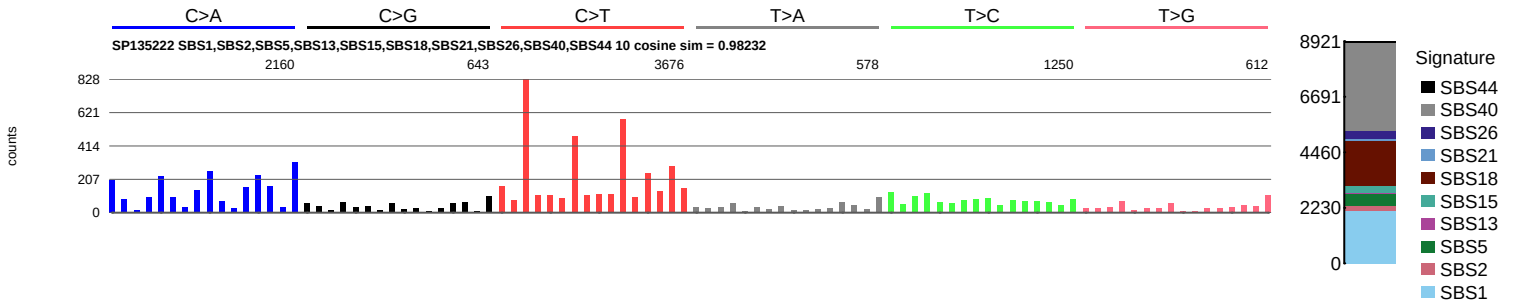
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS18,SBS21,SBS40 6 cosine sim = 0.98233
2186
672
3549
628
1246
644
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS18,SBS41,SBS44 9 cosine sim = 0.98233
2083
543
3687
691
1403
507
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98232
2160
643
3676
578
1250
612
counts
828
621
414
207
0
8921
6691
4460
2230
0
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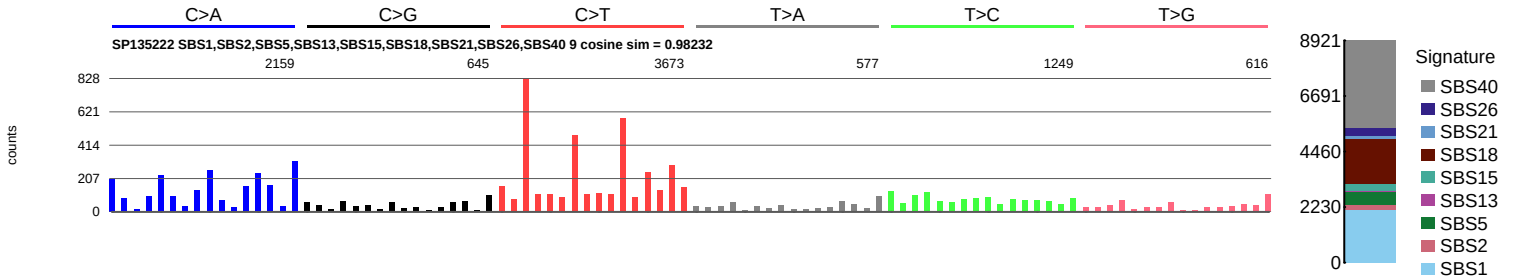
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98232
2159
645
3673
577
1249
616
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
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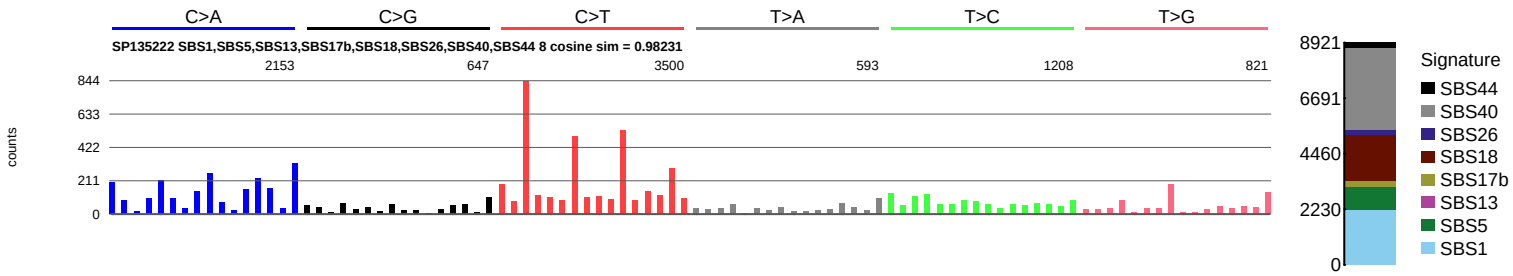
SBS2
SBS1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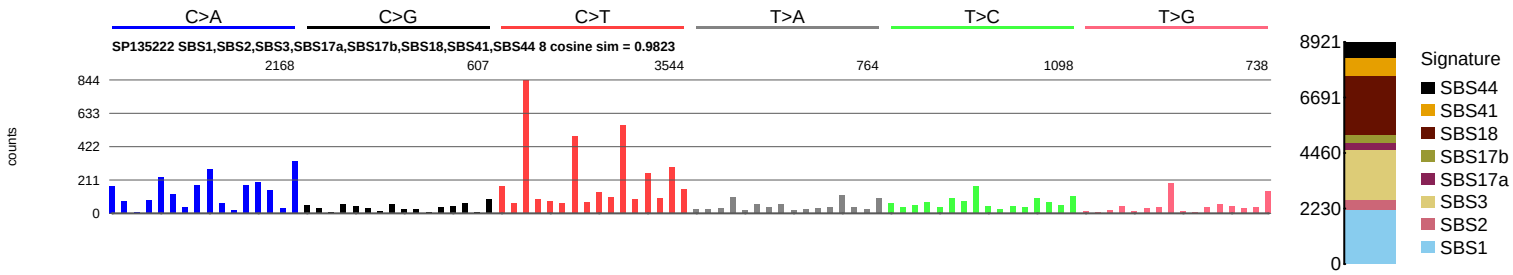
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98231
2153
647
3500
593
1208
821
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS1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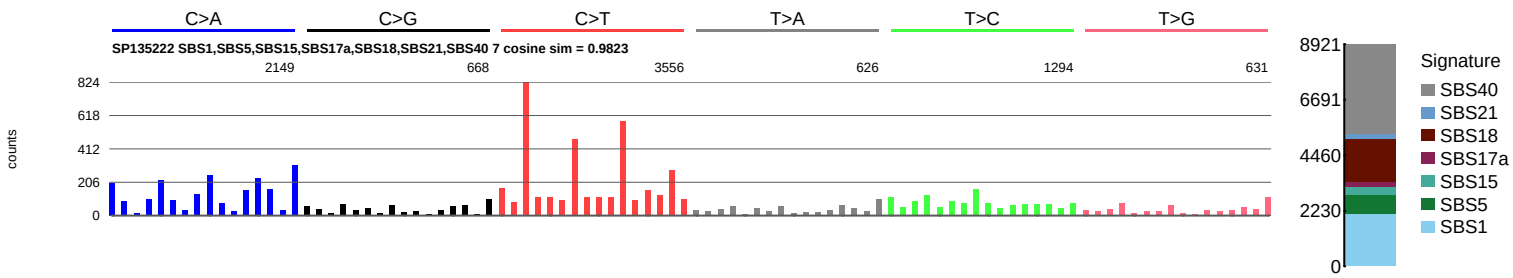
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.9823
2168
607
3544
764
1098
738
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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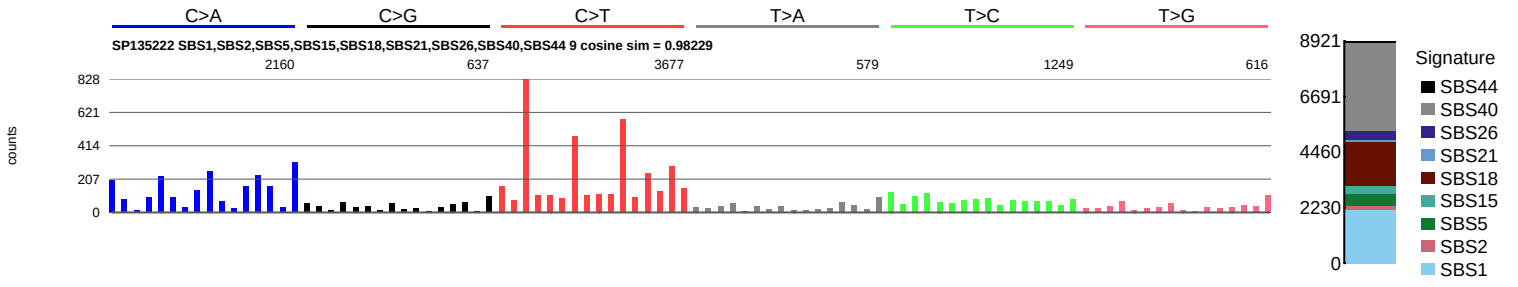
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS40 7 cosine sim = 0.9823
2149
668
3556
626
1294
631
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS1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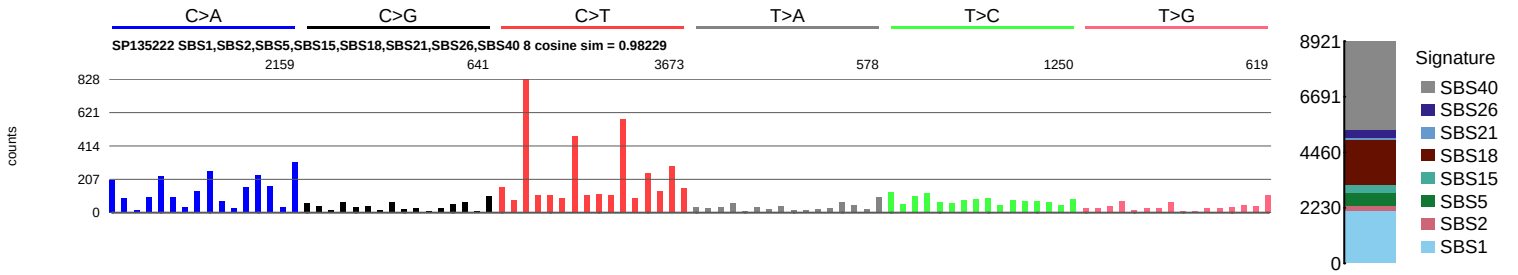
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98229
2160
637
3677
579
1249
616
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS15
SBS5
SBS2
SBS1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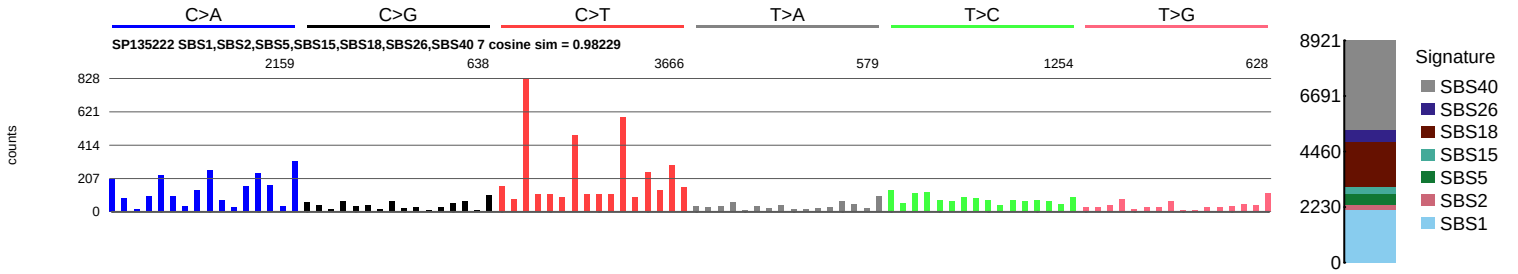
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98229
2159
641
3673
578
1250
619
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS15
SBS5
SBS2
SBS1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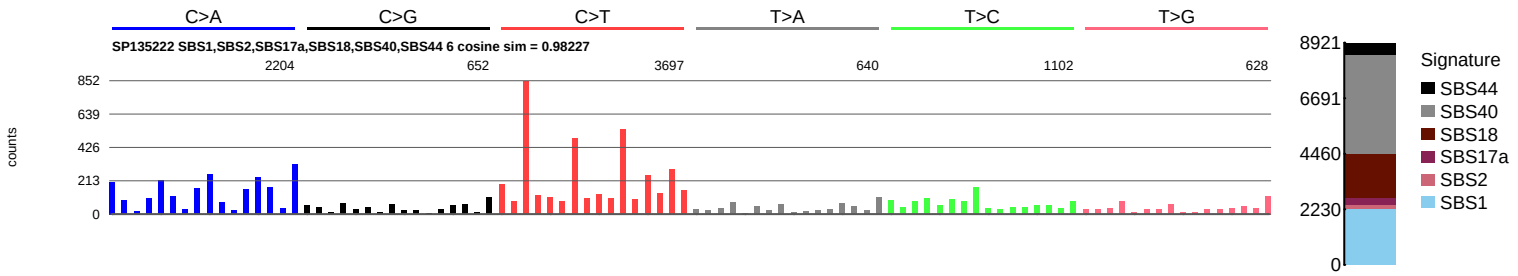
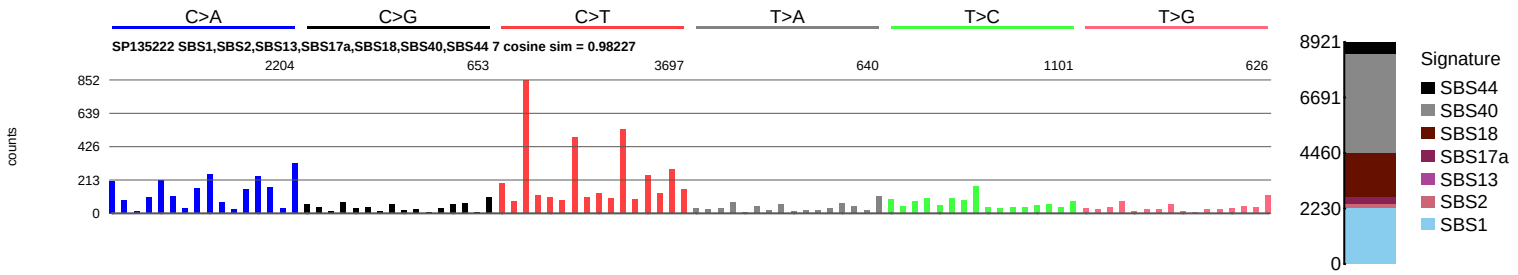
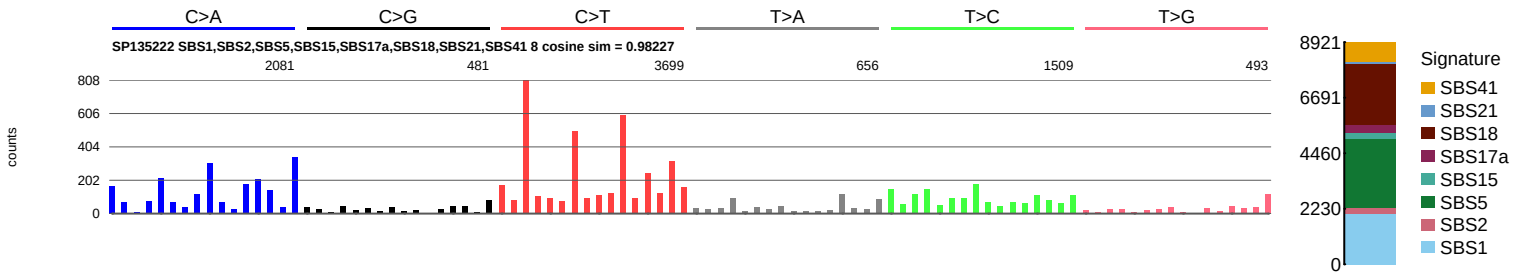
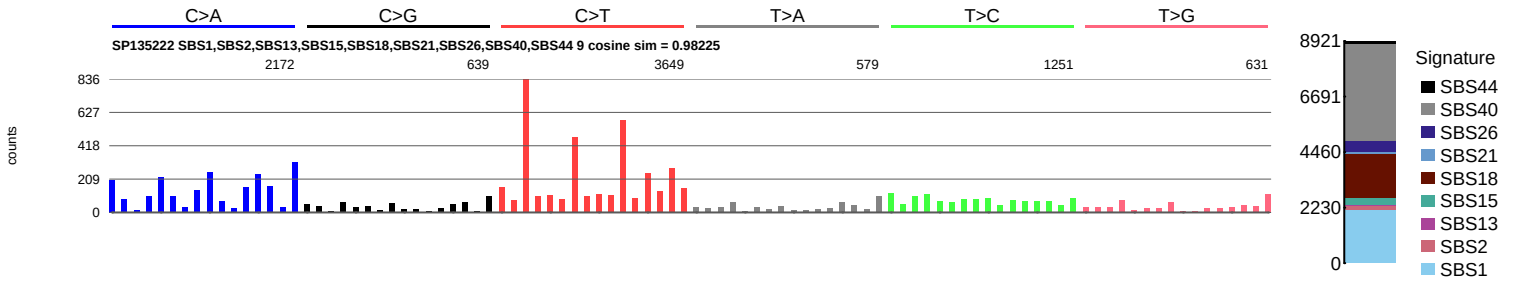
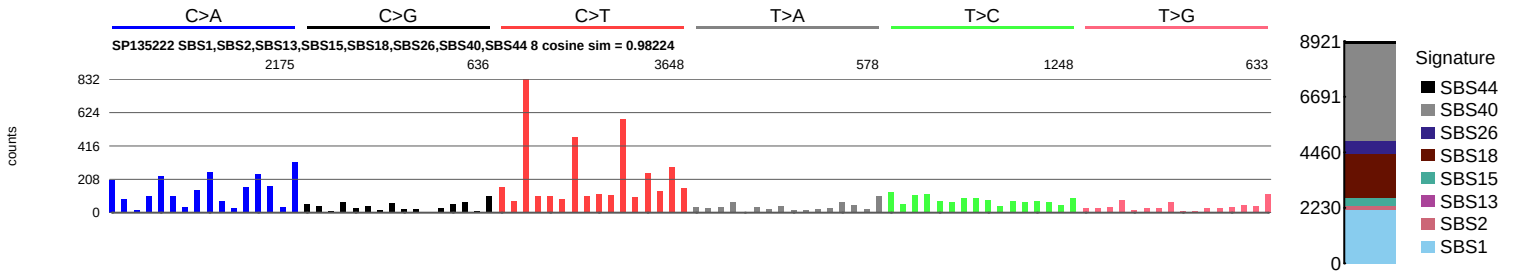
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS26,SBS40 7 cosine sim = 0.98229
2159
638
3666
579
1254
628
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS18,SBS40,SBS44 6 cosine sim = 0.98227
2204
652
3697
640
1102
628
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS17a,SBS18,SBS40,SBS44 7 cosine sim = 0.98227
2204
653
3697
640
1101
626
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS21,SBS41 8 cosine sim = 0.98227
2081
481
3699
656
1509
493
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98225
2172
639
3649
579
1251
631
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98224
2175
636
3648
578
1248
633
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
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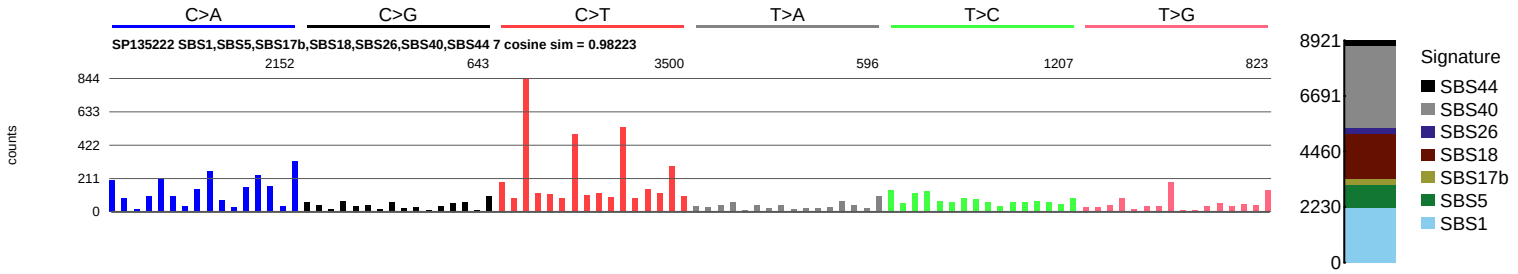
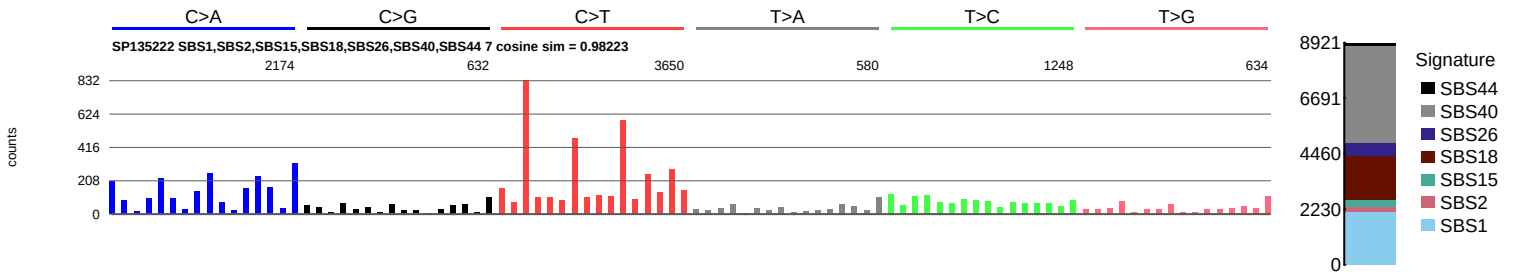
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98223
2152
643
3500
596
1207
823
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS1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C>G
C>T
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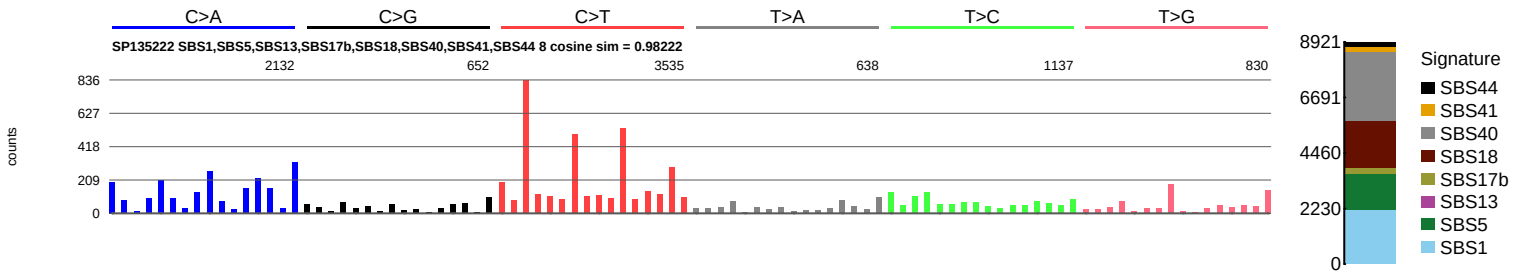
T>G
SP135222 SBS1,SBS2,SBS15,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98223
2174
632
3650
580
1248
634
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS15
SBS2
SBS1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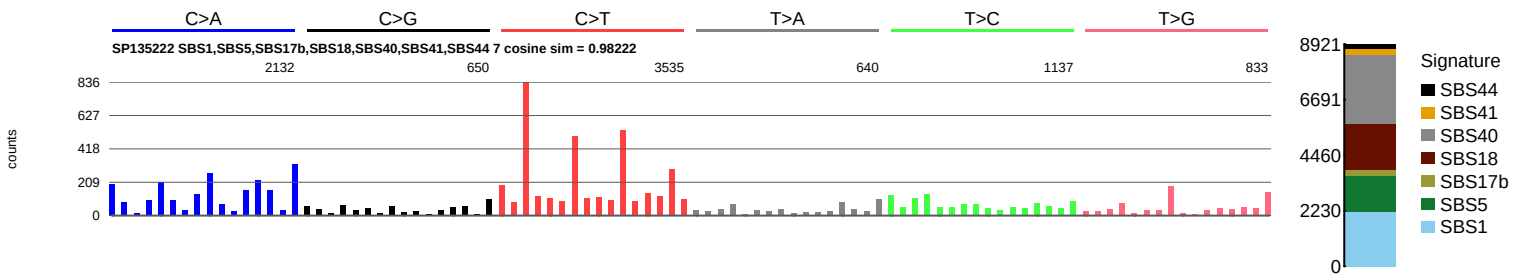
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98222
2132
652
3535
638
1137
830
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS1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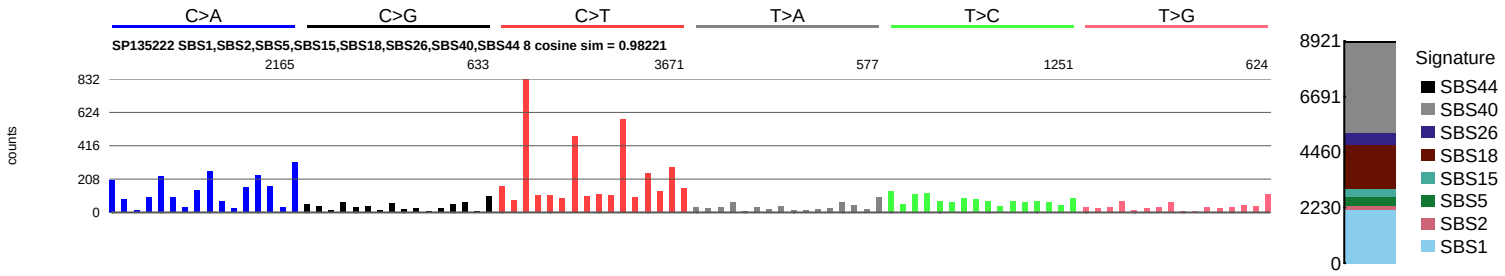
T>G
SP135222 SBS1,SBS5,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98222
2132
650
3535
640
1137
833
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS1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SP135222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98221
2165
633
3671
577
1251
624
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS15
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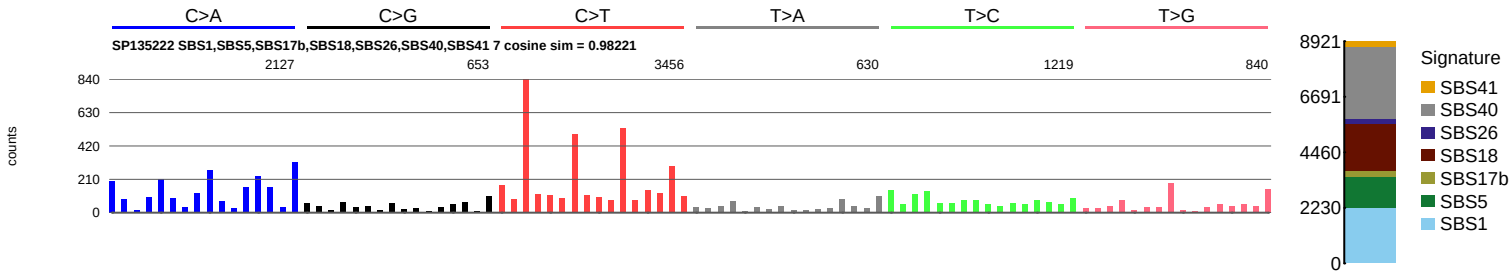
SBS2
SBS1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C>G
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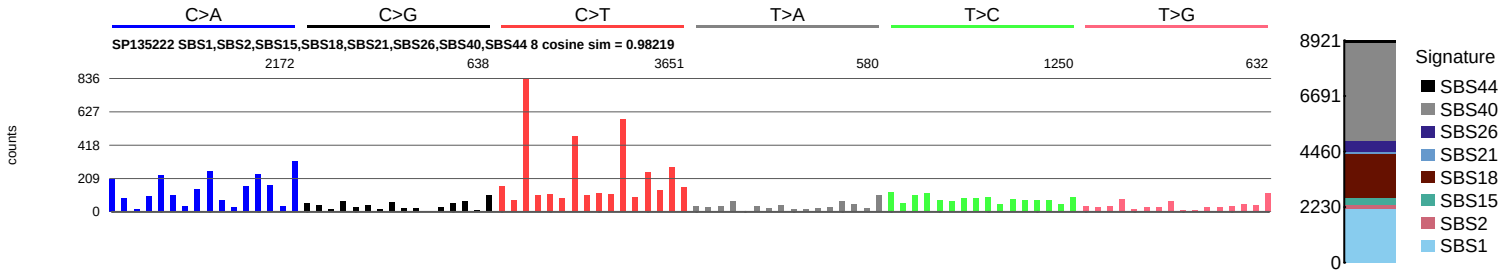
SP135222 SBS1,SBS5,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98221
2127
653
3456
630
1219
840
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS1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2172
638
3651
580
1250
632
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
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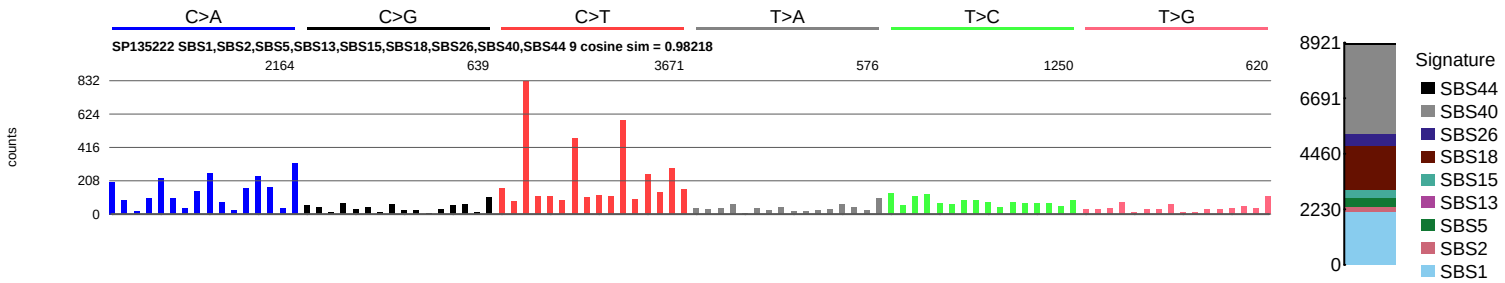
SBS21
SBS18
SBS15
SBS2
SBS1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SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98218
2164
639
3671
576
1250
620
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
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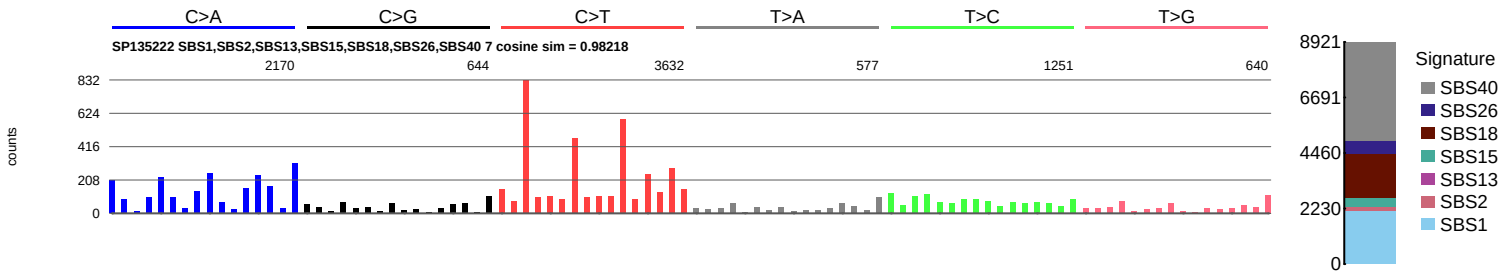
SBS2
SBS1


C>A
C>G
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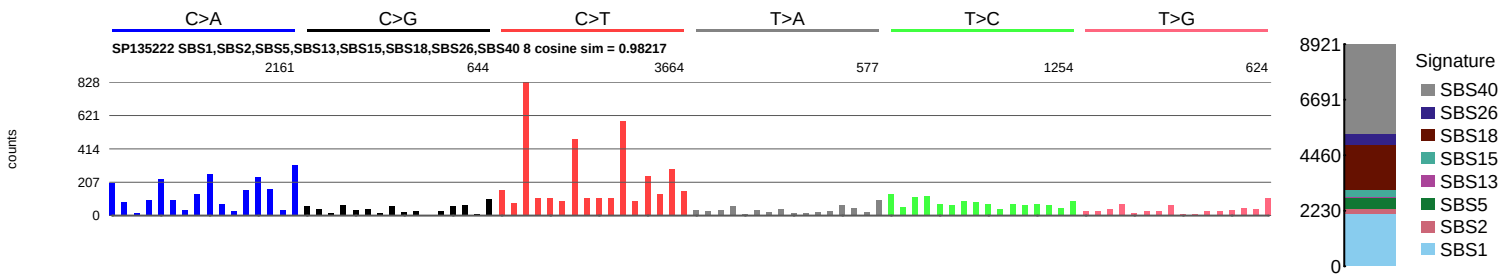
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS26,SBS40 7 cosine sim = 0.98218
2170
644
3632
577
1251
640
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS13
SBS2
SBS1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C>G
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T>C
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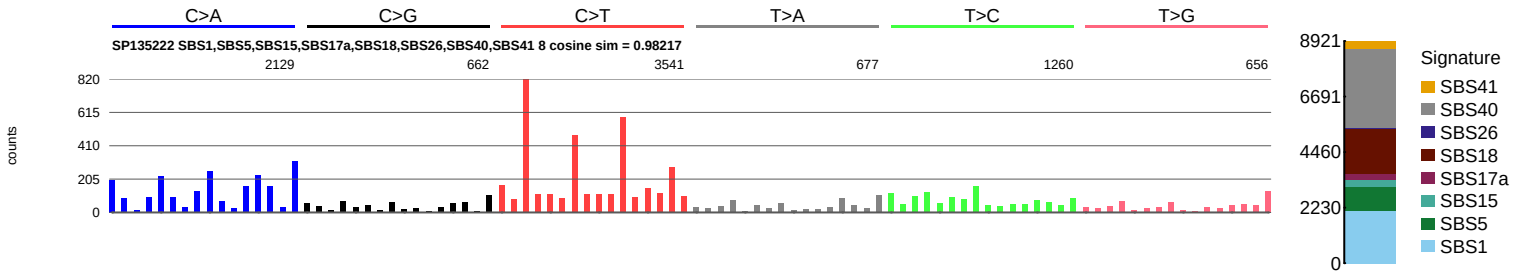
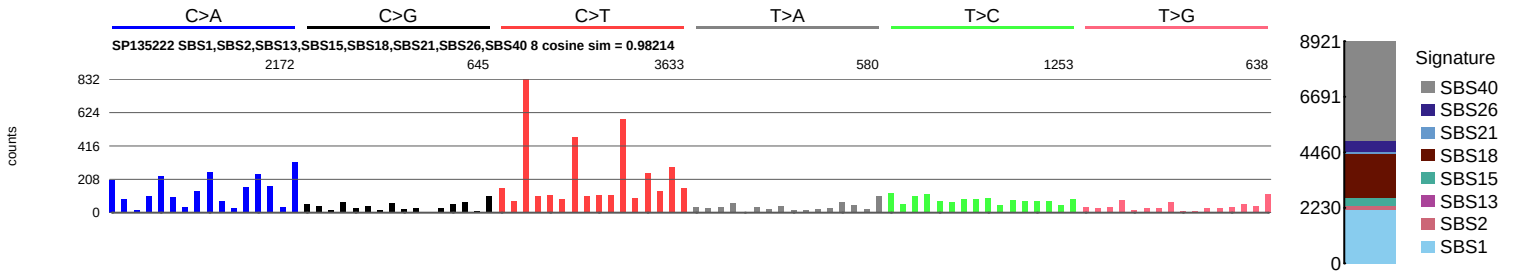
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS26,SBS40 8 cosine sim = 0.98217
2161
644
3664
577
1254
624
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98217
2129
662
3541
677
1260
656
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS15,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98214
2172
645
3633
580
1253
638
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
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SBS18
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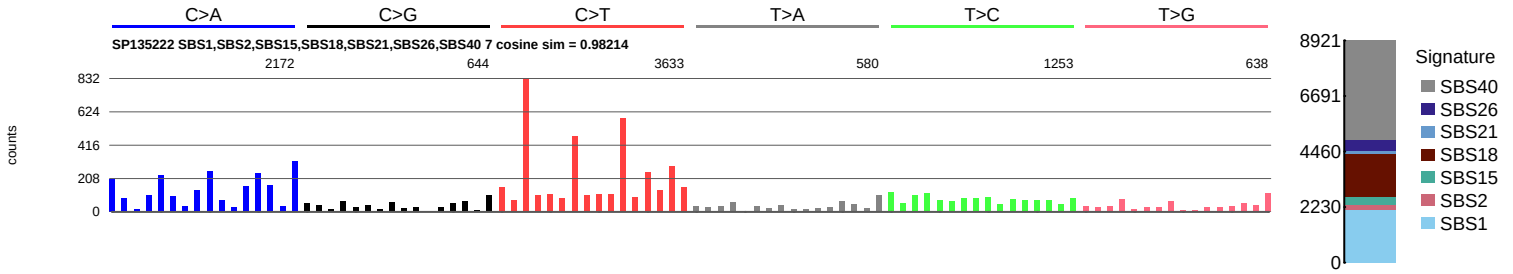
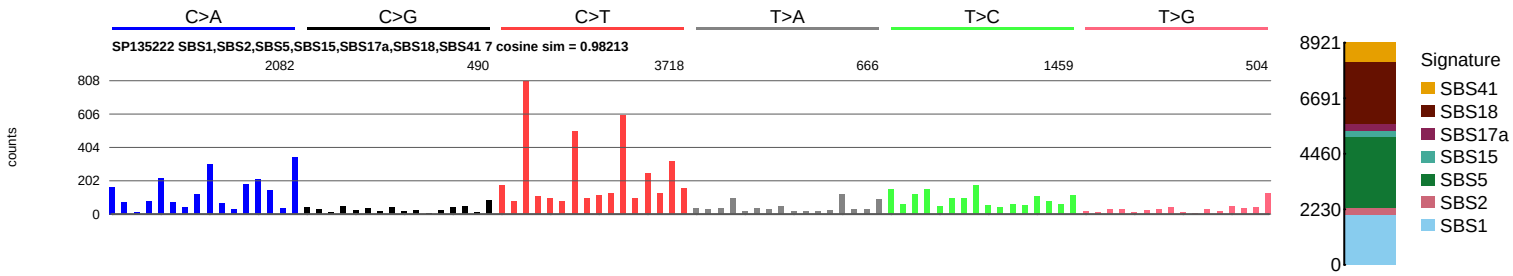
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98214
2172
644
3633
580
1253
638
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS18,SBS41 7 cosine sim = 0.98213
2082
490
3718
666
1459
504
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
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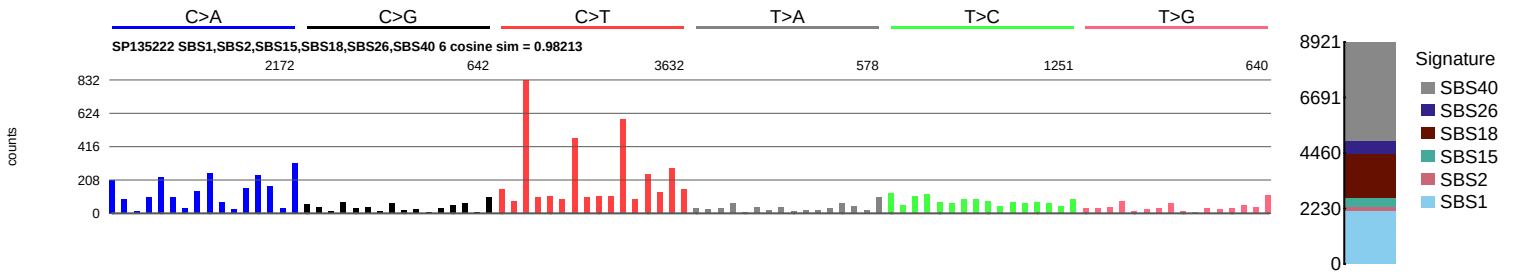
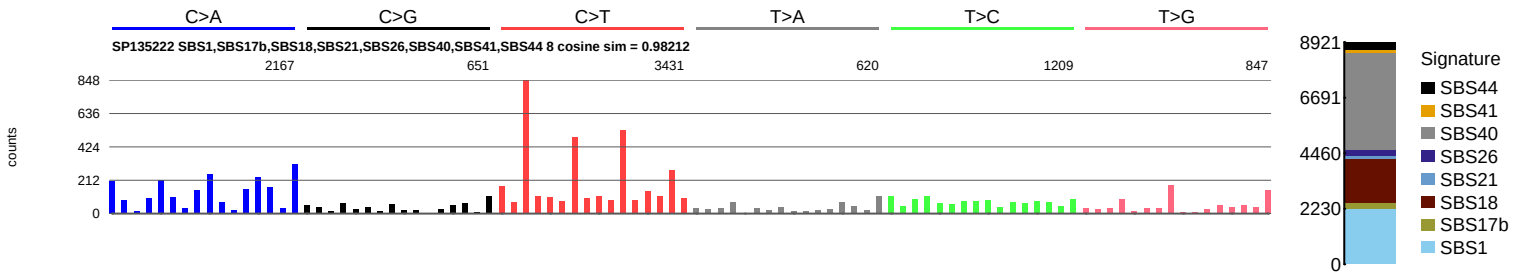
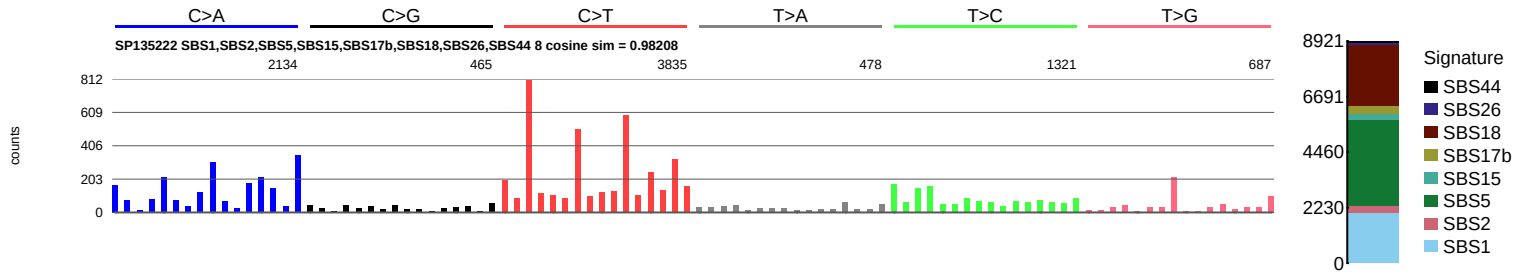
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS18,SBS26,SBS40 6 cosine sim = 0.98213
2172
642
3632
578
1251
640
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98212
2167
651
3431
620
1209
847
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS1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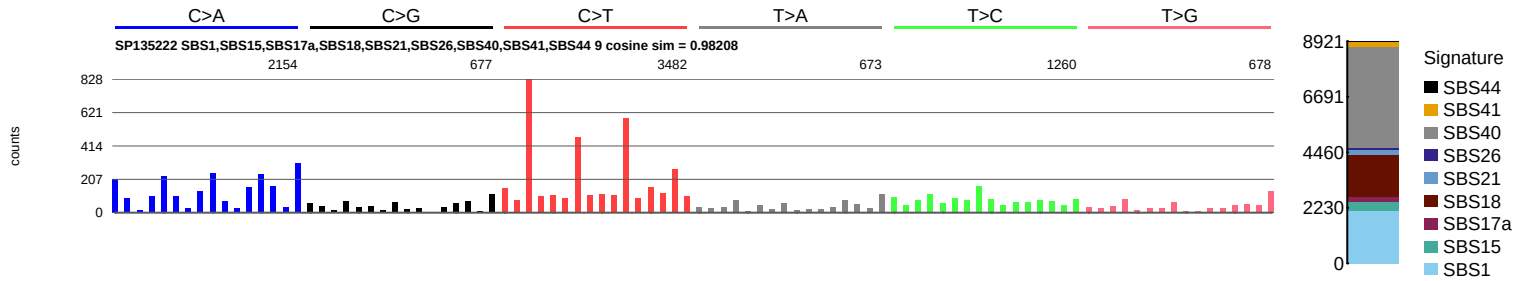
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98208
2134
465
3835
478
1321
687
812
609
406
203
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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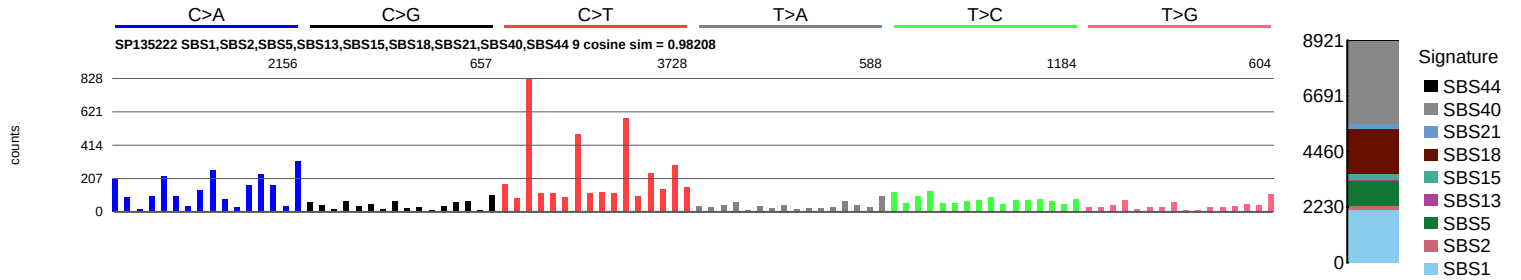
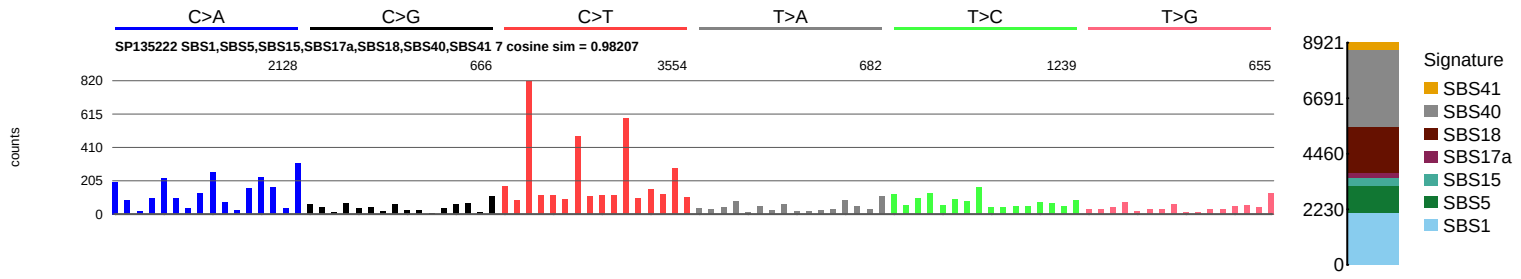
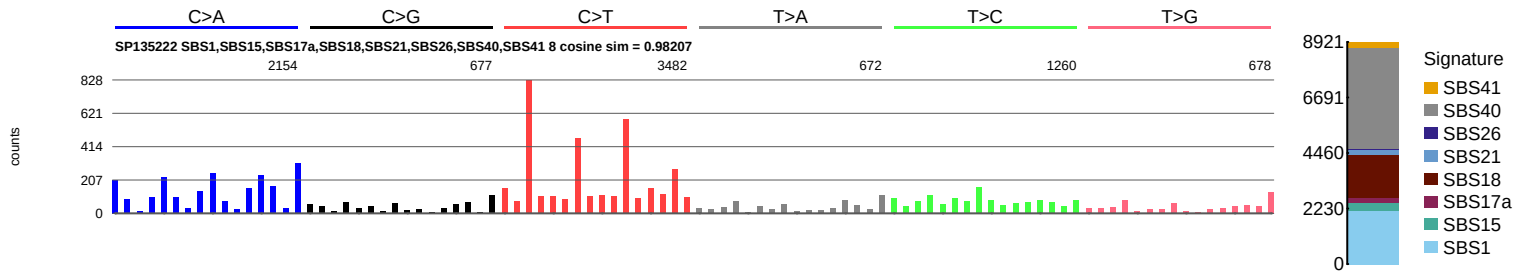
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98208
2154
677
3482
673
1260
678
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98208
2156
657
3728
588
1184
604
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS40,SBS41 7 cosine sim = 0.98207
2128
666
3554
682
1239
655
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS5
SBS1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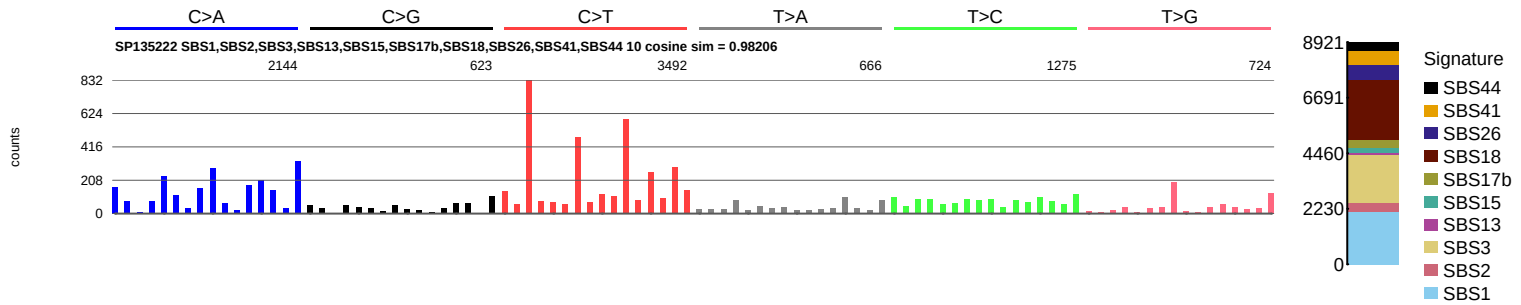
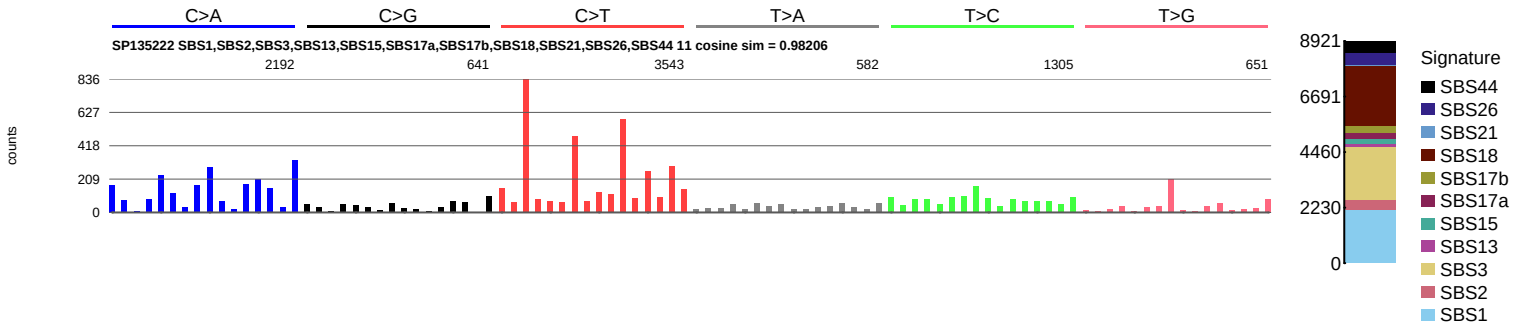
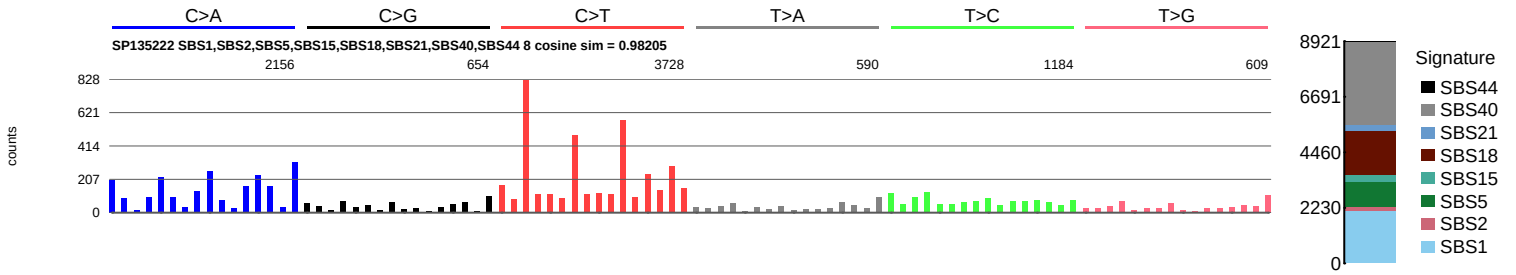
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98207
2154
677
3482
672
1260
678
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98206
2144
623
3492
666
1275
724
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 11 cosine sim = 0.98206
2192
641
3543
582
1305
651
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98205
2156
654
3728
590
1184
609
counts
828
621
414
207
0
8921
6691
4460
2230
0
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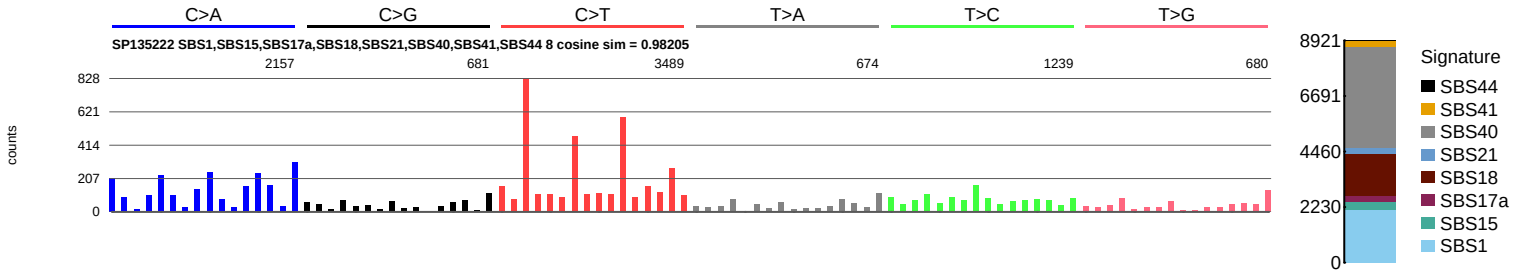
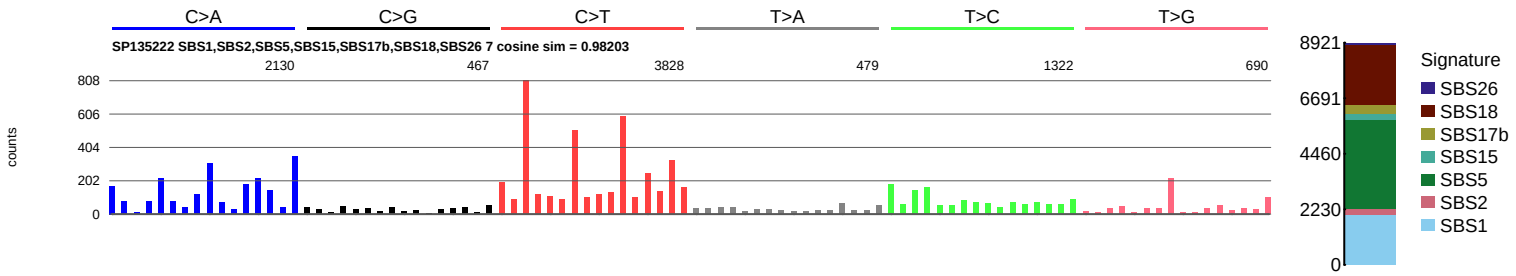
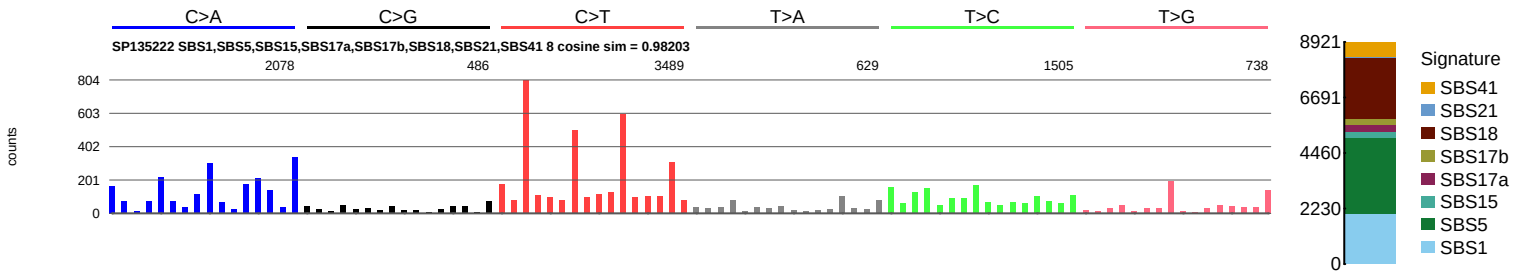
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98205
2157
681
3489
674
1239
680
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS26 7 cosine sim = 0.98203
2130
467
3828
479
1322
690
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS26
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1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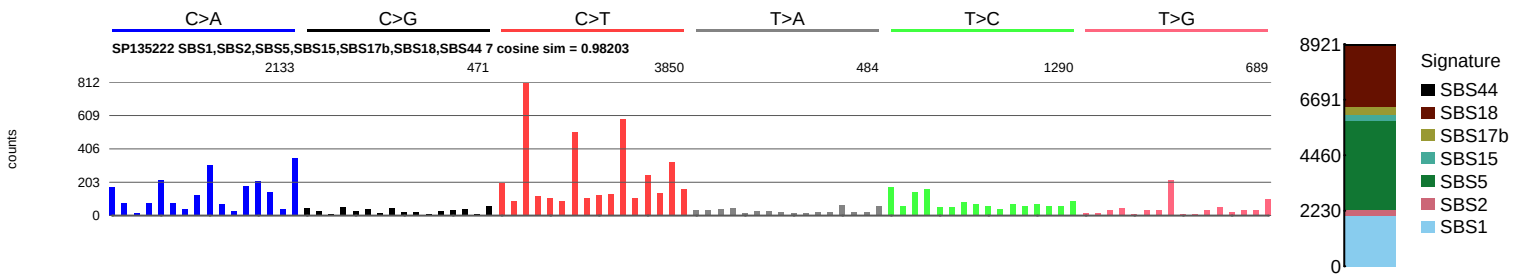
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98203
2078
486
3489
629
1505
738
counts
804
603
402
201
0
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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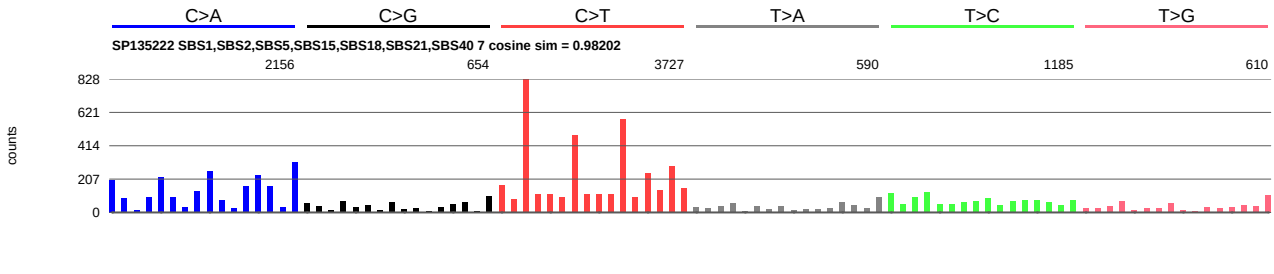
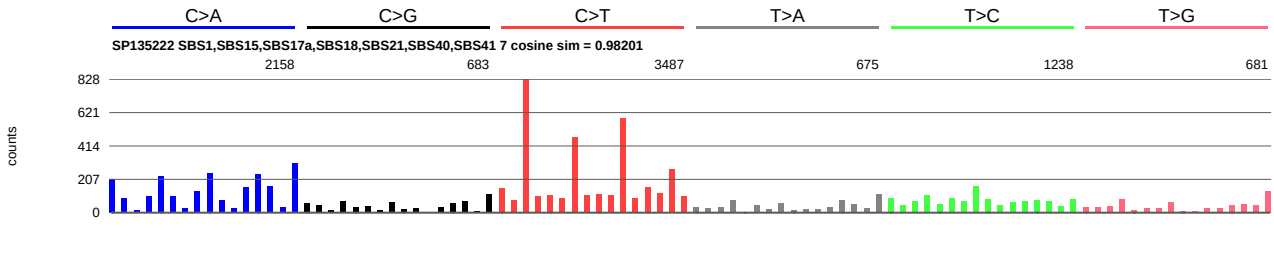
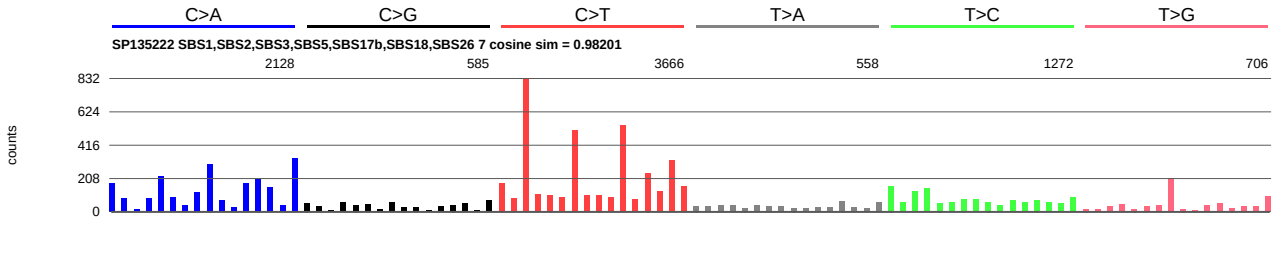
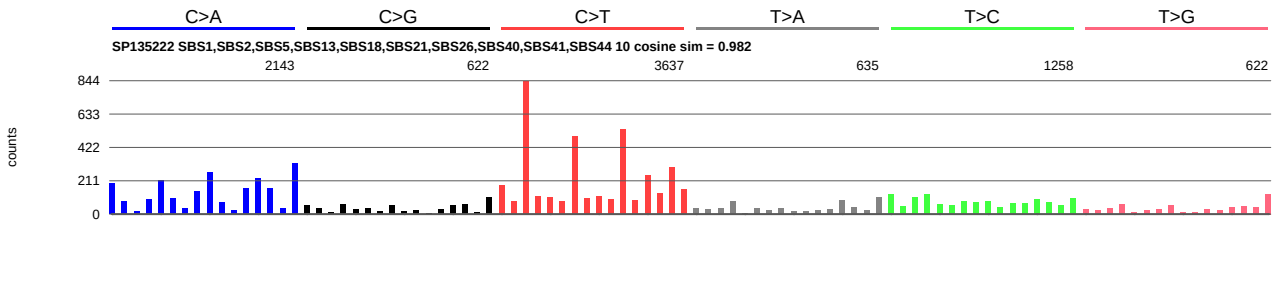
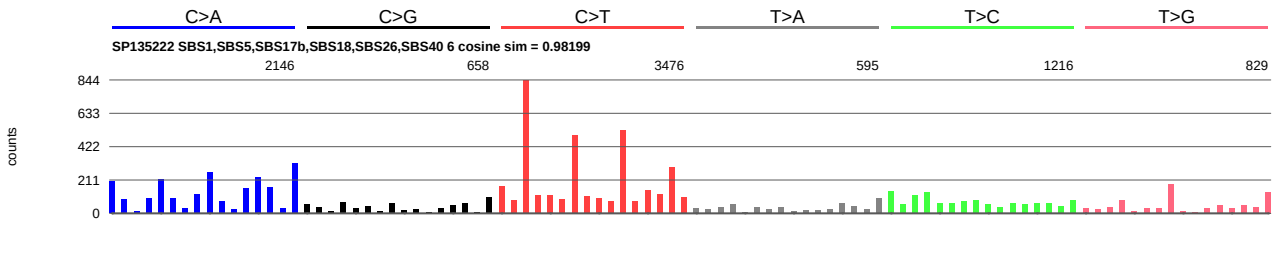
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18,SBS44 7 cosine sim = 0.98203
2133
471
3850
484
1290
689
counts
812
609
406
203
0
8921
6691
4460
2230
0
Signature
SBS44
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

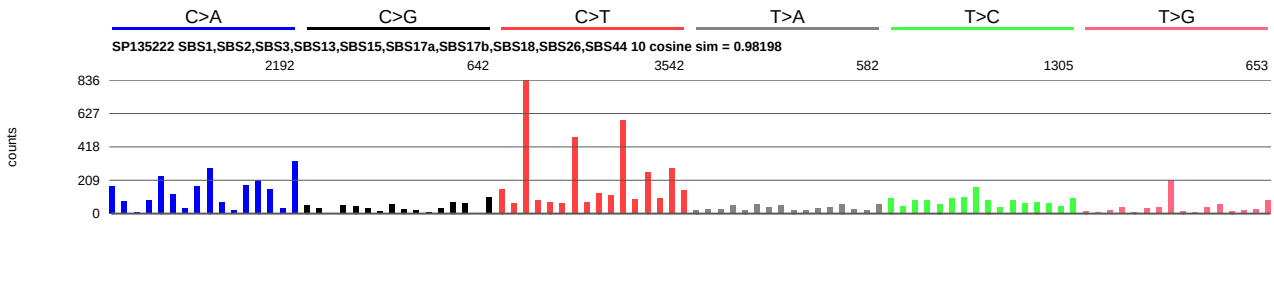

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS21,SBS40 7 cosine sim = 0.98202
2156
654
3727
590
1185
610
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS15
SBS5
SBS2
SBS1

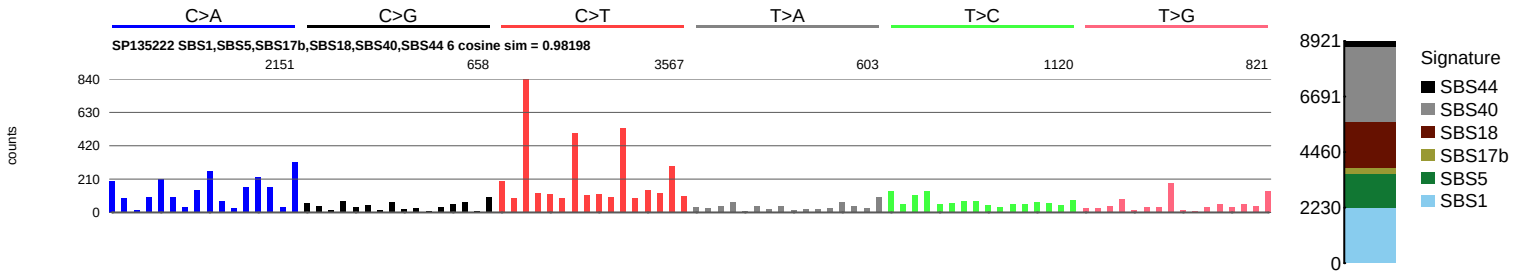

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98201
2158
683
3487
675
1238
681
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS1

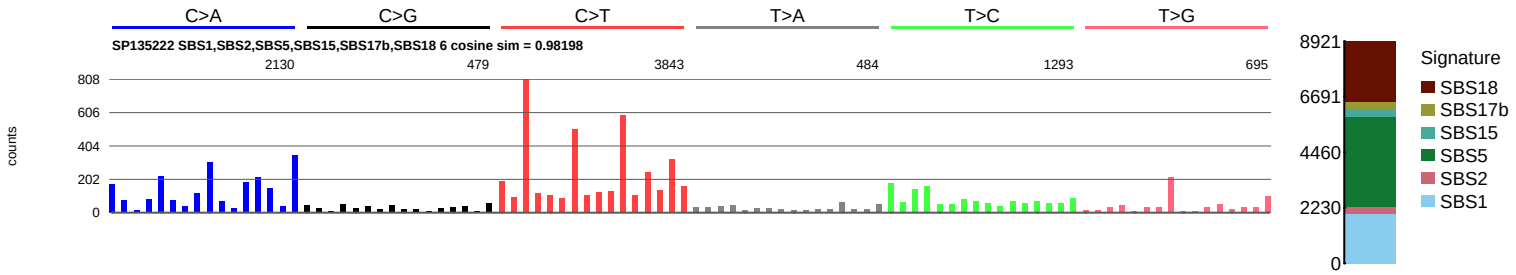

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18,SBS26 7 cosine sim = 0.98201
2128
585
3666
558
1272
706
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS26
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1

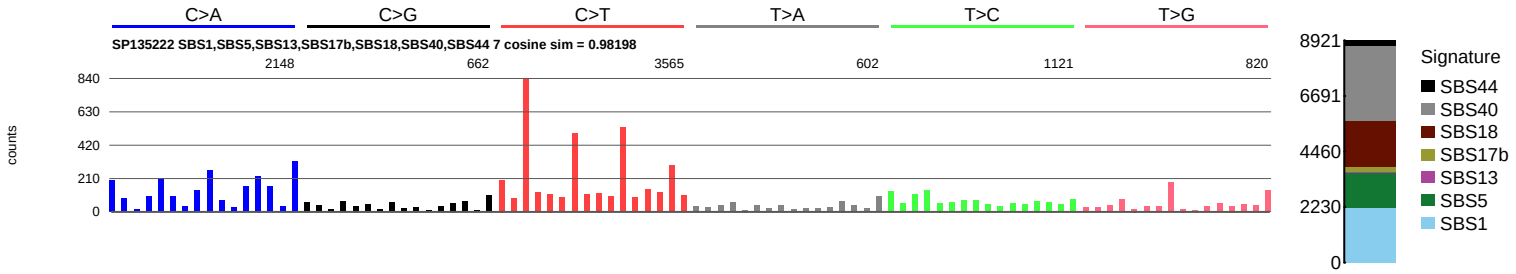

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.982
2143
622
3637
635
1258
622
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1

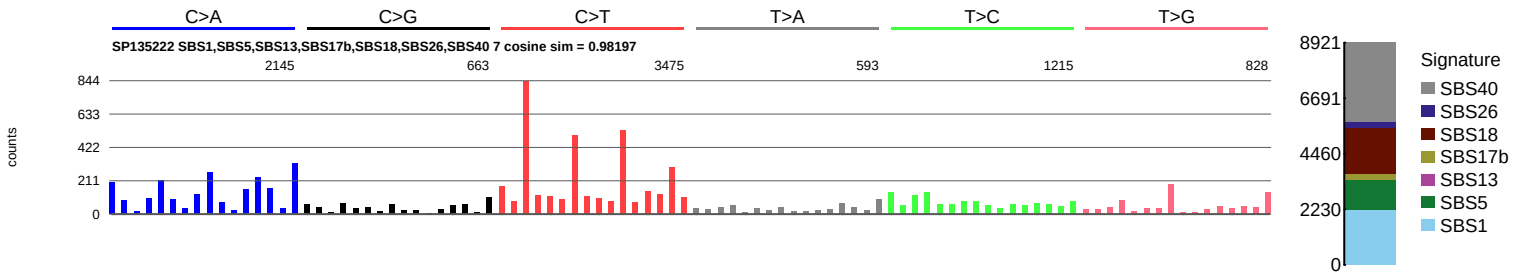

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98199
2146
658
3476
595
1216
829
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS5
SBS1

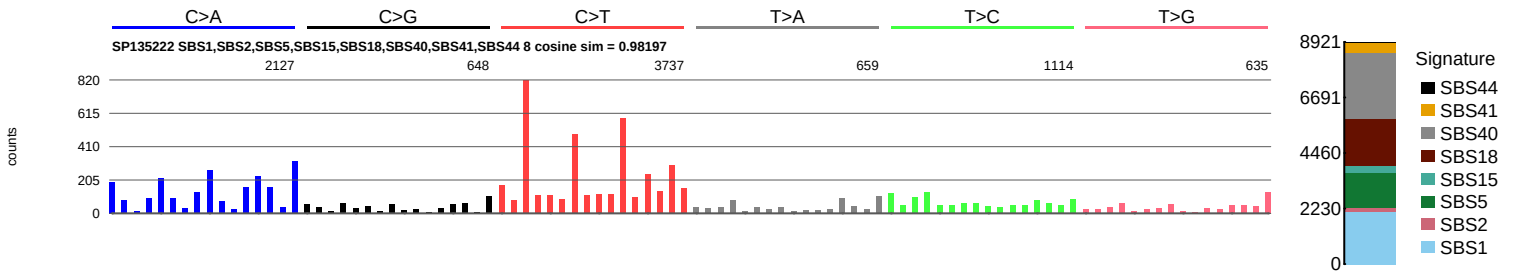

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98198
2192
642
3542
582
1305
653
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

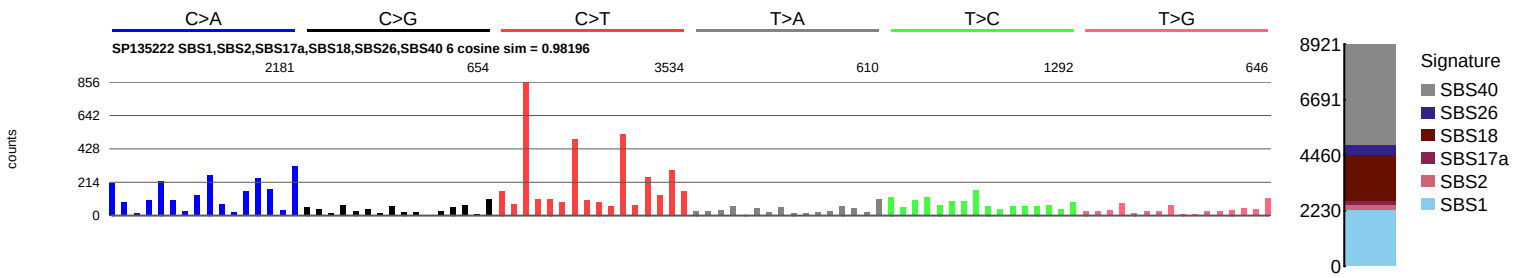

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98198
2151
658
3567
603
1120
821
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS5
SBS1

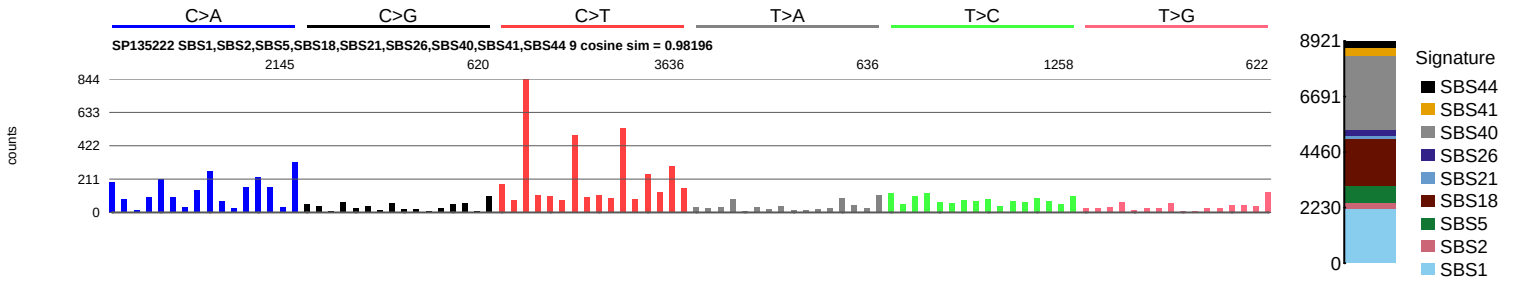

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS17b,SBS18 6 cosine sim = 0.98198
2130
479
3843
484
1293
695
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS15
SBS5
SBS2
SBS1

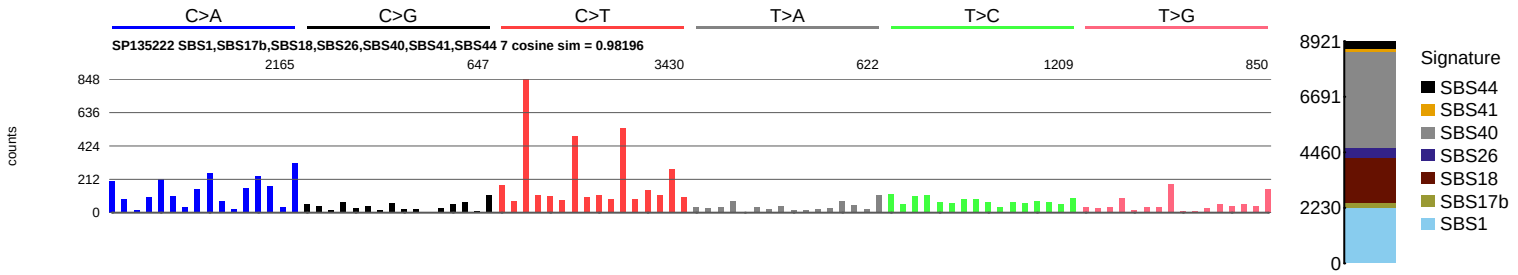

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98198
2148
662
3565
602
1121
820
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS1

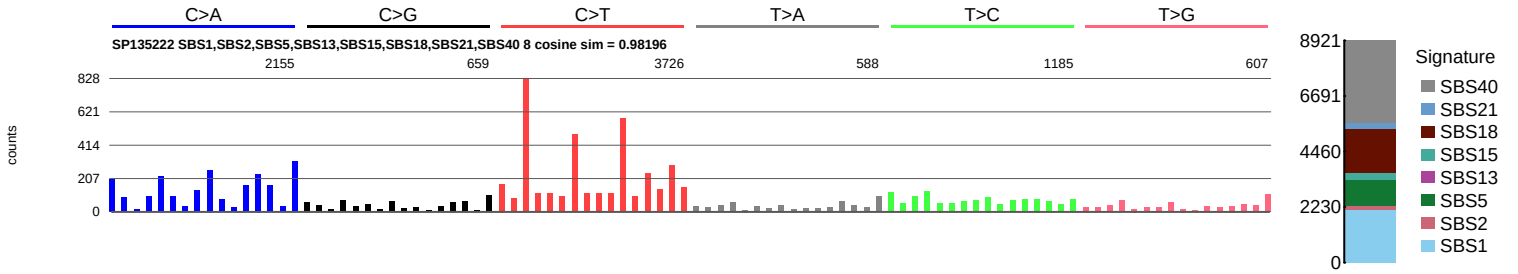

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98197
2145
663
3475
593
1215
828
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98197
2127
648
3737
659
1114
635
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS18,SBS26,SBS40 6 cosine sim = 0.98196
2181
654
3534
610
1292
646
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98196
2145
620
3636
636
1258
622
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98196
2165
647
3430
622
1209
850
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS21,SBS40 8 cosine sim = 0.98196
2155
659
3726
588
1185
607
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
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SBS21
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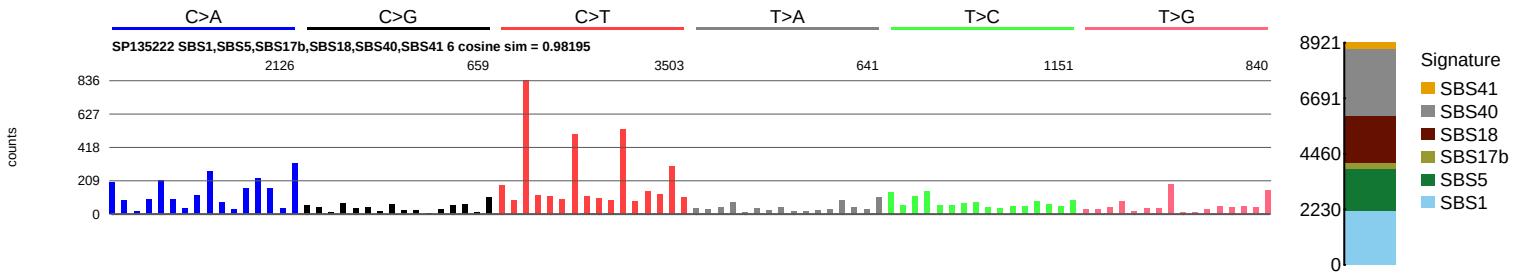
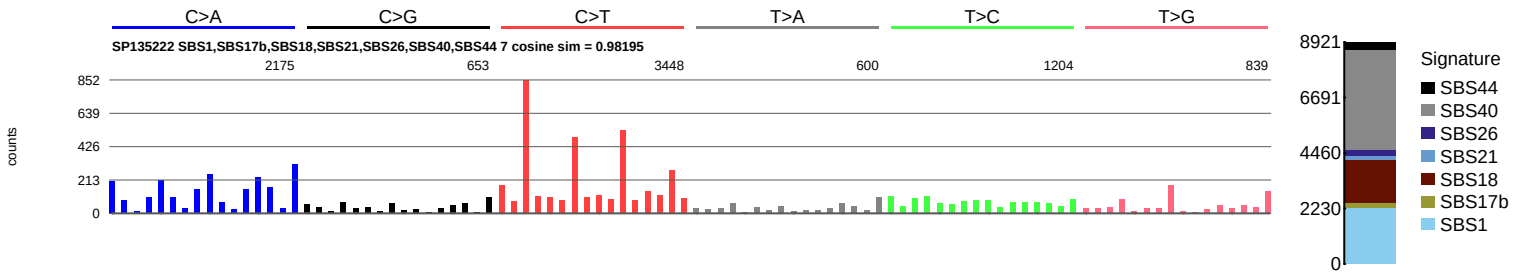
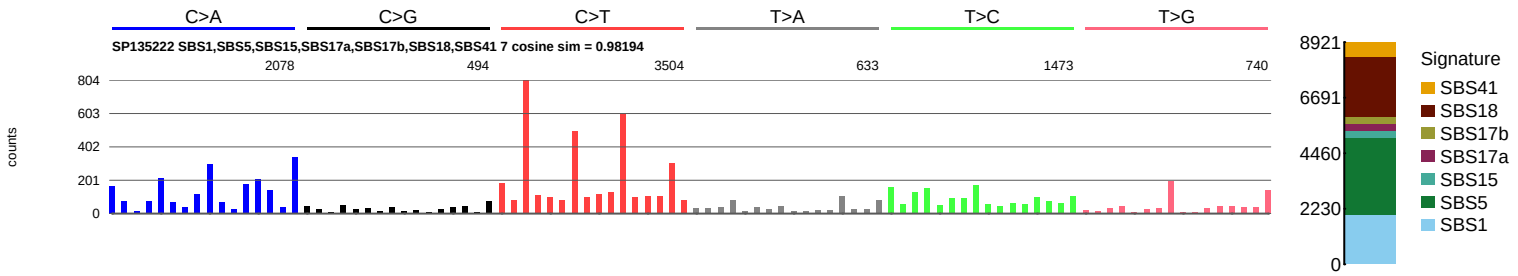
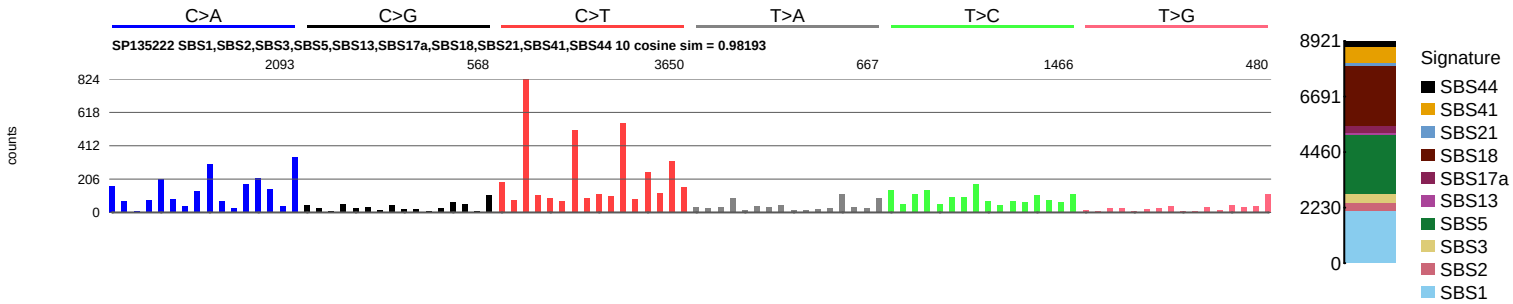
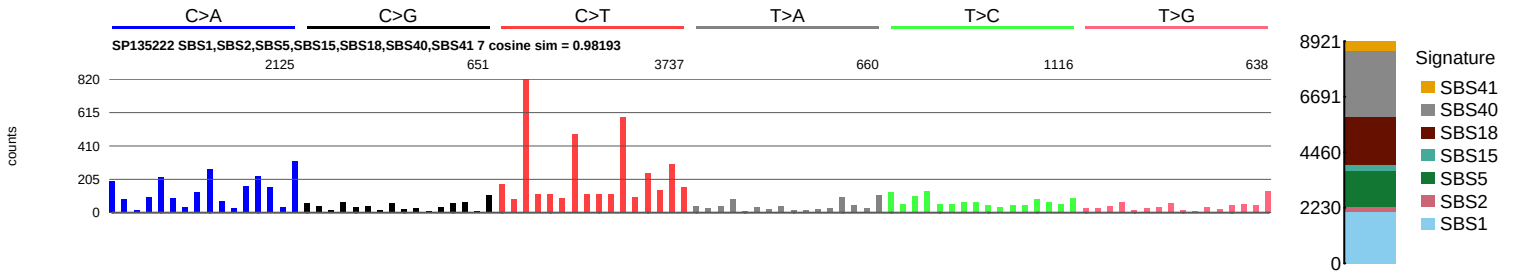
SBS2
SBS1

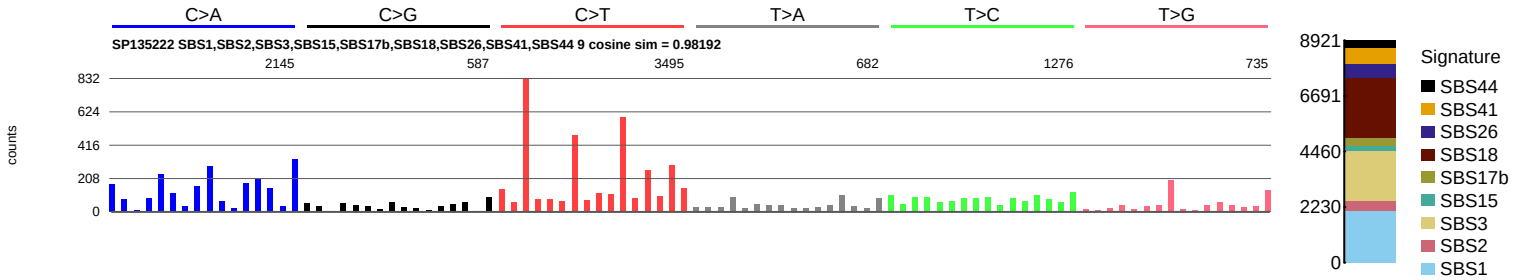

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98195
2126
659
3503
641
1151
840
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS5
SBS1

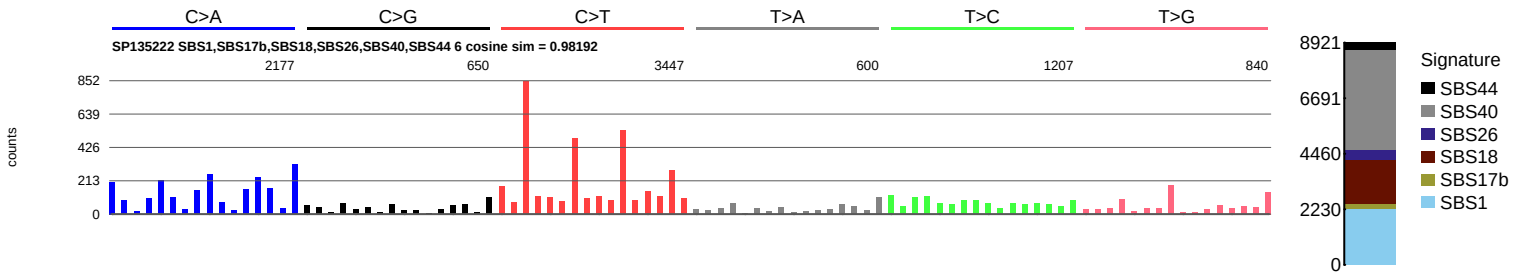

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.98195
2175
653
3448
600
1204
839
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS1

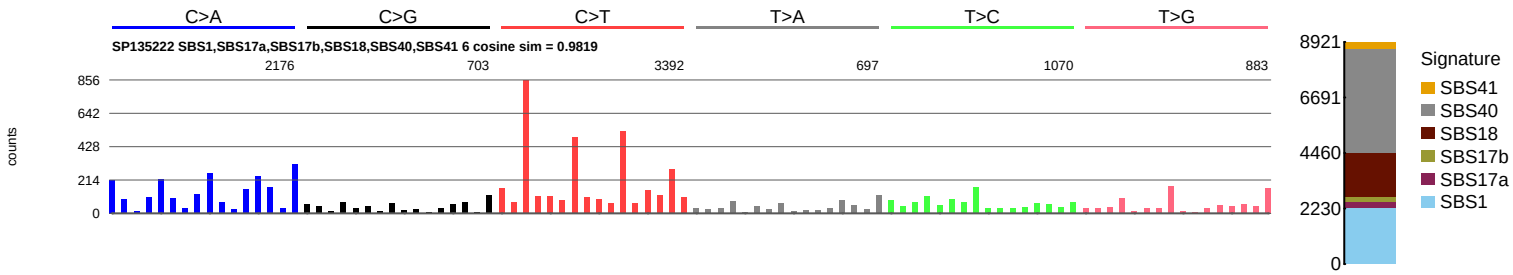

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98194
2078
494
3504
633
1473
740
804
603
402
201
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

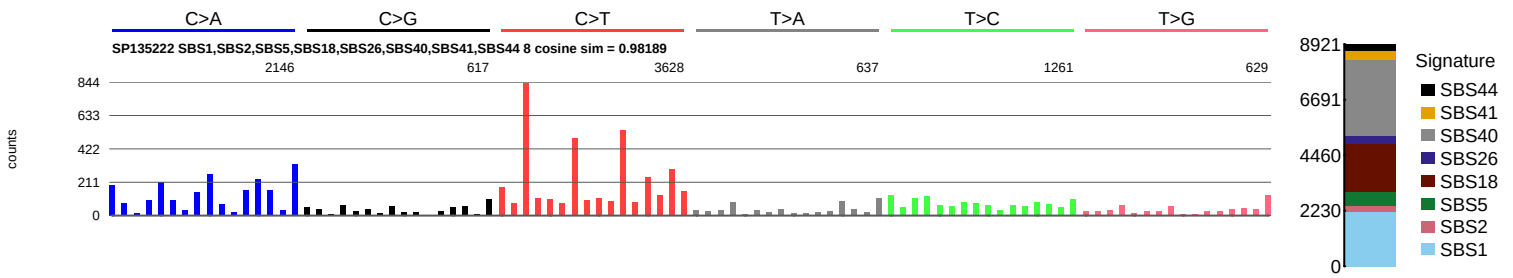

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98193
2093
568
3650
667
1466
480
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

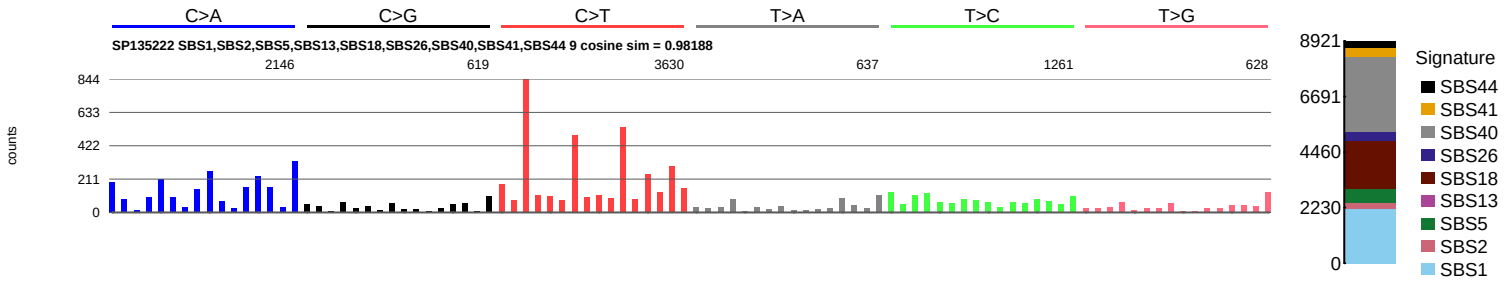

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS40,SBS41 7 cosine sim = 0.98193
2125
651
3737
660
1116
638
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98192
2145
587
3495
682
1276
735
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17b,SBS18,SBS26,SBS40,SBS44 6 cosine sim = 0.98192
2177
650
3447
600
1207
840
counts
852
639
426
213
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.9819
2176
703
3392
697
1070
883
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98189
2146
617
3628
637
1261
629
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS5
SBS2
SBS1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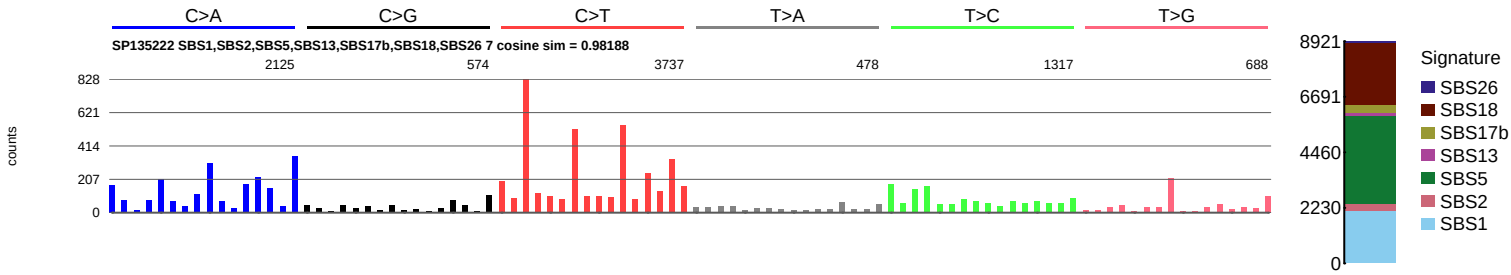
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98188
2146
619
3630
637
1261
628
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18,SBS26 7 cosine sim = 0.98188
2125
574
3737
478
1317
688
counts
828
621
414
207
0
8921
6691
4460
2230
0
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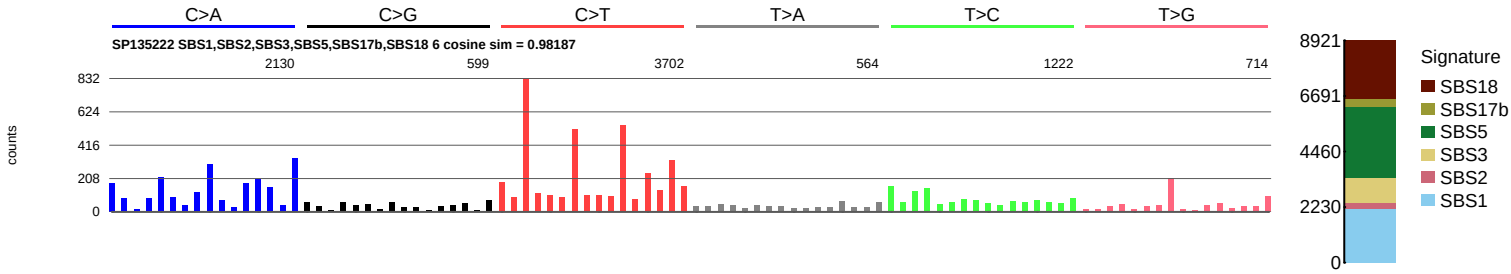
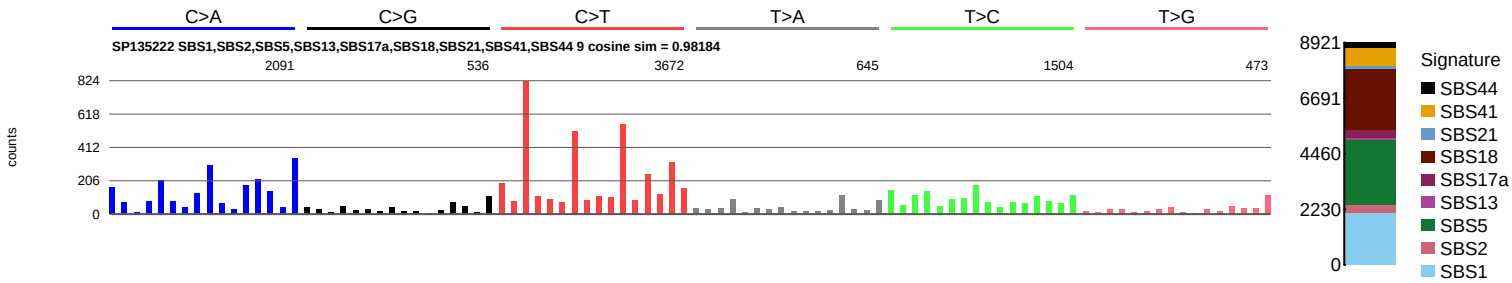
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17b,SBS18 6 cosine sim = 0.98187
2130
599
3702
564
1222
714
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS5
SBS3
SBS2
SBS1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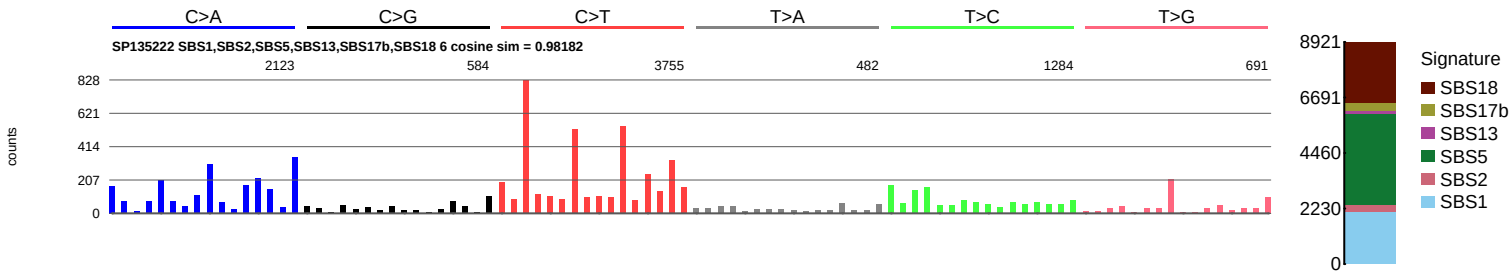
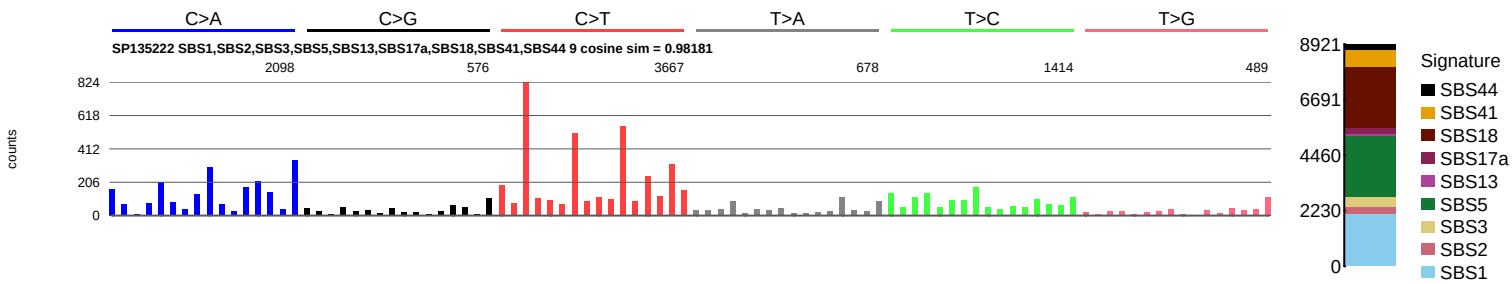
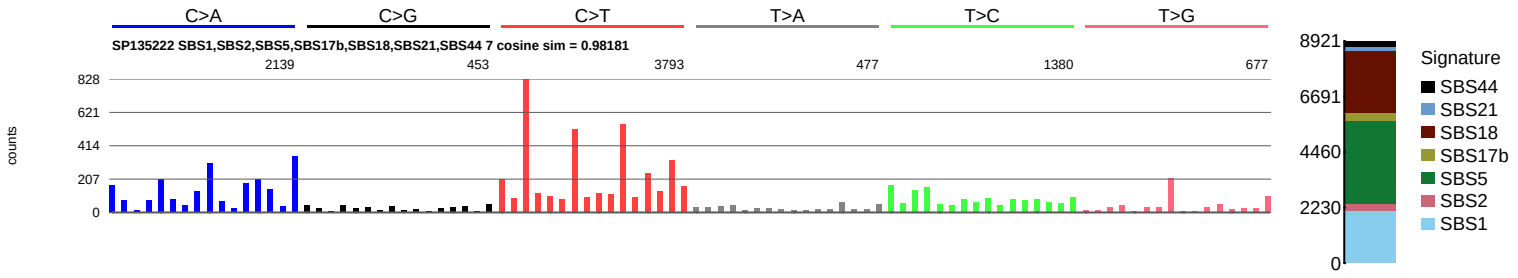
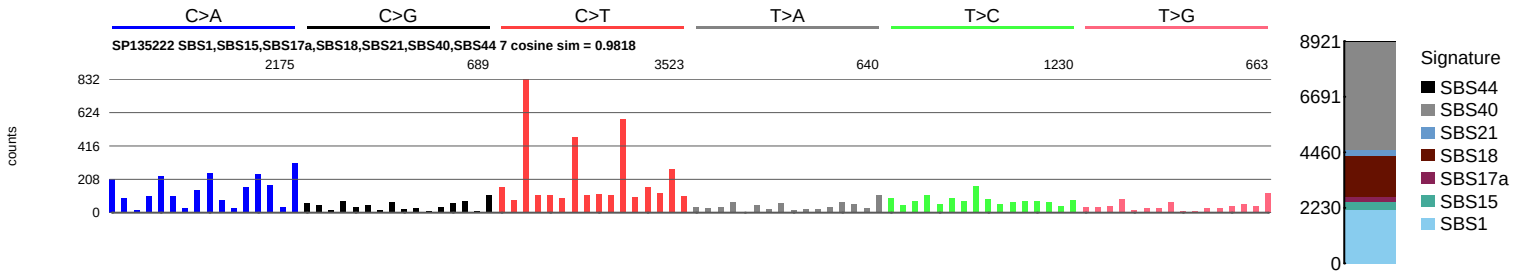
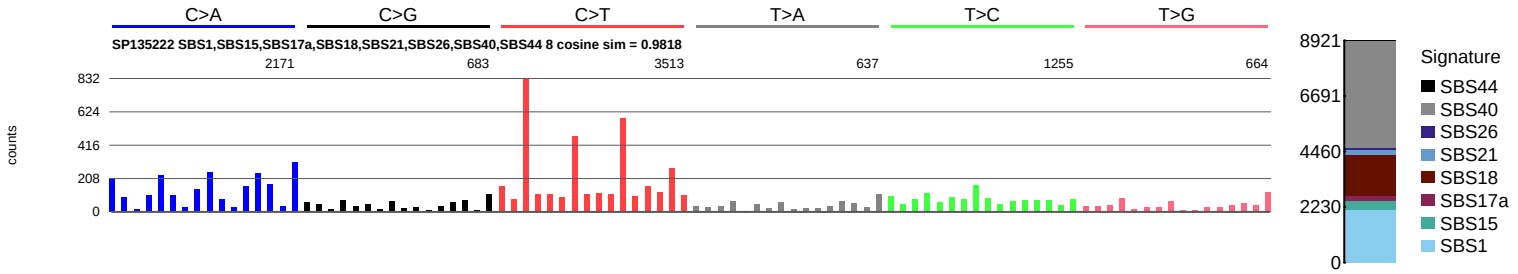
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98184
2091
536
3672
645
1504
473
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1

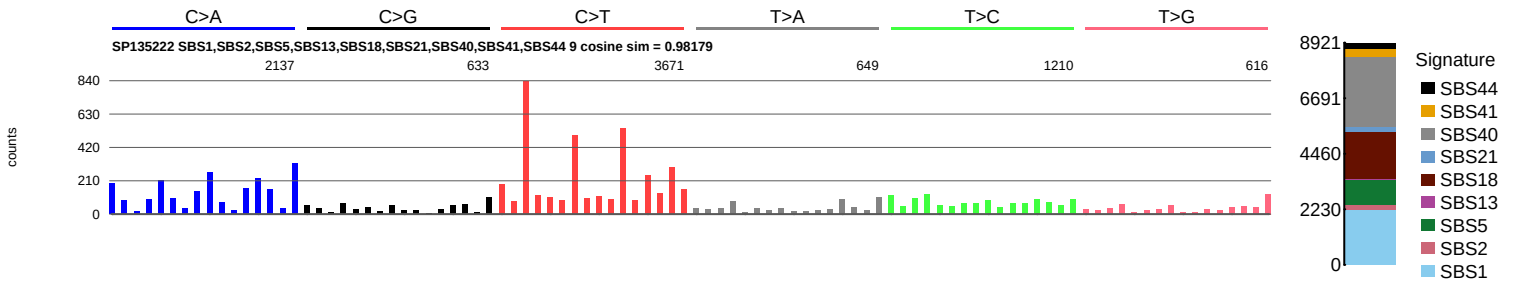

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17b,SBS18 6 cosine sim = 0.98182
2123
584
3755
482
1284
691
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS13
SBS5
SBS2
SBS1

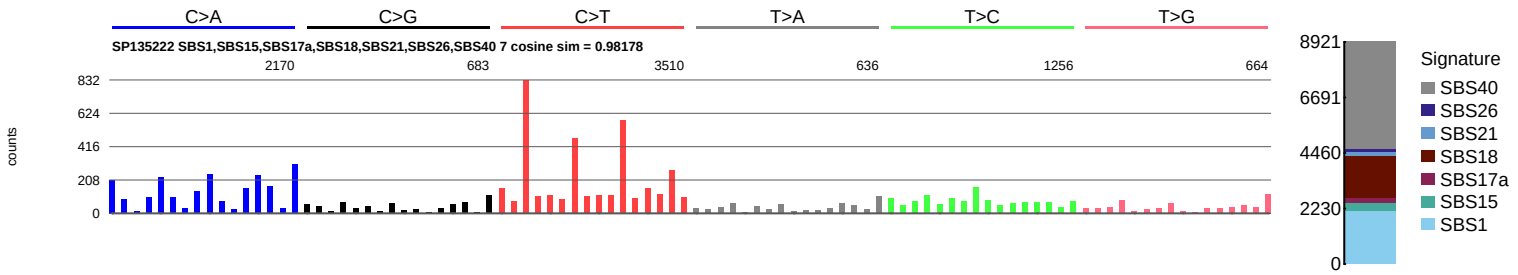

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS41,SBS44 9 cosine sim = 0.98181
2098
576
3667
678
1414
489
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98181
2139
453
3793
477
1380
677
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.9818
2175
689
3523
640
1230
663
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.9818
2171
683
3513
637
1255
664
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98179
2137
633
3671
649
1210
616
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98178
2170
683
3510
636
1256
664
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
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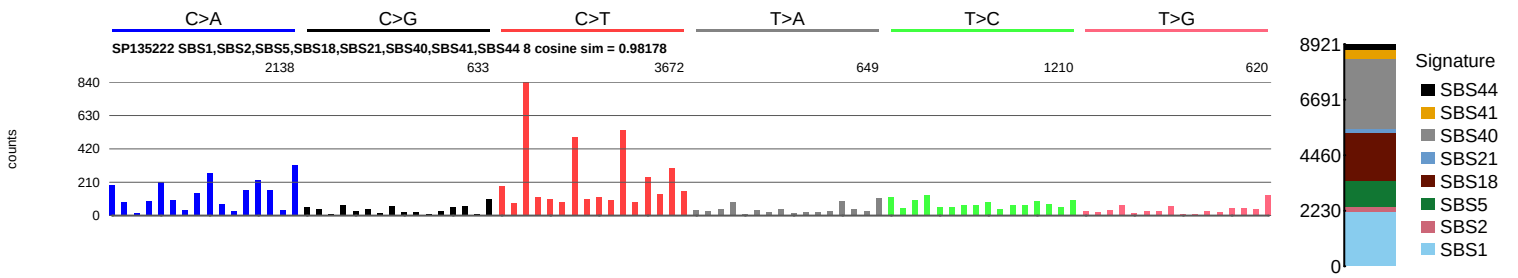
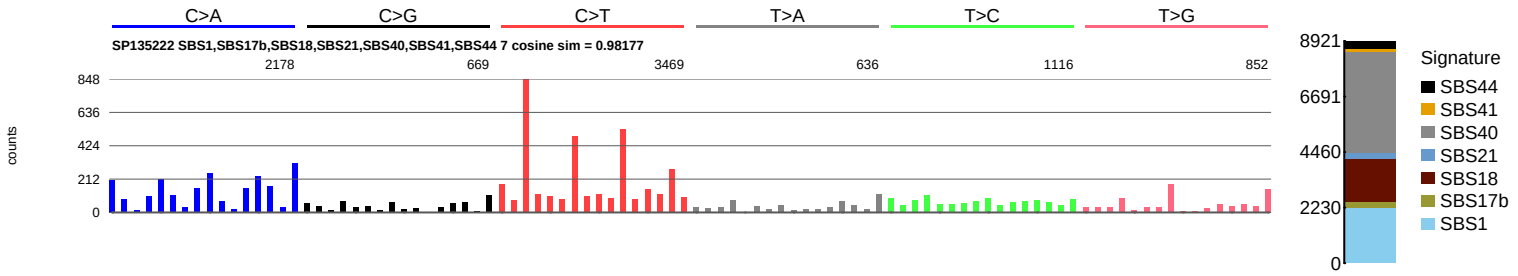
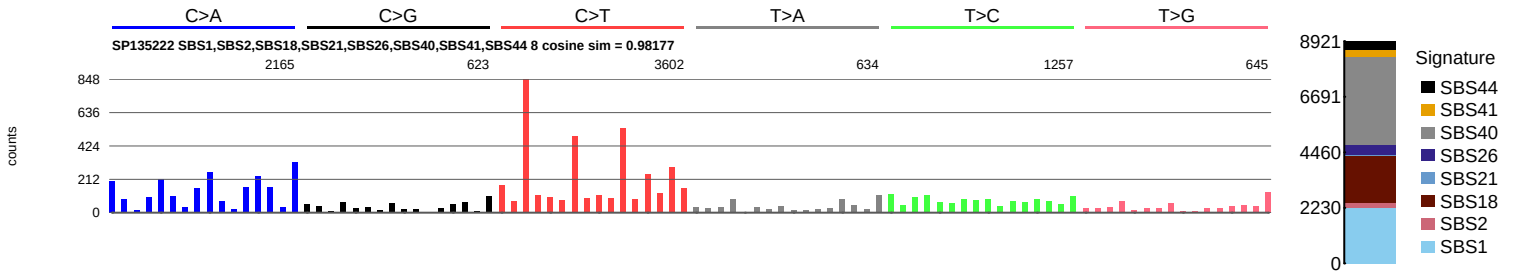
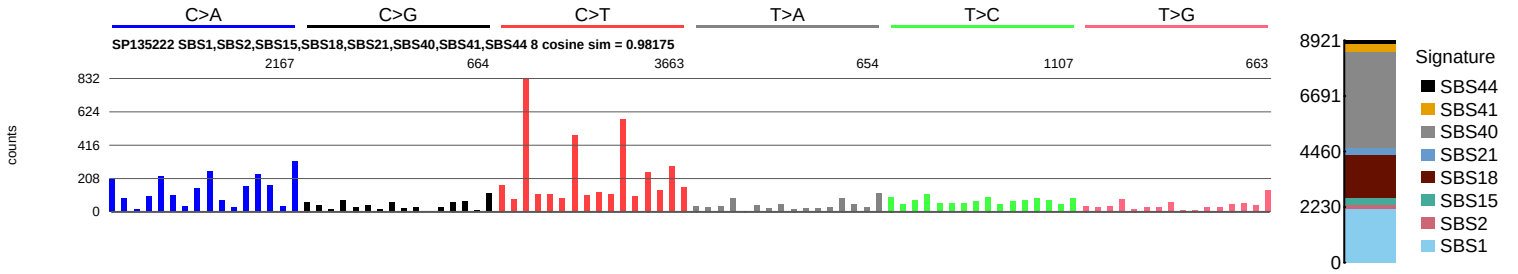
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98178
2138
633
3672
649
1210
620
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98177
2178
669
3469
636
1116
852
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98177
2165
623
3602
634
1257
645
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS2
SBS1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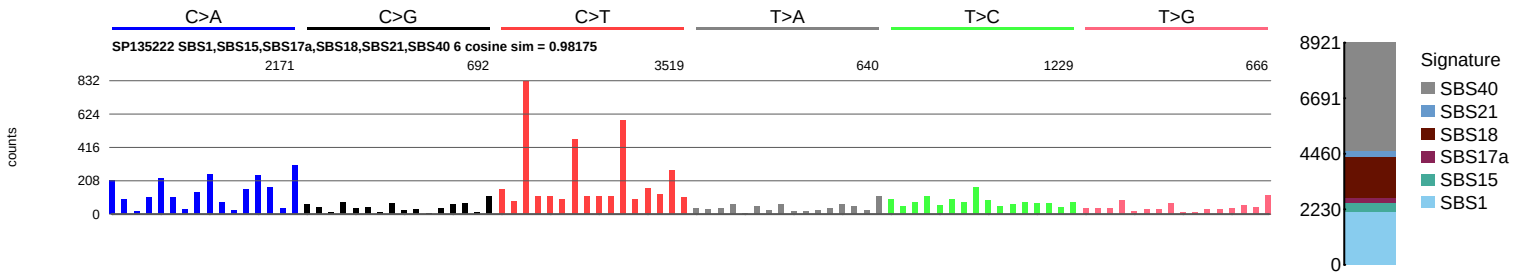
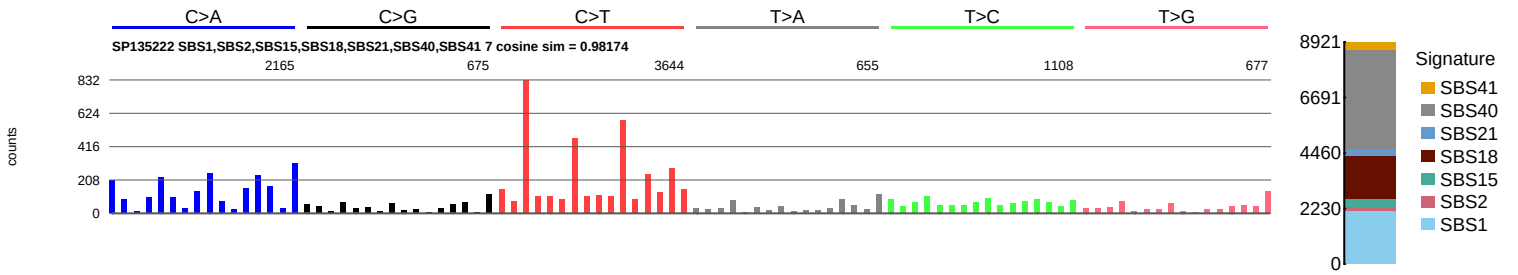
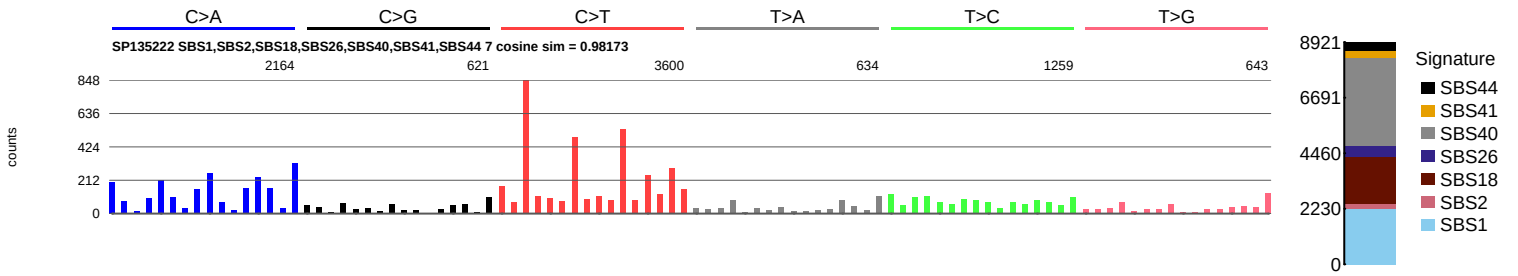
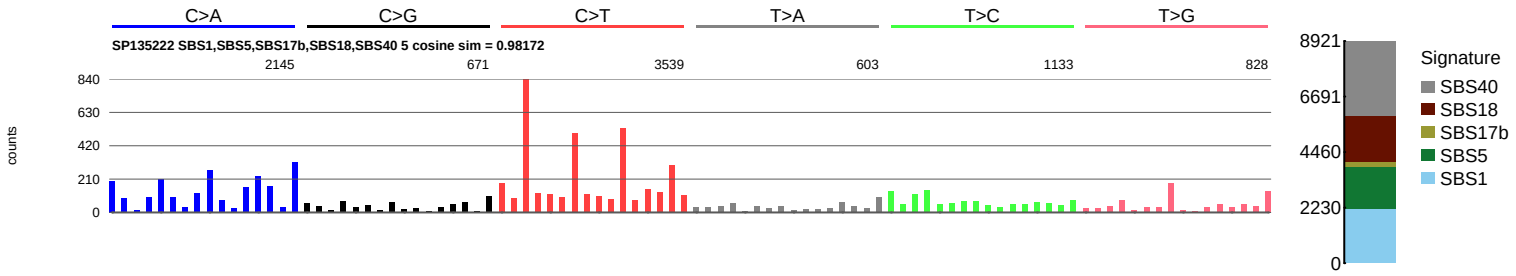
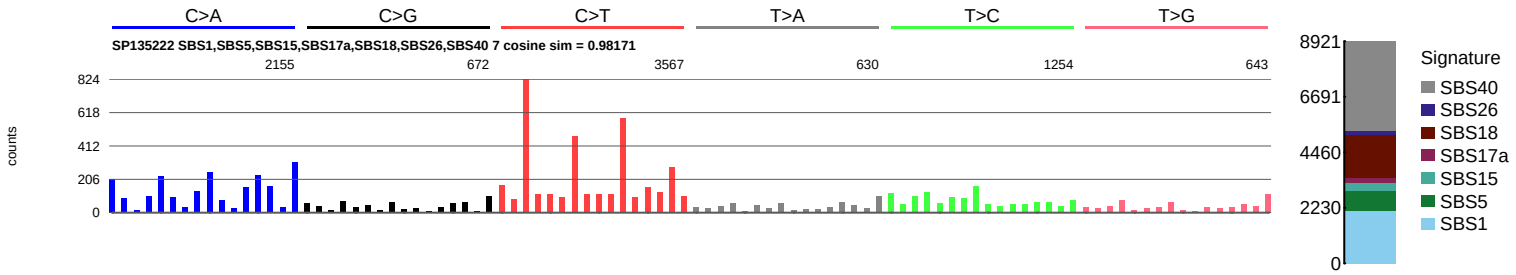
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98175
2167
664
3663
654
1107
663
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS15
SBS2
SBS1

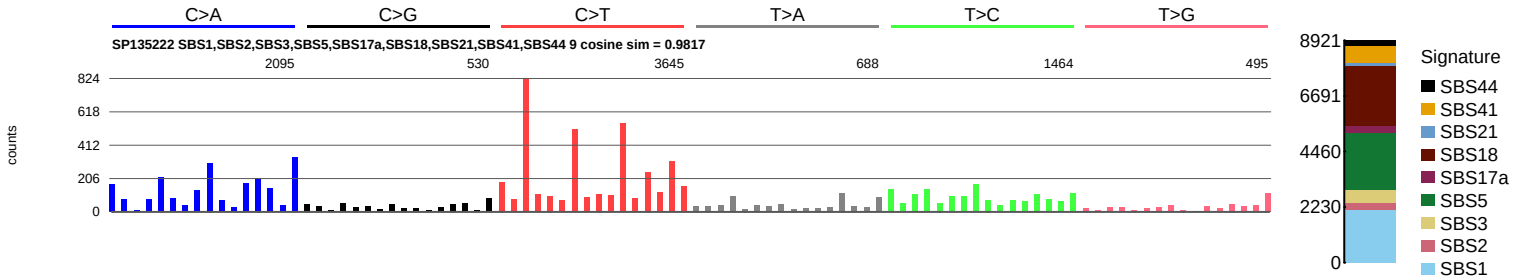

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS18,SBS21,SBS40 6 cosine sim = 0.98175
2171
692
3519
640
1229
666
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS15
SBS1

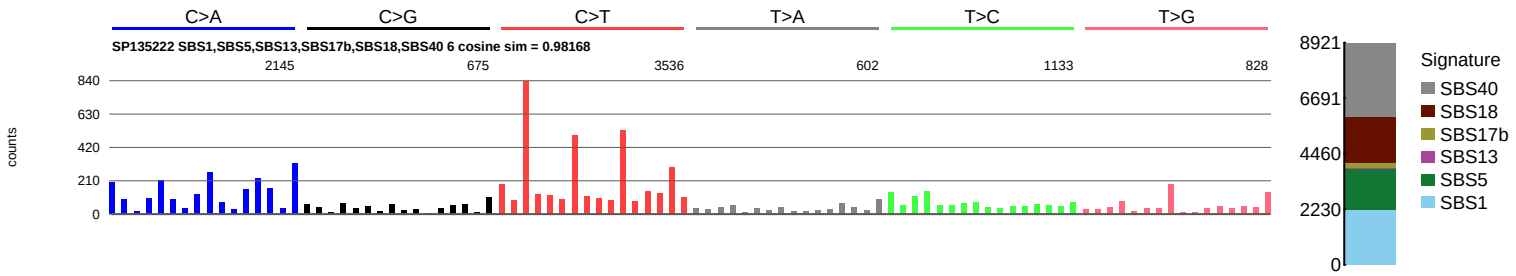

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98174
2165
675
3644
655
1108
677
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS2
SBS1

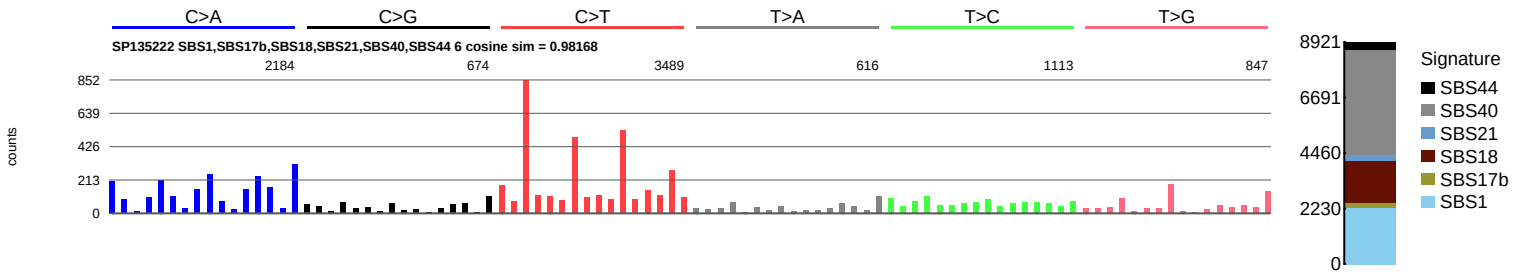

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS18,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98173
2164
621
3600
634
1259
643
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS2
SBS1

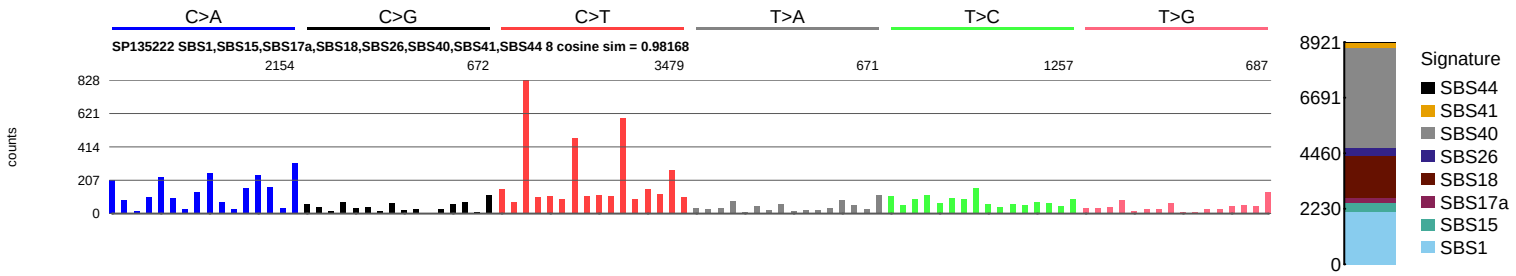

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17b,SBS18,SBS40 5 cosine sim = 0.98172
2145
671
3539
603
1133
828
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS5
SBS1

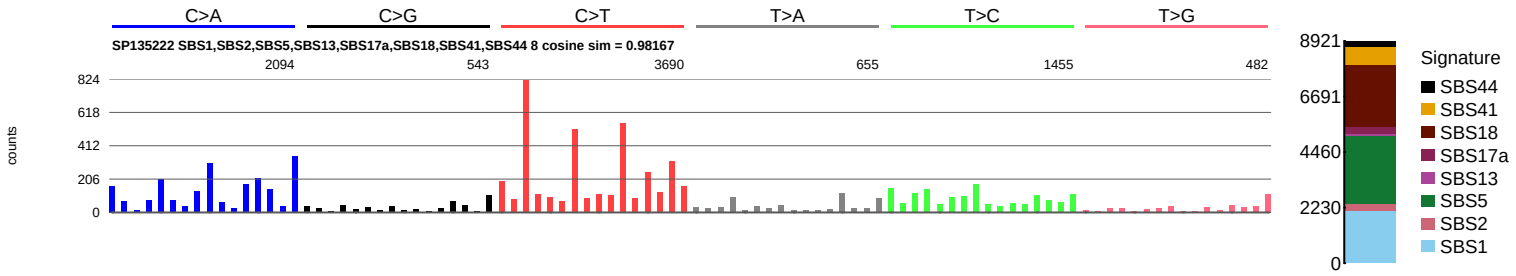

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS26,SBS40 7 cosine sim = 0.98171
2155
672
3567
630
1254
643
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS5
SBS1

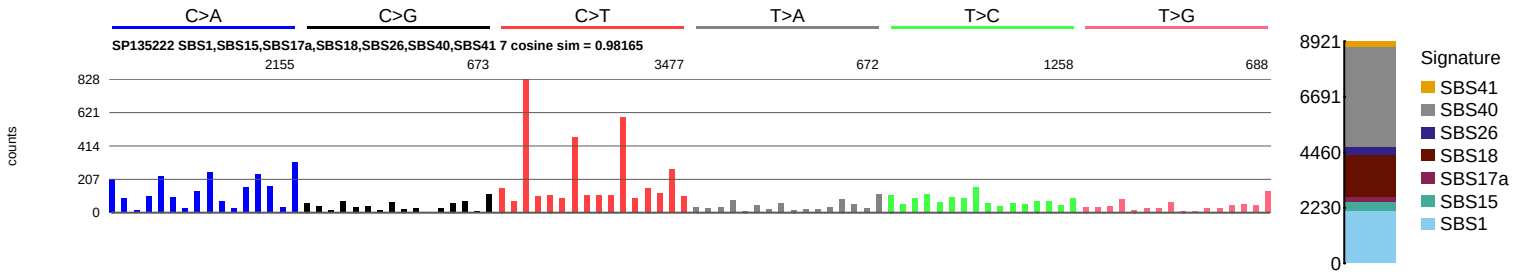

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.9817
2095
530
3645
688
1464
495
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17b,SBS18,SBS40 6 cosine sim = 0.98168
2145
675
3536
602
1133
828
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17b,SBS18,SBS21,SBS40,SBS44 6 cosine sim = 0.98168
2184
674
3489
616
1113
847
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98168
2154
672
3479
671
1257
687
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS41,SBS44 8 cosine sim = 0.98167
2094
543
3690
655
1455
482
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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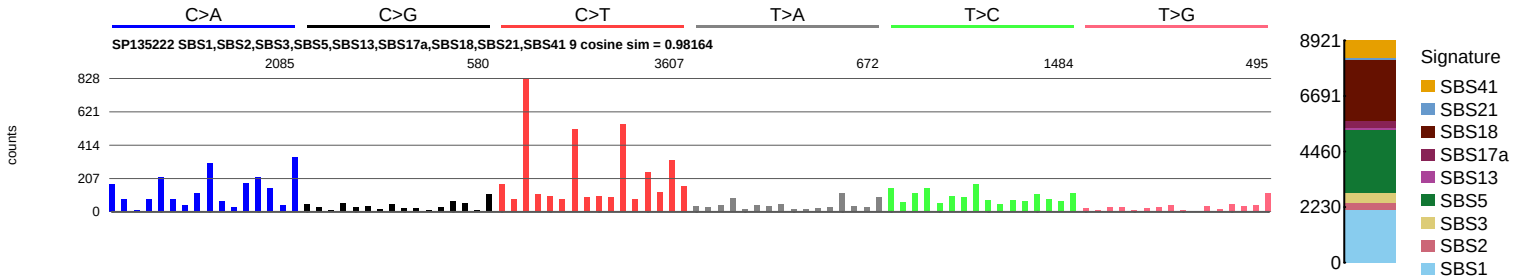
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98165
2155
673
3477
672
1258
688
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS1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T>A
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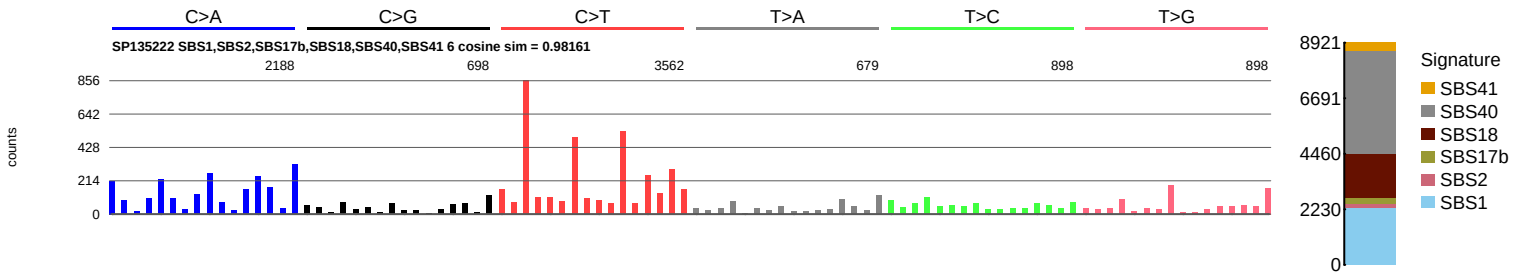
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS41 9 cosine sim = 0.98164
2085
580
3607
672
1484
495
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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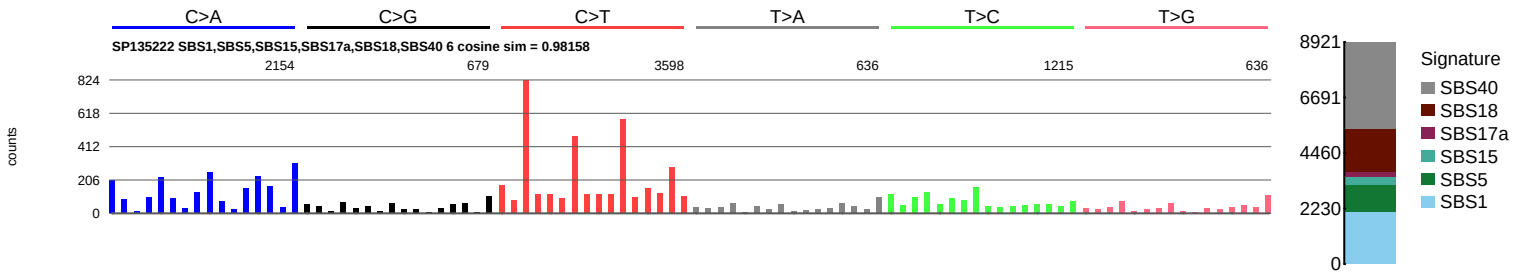
SP135222 SBS1,SBS2,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98161
2188
698
3562
679
898
898
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS2
SBS1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C>G
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T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS17a,SBS18,SBS40 6 cosine sim = 0.98158
2154
679
3598
636
1215
636
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS40
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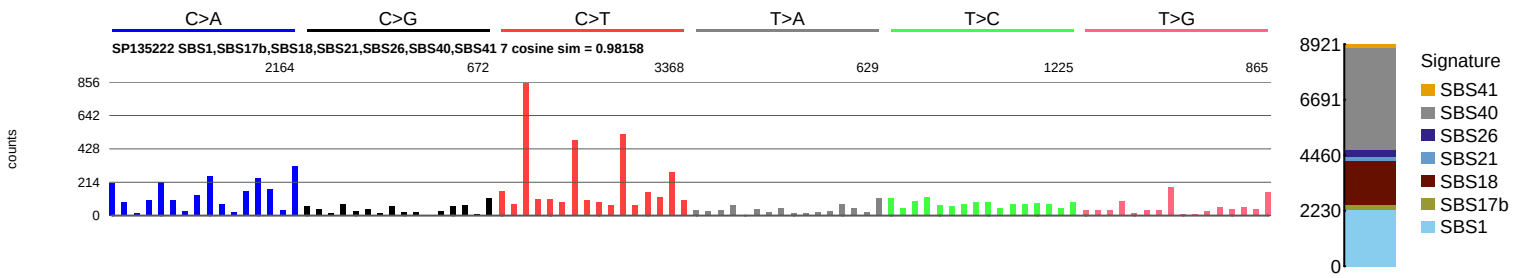
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.98158
2164
672
3368
629
1225
865
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
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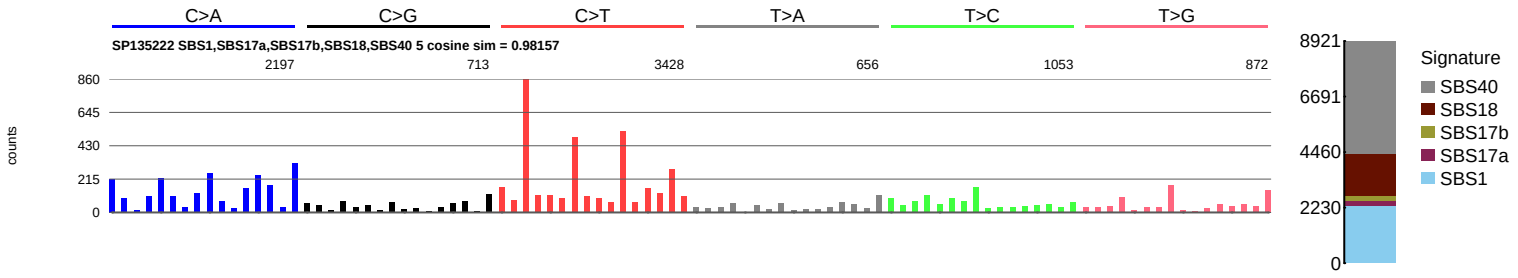
SBS17b
SBS1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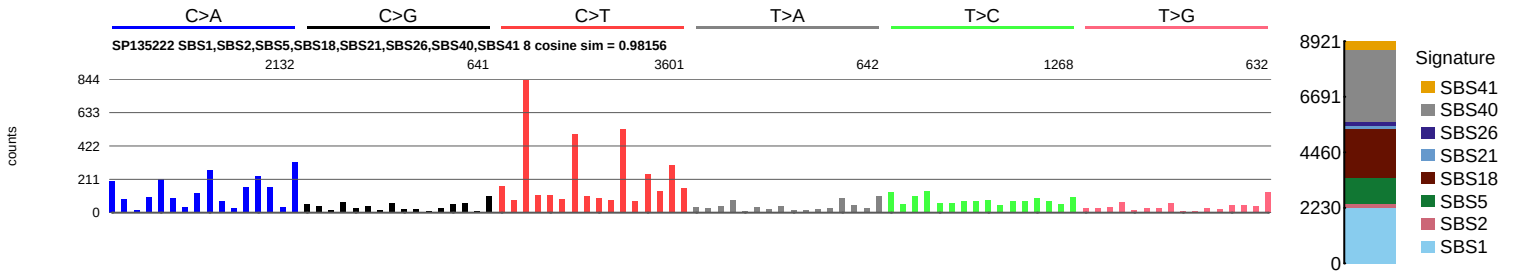
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.98157
2197
713
3428
656
1053
872
860
645
430
215
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS1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C>G
C>T
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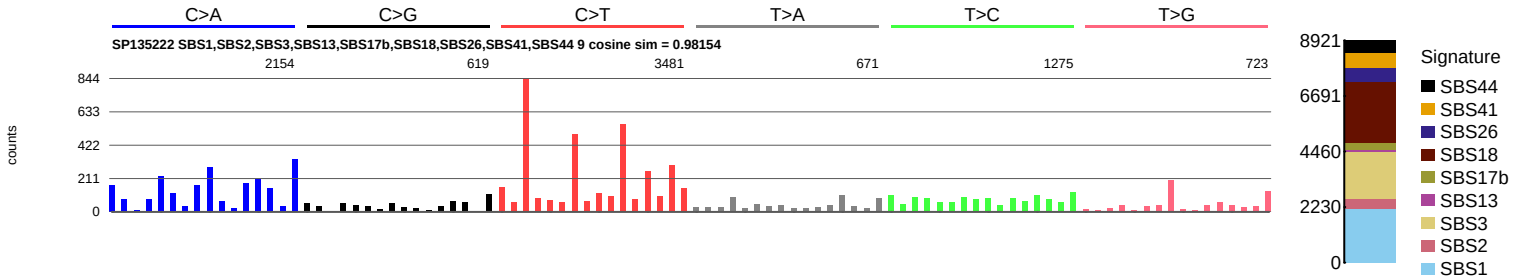
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98156
2132
641
3601
642
1268
632
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS5
SBS2
SBS1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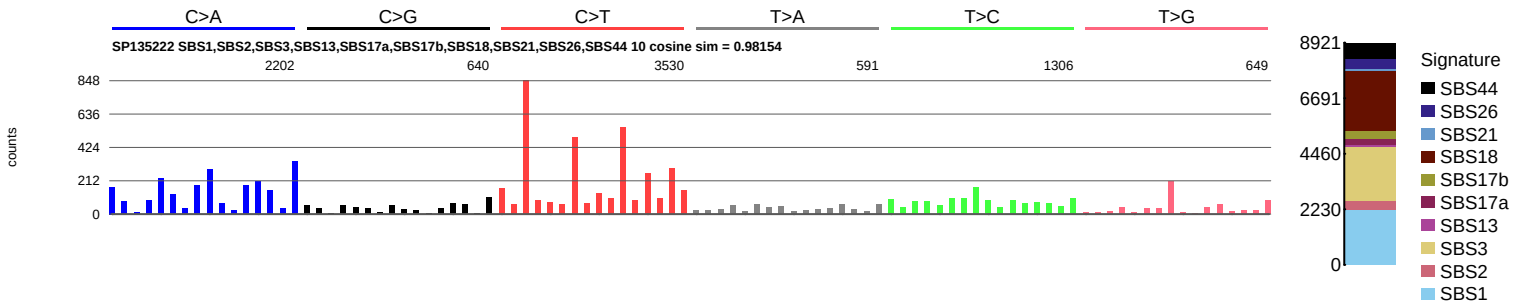
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98154
2154
619
3481
671
1275
723
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS13
SBS3
SBS2
SBS1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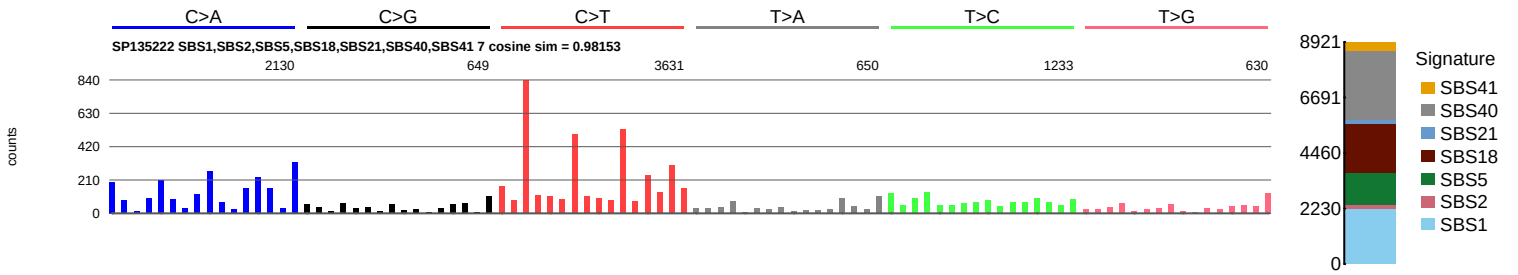
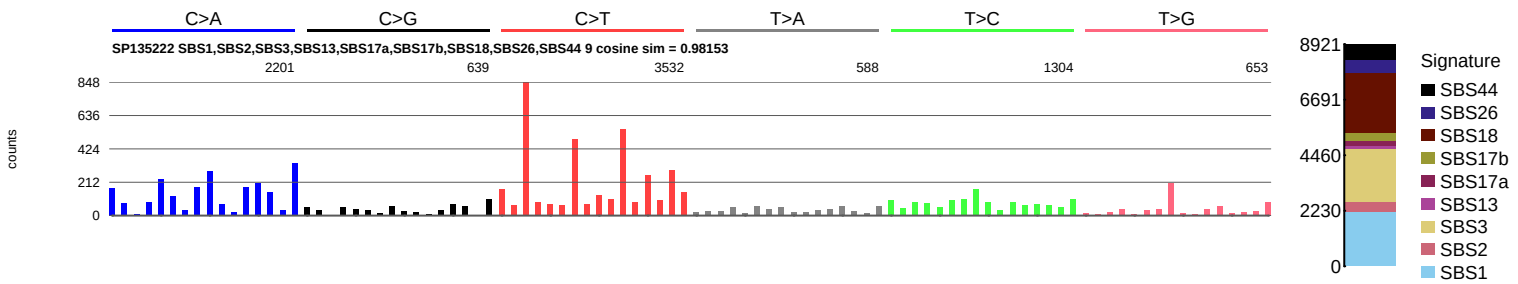
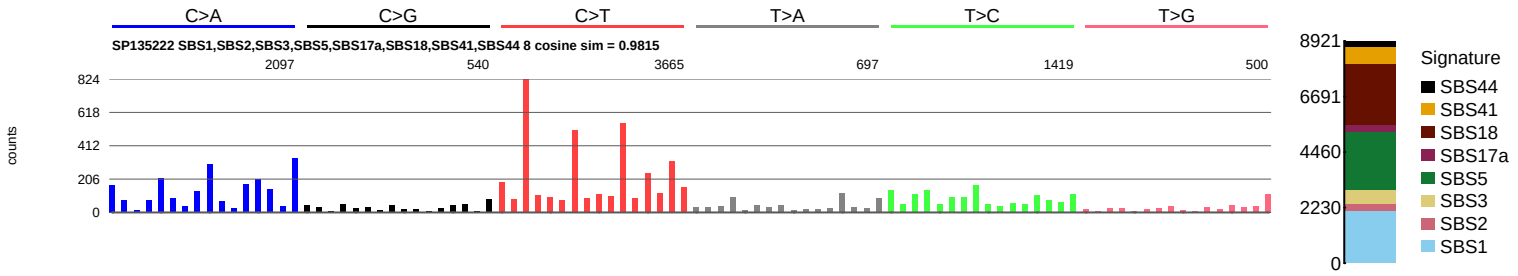
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98154
2202
640
3530
591
1306
649
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98153
2130
649
3631
650
1233
630
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98153
2201
639
3532
588
1304
653
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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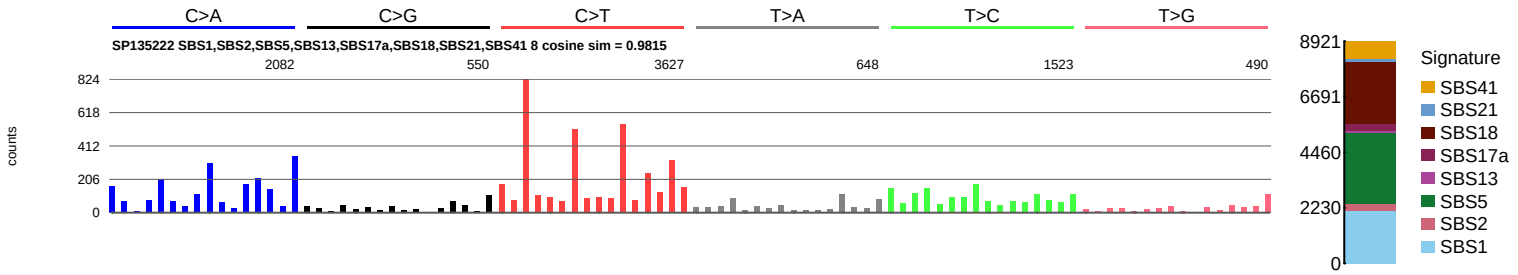
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS41,SBS44 8 cosine sim = 0.9815
2097
540
3665
697
1419
500
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1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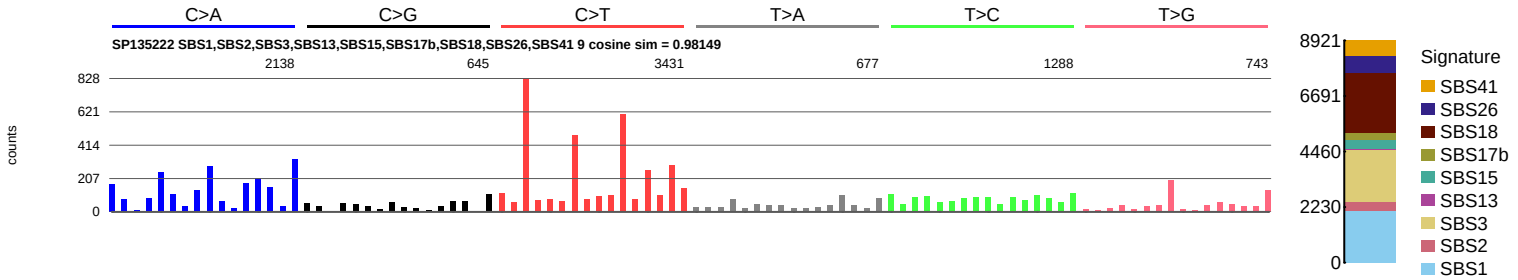
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS21,SBS41 8 cosine sim = 0.9815
2082
550
3627
648
1523
490
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1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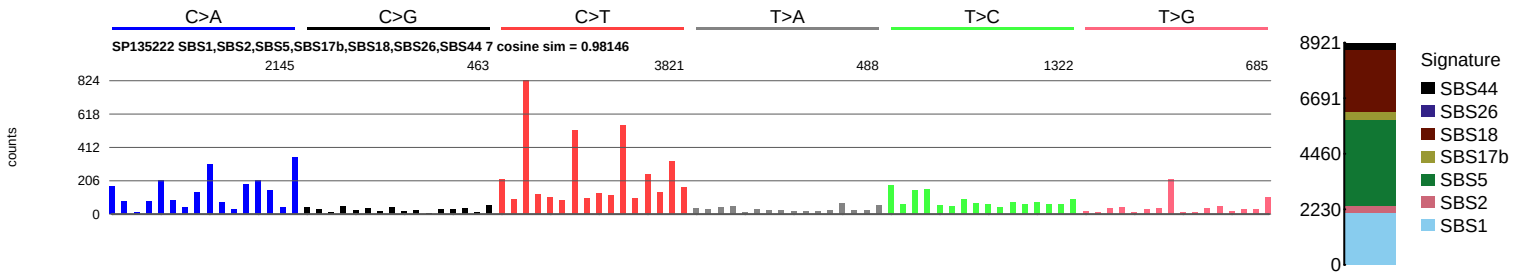
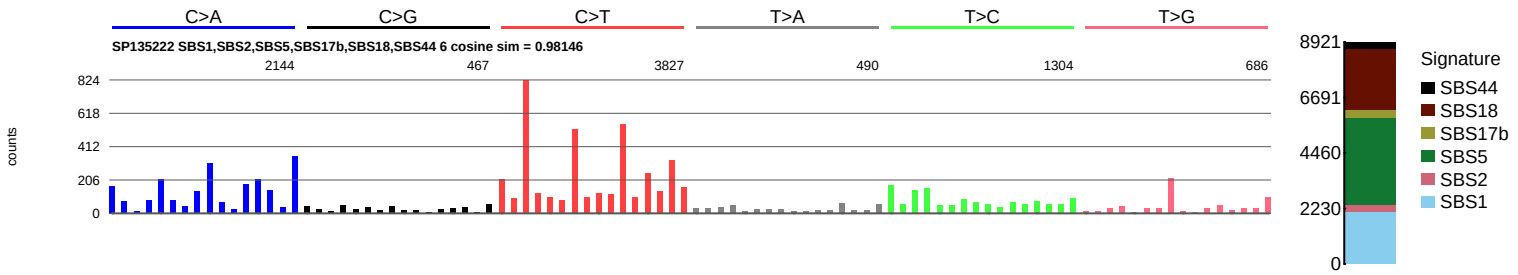
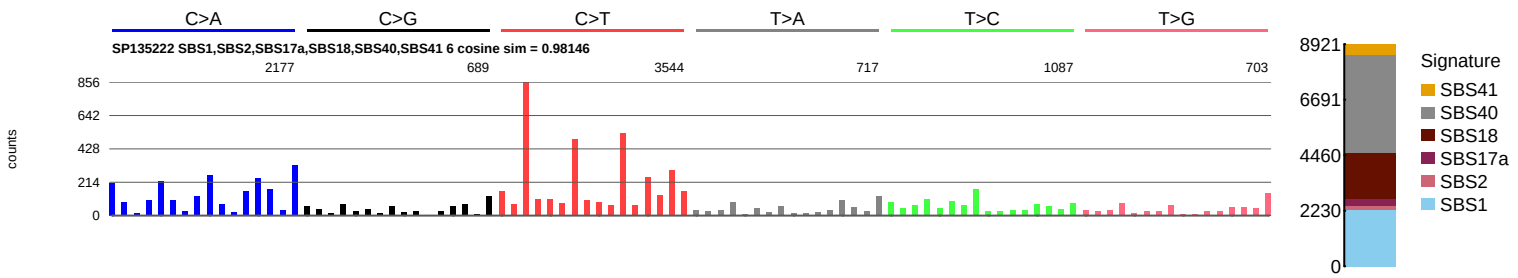
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98149
2138
645
3431
677
1288
743
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98146
2145
463
3821
488
1322
685
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS44 6 cosine sim = 0.98146
2144
467
3827
490
1304
686
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS18
SBS17b
SBS5
SBS2
SBS1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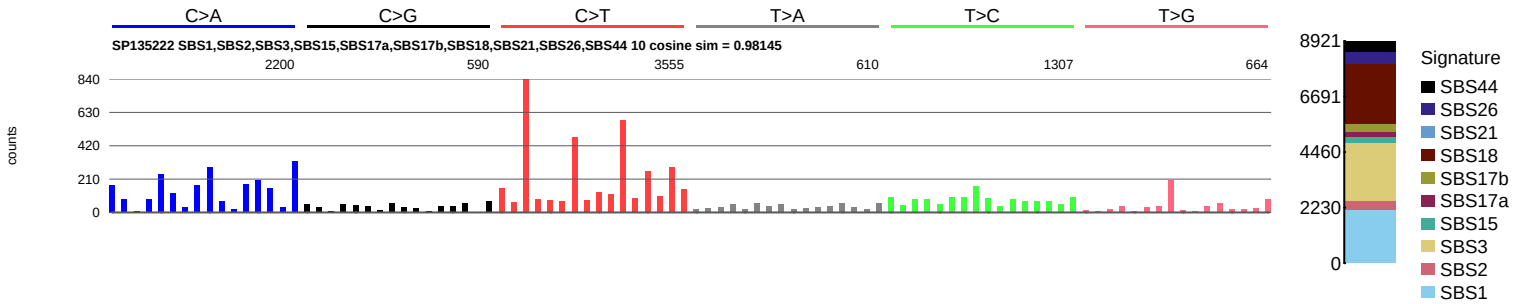
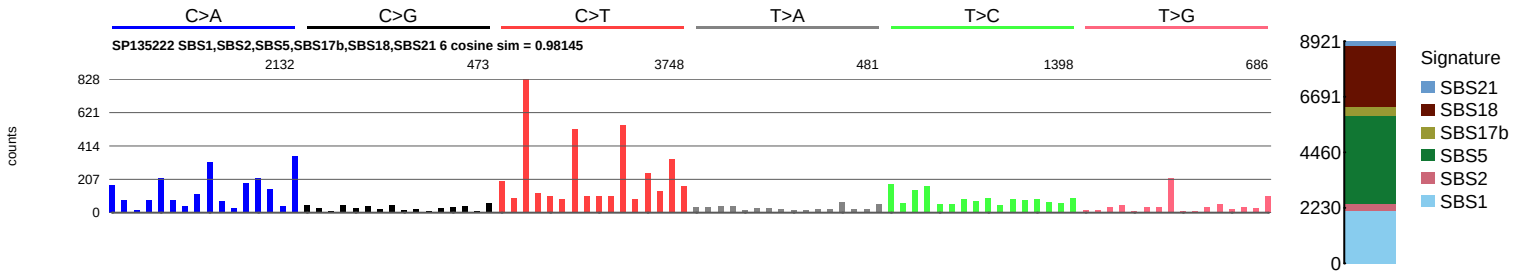
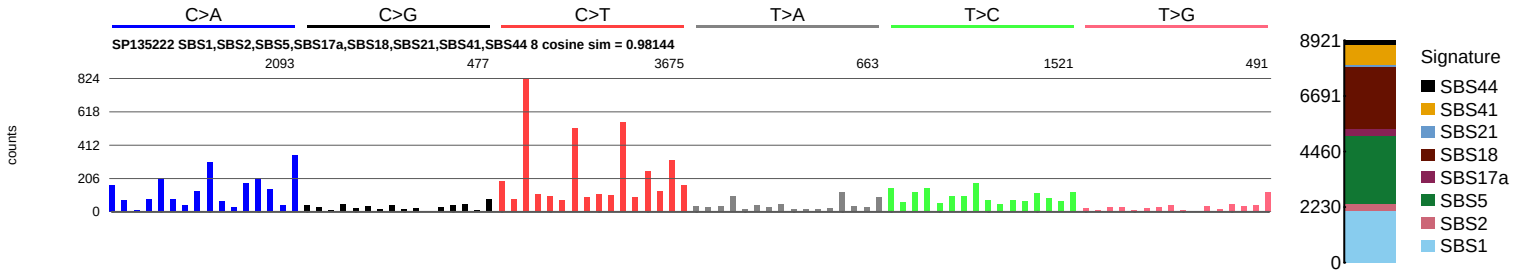
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS18,SBS40,SBS41 6 cosine sim = 0.98146
2177
689
3544
717
1087
703
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98145
2200
590
3555
610
1307
664
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS21 6 cosine sim = 0.98145
2132
473
3748
481
1398
686
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS21
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98144
2093
477
3675
663
1521
491
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
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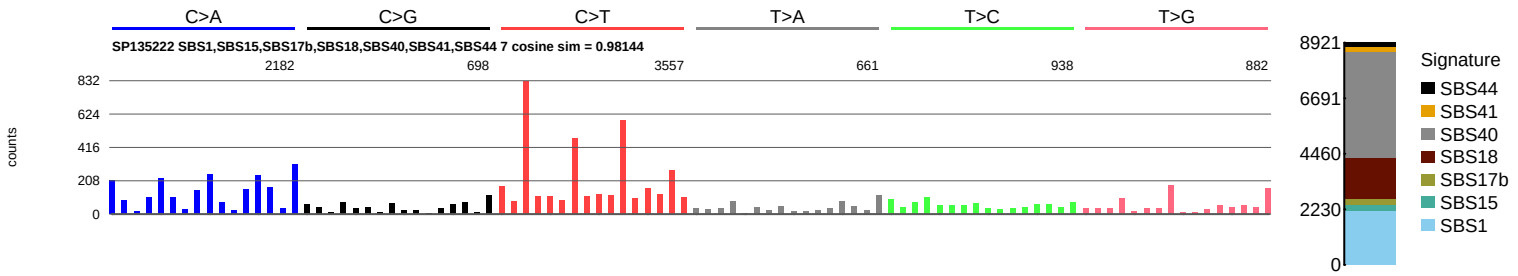
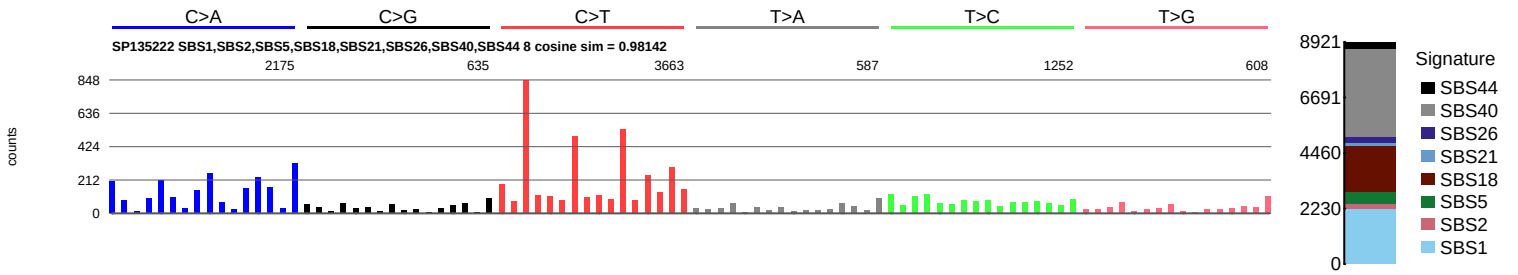
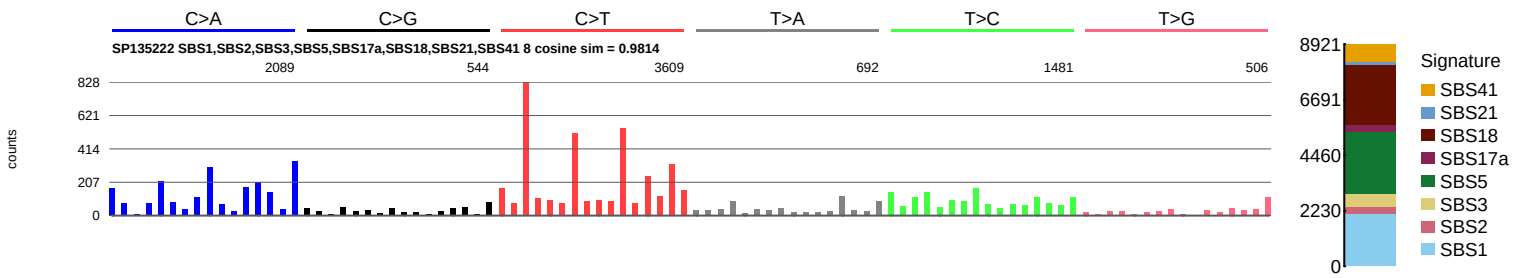
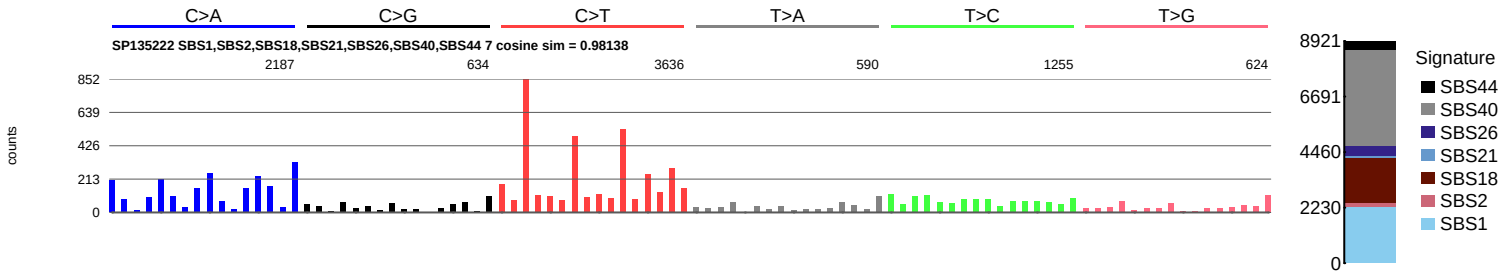
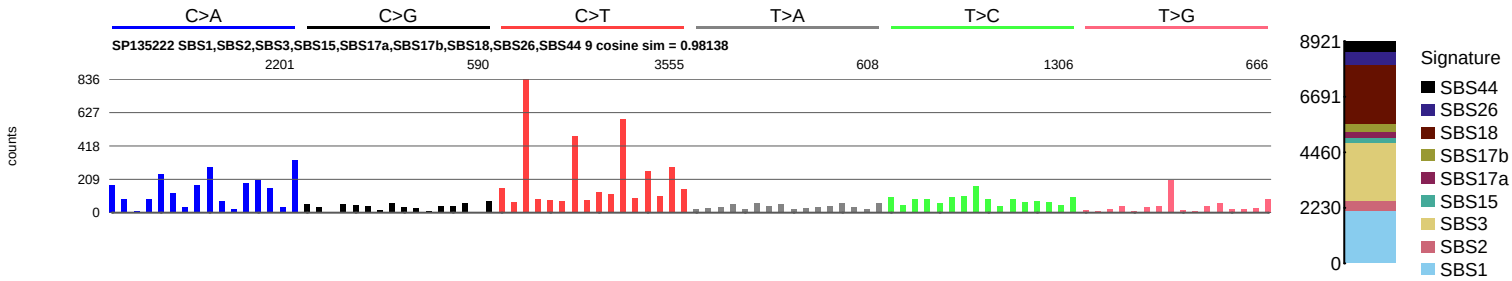
SBS2
SBS1

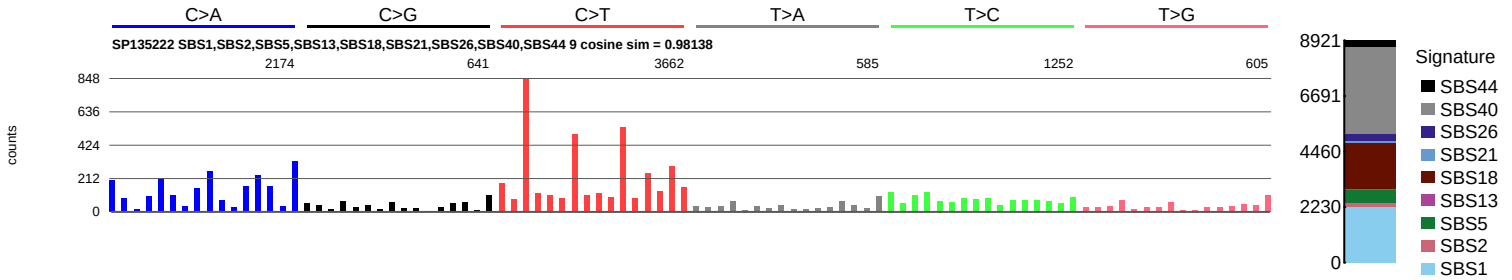

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98144
2182
698
3557
661
938
882
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS1

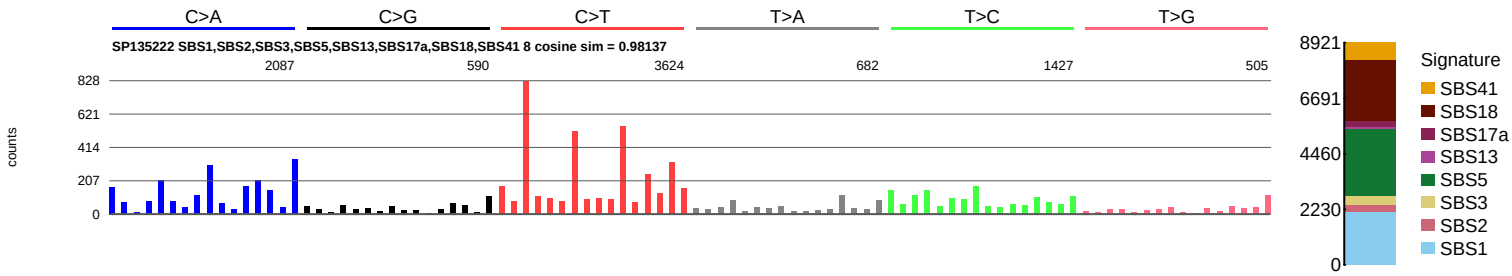

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98142
2175
635
3663
587
1252
608
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS5
SBS2
SBS1

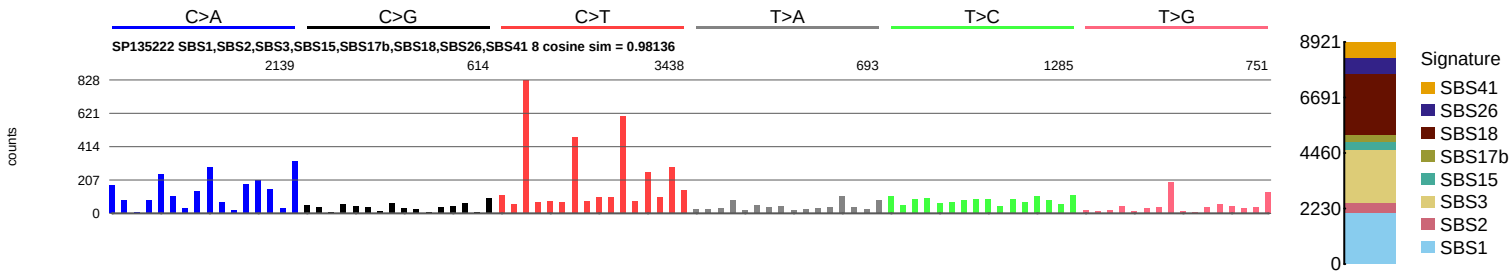

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS21,SBS41 8 cosine sim = 0.9814
2089
544
3609
692
1481
506
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1

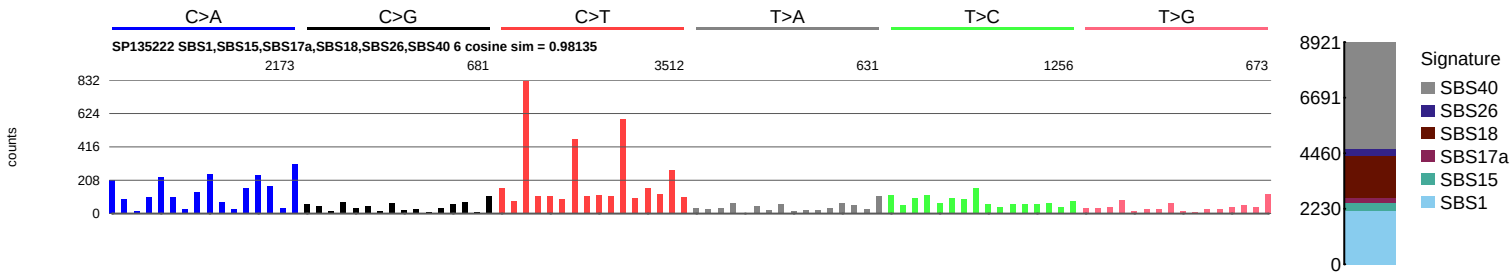

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS18,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.98138
2187
634
3636
590
1255
624
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS2
SBS1

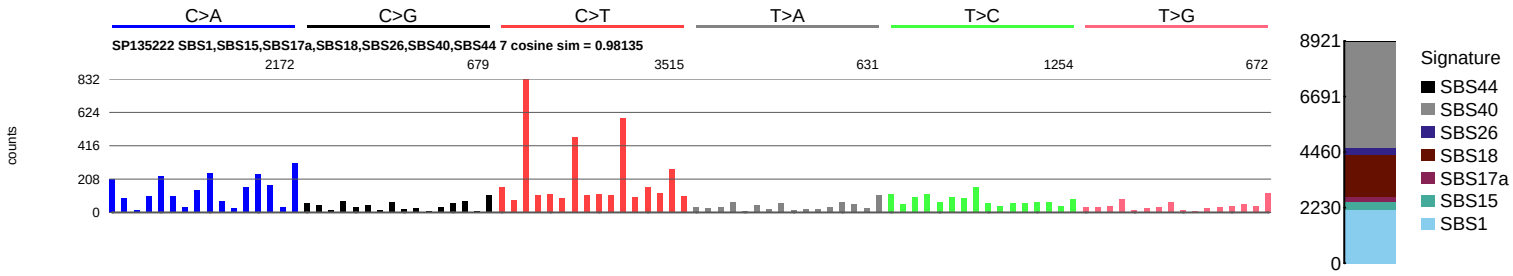

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98138
2201
590
3555
608
1306
666
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

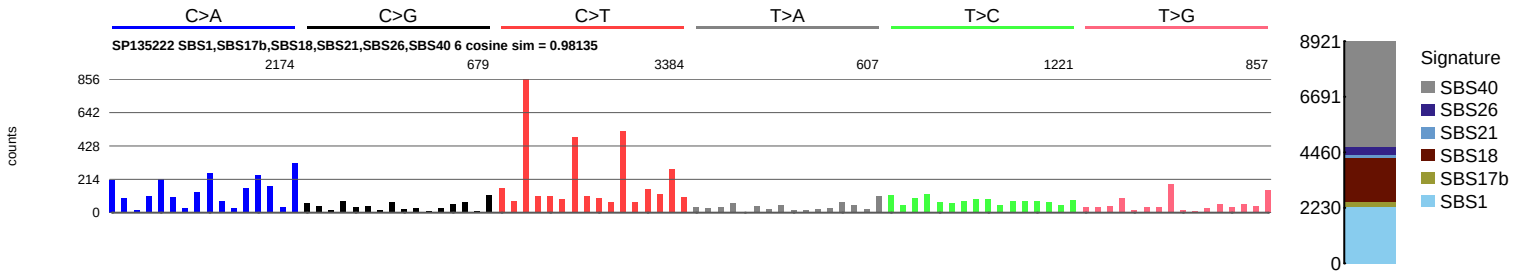

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98138
2174
641
3662
585
1252
605
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1

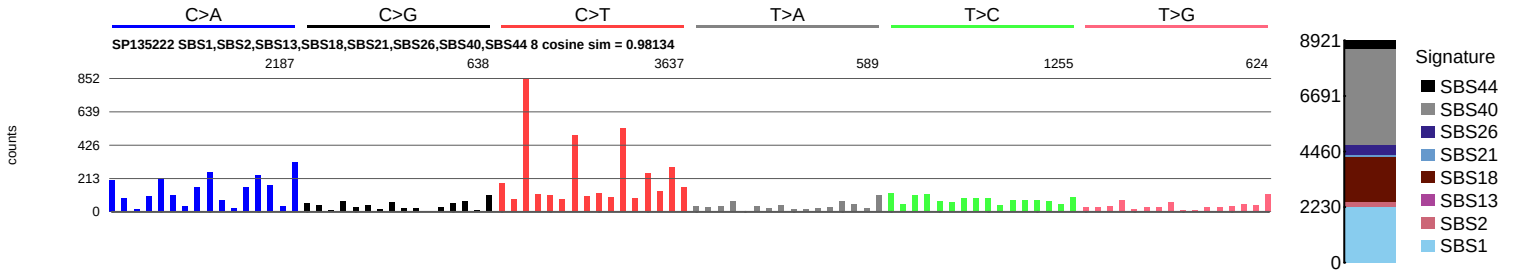

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS18,SBS41 8 cosine sim = 0.98137
2087
590
3624
682
1427
505
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

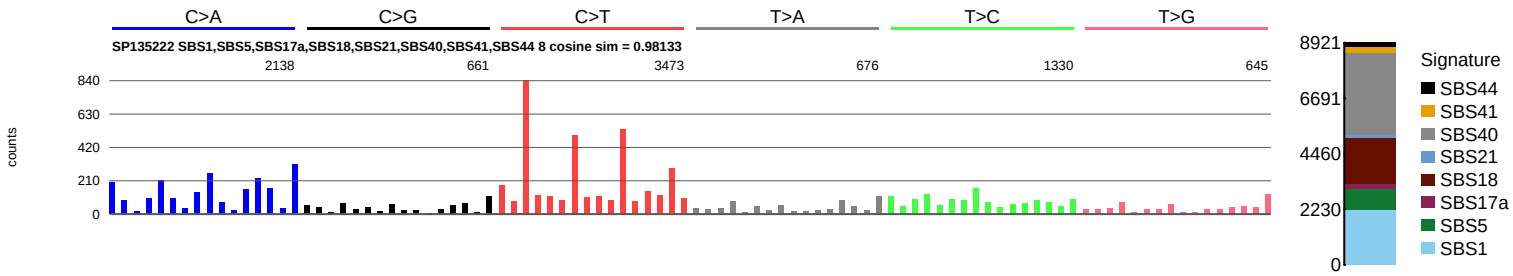

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS15,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98136
2139
614
3438
693
1285
751
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS15
SBS3
SBS2
SBS1

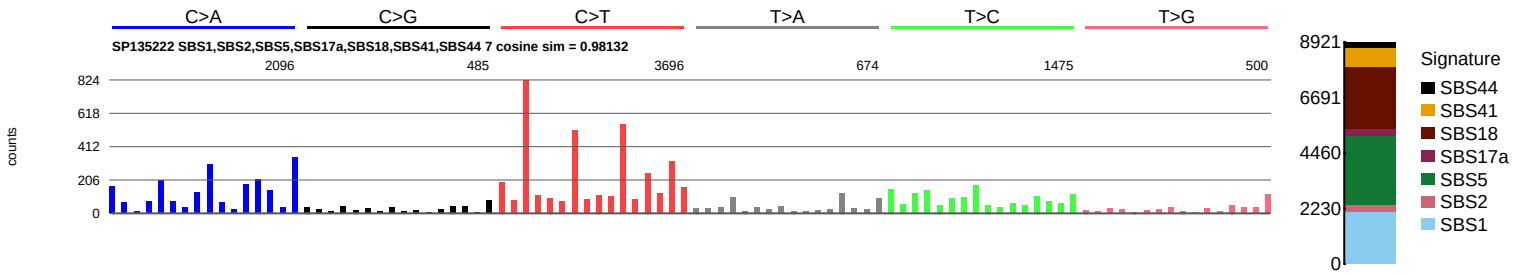

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS18,SBS26,SBS40 6 cosine sim = 0.98135
2173
681
3512
631
1256
673
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS1

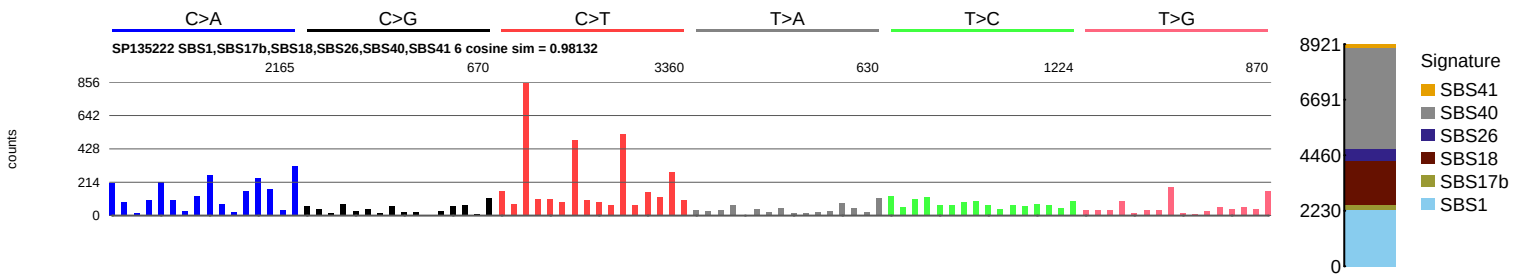

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98135
2172
679
3515
631
1254
672
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS15
SBS1

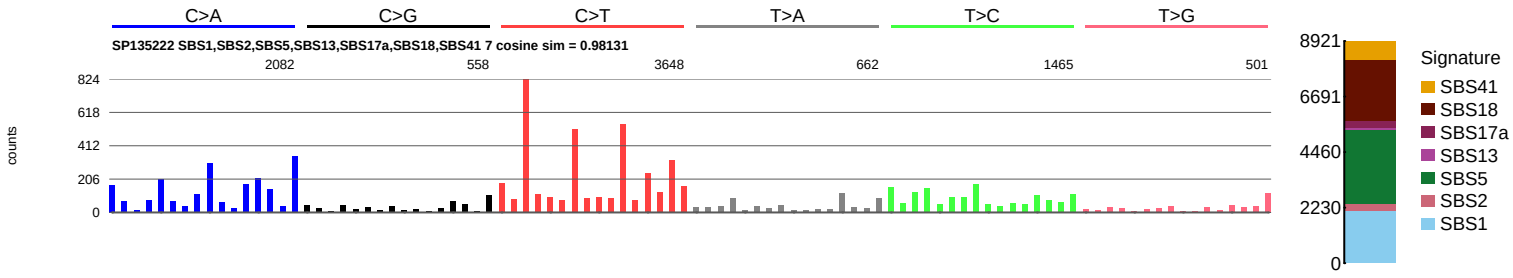

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17b,SBS18,SBS21,SBS26,SBS40 6 cosine sim = 0.98135
2174
679
3384
607
1221
857
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS1

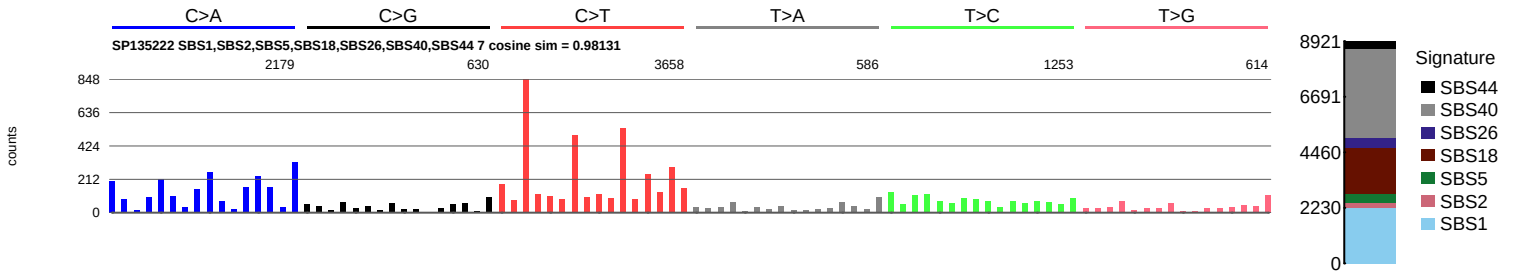

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98134
2187
638
3637
589
1255
624
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS13
SBS2
SBS1

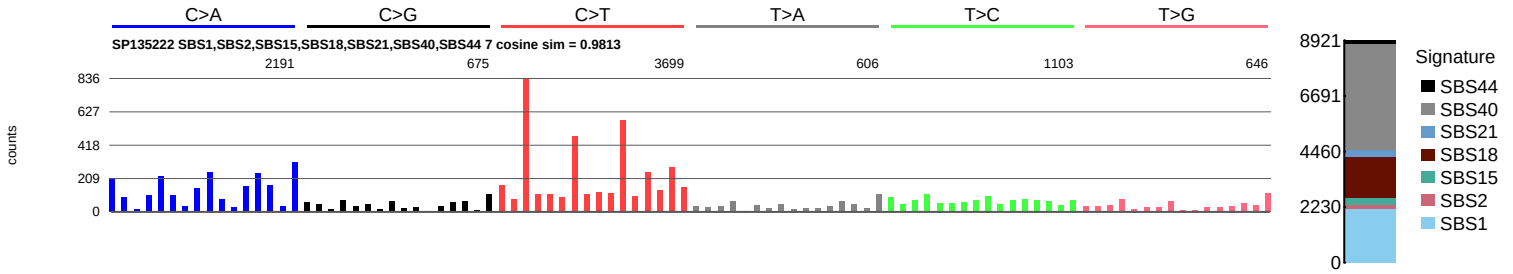

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98133
2138
661
3473
676
1330
645
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS1

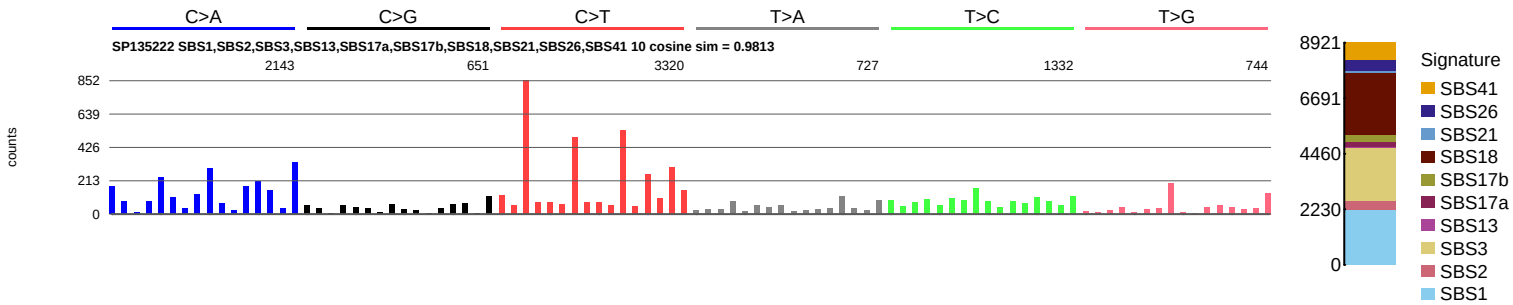

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS41,SBS44 7 cosine sim = 0.98132
2096
485
3696
674
1475
500
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17a
SBS5
SBS2
SBS1

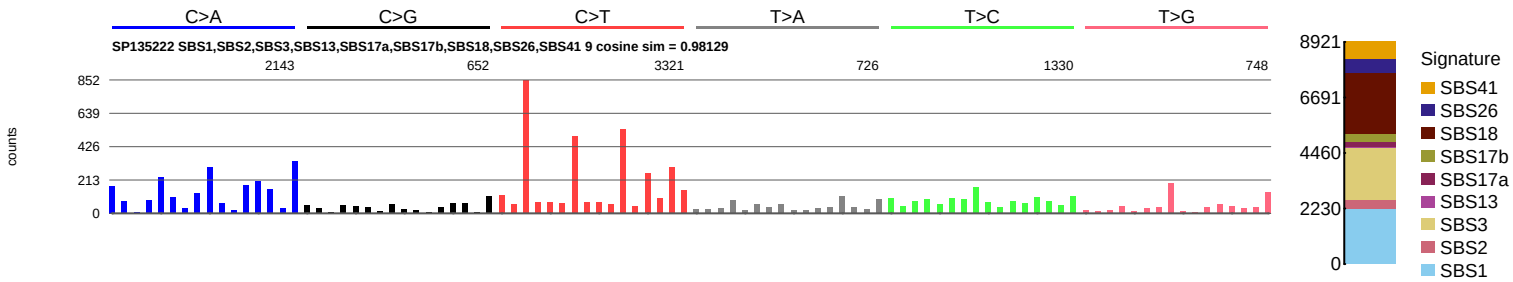

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17b,SBS18,SBS26,SBS40,SBS41 6 cosine sim = 0.98132
2165
670
3360
630
1224
870
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS1

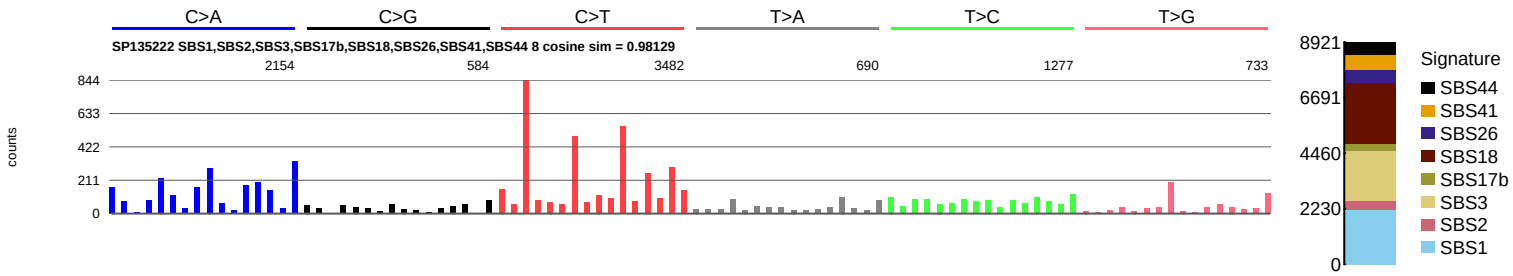

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS18,SBS41 7 cosine sim = 0.98131
2082
558
3648
662
1465
501
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17a
SBS13
SBS5
SBS2
SBS1

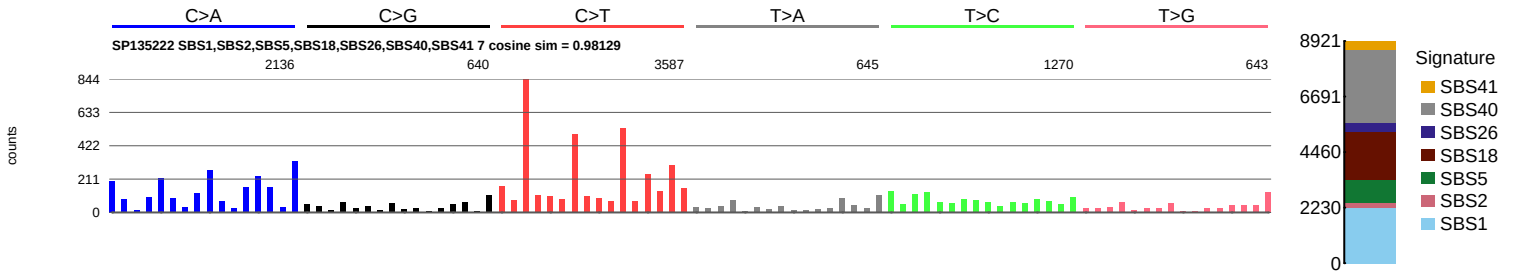

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98131
2179
630
3658
586
1253
614
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS5
SBS2
SBS1

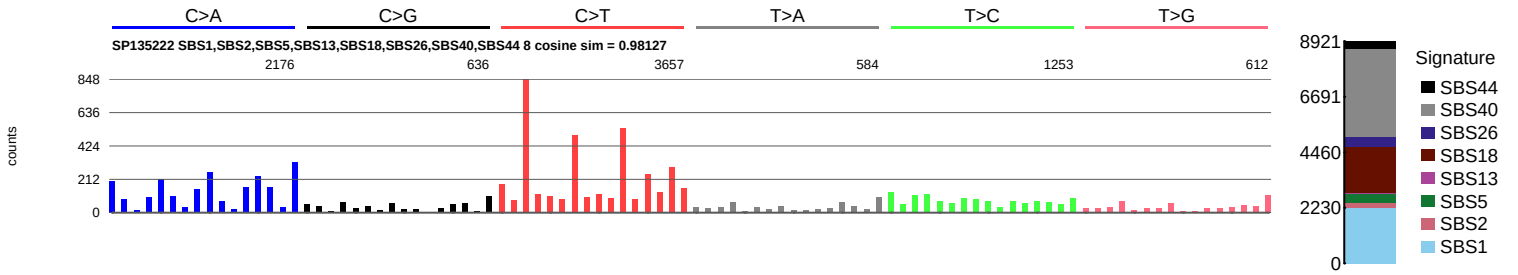

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.9813
2191
675
3699
606
1103
646
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS15
SBS2
SBS1

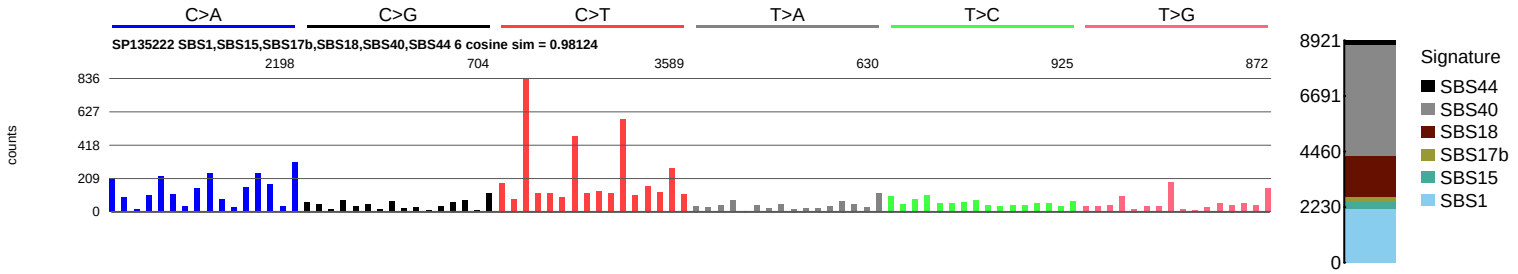

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.9813
2143
651
3320
727
1332
744
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

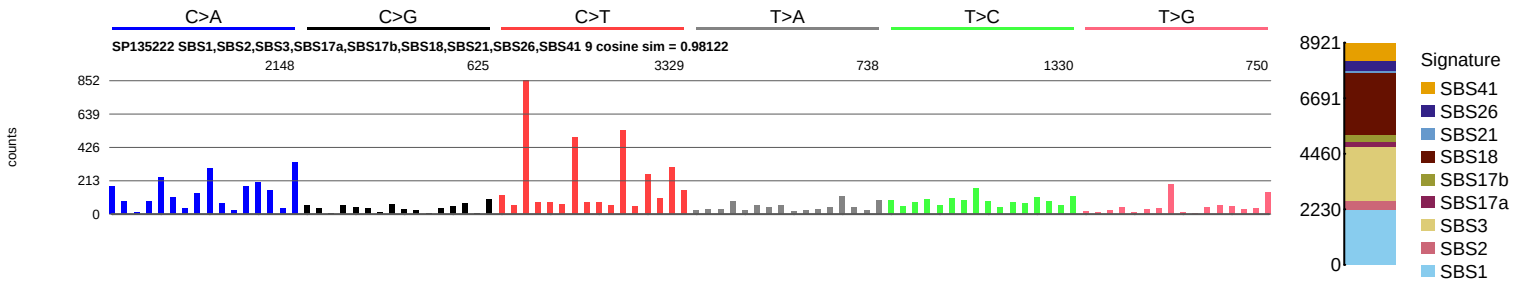

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98129
2143
652
3321
726
1330
748
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

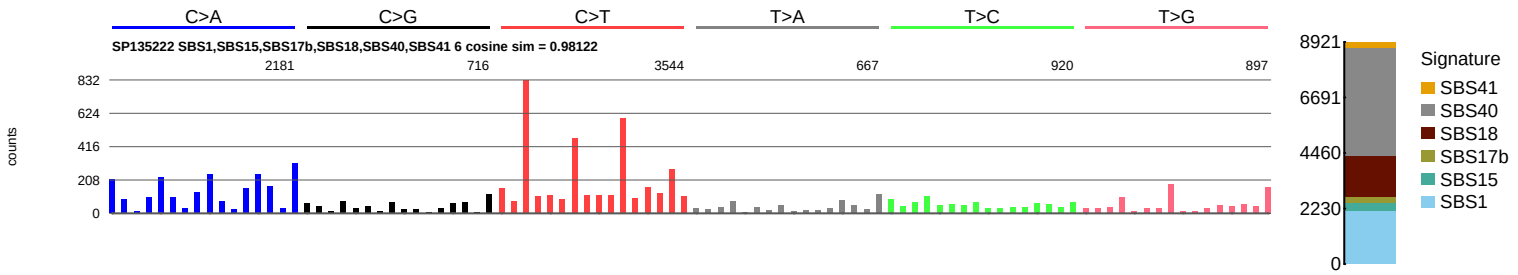

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98129
2154
584
3482
690
1277
733
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS3
SBS2
SBS1

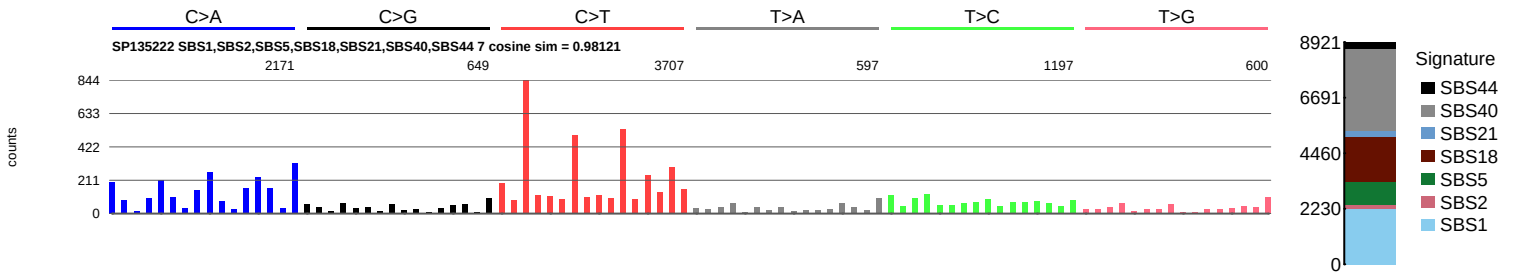

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98129
2136
640
3587
645
1270
643
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS5
SBS2
SBS1

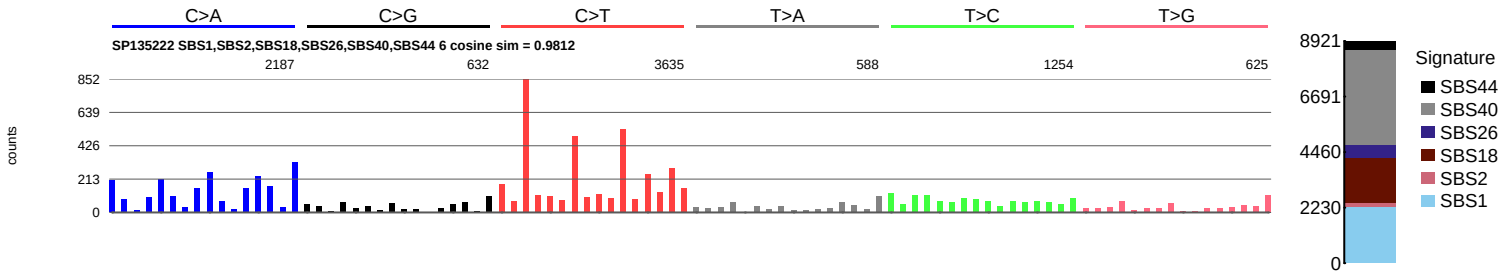

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98127
2176
636
3657
584
1253
612
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS13
SBS5
SBS2
SBS1

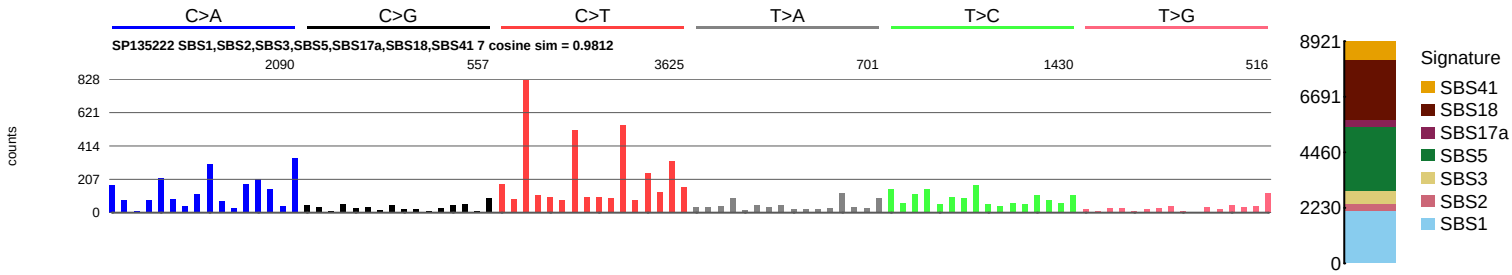

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98124
2198
704
3589
630
925
872
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS15
SBS1

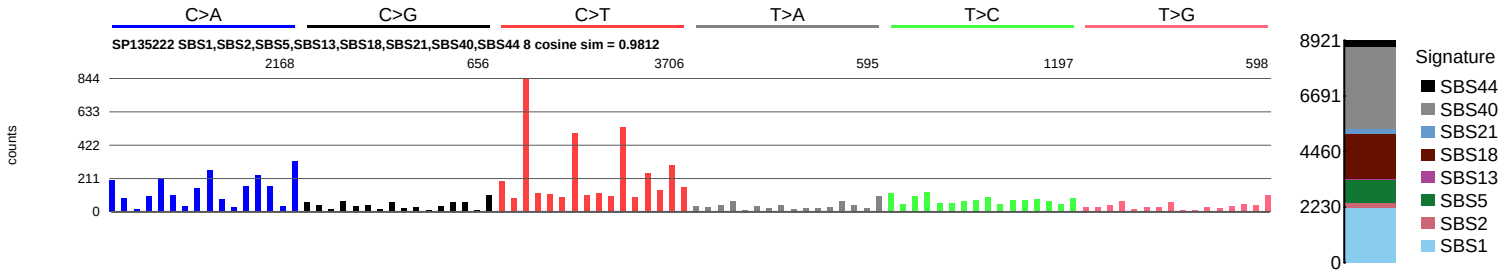

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98122
2148
625
3329
738
1330
750
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

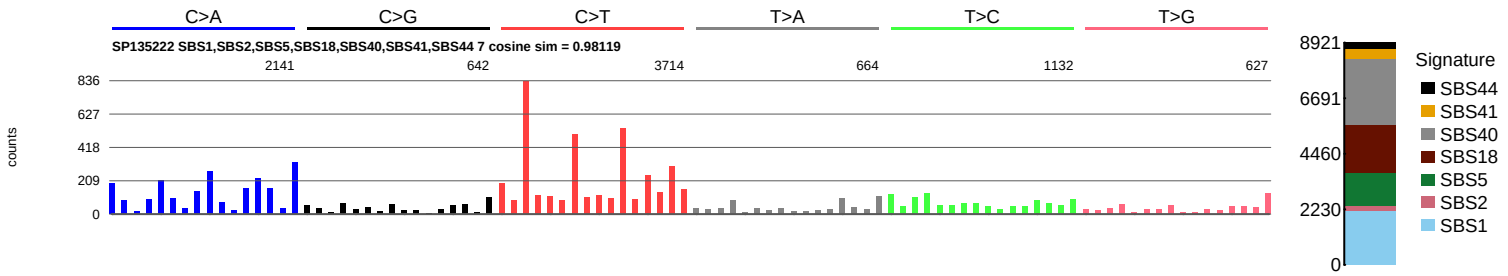

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98122
2181
716
3544
667
920
897
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS15
SBS1

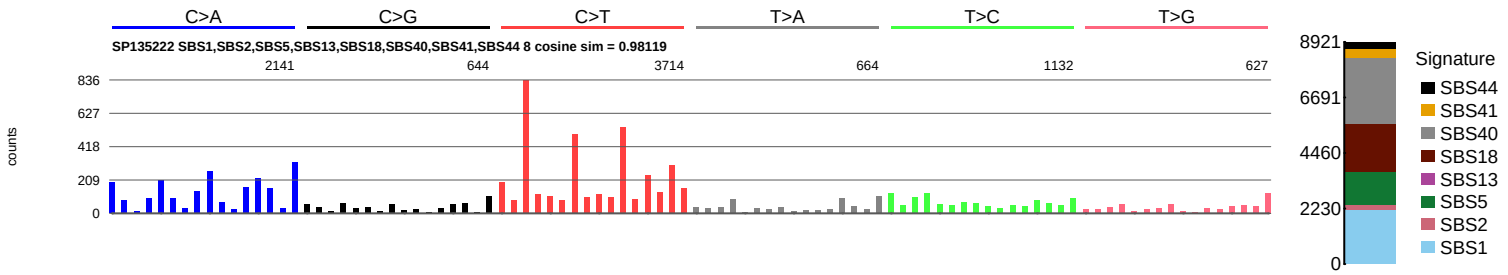

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98121
2171
649
3707
597
1197
600
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS18,SBS26,SBS40,SBS44 6 cosine sim = 0.9812
2187
632
3635
588
1254
625
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS18,SBS41 7 cosine sim = 0.9812
2090
557
3625
701
1430
516
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.9812
2168
656
3706
595
1197
598
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98119
2141
642
3714
664
1132
627
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS5
SBS2
SBS1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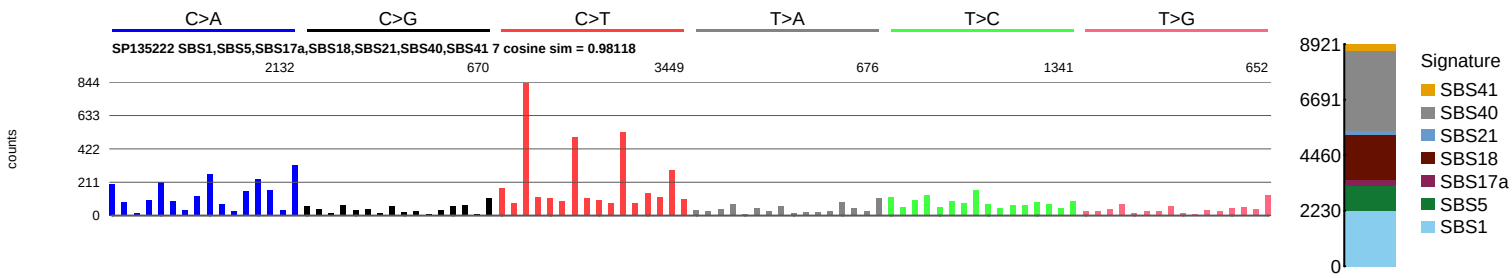
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98119
2141
644
3714
664
1132
627
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1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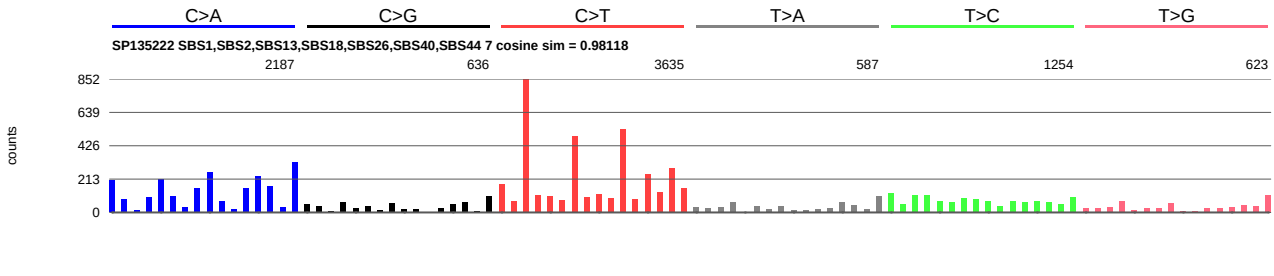
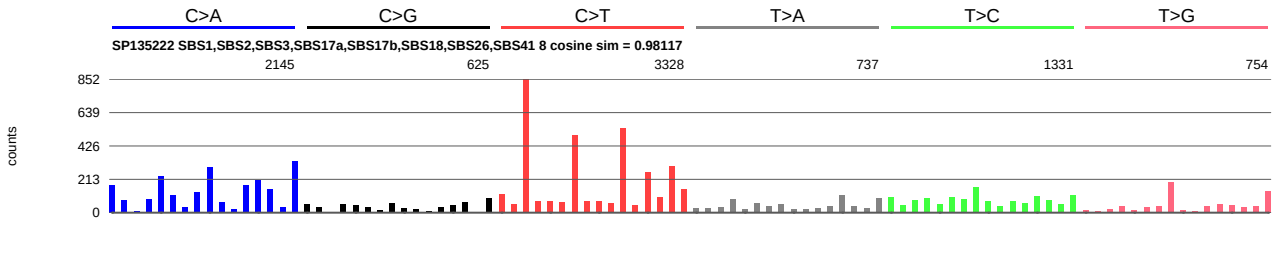
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98118
2132
670
3449
676
1341
652
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS13,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98118
2187
636
3635
587
1254
623
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS13
SBS2
SBS1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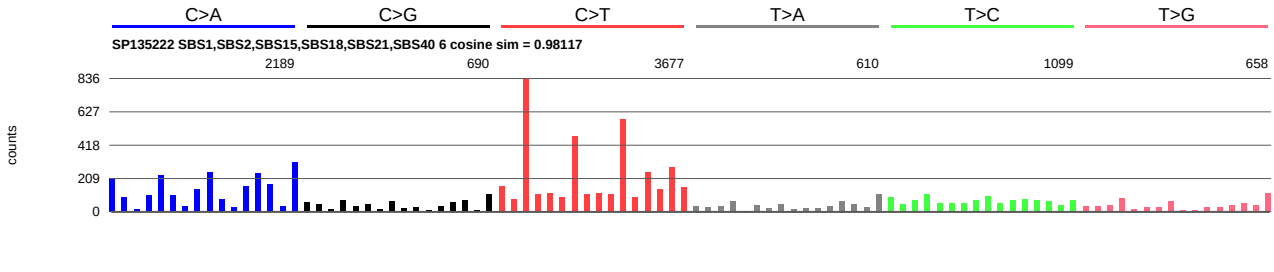
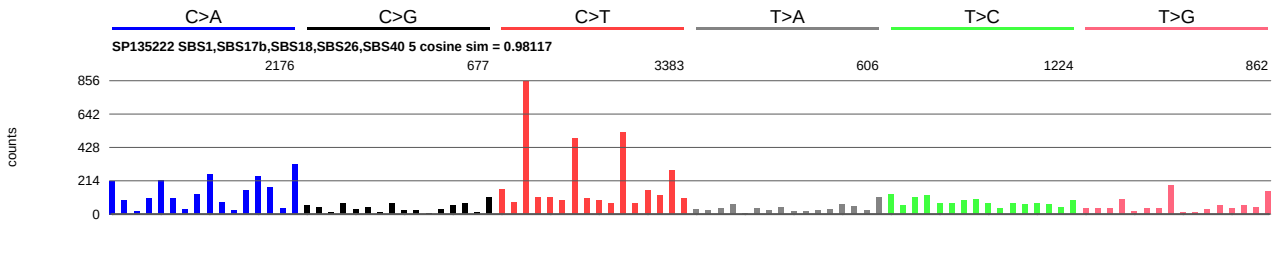
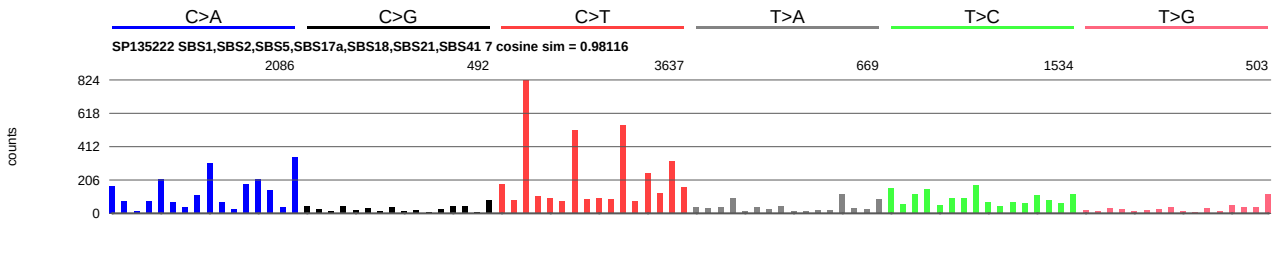
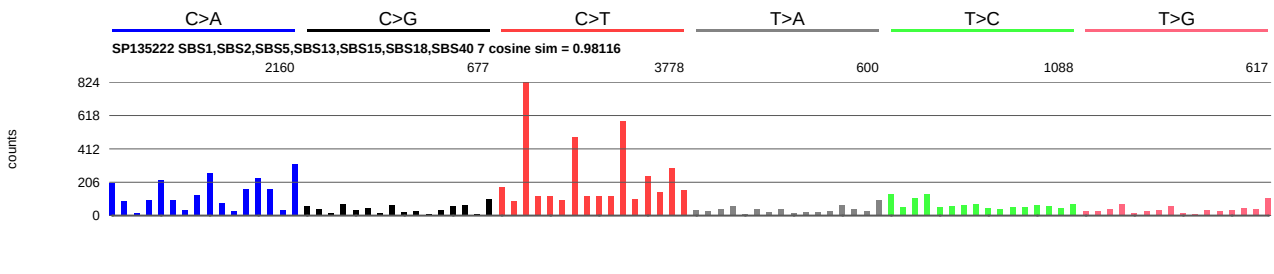
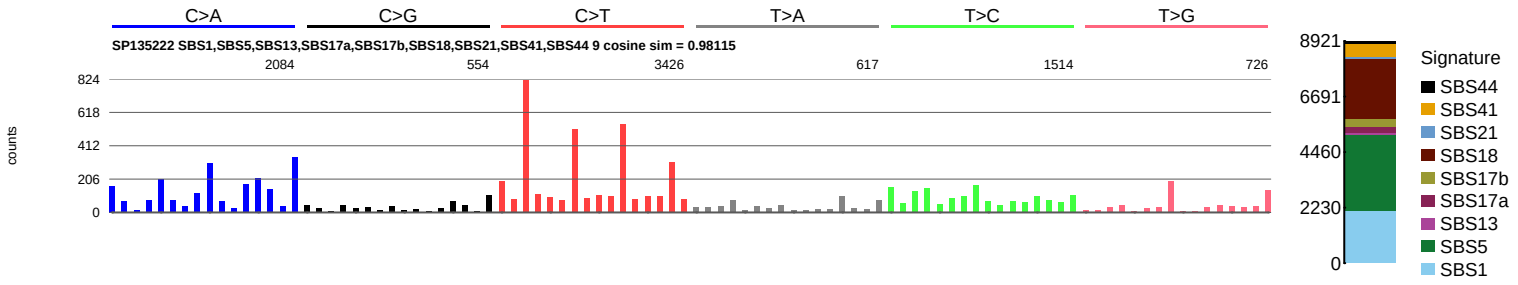
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98117
2145
625
3328
737
1331
754
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS18,SBS21,SBS40 6 cosine sim = 0.98117
2189
690
3677
610
1099
658
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17b,SBS18,SBS26,SBS40 5 cosine sim = 0.98117
2176
677
3383
606
1224
862
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS21,SBS41 7 cosine sim = 0.98116
2086
492
3637
669
1534
503
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40 7 cosine sim = 0.98116
2160
677
3778
600
1088
617
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1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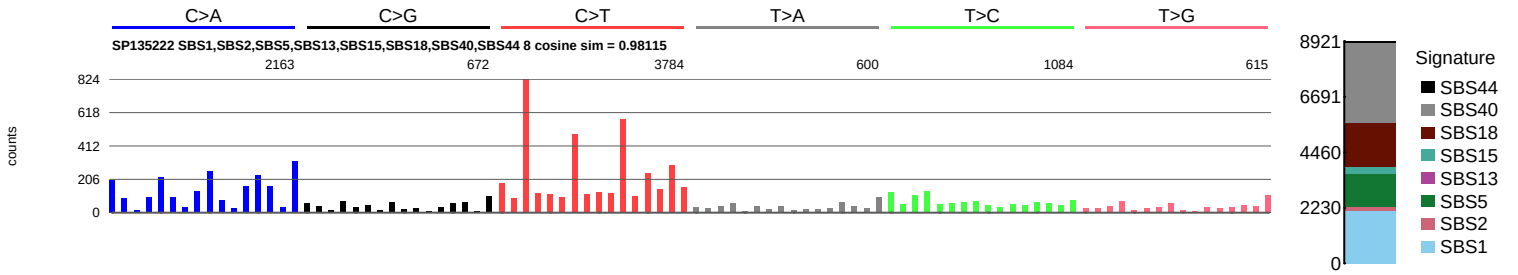
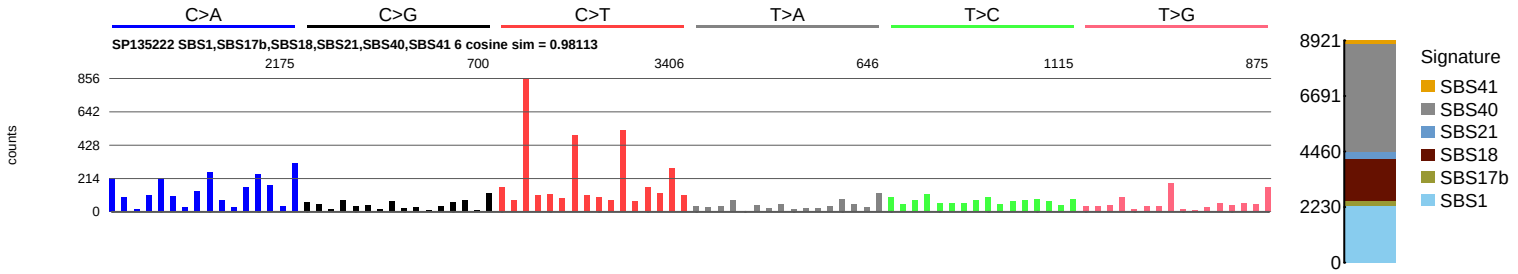
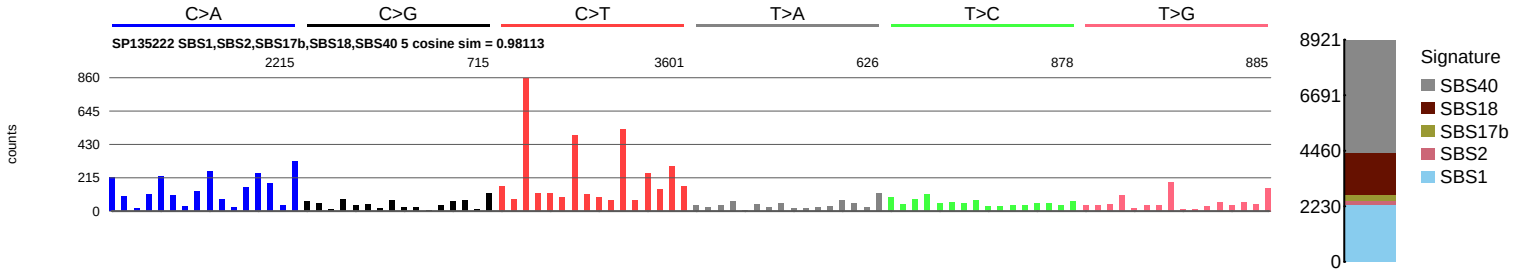
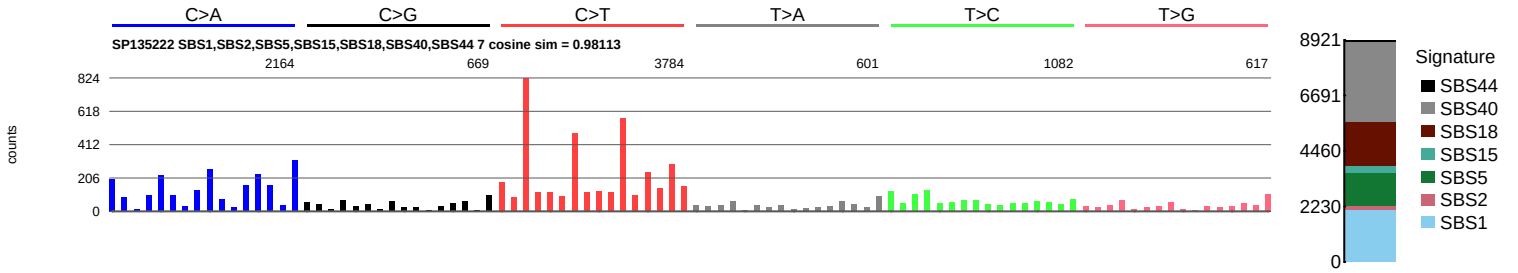
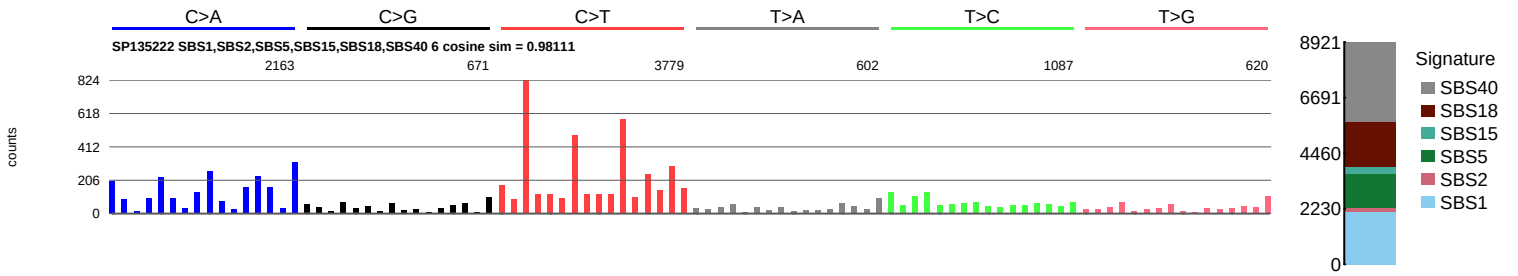
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98115
2084
554
3426
617
1514
726
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

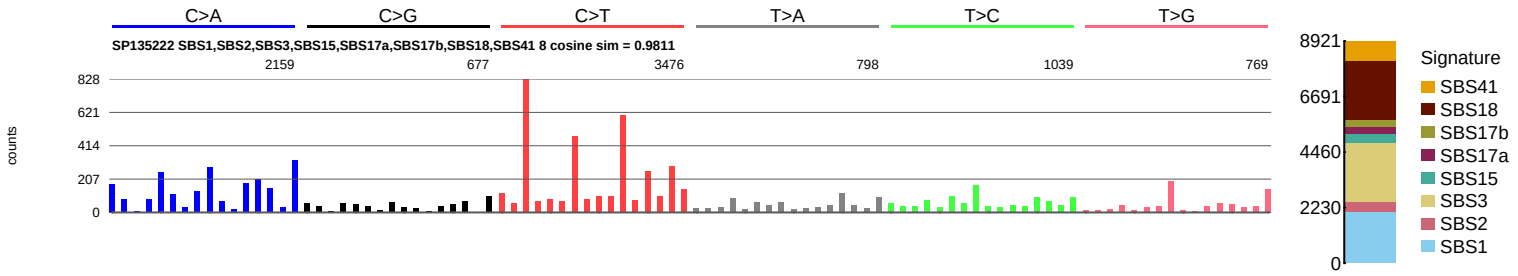

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS15,SBS18,SBS40,SBS44 8 cosine sim = 0.98115
2163
672
3784
600
1084
615
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS13
SBS5
SBS2
SBS1

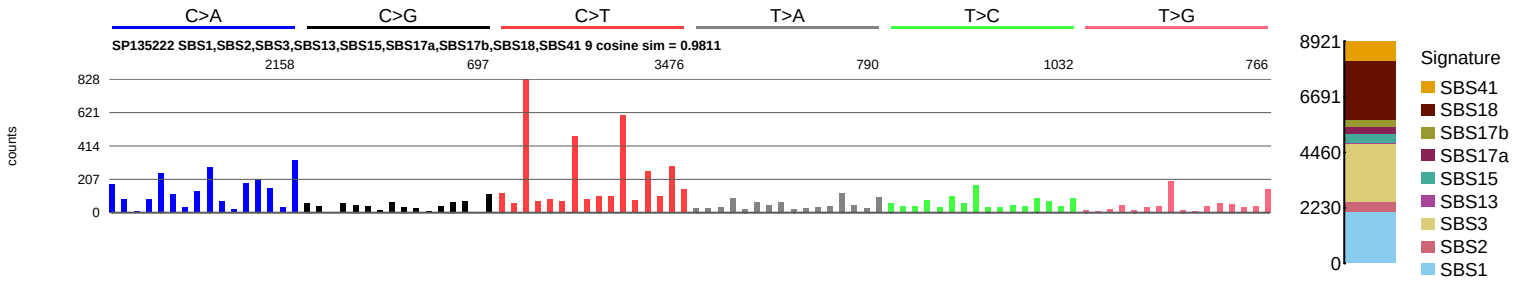

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17b,SBS18,SBS21,SBS40,SBS41 6 cosine sim = 0.98113
2175
700
3406
646
1115
875
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS1

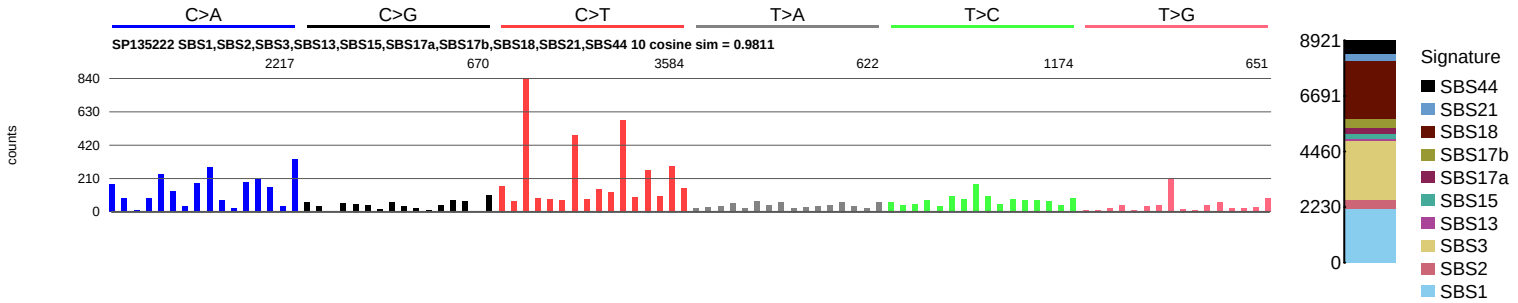

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17b,SBS18,SBS40 5 cosine sim = 0.98113
2215
715
3601
626
878
885
counts
860
645
430
215
0
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS2
SBS1

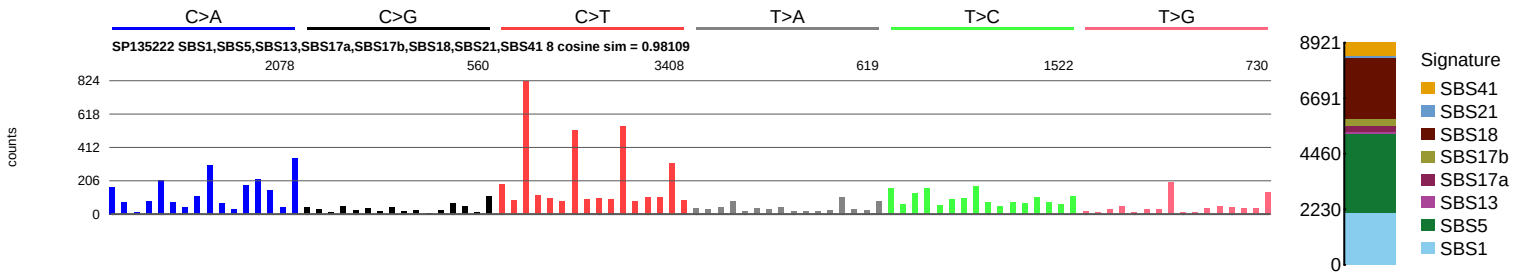

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS40,SBS44 7 cosine sim = 0.98113
2164
669
3784
601
1082
617
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS15,SBS18,SBS40 6 cosine sim = 0.98111
2163
671
3779
602
1087
620
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.9811
2159
677
3476
798
1039
769
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.9811
2158
697
3476
790
1032
766
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.9811
2217
670
3584
622
1174
651
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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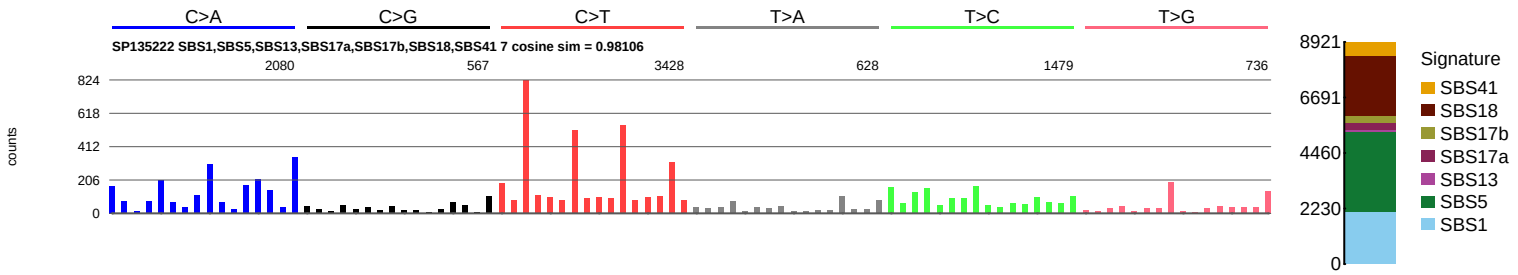
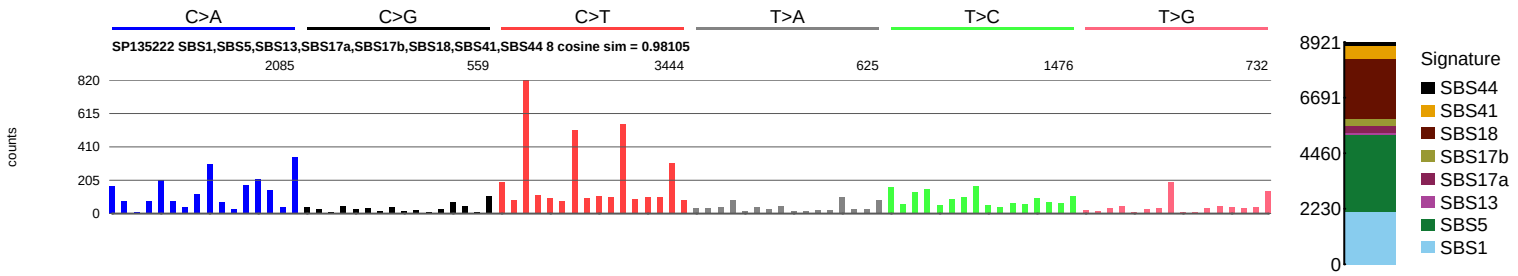
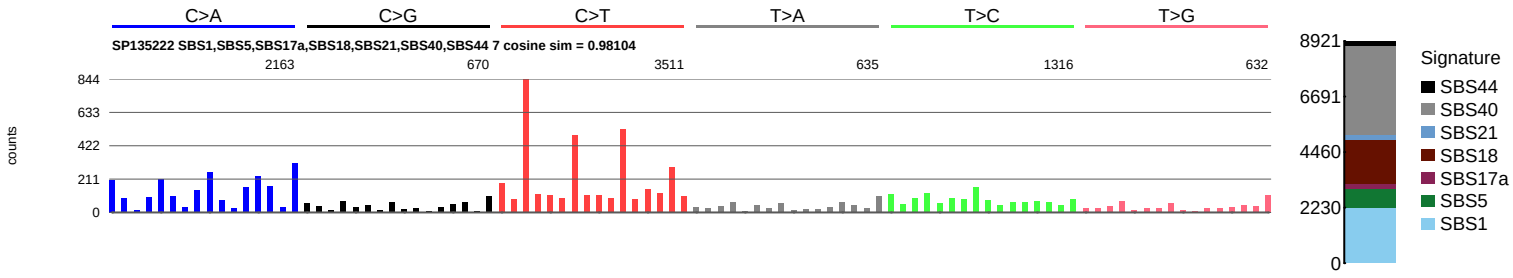
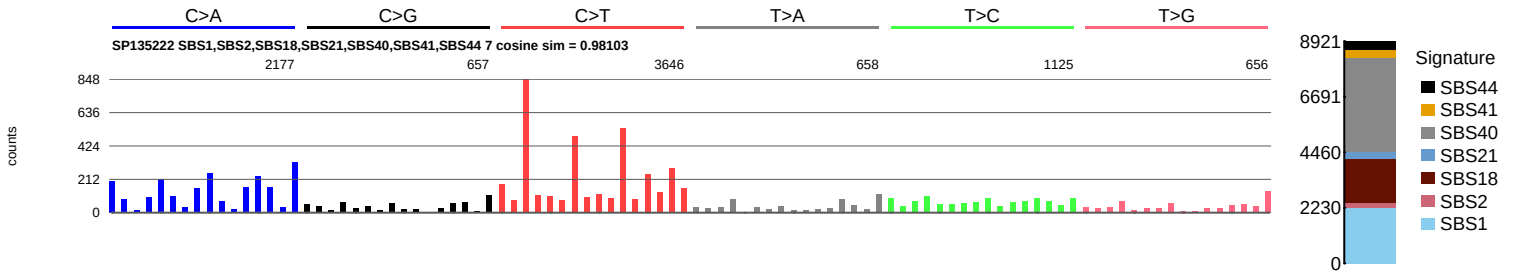
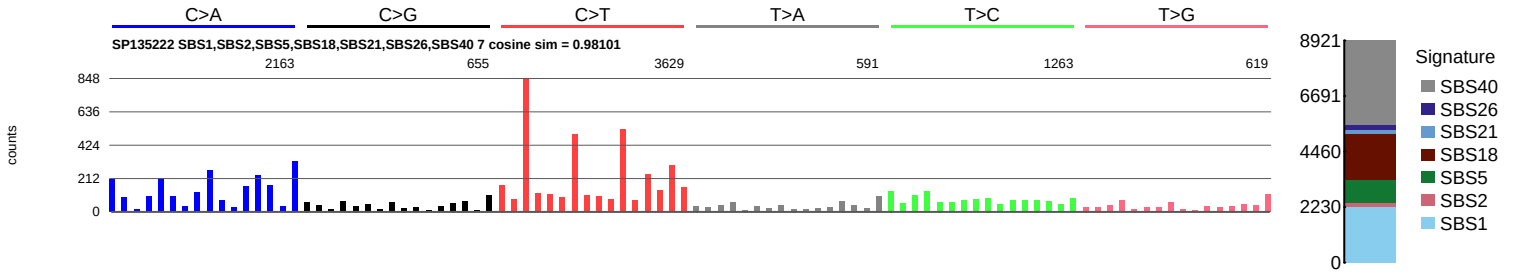
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98109
2078
560
3408
619
1522
730
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

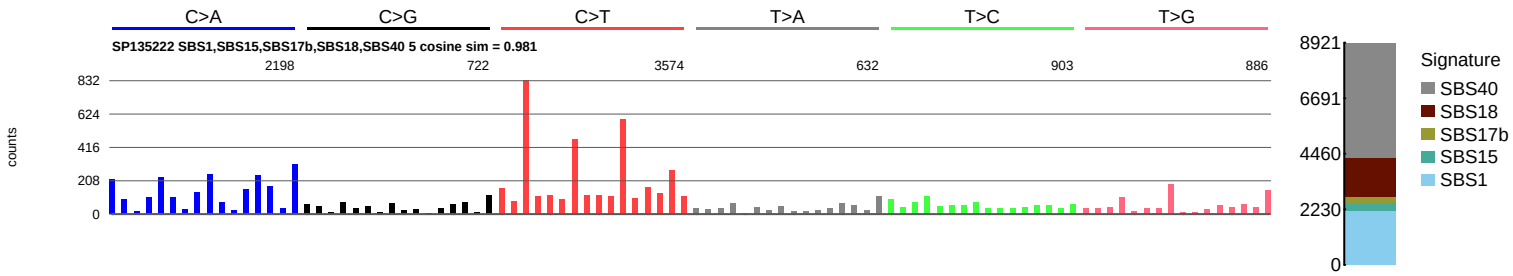

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98106
2080
567
3428
628
1479
736
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

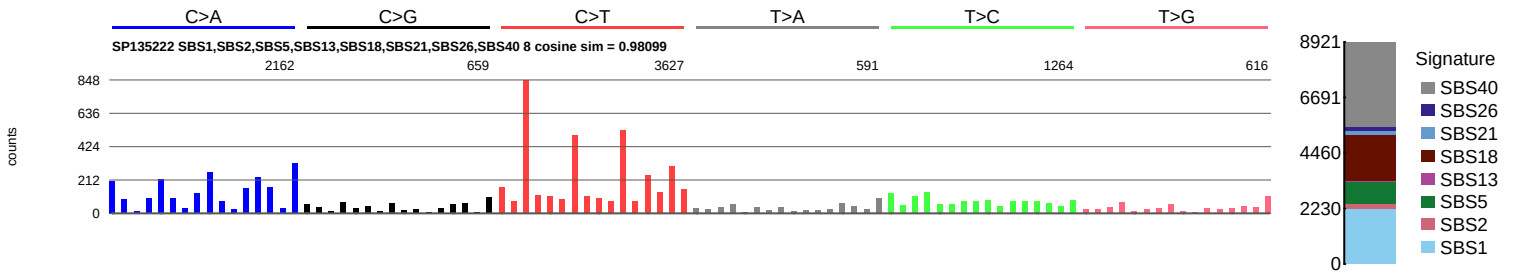

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98105
2085
559
3444
625
1476
732
counts
820
615
410
205
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98104
2163
670
3511
635
1316
632
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS18,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98103
2177
657
3646
658
1125
656
counts
848
636
424
212
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98101
2163
655
3629
591
1263
619
counts
848
636
424
212
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17b,SBS18,SBS40 5 cosine sim = 0.981
2198
722
3574
632
903
886
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17b
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98099
2162
659
3627
591
1264
616
counts
848
636
424
212
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
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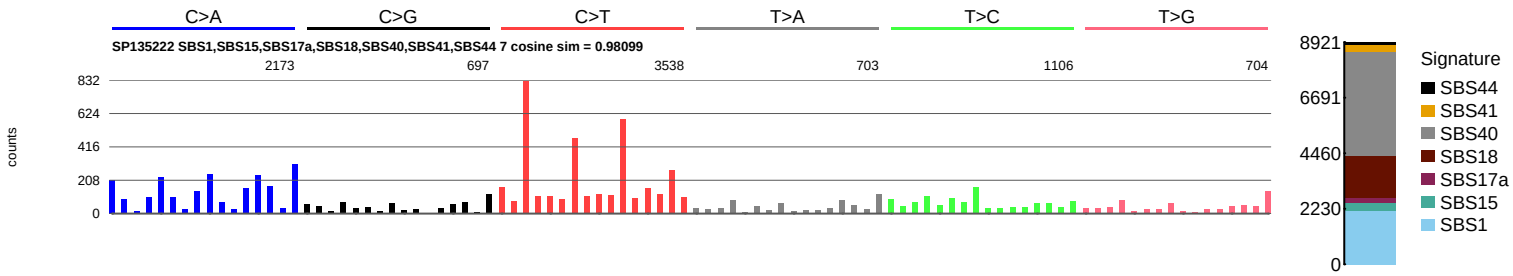
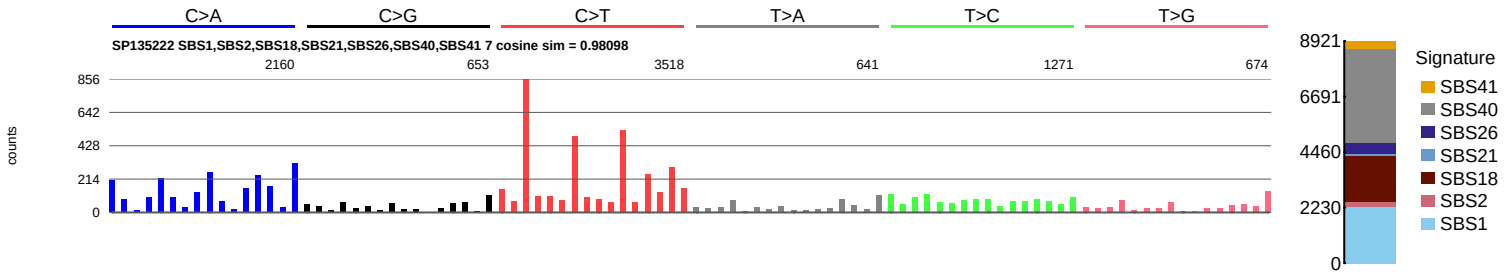
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98099
2173
697
3538
703
1106
704
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS1


C>A
C>G
C>T
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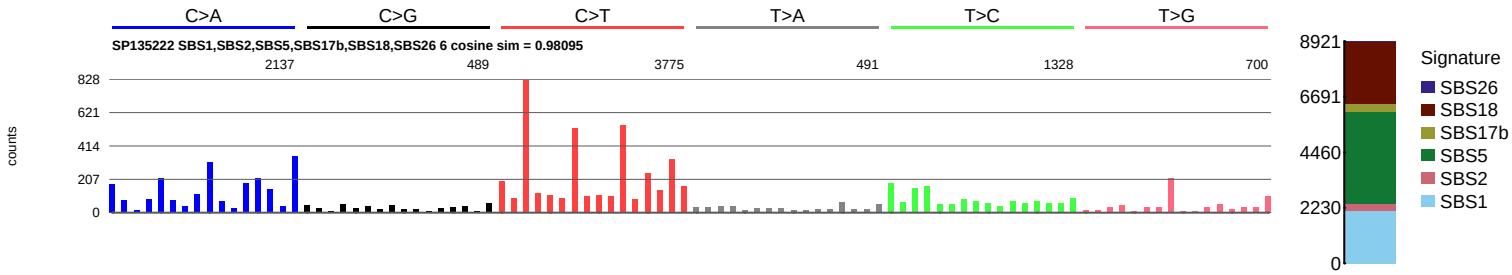
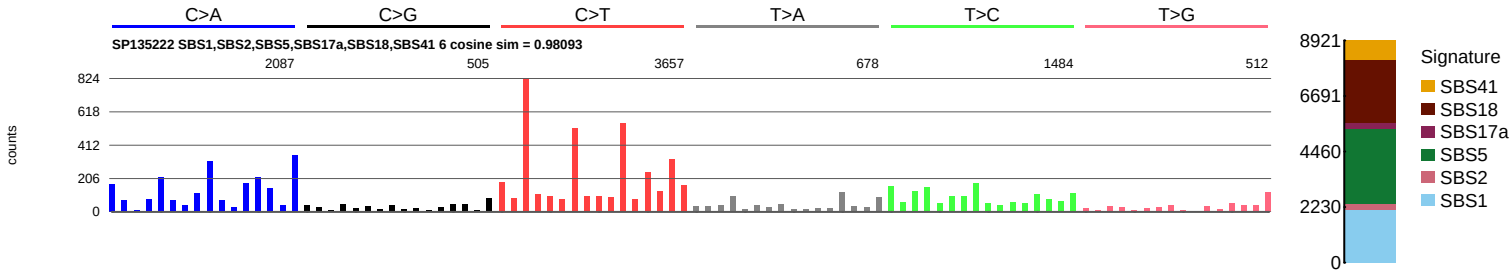
T>C
T>G
SP135222 SBS1,SBS2,SBS18,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.98098
2160
653
3518
641
1271
674
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18,SBS26 6 cosine sim = 0.98095
2137
489
3775
491
1328
700
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS26
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17a,SBS18,SBS41 6 cosine sim = 0.98093
2087
505
3657
678
1484
512
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
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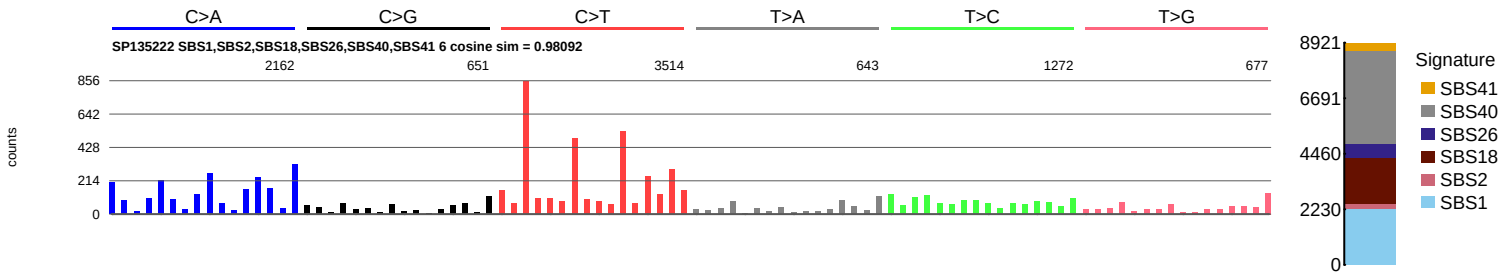
SBS2
SBS1


C>A
C>G
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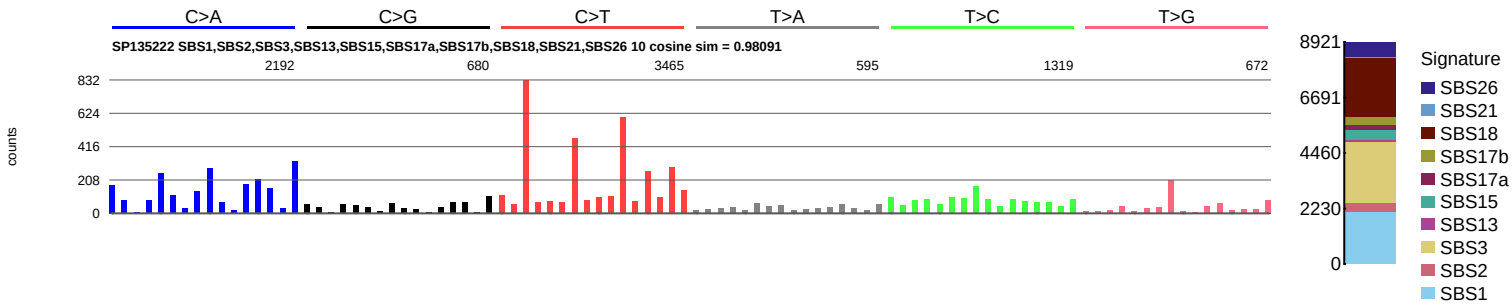
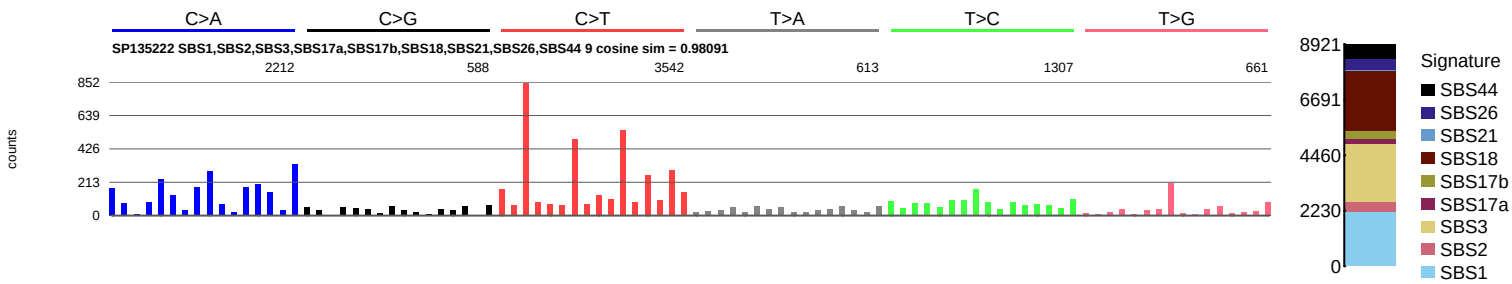
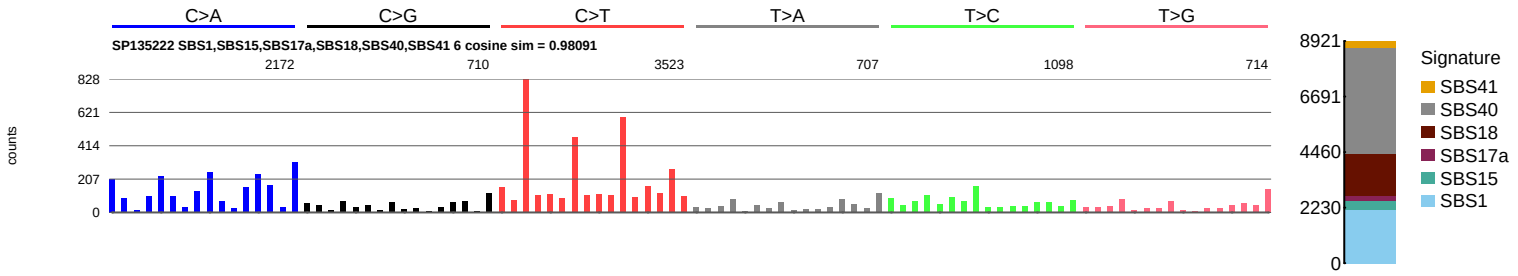
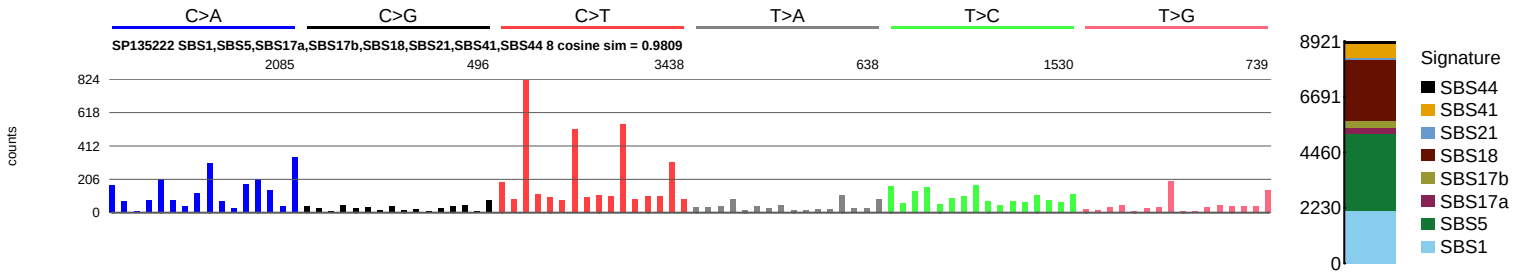
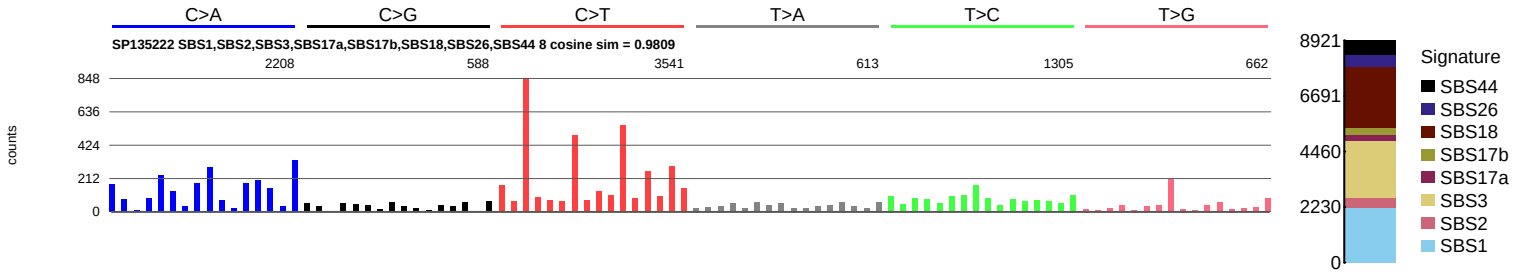
T>C
T>G
SP135222 SBS1,SBS2,SBS18,SBS26,SBS40,SBS41 6 cosine sim = 0.98092
2162
651
3514
643
1272
677
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 10 cosine sim = 0.98091
2192
680
3465
595
1319
672
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98091
2212
588
3542
613
1307
661
counts
852
639
426
213
0
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS18,SBS40,SBS41 6 cosine sim = 0.98091
2172
710
3523
707
1098
714
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.9809
2085
496
3438
638
1530
739
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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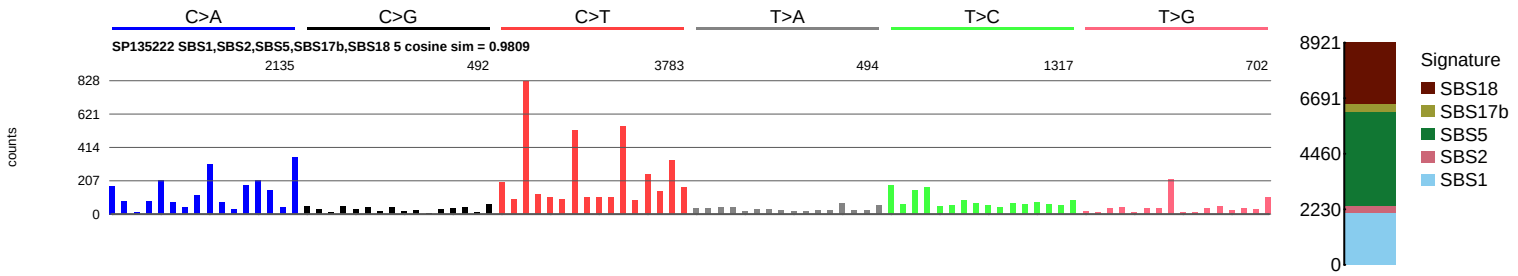
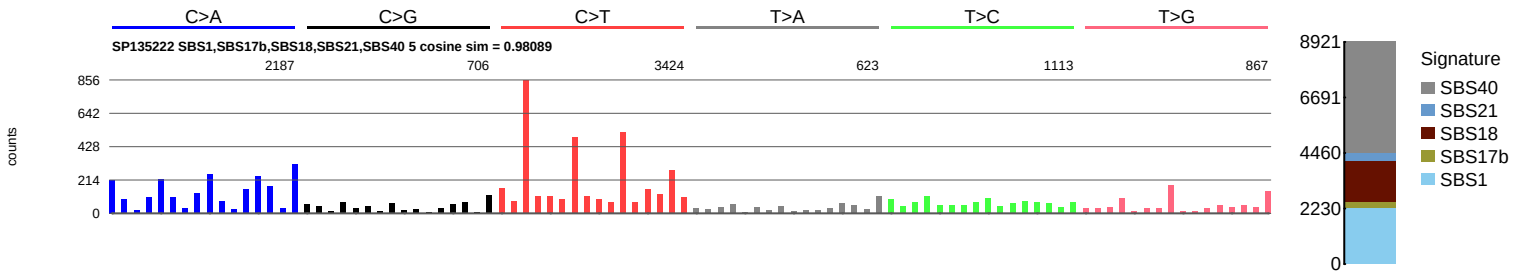
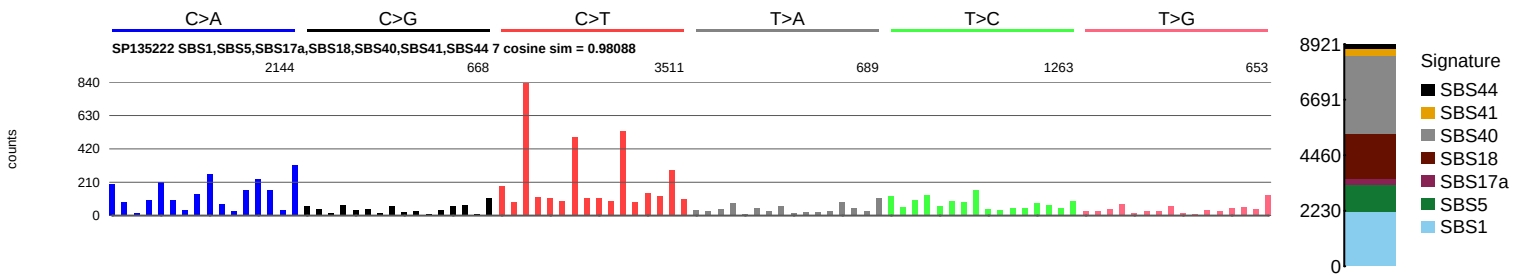
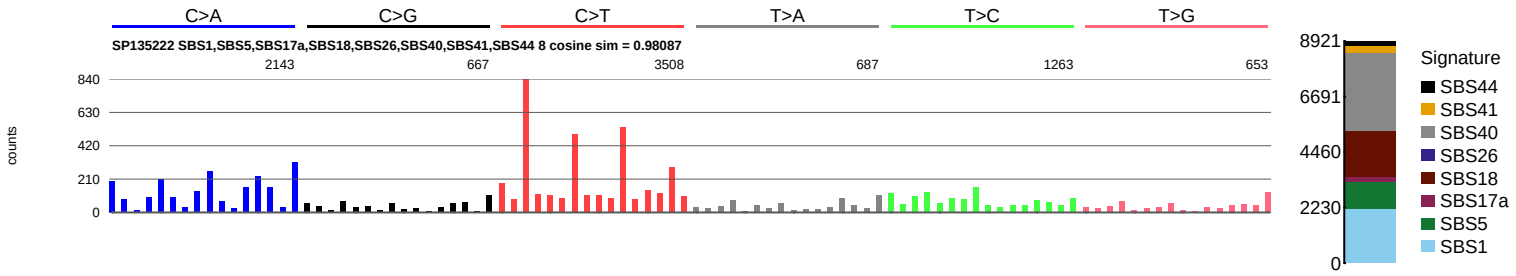
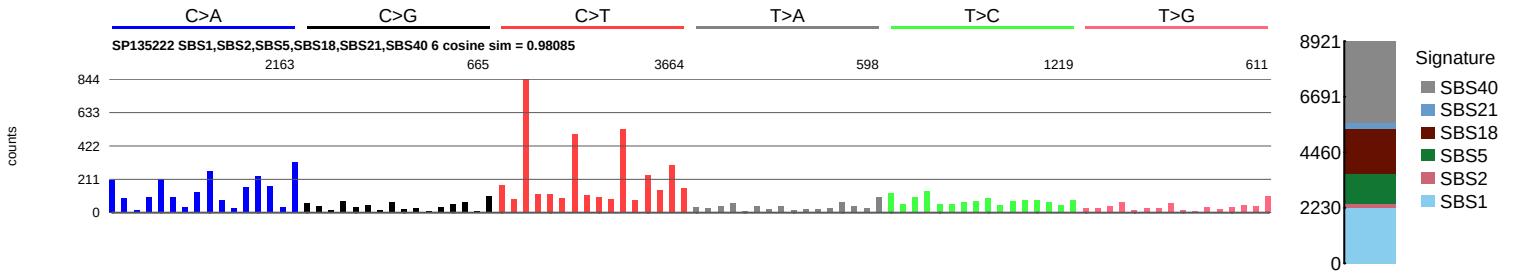
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.9809
2208
588
3541
613
1305
662
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS17b,SBS18 5 cosine sim = 0.9809
2135
492
3783
494
1317
702
828
621
414
207
0
counts
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17b,SBS18,SBS21,SBS40 5 cosine sim = 0.98089
2187
706
3424
623
1113
867
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98088
2144
668
3511
689
1263
653
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98087
2143
667
3508
687
1263
653
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS18,SBS21,SBS40 6 cosine sim = 0.98085
2163
665
3664
598
1219
611
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS21
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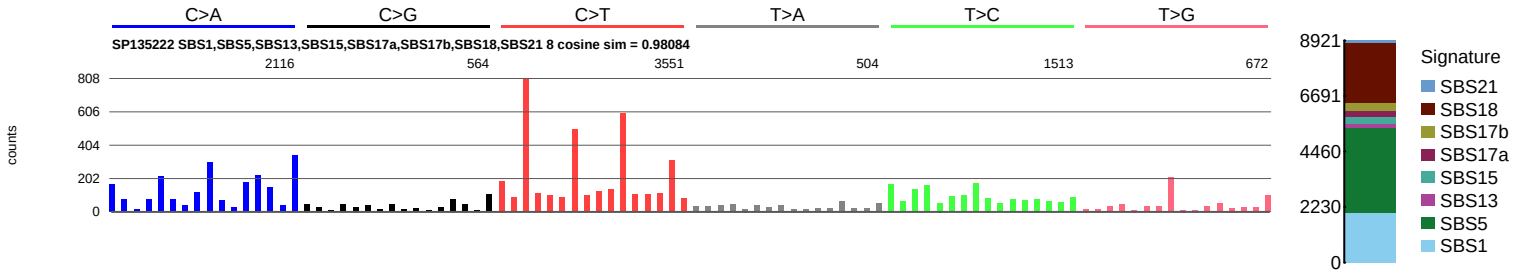
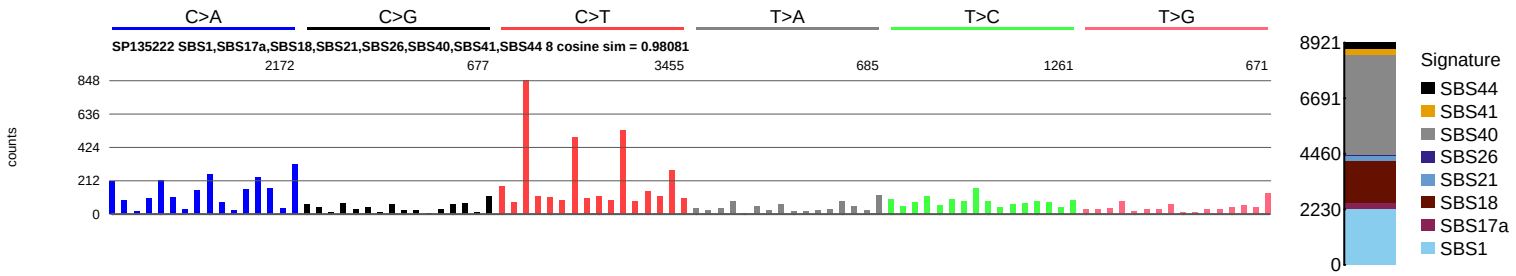
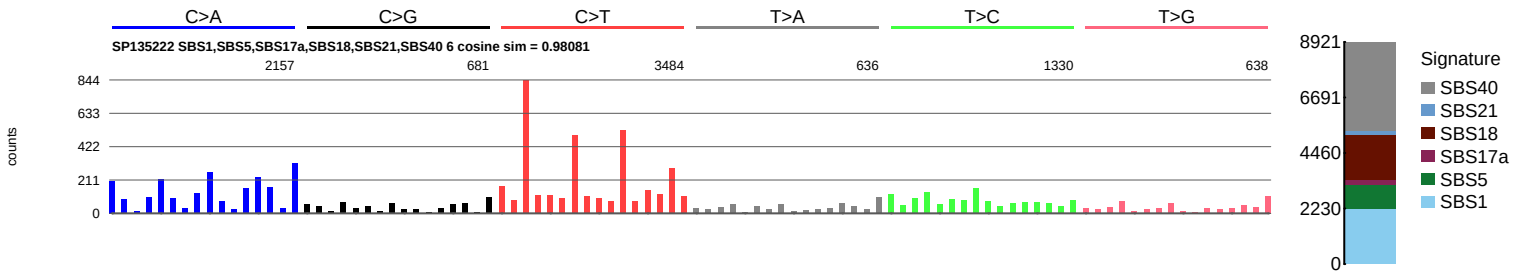
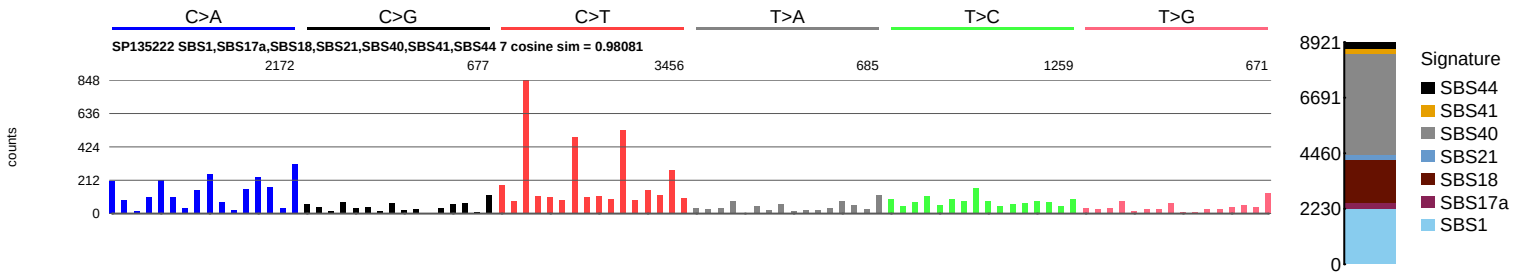
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98084
2116
564
3551
504
1513
672
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98081
2172
677
3455
685
1261
671
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS18,SBS21,SBS40 6 cosine sim = 0.98081
2157
681
3484
636
1330
638
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS21
SBS18
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS18,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98081
2172
677
3456
685
1259
671
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
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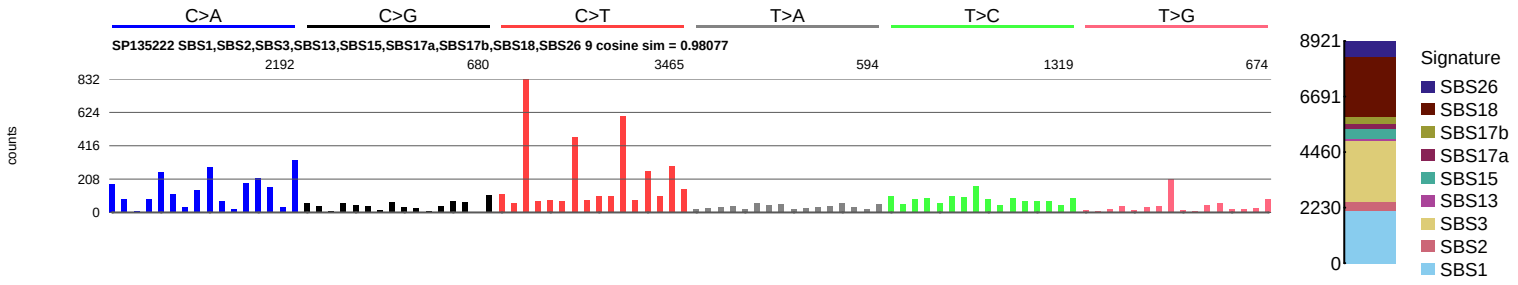
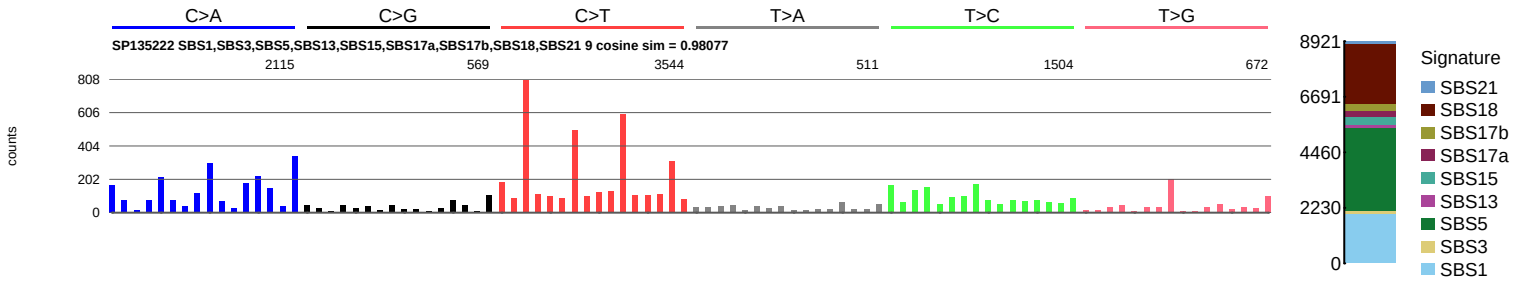
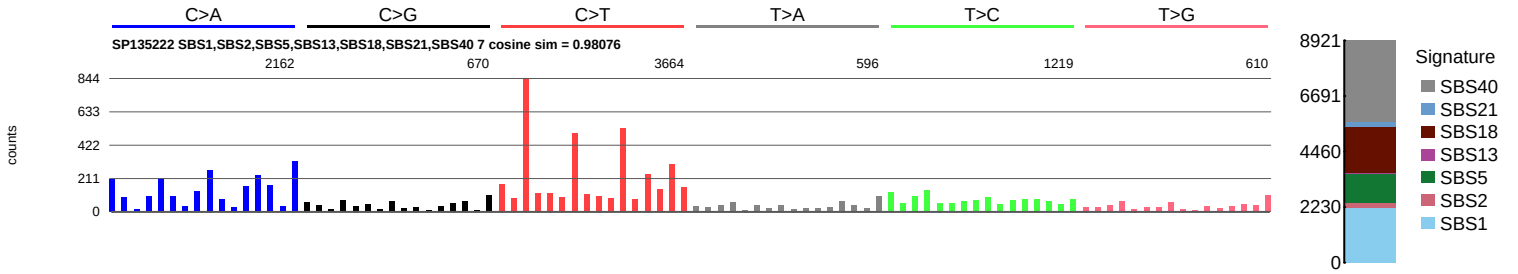
SBS18
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98077
2192
680
3465
594
1319
674
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98077
2115
569
3544
511
1504
672
808
606
404
202
0
counts
8921
6691
4460
2230
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS21,SBS40 7 cosine sim = 0.98076
2162
670
3664
596
1219
610
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
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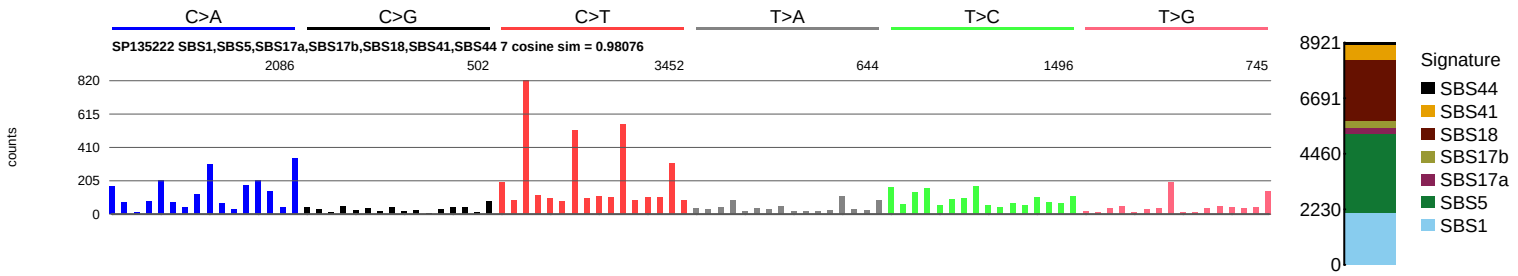
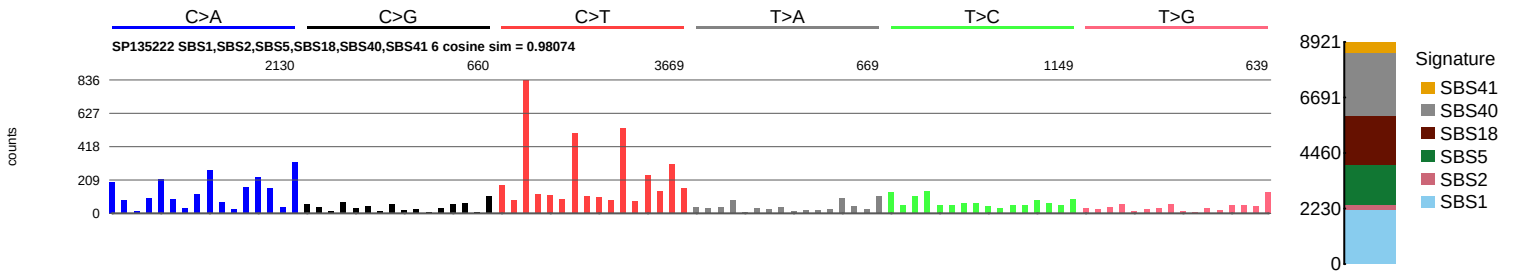
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98076
2086
502
3452
644
1496
745
820
615
410
205
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS18,SBS40,SBS41 6 cosine sim = 0.98074
2130
660
3669
669
1149
639
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
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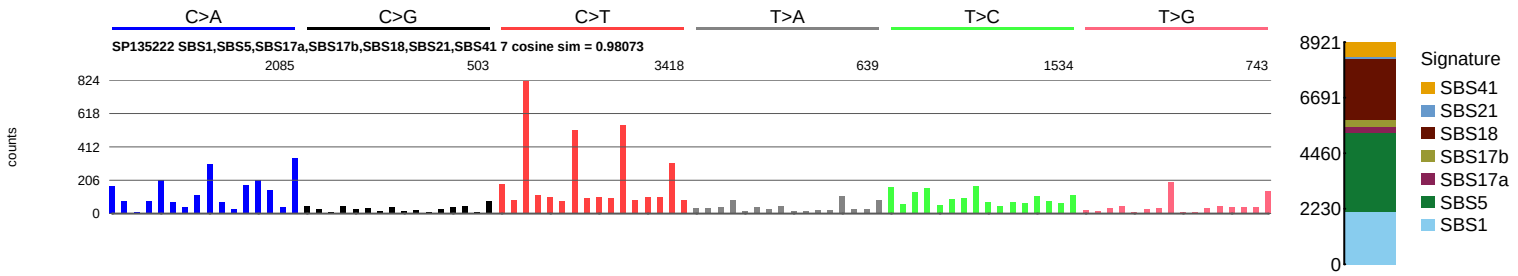
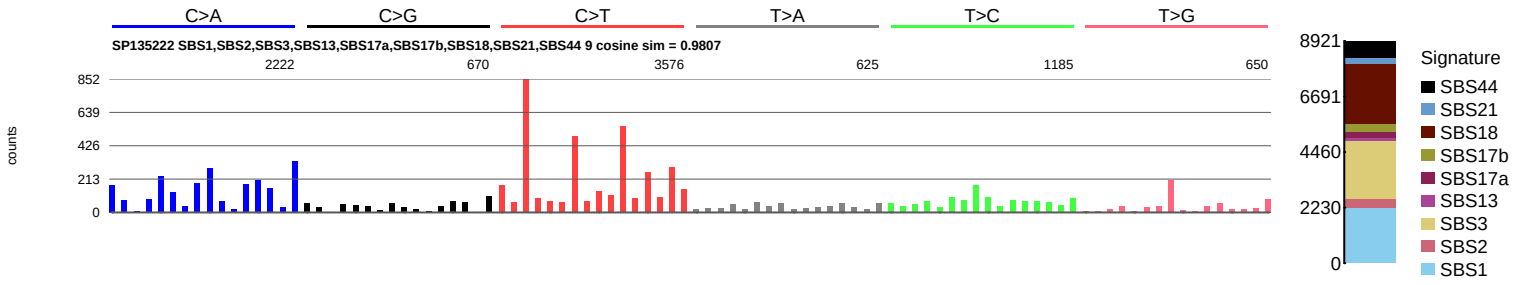
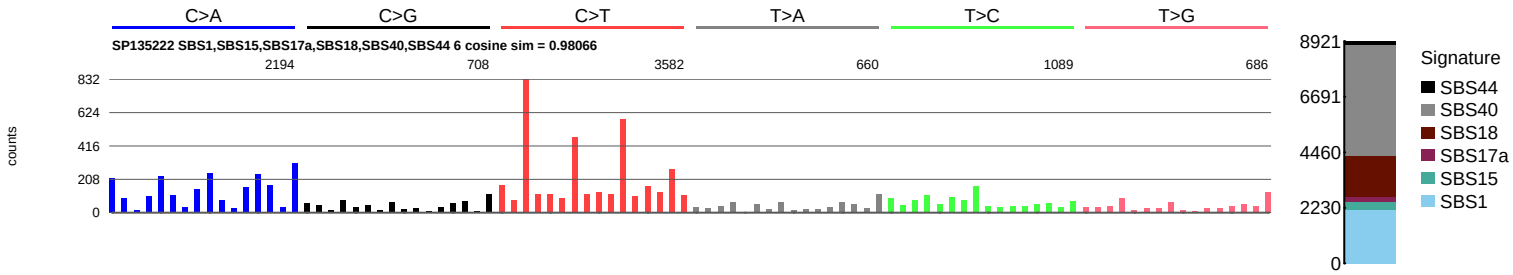
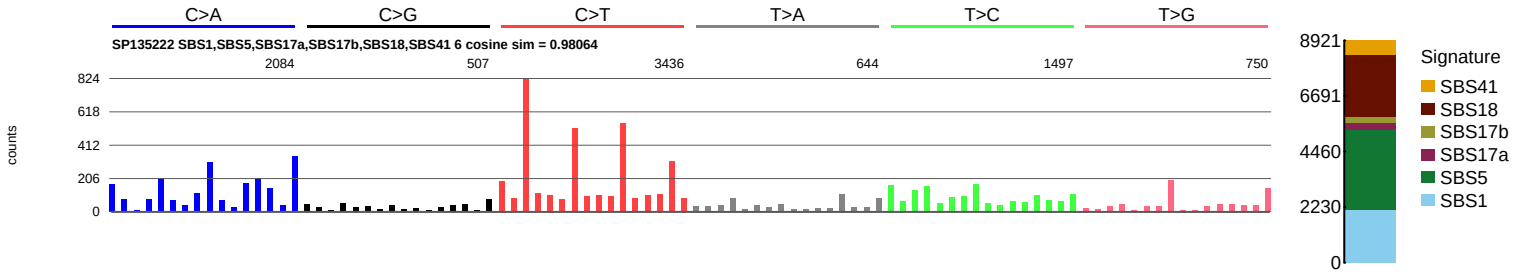
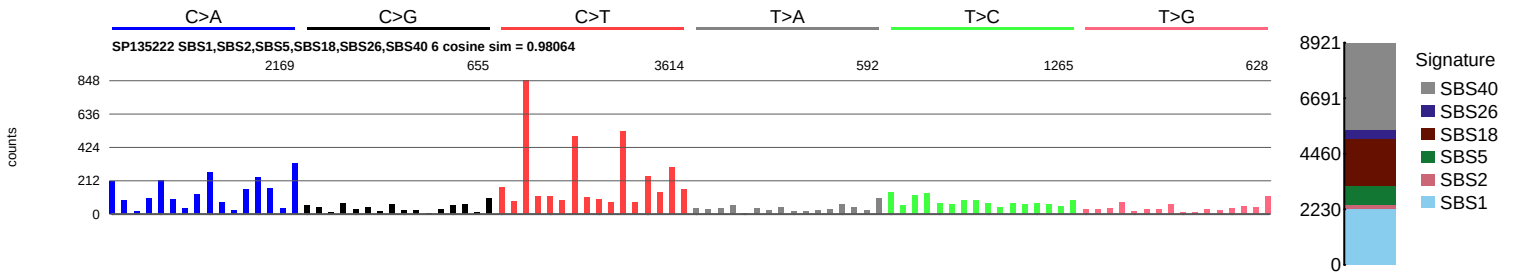
SBS5
SBS2
SBS1

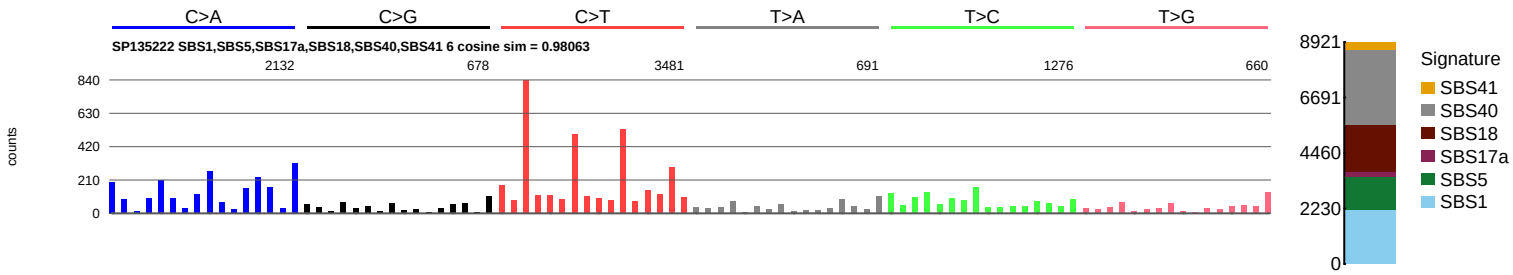

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98073
2085
503
3418
639
1534
743
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

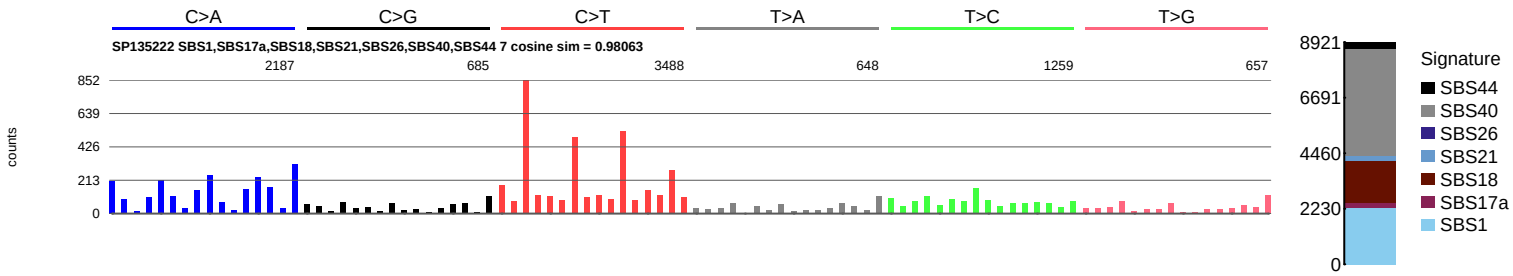

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.9807
2222
670
3576
625
1185
650
counts
852
639
426
213
0
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

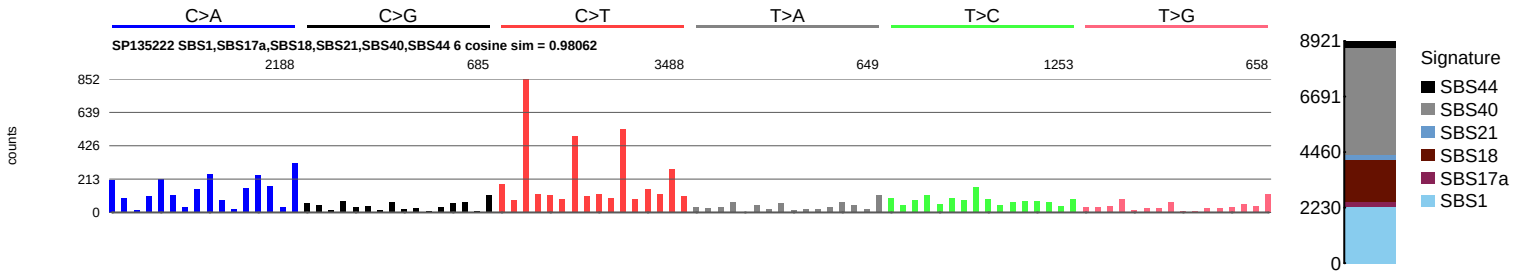

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS18,SBS40,SBS44 6 cosine sim = 0.98066
2194
708
3582
660
1089
686
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS15
SBS1

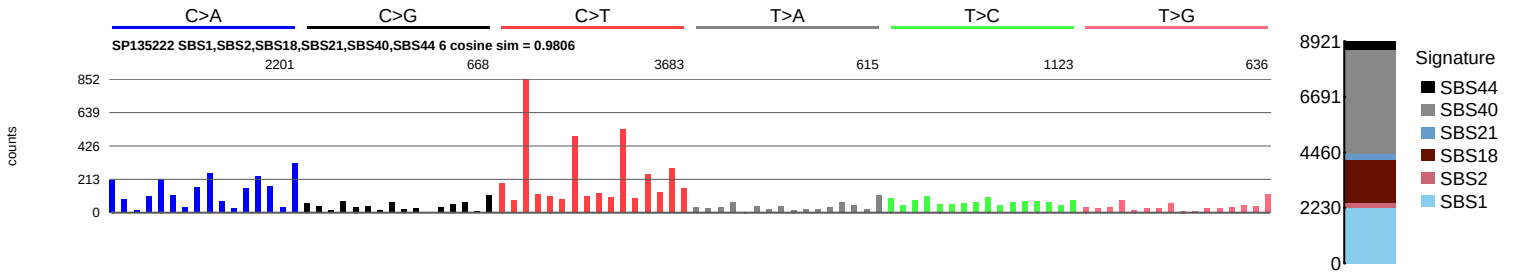

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98064
2084
507
3436
644
1497
750
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1

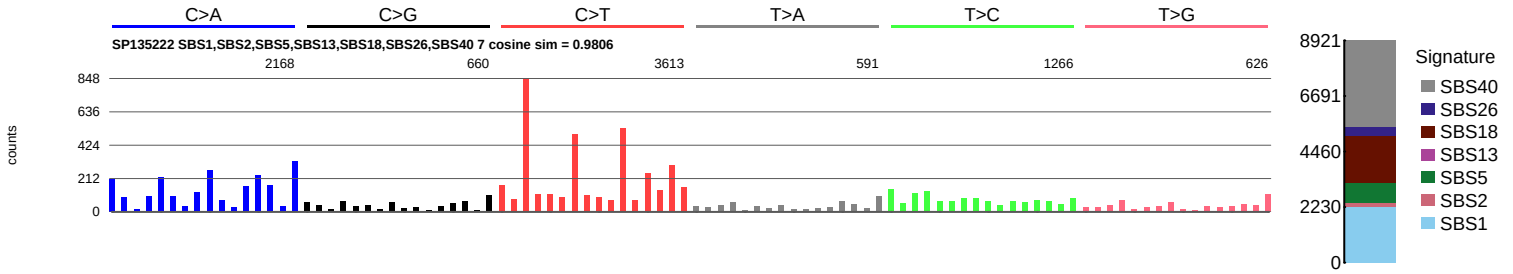

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS18,SBS26,SBS40 6 cosine sim = 0.98064
2169
655
3614
592
1265
628
counts
848
636
424
212
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS5
SBS2
SBS1

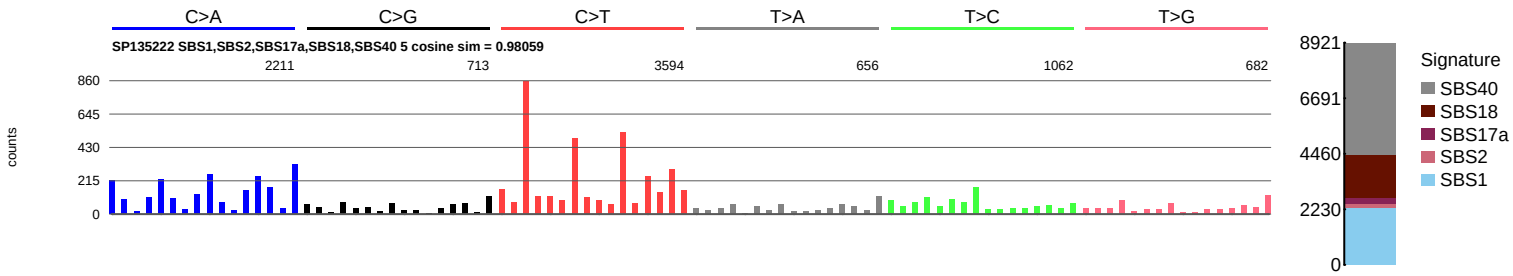

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS18,SBS40,SBS41 6 cosine sim = 0.98063
2132
678
3481
691
1276
660
counts
840
630
420
210
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS17a
SBS5
SBS1

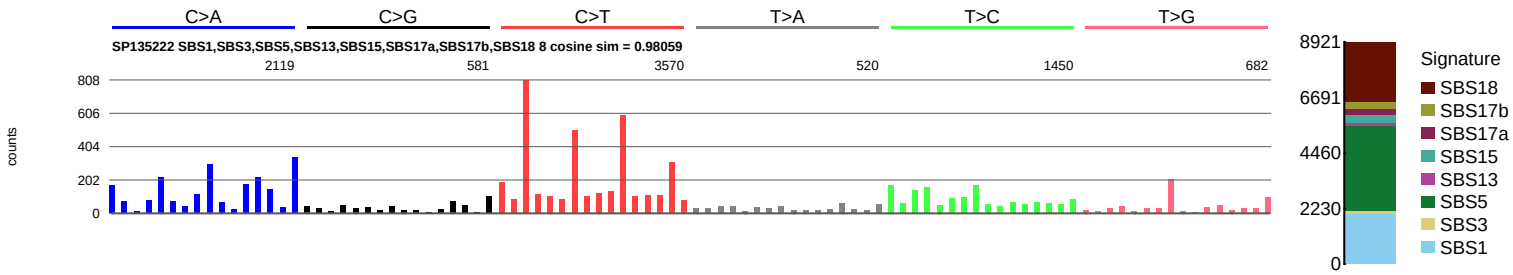

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.98063
2187
685
3488
648
1259
657
counts
852
639
426
213
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS1

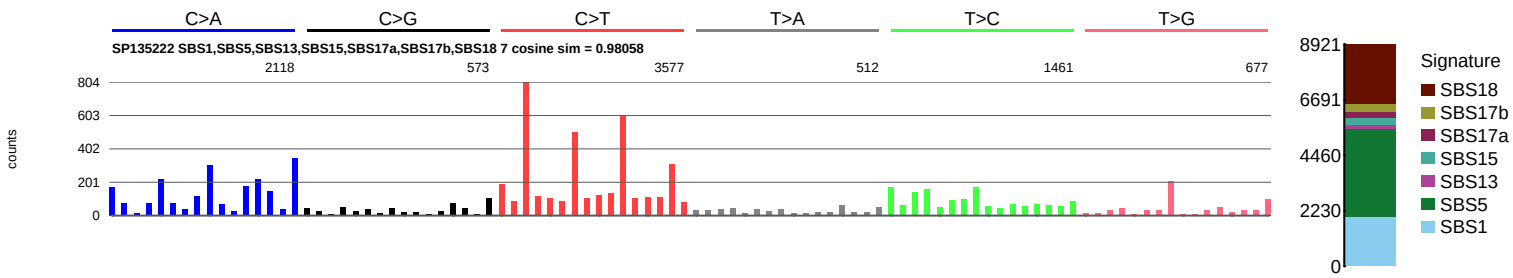

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS18,SBS21,SBS40,SBS44 6 cosine sim = 0.98062
2188
685
3488
649
1253
658
counts
852
639
426
213
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17a
SBS1

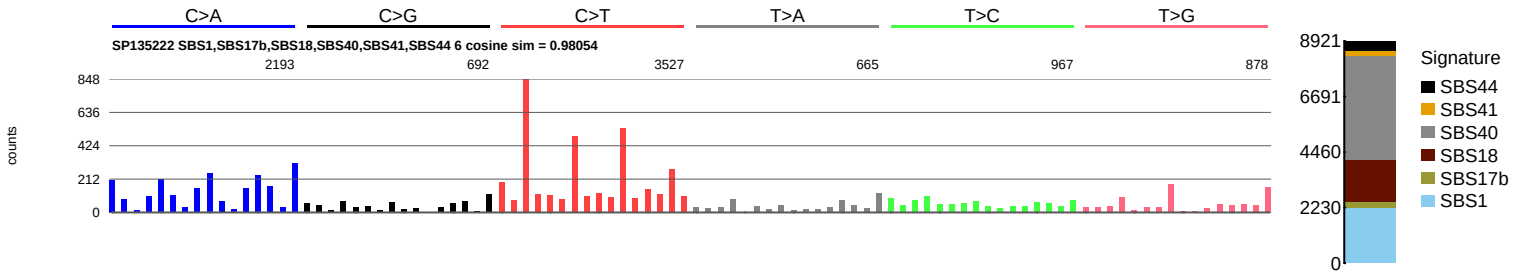

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS18,SBS21,SBS40,SBS44 6 cosine sim = 0.9806
2201
668
3683
615
1123
636
counts
852
639
426
213
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS2
SBS1

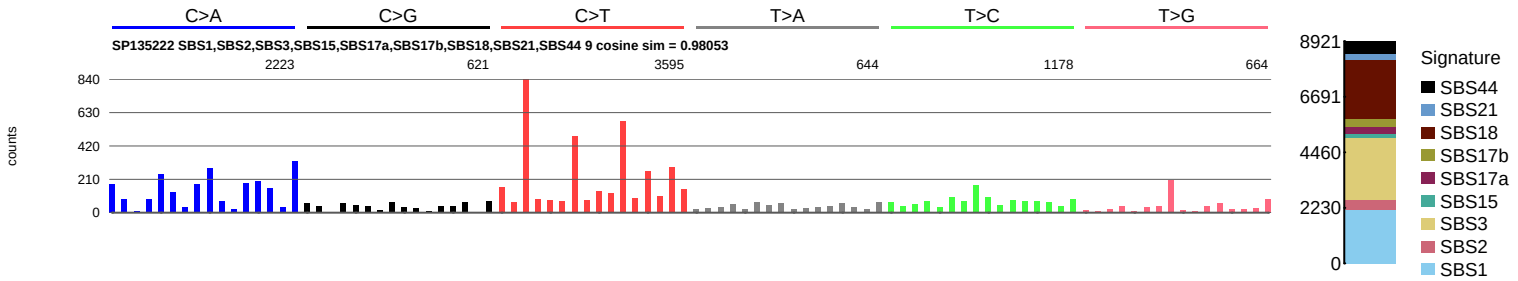

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS26,SBS40 7 cosine sim = 0.9806
2168
660
3613
591
1266
626
counts
848
636
424
212
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS13
SBS5
SBS2
SBS1

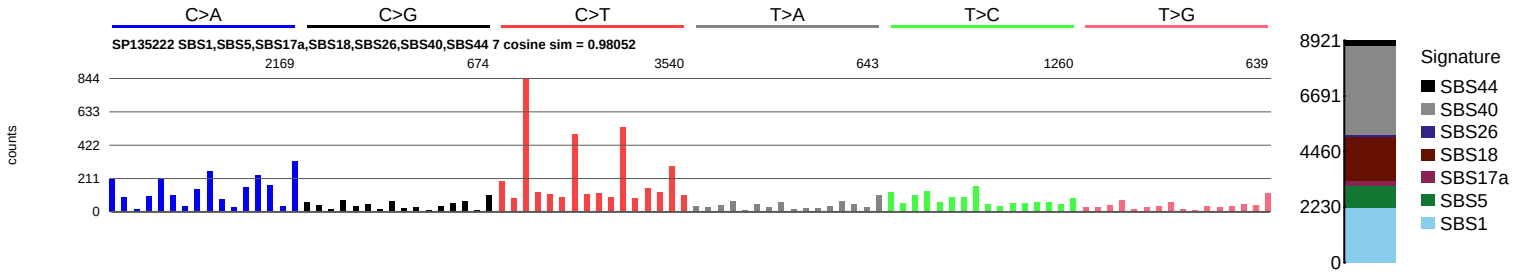

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS17a,SBS18,SBS40 5 cosine sim = 0.98059
2211
713
3594
656
1062
682
counts
860
645
430
215
0
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17a
SBS2
SBS1

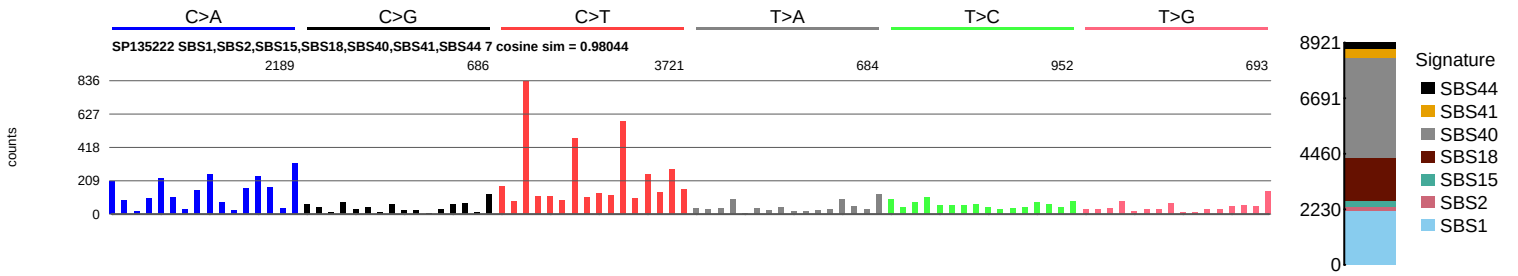

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98059
2119
581
3570
520
1450
682
counts
808
606
404
202
0
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

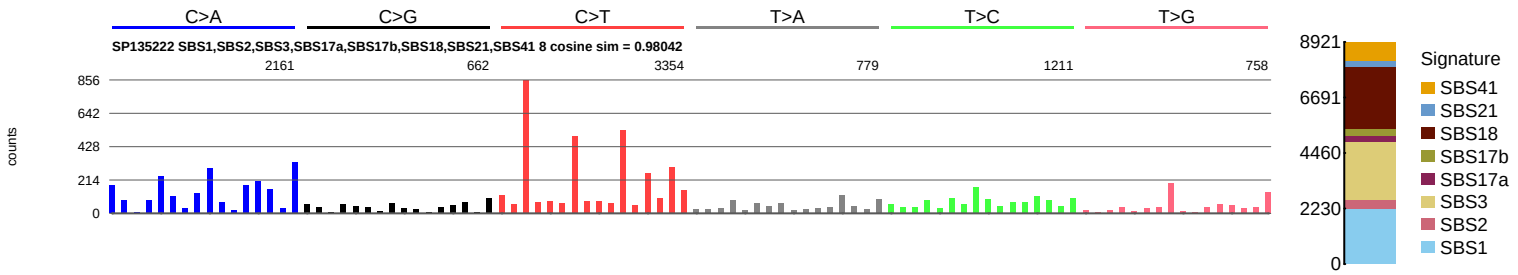

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98058
2118
573
3577
512
1461
677
counts
804
603
402
201
0
8921
6691
4460
2230
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

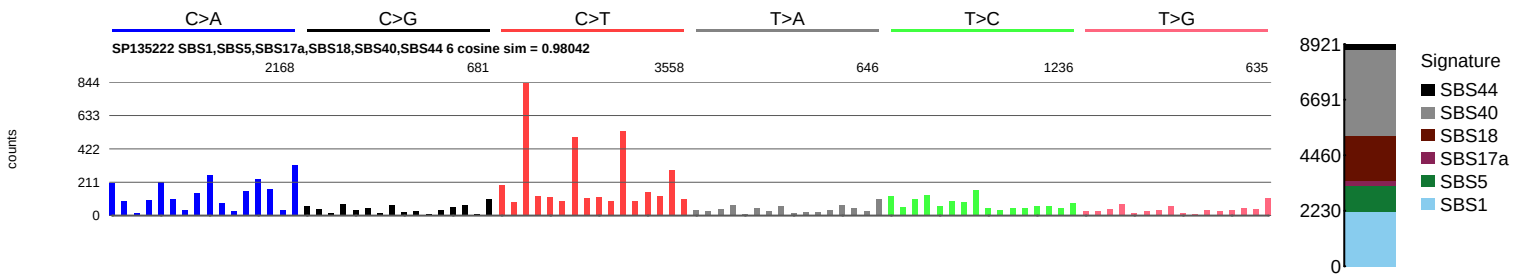

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17b,SBS18,SBS40,SBS41,SBS44 6 cosine sim = 0.98054
2193
692
3527
665
967
878
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS1

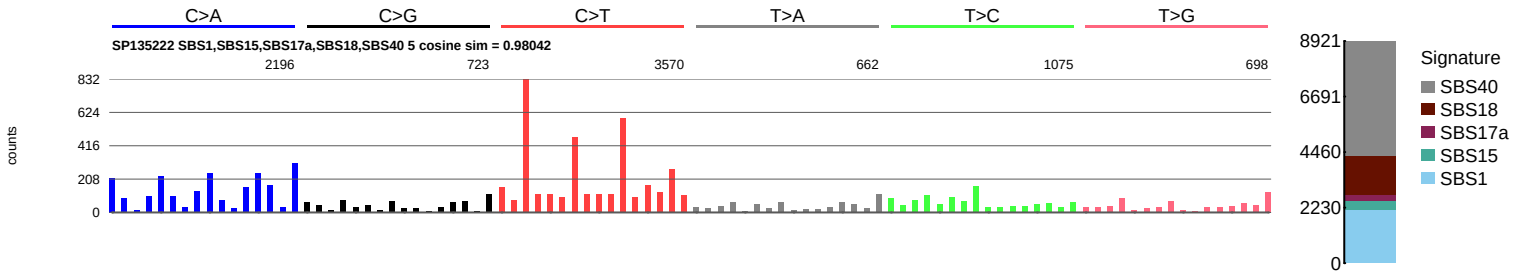

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98053
2223
621
3595
644
1178
664
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

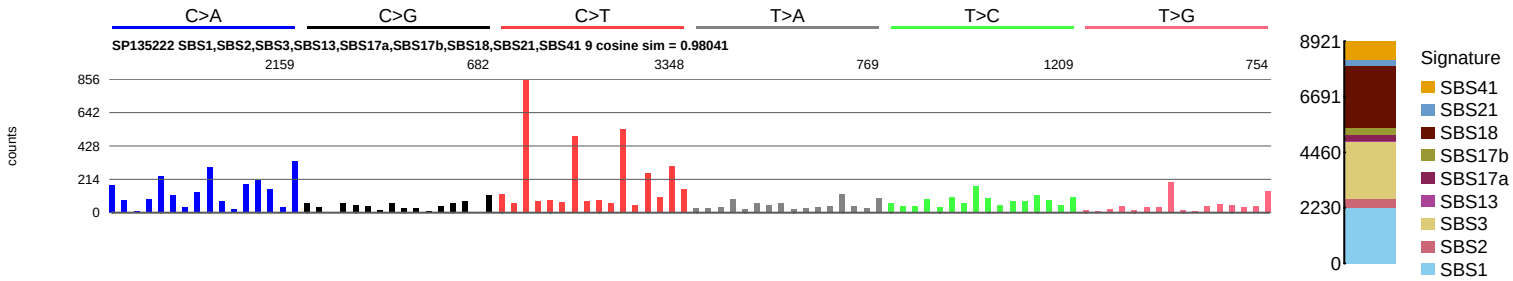

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98052
2169
674
3540
643
1260
639
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS5
SBS1

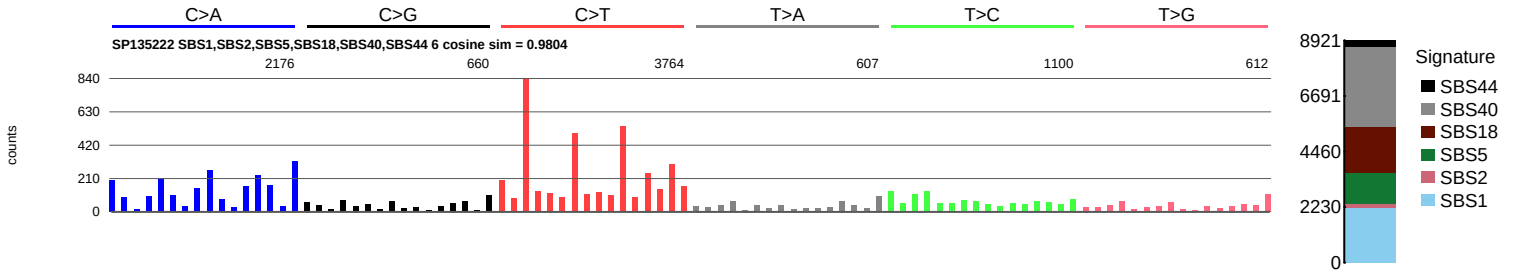

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98044
2189
686
3721
684
952
693
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS15
SBS2
SBS1

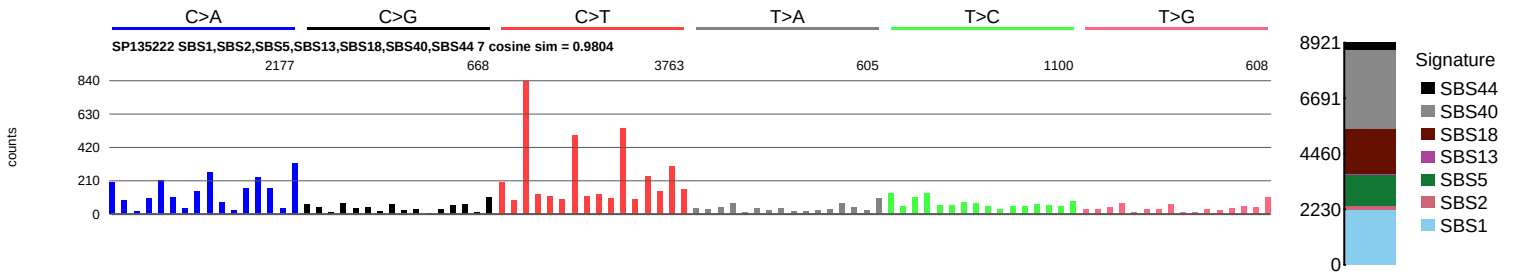

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98042
2161
662
3354
779
1211
758
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

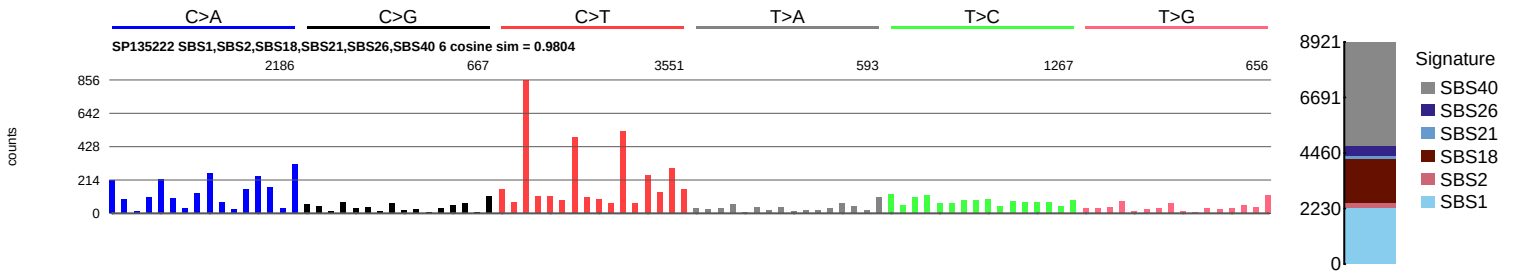

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS18,SBS40,SBS44 6 cosine sim = 0.98042
2168
681
3558
646
1236
635
844
633
422
211
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17a
SBS5
SBS1

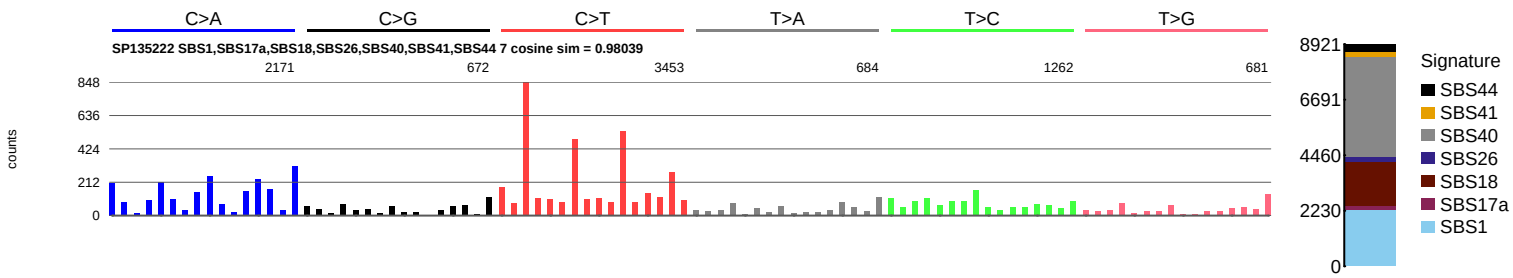

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS17a,SBS18,SBS40 5 cosine sim = 0.98042
2196
723
3570
662
1075
698
832
624
416
208
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17a
SBS15
SBS1

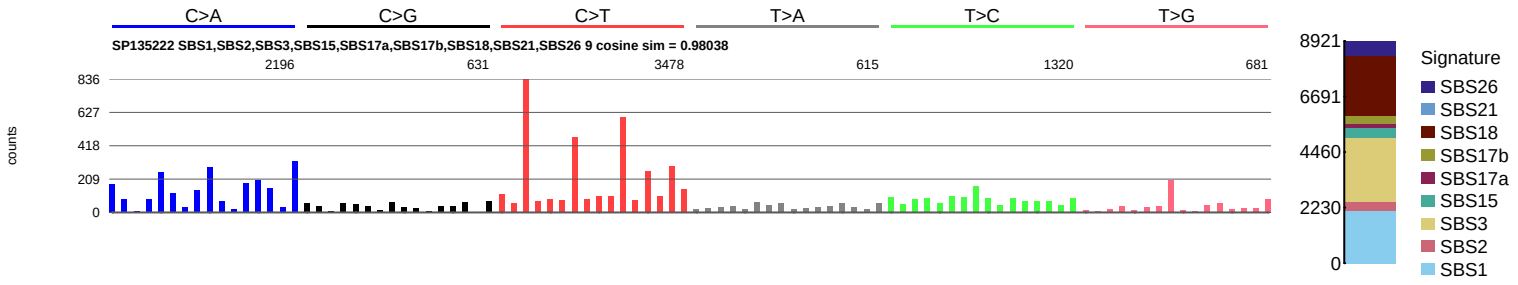

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98041
2159
682
3348
769
1209
754
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

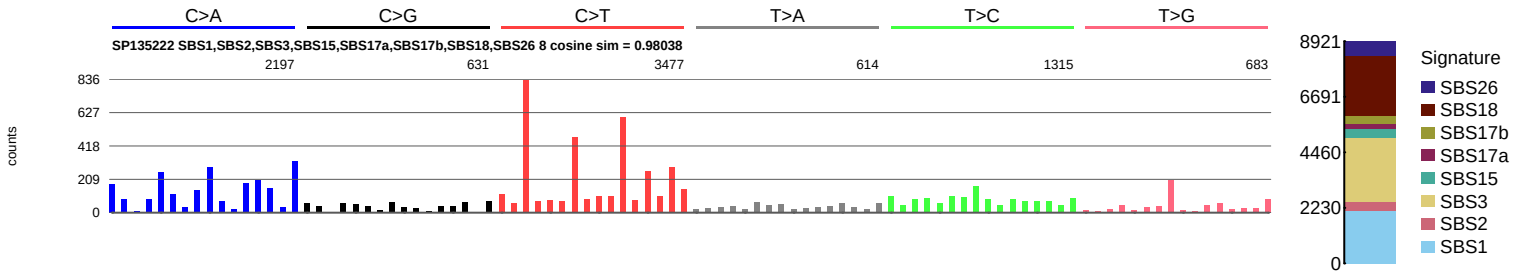

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS18,SBS40,SBS44 6 cosine sim = 0.9804
2176
660
3764
607
1100
612
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS5
SBS2
SBS1

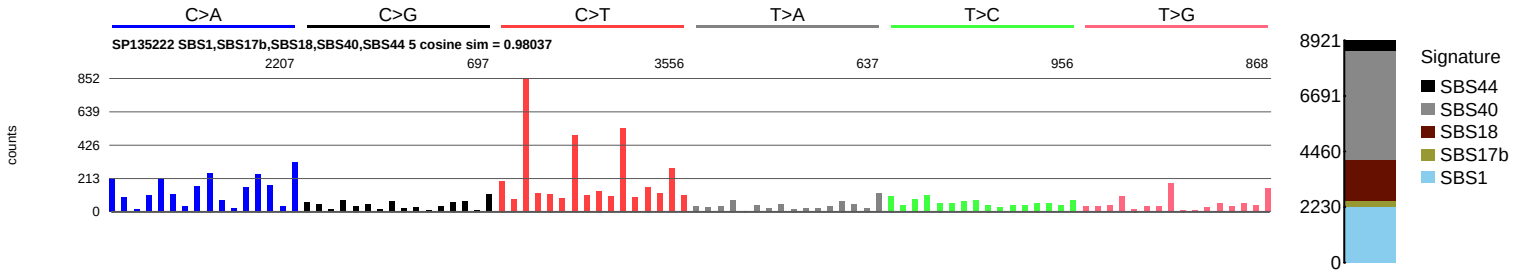

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS5,SBS13,SBS18,SBS40,SBS44 7 cosine sim = 0.9804
2177
668
3763
605
1100
608
840
630
420
210
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS13
SBS5
SBS2
SBS1

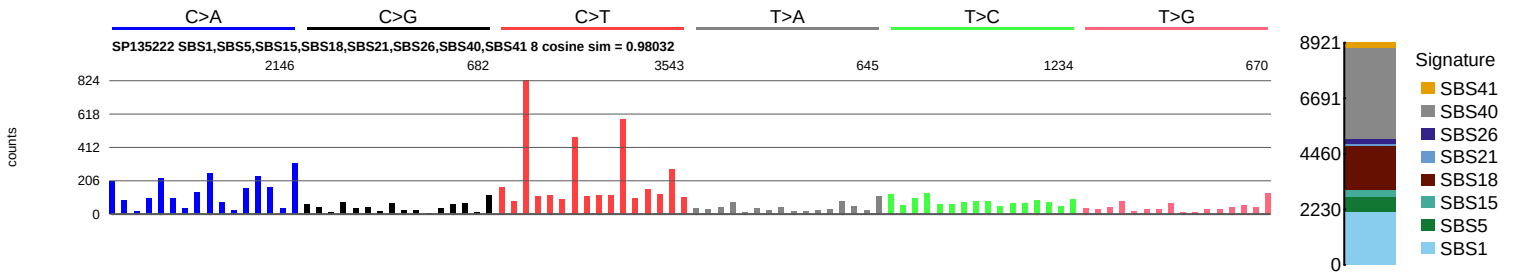

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS18,SBS21,SBS26,SBS40 6 cosine sim = 0.9804
2186
667
3551
593
1267
656
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS2
SBS1

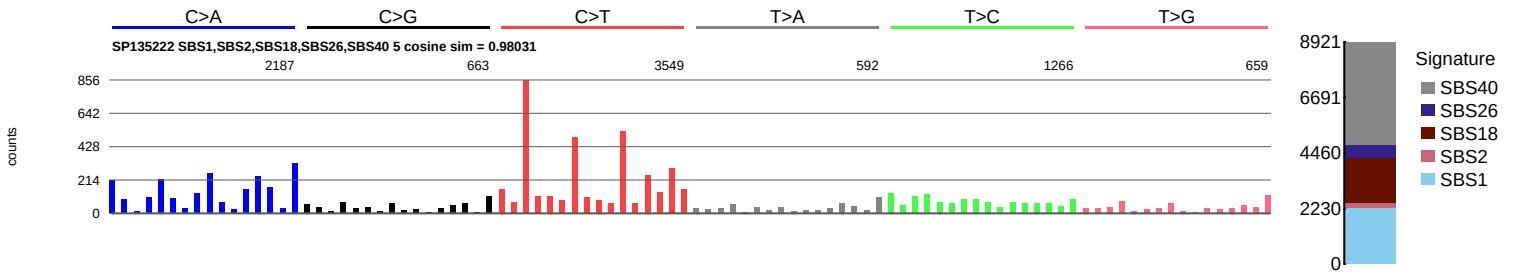

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS18,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98039
2171
672
3453
684
1262
681
848
636
424
212
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17a
SBS1

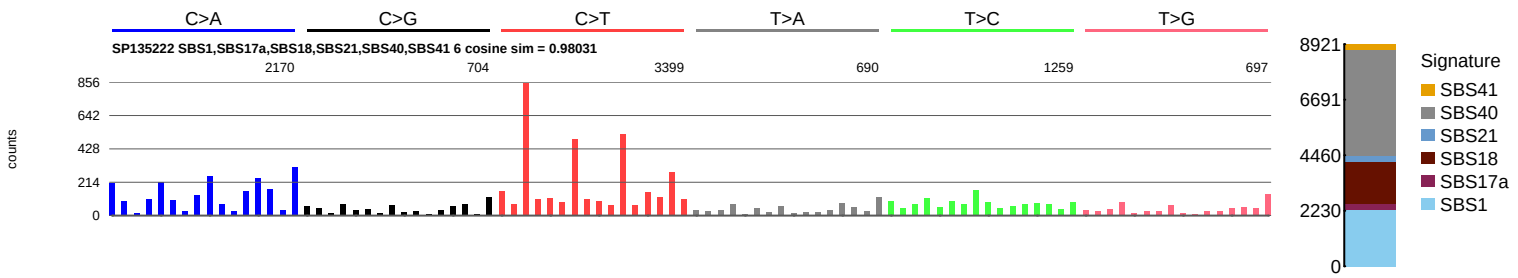

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98038
2196
631
3478
615
1320
681
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

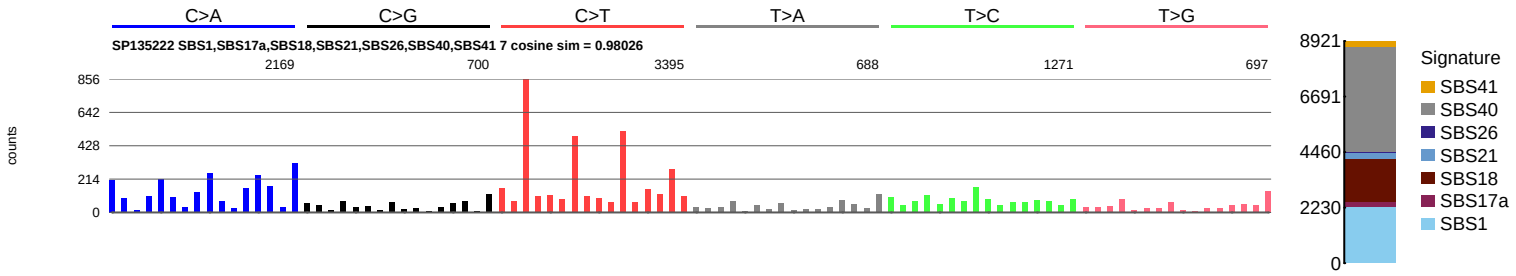

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98038
2197
631
3477
614
1315
683
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

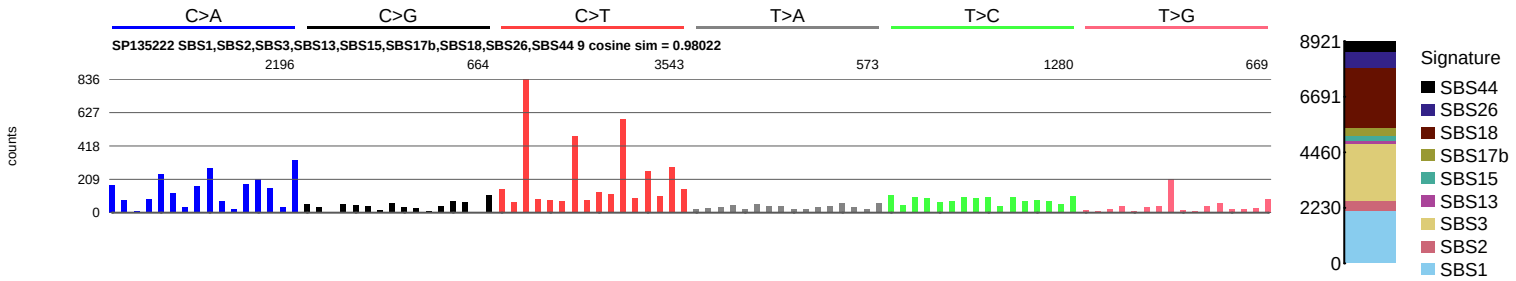

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17b,SBS18,SBS40,SBS44 5 cosine sim = 0.98037
2207
697
3556
637
956
868
counts
852
639
426
213
0
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS1

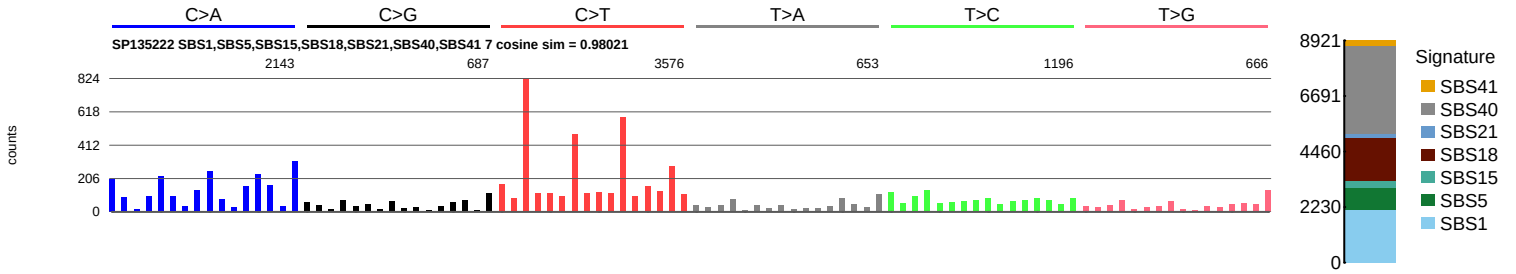

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.98032
2146
682
3543
645
1234
670
counts
824
618
412
206
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS5
SBS1

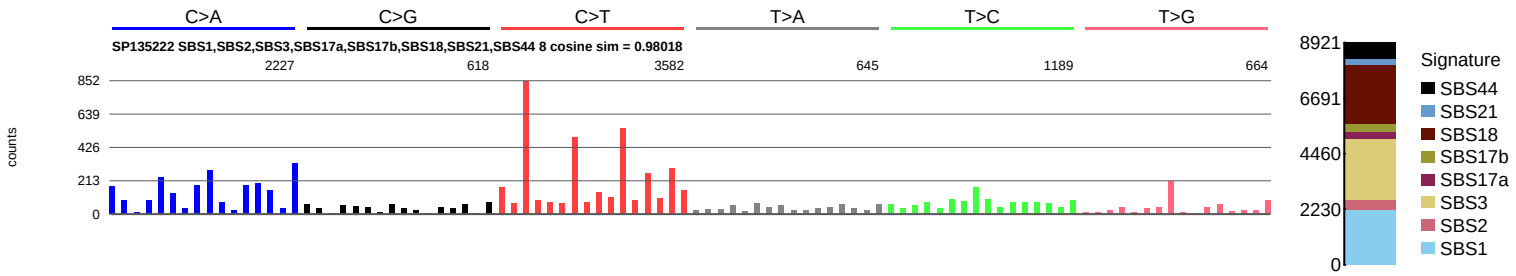

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS18,SBS26,SBS40 5 cosine sim = 0.98031
2187
663
3549
592
1266
659
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS2
SBS1

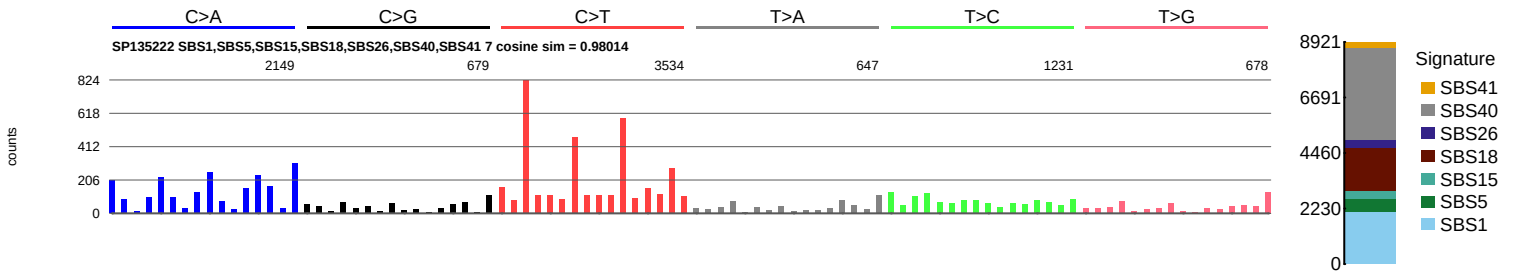

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS18,SBS21,SBS40,SBS41 6 cosine sim = 0.98031
2170
704
3399
690
1259
697
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17a
SBS1

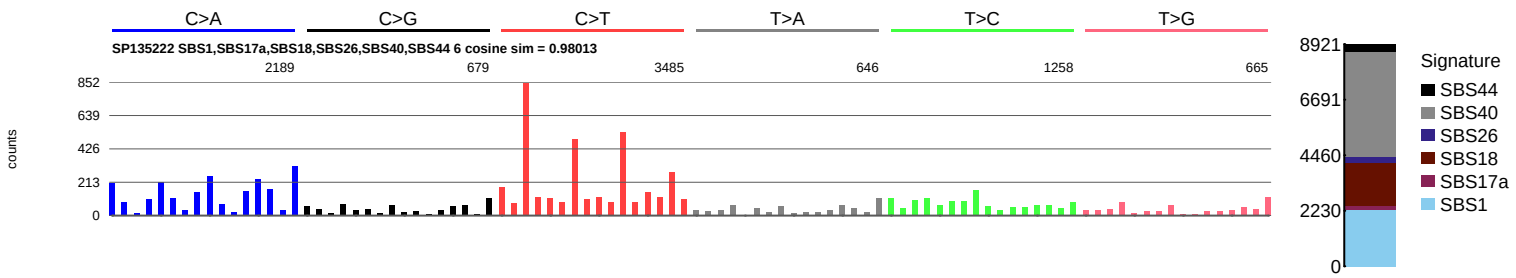

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS18,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.98026
2169
700
3395
688
1271
697
856
642
428
214
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17a
SBS1

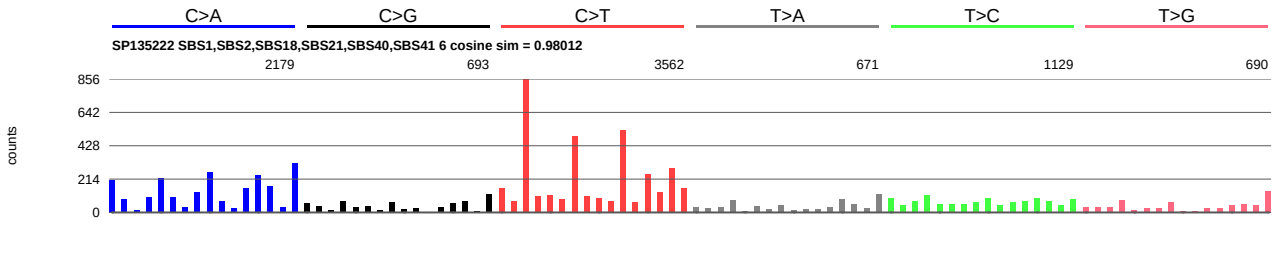

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98022
2196
664
3543
573
1280
669
836
627
418
209
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1

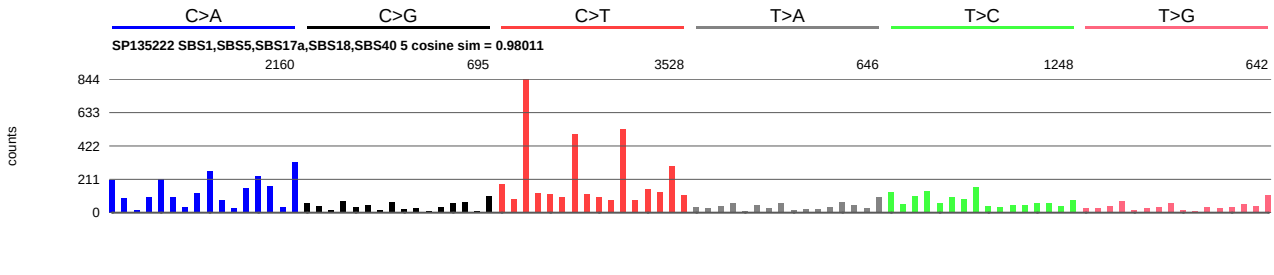

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98021
2143
687
3576
653
1196
666
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS15
SBS5
SBS1

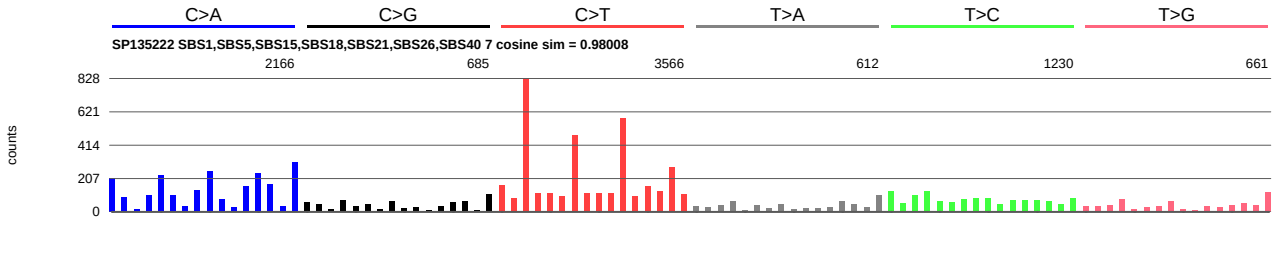

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98018
2227
618
3582
645
1189
664
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

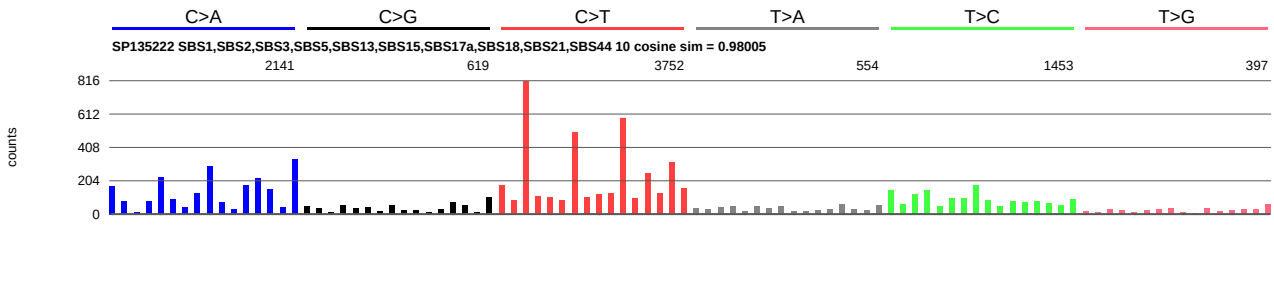

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98014
2149
679
3534
647
1231
678
824
618
412
206
0
counts
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS15
SBS5
SBS1

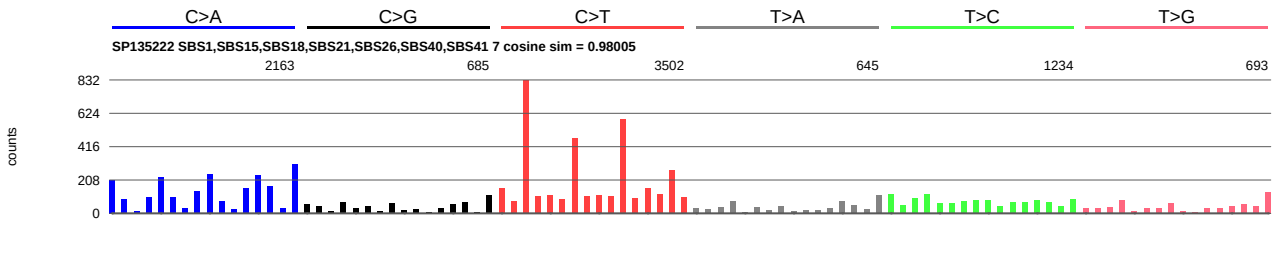

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS17a,SBS18,SBS26,SBS40,SBS44 6 cosine sim = 0.98013
2189
679
3485
646
1258
665
852
639
426
213
0
counts
8921
6691
4460
2230
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17a
SBS1

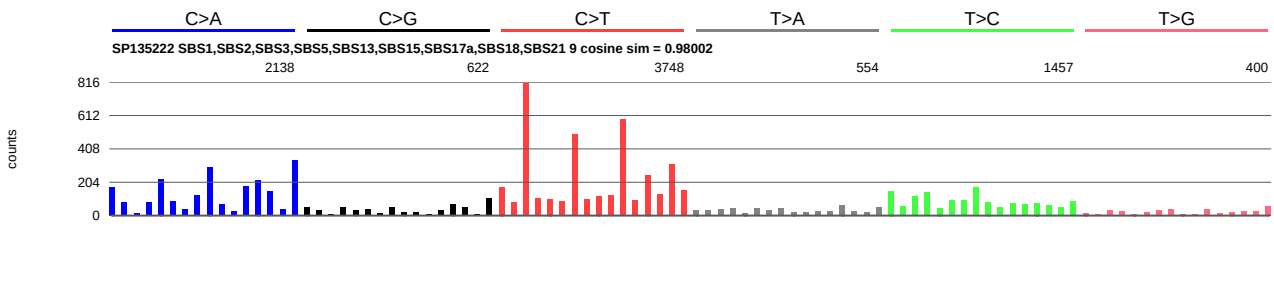

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS18,SBS21,SBS40,SBS41 6 cosine sim = 0.98012
2179
693
3562
671
1129
690
counts
856
642
428
214
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS2
SBS1

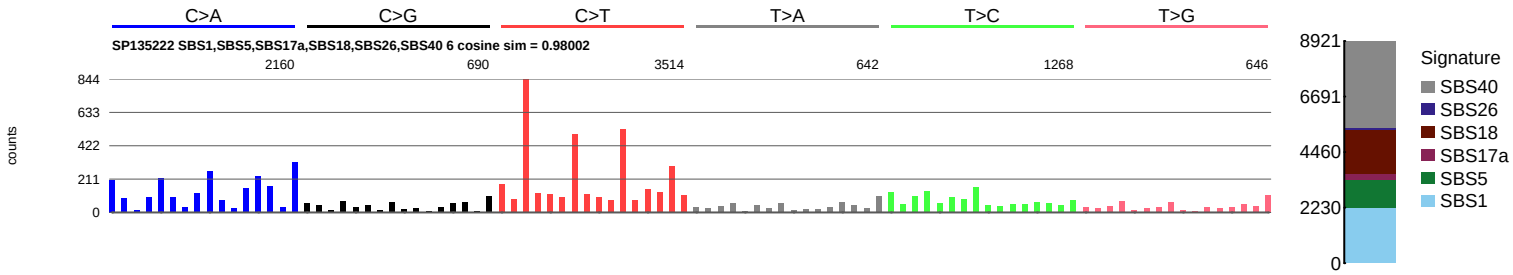

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS18,SBS40 5 cosine sim = 0.98011
2160
695
3528
646
1248
642
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS40
SBS18
SBS17a
SBS5
SBS1

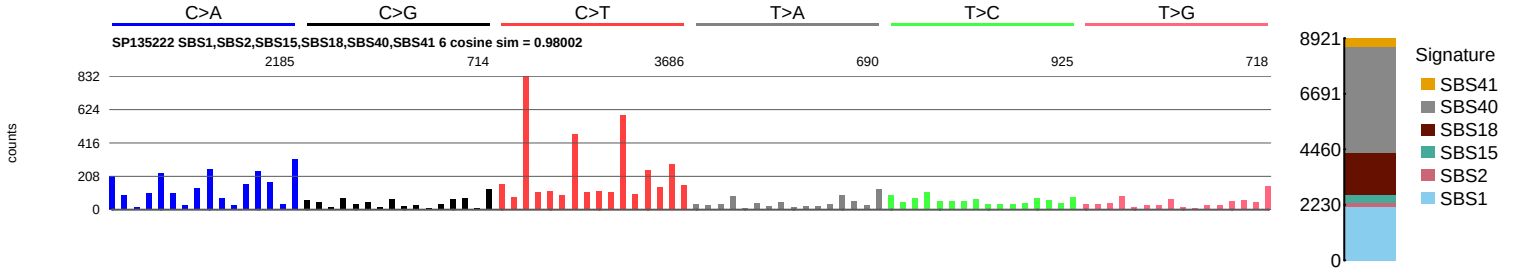

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS15,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98008
2166
685
3566
612
1230
661
counts
828
621
414
207
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS15
SBS5
SBS1

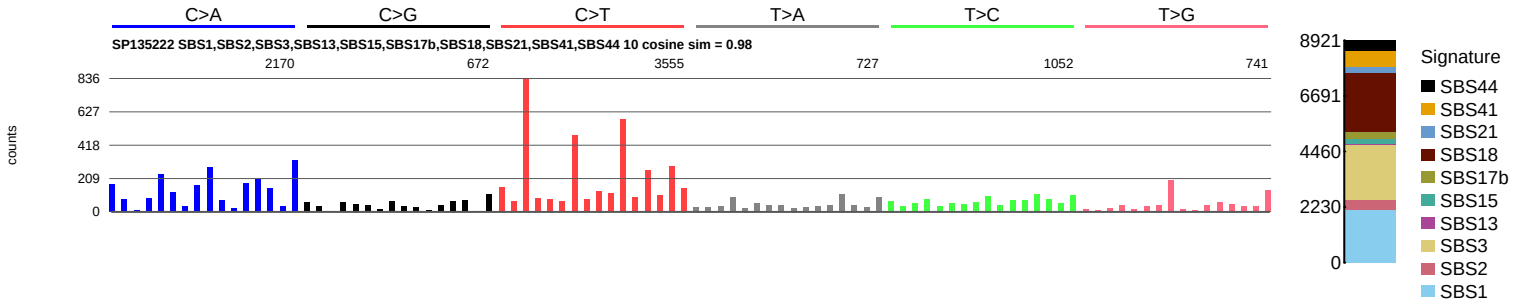

C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21,SBS44 10 cosine sim = 0.98005
2141
619
3752
554
1453
397
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS44
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS15,SBS18,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.98005
2163
685
3502
645
1234
693
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS18,SBS21 9 cosine sim = 0.98002
2138
622
3748
554
1457
400
counts
816
612
408
204
0
8921
6691
4460
2230
0
Signature
SBS21
SBS18
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS5,SBS17a,SBS18,SBS26,SBS40 6 cosine sim = 0.98002
2160
690
3514
642
1268
646
counts
844
633
422
211
0
8921
6691
4460
2230
0
Signature
SBS40
SBS26
SBS18
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS15,SBS18,SBS40,SBS41 6 cosine sim = 0.98002
2185
714
3686
690
925
718
counts
832
624
416
208
0
8921
6691
4460
2230
0
Signature
SBS41
SBS40
SBS18
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135222 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98
2170
672
3555
727
1052
741
counts
836
627
418
209
0
8921
6691
4460
2230
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS15
SBS13
SBS3
SBS2
SBS1